

GOLIATH'S CURSE

THE HISTORY
AND FUTURE
OF SOCIETAL
COLLAPSE

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For my brother David

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Introduction: A People's History of Collapse

Unravelling History Through Collapse – Goliath's Curse – Our Fragile Future – A Blessing in Disguise – A 1 per cent View of History – Is History Relevant for the Future? – A Historical Autopsy

UNRAVELLING HISTORY THROUGH COLLAPSE

Human history is the story of a power struggle. Power comes in four different forms: control of decision-making, including by setting up a centralized government; control of the resources that others depend on, such as the wheat and rice that most people eat every day; control through threats and violence; and control of information to understand and manipulate others, whether by a bureaucracy, a priesthood, or a big tech company. Then there is the great amplifier: power depends on who you are controlling – the size, skills, and health of the population.¹

Each of these power structures can fall apart. When a government – whether in ancient Rome or modern-day Somalia – falls apart, we call it a state collapse or ‘state failure’. When the economy breaks down, as with the Great Depression in the 1930s, we call it an economic collapse. When a large number of people die, we call it a ‘bust’ or population collapse. When all these power systems crumble together in a relatively rapid and enduring manner we call it a societal collapse.² Societal collapse has shaped our history and will determine our future.

Around 500 kilometres south-west of modern-day Chicago, a city of flat-topped earthen pyramids rises above the Mississippi River. At one stage

there were almost 200 of these mounds covering around thirteen square kilometres. The largest pyramid – Monk's Mound – stands at almost thirty metres, making it taller than the White House. These flat-topped pyramids were made by hand. Villagers would dig up the rich soil, transport it with baskets, and layer it carefully to form miniature mountains. They surrounded a grand plaza 490 metres long and 265 metres wide. The layout was eerily similar to those found over 2,000 kilometres south-west, in the pyramid cities of Mesoamerica.³

This is Cahokia, the first true city of the US ([Figure 1](#)).⁴ Around 1,000 years ago (1050–1100 CE), the settlement grew into a city of 10,000–15,000 people.⁵ Just a century or so earlier in 900 CE corn had been introduced to the area. The boom in agriculture, construction, and population was accompanied by more disturbing practices. Within the abandoned pyramids lie the remains of hundreds of men and women. One of the mounds (Mound 72) contains 272 burials, many of which were ritual sacrifices. They were killed and then tossed into graves as part of an elite funeral.

One burial site alone has the skeletons of fifty-three women, all of which are of reproducing age: fifty-two are fifteen to twenty-five years old and one was around thirty-five. The women may have been virgins, or a harem of slaves. Another pit contains four men with their heads and hands cut off. Other victims kept their hands but suffered slower deaths, with their fingerbones left digging into the sand as they tried to claw their way out. At the top of the burial are two men who were buried lavishly, with one sporting a 5cm-thick blanket of 20,000 shell beads. Cahokia had become unequal and centralized, with a priestly ruling class who used human sacrifice to legitimize their reign. It was on course to become the first state of North America centuries before Europeans would arrive.⁶

A few decades later, the first signs of instability emerged. A farming complex supplying the city with grain was abandoned, a palisade was erected around the central settlement, and people began to exit the city. Within fifty years its population had halved. Just over a century after that, the entire site, and the area extending hundreds of kilometres to the south-east of Cahokia – once full of similar pyramid-centred towns – had been abandoned. It became known as 'the Vacant Quarter', a vast tract of swampy land that was avoided by the indigenous people. Remarkably, the Native Americans of the area have no stories or oral traditions about

Cahokia. It was as if they had intentionally forgotten it. This pioneering civilization-building project was over and would never be resurrected.⁷



[Figure 1](#): An artistic depiction of Cahokia around 1050 CE

Over 8,000 kilometres away, across the Atlantic, lies the city of Rome. It was once the beating heart of Europe's largest, most enduring empire. While relatively few know of or visit the ruins of Cahokia, millions flock to see the remains of Rome: the stone and marble skeletons of the Forum and the Colosseum. Rome collapsed, the western half of the empire fragmented, and the population of the city shrank from around 1 million people to 30,000. Unlike Cahokia, Rome was never entirely abandoned. It remains a thriving metropolis of 2.8 million people. And unlike Cahokia, the legacy of Rome would be kept alive by dozens of later kingdoms and empires, including the British. Cahokia's collapse was terminal, while the city of Rome, its culture and institutions, and the eastern half of its empire headquartered in Constantinople (now modern-day Istanbul) lived on.

Both Rome and Cahokia collapsed. How they fell and the legacy they left varies dramatically. Rome was revered, while Cahokia was left as a bad memory to rot in the swamps of the Mississippi.

To understand the similarities and differences of Rome and Cahokia we need to unravel some of the biggest questions of history. Why did people build these states and civilizations? Why did some of these first attempts to

build great power structures fall apart? Why was Rome replaced by dozens of imitators while Cahokia's collapse was seemingly permanent? What did these systems of power mean for the health and wealth of the people living in them? What do these past collapses mean for us in the modern world? Once you pull on the thread of collapse, the entire tapestry of history begins to unravel.

GOLIATH'S CURSE

A state is a set of centralized institutions that imposes rules on and extracts resources from a population in a territory, whether that be ancient Egypt or the US today.⁸ All states throughout history have eventually ended. The average lifespan of a state is 326 years. The largest states – mega-empires covering over a million square kilometres – are more fragile, lasting on average just 155 years. But these averages obscure a staggering range, from the fleeting four-year reign of the Later Han Dynasty in China to the Byzantine Empire of the Mediterranean, which lasted over 1,000 years. While all states have ended, most did not end in societal collapse. Societal collapse is a rarer phenomenon. A phenomenon that afflicts not only states but also broader power structures. Structures that many would call 'civilization'.⁹

When most people speak of civilization they are referring to a few basic ingredients: hierarchy (government, bureaucracy), the capture and use of great amounts of energy (agriculture and monuments), and dense populations living in close quarters (cities and urbanism). Collapse involves a rapid and enduring loss of each of these ingredients of civilization.

Yet civilization seems to be a wholly inappropriate term for what we see in Cahokia and Rome. It is derived from the Latin *civilitas*, which implies restraint, moderation, and good political conduct. Slavery, war, patriarchy, sacrificing teenage girls as the priests of Cahokia did, punishing slaves with crucifixion as in Rome, or dropping atomic bombs on cities as the US did less than a century ago, are not what most of us would consider civilized conduct. Orchestrating mass sacrifices or building bombs that vaporize cities does not require altruism, civility, or democracy. It requires top-down obedience enforced through the threat of violence.

While the term ‘civilization’ is still commonly used, scholars have always struggled to agree on a sensible definition. Some suggestions include an advanced culture (an idea as vague as it is prejudiced against indigenous peoples), or a random checklist of factors such as writing (which neither Cahokia nor the Incan Empire had) and long-distance trade (which essentially all hunter-gatherers conducted). The problem is that most of us are uncomfortable in recognizing the most common element of civilization: rule through domination.

A more apt label for these systems of violence is ‘Goliath’. A Goliath is a collection of hierarchies in which some individuals dominate others to control energy and labour. Goliath refers to a figure from the Old Testament (1 Samuel 17), a towering Bronze Age warrior who tormented the Israelites. Despite his size he was eventually brought down by David, the future King of Israel, who slung one smooth stone directly between his eyes. Similarly, Goliath societies began in the Bronze Age, are imposing in size, rely on violence, and can often be surprisingly fragile.

It was Goliath, not civilizations, that we saw arise in places such as Egypt, China, and Cahokia thousands of years ago. ‘Goliath’ is a far more useful term since it pinpoints what exactly changed in the lands around the Nile, the Yellow River Valley, and the Mississippi. It was a move towards human societies organizing, like many of our chimp cousins do, as a dominance hierarchy. That is, a social-ranking system in which one group or individual is placed above others owing to their ability to impose penalties, including violence. It was the emergence of Goliath.

A Goliath is not just a state. While the Roman Empire was the political heart of the Roman Goliath, there were other hierarchies that preceded it: rich and poor, master and slave, man and woman. A Goliath is a collection of dominance hierarchies organized primarily through authority and violence.

We can use the term ‘Goliath’ in a similar way to civilization. We use civilization to refer both to a general way of organizing society and to specific societies. We speak of ‘the rise of civilization’, or of the ‘global civilization’ of today, yet we also refer to specific civilizations, such as China and Egypt. I’ll be using Goliath in a similar manner. Goliath can mean a general template of organizing society into dominance hierarchies that control people and energy. It can also mean a specific set of dominance hierarchies that often coincide with an empire, such as Rome or China.

Goliaths contain the seeds of their own demise: they are cursed. This is why they have collapsed repeatedly throughout history.

Sometimes it is not just a single Goliath that is toppled, but an entire group of them. Approximately three millennia ago, the Mediterranean and Near East were ruled by a network of Bronze Age empires and states. This Late Bronze Age world-system (around 1700–1200 BCE) included the Egyptians, the Mycenaeans in ancient Greece, the Minoans in Crete, and the Assyrians and Babylonians of Mesopotamia. In about 1200 BCE this entire system began to unravel.

Around a millennium later, Europe and Asia were dominated by two empires: the Roman Empire in western Europe and the Han Dynasty in China. Each at their peak ruled over some 5–6 million square kilometres of land and 60–70 million people.¹⁰

Collapses can be abrupt and unforeseen. The Han faced a small popular uprising that snowballed into the capture of the capital and the end of their dynasty within five years. It took under a century for Rome to go from rule by a single emperor to being a fractured chessboard of competing kingdoms. The fall of the Bronze Age system also took under a hundred years to unfold. Even the most entrenched power structures can be torn apart quickly and unpredictably.

As far as we can tell, no one living at the time anticipated the demise of these empires. Apart from individual disasters, famines, and sacked cities, they likely did not know they were living through a collapse. While a historian may see collapse as a graph of falling population or territory, the first-hand experience is far different. People living through collapse tasted the fear on their tongue while fleeing a city, they felt the heady rush of anger while raiding a granary, and they heard the screams of conflict coming closer. But we can see clearly only when looking back. Collapse may be invisible until after it has occurred. We may be today living through a collapse that is for now slow and imperceptible.

Gazing up at the faded marble columns of Rome or the abandoned granite bones of Machu Picchu often brings a sense of tragic inevitability. A reminder that death comes not just for every individual, but also for every great power structure. Percy Shelley caught that feeling in his sonnet ‘Ozymandias’:

And on the pedestal, these words appear:

‘My name is Ozymandias, King of Kings,
Look on my Works ye Mighty, and despair!’
Nothing beside remains. Round the decay
Of that colossal Wreck, boundless and bare
The lone and level sands stretch far away.

Both Ozymandias and real-world examples, such as ancient Rome, evoke a certain ominous uncertainty. Will we face the same fate? Collapse is the spectre that haunts the victors of history.

Collapses throughout history can offer an insight into how and why different power structures unravel, and whether there are any patterns across the millennia. History is a vast database we can learn from, but it must be done soberly. Bear in mind that Shelley’s Ozymandias was modelled on the Egyptian pharaoh Ramesses II (Ozymandias was the Greek version of his name) and it was not a lament for pharaohs or for lost civilizations. Shelley was a staunch believer in republican rule, and the poem was a reminder about the ephemeral nature of tyrants such as Ramesses II.

Historical collapses are neither uniform nor apocalyptic. Many historians and archaeologists now prefer to speak of ‘transformation’ rather than collapse. While some were quick, others were protracted, taking entire centuries to play out. Cahokia lost half its population in half a century, while the fall of the Western Roman Empire took multiple generations. The effects also varied depending on who and where you were. A woman born in Roman Britain in 360 CE could have grown up in a prosperous market town using currency, speaking Latin, seeing columns of Roman soldiers, and glimpsing the bathhouses of the elite. If she lived to sixty, she would have inhabited a Britain with no large towns, no currency, no Roman soldiers or administrators, abandoned bathhouses and monuments, and surrounded by pagan immigrants who didn’t speak a word of Latin. By contrast, an aristocrat in southern France during the same period would have experienced little change, apart from who they paid their taxes to. To a historian, the fall of Rome may look like an obvious event in global history. To a peasant in Spain, it may have barely been perceptible. Collapse is partly in the eye of the beholder.^{[11](#)}

The people and culture of collapsed states often persist after the fall. The language, customs, and culture of Rome lived on through new Germanic

rulers, and the empire itself persisted for another millennium through the Eastern Roman Empire. The US Senate was inspired by Rome, and the Roman language of Latin is still used in institutions today. The Incan language of Quechua is still spoken by around 10 million people today in Peru and surrounding areas. Much of the Mississippian culture that Cahokia pioneered lived on in future Native American communities for centuries, even if the experiment with cities and rulers was forgotten.¹² Collapse reveals the fragility of empires, but also the resilience of culture and communities.

Collapse doesn't just change across space, but also over time. And the future of collapse looks far grimmer than the past.

OUR FRAGILE FUTURE

We live in a uniquely dangerous time. Our world is scarred by a pandemic, beset by unprecedented global heating, riven by inequality, dizzied by rapid technological change, and living under the shadow of around 10,000 stockpiled nuclear warheads. Since the invention of the atom bomb the world has come frighteningly close to nuclear war at least a dozen times. The climate change we face is an order of magnitude (ten-fold) faster than the heating that triggered the world's greatest mass extinction event, the Great Permian Dying, which wiped away 80–90 per cent of life on earth 252 million years ago. Viruses can now spread at the speed of a jet plane, and computer viruses at the speed of an internet connection.¹³

The better-known and more deeply studied threats of nuclear war and climate change are joined by new, more hypothetical technological terrors. In 2023, hundreds of AI scientists and other luminaries (strangely including the CEOs of the main companies building these new AI systems, such as Sam Altman of OpenAI and Demis Hassabis of Google DeepMind) released a statement warning against the risk of human extinction from AI. They fear that an uncontrolled, ultra-intelligent machine whose interests are not aligned with those of humanity (or at least most of us) will either destroy or enslave us. Other scholars, including biological scientists, have warned of advances in bioengineering that could create doomsday diseases – far more contagious and deadly than anything that has ever existed. And

those are just the present threats. Who knows what new hazards rapid technological change could conjure in the coming decades. The confluence of these different threats has led some to call our current predicament the ‘metacrisis’, ‘the precipice’, or a ‘global polycrisis’.¹⁴

Such apocalyptic angst is not new. Every age has its doomsayers and extinction panics. In the 1920s Winston Churchill warned that humankind was in the unprecedented situation of being able to ‘unfailingly accomplish its own extermination’ in the morbidly titled essay ‘Shall We All Commit Suicide?’ The Czech play *R.U.R.* (which coined the term ‘robot’, derived from the Slavonic *robota*, meaning servitude or forced labour) told the story of an uprising of intelligent machines that wiped out their human masters. This occurred in the 1920s, a time still reeling from a global pandemic, the First World War, and rapid industrialization. This doesn’t mean that either current concerns, or those from a century ago, are invalid. It means that we need a reliable way to put the current threats we face into perspective.¹⁵

The history of collapse is the most relevant historical phenomenon we can learn from. The present is an extension of the past. The past can tell us how catastrophes have unfolded, how people have proved resilient to them, and why we find ourselves in our own acutely dangerous predicament. Looking at collapse also forces us to confront how societies can destroy each other, as well as produce threats that undo themselves. Collapse isn’t just about how great power structures die, but also about how they kill.¹⁶

History can also help us understand the origins of the risk we are facing today. Studies of current global risk tend to become a checklist of different threats: nuclear weapons, climate change, pandemics, and AI. Focusing on each in isolation and looking for technical solutions – such as building benevolent, controllable AI systems – is useful yet short-termist. In 1950 there was only one well-known global risk: nuclear weapons. Diplomatic efforts in the 1980s to reduce nuclear stockpiles by around 80 per cent were welcome, yet they haven’t stopped the world from spiralling into new arms races with algorithms or killer drones, or stockpiles mushrooming again. If we don’t find a way to address the underlying drivers of arms races, then we are doomed in the long term. We need to reverse the root causes of global risk, not just build new technology to cope with the symptoms.

Competing companies and countries didn’t just appear overnight; they have a history that can be traced back to human psychology. Fundamental

psychological factors and the environmental changes that allowed them to shape the world are the root causes.¹⁷ Excavating these root causes can help us determine what we can do next to safely navigate the coming decades and centuries. This excavation of root causes will take us all the way back to 300,000 years ago, when we first developed as a species.

Yet while a future clouded by threats such as nuclear weapons, climate change, and uncontrolled technological change may seem dire, collapse has not always been a bad thing. Throughout history it has been a blessing for many.

A BLESSING IN DISGUISE

Somalia is often used as an example of a modern-day collapse. In 1991, the small sub-Saharan African country fell into civil war. It had been ruled by the brutal dictator Mohamed Siad Barre since a bloodless coup in 1969 put him in power. Armed rebel groups began to surface in the 1980s, before overthrowing the Barre regime. Centralized government was lost, a power vacuum opened, and Somalia became what many would call a ‘failed state’, or even a collapsed one.

But the loss of the Somali state was a blessing for its citizens. A decade later almost every indicator of welfare and development had improved compared to life under the Barre regime. The number of mothers dying during childbirth had decreased by 30 per cent, infant mortality had been reduced by 24 per cent, and extreme poverty had dropped by almost 20 per cent.

Many difficult-to-measure aspects of life also flourished. Under the Barre regime, media was censored, foreign travel restricted, and many forms of free speech (including ‘gossip’) and free association could be punished by death. After the state collapsed, people spoke more openly and moved freely across borders, while numerous media outlets thrived.

Yet these advances cannot be explained purely by the collapse of the brutal Barre government. Somalis’ lives improved in most development indicators even relative to their more stable neighbours in Kenya, Djibouti, and Ethiopia. The only areas of life that got worse were school enrolment and adult literacy, mainly because Western donors withdrew their aid from

education programmes. Local militias and governments proved to be more responsive and less extractive than the UN-recognized (and US-backed) reign of Barre. Somalia was better off stateless.^{[18](#)}

Somalia's experience is not unique.^{[19](#)} Many historical cases of collapse may have been a net benefit for many citizens. This should not be too surprising: most states of the past were predatory constructs not dissimilar to Barre's government.

Despite their splendid ruins, we need to keep in mind the ugly reality of past empires. Rome was an autocratic machine for turning grain into swords. Between 410 and 101 BCE the Roman Republic was at war for well over 90 per cent of the time.^{[20](#)} The Roman Empire was the largest slave system ever to have existed, enslaving around 100–200 million people during its reign.^{[21](#)} The Incan Empire was an ethnic pyramid scheme with the ruling Sapa Inca regularly and forcibly displacing hundreds of thousands of people (perhaps up to a third of the empire's population).^{[22](#)} The different Egyptian kingdoms were national economies organized around a royal burial cult. Much of the workforce was mobilized to build grand cemetery monuments, such as pyramids, to ensure a few leaders' safe passage into the afterlife. Today, many look at the Great Wall of China as an impressive achievement, a wonder of the world. It's easy to forget that it was built by prisoners and slave labour, and was intended not just to keep mounted raiders out but to keep citizens in.^{[23](#)}

Most empires strove to be endless. The poet Virgil wrote of how the god Jupiter gifted to Romulus, the founder of Rome, an 'empire without end'. The Nazi regime dreamed of a thousand-year Reich. We should be grateful that such quests for imperial immortality have been repeatedly unsuccessful.

The collapse of a Goliath is not always a largely beneficial affair, as it was with Somalia. It can involve mass displacement, the breakdown of long-distance trade and communication, an end to government services, increased violence, the abandonment of writing and loss of some technologies, and large-scale loss of life. The more typical view of collapse as a historical tragedy is not entirely unwarranted. Collapse is a process that has costs and benefits, winners and losers.

This more complicated picture is often missing in popular depictions of collapse. The experience of Somalia and the blessings of collapse may seem unexpected because the entire idea of collapse is shrouded in myth. It is rooted in deep beliefs about human nature and our apparent need for hierarchy. The most popular image of collapse is that, in the absence of a powerful authority, chaos and disorder reign; people turn to violence in a panicked scramble for resources. Hence, the misguided doomsday ‘prepper’ strategy of gathering guns, ammunition, and tinned foods into remote bunkers. Such myths inform how we think about history. It’s common to envision the past as a series of rises into golden ages during the peak of an empire and falls into post-collapse dark ages. These stories are the hallmarks of disaster movies, fictional post-apocalyptic stories, and bestselling non-fiction books. And these stories persist because the history we have is, of course, not an objective account of the past. It was written on parchment and engraved in stone by the conquerors and slave-owners of the past.

A 1 PER CENT VIEW OF HISTORY

‘The whole of Upper Egypt died of hunger and each individual had reached such a state of hunger that he ate his own children.’ This inscription comes from the autobiography of Ankhtifi, a southern provincial governor of Egypt during the end of the Old Egyptian Kingdom c. 2181–2055 BCE. The story etched into his tomb depicts an almost cosmic breakdown of law and order.

Ankhtifi was not alone in describing the loss of the Old Egyptian Kingdom as a tragedy. The *Admonitions of Ipuwer*, a collection of poems written on papyrus scrolls decades later, also paints an ugly picture. It is a typical example of what historians call ‘lamentation literature’, which gazes mournfully back at a fallen empire and lists a series of evils that befell it, ranging from civil war to famine. The *Admonitions of Ipuwer* described what is known in Egyptian history as the ‘first intermediate period’, or what most of us know as a dark age.²⁴

Yet the fall of the Old Egyptian Kingdom, as told through Ankhtifi’s autobiography, was deceptive. It seems that there was a drought, but there is

little evidence of mass death, starvation, or cannibalism as reported in his inscription. The pharaoh lost power, and there does appear to have been increased conflict, but it was not a complete catastrophe. Instead, non-elite tombs became richer and more common. Grave goods, monuments, and amuletic symbols that had previously been used only by an elite became more widely available. Egyptologists refer to it as a 'democratization of religion'. Government decentralized to provincial levels. Nobles became more impoverished, while commoners appear to have become richer and more socially mobile.²⁵

The *Admonitions of Ipuwer* confirms this. Most of the text is a tirade against the empowerment of the peasants and an inversion of the social order:

The corn of Egypt is common property ...
Indeed, the poor man has attained to the state of the Nine Gods,
he who could not make a sarcophagus for himself is now the
possessor of a tomb ... Behold, noble ladies are now on rafts,
and magnates are in the labour establishment,
while he who could not sleep even on walls is now the possessor of
a bed.
Servants spoke freely. The nobles lamented. And the poor
rejoiced.²⁶

The horror of servants gaining free speech and the collective ownership of collectively produced food is not usually what comes to mind when one thinks of collapse. Though clearly it was for the writers of the *Admonitions of Ipuwer*. The poems expend far more energy fretting about this reversal of fortunes than on war and starvation.

Many people benefited from the fall of the pharaoh and the Old Egyptian Kingdom. As a nomarch (provincial governor) of Upper Egypt, Ankhtifi himself may have been a beneficiary. When the authority of the pharaoh disappeared, nomarchs took on more power. Ankhtifi was buried in a lavishly decorated tomb, which portrays him as a great saviour during this period of instability. His autobiography was mainly an exercise in self-aggrandizement.

Ankhtifi and lamentation literature more generally are emblematic of a key problem: the 1 per cent view of history. Most of the archaeological

evidence we have – palaces, pyramids, and written records – is all from the smallest, wealthiest class.

We judge historical collapse based on what archaeologists have dug up. The easiest remains to find are monumental structures such as palaces, temples, and fortresses, as well as grand burials with a few prized individuals buried with their weapons and jewellery. In short, the more stuff you produced and left behind, the larger your entry in the archaeological record. Unfortunately, the vast majority of people in the past left faint, biodegradable fingerprints. For over 99 per cent of humanity's time on earth people have not lived in cities but as foragers or as peasant farmers.²⁷ Even at the height of the Roman Empire around 90 per cent of the population were rural farmers. And yet, we know most about what happened to a few palaces and city centres, not the thousands of farms and villages.²⁸

This leads to problems when studying ancient collapses. Did the fall of the Classic Lowland Maya and the Western Roman Empire lead to large-scale losses of life or mass displacement? People who leave cities and move back to the land leave behind fewer traces. They become archaeologically invisible. This migration from urban to rural areas can make it difficult to tell whether much of the populace perished or simply moved away. It can also make it difficult to know how their lifestyles changed, giving us a skewed picture of how collapse unfolded.²⁹

The elite bias is most evident in writing. Writing and other forms of documentation were an exclusive art for government bureaucrats and aristocratic representatives up until the invention of the printing press. The earliest forms of fully fledged writing, dating back to around 3000 BCE in Mesopotamia, were not letters or poems but tax records, administrative documents, and government propaganda. The first writer was probably an accountant.

Writing, for most of human history, has not been a widespread practice. Most people in the world couldn't read or write until the nineteenth or twentieth century. This was because the average individual faced numerous impediments to literacy. Writing materials were expensive and hard to procure, writing is often difficult to learn (it takes years of training for children even today), there were frequently restrictions (particularly for women) on who could learn, and there was usually little to no benefit if you were a farmer or peasant. Before the advent of the printing press, writing

was dominated by a small, select number of scribes employed by rulers and the richest families.³⁰

Records are also mainly penned by the victors of history. We have a deep insight into the lives of the Roman elite, but little knowledge of the non-state peoples who covered most of the world.³¹

Most of our sources tend to show collapse through the eyes of its greatest victims: enriched elites. ‘Elite’ does not mean individuals who are particularly skilled or high-performing, but rather the top 1–10 per cent of decision-makers and wealth-holders in a society. We need to go beyond the elite view and provide a more realistic analysis of history and collapse. What did it mean for the health and wealth of the slaves, peasants, and farmers?³² We need something closer to a people’s history of collapse.

IS HISTORY RELEVANT FOR THE FUTURE?

A common question is whether the history of crisis and collapse in agricultural empires can provide lessons for the modern, globalized, and industrialized world. While many factors distinguish the modern world from ancient history, they don’t make collapse less relevant. In each of the key dimensions of collapse – energy capture, population density, and hierarchy – the modern world is simply an intensification of the past. The study of collapse is more relevant than ever.

Globalized society is dependent on unfathomable amounts of energy. In just the last 200 years energy consumption has risen ten-fold.³³ The surge in energy use can also be seen in how much energy is ‘captured’ per person. That is, how much energy – calculated in terms of calories – is taken from the environment and used by humans, whether that be for food, heating, electricity, or construction. The average ice-age hunter-gatherer in Europe would have captured about 4,000 calories per day. The average European now uses around 230,000 calories.³⁴

While fossil fuels are a new source of energy, we are still dependent on the same few types of grains to feed almost the entire world. About two-thirds of the world’s food energy intake comes from three staple grasses: wheat, rice, and maize. Wheat alone provides about one-fifth of global

dietary calories. We are still agrarian in our tastes, even if we now rely on fossil-fuel-devouring production.³⁵

Collapse tended to hit urban areas far harder than rural ones. Today we also live in an unprecedentedly urban age. For most of human history, the majority of humanity has lived in rural settings. It was not until 2007–8 that over half of the global population resided in urban areas. By 2050, around two-thirds of the global population (6 billion) are expected to live in urban areas.³⁶

Societal collapse is mainly about the fall of great power structures. The world is dominated not just by cities, but by hierarchy. It is covered by a single political type: political states. If you are reading this, you are almost certainly a citizen of one of the 205 states that cover the world. Modern nation-states have their roots in far more ancient institutions. The first state was the First Dynasty of Egypt (around 3100 BCE). Now, almost every square inch of habitable land is under sovereign rule. Previously, large states shaped how people and disease spilled across the land. Now, they can unleash nuclear war across continents and change the global climate.

Perhaps most importantly, the basic human psychology that has driven war and collapse throughout human history has not changed.

Modern technology may seem like a saviour, but it also brings unprecedented threats, whether it be nuclear weapons or fossil-fuel-driven climate disruption. Remember that it was innovations in gunpowder, steel, and navigation that provided the basis for the transatlantic slave trade.

A HISTORICAL AUTOPSY

We'll examine dozens of cases of collapse, from the first cities to modern state failure, as well as a novel database of over 300 states spanning the past 5,000 years. This is the Mortality of States (MOROS) dataset, named after the Greek god of doom. It also includes the leading work of others, such as the Seshat Global History Databank (named after the Egyptian goddess of knowledge), the world's largest dataset on global history, maintained by a group of archaeologists, historians, and others.

This broad-sweeping approach diverges from how others have analysed collapse. Most popular books on collapse have tended to focus on a few

select examples. Jared Diamond's well-known *Collapse: How Societies Choose to Fail or Survive* covers five historical cases: the Pitcairn and Henderson Islands, the Rapa Nui, the Greenland Norse, the Anasazi, and the Lowland Maya. Four of these are small, isolated communities and three were islands. The Lowland Maya were a collection of competing city-states with a common culture living in a unique environment with a tropical climate and a fragile topsoil. While these are helpful as isolated examples, they are not necessarily the most instructive cases for the globalized, interconnected world of the twenty-first century.

Eric Cline's magisterial *1177 B.C.: The Year Civilization Collapsed* surveys a more relevant case: the Late Bronze Age (1200–1150). The Late Bronze Age was, in many ways, a Mediterranean microcosm of our contemporary world: a group of diverse empires, kingdoms, and cities with close political and trade ties. The splintering of the Bronze Age world is informative, but there are still thousands of years and hundreds of other cases that need to be considered. We can't discern any general patterns, and absorb their lessons, without taking a broader view across time and space, one that reaches back to our very beginning as a species.³⁷ [Part One](#) examines collapse from the dawn of our species to the emergence of the first states. [Part Two](#) surveys the rise and fall of empires over the past five millennia. [Part Three](#) concludes by looking at the future and the prospect of a modern global societal collapse.

Part One

DAWNS AND ENDS

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1

Hobbes's Delusion

Collapsing Back into Human Nature – Fluid Civilizations – Neither Nasty nor Brutish – The Myth of Mass Panic – Egalitarian Origins – Killing in the Name of Equality – Status Competition

COLLAPSING BACK INTO HUMAN NATURE

Our views of human nature and collapse have been profoundly shaped by one seventeenth-century philosopher: Thomas Hobbes. Hobbes's mother reportedly entered labour in 1588 when she heard that a Spanish armada was approaching the English coast. Later Hobbes remarked that 'my mother gave birth to twins: myself and fear', and Hobbes's life was scarred with fear and insecurity. He lived through repeated periods of political turmoil, including the English Civil War, during which he was exiled for eleven years to France. His pessimistic philosophy about human nature reflected the chaotic life he endured.

Hobbes's most lasting idea was the 'social contract': a political bargain in which citizens give up some liberty to a supreme ruler who can provide them with security. The monarch protects the people from each other. Hobbes saw 'the state of nature' – life without a central political authority – as one of perpetual war of all against all. In a much-quoted passage Hobbes decries how this 'state of nature' would entail no navigation, no industry, and 'no commodious building; no instruments of moving, and removing, such things as require much force; no knowledge of the face of the earth; no account of time; no arts; no letters; no society; and which is worst of all,

continual fear, and danger of violent death; and the life of man, solitary, poor, nasty, brutish, and short.’ Without government, chaos would rule instead.¹

Hobbes had no access to archaeological finds, psychological experiments, or anthropological ethnographies to verify these theories. So instead, he constructed a list of assumptions about human psychology and used them to paint a dark picture of life without rulers. People lived in the deadly and destitute ‘state of nature’ until they managed to agree on a social contract and appoint a sovereign ruler to keep them from each other’s throats.

His views are also representative of why so many people fear collapse: once the social contract breaks, the state of nature re-emerges with dire consequences. This bleak vision is known as veneer theory: ‘a thin veneer hiding an otherwise selfish and brutish nature’. Once civilization is peeled away, chaos spreads like brushfire. Whether it be in post-apocalyptic fiction, disaster movies, or popular history books, collapse is often portrayed as a Hobbesian nightmare.²

None of this originated with Hobbes. The story of kings rising up to tame a primordial chaos is thousands of years old. One of the foundational documents of Hinduism, the Mahabharata of the first millennium BCE, has a warrior reflect, ‘We have learned that peoples without kings have vanished in the past, devouring each other, the way fishes in the water eat the smaller ones.’ The Buddhist scriptures known as the Dīgha Nikāya recount how during a mythical time of discord and theft a group of people went to the one ‘who was most handsome and good-looking, most charismatic, and with the greatest authority’ and gifted them the powers of a ruler such as the ability to accuse and banish others. In exchange, the people offered a portion of rice to their new leader: an early, edible tax. Similar fables of mutual dependence between the rabble and a necessary ruler are retold in literature throughout the ages, from Aristotle’s writings about an ‘elected dictator’ through to Zhou-era songs in China. Hobbes merely gave the best-known philosophical articulation to a story that has been told for thousands of years.³

Hobbes’s ideas weren’t even novel for his own time. They reflected the Puritan Christian belief that the natural state of humanity was one of perpetual war against each other, God, and all creation. It was only through

God's grace that they could be peaceful with each other and eventually reach heaven. Hobbes's contemporary John Milton claimed that states were formed 'to avoid the discord and violence that sprang from Adam's transgression'. Earlier leaders of the Church had similarly used the supposed sinfulness of man since the fall of Adam and Eve from the Garden of Eden to justify the establishment of governments. Church and state became an extension of the heavenly hierarchy on earth. Hobbes borrowed an idea from theology and applied it to political philosophy.⁴

The Hobbesian worldview is closely entwined with another collapse-related stereotype: mass panic. The story goes that as soon as a crisis hits, whether it be a war, pandemic, earthquake, volcanic eruption, asteroid strike, or a plague of zombies, people will react with a selfish and irrational rush for resources. This is related to what some call 'civil disorder'. Disasters bring out the worst in people, leading to rioting, looting, robbery, and violence.⁵ Scenes of mass panic can be seen in almost every disaster film, including *Independence Day*, *Armageddon*, and *Contagion*. One survey of 448 UK citizens and emergency-response professionals suggests that this assumption about mass discord is common.⁶ The Hobbesian nightmare of panic, disorder, and bloodshed colours how we think people act in crises.

This Hobbesian nightmare of violence and mass panic is a theory not just about life outside government but also about human nature. Hobbes built his image of humanity on a mythical 'state of nature', a time before government. This is a kind of fictional, primordial situation when true human nature shone through. There is, of course, a non-fictional parallel we can draw on, the period from around 300,000 years ago when humans evolved in a world without governments or great power structures. This is commonly called the 'Palaeolithic' (see [Table 1](#); I'll be using 'Palaeolithic' or 'ice age' throughout the rest of the book), a time before the unusually warm and stable climate known as the 'Holocene' began 12,000 years ago. When scholars claim that humans are innately violent or antisocial it is built on how humans evolved during this stateless time. The Palaeolithic can indeed help us see our evolutionary inclinations (what is usually referred to as human nature). That is, the innate behaviours we tend to adopt unless culture strongly compels us not to.⁷

Table 1: Human epochs (BP = Before Present)

Lower Palaeolithic	Middle Palaeolithic	Upper Palaeolithic	Holocene
3.3 million – 300,000 BP	300, 000 – 50,000 BP	50, 000 – 12,000 BP	12, 000 – 80 BP

Everyone has assumptions about our evolutionary inclinations, Hobbesian or otherwise. With so many blank pieces in the puzzle of history, we fall back on our view of innate human drives to fill in the gaps. Even the most careful historian will still end up stitching together any loose threads using their own unspoken assumptions about humanity.

Rather than silent assumptions or old philosophers, we are better off relying on evolutionary biology, studies of modern foragers, and the thousands of archaeological sites and artefacts from the Palaeolithic to construct a picture of how human societies organized during the ice age.⁸ That can give us an idea of our evolutionary inclinations: the behaviours that shape the causes of collapse and our response to disaster. The picture that is emerging is far more interesting, and hopeful, than a Hobbesian chaos.

FLUID CIVILIZATIONS

The Palaeolithic was a challenging time to survive. The world went through a period of glacial cycles. During the coldest periods global temperatures were around 5°C lower than today and glaciers covered up to 30 per cent of the earth's surface. This would have made prolonged, intensive agriculture impossible.⁹ Africa was far drier than today and littered with around eighty-six active volcanoes.¹⁰

During this time our ancestors wandered out of Africa into the wider world in sociable, well-connected, egalitarian, nomadic bands. This was no random choice of social structure: it was a winning strategy to survive the ice age.¹¹

These egalitarian bands were cosmopolitan and varied in size. The average was likely to have been small, around twenty to forty individuals, but may have ranged up to 150–200 people. They were not restricted to family members. Recent genetic evidence suggests that in many hunter-gatherer groups less than 10 per cent of individuals in a band were closely related.¹² Each group drew on a wider pool of non-genetically close individuals, often from far-flung territories, who may not have even spoken the same mother tongue.¹³ These unrelated friends are vital. Studies of the Agta hunter-gatherers of the Philippines and the Bayaka of central Africa show that some types of knowledge, such as information about plants, are more commonly shared among close friends than family members. When it comes to keeping groups together and progressing culture, friendship is as thick as blood.¹⁴

This almost certainly held true for our ice-age ancestors. Without webs of friendships, we also wouldn't have seen new stone tools and rituals develop and spread. Without intermarriage across groups to encourage genetic diversity, we would have fallen prey to inbreeding and other genetic problems.

Our Palaeolithic ancestors were intensely social. Not just among their bands, but also across far broader networks. For a long time, people believed that close human connections had a ceiling of roughly 150 people. This is called Dunbar's number and crops up in areas ranging from the size of church congregations to Facebook groups.¹⁵ Dunbar estimated this by looking at the size of the neocortex (a part of the brain responsible for higher cognitive functions) of primates relative to the size of their groups and then extrapolating this to humans. Whether this extends to humans is questionable, and numerous studies using updated data and multiple methods have tended to come up with larger numbers across a wider range, the estimates varying from 2 to 520. There is no easy, clear ceiling to our social connections.¹⁶ It is also a little misleading. While we might have a flexible cap on how many people we can closely know at a given time, this doesn't stop us from frequently changing and updating our connections to create a far larger web. Hunter-gatherer groups of today have fluid, flexible social networks with thousands of connections. Individuals are constantly mobile and moving between groups, picking up new friends, family, and sometimes partners along the way. They are more like a city dweller than an

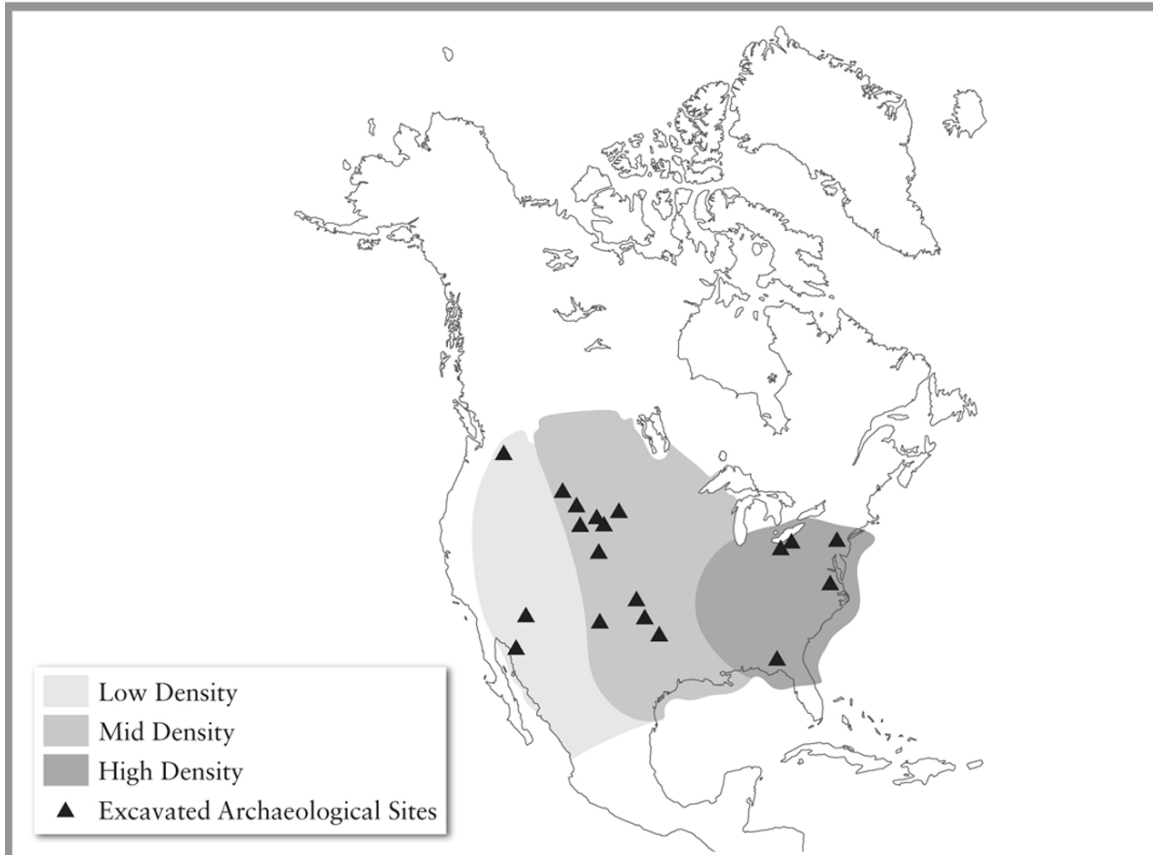
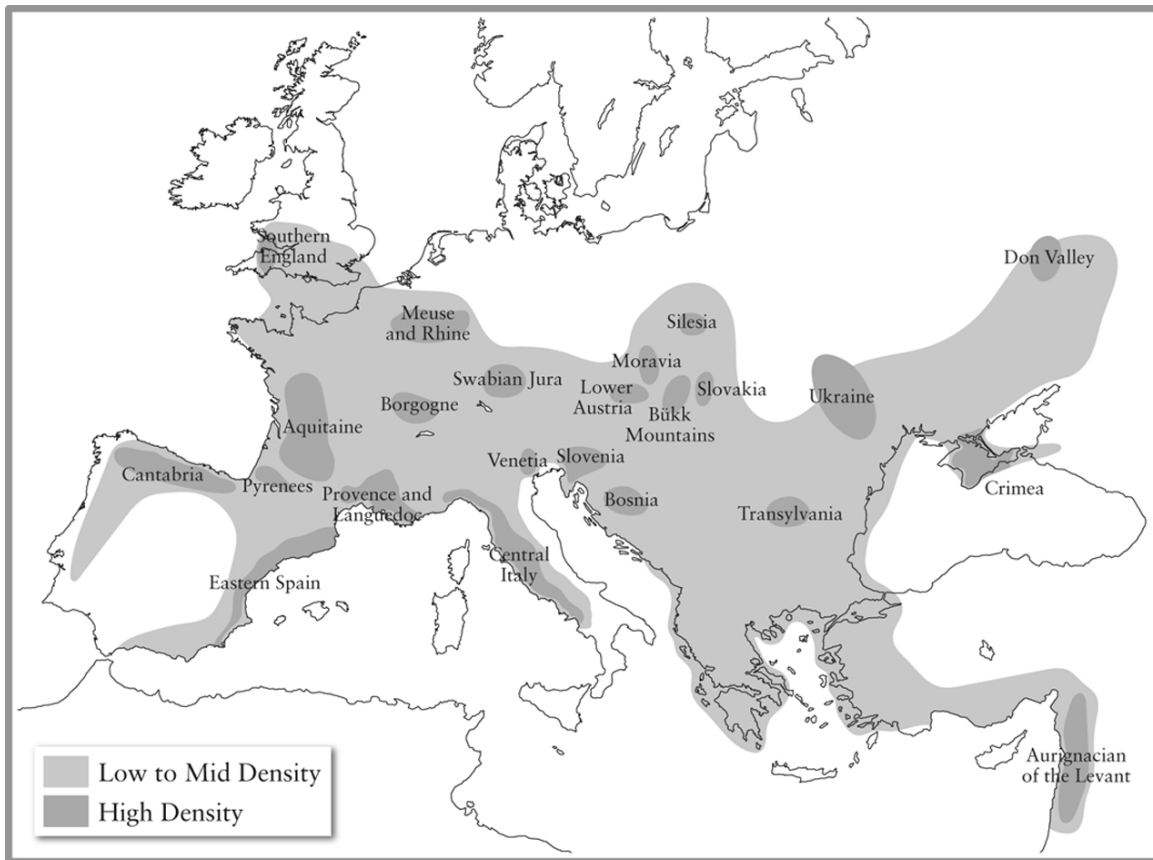
isolated family in the countryside. There is no easily numbered ceiling on human social group size.¹⁷

Our Palaeolithic predecessors were never antisocial loners focused solely on their families and distrustful of strangers. They were well-travelled, well-connected socialites who usually lived in vast societies. We inherited their innate sociability. There is a good reason why solitary confinement is the worst punishment that can be handed out, even in prisons. Our love of other people led to vast webs of shared culture and trade.

Our ancestors constructed a regional microcosm of the current globalized world. They were a people on the move who traded raw materials and shell beads over distances of about 200 kilometres per year (as egalitarian hunter-gatherers do today). Many of these networks are truly ancient. Obsidian – a black, volcanic glass that can be used as a tool, decoration, or blade – was traded over 160 kilometres around 200,000 years ago while ostrich eggs were exchanged across eastern and southern Africa at least 50,000 years ago. Much of this was built on gift-giving and reciprocity: the impulse to give back whenever we receive, whether that be gifts, help, or favours.¹⁸

These trade networks could overlap to spin webs of shared culture and technology. Across the world we see that Palaeolithic foragers crafted large, continent-spanning zones of common tools, and rituals. We may have the African Union today, but 120,000 years ago hunter-gatherers were exchanging genes, tools and musical instruments from one side of the continent to the other.¹⁹ Around 43,000–28,000 years ago, across Europe and Asia, from Siberia to France, people left similarly crafted tools, weapons, figurines, and even artwork in what archaeologists have called the ‘Aurignacian’ cultural tradition. It was a zone of common culture that surpassed the European Union in size. Similarly, people across North America 13,050–12,750 years ago all used the same ‘Clovis’ tipped projectiles, stone blades, bone rods, and other tools. Mobility was the connective tissue of these giant cultural bodies.²⁰ As the Cambridge anthropologist Cecilia Padilla-Iglesias observed, ‘movement was never just a means of finding food, but of finding one another across entire continents’.²¹ Think of these as *fluid civilizations*: networks of cooperation built on foot by people moving and intermixing ([Figure 2](#)).²²

The lives of those within these fluid civilizations were not poor, unhealthy, or doomed to be 'short' and 'nasty'. Ice-age foragers were taller and in better health than the farmers who took over the world. Today, foragers are less likely to face famine than non-industrial farmers.²³ The modern forager-horticulturalist Tsimané of the Bolivian Amazon have the lowest rate of atherosclerosis (hardening and thickening of the arteries, a key precursor to stroke and heart disease) of any recorded group. The Tsimané also experience less brain atrophy than their industrial counterparts. They lose 70 per cent less brain volume as they age than their peers in Europe and the US.²⁴ The Tsimané are not an outlier. Reviews of hunter-gatherer populations have found that they have exceptional health, with low rates of obesity, diabetes, cancer, stroke, heart disease, and other cardiovascular and metabolic health problems.²⁵ These findings are even more impressive given that modern foragers are often marginalized groups.²⁶



[Figure 2](#): Fluid civilizations in Europe and North America

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Top is the Aurignacian Culture (41,000–26,000 BCE) and bottom is the Clovis Culture (around 11,000–10,800 BCE). For both, heavier shading shows key areas of dense and intense occupation, while lighter shading shows the general reach of the cultural zone.

The working hours of foragers also compare favourably to both agriculturalists and modern workers. One study compared the Agta foragers of the Philippines to an Agta group that had been paid by governments to settle and take up rice farming. The foragers had about ten hours more leisure time per week than their farming neighbours. While estimates vary, most studies suggest that hunter-gatherers work around forty hours (with a range of approximately twenty to fifty) per week. This ‘work’ includes hunting, fishing, gardening, foraging, cooking, and walking through natural landscapes. Most people would not necessarily consider this work and go hiking or hunting for holidays. Many of these studies also classify walking, childcare, and domestic chores as work. If we applied that same definition to the average American, they would spend around fifty-five to seventy-seven hours a week working.²⁷

There is a vast difference between spending your days following orders in an office or factory, and fishing with friends. Critically, the average person in the world today is also not a middle-class worker in Australia or Norway with paid holiday leave. Instead, they are more like poor farmers or sweatshop workers in India or China.

The lives of both our prehistoric ancestors and non-state foragers were still far from ideal. Roughly a quarter of children died before reaching the age of one, and only half made it to puberty. Literacy was non-existent, and many tasks involved hard labour. There are reasons to pick being a citizen of modern Denmark over being a Palaeolithic forager.

Still, hunter-gatherers who made it past childhood generally led healthy, happy, relatively long, and deeply social lives. A centralized authority was not necessary to ensure coordination or prevent deprivation. Instead, our mobility and sociality allowed us to create great webs of shared culture and technology. All of that would have been difficult if humans had been trapped in a state of perpetual war against each other.

NEITHER NASTY NOR BRUTISH

The key to the Hobbesian story about human nature and collapse is violence. Without an overseer, people allegedly turn on each other. An orgy of disorder and bloodshed is not conducive to trade, constructing buildings, or doing anything else that involves cooperation.

It is worth pausing here to specify what we mean by violence. In this case we can focus narrowly on direct acts of lethal aggression. This includes both interpersonal violence (between individuals) and warfare (between groups). Just how violent are stateless foragers?²⁸

Foragers without a government in the modern world vary dramatically in their rates of violence. One study across fifteen hunter-gatherer groups found homicide rates per 100,000 person-years of life ranging from 1 (in the Batek of Malaysia) to 1,018 (in the Hiwi of Venezuela and Colombia). In other words, for a population of 100,000, every year there would be one murder among the Batek and 1,018 among the Hiwi. Another found a spectrum from the peaceable Bakairi (of Brazil) who had 0 per cent of deaths from violent causes to the Ache (of Paraguay) who had a chilling 55.5 per cent. We thus have a range from the safest people on earth to people whose daily existence is more dangerous than in most war zones.

Rates of violence can also change substantially over time.²⁹

Looking across all modern foragers isn't always the most reliable method, since these groups have been shaped by their interactions with farmers and colonizers, including invasions and confrontations. These rates of lethal violence among non-state peoples also misleadingly include massacres and murders inflicted by farmers, miners, and other intruders. This makes the high estimates of violence among foragers entirely misleading. For instance, all the cases of large numbers of Ache being killed (often mistakenly reported as war-deaths) were due to slave-traders and Paraguayan frontiersmen. Similarly, many of the instances of lethal violence among the Hiwi (and all reported war-deaths) were caused by invading ranchers.³⁰ These statistics also include homicides fuelled by alcohol or drugs, which were usually introduced by colonizers. Many of the deaths are better attributed to colonization than to our immutable ancient instincts.³¹

Rather than just looking at living examples of stateless hunter-gatherers, we can turn to archaeological evidence. In 2013, the anthropologists Jonathan Haas and Matthew Piscitelli of Chicago's Field Museum conducted the most comprehensive analysis of violence in pre-history. They looked at almost 3,000 skeletons from over 400 sites before 10,000 BCE. They concluded that interpersonal violence in the Palaeolithic was rare and warfare absent. Only six sites contained any indications of violence at all. One was a triple burial in the Czech Republic, although none of the skeletons bore any of the telltale signs of a grisly death. Four more sites had one or more individuals with projectile points embedded into their bones. However, these could have been accidents, perhaps a mishap while hunting. The only legitimate site that indisputably shows interpersonal violence is as famous (at least among archaeologists) as it is exceptional: Jebel Sahaba in Sudan. At this site, fifty-eight skeletons were buried, twenty-four of which had signs of violence etched into their remains. Yet the injuries seemed to be repetitive, and some were healed. This may suggest occasional raids and ambushes rather than a single bloody war. The site also dates to 12,000–10,000 BCE, at the very end of the Palaeolithic during a time of environmental upheaval when the planet warmed and exited the ice age.^{[32](#)}

Another systematic study produced similar results to Haas and Piscitelli: the observed deaths in the archaeological record equate to a 1.3 per cent violent death rate. In other words, about one in every hundred people died due to some form of physical violence, whether that be a hunting accident or interpersonal conflict. This makes the Palaeolithic more or less as peaceful as the modern world. Today, about 0.9 per cent of deaths globally are due to homicides or war. Another 1.3 per cent are today due to suicide, putting total violent deaths at 2.2 per cent.^{[33](#)}

These numbers may seem surprising. Many famous accounts suggest that the Palaeolithic was a war zone with homicide rates of 15–25 per cent. Perhaps the most well-known is Steven Pinker's *The Better Angels of Our Nature*, which estimates a Palaeolithic violence rate of 15 per cent. These numbers don't stand up to scrutiny. The anthropologist Brian Ferguson has shredded the list of twenty-one archaeological sites that Pinker draws from: three are duplicates, three have only one case of violent death (no reliable indication of war), and another has no war fatalities at all. Of the remaining two-thirds, only one case dates to the Stone Age. The systematic studies of

Hass, Piscitelli, and others, which found a much lower rate, are a far better guide to violence before the Holocene.^{[34](#)}

The skeletal record of low rates of violence is reinforced by the history of cave art. Before 8000 BCE, there are thousands of depictions of humans hunting and butchering animals. There are only three caves with art that can be interpreted as depicting human-on-human killing. Cosquer, Cougnac, and Pech Merle in France collectively contain four images that could be construed as a figure punctured with spears. Yet even that reading seems suspect: two of the figures have tails (see [Figure 3](#)) and the other two more closely resemble a goat antelope frequently drawn in other caves.^{[35](#)}

Phylogenetic evidence (studying the evolutionary relationships between species using genetics) can also shed light on our proclivities for both war and peace. One study surveyed the rates of violence within species for 1,024 mammalian species to predict how violent humans should be. Species that are genetically closer to us, such as chimpanzees, should be a better predictor than mammals generally of how murderous humans are. Their analysis suggests that prehistoric hunter-gatherers have an interpersonal kill rate of about 2 per cent, similar to other great apes. In other words, for every hundred ancient foragers, two would have died from a lethal attack from another human. That is hardly evidence of a belligerent nature or of Palaeolithic war. And it is ten times or so lower than the 15–25 per cent kill rates that more pessimistic scholars have claimed. There is also evidence that we are *less* violent than some of our primate relatives.^{[36](#)}



[Figure 3](#): Human or horse?

The figure on the left is from Cougnac cave and the one on the right is from Pech Merle cave, both in France. Both have been interpreted by some as humans riddled with spears and arrows. Ask yourself, do these look like unfortunate victims of war, or a successfully hunted horse, buffalo, or antelope goat, complete with tail?

Scholars frequently cite examples of warfare and fighting among chimpanzees as evidence that we are genetically predisposed towards violence. Yet chimps and humans are quite different. Chimpanzees have developed several genetic changes that allow them to endure more pain, heal more rapidly, and stay calm during high levels of stress. These are adaptations to higher levels of interpersonal violence. They were also all genetic changes that appeared to have happened after humans and chimpanzees diverged from a common ancestor.^{[37](#)} This is evidence of our relative peacefulness, not aggression. And those few examples of chimp ‘warfare’? Most of them seem to be caused by human interference, whether that be the destruction of chimpanzee habitats causing territorial conflicts or anthropologists feeding chimps.^{[38](#)} It’s important to bear in mind that chimpanzees are just one of our primate relatives which is often cherry-picked to cast humans in a bloody light. We have other relatives, such as bonobos, among whom violence between groups is virtually absent.

Perhaps the most compelling proof of our relative pacifism comes from our simple reluctance to hurt each other. Interviews with soldiers, examinations of weapons, analysis of combat photos, and re-enactments of battles all suggest that in past conflicts weapons were often unused by well

over half the combatants, and that soldiers often misaimed intentionally to avoid hurting their enemies.³⁹

It wasn't until the Vietnam War that the US developed more immersive training experiences to overcome this squeamishness and brainwash men into being better fighters.⁴⁰ The truth is that most of us are not hardwired to hurt or kill each other. We need to be programmed to commit murder.

This doesn't match the Hobbesian story. If our deep past was marked by persistent battles, then shouldn't we be aggressive killers? The pacifists and peaceniks would have been killed off before they could reproduce. Soldiers today shouldn't need so much training and moulding. They certainly shouldn't be psychologically scarred by being on the battlefield.

This scenario of pre-history as a war zone also doesn't fit with the presence of fluid civilizations. If there were such high rates of violence, especially between groups, then long-distance trade and constant migration across groups would have been offputtingly risky. No one wants to walk across a war zone. Yet we see ample evidence of trade, intermingling, and the sharing of culture and technologies across vast areas of the world. None of this is to say that there wasn't competition between groups or that violence didn't exist in the Palaeolithic. Rather, lethal conflict was muted with a maximum of around two out of every hundred people dying owing to interpersonal violence (if the genetic evidence proves reliable). That makes the Palaeolithic a little safer than the modern world (if we include suicide) and far safer than most other historical periods. Small-scale war may still have existed, but we simply have no evidence for it.⁴¹

Palaeolithic violence was probably highly personal. With little inequality, there would have been few material rewards for killing others. One study of 148 deaths in modern nomadic hunter-gatherers found that 55 per cent were caused by violence between individuals. Two-thirds were due to interfamily disputes, accidents, in-group executions (usually of an aspiring tyrant or violent individual), and sexual jealousy.⁴² We should take evidence from more recent hunter-gatherers with a dose of scepticism, but in this case it is limited to nomadic groups who have lifestyles and food-gathering strategies similar to those of our Palaeolithic ancestors. It is also simply logical that violence in the Palaeolithic would be similarly personal, given the lack of wealth to fight over. Crimes of passion will be as old as human emotions; war, however, is a modern phenomenon.

At this point a sceptic might still argue that, while we can cooperate, our true violent, disorderly tendencies emerge during a time of scarcity and crisis. Disasters reveal our true nature. Palaeolithic foragers may overturn Hobbes's view of human nature in general, but what about during a collapse?

THE MYTH OF MASS PANIC

Scholars of modern disasters refer to the idea of 'mass panic' as a myth. Groups panicking and devolving into a directionless rabble is ubiquitous in Hollywood films and the warnings of politicians but is rarely spotted among actual crises. Instead, people rapidly self-organize to provide medical care, support, and essential services.⁴³ Fifty years of sociological, psychological, and documentary evidence has converged on this simple yet profound finding. It has been reported in study after study, including earthquakes, hurricanes, terrorist attacks such as 9/11, and the mass bombings of cities.⁴⁴ The world crumbles away, often revealing the best in us – our cooperative, civil side.⁴⁵

People not only tend to respond quickly, creating alternative arrangements for services that may have been disrupted, but they also tend to remember these moments of disruption with fondness. They describe a peak experience where the fabric of reality split open and revealed not chaos, but the possibility of a more intimate and meaningful community. Rebecca Solnit explores this beautifully in *A Paradise Built in Hell*, showing that, whether it be 9/11 or the 1985 Mexico City earthquake, people usually respond with empathy, generosity, and resourcefulness.⁴⁶

It makes sense when we think of this from an evolutionary perspective. The winning strategy to surviving in the Palaeolithic was to use our mobility, extreme sociality, and broad connections to our advantage during calamity. If there was extreme weather in one area, rather than panic, you could migrate to a set of friends or relatives elsewhere as a social safety net. We still see this among the nomadic Ju/'hoansi of South Africa today. They maintain long-distance gift-giving and sharing relationships (Hxaro exchange partnerships) up to 200 kilometres away as a kind of insurance

policy. When disaster strikes, they can rely on these distant friends for shelter, food, and companionship. Many other egalitarian foragers have similar arrangements. This is logical: individuals who could rely on their social networks during disaster were more likely to survive and pass on their genes (as were their friends whom they would have supported as well).

That same evolutionary logic also applies to groups. If a Palaeolithic group of hunter-gatherers were hit by a drought, a storm, or a volcanic eruption, then responding with panic would have only made things worse. Groups who reacted with disorder would have been less capable of an effective emergency response, such as sharing food and migrating to friends, while those who worked together would have survived and passed their genes and culture on.

The prevalence of cultures of hospitality towards guests may even be a relic of this ancient approach to handling disasters. The Greeks and Romans had the idea of *hospitium*, the divine duty of the host to help, entertain, and even give gifts to guests. *Hospitium* was underpinned by cautionary tales of the god Zeus disguising himself as a beggar to punish those who abused their guests and reward the hospitable hosts. The Norse too had an emphasis on guest rites with similar stories of Odin dressing as a traveller to test different households; the Indian Upanishads advise *Atithi Devo Bhava*: the guest is akin to god. Being kind to visitors and strangers in need is perhaps the lingering memory of how we helped each other through the ice age.

Fluid civilizations may have even been partly driven by this friendly solution to managing disasters. Treating guests with great kindness would have encouraged mobility and sharing, especially when faced with disasters. That, of course, wouldn't have happened if mass panic had been our automatic response.

It would be strange if people's response to historical collapse was contrary to both modern behaviour and evolutionary logic. Rather, historians see the same displays of communal resilience and solidarity throughout cases of societal collapse from antiquity to the modern world.⁴⁷ Collapse is not a matter of reverting to instinctual violence and civil disorder. Whether it be the sack of Rome, or famine in Egypt, or Covid-19, the response of most people is not mass hysteria, selfish struggle, or pervasive conflict. It is instead often fairly egalitarian altruism.

EGALITARIAN ORIGINS

For the vast majority of human history there were no chiefs, commanders, or aristocrats. There were none of the usual signs of disparities in power, whether that be unequally sized dwellings or an individual or group buried with especially extravagant goods. These indicators didn't unambiguously appear until around 11,000 years ago, once we exited the ice age. Our Palaeolithic ancestors were egalitarian.

Part of this was simply due to the environment they lived in. In a world of volatile weather and scarce resources, being constantly on the move was a necessity. That mobility also meant that few resources could be carried around or hoarded. This prevented the build-up of economic inequality. Human numbers were also incredibly low and widely dispersed. By the Late Palaeolithic, the combined populations of Europe, Africa, Asia, and Australia totalled roughly half a million people. That's about the population of a small city such as Bristol in the UK or Atlanta in the US. It would have been easy to escape anyone who tried to dominate a group.

Yet this egalitarianism was not a passive condition but an active choice. Modern egalitarian foragers have an array of customs and traditions to strictly enforce sharing. The Ju/'hoansi swap arrows before a hunt and it is the original owner of the arrow – not the hunter who fires it – who allocates the meat from a kill. Thus no single hunter is ever able to amass a resource surplus over the rest of the group. Such food sharing is common across egalitarian foragers where any refusal to share food or resources is met with opprobrium (there is evidence of meat sharing from Palaeolithic carcasses too).⁴⁸ This makes good evolutionary sense: no individual could rely on being the most successful hunter or gatherer every day. Sharing was both an insurance mechanism and a bonding method. That love of sharing is deeply ingrained in us. For most of us, the best activities are ones that are shared, whether food, jokes, beds, or lives.⁴⁹

Sharing also meant sharing political power. Democracy comes from the Greek words *demos* ('people') and *kratos* ('power'). Democracy is people power: how equally shared is the ability to govern and make decisions. It is a spectrum of (more or less) equal and inclusive political practices. Any endeavours that require collective decision-making can be democratic, whether it be workplaces, families, or organizations. Across the board,

hunter-gatherers tended to be deliberative and inclusive, often more so than even modern democracies.^{[50](#)}

In many modern forager communities, decisions are made either by individuals or by consensus. Many foragers, from the Huron peoples of North America to foragers in central, southern and eastern Africa, regularly practise small-scale consensus deliberations. There are no permanent chiefs or commanders. When leaders do appear – individuals who are more outspoken, persuasive, or respected and heeded by the community – they are shackled by expectations. They are expected to be modest, generous, and egalitarian, and are kept politically toothless. The anthropologist Robert H. Lowie in his studies of numerous hunter-gatherer groups across the Americas, such as the Ojibwa, the Dakota, and the Nambikuára, found that chiefs had no power to impose decisions on others. They were more like unpaid facilitators. Again, while we can't easily extrapolate from present hunter-gatherers to ancient ones, the evidence suggests that a similar model was followed in the Palaeolithic. We have no signs of enduring inequality or hierarchy, and modern nomadic-egalitarian groups who have the lifestyles closest to our deep past are especially democratic.

Today, we often make the mistake of thinking that democracy is a Western invention and simply a form of representative government using elected officials. This is just one recent, elitist model of democracy. It can first be seen in the parliaments of bishops, warrior aristocrats, and the rich organized by Alfonso IX in León, Spain, around 1188 CE to achieve support for increased taxes. Britain later adopted a similar parliamentary system, which it exported throughout the world during colonization.

Democracy goes far deeper than this. It can be direct democracy, like an assembly of citizens collectively deciding on whether to go to war or what to build. We see signs of these in ancient Africa, China, and Mesopotamia. It can be sortition, where citizens are randomly selected to represent their peers. Ancient Athenian democracy (from about 508 until 322 BCE) was a combination of both assemblies of free men and magistrates selected randomly by lottery. The democracy of hunter-gatherers, with major decisions decided on by a group of equals, is even more intensely democratic than the Greek version, which was restricted to free adult men. All evidence suggests that political power was widely shared in the

Palaeolithic and group decisions were made democratically. Democracy is humanity's default political system.^{[51](#)}

When we think of this in terms of power, each form (political, economic, violent, and information) was evenly distributed. Perhaps a particularly skilled shaman or warrior may have temporarily gained an advantage over a group, but we have no signs that this ever resulted in a lasting hierarchy. We had a division of labour but this never resulted in a large, lasting division of power. There is also a compelling evolutionary reason behind our democratic-egalitarian origins. One written in blood.^{[52](#)}

KILLING IN THE NAME OF EQUALITY

Those who didn't share, or tried to bully others, were met with harsh sanctions. These 'counter-dominance' strategies were originally documented by the anthropologist Christopher Boehm, for a wide variety of foragers. Such strategies to keep would-be dominators in check included teasing, gossiping, shaming, ridicule, and even expulsion from the group. The most severe cases could result in death. Among the modern Ju/'hoansi there was a skilled hunter named /Twa who was causing problems. He was impulsive and brutal, having killed three others. The Ju/'hoansi took matters into their own hands: 'All the men fired at him with poisoned arrows' until, one informant recalled, 'he looked like a porcupine'. Then, after he was dead, all the men and women stabbed him with spears. It was a symbolic way to share responsibility for his death. /Twa's attempt to dominate his peers cost him his life. The fate of /Twa might explain the few skeletons marked by lethal violence in the Palaeolithic.

This violent edge to egalitarianism began around 2 million years ago. Anatomical changes in our shoulder joint around this time allowed for accurate throwing, whether that be of rocks or spears. (Other primates can, of course, throw objects, but humans can do so with transformative power and precision.)

These physical changes had momentous political implications. Projectiles levelled the playing field, allowing for a single individual or small coalition to easily defeat an aspiring alpha male (as David did Goliath). Even the largest, most imposing fighter could easily be killed by a spear or poison

arrow during an ambush or while they slept. Around the same time, there occurred a significant growth in brain size, with *Homo erectus* totting a cranium with twice the capacity of a chimp. This growing brain seems to have evolved largely for navigating more complex social environments, helping our ancestors to build coalitions against any single individual who wanted to dominate. Together, these were the evolutionary cornerstones of counter-dominance. They led to the ‘first great levelling in human history’.⁵³

Violence was also accompanied by more gentle pressures for cooperation. Evolution selected for communities that could parent collectively. Early humans faced a dilemma. Walking upright on two legs required narrower hips, which was unfortunately a poor match for big-brained babies. Those who had children at a younger age were more likely to survive (as were their offspring). Unlike most mammals, our young are helpless for their first few years of life. Individuals who could rely on help from a community were more likely to have children who survived and thrived (and to return the favour to those who helped them). Children require constant care and attention, and, if the culture is to grow and evolve, education. A group relying on a single male hunter to provide this would quickly fall behind a more dynamic one where the youngsters are receiving better attention, and lessons from the best hunters, fishers, cooks, and storytellers.⁵⁴

Our bodies reflect this egalitarian past. Ape groups where one alpha male dominates, such as gorillas, tend to have much larger males with bigger canine teeth. That is because the most physically dominant male could defend a harem and pass on his genes. We don’t see that in humans. We have comparatively small canine teeth and the difference in stature between men and women is a mere 15 per cent. As history played out, sharp teeth and brawny physiques were of little use against multiple opponents hurling spears and firing poison arrows.

Our eyes also betray our more equal past. Unlike gorillas or chimps, we have ‘cooperative eyes’: you can see the whites of our eyes (the sclera), which gives away what we are looking at. That’s useful for cooperation but isn’t ideal for dominating others. It’s difficult to launch a surprise attack if your intentions are given away by your eyes. Similarly, lying, deceiving, and manipulating others is harder to do with legible eyes. Hence why great

apes have round dark eyes, and why dictators, mafia dons, and poker players prefer sunglasses. One study drawing on fifty-five primatologists found that apes who were more tolerant of a direct gaze were more egalitarian. Primates with dominance hierarchies showed little tolerance for direct eye contact, which is associated with social status and imminent aggression. By contrast, our high tolerance for it suggests that we evolved in egalitarian social settings.⁵⁵

This egalitarianism extended into a balance between the sexes. The fact that hunter-gatherers usually settle in groups that are not based on kin might seem strange from an evolutionary perspective. Surely you would want to help and protect those who are more closely related to you and ensure your genes get passed on? The fact that bands are so mixed and diverse can be explained by men and women having equal choice in where to move to. Modelling by anthropologists suggests that if men and women have the same decision-making power and both want to live closer to their wider family then it results in more mixed bands. Sometimes the men get their way, sometimes the women do. The outcome is diverse, fluid groups, rather than the men or women sticking rigidly to their families over generations. Gender egalitarianism helped create our more mobile, diverse social structure.⁵⁶

Our ancestors did (as far as we can tell) have a rough division of labour with men hunting and women gathering (although women occasionally hunted too). However, a division of labour is not a division of power. Our Palaeolithic ancestors probably had relative equality of the sexes, especially when compared to the agricultural kingdoms that would come later.⁵⁷

STATUS COMPETITION

This is no utopian vision of humanity. There would be no need for counter-dominance strategies if at least some of us did not harbour a desire to rule. That desire to rule stems from status competition. All of us crave status to some degree: to be respected, admired, valued, and deferred to by others.

It's clearly visible throughout history. Emperors and kings regularly kept harems of hundreds or thousands of wives and concubines. King Tamba of Banaras in the sixth century BCE reportedly had a harem of 16,000

(probably an exaggeration). The conqueror Chinggis (often called Genghis) Khan sired so many children across the lands he conquered that around 0.5 per cent of men today are directly related to him.⁵⁸ Higher status through conquest meant more children and a bigger genetic legacy. We see it repeatedly throughout history, including with the Vikings invading eastern England, Ireland, and Iceland, as well as the patriarchal, nomadic Yamnaya of the Steppe sweeping through Europe.⁵⁹ Similarly, genetic evidence suggests that there was a collapse in male genetic diversity 8,000–4,000 years ago, coinciding with the intensification of agriculture and adoption of more top-down social structures.⁶⁰ The leading hypothesis is that the new oligarchs, generals, and kings simply started to monopolize reproduction. A few high-status individuals disproportionately passed on their genes by having more children (and more who survived) than those of lower status.⁶¹

This deeply ingrained desire for status is still central to the human experience. Losing status or being humiliated is one of the worst things that can happen to us. A dominant ethnic group losing status within a country is the single best predictor of civil war today.⁶² The greatest shared common characteristic across mass shooters is not mental illness but rather a sense of frustrated status and a history of being socially humiliated. Because of that, many warn against reporting about the individual killers to avoid giving them the infamy – that is, the status – they crave.⁶³ Falling down the status ladder is also one of the key contributors to people taking their own lives.⁶⁴ People are willing to kill for status, whether that be others or themselves.

Today, seeking to become a CEO, celebrity, billionaire, or world leader is still often a pursuit of status. The Texas oil tycoon Haroldson L. Hunt, one of the richest men in the world during the early twentieth century with an estimated fortune of up to \$700 million, once reflected that ‘for practical purposes, someone who has \$200,000 a year is just as well off as I am. Money’s just a way of keeping score.’ Getting a higher ‘score’ is a quest to win the status game (Hunt had fifteen children from three wives).⁶⁵ The great longing for legacy that many of us feel, to be remembered after our death (or the more modern desire to cheat death), is largely just a continued quest for status beyond the grave. Status competition is hardwired into us.

Status comes in two flavours: prestige and domination. Prestige and dominance have distinct behaviours and neurochemistry, and were selected for by different evolutionary pressures.⁶⁶ Prestige is when others want you to lead based on the skills you can offer or your wisdom. We naturally perceive the difference, with children as young as two readily distinguishing between bullies and leaders.⁶⁷ It's common in hunter-gatherers that an outstanding hunter has a greater say during a hunt. They may even be admired. Yet if the gifted hunter tries to aggrandize themselves or order others around outside of a hunt, they will face a backlash. Prestige existed, but it was limited and couldn't be passed down through generations. Hence anthropologists speak of our ingrained appreciation of 'task-based leadership'.⁶⁸ It's something that continues today: we admire and often try to emulate great athletes, writers, and orators.

Domination is a different, darker strategy for achieving status. It is the ability to gain status through violence, intimidation, or control of resources. Dictators throughout the ages have built a towering social standing because they could threaten to murder or starve much of the populace. We remember Stalin, Hitler, Napoleon, and Alexander the Great because of the great brutality they could and did wield over others. That was an enormous change from our ancestral hyper-egalitarian foragers, who could gift prestige and tolerate task-based leadership but were willing to kill to avoid domination. Because of that, for most of human history, an attempt to dominate others was more likely to get you an early burial than a place in the history books.⁶⁹

Nonetheless, a small fraction of our population persisted in their attempt to dominate. While the desire for status is universal,⁷⁰ it is not evenly distributed. The need to compete for status varies across individuals; men are more likely to be status-hungry. Men are also more likely to pursue status through dominating others. In Boehm's work almost every upstart trying to dominate the forager group was a man.⁷¹

Seeking status by dominating others isn't just more common among men. It's also more common among certain types of people. That is, those who rank high in what psychologists call the 'dark triad': psychopathy (callousness, and a lack of empathy and remorse), narcissism (an inflated sense of entitlement and self-importance), and Machiavellianism

(manipulating others for personal gain).⁷² It is unsurprising that the dark triad leads to a greater tendency to try to dominate others. Psychopathy and Machiavellianism are characterized by a disregard for others, while narcissism is a desperation to have others recognize an overvalued ego.

The desire for dominance is not just a darker path, but also an older one. It is one that we inherited from our primate ancestors. While higher status does lead to greater reproductive success in humans, the effect is four-fold higher among primates. Achieving status through dominance is also stronger among primates. Chimps build intimidating coalitions to become alpha males, while among gorillas a single male silverback monopolizes a harem with size and strength.⁷³ There is a clear evolutionary picture here: status competition (especially through dominance) was more intense in our hominid ancestors before *Homo sapiens* and then counter-dominance strategies over the last 2 million years selected against would-be tyrants and overly eager status seekers. Status competition – especially through domination – is lower in modern humans but was never eliminated.⁷⁴

We are stuck in this eternal bind between two evolved traits: a desire for status and an aversion to dominance. Counter-dominance strategies and more leadership based on prestige were reasonable outcomes, since only a small proportion of us seek status through domination, while no one likes being dominated. Some of us seek to rule, but if we can't then we prefer to be equal. For 300,000 years the balance was maintained against dominance. Not because we are innately good in our hearts, but rather because egalitarianism was a winning strategy in the challenging conditions of the Palaeolithic.

We can be sure that this view of our deep past is accurate because it is supported by multiple lines of evidence, including archaeological finds, studies in evolutionary biology, and observations of modern egalitarian foragers. These groups include the Hadza of East Africa, the foragers of central Africa (such as the Aka, Efe, and Mbuti), and the Khoisan of the Kalahari Desert (especially the well-studied Ju/'hoansi of southern Africa). These foragers are not relics from the Palaeolithic, but they are the closest analogues of our Palaeolithic ancestors. Most live in Africa, the cradle of humanity, and practise similar kinds of hunting and gathering to those of our ice-age ancestors. All the archaeological sites we have from the Palaeolithic left the footprint of egalitarian foragers with residential base

camps, hunting stands, lithic quarries, and evidence of food sharing. Modern egalitarian foragers and ancient ice-age hunter-gatherers also appear to have travelled and traded over similar distances.

Importantly, many of these groups appear to have had a continuous cultural and genetic presence in their regions for over 100,000 years. Genetic evidence for the Khoisan suggests that they diverged from other groups around 300,000–200,000 years ago, and now represent the most genetically diverse and divergent lineage of all living peoples in the world. Similarly, foragers of central Africa have distinct lineages that are hundreds of thousands of years old, and the Hadza diverged at least tens of thousands of years ago. That is not to say that their culture has remained unchanged, but we have no evidence that there have been large changes to their basic egalitarian, democratic way of life.⁷⁵ Indeed, remarkably they all seem to share several characteristics; including a polyphonic song style, use of body paint, similar creation myths and cosmologies, and an egalitarian way of organizing.⁷⁶ That is more likely to result from a shared origin, rather than a mere coincidence. We'll never have a completely clear window into the life of our ancestors in the Palaeolithic, but we have more than enough evidence to be confident in this account of our egalitarian origins.

This account of our species is radically different from what Hobbes envisaged and is one that has momentous implications. Domsday preppers today (and most post-apocalyptic fiction) imagine the survivors of a collapse as being self-reliant individuals ready to deal death to stay alive. It's a strategy that might work well in a Hobbesian fiction but bears little resemblance to our shared history. For most of history people coped with disasters and catastrophes by moving, giving and receiving aid, and cooperating as equals.

We are a species that survived the ice age by being intensely social, cooperative across groups, and equal, even during times of need and danger. There appears to have been peace across groups, but individuals were willing to kill, especially to avoid oppression. Those aggressively seeking status or wanting to dominate others were both figuratively and literally cut down. That egalitarianism is what made us human. Yet, deep down, our primal heritage still hungers for status. Some of us are willing to lie, cheat, threaten, and even murder for it. These traits – mobility, collaboration

(especially through reciprocity), status competition, the dark triad, and an aversion to dominance – have shaped the course of collapse and history.

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Collapse for 99 per cent of Human History

Homo solus – The History of Human Extinction – Collapse During the Palaeolithic – Coming Together and Breaking Apart – Seasonal Cycles

HOMO SOLUS

Recorded human history is less than the tip of an iceberg. Just 10 per cent of an iceberg is visible, while the remaining 90 per cent of its mass is hidden beneath the surface of the water. Just the most recent ~1 per cent of human history is visible through the lens of written records. Ninety-nine per cent of human history occurred before the first symbols were stamped onto clay; before states and Goliaths – hierarchies in which some individuals dominate others to control energy and labour – emerged.

The first states arose approximately 5,100 years ago, with the First Dynasty of Egypt. This represents around 1.7 per cent of the roughly 300,000 years that anatomically modern humans have wandered the earth. The rise of agriculture and states across the world was unsteady and uneven. We don't know when exactly empires and states came to capture most of the human population. It was still an arduous process, and many areas, such as North America, were free from permanent states until around the eighteenth century. At best, states dominate less than 0.5 per cent of the human timeline. What did collapse look like for non-state peoples, that is for over 99 per cent of human history?^{[1](#)}

For most of this time we weren't the only humans wandering the planet. We once shared the earth with at least eight other human species including *Homo neanderthalensis* (the Neanderthals). We shared territories with them, bred with them, and likely even played and travelled with them. Our deceased hominin ancestors left a lingering mark on most of us, with modern non-African humans having inherited 1–3 per cent of their genome from Neanderthals (since it was only the groups that emigrated to Europe and Asia that met and interbred with the Neanderthals).²

It was a perilous existence for our ancestors. All our hominin cousins died out by 40,000–30,000 years ago. Human extinction has happened before numerous times, just not to us. We, the survivors, are *Homo sapiens*: *sapiens* means 'one who knows' in Latin, and *Homo sapiens* means 'wise man'. Perhaps we would better be known as *Homo solus*: the solo human, or human alone.

THE HISTORY OF HUMAN EXTINCTION

Why did we end up alone? Extinction is the most extreme form of collapse: energy capture, population, and hierarchy permanently reach zero.³ Such complete ends to a social order and its people have very rarely happened throughout history. Yet, understanding why our hominin relatives died out can help us understand why we persisted, and what pressures ancient *sapiens* would have faced. Unfortunately, apart from our genes, most of our hominin cousins have left scant archaeological remains. This gives us few clues as to why they perished. One exception is the Neanderthals. They left enough evidence to permit some informed speculation about their extinction.

It is easy to imagine that we outlasted the Neanderthals and came to dominate the world owing to our intellectual superiority. That's part of the story; we are remarkably intelligent but we often overestimate our uniqueness and intelligence. For instance, chimpanzees have a better short-term working memory than us. What is more important than our individual intelligence is our ability to cooperate with one another.

We can cooperate flexibly and in large numbers partly because we can share ideas to create imagined communities. It's a fundamental insight, one

that dates back at least as far as the sociologist Émile Durkheim in the nineteenth century, before being summarized by the historian Benedict Anderson, and later eloquently popularized by Yuval Noah Harari. A monkey cannot identify itself as part of a mythical construct like a nation or a company, but we humans can. Storytelling is key to our existence. Among modern hunter-gatherers such as the Agta, storytelling is still the most desired skill, and the best storytellers have the most children. Stories that encourage cooperation and a shared identity also create stronger groups that are more likely to survive. Good storytelling is evolutionarily advantageous for the storytellers and their group.⁴

Stories are important not only because they create imaginary communities but also because they can communicate ideas. Our great superpower is ‘social learning’: our capacity to imitate and learn from each other. Chimpanzees and other primates can create and use tools, such as sticks for digging out termites, and they imitate each other. Of course, we excel at this and can pass our ideas along to each other and down through the generations. Tests between chimpanzees and toddlers show that, while the chimps are comparable (and sometimes superior) in cognitive abilities, toddlers completely outpace them when it comes to imitation and social learning. If you and a chimpanzee were placed in a random patch of wilderness with just your wits to survive, the chimp would outlast you. If you had a few forager mentors who had generations of accumulated knowledge of how to survive in that environment, you would beat the chimp. This accumulation of selected knowledge is known as ‘cumulative cultural evolution’ (for ease, I’ll be calling it ‘cultural evolution’). Culture can evolve far more quickly than our bodies, which is what allows us to adapt to a wide variety of environments, from the deserts of the Sahara to the tundra of the Arctic. Cultural evolution, whether it be in language, tools, technology, or traditions, is a human superpower.⁵

Cultural evolution is made possible by our egalitarianism and our wide social networks. Chimpanzees lower in the pecking order are less likely to be copied or show behaviours they learned from others if they are in the vicinity of a dominant figure. The world is siloed in these hierarchies, and, with fewer close interactions, juvenile chimpanzees have fewer opportunities to master new tools and technologies. In contrast, in egalitarian hunter-gatherers, children regularly observe and learn from

scores of individuals – individuals who walk hundreds of kilometres to reach new groups, where they in turn share and learn. The breakdown of social hierarchies enabled our superpower of cultural evolution.⁶

It was once believed that many of these traits were unique to us. The old hypothesis was that we had a rapid ‘cognitive revolution’ 70,000 years ago and acquired the magical gifts of language, imagination, and cultural learning. The modern human brain was born. That idea is no longer tenable. We now know that 600,000 years ago there was a radical increase in the complexity of stone tools. In other words, hominids before us could already learn from each other. Our ancestors were using the red pigment ochre 500,000 years ago. Ochre is commonly used for decoration during rituals. There is evidence of abstract artwork on rocks dating from up to 158,000 years ago. These are signs that hominids already had imagination, as well as shared ideas and rituals that were passed down across generations, well before the so-called ‘cognitive revolution’.⁷

We also overestimate how unique our brains are. Many of our hominid cousins, especially the Neanderthals, were remarkably similar to us. Neanderthals had brains that were larger than ours in size (cranial capacity). They also had complex burial practices with their graves containing tools, weapons, flowers, and sacrificed animals; they nursed their ill and injured; they even appear to have adorned themselves with jewellery and feathered decorations. All this suggests some capacity for imaginative thought, perhaps even a belief in the afterlife, and an ability to share those ideas.⁸

Neanderthals even beat us to painting cave art. In three caves across Spain are the earliest examples of cave art, which date back 64,000 years. That is 20,000 years before we arrived in Europe. These collections of sketched animals, handprints, and geometric shapes were created by Neanderthal artists. The earliest cave art from modern humans – a depiction of a human-like figure and a pig in Indonesia – is 51,200 years old.⁹

Neanderthals were not just similar to us mentally, but also physically. If you walked past a Neanderthal in the street dressed in modern garb you might not look twice (see [Figure 4](#)).¹⁰ Overall, it appears that the cognitive differences between modern humans and Neanderthals are minor. This is a radical change from the nineteenth and twentieth centuries, when Neanderthals were regularly depicted as club-toting brutes, to the extent that the German Darwinist Ernst Haeckel suggested renaming them *Homo*

stupidus. Turns out they're just about as smart (or stupid) as we are, and there is much to learn from their extinction.



[Figure 4](#): 'The Neanderthal Businessman'

A model of a Neanderthal dressed as a modern businessman, contemplating what a freeze in the mammoth market might mean for his investment portfolio.

Three possible explanations are usually offered for Neanderthal extinction: competition, environmental change, or population vulnerability (small, disconnected populations that could easily collapse). The competition hypothesis posits that Neanderthals were either killed off by us, or we were simply better hunters than they were, leaving them to starve. Environmental factors include a cold snap that pushed Europe into subarctic conditions 40,000 years ago. The demographic argument is that Neanderthals were doomed in the long run owing to poorly connected, small population numbers. This could have led to widespread problems with fertility and inbreeding.^{[11](#)}

One recent survey of 216 palaeoanthropologists who've published on Neanderthal extinction shows a surprising consensus emerging:

demography was the principal cause.¹² Neanderthals lived in small, fragmented communities, making them less genetically diverse, more prone to inbreeding, and vulnerable to population collapse owing to a small increase in deaths (or decrease in births). An analysis of thirteen Neanderthal remains in the Altai Mountains of southern Siberia found that six of the members were relatives and all showed low levels of genetic diversity. These levels are as low as those found in endangered mountain gorillas, which live in tiny groups of four to twenty individuals.¹³ In contrast, Palaeolithic humans are notable for having far higher levels of genetic diversity and for usually being buried with distant strangers and non-kin.

The Neanderthals' comparative lack of diversity and connection, as well as high mortality rates, meant that all it might have taken was a few bad winters, or an encounter with a new disease, to cause entire groups to fall apart. With so few members, recovering without inbreeding and genetic issues would have been difficult. Indeed, one calculation suggests that just a 1.5 per cent drop in how many Neanderthal children made it to adulthood could have resulted in their extinction in just 2,000 years.¹⁴

These explanations are not mutually exclusive. Climate change, new diseases, and competition with *Homo sapiens* could have all interacted and led to population loss. The population vulnerability of Neanderthals made these cuts in their numbers deeper and recovery less likely. The exact contribution of demography, environment, and competition with humans would have varied across time and place. Modelling by anthropologists and physicists supports this idea that Neanderthals went extinct owing to diverse, local causes rather than one single factor. Over centuries their communities fragmented and diminished until the last breath of the last Neanderthal was drawn in the cold.¹⁵

In the field of disaster risk, we refer to four determinants: threat, vulnerability, exposure, and response ([Figure 5](#)).¹⁶ Take the case of a tsunami. The tsunami (threat) is only dangerous if you lie in its way (exposure), and you lack robust shelter that could withstand the wave (vulnerability) or the ability to escape (response). For Neanderthals, climate change and competition with us were hazards, while small, isolated populations were their underlying vulnerability. We have little idea about how they responded. We should think of the overall risk and how each of

these determinants contributed to the extinction of our Neanderthal cousins, rather than bickering about whether it was just one factor.

$$\text{Hazard} \times \text{Vulnerability} \times \text{Exposure} \times \text{Response} \\ = \text{Risk}$$

[Figure 5](#): The determinants of risk

Just as there was no single cause behind the extinction of the Neanderthals, there was also no single extinction event. There were instead a series of population collapses that eventually, cut by cut, drained the species out of existence. Each of these local disasters would have had a slightly different cause; one may have been triggered by disease and another by a particularly bitter winter.

Competition with us may have been part of it, but there are no signs of massacres or conflict. The strongest signs of violence we have are two Neanderthal skeletons – one with a stabbed (and healed) rib and the other with a fractured skull – but we have no idea whether these injuries were accidental, intentional, or even caused by humans.¹⁷ No matter the threat, the underlying vulnerability was the same: the disconnected nature of Neanderthal communities.

The vulnerability for the Neanderthals was not their wits. Neanderthals, too, were intelligent, and capable of making and sharing ideas and stories. Their weakness was their fragmented, isolated groupings. In contrast, humans had sprawling networks that formed a constant cultural and genetic melting pot. Our fluid civilizations provided us with a safety net during lean times, which Neanderthals lacked. We survived and became *Homo solus* not (just) because of our intelligence but because of our interconnectedness. Yet, like the Neanderthals, when those linkages were broken, calamity wasn't far away.

COLLAPSE DURING THE PALAEOOLITHIC

It was long believed that humans first left Africa in a 'Great Exit' 70,000–55,000 years ago. The Apidima Cave in southern Greece tells a different

story. Archaeologists there uncovered a skull dating to 210,000 years ago. That skull was of an anatomically modern human. In the same cave they found a skull of a 170,000-year-old Neanderthal.¹⁸ Humans left Africa far earlier than the Great Exit, spreading out to the Mediterranean and the Levant. (However, it is likely that all non-Africans today are descended just from that one Great Exit.)

Something clearly happened to these earlier scouting parties that departed Africa. What fate befell them is unclear. Some may have gone extinct. Cut off from the wider fluid civilization of humanity, they may have lacked the genetic diversity and cultural know-how to survive in a new, alien world. As the Neanderthals had learned, a small, isolated population is a vulnerable one. Perhaps they fell apart in the face of ash and cold from one of the volcanic eruptions that characterized the ice age. Maybe they were outcompeted by their Neanderthal cousins. We like to think of ourselves as the more successful human variant that always won out, yet there may have been countless times when we were the losers.¹⁹

Perhaps some *Homo sapiens* did persist as small enclaves outside Africa but left little to no archaeological footprint. They may have then been subsumed by the larger wave of migrants out of the African continent during the Great Exit. After tens of thousands of years of isolation, it must have been quite the reunion.

Climatic swings, particularly those caused by volcanoes, can be blamed for many of the failed migrations and losses of regional populations. Variations in climate and hunter-gatherer populations were closely entwined. Studies suggest that for Britain and Ireland during the Last Glacial Maximum (27,000–19,000 years ago) climate was one of the key drivers of population change. These changes could be dramatic.²⁰

One event 73,000 years ago could have brought humanity close to extinction: the massive volcanic eruption of Mount Toba, in what is now Indonesia. This would have been the largest eruption in the past 2 million years. It created a volcanic winter, dropping temperatures sharply in many areas across the world and blanketing India with two and a half metres of ash.²¹ The eruption seems to have created a genetic bottleneck across *Homo sapiens* – a significant reduction in genetic diversity, suggesting a large loss in population numbers. However, genetic bottlenecks don't always mean large losses in people and many human settlements in

northern India and southern Africa continued relatively unfazed.²² North America, Europe, and central Asia suffered far more severely than the southern hemisphere.²³ The global climate was disrupted, but regional shelters persisted. We got lucky. It seems likely that Toba led to the die-off of more isolated groups, such as those in the Levant, while many others in India and elsewhere escaped relatively unscathed.

Dips in numbers, dispersal, and changes in culture were also found by archaeologists across three large volcanic events in the northern hemisphere: the Aniakchak eruption (~1645 BCE) in Alaska, the Mazama eruption (~5630 BCE) in western North America, and the Laacher See eruption (*See* is German for ‘lake’, and the explosion left a two-kilometre-long lake in central Germany) (~11,000 BCE). The Aniakchak eruption separated the Aleut and Inuit peoples in the Alaskan Peninsula, and may have caused an ethnic and language divide between them that persists to this day.²⁴

Throughout the Palaeolithic the closest we have to a societal collapse is the extinction of communities, largely due to environmental shocks, especially volcanic eruptions. There were no Goliath structures like empires to collapse. Ice-age foragers captured very little energy beyond what was necessary for subsistence: the basic food, shelter, clothing, and technology needed to survive and reproduce. Any large drops in energy capture meant starvation and death. Reductions in the density of communities was one of the few things that could happen. And it seemed to occur with surprising frequency.

COMING TOGETHER AND BREAKING APART

Our ancestors led lives of constant flux and movement. They would break apart during the day to hunt and gather before coming together in the evening to share food, stories, and beds. In modern hunter-gatherers we see groups can break off to join others, or even form their own enclave, for a range of reasons from bickering to falling in and out of love. Sometimes they rejoin the original community and sometimes they remain estranged. (This could even have been a counter-dominance strategy to flee from a megalomaniac.) These fission-fusion movements involve several smaller

groups briefly fusing together into a larger community, before dissolving again. These patterns are common across cognitively advanced species such as dolphins and elephants, but especially humans.²⁵

While abandoning a city or town is a recurring feature of collapse, such reductions in the concentration of communities were common and advantageous for our ancestors. Moving, travelling, coming together, and then breaking away have always been part of our human heritage. These things helped make us human.

Constant movement across vast networks, and fission-fusion cycles, were central to our Palaeolithic past.²⁶ This was probably done to exchange tools, gifts and knowledge, to build new relationships and search for seasonal resources, and to use rituals to reaffirm the bonds that held together scattered communities. It also helped create the social safety nets and fluid civilizations that allowed us to survive the ice age.

There would have been the constant churn of individuals, couples, and groups leaving one location to join friends and relatives elsewhere. There may have been regular ceremonies throughout a year (or special events such as marriages and funerals), which brought groups together as we see in foragers today. For instance, the longest trips taken by the Mbendjele Bayaka of central Africa are for Eboka ceremonies, in which they sing together and conduct rituals to connect with forest spirits. Other times, groups fleeing a catastrophe such as a drought would have joined far-flung neighbours for safety. Our mobility and willingness to regularly change groups may have even given us the unique human ability to gossip – to talk about others who aren't currently present. Gossip was partly developed as one way to keep a mental grasp on people who had temporarily left. It now makes up around two-thirds of human conversation.²⁷

Our genetics, as well as archaeological finds, suggest we lived through cycles of isolation and connectivity. The failed early migrations out of Africa are one example, but there are others. Researchers have found fossils in the Democratic Republic of Congo that resemble 300,000-year-old hominids but are only 22,000 years old.²⁸ This seems to have been a group that became cut-off from the broader fluid civilization of humanity. Then, with the change of climate, they may have died off or rejoined the wider web of people. They are just one indication of groups that became

environmentally dislocated yet persisted and developed their own unique culture.

When these wayward communities rejoined the great human highway of shared culture and genes, it could cause a renaissance. What have typically been mistaken as ‘cognitive revolutions’ in tools and thinking are more likely to have been the merging of previously separated groups, unions that spark vast changes, much as the combining of cultures or technologies does today.²⁹

Such changes were not, of course, collapses. Yet they do show that humans have always been mobile and prone to leaving arrangements that were dangerous or no longer desirable. Leaving a city that is becoming poorer with every passing year or departing an empire that is deteriorating is not necessarily a bad thing. It is, in many ways, an entirely normal and sensible adaptation. This was a precursor to collapse: a deeply encoded survival strategy that can help us make sense of later collapse events. There is another such precursor: our ability rapidly to build and dismantle entire social structures.

SEASONAL CYCLES

Some modern hunter-gatherers don’t just change the size of their communities. At times they also shift their social structures.

We see this in the accounts of anthropologists over the past centuries from North America to Australia. In the Great Plains, an expanse of flatlands across the central United States and western Canada, the Lakota and Cheyenne would change their societies with the rhythm of the seasons. Typically, they existed in small groups reminiscent of the nomadic-egalitarian model. That changed during the large bison hunts and ‘sun dance’ rituals of the summer, when they congregated and organized into a far more authoritarian society. A quasi-police force would form based on clan affiliation, membership in a military club, or ad hoc appointment. These soldiers could confiscate game, whip offenders, destroy their property, and even kill those who disobeyed orders.³⁰

This is not an isolated example. The Shoshone of the Great Plains of the US lived most of the year in egalitarian, unranked families. Periodically

some would adopt ‘rabbit bosses’ and ‘antelope shamans’ to lead and coordinate rabbit drives. In Australia, far north on the Cape York Peninsula, the Wik-Mungkan lived nomadically during the dry season before settling down into villages during the wet season when the plains were flooded.³¹ Far further north, the Inuit would hunt as small family groups under a patriarchal figure in the summer, and then in the winter come together in more communal, egalitarian settings where partner sharing was practised.³² Across the world, people came together and broke apart, fusing and splitting, changing their customs, politics, and even sexual norms in the process.

These seasonal gatherings also disproportionately resulted in the building of impressive new ritual sites and temples. Stonehenge in the UK was built between 3000 and 2600 BCE, probably by herders practising seasonal ceremonies. For migrating pastoralists, such gatherings around great monuments could have provided a time to cross-breed animals (to ensure better genetic diversity and health), find partners, and host great feasts and alliance-building marriages. Similarly, Göbleki Tepe in eastern Turkey was seemingly built by foragers 9,000 years ago as a seasonal settlement near a point where migrations of gazelle passed. It is a vast site of twenty megalithic structures, including 5.5-metre-high, eight-tonne T-shaped pillars sculpted with images of predators, prey, headless men, and a surprisingly high number of phalluses.³³

These seasonal cycles are, like fission-fusion patterns, another precursor to collapse, a sign that groups can and do easily change their power structures and populations. It should give us pause in thinking of collapse as an aberration. These routine social shifts instead suggest that disassembling a hierarchy is neither scary nor remarkable if that hierarchy was never meant to be permanent. This precursor followed from our fission-fusion practices and is a more recent phenomenon. We have little evidence that these seasonal cycles occurred through most of human history.

The oldest evidence of such cycles comes from Europe during the Upper Palaeolithic 36,000–30,000 years ago. The main signs come in the form of grand burials. These include remains from Sungir, northern Russia, which were adorned with thousands of perforated fox canine teeth and mammoth-ivory beads. The fox teeth alone are estimated to have required 10,000 hours of work. The most lavish decoration is for a boy and girl who are

flanked by mammoth tusks. Then there is the burial of the Lady of Saint-Germain-la-Rivière, in south-western France. An adult woman was delicately placed on ornaments, shells, and stag teeth sourced from 300 kilometres away. Grand burials were also accompanied by grand buildings: 18,000–12,000 years ago, foragers across eastern Europe, from Kraków in Poland to Kyiv in Ukraine, constructed small buildings from mammoth tusks and bones. It was during this time of mammoth-bone houses and opulent burials that the first examples of European cave art also appear.³⁴

Most of these impressive and anomalous remains from Upper Palaeolithic Europe occurred near points of rich, periodic resources, such as floodplains or the choke points of migrating herds of deer, bison, or mammoth. Just like the Shoshone, Lakota, and Cheyenne, ice-age hunters adapted their social structures to their environments. Large collective hunts brought together great populations of people, triggering artistic and architectural booms.³⁵ Then, once the season had changed or the hunt had ended, people disbanded across the landscape once again.

These dynamics should be unsurprising since migrating animals were central to the European way of life during the late ice age; so much so that the very first stirrings of writing were created to help track them. At least 400 European caves contain hauntingly beautiful painted images of animals and figures alongside strange dots, lines, and Y-marks (see [Figure 6](#)). They range across France and Spain, in Lascaux, Chauvet, and Altamira. The dot and line symbols began at least 42,000 years ago, and the depictions of animals such as horses and aurochs date to 37,000 years ago. Analysing a database of hundreds of images, researchers have proposed that this was a kind of calendar and proto-writing system. The Y-marks show when an animal gives birth, while the dots and lines represent lunar months after the start of spring, when animal mating seasons began.³⁶ These peoples were tracking the birthing and mating of their prey. Seasons, and lunar months, were so critical to Palaeolithic Europeans that they enshrined them into the world's first recorded calendar and rudimentary writing system.



[Figure 6](#): Examples of Palaeolithic proto-writing

The aurochs and deer are both from cave sites in Lascaux (France) and the salmon from Abri du Poisson (also France).

The new relics we see during the seasonal cycles of Upper Palaeolithic Europe – the artisan jewellery, large buildings, and rich burials – could have been either celebrations of successful hunts or perhaps even temporary bursts of inequality as a small group or individual used the surplus to establish a fleeting ice-age nobility. The latter seems unlikely. Most of the graves do not contain alpha-male leaders or their children but, rather perplexingly, individuals who were disabled or disfigured. The finds are also rare; humans lived in these areas for tens of thousands of years but left only a handful of such relics.^{[37](#)}

These seasonal cycles are highly unlikely to have occurred earlier than the Upper Palaeolithic (c.50,000–10,000 BCE) in Europe. We have no evidence of opulent burials or grand structures in Africa or elsewhere at any earlier periods. That makes sense since ice-age Africa didn't experience the same kind of seasons as we have today. The cold, arid, volatile climate of the Palaeolithic also would have made seasonal changes more muted.^{[38](#)}

What triggered this precursor to collapse? Understanding why people propped up and then took down social structures can help us glean some lessons for why people in more established kingdoms may have let them fall (or even overturned them). The obvious answer is that they are adapting to new material circumstances. The Wig-Mungkan dispersed when water was scarce and came together when it was concentrated into a flood, and the Inuit settled down into warmer, closer communities when faced with the harsh, cold darkness of the Arctic winter. The Inuit 'are governed by environmental circumstances', observed the famed anthropologist Marcel Mauss, who wrote some of the most seminal studies of the Inuit.^{[39](#)} For the Shoshone, Lakota, and Cheyenne people, migratory herds presented an

enviable, enormous source of food. To capitalize on that they needed a larger group of swiftly coordinated hunters. It was task-based leadership on steroids: good hunters and leaders were given temporary powers to make the most of the situation. Once the emergency ended, so did their powers.⁴⁰

Some scholars have suggested that these seasonal cycles were playful political choices rather than savvy adaptations. Foragers had the political superpower of being able to experiment with new social structures, erecting and collapsing them for fun.⁴¹ But these shifts didn't happen at random: they followed the seasons. Nor did the tribes sit down before each hunt or season and agree on what new political structure to use that year (which would give the young a say and help refresh old traditions). It's a tradition they built over time to fit environmental conditions. This doesn't mean they lacked choice or agency. It means that, like us, they were practical thinkers who were shaped by their circumstances.

Fission-fusion dynamics and seasonal cycles are a happier alternative to full-blown collapse. They suggest that collapse need not be seen as an unintentional or undesired fall from grace. Instead, it was always part of our adaptive strategy. Whether it be to prevent individuals from dominating the group, to coordinate hunts, resolve conflicts, move with the availability of water, or encourage marriage across groups, regular changes to population concentration and sometimes culture were not something to be feared. They were part of what it meant to be human. They were also precursors to collapse. Given that we are a mobile, socially fluid people, the biggest question is perhaps not why later empires and kingdoms fell apart, but how they contained so many of us in the first place.

For the Palaeolithic, collapse usually meant local extinctions. For the Neanderthals their disconnected groups were their weakness, and climatic change was the threat. Similarly, for us, communities were vulnerable when they were cut off from the wider fluid civilization and faced threats such as volcanic eruptions. While the small-scale extinction of Neanderthal communities eventually snowballed into the loss of their species, our local failures never did. That is because we were flexible, able regularly to break apart and come back together to find food, explore, trade and share, and renew our connections. That connectivity is what allowed us to become *Homo solus*, the sole human survivor of the ice age. A surviving species that emerged into a warmer, different world.

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3

From Hunting and Gathering to Being Hunted and Gathered

*When People Became Fierce – Goliath Fuel – The Origins of War –
Evolutionary Backsliding, or Dropping the Progress Narrative – The
Instability of Domination*

WHEN PEOPLE BECAME FIERCE

In 1549 Hernando de Escalante Fontaneda was shipwrecked on the coast of Florida. Hernando was a thirteen-year-old student, born in Colombia to wealthy Spanish parents and on his way to Salamanca for his studies. He never made it there. Instead, he became a captive among the indigenous hunters of Florida. The native foragers he met there resembled Spanish society more than the Hadza or Khoisan.

Fontaneda was taken captive by the Calusa, a name that meant ‘fierce people’ in their own language. His brother and most of the other shipwreck survivors were sacrificed. This was a normal practice for the Calusa, who also used their captives as servants. The Calusa operated like a mini-kingdom. It was a hereditary monarchy of sorts, with a ruler, elites, and around 300 full-time warriors. They had a fleet of war canoes, which were used to raid surrounding lands. The ruler of the Calusa was a supreme chief who claimed to own the lands and renewed their fertility by undertaking regular, secret rituals. He was so powerful that when he died children were often sacrificed.^{[1](#)}

The Calusa were not farmers; they were fisherfolk. They built great houses that could hold up to 2,000 people and made watercourts covering 6,000 square kilometres to store fish. They controlled around fifty villages that together contained some 20,000 people, including farmers who grew corn and squashes. They demanded 'tribute' – irregular payments designed to show submission, respect, or allegiance – from these surrounding peoples. The Calusa exacted their tribute in the form of food, gold, and European or African captives. The stratification of the Calusa resembled Fontaneda's original destination of Spain: society split between nobles and commoners, the rulers and the ruled. The Calusa were not an anomaly.²

If Fontaneda had been shipwrecked on the other side of the continent further north he would have encountered the north-west coast Native Americans. Some were egalitarian and subsisted on acorns and nuts as staple crops. Others, such as the Kwakwaka'wakw and Tsimshian, more closely resembled the Calusa. These north-west coast Native Americans built long houses, had hereditary chiefs who waged war and took captives, and even kept slaves. During the winter months they dwelled in large coastal villages.

Most of these hierarchical foragers relied on great salmon runs that passed through the rivers, which they caught, smoked, and stored. Elites would also host decadent banquets. During these ceremonies one chief would host another, as well as many onlookers, and either give away or destroy a large amount of wealth in the form of food, canoes, and slaves. These ceremonies served many purposes, increasing prestige, cementing alliances, and destroying competitors. The guest chief was expected to host an equally lavish feast in return. A failure to do so could result in a loss of status and followers. The competing chief could even go into debt to host an adequately opulent feast. For some groups, anyone who coughed or laughed during the celebration would be coerced into throwing a feast themselves. The chiefs who oversaw these feasts and ceremonies claimed exclusive access to ritual knowledge and sometimes even heavenly beings. Some ethnographers have referred to them as 'terrorist organizations' that ruled through violence and black magic (which wouldn't put them out of place among empires, with the Roman Republic once being described as a mafia cartel). A leisure class of hereditary nobles with slaves competing through

banquets and dazzling aesthetic ceremonies looks more like the courtly estates of medieval Europe than hyper-egalitarian foragers.³

Though far rarer than mobile, egalitarian groups, these unequal, hierarchical foragers existed (and some still do) around the world. There were the Chinchorro (7000 to 1500 BCE), violent fishermen of modern-day Chile who mummified some of their dead, and the Jarildekald Aboriginal people of Australia, who fished along the Murray River. Both were sedentary, with bigger, more unequal populations than egalitarian foragers. One review of past and present foragers found seventeen examples of unequal, stratified hunter-gatherers, all of which were either semi- or fully sedentary.⁴ Anthropologists imagine a spectrum of foragers, from nomadic to sedentary, peaceful to warlike, equal to unequal. Nomadic egalitarian foragers are at one end, hierarchical sedentary hunter-gatherers at the other.⁵

Hierarchical hunter-gatherers are ancient. The first conclusive evidence for them is around 12,500–10,800 BCE. This is when the earth began to warm and exit the ice age. These were the early Natufians, who hunted and gathered across modern-day Syria, Palestine, and Israel. Their descendants would eventually be the first farmers and state-builders in the world.

As the climate changed, so too did the Natufians. They began to develop more permanent settlements, including large stone huts. They wore personal adornments such as shell, bone, and stone pendants, systematically buried their dead, produced large amounts of refined stone tools, and domesticated the dog. They may have also been the first to practise horticulture and animal husbandry. They collected wild cereals, legumes, and nuts, and hunted deer, aurochs, and gazelle.

The Natufians were also unequal. Some of the burials were lavish, with 8 per cent sporting pendants and beads imported from as far as 400 kilometres away. Some houses were larger than others and possessed larger stone mortars that are typically used for feasting. The original 1 per cent were foragers, not farmers.

Yet the Natufians were not purely foragers or farmers. They were on the path to farming through the collection of wild cereals. While we often say ‘farmers’ or ‘hunter-gatherers’ for simplicity, in reality most people after the Holocene almost always practised a combination of hunting, gathering, foraging, pastoralism, fishing, horticulture, and agriculture. The Calusa, for

instance, were fishers and foragers who also demanded tribute from part-time farmers. The first ‘farmers’ of the world in Mesopotamia, who followed on from the Natufians, spent thousands of years still supplementing their diet with fish, herded cattle, and hunted gazelle. Even English peasants in the thirteenth century relied on foraging and hunting in shared woodlands alongside farming. Nonetheless, the main resources a group relied on tended to shape how they organized themselves. And if the climate or availability of certain resources changed, societies usually did too.

This happened with the Natufians. Those first stirrings of a rich class – feasting and unequal burials – disappeared as the world temporarily re-entered an ice age around 10,900–9700 BCE (known as the ‘Younger Dryas’ period) and the Natufians returned to their more mobile ways. In the eyes of some archaeologists, this represents the first collapse in history. That is because they see the Natufians as the first stirrings of Goliath.⁶ The Natufians were just the forerunner of what was to come.

GOLIATH FUEL

Entering the Holocene (around 9700 BCE) meant entering a new world (see [Figure 7](#)).⁷ People began to settle into larger villages and become sedentary. Agriculture was adopted and intensified, as was the rearing of domesticated animals. Wealth inequality between people intensified. Goliaths – whether they be hierarchical hunter-gatherers or agrarian states – arose. This period after the ice age but before the first states is called ‘the Neolithic’.⁸

The usual story is that groups who developed a ‘surplus’ of resources through agriculture could afford to support administrators and a ruling class. There is undoubtedly some truth to this, but it isn’t the full story. Hierarchical foragers show that inequality, slavery, war, and hierarchy can all occur without agriculture. Many of these hierarchical hunter-gatherers also had only modest surpluses yet became mini-kingdoms. What mattered was the type of resource they used.

Hierarchical foragers, violent pastoralists, and farming kingdoms all gained access to resources that could be easily seen, stolen, and stored. In

other words, ‘lootable resources’. Large stocks of these lootable resources became available in the Holocene as the planet warmed and intensive agriculture and animal rearing became possible.⁹

The Calusa ate shellfish, molluscs, and different types of fish. Similarly, north-west Pacific coast Native Americans relied on salmon runs. The Natufians had access to patches of wild cereals, which they harvested and stored. Horseback kingdoms and hierarchical pastoralists prized horses and livestock (as well as the grain of the communities they conquered). Great runs of salmon, farms of shellfish, patches of wild cereals, domesticated animals, and (more ideally) tilled fields of wheat, rice, or corn, could all be easily monitored and controlled. When the resources were ready to harvest, a tax collector would know and could easily take a share, transport it elsewhere and store it. A commoner could not easily hide their flock or their harvest of cereals, salmon, or shellfish. One review of eighty-nine forager societies along the Pacific coast of North America found that clumped, defensible (lootable) resources, especially aquatic ones such as salmon, were associated with slavery, social ranking, and unequal access to food. It is a trend we see across the world: certain resources tend to lead to dominance hierarchies.¹⁰

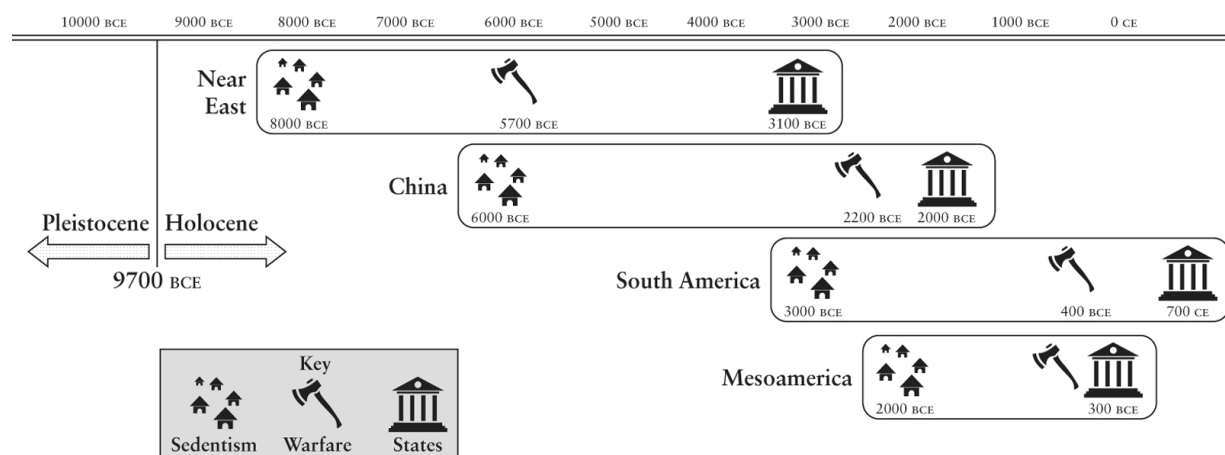


Figure 7: Post-Holocene social trajectories in Eurasia and the Americas

While the timing varies, most world regions followed a similar post-Holocene trajectory: sedentism (and more intensive agriculture) was followed thousands of years later by war and then the creation of states.

Lootable resources didn’t have to be edible. Some kingdoms could be built solely on controlling trade and strategic resources. The Nabataeans

were originally Arabic nomads who built a kingdom by taxing the traders who passed through their territory, which was on a popular route for transporting incense. Other early states and hierarchical hunter-gatherers have arisen in areas where they could control an unavoidable trade route (like the Nabataeans) or around a highly prized and concentrated resource like obsidian. Trade, like salmon, often had to pass through a choke point, whether that be a desert, mine, or mountain pass, making it easy for someone to establish an armed monopoly over it.¹¹

That monopolized wealth could then be passed down through generations. Before this moment, it was almost impossible for parents to bequeath their own status or advantages to their children. A study of twenty-one different populations across the world found that neither physical ability nor social connections could be easily transmitted across generations from parents to children, but material wealth such as land could.¹²

Lootable resources allowed for individuals and groups to become not just rich but also powerful. The four fundamental ways to gain power over others is to control valuable information (information power), to control by threat or force (violent power), to control the decision-making (political power), or to control the critical resources that others need (economic power). In an egalitarian forager group with few status differences, an ambitious status seeker could pursue one of these, but it was difficult if not impossible to achieve all of them. You could become a shaman and claim exclusive access to the spirits. You could try to bully others. You could use charisma to build a coalition of supporters. But having the charisma, brawn, *and* intelligence to be a master priest, negotiator, *and* hunter is rare, especially since each requires dedicated time, practice, and innate talent. The magnetic genius-prophet-warrior may be common in movies and comic books, but not in reality. Even if some preternaturally talented figure did emerge, they could not easily pass down their status and skills to their progeny. Being rich was essentially impossible since critical resources were spread far and wide across the landscape.

What changed between the egalitarian ice age and the rise of inequality in the Holocene was the use of loutable resources. Once you had resources that others depended on, you could leverage them for other forms of power. The economic power of access to stores of fish or cereal could be converted to other forms of power by throwing feasts, such as with the north-west

coast Native Americans, making others work for them in return for resources, or gifting them to create debts and obligations. The Calusa king was at once a religious head, a military commander, the supreme authority, and disproportionately rich.

Lootable resources were, in the words of economists, ‘fungible’: economic power could be transferred into other forms of power. Yet, salmon, livestock, and grain did not automatically create dominance hierarchies. A tyrant needed force to both take lootable resources and defend their stockpiles.¹³

Taking and defending resources required weaponry – ideally these new weapons or tactics would be concentrated and give one group an advantage over competitors. For instance, the few who had access to bronze swords and armour had a significant advantage over those fighting with wooden spears. Warriors on horseback could outrun or run over those on foot. Think of these as monopolizable weapons. Sometimes even minor changes could provide a competitive edge over others. The Chumash of central and southern California show no signs of inequality or warfare until after 600 CE, when they developed ocean-going canoes. This enabled them to hunt not just marine mammals but also their neighbours along the coast.¹⁴

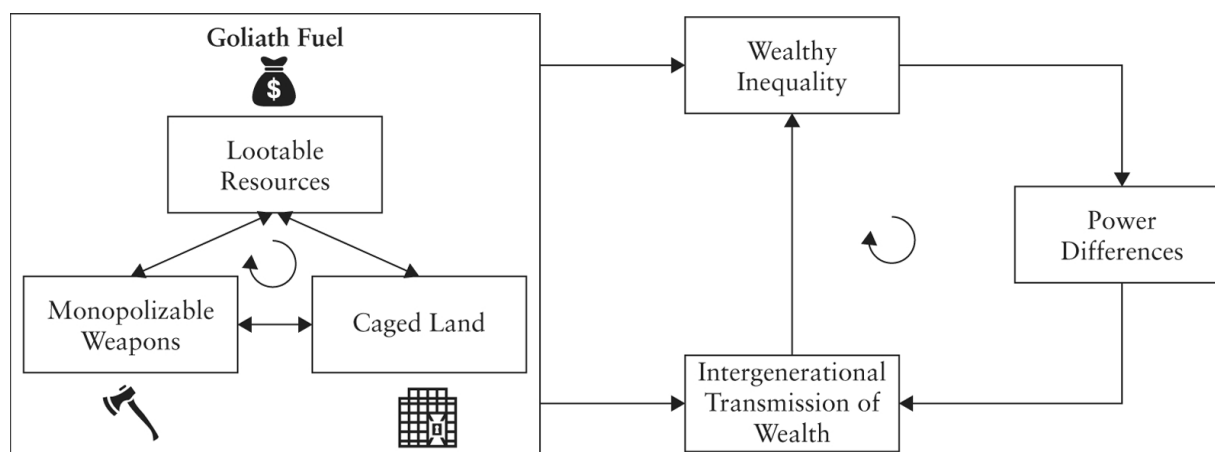
We see a similar story with the Comanche, a native American confederation of nomadic peoples who practised no agriculture. Instead, they relied on horses to raid their neighbours. Horses were introduced into South America by European invaders, and by the early seventeenth century had spread to North America via indigenous trade networks.¹⁵ Horses are not just great for transport; when combined with projectiles and the right tactics they become four-legged war machines. The Comanche put horses to good use, becoming hierarchical raiders who took hold of the southern plains and travelled as far south as Mexico.¹⁶ Monopolizable weaponry allowed the few to conquer and hold the resources of the many (and sometimes the many themselves as slaves). Monopolizable weapons were also a post-Holocene invention: horses were domesticated in central Asia around 3500 BCE and bronze weapons appear around 3300 BCE in the Near East.¹⁷ Yet neither lootable resources nor military might would lead to oppressive conditions if people could easily escape.

The geography of a Goliath is caged land: territory with few exit options. Exit options are people’s ability to vote with their feet and easily leave an

area. A bully or aspiring tyrant will be of little threat if people can easily flee in the middle of the night. Even if a dictator can lay claim to a set of lootable resources, they will be forced to negotiate with their subjects *if* the subjects can simply walk away, taking their labour with them. This is evident in the wider animal kingdom. Species with greater ‘dispersal’ options usually develop weaker hierarchies.¹⁸ The same is true of humans. For both the Calusa and north-west Pacific coast Native Americans it was difficult for people to move away easily. The salmon runs and watercourts were concentrated and not readily available elsewhere. People were trapped and tethered to the rivers and coastline if they wanted easily available food. The rich spots of cereals in the forests of the Natufians were scarce. When people couldn’t escape and were tied to their lands or herds, they were more likely to accept subordination.¹⁹

Just growing wheat did not mean a society would automatically become hierarchical, but more abundant lootable resources, monopolizable weapons, and fewer exit options together pushed people towards hierarchy ([Figure 8](#)).²⁰ Culture and choice played an important role too, but over the long term environmental conditions encouraged particular social arrangements and made them easier to sustain.²¹

Lootable resources, monopolizable weapons, and caged land are all sources of *Goliath fuel*. These are different conditions that allow one group to control another; that allowed a group or individual to gather the greatest resource of all: status. It took thousands of years, but our social arrangements started to look more like the dominance hierarchies of gorillas and chimpanzees. But while our social arrangements began to seem more primal, there was one interesting, unique twist: power became hereditary. Across other primates, status cannot be easily passed down from the parents to their offspring. The son of a powerful silverback is not gifted the harem once his father passes away. For us, resources and claims to a throne can be inherited. In 2024, all billionaires under the age of thirty had inherited their fortunes and all the status and power that accompany them.²²



[Figure 8](#): From Goliath fuel to Goliaths

Goliath fuel allowed for large differences in wealth and power that could be accumulated and passed down through generations.

We were democratic, egalitarian foragers until Goliath fuel appeared in the Holocene. The transition from egalitarianism to dominance hierarchies thereafter was not smooth or straightforward. It was a bloody affair.

THE ORIGINS OF WAR

Warfare emerges as Goliaths develop. Violence develops from one-on-one fights, through small groups raiding and skirmishing, to organized armies waging mass warfare as societies become more unequal and centralized.

We see this development of war across the world. In the Oaxaca Valley in Mexico, groups appear to have lived in peace for 6,000 years (8000–2000 BCE) before the first villages and settlements began to appear. Just over a millennium later (800–450 BCE) signs of violent raids appear alongside the first hierarchical societies. War arrives alongside the first state, the

Zapotec.^{[23](#)} In Japan during an earlier part (13,000–800 BCE) of the Jomon Period there are no fortifications or other signs of warfare. Then weapons and defensive structures begin to appear alongside the first signs of

agriculture.^{[24](#)} In eastern North America there were six millennia (11,000–5000 BCE) of just interpersonal homicides. This was a time of egalitarian, mobile foragers. It was then another four millennia (5000–1000 BCE) of small-scale raiding as groups became stratified and settled into towns. Then

war took root alongside agriculture, large villages, and societies based on domination and tribute.²⁵

In Europe, there are no signs of war during the Palaeolithic. Around 9500 BCE, after the entry into the Holocene, evidence of lethal violence begins to become more common. After around 5500 bce, as agriculture spreads across Europe, people settle down, and inequality intensifies, clearer indications of warfare, including fortifications, towns surrounded by ditches, and evidence of massacres, all creep upwards.²⁶

The Near East follows a similar trajectory. The Natufians killed each other at low rates and didn't conduct war. For thousands of years we find just a few skeletons with fractured skulls and embedded projectiles. Then in the seventh century BCE, perhaps the first fortification in the world – Tell Maghzaliyah in northern Mesopotamia – was constructed near a node of long-distance obsidian trade. Thereafter, trade began to shift from obsidian to flint, probably a reaction to someone restricting the supply of obsidian. Around 5700–5600 BCE the Halafians, a culture from northern Mesopotamia with a keen interest in the obsidian trade, began to leave behind mass burials. The Halafians then moved south and built fortified settlements along an east–west overland trade route (mainly for obsidian). They razed pre-existing towns and built over the top of them.²⁷ The war for lootable resources had begun.

Societies, in different environments and even on opposite sides of the world, all follow the same pattern. Organized warfare arises as groups become more unequal, sedentary, and hierarchical.

This pattern is supported by artwork. After 8000 BCE, as settlements, hierarchy, and farming spread across Europe, China, and the Near East, cave art becomes increasingly populated with images of spear- and bow-wielding groups ([Figure 9](#)).²⁸

The relationship between organized conflict and hierarchy is also clear in modern hunter-gatherers. One systematic analysis of thirty contemporary forager groups (including twenty-one mobile, egalitarian ones and nine settled, non-egalitarian ones) found that group-on-group violence was correlated with bigger, denser, and more sedentary groups.²⁹ This is just one study in a larger literature on the correlation between violence and inequality.³⁰

This was not just about more people being involved in conflict. It was a transformation of how and why we killed each other. Among egalitarian hunter-gatherers, murdering another human is a personal affair: it usually results from a quarrel or sexual jealousy, or perhaps the solemn execution of a budding tyrant. In contrast, the larger-scale massacres we see after the end of the ice age often involve the indiscriminate killing of entire communities, including women and children. These atrocities also frequently involved torture and mutilation. This is noticeable in a range of cases, including the first farming communities in Europe.³¹ The target of violence shifts from an individual wrongdoer to an entire group. It doesn't matter whether you kill a particular individual since you can replace them with another member of their group. The victim becomes interchangeable.³²



[Figure 9](#): A cave art battle scene (5000–3500 BCE) found in Sefar, southern Algeria

War transformed the very character of violence because the motivations for violence changed. Conflict was no longer about personal grievances and disagreements but about eliminating a threat, procuring resources, or, most importantly, expanding the power and status of a group and its leader(s).

Who you killed became less important and violence became more indiscriminate. As we can see with the early signs of torture, mutilation, and genocide, it also became much more brutal.

War, or any large-scale act of violence, has three key ingredients, all of which increased in the Neolithic. First, it needs a large-scale organization to corral and coordinate troops. Second, it needs an ideology to inspire them to fight, whether it be communism or nationalism. Third, it needs to create tight bonds between groups of fighters (usually young men). Ideologies that justify bloodshed and tightly connected groups of fighters don't come naturally. They require extensive organization, training, and indoctrination. That is something the first hierarchies and states started to offer. It is something that the militaries of today provide even more successfully. Rather than violence decreasing across history, the ingredients for warfare have steadily increased, starting with the warmer, bloodier world of the Neolithic.³³

It's easy to see why the Palaeolithic was warless. There weren't the large-scale organizations to coordinate mass killings, to craft ideologies encouraging battle, and to train small groups of young men to become efficient murder squads. An ambush, raid, or full campaign requires planning, commitment, and the motivation to risk one's life. It wasn't possible, nor was it worth risking death to strike another group that had no accumulated riches and with whom you were probably trading. War was pointless without lootable resources to battle over, or the advantage provided by new battle aids such as horses and Bronze Age swords.

Some have suggested that we may have gone to war during the Palaeolithic over women. One group would assault another to capture and carry off new wives and partners. But intermarriage and continuous migration across groups would have prevented the need for bloody raids to capture women. We also simply don't see modern egalitarian foragers commit these kinds of raids.³⁴ One study of twenty-one nomadic forager societies found that only 0.7 per cent of the cases of lethal violence were due to the theft of women between neighbouring groups (and only 1.4 per cent were due to disputes over resources or territory).³⁵

Naturally, the skeletons we have dug up from the Palaeolithic are only a small fraction of the people who lived and died. Nonetheless all the evidence – skeletons, cave paintings, and cultural zones of open trade –

points to one conclusion: organized warfare came only after we entered the Holocene.³⁶

EVOLUTIONARY BACKSLIDING, OR DROPPING THE PROGRESS NARRATIVE

War, inequality, patriarchy, and slavery are probably not what come to mind when you hear ‘progress’. Nor is it easy to see the move from nomadic egalitarian hunter-gatherers like the Khoisan towards groups like the Calusa as a social advance. Yet this is how these developments have historically been framed.

Our ideas of progress have been shaped by nineteenth-century thinkers such as Lewis Henry Morgan. In 1877 Morgan pioneered an influential three-step blueprint in which societies moved up the staircase of progress, with abrupt changes in technology and society marking the steps from one stage to another. ‘Savagery’ was marked by a reliance on hunting, gathering, fishing, and simple technology such as the bow and arrow. ‘Barbarism’ involved horticulture, domesticated plants and animals, as well as new practices such as pottery. ‘Civilization’ was a stage where groups adopted intensive agriculture, towns and cities, and innovations such as writing.³⁷

We now know that things are not that simple. Hierarchical hunter-gatherers had many of the hallmarks of what most would see as civilization (sedentism, inequality, rulers, war, and more monuments) yet relied on foraging.³⁸ They were denser, captured more energy, and had more hierarchy but didn’t farm. Some groups jumped straight from being mobile foragers or herders to being states. Other groups, such as the north-west coast Native Americans, stayed hierarchical without ever becoming a state. Many regularly collapsed in a continuous process of fission-fusion, coming together under a leader and then splintering apart, on either a seasonal or more random basis.³⁹ There is no stepladder.⁴⁰

Because of these problems with the stepladder approach many historical scientists don’t talk about progress, savagery, barbarism, or civilization. Instead, many now talk about an increase in ‘complexity’. In the eyes of

many archaeologists the Calusa and north-west coast Native Americans would be classified as ‘complex hunter-gatherers’, while the Khoisan would be less complex or ‘simple’ hunter-gatherers. On that scale of complexity, modern, industrialized societies are – of course – the most complex. Most of the well-known books on collapse, whether Jared Diamond’s *Collapse* or Joseph Tainter’s *The Collapse of Complex Societies*, define collapse as a sudden and enduring loss of social complexity. It is, in essence, falling down the stepladder of progress.⁴¹

While complexity may seem like a more savoury alternative to talking about progress and civilization, the two are still largely interchangeable. One of the most accepted and common definitions of early civilization is ‘the earliest and simplest form of class-based society’.⁴² Similarly, many archaeological definitions of complexity focus on hierarchy, inequality, and centralization. It is about a small group controlling the labour and resources of others. For many archaeologists, the first signs of rulers, a military, bureaucracy, and inequality are the first signs of complexity.⁴³

Characterizing the move towards hierarchical hunter-gatherers and states as an increase in complexity doesn’t make sense. Many democratic foragers have more complex politics than the Calusa or even the Roman Empire. Hierarchy, for example top-down orders from a Roman emperor, is intended to simplify decision-making. By contrast, decisions among democratic foragers require extensive debate, negotiation, and compromise among equals. Something is more complex when it has more diverse and interconnected parts, not when one part controls the others through violence.

This archaeological idea of complexity is also oddly selective and leaves out most of what makes up a society. For instance, look at the Australian Aboriginal peoples. These groups possessed at least 250 languages prior to European invasion, practised ingenious environmental management tools such as firestick farming (deliberate, controlled bush-fires), and have a varied and rich ‘dreamtime’ set of beliefs that aid coordination, navigation, and their understanding of the land. Yet if they dropped all the languages and started speaking English, adopted a few staple crops for farming, built a big bureaucracy rather than relying on subtle cultural practices, and dropped their rich cosmology in favour of worshipping a single god-king, they would apparently be becoming more ‘complex’. They are clearly not: they

are getting more hierarchical, capturing more energy, and living more densely. Yes, bureaucracies and monument-building might get bigger and more complicated, but much of life, from diets to culture, is simplified. The use of complexity is not just inaccurate; calling a society simple is akin to accusing its people of backwardness.⁴⁴

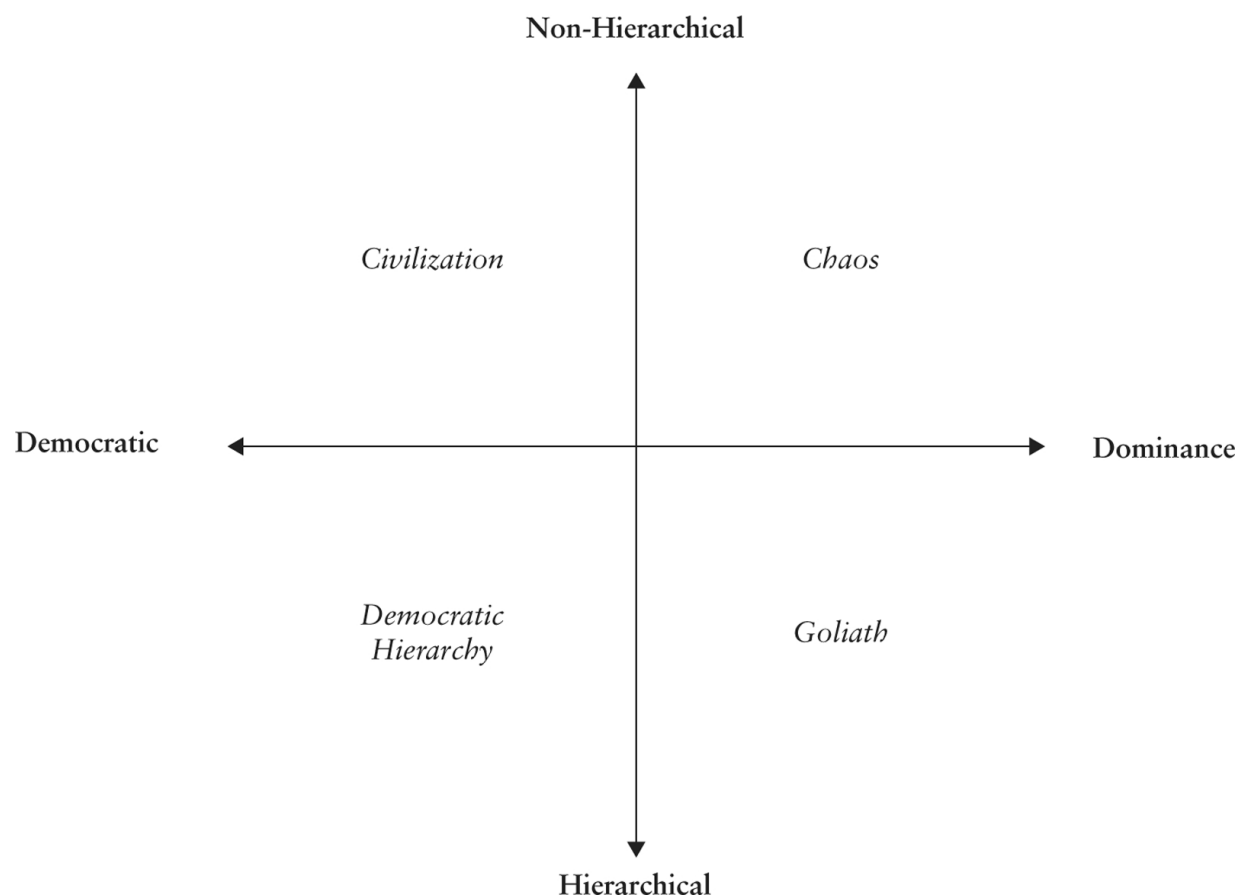
What we are seeing here during the Neolithic is not progress or increasing complexity but rather the painful growth of a Goliath. A slow, reversible, and messy shift towards social relations built on dominance hierarchies. Hierarchy is when some are ranked above others in status and authority. We are surrounded by these: CEOs are above managers who are above workers.⁴⁵ A hierarchy can be more democratic (with more people sharing decision-making power), or more dominance-based (with just a few wielding power backed by violence). Dominance hierarchies are imposed by the powerful for their own benefit, usually through violence.

Such dominance hierarchies are essentially the opposite of the fluid civilizations of the Palaeolithic, which had non-hierarchical, democratic cooperation. In the Palaeolithic our hunter-gatherer ancestors relied on trade, hospitality, reciprocity, and mutual aid to cooperate, maintain the peace, and survive the ice age. They were practising what we usually think of as ‘civilized conduct’: restraint, political wisdom, and cooperation without coercion. Violence is still used in civilizations (remember the counter-dominance executions of the Khoisan) but is only legitimate when it’s done collectively as a last resort, as opposed to being regularly imposed top-down.⁴⁶

Rather than a stepladder of progress, this movement from civilization to Goliath is better described as evolutionary backsliding. Our egalitarianism and counter-dominance shaped our bodies and minds and supercharged our cultural evolution. They made us unique and helped us navigate the Palaeolithic. The move towards Goliath – whether it be through class or patriarchy – made us look more like the harems of patriarchal gorillas and the chimpanzee hierarchies built on violence and politicking.

Goliath and civilization are not air-tight categories; they are a spectrum. It is about the main way in which cooperation is organized in society: hierarchy, violence, and rule by the few; or democracy and voluntary cooperation. Whether we work together at the end of a handshake or the barrel of a gun.

There are other types of social forms beyond the civilization–Goliath spectrum. The temporary hierarchies we see in egalitarian hunter-gatherers, such as in hunting parties, are democratic hierarchies: the members choose a temporary, restrained ruler to achieve a common goal. Leaders exist for the public good, rarely have violent power, and are always accountable to their followers. A good way to distinguish between democratic and dominance hierarchies is the consent test: if you say stop and the hierarchy continues regardless and any resistance is met with force, then you are existing in a dominance hierarchy.⁴⁷ One could also have an arrangement in which everyone is trying to relentlessly and violently dominate, but no one ever succeeds. No dominance hierarchy is ever achieved: think of this as ‘chaos’. It’s what Hobbes was envisaging in his fictional ‘state of nature’. For a pictorial overview, see [Figure 10](#).⁴⁸



[Figure 10](#): Forms of human society

Some countries, such as the liberal democracies of Scandinavia, are more like democratic hierarchies. Yet these democracies rely on the flimsy idea that choosing between a few different potential rulers every four years or so is a sufficient degree of power sharing. The party with the most funding tends to win and governments rarely represent popular opinion or implement policies with the strongest popular support. The ancient Athenians would have recognized our current systems not as democracies, but as oligarchies.⁴⁹

This is at best a broad yet weak version of democracy, one in which the rules are still applied using brute force. All other areas of life are still organized into dominance hierarchies. Most of us work nine-to-five in mini-dictatorships (where one individual or a small group can hire, fire, and direct most others), pray in top-down churches, and invest in markets controlled overwhelmingly by a few big firms. Even the citizens of Norway, the US, or Australia still live in a Goliath. Indeed, humankind now lives in a single Global Goliath.

The Goliath of today is the product of a 5,000-year process. One that was marked by setbacks, reversals, and collapses.

THE INSTABILITY OF DOMINATION

In northern Iraq lies a hill jutting up from the desert, and inside it are dozens of layers of settlements. When one was destroyed or decayed, another would be built over it. The layered villages of Tepe Gawra can give us a glimpse into just how dangerous the early Holocene was.

The first village was created 6,500 years ago, and over time became increasingly unequal. Around a thousand years after its creation, three great mudbrick temples painted red and purple were erected. Then they were abandoned. Thereafter, the site was turned into a fortified village with a single narrow entry road overlooked by a watchtower. Its defences didn't work. The village was raided and burned, with its largest elite house hit hardest. Inside, a baby and a twelve-to-fourteen-year-old child were buried under 15cm of ash. The town remained abandoned for a century. It was eventually reoccupied. And then burned down again. The cycle repeated itself in the centuries thereafter.

Tepe Gawra is emblematic of the fate that befell many settlements in the early Holocene. Inequality encouraged more people to turn to plundering their neighbours. Villages, towns, and cities became prime targets. Other times, there were no signs of invaders. Sites were constructed, grew more unequal, and were simply abandoned. Why would people just walk away en masse from a site that they had tied their lives to?

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Abandoned Experiments I: Failed Farmers

A Slow and Resisted Revolution – Why Even Bother? – Growth Fetish – Reversals of Agriculture – Cursed by Grain

A SLOW AND RESISTED REVOLUTION

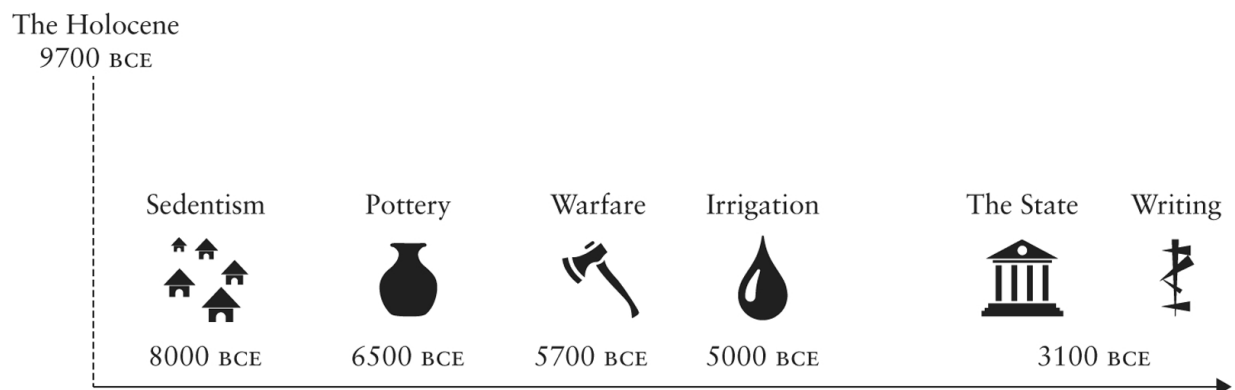
While the ‘agricultural revolution’ was revolutionary in its consequences, it was not so in its pace. The transition from agriculture and stationary living, through cities and inequality to states, was a long, protracted affair. Each step took thousands of years and there were many reversals.

Mesopotamia is the earliest and best-studied example of this transition. Twelve thousand years ago, the Late Natufians started experimenting with seasonal settlements. Then they collapsed and dispersed as the climate changed. Around two millennia later the first signs of full-time sedentary living begin. Pottery becomes commonplace 1,500 years after that. About another 1,500 years later there is the first undisputed evidence of warfare.¹ Then, two millennia later, the first state – Uruk (3200 BCE) – is built and the first cuneiform documents are written. This is the slow, arduous development of a Goliath ([Figure 11](#)).²

Across the world this sequence repeats itself again and again, though the details and timing vary. For instance, South America was a late bloomer but experienced a more rapid transition. It didn’t see sedentism until around 4,000 years ago and states emerged 2,600 years later. On average, the lag between the adoption of intensive agriculture and states was approximately

3,000 years – a period greater than that which separates today from the beginnings of Classical Greece.³

Merely moving from part-time to full-time farming was a lengthy process.⁴ In most fields in south-west Asia, it took over 5,000 years for the share of grasses (such as wheat, emmer, and einkorn) to increase from around 20–30 per cent to approximately 50–60 per cent of crops grown.⁵ That is roughly the same period that separates today from the first states in Egypt and Mesopotamia.



[Figure 11](#): The Mesopotamian post-Holocene social trajectory

While the exact sequence varied, sedentism and agriculture often went hand in hand. As one became more dependent on tilled, fixed-field agriculture, it usually became more sensible to stay put as the fields needed the full-time work of sowing, weeding, and fending off pests. An agrarian community was more likely to develop a larger population and eventually become a city. In most of Mesopotamia and Egypt, people started to settle down in sedentary communities while beginning the slow process of intensifying crop cultivation. Still, it was entirely possible to be sedentary without having agriculture, as we've seen with numerous cases of hierarchical foragers.⁶

The adoption of agriculture was not always an inevitability. Numerous areas with suitable ecological conditions and foragers who were almost certainly aware of cereal farming never adopted it, such as in Native American California. When farming did spread, whether it be from Turkey into Europe or Mesoamerica into the south-west US, it was often a stop-start affair. Arrival into a new region would often lead to a pause of a few centuries before it spread once again. The lack of uptake wasn't due to the

ignorance of the hunter-gatherers surrounding farmer communities. The farmers and their fields would have been hard to miss, and in some cases hunter-gatherers were even having children with them. The picture is one of foragers who were reluctant to adopt full-time farming.⁷ Indeed, the strangest conundrum is why our ancestors chose to adopt farming and sedentary living at all.

WHY EVEN BOTHER?

From our perspective, it's easy to assume that agriculture and cities were steps forward in the march of progress. But in the beginning at least, they were a disaster for health and well-being.

If living in villages and cities meant more disease, noisier neighbours, and a greater threat of being raided, then why bother? We don't fully know the answer, and multiple factors must have played into the decision to stay put. Over the space of centuries and millennia people started spending more time in seasonal camps. They may have used these temporary camps for trade, rituals, or better access to seasonal food and migratory herds. People then became tied to the land as populations grew and they became more dependent on agriculture.⁸

The vast intensification of agriculture during the Neolithic is more difficult to explain. Hunter-gatherer diets are substantially healthier and involved more enjoyable labour than those of farmers. This decline in health and diet is reflected in the height of people across the world. Height is a decent indicator of general physical health. Eating more calories, more protein, and a wider variety of foods tends to lead to a taller, healthier population. What we see in skeletons during the transition to agriculture is a decrease in height. While studies vary, most see a reduction of a few inches in height between our ice-age ancestors and those who took up farming. This shrinkage in stature is not the only marker of a decline in general health.⁹

Agriculture meant a more restricted diet that was lower in protein and nutritional quality. As people became increasingly sedentary, exercise and physical activity were replaced by the repetitive physical stress of a farmer's lifestyle. A telltale indicator of an early agriculturalist is deformed

bones in the knees and toes of women. These are the results of countless hours spent rocking back and forth on one's knees to grind grain. Sedentary communities that became increasingly reliant on grain also suffered from substantial enamel defects, dental cavities, and oral infections. Agriculture was a wrecking ball to human health and height, and it also meant more work.^{[10](#)}

Hunter-gatherers are also renowned for being work-shy. They produce far less food than they are capable of producing and choose to maximize leisure time instead of working harder to produce a surplus. They also tend to view farming as unnecessary drudgery. As one Ju/'hoansi told an anthropologist, 'Why should I farm when there are so many mongongo nuts?' Australian Aborigines have a similar ethos: 'You people go to all that trouble, working and planting seeds, but we don't have to do that. All these things are there for us.'^{[11](#)}

Across the world, foragers tend to act in a way that minimizes the expenditure of energy. For instance, when population density is low, they hunt big game, which often provides the biggest caloric return, before moving on to collect tubers, plants, and animals that offer fewer calories for the energy expended. We can still see this in farmers who tend to practise less laborious forms of agriculture, such as shifting agriculture (in which a field is used, burned, left fallow and then returned to later), unless they are compelled to produce more by an external force, such as a market or master. Similarly, pre-capitalist peasants would often take up wage work to meet a certain target (for instance, to pay for a wedding) and would stop working as soon as they reached it. If they were offered higher wages they would simply work less, because their target would be reached sooner.^{[12](#)}

It is a pattern that completely contradicts conventional economic logic (you *should* want to work more when the wage is higher). Farmers also often tend to return to these less onerous ways of producing food whenever a collapse occurs, or when they are trying to evade a state. This is known as 'the principle of least effort'. Moving from this to intensified agriculture is a radical change, not just in how one produces food but also in priorities. It's unlikely they made such a change unless compelled by great benefits or penalties.^{[13](#)}

Farming does offer benefits. It can support denser populations and sometimes provide more calories per hour of work. But it's hard to see why

the average hunter-gatherer would opt for more calories and bigger families over better work, food, health and more leisure time. Even if they wanted to, it wouldn't be an easy switch. Farming requires highly specialized knowledge about seasons, crops, fertilizers, and techniques that take painstaking time and effort to accumulate. It is unsurprising that we rarely see hunter-gatherers voluntarily adopt farming. Instead, they are usually displaced or killed off by, or intermingle and marry with, the intruding farmers.¹⁴

Still, there clearly remained some populations that did adopt and expand agriculture, even if slowly and reluctantly. Why? One hypothesis is that population growth led to overhunting in some areas, such as the Near East. Around Mesopotamia, big game dwindled and even became extinct during the early Neolithic. Bigger communities with less protein-rich food had to rely more and more on calorie-dense and predictable agriculture, as well as on domesticated animals. They probably started by adopting easier farming methods, such as flood-retreat agriculture. This entails waiting by a floodplain, casting some seeds in the soil after a flood, and reaping the harvest with little effort. That helps to explain why the earliest Goliaths arose around flood-prone rivers, such as the Nile in Egypt, the Yellow River in China, and the Tigris and Euphrates in Mesopotamia. Something then gave them a gentle push towards relying more on agriculture. It could have been a change in the climate or the collapse of other food sources, such as large game.¹⁵ From there, population and agriculture reinforced each other with farmers generally being more fertile and agriculture offering more support for big, dense populations.¹⁶

Population may have been one factor that pushed us towards farming, but it's not a complete explanation. One problem is that most areas that developed intensive agriculture were areas of abundance, not scarcity. For instance, the earliest Goliaths in Mesopotamia, as well as the cities of Tiwanaku and Teotihuacan sprang up in rich wetland areas. Another issue is that many early cultivated crops are not staples but more luxury foods, such as avocados and chilli peppers in central America.¹⁷ Clearly, something more than just population growth and a lack of big game was behind the intensification of agriculture.

GROWTH FETISH

Status competition can help provide a more complete explanation of the great intensification of food production. Signs of feasts, such as the potlatches of north-west coast Native Americans, appear all over the world after the Holocene. Neither avocados nor chilli peppers are commonly eaten, but they're regular foods for feasts among the highland Maya and others. That was the allure of these foods: avocados and chillis were all scarce, high-status foods fit for a feast. Dogs, too, may have been domesticated initially as high-status objects rather than four-legged helpers. Throwing banquets is primarily an act to gain status and power: it can aggrandize an individual or family, cement alliances, and even create debt peonage. The catch is that feasts require more food and more work for everyone in a community. Hierarchical hunter-gatherers across the world marginalize those who refuse to work harder to support banquets. Dissenters are harassed, ostracized, ridiculed as 'lazy moochers', 'rubbish men', and 'abominable people' and are even physically attacked.^{[18](#)}

The archaeologist Brian Hayden has compiled decades of work to show how the transition to agriculture wasn't (just) a matter of more people demanding more food, but rather a few status seekers cajoling, corralling, and coercing others into producing a surplus. Once lootable resources became accessible, cycles of feasting both exacerbated inequality and demanded ever-greater quantities of food be produced. Nomadic-egalitarian groups, too, often hold collective banquets, but these seem to be more for group bonding rather than the aggrandizement of a single individual or family. For many, feasting morphed into a tool of politics and status competition.^{[19](#)}

Status competition can help us make sense of not only the adoption and intensification of farming, but also of many of the other trends we see in the earliest farming communities and cities. When archaeologists look for signs of inequality they search for feasts, human sacrifices, items imported from vast distances (even further than hunter-gatherer trade), unequally sized houses, great monuments such as palaces, and grand burials where an individual is left with a disproportionate number of luxury items such as jewellery, weaponry and even sacrificial victims.

Two things unite these indicators of inequality: they all signal distinction and are of little practical use. More than being unnecessary, they are detrimental for a group trying to live in a landscape with a paucity of food or an abundance of enemies. These activities are now what we call ‘conspicuous consumption’: wasteful spending to enhance social status. It is the inverse of the principle of least effort. If conserving energy is a common intuition, then wastefully using resources and the labour of others is the ultimate display of power. Having others expend time and energy building a palace or a grand burial also helps to reinforce social divides. Nothing says ‘I’m more important than you’ like having masses of people spend their living days toiling to create a tomb for a dead leader.²⁰

That wasteful consumption to gain and signal status easily becomes a preoccupation with growth. If you can improve your status through throwing a feast, then an even larger feast with a greater variety and abundance of foods will be even better. If you are competing with another chief, then there is an even stronger compulsion to grow a bigger crop to outdo them. The same goes for importing exotic jewellery from thousands of kilometres away, conquering enemies, or building a great tomb. The bigger the waste, the more status it signals, the wider the gap between you and the others it creates and the greater the chance you will be remembered. Today we still tend disproportionately to know of and value the biggest monuments of the past: the Colosseum, the Great Wall of China, and the pyramids of Giza. Unfortunately, status is an inexhaustible resource and competition is endless. Hence, wasteful consumption becomes a fixation on material growth and accumulation: a growth fetish.

Wasteful consumption and inequality were often not warmly welcomed, as we see in the first farming communities across the world.

REVERSALS OF AGRICULTURE

Perhaps unsurprisingly, settling down and eating grain was far from universally accepted – or irreversible. Fixed-field farming and sedentary living were adopted and abandoned on multiple occasions throughout the early Holocene. People rebelled, left, and went to war as the social arrangements accompanying farming became more lopsided and

exploitative. Nowhere does the transition towards agriculture look more chaotic and bloodier than in Europe.

Agriculture did not begin in Europe. Instead, it was imported from modern-day Turkey through the Mediterranean and further north into Britain and Scandinavia. Grain-based lifestyles appear to have first arrived in south-eastern Europe around 6500 BCE. Genetic data suggests that native Europeans rarely adopted farming practices. Instead, the intrusion of farming came via migrants originating in modern-day Turkey and Greece.²¹ Farming practices had expanded into the area of modern-day Paris by approximately 5200–5100 BCE and took another millennium before they reached Ireland, Britain, and Scandinavia.²² The time it took for farming to spread from northern Turkey to Ireland was more than the same span of time that separates us from the time of Jesus of Nazareth. This was not a simple success story of the inexorable march of grain (see [Figure 12](#)).²³

What followed was a boom-and-bust pattern of population growth and collapse across Europe. Populations spiked, before dropping back to preceding levels, or even dramatically below them.²⁴ Farmers were arriving, establishing a community, flourishing in numbers over several generations, and then suddenly dying off or dispersing. Some settlements just declined, while others were completely abandoned.

How long it took to go from boom to bust varied. For Ireland and Britain, the arrival of farming (around 4000 BCE) was followed by a drop-off in population after 300–400 or so years. Similarly, in Scandinavia a peak was reached after 400 years, then populations dropped by some 70 per cent in the next 300 years. In regions across central Germany there was a slower population decline that took almost a thousand years.²⁵

Despite differences in regions, all of them experienced some depopulation.²⁶ This was often accompanied by changes in diet and agriculture. In Britain and Ireland, the initial population surge is associated with increased use of wheat, while the decline sees wheat replaced with hardier barley and hazelnuts.²⁷ Eventually almost all the grain was phased out in favour of hazelnuts.

Changes in population and diet were accompanied by another, more disturbing, trend. As farmers settled down across Europe, their lives

changed. Prior to the arrival of farming, there are no signs of large-scale violence or fortifications. Across the different communities there is then evidence of wealth accumulation, inequality, and defensive structures such as walls and palisades. In southern Britain we see the slow build-up of defensive structures such as enclosed causeways. These are torn down or abandoned once the population falls.

People didn't just pick up the plough; they also picked up the sword. While interpersonal violence wasn't absent from Neolithic Europe before farming, it was low, with no signs of warfare. This changed as farming progressed. Around 11 per cent of skeletons from a sample of 2,363 individuals in Neolithic Europe have some form of cranial trauma. As signs of inequality and stratification increased, violence intensified. Mass burials begin to appear. Demographic collapses in Germany and Austria saw at least three massacres numbering twenty-six, thirty-four, and sixty-seven victims. In some areas, such as Hesse in central Germany, there were signs of torture and mutilation.²⁸ The victims included women and children, suggesting it didn't matter whom you killed so long as they were from the opposing group. If this was a battle for scarce productive land then mass killings would help to dislodge existing communities, making it easier to colonize. In other mass burial sites across Germany and Europe there is a conspicuous lack of remains from adolescents or women. Many archaeologists interpret this as evidence that they were taken as slaves and spoils of war. The massacres were sometimes brutal and systematic enough to have wiped out entire communities. This period may represent a 'high watermark of human violence'.²⁹ Farmers from Turkey and Greece didn't just bring grain to Europe: they brought war and Goliath.

These bloody agrarian cycles happened not just in Europe. Across the world in North America before European colonization we see the same pattern unfolding in an entirely different landscape, with an utterly different culture. In the arid deserts of the south-west United States lie the ruins of clusters of sandstone houses. They seem impossible: great houses jutting out from such an inhospitable terrain. They were the 'pueblos' (Spanish for 'village') of the Ancestral Puebloan people.³⁰ Chaco Canyon, one of the grandest locales, houses buildings that appear to be aligned to capture solar and lunar cycles. They were the largest non-earthen buildings constructed in North America until the nineteenth century. These were a people founded

on a trio of crops that would become known as ‘the three sisters’: beans, squash, and maize. All of these were imported from the southern lands of Mesoamerica. Maize was especially important, constituting up to 90 per cent of their caloric intake. These resilient peoples settled down in villages, took up increasingly ritualized, hierarchical arrangements, fell into infighting then frequently left their increasingly unequal settlements.

Puebloan settlements in key parts of the Mesa Verde experienced four boom-and-bust cycles. The busts were marked by drops in wood cuttings, maize harvests, and population. These periods of crisis occurred in 700, 890, 1145, and 1285 CE. Each crisis saw a preceding increase in inequality, followed by spikes in violence, depopulation, and then declines in inequality. Each of these crises also coincides with a drying of the climate. The final termination of 1285 was catastrophic.

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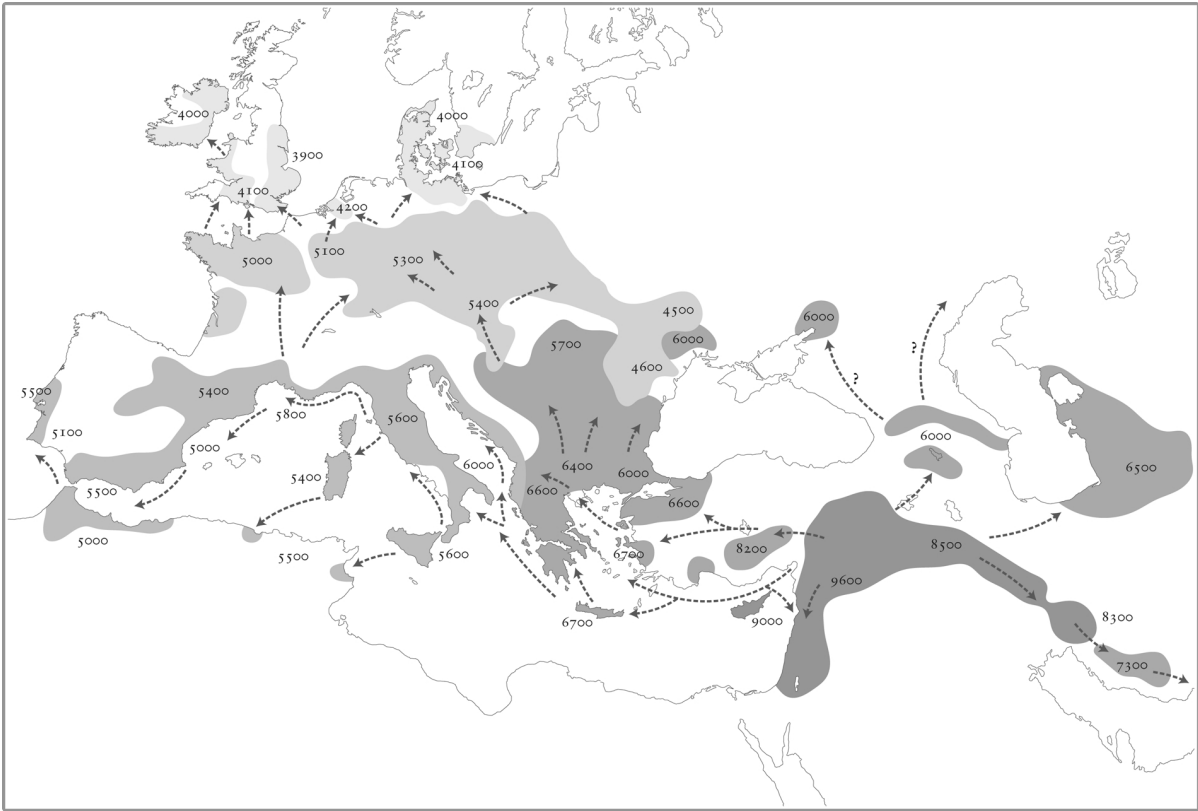


Figure 12: The spread of the first farmers of Europe

It took over 4,000 years for farming to spread from Turkey to the British Isles. The map shows the approximate dates of farmers' arrival into different areas of the continent (all dates are BCE).

In 1245 there were around 25,000 farmers in the northern south-west of the US. Forty years later there were none. Climate change, in the form of drought, played a key role in cutting maize production. In the first three downturns, the droughts were not decisive; indeed, there were many similar climatic fluctuations that the Puebloans had previously weathered with only a few losses. The fourth drought was more severe. Across these boom–bust cycles is another recurring feature. In each case the Puebloan communities saw rising wealth inequality, more exploitative conditions, and falling social cohesion. These deteriorating social conditions were the vulnerability and climate change was the threat. When the two coincided, things fell apart.³¹

After the final depopulation of the northern south-west US by farmers in 1285 CE, people moved south and east to occupy larger, more peaceful towns. These towns were placed in areas that were less reliant on variable rainfall for their farming. This way of life proved to be far more resilient to swings in the climate. Despite Spanish and then American colonization, the Puebloans remain in their ancestral lands today.³²

CURSED BY GRAIN

There are a few different reasons why the first experiments with farming in North America, Europe, and elsewhere experienced such cycles of collapse. All of them hinge on grain creating larger populations and bigger problems.

It should come as no surprise that climate change played a recurring role across so many of these early collapses. Bigger populations with less mobility and more restricted diets made these first farmers more vulnerable and reliant on good weather. When that weather didn't come, disaster could follow. The termination periods of the Ancestral Puebloan peoples coincided with drought, and the swings in population among the first farmers of Europe also show a loose relationship to the climate. The onset of wetter and cooler conditions in Britain and Ireland had a deleterious impact on food production and coincided with the population collapses.³³

Farming communities were also the first in which humans, animals, and crops were packed together along with commensals including mice, rats, and crows, as well as a host of microparasites such as ticks, fleas, and mosquitoes. This cramming together of species created a hotbed for zoonotic infections. These are diseases that jump from animals to humans. Many of the plagues of the world today, such as influenza, measles, mumps, cholera, smallpox, chicken pox, and, of course, Covid-19 are all recent developments from the past ten thousand years. Our hunter-gatherer ancestors never had to endure the throaty cough of influenza or the painful and disfiguring scabs of chicken pox or smallpox, which emerged only with the advent of urbanism and agriculture. These ‘density-dependent’ diseases require intense interaction between humans and animals, something many of the earliest cities provided in abundance.³⁴

Such diseases also need a sufficiently large population so they don’t burn themselves out by killing all their hosts. If a new infection passes through the population, killing half, then the remaining half will be immune. A large population and high birth rate are needed for the disease to continue to infect and mutate. For smallpox, this is estimated to be a population of around 300,000. Early farming settlements would have provided enough of a crowd of people for some diseases to become self-perpetuating.³⁵

Disease has been the scourge of cities throughout history. In London, for every year until 1800, there were more deaths than births. The city required a constant infusion of new people to survive. This was due in large part to the spread of typhus, syphilis, cholera, and the plague (caused by *Yersinia pestis* bacteria). And the same is true for many cities throughout history.³⁶

Bigger groups also meant that there was more chance of a disease both sparking in a population and spreading like fire. There is evidence this occurred. An analysis of 108 Scandinavian skeletons from the Neolithic found that 17 per cent contained the plague. It appears that 5,200 years ago three waves of plague swept through at least some of these communities, with the first two being smaller and the final one more severe. None contained the gene that allows for transmission through fleas, meaning that these would have been far less widespread than the later Black Death that ravaged Europe. Such a horrific disease could have played a role in the collapse of Europe’s first farming communities. Yet the outbreaks can’t quite explain the boom-bust cycle, and don’t appear to have played a role in

the similar cycles that the Puebloans experienced.³⁷ Disease could have also contributed to the abandonment of many other early settlements, although we lack evidence for now.³⁸

Larger densities of people can also cause conflict. Urban archaeologists use the term ‘scalar stress’ to refer to how more people packed together means more issues and disputes. Even today some crime increases more than proportionally to population. For example, if you double the population of a city then instances of theft more than double. Crime in this case is just one example of the underlying increase in conflict and disagreements. The theory of scalar stress suggests that people needed to deal with this increased social friction. This included old ways, such as breaking off to form new communities (fission) or using new or larger rituals to give people a renewed sense of unity. Others suggest they took the novel approach of organizing hierarchically.³⁹

Scalar stress provides one potential explanation for the boom-bust cycles we see with early farmers. Indeed, one study of the first farmers of Europe found that a simple density-dependent conflict model matched the boom-bust record well. Yet it isn’t a convincing explanation as to why people left the early European farms or Puebloan settlements just as inequality and hierarchy were becoming entrenched. Moreover, coming from fluid civilizations of hunter-gatherers, we were already evolved to deal with strangers all the time. The bigger change we now faced was that we were no longer equals.⁴⁰

We see the same pattern in many of these first large settlements and cities of deepening inequality preceding social breakdown.

Abandoned Experiments II: Voting With Feet and Fire

Aborted Goliaths in Eurasia – Abandoning the First Goliaths of the Americas – The Forgotten Pyramid Cities of North America – Forgetting the Experiments – Walking Away from History

ABORTED GOLIATHS IN EURASIA

It wasn't just agriculture that was a slow, staccato process. The same can be seen with the adoption of Goliaths and cities across the world. Cities are permanent settlements with large, dense populations who have different social and economic roles (such as traders, bakers, and soldiers). Both the earliest large farming settlements and the earliest cities were slow to form and often collapsed in a similar fashion. And this wasn't the case of the occasional disaster pushing back the tide of progress. Rather, these early collapses appear to have been a deliberate abortion of an experiment gone wrong.^{[1](#)}

*

Man-made mounds are scattered across the world, lumps rising out of deserts and plains. Within them lies a trove of historical treasure. In archaeology, these are called 'tells': hills built from previous settlements. Layer after layer of refuse, buildings, and sediment are piled together into

one rounded legacy of a past community as at Tepe Gawra. There are thousands of tells in the Near East alone. Some have never been excavated. Others have been built over. Many are the remnants of settlements, communities, cities, and villages built before the world's first states ([Figure 13](#)).

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[Figure 13](#): Abandoned experiments across the world

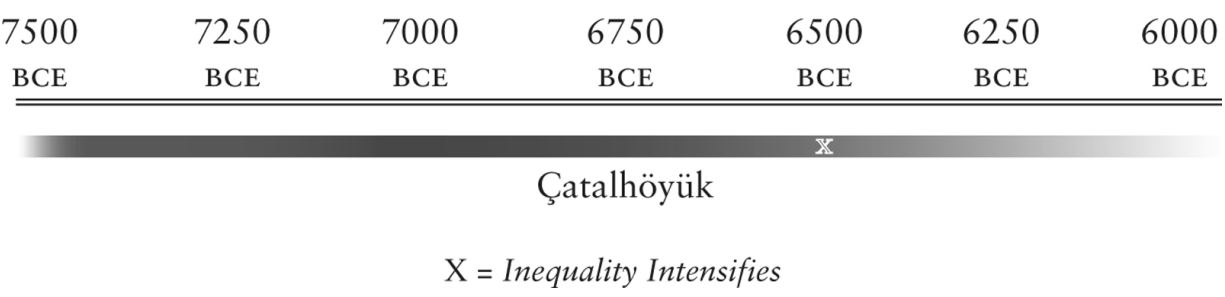
One of the best known is Çatalhöyük, in modern south-central Turkey. Çatalhöyük is renowned as having been the largest village in the world at the time. Around 9,000 years ago, 7–8,000 people lived there, packed together in just ten hectares of land. It was a fascinating fractal of crowded, near-identical houses. Each mud-brick residence acted as a house, temple, and mausoleum. These were multifunctional houses where families slept on platforms that contained the remains of their ancestors, often their preserved skulls, which were modelled to resemble the living. The houses were crammed together so closely that for many the only means of entry was through holes in the rooftops and down ladders that descended directly into the houses. They practised agriculture alongside foraging, with a diet of wheat, barley, lentils, peas, goat, and mutton. Violating the usual expectations about agriculture, men and women appear to have had the same social status, and there were no walls, fortifications, or signs of warfare.²

Perhaps the most striking feature of Çatalhöyük is its equality. The houses are of similar size and there are almost no signs of large differences in wealth, nor any administrative or government buildings. Ian Hodder, the lead archaeologist of the Çatalhöyük excavation, once called it ‘just a very, very large village’ that ‘pushed the idea of an egalitarian village to its extreme’.³ It was not a city, but it was long-lived. It persisted for around 1,000 years before something changed.

In the middle of the seventh millennium BCE, the equality between houses came to an end. Larger complexes, dwarfing those of the preceding era and of their neighbours, began to be built. The differences in the size of houses show a dramatic increase in inequality. The Gini coefficient (a 0–1 measure of inequality) more than doubled from 0.206 in 6650 BCE to 0.499 in around 6300 BCE. That is a larger jump than moving from being as equal as the Netherlands to as lopsided as Brazil. There are fewer signs of communal feasting and other collective activities. There are indications that animals that may have been collectively stewarded previously were increasingly under private ownership.⁴

This does not look to have been a welcome change. Around 6400–6300 BCE, the northern segment of Çatalhöyük was suddenly abandoned. The southern side continued for another 400 years or so, although it shrank as well. During this period, violence also increased. Many skulls with healed

fractures have been found across the site, alongside clay balls that were hurled by slingshot.⁵ Many other factors could have contributed to the abandonment, ranging from a drying climate to environmental destruction. There was no single great sacking, riot, or terminating blaze. Instead, there was a slow drain of occupants until this settlement was abandoned in around 6000 BCE ([Figure 14](#)).⁶



[Figure 14](#): A timeline of Çatalhöyük

The slow fade-out from around 6250 BCE marks a decline over time. It is used when there are competing dates over exactly when a society collapsed and/or clear indicators of a strong deterioration prior to the collapse.

A similar pattern can be seen further afield in Africa. Jenne-Jeno is one of the oldest urban complexes in all of that continent. It was an ancient city, embedded in the rich floodplain of the Niger Delta in modern Mali. It was first settled in 250 BCE and over the course of centuries grew into an ancient African metropolis. By 800 CE, 50,000 people lived within an afternoon’s walk of each other, all revolving around a forty-two-hectare urban centre. That complex had no palace, plaza, or public buildings, but rather was a maze of interwoven alleys and interlocking compounds. Like Çatalhöyük, there are no signs of stratification or inequality. Like Çatalhöyük, the Jenne-Jeno people were neither solely farmers, herders, nor foragers: they relied on a cuisine of fish, cattle, and African rice. The site appears to have been originally occupied during specific seasons, before groups began to settle permanently and become more specialized. Potters, blacksmiths, fishers, crafters, and herders all lived cheek-by-jowl.⁷

For over a thousand years Jenne-Jeno remained equal. Then, around 900 CE, the first monument was constructed. It was a wall around the urban centre, erected at a time when trans-Saharan trade routes began to offer novel opportunities for lavish wealth accumulation. The first signs of

increased wealth and power also appear to signal the decline of Jenne-Jeno.⁸

About a century later, the abandonment of Jenne-Jeno began. Nearly two-thirds of the settlement around the main mound was deserted. Some families moved into the walled compound, but most dispersed into smaller, self-sufficient villages. A chief took over the central mound and erected a palace. In 1180 CE he converted to Islam and shifted his palace to the nearby site of Djenné. The move coincided with flooding of the delta, which may have spoiled the rice harvests. But it also gave the chief an opportunity to break from the egalitarian tradition of Jenne-Jeno, or what remained of it. Some of the families loyally followed the chief, while the rest disappeared into the countryside.⁹

A similar yet subtly different trend unfolds some 11,000 kilometres away in northern China. There, the city of Taosi is the site of one of the earliest rebellions. In 2300 BCE, centuries before the earliest states of China, a fortified sixty-hectare town grew into a city nearly five times as large. Both the residences and burials were segregated, with the poor split off from the wealthy. Wealthy burials were replete with hundreds of lacquered vessels, jade axes, and traces of luxurious feasts. The city was ringed with walls, criss-crossed with roads, and studded with monuments.¹⁰

Around 2000 BCE the city wall was razed to the ground. Ritual sites to the south were abandoned, elite quarters were retrofitted to make stone-tool workshops, royal tombs and monuments were destroyed, and the former palace became occupied by commoners and surrounded by rubbish pits. It was a clear act of rebellion.¹¹ Now unwallled, the city swelled in size and population. Some archaeologists lament that ‘the city had clearly lost its status as a capital and was in a state of anarchy’ (described elsewhere as ‘collapse and chaos’).¹² Yet it grew and flourished for another two centuries in this apparently perilous state.

Taosi started as a Goliath before being overthrown. The city was reborn in the ruins of the ruling class and lived on for two hundred years. Life did, of course, change: the monuments, mansions, and palaces were no longer maintained. This didn’t mean that people’s lives were worse. The palaces were now dilapidated whispers of their former glory, but they housed more people, while the old mansions produced practical stone tools rather than richly dressed nobles.

Each of these cases is different. Jenne-Jeno and Çatalhöyük had egalitarian origins, while Taosi started as a Goliath. There are also similarities, with each one showing signs of either rebellion or abandonment in reaction to inequality and centralization. In some ways, this is not dissimilar to the counter-dominance strategies used by foragers. Ostracizing an aspiring ruler develops into walking away from established rulers, while execution of a domineering individual is scaled up to mass rebellion. In Jenne-Jeno and Çatalhöyük it was accomplished by the people voting with their feet, while the residents of Taosi voted with fire.

These abandonments were not rare. We see it not just in Eurasia but across the Atlantic in the Americas, too.

ABANDONING THE FIRST GOLIATHS OF THE AMERICAS

Stone heads engraved with eerie, lidless eyes mark the land of Mesoamerica's oldest city. Now, it is a graveyard of basalt faces. The largest of these weighs 40–50 tonnes. They were truly colossal undertakings.

These heads were the work of the 'Olmecs', Mesoamerica's first Goliath. San Lorenzo (we don't know what the Olmecs called it) was their capital and it sat on a salt hill that rose almost fifty metres above the surrounding plains of east-central Mexico. The ruby jewel atop this mountain was the 'Red Palace', a vast earthen temple that was stained rust-red by haematite and held up by giant basalt columns. From here, the Olmec rulers held feasts with copious quantities of maize-beer to legitimize and expand their rule over the commoners who lived on the slopes and plains below.¹³

San Lorenzo didn't begin with a palace, colossal stone heads, and corn-beer-keg parties, however. Instead, 800 years earlier, the site was occupied by semi-sedentary hunter-gatherers who built artificial mounds on the floodplains to act as island homes during inundations. The city itself appears to have been a ceremonial centre. As the population grew, these ceremonies would have become more important as a way of fostering a sense of connection between larger groups. More rituals meant that more maize was grown. It was during this period, which archaeologists call the

‘Chicharras phase’ (1250–1150 BCE), that an elite class began to emerge, and the construction of the Red Palace commenced. The demand for commoners to grow more maize for the rituals appears to have run contrary to their hunter-gatherer traditions, stoking tension between them and the elites. Shortly after its peak in power, around 1000 BCE, residents began to drift away. Some scattered to the countryside while others emigrated to the rival city of La Venta. As San Lorenzo declined, instability grew. Many of the great stone heads were mutilated and buried, indicating a violent rejection of elite rule and symbols. By 400 BCE, the city was a mausoleum of basalt decapitations sinking into the mud.

This was not uncommon in Mesoamerica, although some early settlements were much longer-lived.¹⁴ Take, for instance, one of San Lorenzo’s neighbours, Monte Albán. Around 1500 BCE farmers began to settle permanently into the different arms of the Oaxaca Valley in southern Mexico. A millennium later, a new settlement appeared on a central hill, almost two kilometres above sea level, overlooking the surrounding plains. Whoever built it levelled the top of the hill, creating a vast ceremonial plaza. At 150 by 300 metres, this was large enough to host a small city for public rituals. It also had a long, narrow stone ballcourt, used for playing a popular Mesoamerican game in which opposing teams tried to strike a ball through their opponent’s hoops using their hips.¹⁵ Over the centuries, thousands of people flocked to this new, growing city of Monte Albán. They lived in dried mud-brick (called ‘adobe’) houses high above the valley, ate hot tortillas made from corn, used tools made from imported obsidian, and increasingly warred with their neighbours. By 150 BCE it was a city of 17,000, with irrigated canals and monuments celebrating their successful raids.

It was a surprisingly equal city. There are no signs of a central palace, no hoards of personal wealth, no centralized food storage, no large differences in housing size, and no aggrandizing depictions of rulers. From 500 to 100 BCE the closest we have to a glorified advertisement for a ruler is a stone carving of a masked individual leading a ritual procession. There may have been some rulers and elites, but they were largely faceless. Then things slowly began to change.¹⁶

In the seventh century CE, inequality began to rise.¹⁷ Rulers began to erect carved stone artworks. These represent genealogies and marriages, a

way to flaunt hereditary prestige. Monte Albán was losing its collective character. Eventually, the populace began to abandon the city. Two of the most established archaeologists of Monte Albán, Gary Feinman and Linda Nicholas, observe that commoners may have simply ‘voted with their feet’.¹⁸

There was an ancient conurbation in Mesoamerica even greater than Monte Albán. On the outskirts of Mexico City lie the remains of a city dating from around 100 BCE, crowned with colossal pyramids. We don’t know what the city was originally called, but 700 years after it fell apart the Aztecs named it Teotihuacan, meaning ‘place of the gods’. It was a pre-Colombian metropolis, which held perhaps up to 100,000 people and covered twenty square kilometres.

Between 100 BCE and 550 CE Teotihuacan appears to have been a planned city. It was organized into quadrants, its north–south axis known as the Street of the Dead and its east–west axis marked more prosaically by two avenues named East and West. The settlement is framed by three gigantic monuments: the Pyramid of the Sun, the Pyramid of the Moon, and the Feathered Serpent Pyramid. Despite the glut of monuments, there are no clear signs of centralized hierarchy, nor any artwork that shows one human subjugating another. Its inhabitants appear to have known about the writing systems of the Maya and the Zapotec, but rarely used their own, apart from a few scant glyphs on murals.¹⁹

Whether Teotihuacan began with an oligarchy of kings or as some kind of collective arrangement is unclear. What is clear, however, is that it changed dramatically. That change occurred shortly after the Feathered Serpent Pyramid was built, around 200 CE, with the sacrifice of some 200 victims. Most were from outside the city, and had their hands tied behind their backs before being killed. Despite this opening massacre, the pyramid didn’t last long. About fifty years later it was desecrated, its offerings looted, and the plumed serpent statues that adorned it pulled down and smashed to the ground. What was left of it was covered by a raised platform. This practice of defacing or smashing items and then covering or burning them is common in Mesoamerica, and is known as a ‘termination ritual’. It is designed to sever the cosmic connection between a site or object and a deity.

After the termination of the Feathered Serpent Pyramid, no new pyramids were built and no more people were sacrificed. Instead, wealth and energy were channelled into an urban renewal. New great stone multi-occupancy apartment buildings were erected across the city. It was not just the rich who lived in these dwellings, but also the broader populace. Each dwelling contained well-plastered walls and ceiling, internal drainage systems, and around a hundred people. We don't know how they were governed, but it does not appear to have been in a top-down manner (at least not one that turns up in their art, writing, or buildings). The most likely reality is some kind of decentralized management across twenty or so 'three-temple complexes', with each overseeing around a hundred apartment blocks. While there were probably still nobles and elites, the city was remarkably equal in wealth. Some tentative calculations based on the size of houses suggest that income inequality was under half that of modern-day Denmark.^{[20](#)}

It was not to last. Towards the final periods of Teotihuacan, the walled compounds begin to become more isolated. Each developed its own trade routes to foreign locations and controlled access points. Teotihuacan had become a cluster of gated communities. More signs of wealth inequality emerge. In about 500 CE, the city descended into flames. Over 150 buildings across the city were burned, including almost all of those on the Street of the Dead. All the temples, pyramids, and public buildings were demolished, the images of gods destroyed, and people butchered in the palaces. There appear to be no signs of foreign invaders. There is evidence of drought, but this appears to occur after the great burning of Teotihuacan. Instead, this appears to have been a case of popular rebellion and elite conflict. We can't see the details, but archaeologists suggest that it could have been 'intermediate elites' (the rich or neighbourhood leaders) attempting to topple the ruling class, or factional strife among elites, or a mass revolt, or a bit of all three. We simply don't know. What we do know is that there was a systematic campaign to incinerate the ritual heart of the city.^{[21](#)}

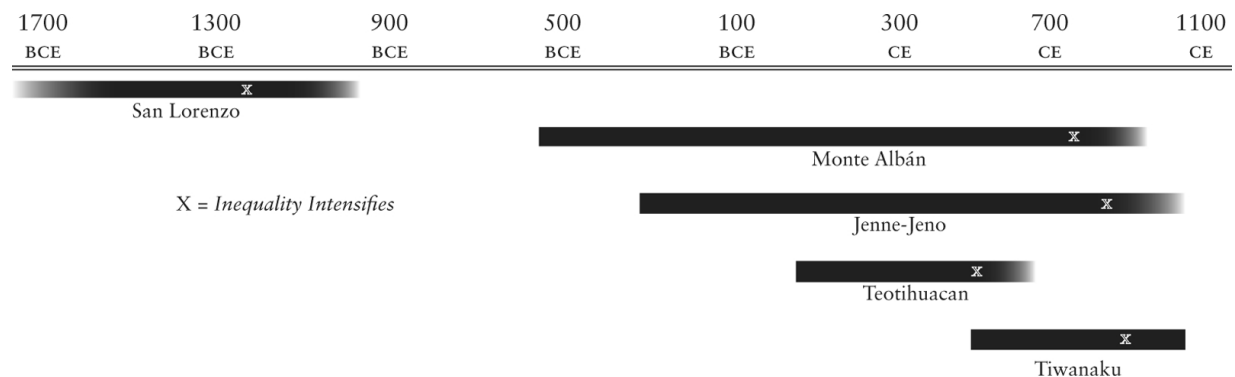
Within a century the population of the city had dropped by half. It was essentially abandoned but lingered. In the epi-classic period (550–850 CE) the population dropped to 30,000 or lower as people retreated into the surrounding hills and mountains. Teotihuacan never recovered.

Evidence of emigration and revolution can be seen in other areas, such as the Andes of South America. Take, for instance, the collapse of the later city of Tiwanaku (c.500–1000 CE).²² Tiwanaku, founded on the shores of Lake Titicaca in western Bolivia, begins as an egalitarian site. After it grows into a city of tens of thousands, inequality and hierarchy creep up. Massive stone monoliths of high-status individuals are erected. Public ceremonial centres begin to be cordoned off. The locally managed countryside that fed the city is grabbed and remade into a top-down production system.²³ It results in over-exploitation of land, resources, and people.²⁴ By 800 CE Tiwanaku is a Goliath. It continues in this unequal state for around another two centuries. Then, within a few decades, it falls apart. The monolithic Gateway of the Sun is toppled, and storage jars smashed, in an act that bears all the hallmarks of a revolution. The peoples who left the city disregard the previous elite culture. Post-Tiwanaku pottery styles are bereft of elite-imperial iconography such as the ‘Front Face Deity’ and ‘Puma Head’. The new, smaller, rural communities replicated many of the old ceremonies performed at Tiwanaku, but these now took place in open, collective spaces. Those who left Tiwanaku seemed intent on rejecting the power and authority of its previous elite.²⁵

Tiwanaku and San Lorenzo faced additional pressure in the form of climate change: these highland peoples were reliant on an agricultural system that was susceptible to drought. The abandonment of San Lorenzo coincided with a prolonged phase of drought. And the flight from Tiwanaku and reorganization of its politics also seems to have coincided with a period of drought, indicated by the fluctuating levels of Lake Titicaca. Notably, as they move away from Tiwanaku, they become reliant on a more flexible combination of fishing, hunting, gathering, and herding.²⁶

This is not a comprehensive account of the different sites that were abandoned in a similar way across southern and central America. There are signs of revolt across the Andes, largely in modern-day Peru, including the Wari culture (c.600–1000 CE), the Moche (100–700 CE), and the ritual site of Pukura.²⁷ What we see across all of them is a pattern (see [Figure 15](#)).²⁸ People revolted and abandoned urban settlements as they became increasingly unequal and hierarchical. It looks similar to what happened thousands of kilometres away with Çatalhöyük, Jenne-Jeno, and Taosi.²⁹

These weren't just abandonments or rebellions. People were choosing to walk away from what we tend to see as the natural course of history: the adoption of cities, agriculture, and states.



[Figure 15](#): Inequality and collapse in early Goliaths

THE FORGOTTEN PYRAMID CITIES OF NORTH AMERICA

The first European explorers often portrayed the land they invaded as an empty wilderness filled with wild animals and even wilder humans. In North America, they overlooked the signs of more urban, centralized peoples who farmed and built monuments. The Puebloans of the south-west US were not the only pre-European farmers and builders of North America. European colonizers came across the remains of numerous old cities and towns built around great flat-topped pyramids made from soil. They dismissed these as being either inspired by the Spanish or built by a more advanced and civilized people who preceded the indigenous Americans. They ignored the fact that the modern-day United States was once littered with indigenous farming cities. All of which collapsed prior to European arrival.

A millennium ago, in a land of dispersed hunter-gatherers, arose a city of vast earthen temples and powerful priests. This was Cahokia, the first city of the US. Before Cahokia there had been villages with no signs of war and some indications of democracy, such as longhouses that could have held councils of two to three dozen people. Domesticated corn arrived in the area in 900 CE and took an increasingly prominent role in the diet of the Native

Americans there. Just over a century later the ‘big bang’ of Cahokia occurred. Neighbouring villages were drained of their populaces, who resettled into Cahokia. Cahokia appears to have been a ceremonial centre and the sacrifices were one element of its ritualistic practices. Whatever services it did provide, they didn’t last long.³⁰ As we’ve already seen, Cahokia was abandoned and became a haunted marsh of derelict pyramids.³¹

A thousand years ago, the US was dotted with urban centres and emerging Goliaths built around great piles of earth. In northern Alabama there is ‘Moundville’, a collection of twenty-eight flat-topped mounds housing perhaps 10,000 people. Stretching across the southern Appalachians (northern Georgia and some of the surrounding areas of Alabama, Tennessee, and South Carolina) are dozens of similar sites. Bordering the Puebloan farming communities of the south-western US were the Huhugam (previously called the Hohokam by archaeologists), who had flat-topped pyramids for public rituals. They also had plazas and ballcourts, much like Mesoamerican cities such as San Lorenzo. The Huhugam built all this in the desert of modern-day Arizona using vast networks of irrigation canals. Hiding in plain sight are a host of ancient North American Goliaths.³²

Despite being thousands of kilometres apart, these different sites have much in common. All were built at similar times, with initial construction for both Moundville and Cahokia beginning in 1000 CE, and 1050–1150 CE for a range of sites across the Huhugam culture, due probably to favourable climatic conditions. During 950–1250 CE the northern hemisphere underwent an unusual period of warmth known as the Medieval Warm Period. This would have provided better conditions for growing crops and supporting bigger populations. Like the Puebloans, all these sites and cities were built on the three sisters of corn, beans, and squash. The Huhugam, Moundville, and the mounds of northern Georgia all also look strikingly similar to the cities of the Maya and the Aztecs, with flat-topped pyramids and grand plazas (the ancestral Puebloans are a notable exception to this architectural trend). The Huhugam even had a similar ballgame to the one played in San Lorenzo and Monte Albán.

Many of the ingredients for cities and Goliaths seem to have been imports from Mesoamerican empires. The three sister crops were first

grown in Mesoamerica and then spread via indigenous trade routes through the deserts of the south-western US (the Puebloans and Huhugam) before disseminating across North America and even into Canada. They flowed into Cahokia, Moundville, and northern Georgia. These routes that snaked up and across the Americas may have carried not just food, crops, and jewels, but also ideas, stories of cities, great rectangular ballcourts, and imposing religious rulers who cemented their reigns through the blood of sacrificed men, women, and children. Some activities, like mound building, preceded the influence of Mesoamerica. For instance, the first earthen pyramids in the Americas dating back to 3500 BCE can be found in north-eastern Louisiana, not in Mexico or the Yucatán. Other practices among the first Goliaths of North America, such as growing the three sisters, ballcourts, using stone-chipped daggers that represented gods, and much of the cosmology, seem to have been inspired by Mesoamerica. Nonetheless, in North America these were combined to make new, unique societies.³³

All these sites progressed in a fashion that will by now be familiar. Moundville began as a small settlement of dispersed mounds. Two centuries later the two largest temples, a plaza, and a surrounding palisade were built. Shortly thereafter there are signs of a religious elite receiving tribute in the form of food from the surrounding villages. Elite burials adorned with costumes and jewellery begin to turn up in the pyramids.³⁴ Over the course of two centuries people began to leave Moundville, returning to the surrounding villages. The palisade was no longer repaired or replaced and many of the pyramids became unused shells. What was once a city of up to 10,000 people became a purely ceremonial centre populated by a few elite families. By the end of the sixteenth century, all the pyramids had been abandoned. The exact reason for the slow drain away from Moundville is unclear. However, it's hard to ignore the pattern of inequality rising and then the commoners making an exodus from the city.³⁵

Around 2,500 kilometres across the green ridges and valleys, and the burnt-gold deserts of southern North America, lies the Sonoran Desert. There the people known as the Huhugam built open, small, rounded platforms that were used for public events and ceremonies. These platforms were accompanied by open ballcourts for public games. Dozens of stadium-mound-centred villages were sprinkled across the Gila River to form a hive

of thriving towns, often within a day's round trip of each other. The indigenous peoples made an oasis in the deserts of the south-western US.³⁶

Then things began to change. Many of the towns were consolidated into cities as populations surged. The rounded mounds became taller, steeper, flat-topped pyramids. The public stadiums were vacated. The games were clearly over. Rich residences began to be placed atop these stepped monuments, which now leered over the city. Elites began to be buried differently, in their own adobe sarcophagi.

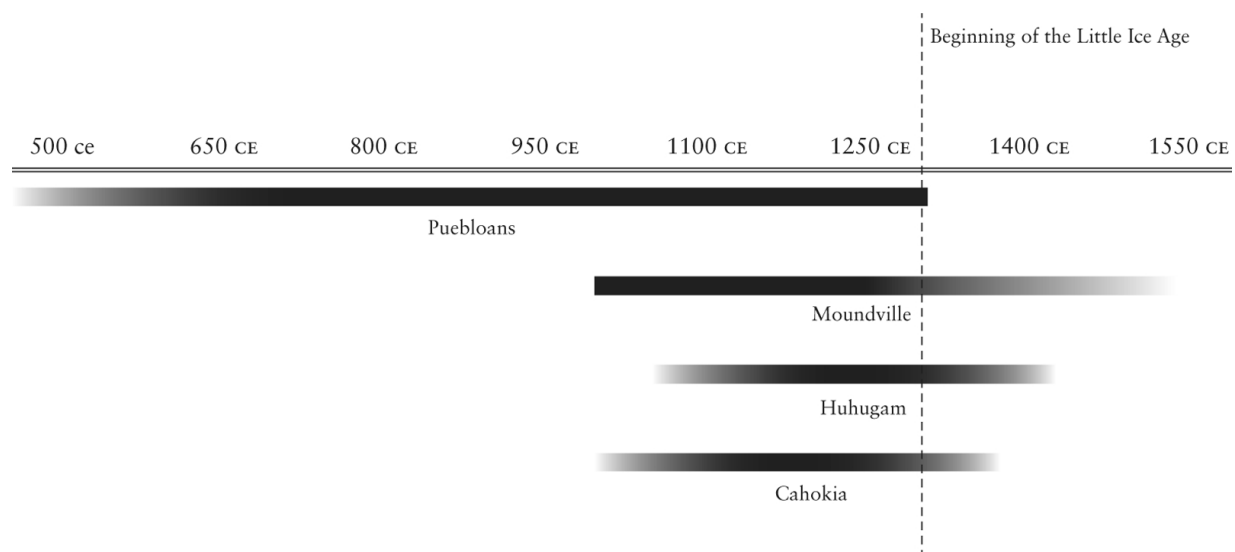
Around the year 1300 CE the Huhugam chief Siwani would have sat over an ancient desert metropolis full of pyramids and a sixty-square-kilometre irrigation system that fed another five cities and a castle that the Spanish would later call Casa Grande. By the end of the century the cities and villages lay largely empty. The irrigation systems fell into disuse and the canals were not repaired. Specialists have pointed to numerous reasons for this decline, including disease, drought, conflict, overpopulation, and environmental degradation.³⁷ The Huhugam ditched their cities and powerful chief in favour of more resilient smaller communities.³⁸

Other areas seem to have undergone periodic collapses that look more like the Puebloans. The mini-Goliaths throughout the Appalachians would rise up, last a century or two, and then contract and collapse. Accounts from the first Spanish invaders, archaeological remains of palisades and warfare, and studies of similar societies can give us some insights. Across these mini-kingdoms individual warriors and leaders would compete for status, leading not only to infighting among elites, but also to warfare between neighbours. In some ways, war was an extension of elite conflict: a pursuit of power and status through violence scaled up from individuals to larger groups. A lack of control over citizens, who regularly left, and uprisings also seem to have been factors. The Appalachian mini-Goliaths are not remarkable: these problems of elite competition, warfare, instability, and abandonment were often endemic in chiefdoms, hierarchical hunter-gatherers, and Goliaths that had yet to become states.³⁹

In each of these cases (as well as those of the Puebloans, the Calusa, and north-west coast Native Americans) we see a similar phenomenon. A small priestly class emerges and begins to boss others around. They display their power, sometimes through innocent means such as feasts and opulent burials, and sometimes through more unsettling methods, such as slavery

and human sacrifice. Many archaeologists have pointed to these ‘secret societies’ as being one of the original blueprints for gaining power. Exclusive groups claiming access to special knowledge and skills would use their religious advantage to threaten others, gain political power, and amass more resources. Once lootable resources appeared, these covert cabals could fuse together all the sources of power: control of information, violence, authority, and resources.⁴⁰

All these sites across North America share another trait. All of them collapsed around the same time. Warmer conditions helped to create larger surpluses that these cities and pyramids were built on. Similarly, cooler conditions seem to have played a key role in their demise. The ends of the cities and hierarchies of Cahokia, Moundville, the Pueblos, and the Huhugam all came with the end of the Medieval Warm Period and the onset of the ‘Little Ice Age’, c.1250–1850 CE ([Figure 16](#)). There were a series of pulses of cooling across the northern hemisphere, with some areas cooling by around 1°C. In some cases, a changing climate seems to have played a central role. The big bang of Cahokia coincided with one of the wettest fifty-year periods in the last 1,000 years, while the abandonment of upland farming complexes occurred during a series of severe droughts.⁴¹



[Figure 16](#): A timeline of the first Goliaths of North America

FORGETTING THE EXPERIMENTS

What is most telling is how this period of nascent cities and urbanism was remembered by later generations. It was not looked on fondly. Indigenous people today across these regions have oral traditions, stories passed down by word of mouth, which recount this as a time of rebellions against overbearing despots. The modern Puebloans recall the Chacoan collapse as a reaction to the concentration of power in leaders. In the Navajo rendition, a stranger called the 'Great Gambler' (or Noqoilpi) descended to the Puebloans. He challenged them to games, beating them so consistently that he came to own much of their property, as well as their men, children, and women. He then tricked the rest of the Puebloans into building Chaco Canyon on the false promise that he would release their property and people. Eventually the people aligned with a group of gods to overthrow him.⁴²

The Cherokee, whose nation covers much of the mounded territory of northern Georgia, Alabama, and Tennessee, recall once being ruled by the Ani-Kutani, a hereditary secret society who built the pyramids that the abandoned towns were once centred around. These rulers grew arrogant and oppressive and, in some versions, even became cannibals. The people despised these tyrants, but also feared them, and so endured their fate in silence. Eventually, the people rose up in revolt and destroyed the mound-builders.⁴³

The O'odham descendants of the Huhugam pyramid builders see the collapse of these earlier cities and pyramids as stemming from a fight between the cultural hero Elder Brother and the ruling priest class at the time. The ruling priests grew arrogant, suggesting that they knew more than Elder Brother. They offended him, and rather than apologize, they tried to murder him on multiple occasions. Elder Brother retaliated and assembled an army of the disenfranchised. As they approached Casa Grande the rebels sang 'Yonder stands the doomed habitation. About the pueblo runs its frightened chieftain.' The chieftain ran, never to return.⁴⁴

Then there is of course Cahokia, which the Native Americans chose not to remember at all.

The Native Americans didn't just drop their elites and cities; they organized themselves in such a way as to prevent them from re-emerging. The peoples of Moundville, Cahokia, and the Huhugam dispersed into smaller towns and farms. Across the Mississippi, these towns still often contained mounds, but they were usually open, smaller, and used for public ceremonial celebrations. The O'odham descendants of the Huhugam started to build new stadiums with a ballcourt game played by women. Some mound-based mini-kingdoms remained around the Mississippi, such as the Natchez and Tunicas in the Lower Mississippi Valley (crossing Arkansas, Louisiana, and Mississippi), but these were smaller and fewer than before.

Religion became decentralized: priests no longer ruled towns and ceremonies. Spirituality and medicine became more shared, public practices. The Cherokee replaced the fabled Ani-Kutani with individual healers and conjurers. The Huhugam stopped burying elites in adobe coffins. Instead, they cremated the remains of all the dead. The priests of Cahokia, Moundville, and Huhugam likely justified their rule through exclusive access to rituals and the supernatural. These peoples were rearranging religious practices to ensure that no such imbalance could form and be exploited again.

Leaders still existed across North America in the centuries that followed the fall of the Goliaths. Yet these communities tended to be democratic hierarchies with strict checks and balances. Each Cherokee town had both a council of men and a council of women, while family-based clans spread their members across towns as a check against the councils and chiefs. Most Native Americans (both men and women) contributed to major decisions, such as whether to go war. Sometimes great assemblies convened to seek a consensus decision. These decentralized political webs were held together by threads of reciprocal gift-giving, trade, and, in some cases, food sharing. Diplomatic arrangements, like confederacies of different nations (such as the Haudenosaunee, or Iroquois people) helped to bring people into larger groups without over-centralizing power.

Most of these dispersed, egalitarian peoples began to rely more on hunting and gathering and less on farming. After the Huhugam, the Tohono O'odham (desert people) derived less than 20 per cent of their calories from agriculture, while the Hia-Ced O'odham got even less than that. The arrangements were not perfect: there was still war and suffering, as well as some inequality and hierarchy. Yet it is hard to see this as a worse

arrangement than their world before the collapse. For most people it meant healthier diets, more movement, and freedom from the threat of having a loved one, or yourself, beheaded and thrown into a shallow grave beneath a pyramid of dirt. There is a good reason why the later generations organized themselves to prevent a Goliath from being reborn.^{[45](#)}

This pattern of organizing to prevent Goliath rebirth was aided by the climate. The cooler, more drought-prone conditions of the Little Ice Age would have made creating large agricultural surpluses much more difficult. This is perhaps why the Calusa and other hierarchical foragers (who relied on fish rather than grain) persisted until European contact, while the agrarian pyramid builders of Cahokia and Moundville fell apart and never returned. The climate conspired against Goliath.^{[46](#)}

WALKING AWAY FROM HISTORY

Twelve thousand years ago, our communities changed alongside the climate. Yet our psychology did not. As lootable resources were gathered, some came to try to dominate others in a slow, protracted struggle. Some chose to accept the new arrangement, while many suffered, rebelled, or simply walked away. The adoption of agriculture, cities, and states took thousands of years partly because it was resisted by so many of us. That was due to our deeply ingrained psychological traits.

One was our aversion to being dominated. Many of the cases here are not just collapses but also examples of successful counter-dominance strategies. Intentionally destroying elite monuments and exiting (voting with both feet and fire) are a larger-scale version of ostracizing or eliminating an upstart hunter. Groups coming together and then breaking apart when conflict arises are also not dissimilar to earlier fission-fusion patterns. Indeed, some scholars have even described the rise and fall of mini-Goliaths across northern Georgia as exactly this kind of process.^{[47](#)}

Interestingly, many of the cases, such as Tiwanaku, San Lorenzo, and Jenne-Jeno, cut against the conventional wisdom that states allowed for the creation of large-scale farming and eventually cities. The logic was that people needed rulers and bureaucracy to create stable food production to sustain a city's large population. This clearly wasn't necessary: across the

world cities and large-scale irrigation *preceded* centralized, hierarchical arrangements. And they often fell apart once a domineering Goliath emerged.

Rather than people willingly and logically choosing to adopt rulers and then cities, they were developing agriculture and cities (in different orders) and sometimes aborting them once these social experiments became infected with dominance hierarchies. Rather than states being constructed to allow for irrigation and cities, we often see cities arising first, elite groups hijacking and repurposing pre-existing food production, then people often resisting an emerging hierarchy. It's not a universal pattern. Other areas of the world see a smoother rise from inequality to hierarchical cities. Yet, history and the march towards states was far slower, messier, and more resisted than is commonly assumed.^{[48](#)}

Cahokia, and the other cases covered here, are indicative of the fate that befell many infant Goliaths, not just in the Americas but across the world. Cities and settlements often began in an egalitarian fashion with no signs of leaders and roughly equal-sized houses. These more collective communities often persisted for an extraordinarily long time: Çatalhöyük for some two millennia, and both Jenne-Jeno and Monte Albán for around a thousand years. (Only a few exceptional empires, such as the Byzantine, persisted for a thousand years or more.) Then, inequality slowly, haltingly grew into more entrenched forms of rulership, a process that was often accompanied by war, fortifications, centralized agriculture, and human sacrifice. Then, people left. Sometimes it was a more peaceful affair; at other times their exit was celebrated with smashed gates and monuments and cities set ablaze.

This trend would continue for millennia. The boundary between Goliaths and surrounding non-state peoples was porous. Whether it was Chinese peasants joining the nomadic raiders to the north or Romans defecting to the Huns, citizens regularly and voluntarily left Goliaths. When the walls were weak and social control underdeveloped, fading back into the periphery was a tempting option.^{[49](#)}

Despite this general pattern, we also need to be aware of the differences. The collapse of Tiwanaku and Cahokia took a generation or two, while the draining of Çatalhöyük and San Lorenzo played out over centuries. The aftermath of the collapse also varied. While swathes of land around Cahokia

and the Mississippi Basin were left vacant, the communities that followed Monte Albán in the Oaxaca Valley were even more unequal and autocratic with bigger, richer palaces. These differences, while important, should still not distract us from the underlying picture that early Goliaths were frequently fragile and rejected. It's a trend we can see beyond the case studies covered here.⁵⁰

There were thousands of villages and settlements during the Neolithic. While we can't carry out a systematic investigation, there is some evidence of a similar pattern. In 2023, the archaeologist Dan Lawrence and his colleagues examined the long-term trends of several hundred urban sites across south-west Asia, from 12,000 to 2,000 years ago. They initially expected that the longest-lived settlements would be those with the most productive agricultural land. That wasn't the case. Many of the most persistent settlements were in marginal environments. Instead, settlement longevity was negatively correlated with more centralized arrangements. Whether due to war or abandonment, longevity declined as hierarchy increased. The earlier settlements were astonishingly persistent. Their average age during the period 12,000–10,000 years before today is over 3,000 years. This drops in subsequent periods as Goliaths grow and take over.⁵¹ The earlier settlements were perhaps more egalitarian, but also much more resilient. We'll need to wait for more evidence and bigger studies to see whether or not this trend is a statistically significant one across the world.⁵²

This pattern, if robust, could even extend to more recent finds. Laser scanning across the Amazon has revealed cities and settlements with advanced irrigation systems hidden beneath the lush canopy. In the Ecuadorian Amazon 2,500 years ago, the inhabitants crafted a city of 6,000 earthen platforms that would have held tens of thousands of people. Those platforms, like those in Cahokia and the Maya, were organized around a central structure that itself had a raised platform. The platforms were connected by roads, the longest of which is 25 kilometres. In the south-west Amazon are several urban sites dotted with rectangular platform mounds, stepped platforms, and conical pyramids standing up to twenty-two metres tall. The Ecuadorian sites were inhabited for around a millennium. Yet eventually all these cities and settlements were abandoned in favour of more dispersed lifestyles. For now, we don't know why. But it is yet another

reminder that the path towards urbanism, and often the inequality that accompanied it, was reversible.^{[53](#)}

Not all early Goliaths were aborted or abandoned, however. Some became far larger, more permanent structures that spread across the world.

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Goliath Rises

New Fuels, New Goliaths – How to Escape a Goliath – Goliath Evolution – Our Sacrifice – Stories of Subjugation – A Gendered Goliath – The Losers of Evolution

NEW FUELS, NEW GOLIATHS

Sixteenth-century New Guinea had practised agriculture for as long as Egypt (since 5000–4500 BCE), growing yam, taro, and bananas. Despite this, New Guinea looked rather different from Egypt. It never developed a large, stratified society, let alone a pharaonic ruler and towering pyramids.

In the seventeenth century a new crop was introduced to the archipelago of New Guinea: the starchy orange tuber that we call sweet potato. Portuguese colonizers took it into their Indonesian provinces, and it travelled from there through the hands of local traders. For a time, it appeared that the introduction of the sweet potato might lead to an indigenous state arising in New Guinea. This new staple led to a surge in agricultural productivity and a boom in population. People began to craft and trade more prestige goods. Yet, no group ever went the way of Egypt. A state never arose, and the politics remained fragmented in competing communities.¹

Understanding why some areas of the world established enduring empires and kingdoms while others didn't means understanding not just how

Goliaths were created, but how they evolved the right structures to persevere and avoid collapse.

The famed economist Adam Smith, the forefathers of communism Karl Marx and Friedrich Engels, and many big history books have all theorized that agriculture created a surplus of resources, thus enabling full-time rulers to manage everyone else without wasting time on producing food. It's clear that this isn't the full picture. We have already encountered hunter-gatherers who produced hierarchies and a surplus but never tilled a field. Then there is New Guinea, which developed large agricultural surpluses yet never produced a state or a large-scale Goliath.

There were ten centres of independent agricultural production around the world, and only around four of them (the Near East, the Yellow River Basin in China, the Andes, and Mesoamerica) developed states (see [Figure 17](#)).² These were the five original states (initial experiments that did not emulate a neighbour or competitor): the First Dynasty of Egypt, Uruk in Mesopotamia, the Xia Dynasty in China, Monte Albán in Mesoamerica, and the Wari (or potentially Tiwanaku) in the Andes. Original states were only ever created in these centres of food production. States became the political scaffolding for Goliaths, creating far more organized, stable, and large-scale societies. All the abandoned experiments we've covered, from the first farmers of Europe to the pyramid cities of North America, also relied on agriculture. Clearly, agriculture is a necessary but not sufficient condition for creating a state.³

Goliath fuel (caged land, lootable resources, and monopolizable weapons) can help explain why states arose in these areas yet not in Japan or New Guinea. They all required restricted exit options, monopolizable weapons, and new lootable resources in the form of domesticated crops and animals. Growing populations and a reliance on agriculture shrank exit options. Mesopotamia was a land cordoned off by two rivers (the name quite literally means 'the land between rivers' in Greek), while Egypt was trapped between the Red Sea and Sahara Desert.⁴ New weapons were still needed to capitalize on greater masses of trapped people.⁵ The rise of the earliest macro-states (covering an area of more than 100,000 square kilometres) – such as Akkad in Mesopotamia and the Old Egyptian Kingdom – coincided with the spread of bronze metallurgy in the region and the proliferation of hand-held weapons such as bronze swords (around

3000 BCE).⁶ These probably would not have been enough without the new lootable resources that became available after the end of the ice age: domesticated crops and animals.⁷

Cereals are the ultimate lootable resource. Wheat and barley (in Mesopotamia and Egypt), rice and millet (in China), and maize (in Mesoamerica) all have regular harvest periods, are all calorically dense, and can be easily spotted, confiscated, and carted off to a distant ruler. They can't be easily hidden by a farmer from a tax collector or harvested at unpredictable times. One broad-spanning analysis over several millennia found that the production of cultivated cereals was the strongest predictor for the development of states.⁸ Another, covering Europe in the period 1300–1700, showed that observable calories (easily seen food sources) were highly correlated with stronger states with a greater tax base.⁹ Unlike salmon or other fish, cereal cultivation can also support far larger, denser populations. The world relies on a handful of cereals today largely because they were an ideal tax crop five thousand years ago. As James C. Scott proclaimed, 'grains make states'.¹⁰

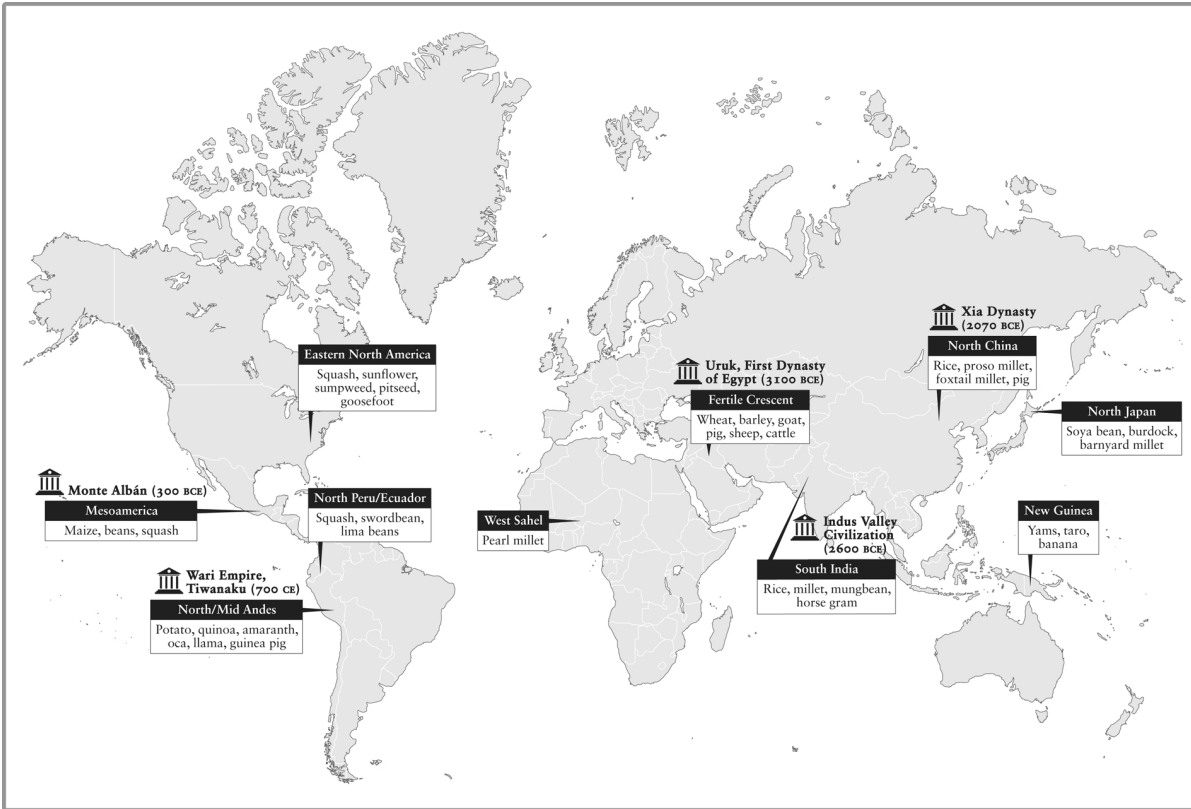


Figure 17: Original states and independent food production centres

Although it wasn't just grains. Domesticated animals helped as well. Cattle, goats, and sheep provided a storable and mobile source of calories and protein, as well as materials such as hide, wool, bone, and sinew. Horses can aid in war and trade. Cattle and horses can both assist and supercharge agriculture. We see this when we compare Mesoamerica and North America to Eurasia. Eurasia developed far greater disparities in wealth than Mesoamerica or North America did. That is largely because Eurasia had horses and other large, domesticated mammals (such as cows, goats, and sheep), while the Americas did not. Ploughs pulled by horse and ox allowed for a greater surplus and disproportionately benefited the few who owned land and could afford them. Eventually, warriors mounted horses to conquer others.¹¹

The new lootable resources of domesticated crops and animals were not evenly distributed. They were largely a matter of geographical fortune.¹² China got pigs, rice, and millet; Mesopotamia and Egypt received cattle, goats, sheep, wheat, barley, and einkorn; Mesoamerica was bequeathed turkey, beans, squash, and maize; and the Andes were handed llamas, alpacas, and potatoes.

Goliath fuel can also explain why some regions of the world had agriculture but never built a state. New Guinea, western Africa, the Sahel, Ecuador, and the eastern US had domesticated crops but few to no lootable ones. New Guinea had bananas that can only be stored for a month or so before rotting. Ecuador and the eastern US had squash that does not have one easy harvest period during which a tax collector could drop by. Rather, they can be harvested over a period of several months. Taro, potatoes, and sweet potatoes are all grown underground and can only be stored for around 1–6 months. They can also often be left underground after their harvest period. These were not crops that were easily captured and monopolized by an elite class. In contrast, wheat, rice, and corn are all highly visible: corn advertises itself with high stalks, while rice paddies and waving golden fields of wheat are difficult to miss. All of them require significant labour, have predictable harvest times, and can be stored for years or decades in the right circumstances. Crops such as corn, rice, wheat, and barley are highly lootable (see [Table 2](#)).¹³

Table 2: The lootability of different crops

Crop		Visibilit y	Post- harvest storabilit y (in optimal conditio ns)	Labour intensity	Lootabili ty
Banana		High	1–2 months	Low	Low
Maize		High	Around a decade	Moderat e	High
Potato		Low	4–5 months	Low	Low
Squash		Moderat e	2–6 months	Moderat e	Low
Sweet Potato		Low	6 months	Low	Low
Taro		Low	1 month	High (moderat e if irrigated)	Medium
Rice		High	Decades	High	High
Wheat		High	Decades	High	High

While western Africa and the Sahel had more lootable resources such as African rice and pearl millet, they lacked domesticated animals and had a

low population density. The geography was made up of flat coastal plains and inland hilly plateaus, providing plenty of exit options. Eventually, the rice-growing Ghana Empire (300–1050 CE) did emerge in the western Sahel, although likely after having interacted with other states.¹⁴ While northern Japan did have millet, it was cultivated late (around 689–1015 CE) and after the introduction of rice.¹⁵ Rice laid the foundation for later elite groups and an empire. When exactly an empire was established in Japan is shrouded in myth, but the current Japanese emperor (Naruhito) appears to have an unbroken lineage from around 500 CE.

Eastern North America was well on its way to building a state with Cahokia before it fell apart. Even then, these pyramid cities arose only after the importation of corn from Mesoamerica. And they struggled to rebuild large Goliaths during the Little Ice Age when the conditions meant less Goliath fuel to support and tax big, dense populations. North America also had bountiful exit options.

However, the fall of Cahokia, Moundville, the Huhugam, and the Pueblos does not mean that North America would never have seen an indigenous empire arise. On the contrary, across the world state-building was a slow process that took centuries or millennia and was littered with failed attempts. It was less than two centuries between the failure of the first Goliaths of North America (about 1250–1350 CE) and the arrival of the first European invaders (around the 1530s). On the coast, European expeditions encountered the Calusa. Further inland, when they arrived at the Mississippi Basin, they still encountered several small, mound-based mini-kingdoms. Imagine if the Europeans hadn't arrived for several more centuries, indigenous populations had continued to grow (decreasing exit options), trade had intensified with Mesoamerica, and North America had had time to emerge from the Little Ice Age. North America would still almost certainly have built an indigenous state eventually, as did all other world regions with Goliath fuel.

Well, all except one. The Indus Valley Civilization is the only remaining case of a centre of agricultural food production that never developed a Goliath. This group (beginning around 3200 BCE) was a collection of commercially connected cities across the Indus Valley in modern Pakistan. The largest city of Mohenjo-daro housed roughly 40,000–60,000 people at its peak. They had plumbing for communal bathhouses and communal halls

that appeared to be open to the public. Despite having lootable resources (namely rice and millet) and being located in a riverine valley, there are no signs of inequality or hierarchy. Here was a community with plenty of Goliath fuel but no Goliath.¹⁶

The Indus perhaps never developed a Goliath owing to low population density. The Indus cities were more spread out than those in Uruk or Egypt, which meant that people had less need to protect their assets or compete for status with neighbours. Or perhaps the Indus simply managed to maintain norms and counter-dominance mechanisms that prevented state formation. Such cultural safeguards would largely be invisible to us today. For now, we don't know. Whatever the reason for its exceptional equality (and longevity, having lasted almost a thousand years) it eventually broke apart. A millennium later South Asia was dominated by kingdoms such as the Magadha Empire.¹⁷

Once a Goliath emerges in a world region – as we see in China, Egypt, and Mesopotamia – it tends to re-emerge despite collapses. Yet, history is not just a story of the endless march of empire. People regularly escaped from Goliaths and tens of millions still do so today.

HOW TO ESCAPE A GOLIATH

In the highlands of Southeast Asia lies an unusual land. It is a 2.5-million-square-kilometre expanse containing 130 million people. If it were a country, it would be the tenth largest in both land area and population. It crosses ten states – China, Taiwan, India, Thailand, Laos, Vietnam, Malaysia, Cambodia, Myanmar, and Bangladesh – yet none of them control it. It is called Zomia and its population is, practically speaking, stateless. It contains a plethora of peoples with hundreds of different languages, ethnicities, and cultures. Many are resolutely egalitarian, but there are also some smaller feudal and hierarchical societies, such as the Shan (in Myanmar), as well as the Tay and Nung (in Vietnam). Yet none are full-blown states, nor have they been incorporated into their larger neighbours. For all intents Zomia is a stateless giant hidden in plain sight.

Zomia appears to be an intentional creation rather than the remnants of ancient 'uncivilized' peoples. For the past few centuries, both the

surrounding states and many scholars believed that the highlands were populated by barbarians who pre-dated the rice-growing lowland kingdoms and empires around them. That's why they lacked kings, writing, and intensive agriculture. But the anthropologist James C. Scott, who spent months living in Zomia, came to the opposite conclusion: Zomia was mainly created in opposition to the states surrounding it. It was a haven of people trying to evade state capture, whether they be refugees, exiles, bandits, rogues, escaped slaves, or oppressed minorities. For instance, the Akha recall their origins as highlanders who moved to the lowlands to farm before being pushed back to the highlands by Tai warriors. Sure enough, their accounts seem to chime with the creation of the first city-state by Tai raiders in the twelfth century, followed by the expansion of the Chinese Yuan Dynasty in the thirteenth century. The lifestyles of the Akha and many others were not archaic relics, but active attempts to escape empire.¹⁸

The violence of building and growing empires in the lowlands, as well as the pandemics that accompanied them, quite literally made people run for the hills. Sometimes what made them flee were the violence and economic losses that accompanied a collapse. There was a constant churn of new identities as more groups migrated into the area, mixing into the cultural soup. Some of those who walked and ran into the hills maintained a semblance of hierarchy. Yet most became dispersed and egalitarian. Zomia is a land that is not ruled by a Goliath. That appears to be an intentional choice rather than a historical accident.

The location of Zomia, the mobile ways in which these groups live, the crops they grow, and even the stories they tell, were all designed to evade being detected and caught by their Goliath neighbours. Not only that, but also to prevent their own societies from generating systems of domination. In many ways they resembled the several Native American groups who turned towards more mobile hunting and gathering, and egalitarian practices and stories, after the collapse of the first Goliaths. Both were practising what Scott called the 'art of not being governed'.¹⁹

This art of not being governed can help us understand how people behave during collapse: how many return to more mobile, rural ways of life, and how many avoid returning to a Goliath.

The people of Zomia, and countless non-state refuges over the past 5,000 years, have relied on their landscape, mobility, agriculture, and culture to

avoid being caught by a Goliath.

Zomia's geography is made up of hilly highlands that range from 300 to 4,000 metres above sea level. The mountains and the hills are difficult for an army to invade, easy to hide in, have plenty of bottlenecks to fight off a far larger foe, and provide ample exit options for fleeing people. It's an inaccessible terrain that is difficult to conquer and even harder to keep hold of. The landscape provides opportunities for hit-and-run tactics, is difficult to operate in, and gives those who have local knowledge a critical advantage. That's partly why the highly trained technologically supreme US military struggled to beat the Viet Cong in the wetland forest of Vietnam and Laos, or Taliban fighters in the mountains of Afghanistan (which has earned the country the moniker of 'the graveyard of empires'). Similarly, governments throughout South America have laboured against insurgents and crime syndicates who hide in the jungle. They are choosing land that favours guerilla warfare.

It's not just rebels, outlaws, and criminals who use geography to their advantage. Escaped slaves (maroons) did too. Many of the slaves who broke away from the cotton-plantations of the American South fled to the 'Great Dismal Swamp' that lies across Virginia and North Carolina. Others stayed closer to the slave plantations, concealed in nearby wilderness. Some started families, built homes, and cultivated farms in the impenetrable marshes that separated them from the grasp and lashes of their former masters. Geography can help create distance and friction between dominators and the oppressed, and people have used that to their advantage perennially.^{[20](#)}

The influence of geography in limiting the spread of Goliath can be seen across the world. One analysis of 1,094 traditional societies found that environmental barriers, such as higher elevations or more hard-to-traverse terrain, led to less diffusion of culture. Interestingly, this was the same for Eurasia and Mesoamerica, undermining the famous idea that Eurasia saw more cultural transmission owing to its east-west axis. Instead, Eurasia likely developed earlier, larger Goliaths simply because it had more Goliath fuel and domesticated its crops and animals earlier.^{[21](#)}

How you get your food can also help evade a Goliath. Crops such as cassava, taro, and potatoes are less lootable since, unlike grain, they can't be stored for very long and can be left in the ground. Colonial

administrators in the Americas deplored these crops and actively tried to paint those who farmed them as lazy. Their fear was that someone with cassava, fishing gear, and a shotgun to defend themselves had no reason to take on wage labour working on the hacienda plantations. Foods such as potatoes also made their way back to Europe, causing similar problems for overseers. Many Europeans adopted the potato during the Dutch Revolt (1556–1648) for the simple reason that it couldn't be easily confiscated. Peasants could leave their crop buried when a marauding army passed through and dig it up after they had left. The soldiers were left hungrier and the poor a little richer. Across Europe, most of the areas that adopted these 'escape crops' tended to experience increases in population without an increase in hierarchy. Escape crops are essentially the opposite of lootable resources: they have low storability once harvested, are less visible, have more unpredictable harvest times, and often require less labour to grow.²²

It wasn't just what you grew, but how you grew it. Shifting agriculture (also known as 'swidden' farming) makes farmers far more mobile. More mobile people are harder to control. The Punan Lusong people in Malaysia left rice farming and returned to the forests as foragers and swiddeners when the Japanese invaded. They returned to their farms in 1961, once the Japanese threat had well and truly subsided. Across Zomia and many other stateless areas we see the same use of these 'escape crops' and a greater reliance on hunting-gathering and swiddening.²³

Like the Native Americans, many of the peoples of Zomia have stories and traditions that warn against rulers. The Akha people have a tale of a thirteenth-century king (Dzjawbang) who tried to create the first census before being executed by his own people. His son is recalled as an Icarus-like figure who flew a shamanic horse with wings glued with beeswax too close to the sun before plummeting back to earth. They are tales warning about hubris and hierarchy. Dzjawbang is slain for trying to control his subjects too closely, and his son for soaring too high into the heavens. The Native Americans remembered their old rulers as evil, oppressive magicians (such as the Ani-Kutani) and malicious trickster gods (such as the 'Great Gambler') who eventually faced justice. The generations who lived after a failed Goliath project tended to remember them very differently from how we often recount history. Westerners still often look back to Rome with nostalgia, many Egyptians take great pride in the sandy tombs of the

pyramids jutting up from the desert, and the Chinese often celebrate their 2,000 years of ‘civilization’. Yet imagine if the earliest Goliaths in the Near East and China had collapsed and never returned. How would we remember them today? Would we reminisce about great kings with advanced civilizations who drove progress, or remember them as mad wizards and corrupt tyrants who were overcome by their people? Or, would we have just forgotten them entirely, as the Native Americans did with Cahokia?

Stateless peoples, whether in the wake of a collapse or fleeing a Goliath, often turn to the same strategies. They limit access to Goliath fuel by dwelling in inaccessible lands and staying mobile to help maximize exit options. They avoid growing lootable resources. They create cautionary tales of how hierarchy went wrong in the past. The exact approach depends on the circumstances. In the Andes, people usually fled to the lowlands and jungles rather than the mountains, simply because these tended to be forested and have less lootable resources. Many escaped slaves in the Great Dismal Swamp in the south-eastern US did grow rice and maize, but only when they were truly camouflaged by the land.^{[24](#)}

While places such as Zomia may have avoided dominance hierarchies, they are still a tiny minority today. That’s because Goliaths have evolved over time, developing better ways to control and coordinate larger groups of people without collapsing.

GOLIATH EVOLUTION

In 1519 a group of Spanish conquistadors led by Hernán Cortés arrived in Mesoamerica and met the Aztec Empire. Like Fontaneda meeting the Calusa, they did not encounter an entirely alien society. Instead, they saw a remarkably familiar one: the Aztec Empire was organized into commoners and elites, with a central government ruled by a king; the commoners gave away resources in tax and tribute; the elites intermarried and threw great feasts; they had a writing system that helped keep records and celebrate leaders; they had cities that were fed by the surrounding towns, as well as a road system for moving troops and trade goods. Out of the myriad ways in which one could organize a society, the Aztec Empire looked astoundingly

like its Spanish counterpart. Two societies separated by 8,000 kilometres and the Atlantic Ocean somehow came up with similar social blueprints.

We see this around the world. Different Goliaths from Cahokia and the Maya in the Americas to Egypt and Benin in Africa, and the earliest states of China, may have had different conceptions of kings and class, but that doesn't change the remarkable fact that they all had them, as well as slavery and ideologies that justified domination. This is what is commonly called 'convergent evolution': different Goliaths independently evolved similar structures.²⁵

Part of this convergent evolution is that early rulers faced similar challenges and often devised similar solutions – solutions that, if they worked, were more likely to survive and be passed down over time. For instance, early states all faced the common problem of controlling lands that were too far away from the capital and the direct grasp of the ruler. Goliaths began to develop internal administrations when they conquered territories that lay beyond a day trip from the capital. This appears to have happened to some extent in Mesoamerica (Monte Albán), Egypt, Mesopotamia (Uruk), and China.²⁶

When an early state expanded too far the leaders had to (begrudgingly) delegate authority to bureaucrats, governors, princes, and local elites. They had to do 'government at a distance'. This included relying on governors while trying to ensure they didn't break away and form their own kingdoms. Every empire thereafter has had to do the same, even today. Bureaucracy and administrations were a case of convergent evolution.

It wasn't just that rulers developed similar strategies to shared problems; they also created more effective ways of spreading them across time and space. Agrarian, state-based communities usually had effective ways of faithfully transmitting culture through generations and regimes, particularly through writing. Leaders and elites could also actively adopt and amplify certain practices and ideas. Coin-based currencies in both China and Lydia (modern-day Turkey) seem to have been set up by private individuals before being embraced and spread by governments. Christianity was already becoming widespread across Rome, but its official adoption by Emperor Constantine (which outlawed the persecution of Christians and supported the Church) aided its ascent. Hence there was sometimes a conscious process of either adopting and spreading innovations or

emulating the successful structure of other groups and empires. It wasn't perfect but, in the long run, evolution selected for larger, more powerful Goliaths.²⁷

Those who were the most successful at violent conflict tended to last longer and grow larger, making it more likely that their religions, warfare tactics, and models of governance would be adopted by others. Through cultural evolution, empires slowly accumulated technologies and practices that let them conquer and control larger groups of people over larger stretches of land. Most of the first states conquered and demanded tribute to siphon off resources, used human sacrifices and ceremonies to entrench their rule, and often had minor part-time bureaucrats and accountants help administer the territory. Scripts, when they did exist, were initially rudimentary, such as cuneiform, hieroglyphs, and Mayan glyphs, rather than fully fledged writing systems. These first states developed roads for shuttling troops and messages. While impressive, these mechanisms still meant that how much territory an empire could closely control was significantly limited. As we'll see, these limits were slowly lifted over time as technology and culture developed. The outcome was not just that Goliaths evolved but also that they often ended up on similar evolutionary trajectories.²⁸

OUR SACRIFICE

We've already seen numerous cases of convergent evolution in the earliest Goliaths. One is human sacrifice, which was practised in Cahokia, Tiwanaku, Teotihuacan, Monte Albán, and by the Olmecs of San Lorenzo. Almost all the earliest states across Mesopotamia, China, the Americas, and western Africa practised it as well. The burial of a ruler alongside dozens, hundreds, or even thousands of human victims occurs across the world in early states from the Shang Dynasty in China, Ur in Mesopotamia, Keber in Nubia, as well as the Wari and Moche in South America (the precursors to the Inca). At Yinxu, the capital of the Shang Dynasty, China's second earliest state, around 13,000 people were sacrificed over two centuries. Many appear to have been slaves and war captives who were brought into

the city, forced to work with a restricted diet, and then executed. Hundreds at a time were sacrificed, with the victims often beheaded and mutilated.²⁹

Human sacrifice was a popular practice partly because it helped to establish early Goliaths. It is a political performance that intimidates challengers and justifies the political order. Most cultures have taboos against in-group killing. A ruler can legitimate their rule by showing that in-group killing is acceptable only when the ruler does it. Studies suggest that the biggest predictors of human sacrifice across cultures are hereditary rulers, bigger populations, and internal conflict over resources.³⁰

Publicly showing that you can legitimately murder others, and that the lives of the many are worth less than one elite individual, can help a budding ruler demonstrate their power to a big crowd of allies (and rivals). It even happens outside states. A study of ninety-seven pre-colonization Austronesian societies found a clear correlation between human sacrifice and social hierarchy. Over two-thirds of the cases with a strong hierarchy practised human sacrifice, while just a quarter with little to no stratification chose to sacrifice people. Those who used human sacrifice were less likely to revert to egalitarianism.³¹

But as hierarchies became entrenched, the practice of human sacrifice fell out of favour. Relying on a bureaucracy and religion was less costly and less prone to backlash than the spectacle of public murders. Administration trumped intimidation. Emerging kings across the world faced a similar problem in trying to prove they had the right to rule. Human sacrifice was a solution that many converged on and then later dropped.³²

Human sacrifice may have provided the grounds for other points of convergence. Sacrificing people was usually a way of giving a gift to superior beings (such as gods or revered spirits and ancestors) to avoid punishment and gain favour. In other words, it was a supernatural form of tribute. You were simply offering human lives to a god or ancestor, rather than slaves or gold to a monarch.

It also hints at one of the most powerful commonalities across every Goliath: the ability to deal out random violence. Over time, leaders have become more constrained by law and expectations against the arbitrary execution of their citizens, but this was once the ultimate primeval display of power. It's an act that puts the ruler above ordinary morality. Politics

across the world have all settled on the importance of violence to make rulers legitimate.³³

STORIES OF SUBJUGATION

We also see convergent evolution when it comes to stories that defend the domination of the few over the many. One version of this propaganda is the Hobbesian story. Hobbes's philosophy is not just about bringing peace to a violent world, but also about bringing order to a supposedly chaotic one. It crops up across the world, from Europe to China and India. Empires have always made similar claims that they bring unity and equilibrium to the cosmos. The Assyrian Empire (2025–609 BCE) depicted their lands outside their borders as an 'outer realm' of chaos. The emperor was morally obliged to expand the frontier and tame the chaos into civilization. It was a divinely sanctioned duty. Like human sacrifice, the Hobbesian story was adopted by many early rulers to help lower the natural aversion of their subjects to dominance.

That Hobbesian tale was part of another common story that almost all Goliaths developed: the battle between the civilized and the uncivilized. It's a near universal myth. The Toltecs and people of Teotihuacan of Mesoamerica labelled non-state foragers as *chichimeca* (dog people). In Imperial China non-state peoples were split into categories of 'raw' (*sheng*) and 'cooked' (*shu*). The cooked were those who had been incorporated into the empire and could become normal (good, or *liang*) subjects. The raw were those who were still too uncivilized to be properly integrated into the empire. The very term 'barbarian' is derived from the Greek noun *barbaros*: an insult, indicating that, to the Greeks, foreigners spoke rough, staccato tongues that sounded like 'ba-ba'. The transatlantic slave trade and mass killings across the Americas and Australia were also defended as fulfilling a sacred duty to bring civilization to savages.

Today, civilization is still largely used as a positive term associated with progress. We have popular stories of how the adoption of agriculture and states has been a march towards less violence and more prosperity. The narrative of civilization makes those outside the state look backward and dangerous, and those within the walls more refined and superior. It's a

useful fiction for making the existing order seem just and to deter anyone from leaving.³⁴

Accompanying these stories of civilization have been stories to justify the inequalities integral to Goliaths. Egyptian pharaohs were supposedly reincarnations of gods. Sumerian kings usually referred to themselves as representatives of gods on earth. In China, the first rulers painted themselves as oracles, and later kings and emperors claimed that heaven had bestowed on them the right to rule for the good of the people (the ‘Mandate of Heaven’). Even the king of the Calusa was a divine figure.³⁵

The historian and archaeologist Ian Morris once defined ideology as ‘a pack of lies that benefits someone’. Most ideologies are ideologies of domination: they benefit those atop a hierarchy. Whether it’s the Hobbesian story, the caste system, racism to justify slavery, kings representing gods, or the myth that progress rests on the indispensable genius of a few billionaires, every society has stories to justify their inequalities.³⁶ Such stories overcame our dominance aversion by justifying hierarchies as a matter of equity, fairness, or simply sheer superiority.

These stories still work today. The modern story for inequality is the myth of meritocracy: the fable that people attain power thanks to individual skill and performance (see the above number of hereditary billionaires). One study of twenty-three modern, rich countries found that those with higher levels of economic inequality had stronger beliefs that wealth was earned meritocratically. Stories and inequality work in a feedback loop: the myth of meritocracy justifies higher inequality and higher levels of inequality result in more powerful people reaffirming the myth.³⁷

Moralizing religions too are an outcome of Goliath evolution. Across the world it appears that people moved from general animist beliefs as hunter-gatherers (a belief that spirits or life-force permeate all the things around us) to a belief in the afterlife, a veneration of ancestors, and eventually the adoption of punitive gods. Gods who constantly watch people and punish or reward moral behaviour are a highly effective force for controlling large groups. These surround us today in the gods of the Abrahamic religions – Islam, Christianity, and Judaism – and reappear across different continents and time periods, such as the Incan supreme god Viracocha. One systematic analysis of hundreds of sites found that in all cases these gods appeared only after large, unequal, hierarchical societies developed. Big kings

preceded big gods. It makes sense: a hierarchy in heaven justifies a hierarchy on earth.^{[38](#)}

What we see in the first Goliaths is a slow fusing of the different forms of social power. All the first states had some form of authoritative hierarchy (political power), backed by force (violent power), with rituals and a story that bequeathed a right to rule (information power), and took resources (economic power) from a community of people (demographic power). There was no single path to statehood, yet they all looked remarkably similar: elites taking wealth and labour from a mass of commoners. They all converged on a similar evolutionary pathway: a pathway that placed men on top.^{[39](#)}

A GENDERED GOLIATH

Patriarchy – a social arrangement in which men hold most power and women are excluded – is one of the most common shared characteristics across states and Goliaths. The Aztec Empire, the Spanish Empire, the imperial dynasties of China, and the kingdoms of Mesopotamia were all dominance hierarchies based on gender. Patriarchy was another point of evolutionary convergence.

Patriarchy had modest beginnings. As covered earlier, men have stronger inclinations for status competition and are far more likely to pursue it through violence and domination. On top of that, men are more capable of achieving status through force. Men are generally larger, stronger, and more aggressive (women on average have 52 per cent of men's upper-body strength and 66 per cent of their lower-body strength).^{[40](#)} There is a reason why men have been the main soldiers and killers across history. While it is not the strongest or meanest who commands the military, mafia, or state today, physical strength and aggression were far more vital in the wars that involved the use of axes, bows, and spears.^{[41](#)}

Within early hunter-gatherer groups these differences did not seem to translate into large-scale male oppression. That is because violence was muted. This changed as lootable resources became available and war arose. Lootable resources are storable but also make a society a prime target for

raiding. Having a close-knit group of male fighters who know and trust each other is advantageous in a conflict. Groups that were regularly engaged in warfare tended to need to keep fighting men together and hence were disproportionately likely to be patrilineal. That is, when marriage across groups occurred, the men stayed in their own group while the women migrated. A migrating woman who was in a new setting and uprooted from her family and friends had far less social power than the man she was now married to. The power balance between the sexes shifts dramatically when this happens.⁴²

The groups that persisted and won battles were more likely to be the patrilocal ones that favoured keeping their male fighters together. Such groups were also more likely to become conquerors and pass on their patriarchal practices and beliefs. That is what we see throughout history. Successful early conquerors tended to have strongly patriarchal cultures with a glorification of male violence and rule. The Akkadian Empire, which conquered much of Mesopotamia, Anatolia [modern-day Turkey] and the Levant, and the Yamnaya raiders from the Pontic Steppe both shaped the early culture of Eurasia, and both were aggressively patriarchal.

Patriarchal societies also tended to become more oppressive over time. Institutions, religions, and legal codes all reinforced and amplified male rule and shrank women's status. The first states were a turning point with large-scale, centralized hierarchies that celebrated and endorsed male rule.⁴³ This was reflected in their legal doctrines. The Code of Ur-Nammu (2100 BCE), the first-known legal code in the world, was obsessed with sex; one of its rules was that if a married woman committed adultery she would be put to death (but her lover would receive no punishment). Such rules tended to spread. For instance, biblical laws on sex, such as cutting off the hand of a woman who injures the testicles of a man (Deuteronomy 25:11–12), seem to be borrowed from the Assyrian Empire, a rival neighbour to the Kingdom of Israel.⁴⁴ Similarly, monotheistic religions – including the Abrahamic faiths – valued sexual segregation, valorized male rule, and praised chastity as well as practices such as veiling. Whether it be states, legal codes, or monotheistic religions, the rules disproportionately benefited men and reinforced their privilege. Such rules may have also helped to quell infighting over women and property in larger societies. Patriarchy is another example of Goliath evolution.⁴⁵

Patriarchy was not (and still isn't) in the interests of all men. As we've already covered, the adoption of lootable resources 8,000–4,000 years ago happened alongside a collapse in the male gene pool.⁴⁶ That's because a small number of men were dominating reproduction. Similarly, harems and polygyny only benefit a sliver of men in a society. Patriarchy was not a social blueprint created by men for men.

Instead, patriarchy was the outcome of a blind process of Goliath evolution. Goliath fuel such as monopolizable weapons and lootable resources also fuelled male domination. A few minor differences in psychology and physiology meant men were more likely to try to seize power by violence, and violent conquerors were more likely to be patriarchal. Over time the power imbalance between the sexes was amplified by new rules and religions, resulting in the deep divides that are carved into history. Domination of women by men became central to the story of Goliath.

THE LOSERS OF EVOLUTION

These different features of convergent evolution were attempts to stabilize a Goliath. In the language of systems, they are 'balancing feedbacks': they serve to resist change and reinforce existing structures. (Think of a thermostat. It monitors the temperature of a room and turns on the heating if it gets too cold or turns it off if it's too hot. Similarly, balancing feedbacks dampen and resist attempts to change the system.) There is evidence that they worked and that a failure to evolve the right mechanisms often led to collapse.

The original states all developed some kind of internal administration and bureaucracy. They formalized their structures of rule and eventually developed state religions. One study using the Seshat databank found a clear correlation between the size of the polity and its 'information processing' tools (mechanisms to order and store information, such as writing and bureaucracy).⁴⁷ Such tools allow for better tracking of citizens (for tax, conscription, and even to identify dissidents), propaganda, and government at a distance. When we look at many of the earliest Goliaths, whether it be the Pueblos, the first farmers of Europe, or the mound-

based kingdoms of the Mississippi, most of them never developed those kinds of tools. Tellingly, they often went through fission-fusion cycles and showed higher levels of hierarchy and inequality before collapsing, fragmenting, and repeating.

Hierarchical hunter-gatherers and other unequal farmers seem to have experienced a similar fate of fission-fusion. Across modern India, Tibet, and Bhutan are different traditional societies that practise shifting agriculture (also known as ‘slash-and-burn’ or ‘swidden’ agriculture), in which a field is used, burned, left fallow, and then returned to later. Societies such as the Konyak Naga of modern north-eastern India frequently cycle between hierarchical societies in which people are ranked – and those social positions are passed down through birth – and more egalitarian settings in which people are unranked, with leaders who are not hereditary and have more limited power. This looks much like the fission-fusion patterns that have been practised for millennia. Part of this may be due to shifting agriculture, whereby lootable resource are created, but the constant movement of people and their homes undermines centralized control. Part of it may be that they have never had the Goliath fuel to develop greater mechanisms for stability and control. These live examples of inequality in the making show that inequality does not occur automatically. It is fought for and often resisted.⁴⁸

The Calusa and north-west coast Native Americans underwent centuries of struggle to create a dominance hierarchy. Trying to gain control over a community with a deep history of egalitarianism is a momentous challenge. These hierarchical foragers would have been preceded by many failed attempts to create a Goliath. Even then they tended to be small, unstable, not highly centralized, and prone to infighting. Some archaeologists have referred to these early Goliaths as ‘fragile’: leaders vie for status and authority; people seek independence; and there is no well-developed bureaucracy to hold everything together.⁴⁹ A fission-fusion cycle tends to ensue. The battle for status, with either words or weapons, made the transition to Goliaths fraught, haphazard, and prone to failure.⁵⁰

This lack of mechanisms to control a populace was one curse that plagued the earliest Goliaths.

Goliath's Curse

Do Societies Age? – Why Inequality Grows With Lootable Resources – Extractive Institutions and Unequal Exchange – How Extraction Creates Fragility – Triggering Collapse – Survival of the Fairest – The Wisdom of Democracy

DO SOCIETIES AGE?

There is a trend that we see over the past ten millennia. Over time many hierarchical societies appear to become more fragile. They seem to be ageing.

This doesn't just happen in the earliest Goliaths, like Cahokia and Tiwanaku; it is common in far more developed empires and kingdoms as well. The Western Roman Empire fell to infighting and invaders but had been successfully coping with both for centuries. The Lowland Maya (neighbours to Monte Albán and Teotihuacan) collapsed during one severe period of drought but had navigated earlier ones.

Does this apparent pattern of ageing hold true for a larger selection of cases? To find out, I and a group of complexity scientists led by Professor Marten Scheffer analysed two databases of state lifespans: the Seshat Global History Databank (the world's largest historical dataset) and the Mortality of States database (MOROS, a collection of 324 states from 2000 BCE to 1800 CE). This study covered fully developed states, not just early Goliaths. We've encountered only a few of these thus far, such as Monte Albán and Tiwanaku, but most collapses are usually centred on states.

We found that states were increasingly more likely to terminate over the first two centuries of their existence. After this point, they had a high but stable rate of termination. Imagine that your likelihood of dying increases as you age up until seventy. Thereafter your odds of perishing each year stay at that high rate until you do eventually go to your grave. We saw a similar pattern in states: a rising risk over the first 200 years followed by a constant high risk of termination. It's not a one-off result either. One database of 168 crisis events (the 'Crisis Database') showed that the average duration for polities was 201 years.¹

As something grows more vulnerable it often begins to recover more slowly from disturbances. We see this in the elderly, who begin to struggle more with recovering from illness or injury, such as a broken hip. This is called 'critical slowing down' and tends to precede a 'tipping point' before a rapid, large-scale shift. That critical slowing down appears to have occurred in cases we've already encountered, such as the first farmers of Europe and the Pueblo societies of south-western America. Analyses of population levels among the first farmers of Europe and construction efforts (measured by the number of trees felled) among the Puebloans suggest that these communities recovered more slowly from shocks such as droughts and population losses just before their collapse.²

The statistics reflect what we've seen in the case studies: many societies do appear to become more fragile over time.

WHY INEQUALITY GROWS WITH LOOTABLE RESOURCES

The first farmers of both Europe and the American south-west, and many of the earliest cities and settlements, faced a similar problem. Inequality and rulers began to creep up from within their communities. It is a persistent trend throughout history: inequality tends to increase once a community becomes hierarchical. That is, until it undergoes a major institutional reform or is hit by a great leveller: a bloody revolution, mass mobilization warfare, a mass-mortality pandemic, or a deep collapse. Fire and sword often leave a level playing field.³

This trend of rising inequality is because of Goliath fuel (lootable resources, caged land, monopolizable weapons). One global review by archaeologists highlighted that lowered mobility (caged land), big populations, and surplus resources that were predictable, dense, and could be transferred across generations (lootable resources), were the key factors that led to inequality. Similarly, a review of the global history of democracy found that the factors favouring early democracy were an abundance of exit options, smaller populations, and a lack of knowledge by rulers about what citizens were producing (lootable resources are easy to observe). In other words, all the sources of Goliath fuel favoured inequality and autocracy.⁴

Once a Goliath was created, inequality tended to increase over time and in turn favour more autocratic arrangements. There are two key reasons for this persistent pattern. First, wealth can easily be translated into other forms of power (and vice versa). Richer families with more land among the first farmers of Europe would have had not just more food, but also more riches to purchase weapons with. The feast-throwing leaders of hierarchical hunter-gatherers were essentially turning edible wealth into political power in the form of debts, prestige, and alliances. We still see this today. It can be as subtle as companies lobbying regulators and politicians for rules that increase their profits, or as blatant as a billionaire becoming president and then introducing tax breaks for the rich. Power begets more power.⁵

Second, the value of capital (assets such as land and stocks) tends to rise more quickly than the value of labour. The economist Thomas Piketty famously showed that the financial returns on capital (land, assets, stocks) have outpaced the returns on labour (wages) for at least the past two centuries. That is despite labour requiring more effort and skill than does charging rent on land or betting on stocks. Simply put: the rich get richer quicker than the poor can get richer, and often the poor simply get poorer. More riches can then be transferred into other forms of power and the cycle continues. Some have taken to calling this a ‘wealth pump’: money is transferred from the bottom of society to the top.⁶

If a Goliath didn’t collapse, the levels of inequality reached were often staggering. One review of twenty-eight pre-modern states, from the time of Rome to British India in 1947, found that they were on average 77 per cent towards the theoretical maximum level of wealth inequality. That maximum means a situation where an individual or small group takes all the excess

produced by a society ('the surplus'), leaving just enough for the rest to survive and reproduce ('subsistence'). Sixteen of those kingdoms and empires were close to being as unequal as possible. Things have improved as most modern states today are around 44 per cent towards the maximum level of inequality (from a low of the Netherlands of 29 per cent to a high of 63 per cent in Brazil).⁷

A few cases even exceeded this maximum level. That could be due to bad measurement, or because a proportion of the population was being worked to death.

Throughout most of history, the top 1 per cent tended to capture 10–40 per cent or so of the total income of a society, and even more of its wealth.

Egypt and Cahokia didn't just build pyramids: they were social pyramids.⁸

There are a few cases that were more equal, such as ancient Athens, and early modern Holland and England, but these were far from democratic, egalitarian arrangements. Athens was a slave-holding society where only 10–15 per cent of the population – just free, male citizens – could participate through voting and speaking at the assembly (*ekklesia*) or being picked for the law courts that oversaw legal decisions (the *dikasteria* that were randomly chosen). Holland and England were aristocratic monarchies. Inequality has been the rule of history since the first Goliaths.

EXTRACTIVE INSTITUTIONS AND UNEQUAL EXCHANGE

Not all Goliaths are the same. They can diverge in how unequal the distribution of power is. More concentrated power (including control of resources, decision-making, violence, and information) within a society tended to lead to elites extracting more wealth, energy, and labour from the rest of society. Hence *extractive institutions* are effectively a measure of how concentrated social power is. A society may be a Goliath but have a broader distribution of power: compare the more egalitarian, semi-democracy of ancient Athens to the highly unequal militarized dictatorship of the Assyrian Empire, or Norway to Saudi Arabia today. All of them technically still organize into dominance hierarchies in which a few make

rules that are enforced with the threat of violence, but there is still an enormous difference in how democratic they are, or how free and diverse their media is. In other words, it's not just wealth inequality, but how unequal the other forms of power are, including how autocratic decision-making is in government, companies, and families.⁹

Increasing wealth inequality tends to spill over into other forms of power and make the overall society more extractive. Tiwanaku and Monte Albán didn't only become more unequal; they also became politically lopsided with rulers rising up and becoming more powerful and self-aggrandizing. Rome fell from republic to dictatorship as inequality rose, and that dictatorship and its bureaucracy became increasingly captured by rival elite families and power-hungry generals. In each case, Goliaths were becoming more unequal and extractive: all four forms of power were becoming more concentrated, and these inequalities in power were used by elites to take more energy, labour, and resources from the rest of the populace.¹⁰

Concentrations of power are extractive because they tend to lead to unequal exchange. Imagine two traders with equal power. Their exchange of goods should be relatively fair. Now imagine one of them suddenly acquires a large arsenal of weapons that they brandish in front of their trading partner. Suddenly one trader may be more likely to accept a bad deal. One actor having more power has changed the bargain and made the exchange more unequal and extractive. We see this in the case studies covered thus far. The maize beer feasts provided by the elites of Monte Albán (or later by the Incan emperor to his subjects) was only a small fraction of the corn that he took from them. Rulers would keep the rest of the surplus for themselves, which of course showed up in the larger houses they dwelled in, the more lavish decorations they adorned themselves with, and the grander monuments they had built to celebrate themselves with (all acts of conspicuous consumption for status).

Often the benefits provided by leaders as part of the exchange were immaterial. The priests of Cahokia might well have claimed that they controlled the weather (and the harvest) but they did not. Nonetheless, in exchange for their ineffective services they were rich and powerful in life, and were gifted thousands of beads and dozens of sacrificed lives in death. This is key to why stories of subjugation – be it weather-controlling priests, kings who represent gods, or billionaires taking us to space – work: they

justify unequal exchange. They give an explanation as to what the leader is providing (as invisible or fake as that might be) and why the exchange is fair. Suddenly giving up a quarter of your wheat harvest may seem reasonable if the ruler is protecting you from evil spirits or a nearby neighbour.^{[11](#)}

Many of these early Goliaths did offer benefits even if they were an unequal exchange. The rituals of Cahokia could have built camaraderie, and the leaders of Monte Albán may have been skilled military commanders (although we have no evidence that early pharaohs or kings were particularly good decision-makers). Yet as the society grew more unequal and extractive, the costs began to pile up and exceed the benefits.

HOW EXTRACTION CREATES FRAGILITY

The people of Tiwanaku, Çatalhöyük, and San Lorenzo did not just walk away from their homes as a way of avoiding domination. We have signs that inequality by itself is corrosive to societies. More unequal countries tend to have poorer physical health, worse mental health, more violence, and suffer from other social ills ranging from more crime to more teenage pregnancy. It's unsurprising that a species evolved in conditions of general equality has a rough time living in unbalanced communities.^{[12](#)}

As inequality rises, the society becomes more extractive. A wealth pump is established in which social arrangements increasingly benefit elites at the expense of the poor. In short, the exchange across societies gets more unequal. That could be due to ancient kings and elites demanding more and more grain from their subjects while offering less and less in return, or lobbying by the rich in the modern world to give themselves tax breaks while more heavily taxing (or eliminating benefits for) the poor. The cost of living gets higher and higher, leading to worsening health and decreasing standards of living. This is what some refer to as 'popular immiseration': the impoverishment of the masses.^{[13](#)}

Inequality also tends to trigger other knock-on effects. One of the most important is elite competition. We see signs of this through the rebellion in the grand city of Teotihuacan to the fission and fusion of mound-based societies across the south-east of the US. It will become even more apparent

in larger empires. As wealth inequality grows and the society becomes more top-heavy, the competition for high-status positions intensifies. The rich and powerful tend to have more resources to cause trouble, whether that be starting a war with a neighbour (as with the first farmers of Europe), stoking a revolution (potentially in the case of Teotihuacan), or creating an outright civil war.

Goliath's Curse is that extractive institutions hold the seeds of their own demise. The abandonment of many early Goliaths is the first manifestation of that curse. Counter-dominance intuitions meeting new rulers, the costs of inequality, and rising elite competition all eroded societies from within. Early Goliaths also lacked the mechanisms to control their populace, leading to either fission-fusion/boom-bust patterns or wholesale abandonment. This meant that even emerging Goliaths with low levels of inequality (such as many of the mini-Goliaths of North America) could prove unstable and fall apart. Sometimes the rejection of a new elite was swift, while at other times it was a slow process of growing fragility. Yet, weak institutions and growing inequality were sometimes not enough to cause a collapse. Something more was needed.

TRIGGERING COLLAPSE

One verbal retelling of the Huhugam collapse involves a great chief (Siwani) and two servants. The servants are the Wind and Storm Cloud. Together they provided great harvests for their people. Then, one day, the Siwani grew irritated with the Wind and Storm Cloud and ordered them both to leave. Naturally, this didn't end well. The harvest came to a halt and then there was a great flood. The Siwani was deposed by an angry, starving mob.^{[14](#)}

Many early rulers, like the Siwani of the Huhugam, seemingly built their authority on an ability to ensure good harvests and avoid the ill will of the spirits. We still see this in religions across the world, which regularly include gods for the harvest, the sun, and even lightning and thunder (notably, the Norse god and now Marvel character Thor). When the weather turned foul and a flood or drought struck, it was a blow to the legitimacy of the ruling class (much as incumbent politicians tend to lose popularity when

the economy tanks or inflation rises today). Even without weather-controlling kings, a few bad harvests could make living in a hierarchical city look far less appealing.

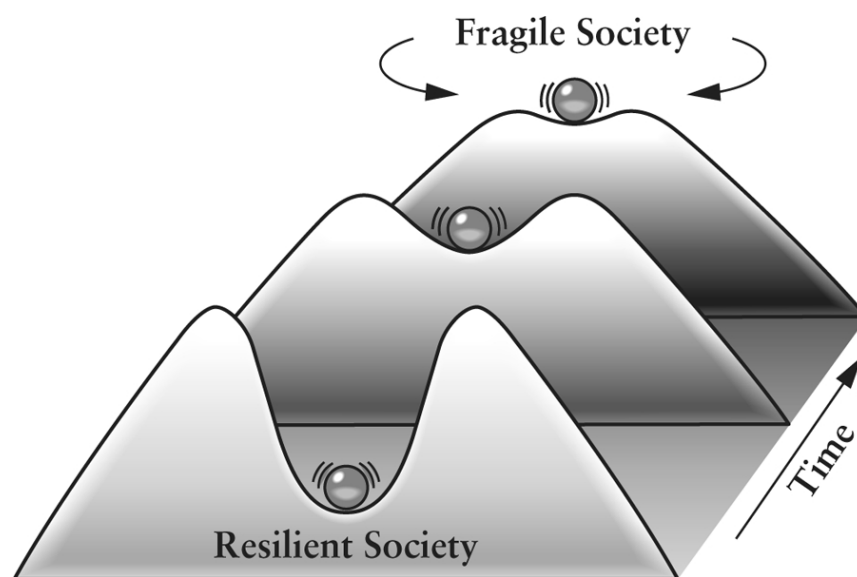
Climate change is a common presence across collapses. Sometimes it played a central role in causing collapse and sometimes it didn't. Not all droughts are created equal, and more severe ones could do far greater damage, as was the case with the final termination of the Pueblos. In Monte Albán, nearby settlements experienced a similar drought but were not abandoned. While the climatic record matches some of the collapses of the first farmers of Europe, it's not a perfect fit and can't explain the boom-bust pattern they experienced. Moreover, different farming communities responded differently to climatic shifts. We see this in the first farmers of Europe. Scotland experienced similar conditions to England and Ireland yet did not see the same downturn in population. Climate change as a threat was most dangerous when it hit an already vulnerable society, one that was already riven with discontent against rulers and rising inequality. Climate change was most often the trigger that exploded a society flammable with social tension.¹⁵

Another trigger could have been disease. Packing people and animals together into cities created a hotbed for new infections. Although the telltale signs of an outbreak, such as mass burials, aren't always in evidence, we can't rule out the possibility that diseases caused by crowded people and animals instigated a slow or rapid exodus from early settlements. Then there is the threat of invaders. Many early European farming communities were wiped out by their neighbours.

When faced with such threats, communities that were increasingly vulnerable often collapsed. Tiwanaku, Cahokia, San Lorenzo, Monte Albán, the Pueblos, and the first farmers of Europe were all becoming more fragile as they became more extractive and then fell apart as they faced shocks, usually in the form of drought. A useful metaphor here is that of a ball (a system or community) rocking in a valley (a particular structure or way of life). More extractive institutions made the valley shallower, while shocks such as war, rebellion, and drought all violently knocked the ball around (see [Figure 18](#)).

When the ball sits on the cusp, ready to roll out of the valley or back, we can think of it as a 'crisis': a tumultuous period that can lead to a collapse,

reform, or an entrenchment of the regime.¹⁶ When the ball is finally knocked out of the valley, we call it a collapse. Sometimes a big shock might knock a ball out, but sometimes the valley is so shallow that just a small nudge might be enough to dislodge the ball. We call this a ‘tipping point’: a critical threshold where a small disturbance can change the entire state and future of the system.¹⁷ The more vulnerabilities a community possessed, the fewer mechanisms it had to control its subjects, and the more shocks it faced, the more likely it was to be knocked out of the valley and collapse.¹⁸



[Figure 18](#): From societal resilience to vulnerability

There is a consistent picture emerging of the collapse of the first Goliaths. Societies became fragile as they became more unequal and hierarchical. Discontent and elite conflict appeared. These societies lacked the mechanisms to control large and increasingly hostile groups of people. Once this increasing stress was hit by a large enough shock, whether that be a drought or an invader, a crisis or even collapse ensued. It fits the details we have of the boom-bust cycles of the Puebloans and first farmers of Europe, the abandonment of many of the earliest cities and settlements, and the synchronized collapse of the pyramid cities of North America. It was often a social tipping point. As we see in the Puebloans and the first farmers of Europe, they were recovering more and more slowly from shocks, which suggests that their valley was getting shallower and shallower. Some could

collapse just from this growing internal vulnerability. Many others fell apart once this growing fragility met a shock: such as a drought, disease, or invaders.¹⁹

Experts in disaster risk management often say that no disaster is ever natural. It always depends on the vulnerability of the community it hits, and how that community responds. Remember that risk is not just a matter of threats and vulnerability, but also of exposure and responses. More extractive arrangements also worsened a society's capacity to respond quickly and effectively to a crisis.

SURVIVAL OF THE FAIREST

Monte Albán fell apart amidst a drought, while settlements around it living under the same dry conditions expanded. Something was clearly different in their responses. We see this in examples across history. Societies could experience similar climatic challenges and fare remarkably differently.

One example is the 'Late Antique Little Ice Age' (536–660 CE). Three different volcanic eruptions (536, 540, and 547 CE) led to several pulses of cooling temperatures across the northern hemisphere. While the exact effects varied (overall cooling may have been up to 1°C) across regions, this was generally a cold time for the north. Thirteen out of the twenty coldest winters in the last 2,000 years occurred during it.²⁰ Not every society experienced the same amount of cooling or transformation. The Merovingian Dynasty that ruled over much of modern France after the fall of Rome experienced a temperature drop of 0.79°C, the pottery-producing hunter-gatherers of Ontario got 0.74°C colder, Jenne-Jeno cooled by 0.26°C, and Monte Albán got just 0.15°C chillier. One study of twenty communities across five world regions and four continents during the Late Antique Little Ice Age found that climate did have some correlation with social change, but the effect varied. The hunter-gatherers of Ontario had the equal lowest amount of social change, despite experiencing the third greatest drop in temperature, while Monte Albán had the equal lowest amount of cooling and highest possible level of change (it both became a full-blown state and then collapsed). Some societies were simply more resilient to temperature swings than others.²¹

The biggest predictor of resilience was how democratic or extractive the society was.²² One study of the Late Antique Little Ice Age found that more inclusive societies had less famine, disease, and conflict, and less change in their rituals or how they were organized. Those results have been verified in another study of thirty-three societies covering twenty-two climate-related disasters over nine world regions, four continents, and a thousand years. More democratic groups tend to respond better to crises.²³

The question is, why? Local people could respond more quickly and effectively to new challenges than distant elites who were often buffered from threats such as a drought. This meant that more dramatic changes (such as collapse) were needed. A second benefit is that shared decision-making means that people in a community had more social connections, which created more effective collaboration (what academics usually call ‘social capital’) during times of stress. This holds true today: acts such as voting, deliberating, and being meaningfully involved in civic institutions (whether that be a sports club or a political group) foster closer and wider social ties. Those ties keep communities together when catastrophe strikes.²⁴

A plethora of studies on modern disasters support the idea that more politically inclusive societies are more resilient ones.²⁵ As one group of disaster-risk experts concluded, ‘there is near-unanimous agreement that people’s participation is essential to reducing the impacts of disasters on society’.²⁶ One indicative study used the Emergency Event Database, an international repository of information on disasters across the globe. This analysis of 150 countries over the period 1995–2009 found that more democratic countries, with more state capacity, had fewer fatalities and fewer people affected during disasters. This was true for a range of disasters, from floods to earthquakes. Strong states have more resources to deal with disasters, while democratic accountability ensures that aid is distributed wisely and not just concentrated on the rich and powerful. Inclusive political institutions create more flexibility, effective responses, and unified societies, and they prevent abuses of power.²⁷

The opposite naturally holds true. Societies with more extractive institutions are likely to have fewer social bonds and respond more poorly to natural disasters, whether that be the droughts that plagued the Pueblos

or the droughts and floods that hit Cahokia. There is, moreover, another reason that extractive institutions tend to be fragile and inclusive institutions resilient: more democracy means better decision-making.

THE WISDOM OF DEMOCRACY

We have ample theoretical and empirical evidence that, over the long run, a large, diverse group of people sharing information leads to better choices being made on average than an ‘enlightened dictator’ or small cadre of technocrats. The logic behind this is the ‘wisdom of crowds’: the knowledge of the many outperforms the knowledge of the few. The wisdom of crowds was first posited by the statistician Francis Galton during a county fair. He observed 800 people guess the weight of an ox. While most were off the mark, the median estimate of 1,207 pounds was, remarkably, less than 1 per cent off the true weight.

Since then, a variety of studies have shown that crowds outperform individuals when it comes to predicting the outcomes of sporting events and geopolitical events and even answering general knowledge questions. Prediction markets, in which a large group of people bet on different outcomes, have also had significant success. The benefit of having large and diverse groups of people is at the cutting edge of forecasting (making numerical predictions about a future event), foresight (creating plausible pictures of the future), and studies of ‘collective intelligence’ in swarms of insects, flocks of birds, and groups of people. Even large, important scientific endeavours, such as the Intergovernmental Panel on Climate Change (IPCC), hinge on the group judgement of a large set of scientists from different backgrounds and disciplines. While most of our modern systems are built on a small number of overly powerful decision-makers making rushed choices – from CEOs to presidents – when it comes to trying to predict the future or reach the best judgements, scientists rely on large diverse groups.^{[28](#)}

What is just as important as having large diverse groups is ensuring that individuals in those groups share information. Think of a complex problem as a jigsaw puzzle with numerous pieces. Everyone brings a different piece of the puzzle: different information, a different perspective. But it is only

when they talk in an open-minded fashion and exchange information freely that they can put their different pieces together and complete the picture.

Deliberation works. When it comes to predicting future events the gold-standard approach is ‘superforecasters’. That is, creating teams of expert forecasters with unique traits and impressive track records (their past accuracy with prediction is calculated as a ‘Brier score’), letting them share information, taking the average of their guesses (weighted by their Brier score), and then ‘extremizing’ (for instance, an estimate below 50 per cent would be nudged further down, and one above 50 per cent would be pushed slightly higher with an algorithm). Despite the details, it really comes down to using the wisdom of democracy within a group of exceptional forecasters. Similar approaches have proved to be most effective at identifying long-term issues in areas such as healthcare and biodiversity conservation.

Experiments with ‘deliberative democracy’ in which a randomly selected group of citizens gather together, are briefed by experts, deliberate, and make policy (either in a small jury of nine to twelve members, or assemblies in the hundreds or thousands) have been shown to reduce political polarization and even aid reconciliation in war-torn countries such as Bosnia. A deliberative assembly even helped Ireland legalize and regulate abortion (2018–19), which had previously been an intractably fraught issue. And of course, the backbone of science, including peer-review publications, is built on open debate and discussion of findings. In both theory and practice, it’s becoming increasingly clear that inclusive institutions make better decisions (and promote social cohesion too).^{[29](#)}

Making good decisions is paramount to ensure that a shock or a crisis doesn’t spiral into a collapse. Notably, most historical polities weren’t particularly democratic; nor are our modern democracies. Modern parliamentary systems rely on aggregating votes to select a few homogeneous leaders (overwhelmingly old and rich) who then engage in little genuine deliberation (often due to being overburdened, time-poor, and lacking expertise). This is on top of the sheer impossibility of expecting just a few people to solve the myriad problems of transport, energy, defence, and welfare. In modern electoral parliaments, people consent to being ruled; ruling itself isn’t widely shared. In other words, most societies have seen only a sliver of the resilience benefits that inclusive institutions have to offer.

The opposite holds true. An oligarchy with smaller, more homogeneous groups of elites who are cushioned from shocks is more likely to make bad decisions and delay necessary change. Writers of collapse history have often puzzled over why leaders and groups seem to make bad decisions in the face of disaster. There is a simple answer: oligarchy. As societies grow more unequal and extractive, decision-making becomes worse.

It's not just that small groups of elites make poorer decisions; they tend to be overinvested in the status quo and buffered from disaster. Even when there is an impending threat, they are more likely to defend their own interests. During the late sixteenth century, the Ottoman Empire was struck by a prolonged drought; the government chose to alleviate tax burdens for local elites rather than for struggling farmers. The latter were more affected and vulnerable, but the former were central to imperial stability. This caused farmers to flee to the cities, eroding the tax base and stirring social conflict. The social burden of a crisis is rarely placed on elites. We don't have enough data to see conclusively whether this was the case in the earliest Goliaths, but history is replete with cases of environmental degradation and climate change compounded by poor elite choices. It's part of why inclusive institutions have tended to handle climatic swings better than extractive ones.^{[30](#)}

More inequality and extractive institutions both increased the vulnerability of societies (through counter-dominance intuitions, inequality, and elite competition) while oligarchy and weak control mechanisms hindered their response to disasters. All it then took was exposure to a threat, such as a sufficiently bad drought, disease outbreak, war, or rebellion, to break the hollowed-out society apart. This increasing risk of collapse due to extractive institutions is Goliath's Curse. Over time, as societies have changed, so has the curse.

Think of this first package of rising inequality, poor decision-making, counter-dominance intuitions, and weak institutions as the 'Primal Curse' that struck the earliest Goliaths. This first manifestation of the curse often meant swifter and deeper collapses (such as Cahokia, Tiwanaku, and San Lorenzo being permanently abandoned) because there were often still many exit options and the first Goliaths had not developed strong mechanisms to control their populace. Worse still, people were moving from egalitarianism into a new, perturbing environment. They were now surrounded by more people with bigger disparities in treasures and even kings who would

behead, cut open, or bury alive others in front of them. It was an unequal, violent world.

Yet that violence could worsen once a Goliath broke apart.

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Part Two

IMPERIAL MARCH AND FALL

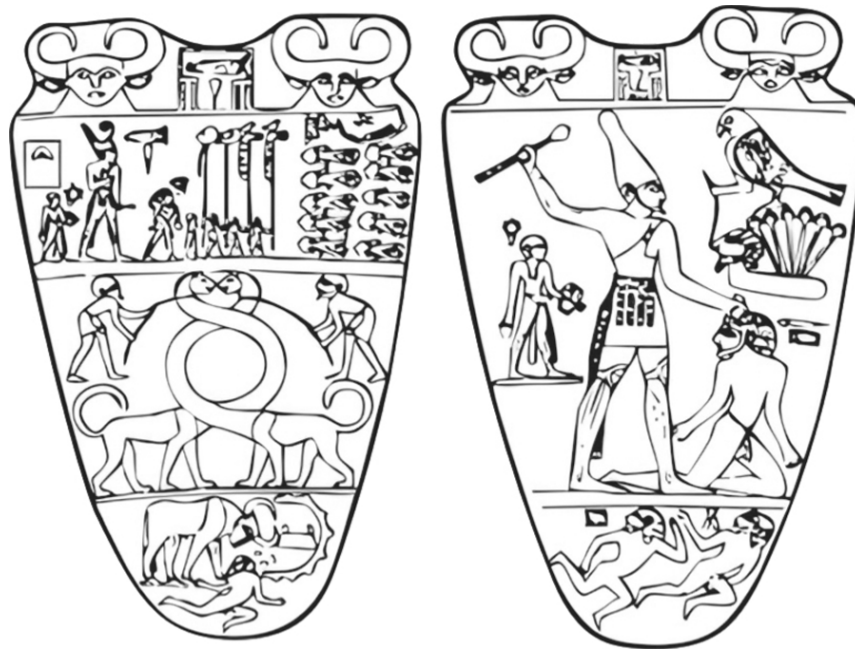
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Gang Warfare

The Warlord Beneath – History as a Story of Organized Crime – A War of the Few Against the Many – A Pyramid of Bodies – Selected by the Sword – The Authoritarian Impulse – The Darker Angels of Our Nature

THE WARLORD BENEATH

In the Egyptian Museum of Cairo sits the Palette of King Narmer, the first king of Egypt. It is over 5,000 years old, stands more than two feet tall, and is made of an eerie grey-green siltstone. It is the earliest depiction of the first known ruler of a unified Egypt – what many consider to be the first state of the world ([Figure 19](#)).¹



[Figure 19](#): A sketch of the Palette of Narmer

While the artwork is beautiful, the scenes it depicts are not. The palette shows Narmer wearing the white crown of Upper Egypt and standing over a kneeling foe. In one hand he holds a raised mace and in the other he grasps the hair of his enemy. He is bludgeoning to death a ruler from the Western Delta: a final bloody victory that sealed his control of all Egypt. Another part of the palette shows Narmer wearing Lower Egypt's red crown in a procession to view decapitated prisoners at Buto (in the Western Delta). They are not only dead: their genitals have been severed and placed atop their heads in a final act of humiliation. These scenes, surrounded by images of approving gods, show Narmer uniting Lower and Upper Egypt, before going on to conquer Canaan. Narmer was a successful warlord, not a freely elected leader.

It seems to be a common motif for state formation. Many of the earliest kings and emperors of Mesopotamia had artworks of themselves maiming and killing their enemies. The artists of Cahokia left a similar image on a flint-clay statue. It shows a warrior crouched over a naked enemy, holding the enemy's hair with one hand and a stone axe or mace in the other. He's just smashed his victim's face in.²

Such scenes are unsettling. To paraphrase the German Chancellor Otto von Bismark: states are like sausages – it is best not to see them being

made. Yet little has changed for much of history right through to the twenty-first century.

In less than a decade Charles Taylor went from being president of Liberia to being sentenced to half a century in jail. A UN-backed special court in 2012 indicted him on eleven charges, including murder, sexual enslavement, and conscription of children under the age of fifteen. Just a few years earlier he had been a legitimate president recognized by the UN.

Before being a president or a prisoner, Taylor was a warlord. He supported Samuel Doe's coup against the Liberian government in 1980 but fled the country after being indicted for embezzling \$1 million. After a prison break in the US and militia training in Libya, he returned to Liberia. He was now at the head of the National Patriotic Front of Liberia (NPFL), an armed militia group that promised revenge and a more favourable order to ethnic groups and wealthy individuals who had been mistreated by Doe's regime.

A civil war lasting almost a decade (1989–97) was unleashed. Several factions fought over the control of Liberia's natural resources such as diamonds and rubber. Taylor controlled a swathe of territory dubbed 'Taylorland', which crossed into neighbouring countries such as Sierra Leone, where he held monopolies over diamonds, timber, and rubber resources. Taylor's forces were partly funded by the tyre company Firestone, which gave him over \$2.3 million in food, cash, and cheques. In return, Taylor's cronies protected a rubber plantation that Firestone owned. (Firestone has said that it did not have a collaborative relationship with Taylor at any time.) The civil war claimed as many as a quarter of a million lives.³

In 1997, Taylor won the post-war general election. His victory was a surprise, given his campaign's motto of 'He killed my ma, he killed my pa, but I will vote for him.' While the election was judged to be fairly run by international observers, Taylor had an advantage in having captured most of the country's radio stations, using them to warn the listeners that the civil war would resume if he lost. Once elected, Taylor aided the smuggling of blood diamonds and weapons and embezzled \$100 million from the Liberian government.

The career trajectories of Narmer and Taylor are by no means unique. There is a glut of leaders who went from warlord or general to head of state through violence. Most (perhaps all) of the original states besides Egypt

were also based on military strongmen successfully conquering their neighbours. In China, there have been at least a dozen cases of dynastic transitions based on a warlord taking over. Chinggis Khan was originally a warlord who subdued and then allied with Mongol tribes to form an empire.

This is the main historical template of becoming a sovereign leader. Take power by force and negotiate with other elites to create a stable new order.⁴ Despite the popular idea of countries being built on a ‘social contract’, there is no observed example of a state being created by a freely negotiated consensus contract between citizens.

The overlap between warlords and state executives should come as no surprise. Warlords are usually defined as individuals who control territory using force and patronage. Organized crime is often defined as the organization of unregulated markets into hierarchical, extractive monopolies ruled through violence. These definitions are, as one scholar put it, ‘virtually interchangeable’ with the description of a state. Warlordism, statehood, and organized crime all have similar ingredients: a hierarchy that coercively extracts resources from a territory and population.⁵

Understanding how and why men like Narmer and Taylor came to power can help us understand both why collapse occurs and why it can involve so much bloodshed.

HISTORY AS A STORY OF ORGANIZED CRIME

From 2009 through to 2012 Medellín, Colombia, was one of the most dangerous places on the planet. Two organized gangs (*razones*) vied for control of the streets. The almost 400 street gangs (*combos*) allied with one of the two *razones*, turning the city into a war zone. The costs were high, both for the public and for the gang leaders, whom the police began to target. Their profits declined as their customers’ fear of being shot overcame their need to buy drugs. In response, the *razones* declared a truce and started governing themselves: they set up a criminal council called La Oficina (The Office). To keep the peace and enforce rules they adopted *El Pacto del Fusil* – The Pact of the Machine Gun. The conflict was quelled as each *razone* started to act like a mini-state in its territory: managing

borders, settling disputes between *combos* (with the threat of the machine gun), and helping to set high prices for drugs.⁶

A similar process occurred in Africa in 2000. The Democratic Republic of Congo (DRC), located in central Africa, is the continent's second largest country by area. It is rich in rare earth metals that have been a source of incessant conflict. Battles over the control of its diamond mines have cost hundreds of thousands of lives, hence the label 'blood diamonds'. In 2000, the price of the rare earth mineral coltan shot up from \$90 to \$590 per kilogram. In response, armed criminals began making permanent bases around coltan mines, 'taxing' the miners in exchange for 'protection'. There was a similar response to a price jump for gold, except the militias began to set up more sophisticated administrations, with a court system and refined tax systems (including poll taxes).

In both cases, armed groups in the DRC responded to a lootable resource becoming more valuable by establishing state-like functions around it, but in the case of gold the tax system and bureaucracy was more elaborate. The difference arises because a coltan miner can extract kilograms of product every day, while gold miners discover only small amounts, which can be easily hidden (even, sometimes, by swallowing them). This makes gold less lootable, meaning the Goliath needs to negotiate more with its citizens, provide more services (such as a court system), and use more bureaucracy to extract taxes (such as a poll tax).⁷

Mafias, warlords, and crime syndicates have a long track record of providing services and sometimes even becoming beloved by the communities they rule. Tajikistan warlords established rackets over lucrative smuggling routes after the collapse of the Soviet Union in the 1990s.⁸ Despite presiding over worsening health, epidemics, and excessive taxation of local farmers, they often enjoyed popular support because they did protect citizens from crime.⁹

Both the *razones* and the militias of the DRC show the process of state creation in real time. They were criminal rackets, but as they gathered more lootable resources and were unified by external threats, they started to provide services to their citizens. It is, after all, in the interests of even the most self-centred despot to ensure that the people they are stealing from have a measure of welfare. A bigger, healthier population means more lootable resources, a bigger defence force, and more status. The basic

services that the DRC militias and the *razones* provided are emblematic of what the first states under Narmer and others usually offered: protection, basic infrastructure (whether that be for mines or farms), and dispute settlement. Most of us are lucky enough to be experiencing a unique, recent period of human history where states provide far more and are far more accountable than they once were.

Seeing states as similar to criminal gangs running protection rackets can help explain what exactly happens during collapse. The best example of trying to do so is the brilliant 2015 book *Gangster States* by Katherine Hirschfeld. Hirschfeld, an associate professor in anthropology, marshals modern and ancient case studies to show that state formation and collapse are best understood as a criminal enterprise being established and then broken apart by infighting or pressure from competitor rackets. Once there is enough of a stationary economic surplus, then establishing a predatory monopoly is a profitable strategy. As we've seen, lootable resources, monopolizable weapons, and a lack of exit options all make it easier to create a racket or state.^{[10](#)}

We see it in the origins of war and the state. War in Mesopotamia originated with the Halafians taking over obsidian trade routes by force. Prior to Narmer, Canaan had been a trading partner of Egypt, undertaking a fair exchange of wine, metals, and olive oil. That changes towards more extractive arrangements after Narmer conquers them.^{[11](#)} Suddenly they were giving more, and in the form of tribute (probably in 'exchange' for protection and perhaps mystical services) instead of trade. Rather than a mutually beneficial negotiation, rackets had transformed fair-trading relationships into unequal exchanges.

The collapse of such rackets involves both their destruction and regeneration. The racket may break down, yet the lootable resources usually remain. That means there is an incentive for either neighbours or internal competitors to violently re-establish the scheme. Hirschfeld shows this occurring in a range of modern cases, such as Yugoslavia, the Soviet Union, Afghanistan, Tajikistan, Sierra Leone, Moldova, and Bosnia.^{[12](#)} It also matches what we saw in the long-term rise in warfare: it rose alongside higher levels of competition for lootable resources.

A WAR OF THE FEW AGAINST THE MANY

Creating states, establishing a racket, is bloody work. The Palette of Narmer is just the tip of the spear. [Figure 20](#) is taken from the systematic study of violence across almost 12,000 years (12,000–400 BCE). The researchers drew on a set of 3,539 skeletons and examined the frequency of cranial trauma and weapon-related wounds such as embedded arrowheads. The story it tells is clear: violence was low throughout the Neolithic; it then spiked drastically during the building of the first states such as Uruk and the First Dynasty of Egypt (4500–3300 BCE, known as the ‘Chalcolithic’). At this high point, somewhere between 20 and 30 per cent of skeletons have signs of conflict on them. That figure then drops throughout the period of empire (the Early and Middle Bronze Age). Violence then spikes back up as the last empires of the Bronze Age come to an end. This rising tide of bloodshed continues during the Iron Age while new empires, such as the Neo-Assyrian Empire, are being forged. We see it elsewhere: increased levels of violence mark the formation of states in the Americas (such as Monte Albán) too.^{[13](#)}

The pattern is that violence peaks when states are being formed. It is not the absence of authority per se that spurs violence, but the frantic efforts of a few to create structures of authority and monopolize lootable resources. In the case of collapse, this would often be accompanied by ‘score-settling’, with previously oppressed peoples seeking vengeance on their erstwhile masters. In any case, this involved only a peculiar sliver of society.^{[14](#)}

Violence, including during collapse, is a gendered phenomenon. Soldiers, mercenaries, brigands, and other violent actors have almost invariably been men. In 2001 99.9 per cent of armed combat forces were men. A conservative estimate is that across history 99 per cent of soldiers have been men.^{[15](#)}



[Figure 20](#): Violence in the Near East: 12,500 to 400 BCE

It's not just men, but also a particular demographic of men: fighting-age men. The Romans conscripted men aged sixteen to forty-six. The *levée en masse*, a policy of mass conscription during the French Revolutionary Wars from 1793, focused on men between the ages of eighteen and twenty-six. Even among fighting-age men, perpetrators of violence are usually a select group. In the 1994 Rwandan genocide, court convictions suggest that less than 3 per cent of the population at the time committed violent crimes. Over 90 per cent of those involved were male, with a median age of thirty-one.^{[16](#)} It is a common finding in criminology that just a minor fraction of people are responsible for a high proportion of violent crimes.^{[17](#)}

During collapse and crisis, most of the combatants tend not to be random young men. They are those who have already been trained. They are soldiers and mercenaries either of foreign powers or of the previous government. This is not a Hobbesian war of all against all. It is a war of the few against the many. It is a concerted effort by a small portion of the population – young-to-middle-aged men, usually with previous training – to take advantage of the situation and establish a new racket. Violence during

the collapse of the Western Roman Empire was dominated by both Germanic soldiers previously employed by the empire and the invading Byzantine (Eastern Roman) Empire. Violence in Saddam Hussein's Iraq was driven mainly by paramilitary groups previously part of his security forces, as well as by the US Army. The recurring image of violence during collapse is one of young, trained, armed men fighting.

This war of the few against the many can explain why so many people die and disappear during a collapse.

A PYRAMID OF BODIES

The Wari (600–1000 CE) were the first empire of the Andes, and they endured a bloody breakdown. The collapse of the Wari saw the incidence of non-lethal head injuries triple from 20 per cent to 60 per cent. The situation becomes more dire centuries later. Fatal head injuries during the collapse were 10 per cent and this quadrupled to 40 per cent among adults and 44 per cent in children in the post-collapse period. Skeletons begin to show peculiar cut marks. These marks suggest that rather than victims having been stabbed or slashed, the blade had sliced close to the bone. It's a tell-tale sign of cannibalism, with survivors carving up the flesh of the deceased (tellingly, this occurred during a drought).^{[18](#)}

We don't need to look back a millennium to the Wari to see the dangers posed by a state collapse. Iraq during the 2003–6 invasion saw around 600,000 deaths and the emergence of ISIS. The overthrow of the Libyan government in 2011 led to the re-emergence of the slave trade. Recurring civil conflict in Congo, after the atrocities of Belgian colonization, have claimed somewhere in the range of 900,000 to 5.4 million lives. Gang warfare during a power vacuum can take a toll.

Behind the dilapidated temples of the Maya or the charred remains of cities from the Bronze Age lay the masses of faces who died or fled. Their names are often lost but their suffering was real. While the numbers are highly uncertain, many collapse events may have led to enormous population declines. [Table 3](#) summarizes the high-end estimates of some of the worst cases from across the world. Each one involves staggering

amounts of displacement and death, with losses ranging between 30 and over 95 per cent of the entire population.¹⁹

While the numbers may seem horrifying, they are exaggerations, or at least misleading. Most of the population losses result from people moving away rather than dying – it’s often difficult to distinguish between the two historically. People who move into more rural areas or melt into a non-state territory often become less archaeologically visible. Their remains and signs of their activities become smaller, more dispersed, and relegated to rural areas that archaeologists don’t excavate as often.²⁰

Collapse is more a story of movement than of an ancient apocalypse. When we look at modern examples, the number of people displaced from their homes vastly outnumbers those lying lifeless in streets and morgues. Take the case of the recent Syrian civil war. Since 2011, 14 million people have left their homes. 6.9 million of these are displaced within Syria while the remainder have fled to other countries. In contrast, estimates of the death toll from the war range from around 500,000 to 600,000, almost thirty times lower than the number of people displaced. Syria is indicative of the norm: most people make the difficult decision to move out of harm’s way before facing the threat of death. Historical collapses involved the same dynamic, with most of these massive declines in population reflecting migration rather than deaths. People usually used the old ways of mobility and fission-fusion rather than face a grisly fate at the end of an arrow.

Table 3: Worst-case population decline during collapse

Case study	Timeframe	Population loss estimate
LBA collapse	1200 – 1000 BCE	30 – 75 per cent
Mayan Lowlands	800 – 1000 CE	Up to 90 per cent
Invasion of the Americas	1500 – 1600	Up to 90 per cent

Case study	Timeframe	Population loss estimate
	CE	
The late Roman imperial system	400 – 600 CE	Up to 50 per cent in some regions. Over 95 per cent for the city of Rome
Rapa Nui	1722 – 1877 CE	Over 95 per cent

Estimates of the number of deaths from collapse are also inflated owing to inaccurate record-keeping. For most of history we have poor, highly uncertain estimates of how many people were alive at any given time. The most reliable data comes from the occasional state censuses taken by large administrations, such as Rome and the Tang Dynasty in China, and even these usually surveyed far below half of the population. The Roman and Tang examples are also exceptional: none of the other cases we've covered, from Cahokia to Tiwanaku, conducted censuses. Even when these numbers exist, they often reflect the bureaucratic capacity of the state, rather than the actual number of people living on the land. This makes them particularly untrustworthy during a collapse. The priority of a state that is falling apart is not to count its citizens. This can lead to enormous problems with trying to tally the lives lost during a period of turmoil. The An Lushan Rebellion was estimated in *The Better Angels of Our Nature* as having cost 36 million lives, a sixth of the world's population at the time. Because of the problems with relying on census information, this estimate was later cut by almost two-thirds, to 13 million.^{[21](#)}

Nonetheless, many did die and entire cities of people moved. Why? The most recurrent culprits are conflict, disease, famine, malnutrition, and the break-up of trade. The loss of a centralized authority frequently involves war or civil strife as groups battle to recreate some version of the state. A review of ten cases studies of collapse I organized with a group of archaeologists and historians found that every case involved some kind of war. In the Crisis Database around three-quarters involve civil wars or revolutions (or both). One-fifth of these are protracted affairs that last a

century or longer. Yet the bigger killer was not from axes or arrows, but rather germs.²²

Throughout history, disease has been the most dangerous foe on the battlefield. Deaths from disease regularly outstrip direct casualties in a war. About three times more people died from the influenza pandemic of 1918 than were killed in the First World War, and twice as many from disease than war wounds during the American Civil War. The squalid conditions of warfare, the splattering of blood and flesh from one soldier onto another, and the weakened immune systems of frequently malnourished and injured fighters living in close quarters, all make the battlefield a dangerous petri dish. In our review of ten case studies of collapse, eight involved diseases and seven involved signs of starvation and malnourishment. This echoes the findings of the Crisis Database in that a third of cases are marked by disease outbreaks. Disease could act as a trigger for a collapse, such as with the invasion of the Aztec Empire by Spanish conquistadors, or simply be spread alongside moving troops or migrants during a crisis. Often it was both.²³

SELECTED BY THE SWORD

Violence lurks beneath the first Goliaths. Across different world regions, more unequal, dense, hierarchical communities were associated with more violence and eventually the onset of warfare. The introduction of new weapons, such as hand-held bronze swords and axes in China, Egypt, and Mesopotamia, were pivotal in the creation of centralized states. The hilltop city of Monte Albán in Mesoamerica seems to be partly a creation of migrants fleeing raided cities and seeking a more defensible location. Monte Albán's transformation into a more organized state (known as the Zapotec) roughly coincides with a major regional conquest (of Cuicatlán), the construction of its first fortress (in Quiotepec), and the building of a skull rack containing the skulls of sixty-one people.²⁴

Then there is the Palette of Narmer and similar records and artwork. Mesopotamian steles (large slabs used as monuments) regularly show great processions of captured prisoners and lines of slain bodies. The inscriptions are usually just as brutal. The Neo-Assyrian King Ashurnasirpal II (910–859 BCE) described his gory celebrations of one of his victories: 'I erected a

pile [and] flayed as many nobles as had rebelled against me [and] draped their skins over the pile; some I spread out within the pile, some I erected on stakes upon the pile, [and] some I placed on stakes around about the pile. I flayed many right through my land [and] draped their skins over the walls.’ Using human skin for decoration is usually the domain of serial killers, but in this case it was a useful way of sending a message: obey and do not rebel.²⁵ Whether it be in politics or religion, violence was central to Mesopotamia where kings promised to win wars abroad and prevent bloodshed at home (well, except when they spilled it). In art and writing they were even given the same weapons and killed their enemies in the same way as the gods. Most early leaders flaunted and deified rather than hid their warlord nature.²⁶

War drove state formation. Every single one of the cases of original state formation involves some kind of conquest.²⁷ We even see it in recent states being formed, such as the creation of the Zulu by Shaka (1816 CE), Madagascar by Andrianampoinimerina (1787 CE), and the unification of Hawaii by Kamehameha (1810 CE) after the introduction of firearms by European traders.²⁸ ‘War made the state and the state made war’, the political scientist Charles Tilly wrote.²⁹ He was referring to states in early modern Europe, a hypothesis that has been supported by studies decades later.³⁰ His theory was too modest. War wasn’t just a central contributor to Europe during that period; it was the key driver of state formation throughout history. Communities rarely give up their autonomy willingly. It’s when they are conquered, or facing conquest by another enemy, that they cede it to a greater power.

Studies using Seshat, the world’s largest historical database, have also pointed to agricultural productivity and military innovations as being the key drivers of bigger, more hierarchical societies. The adoption of bronze weapons paved the way for the first states in the Near East, while iron weapons and cavalry laid the foundation for far larger empires such as the Western Roman Empire, the Han Dynasty, and the Achaemenid Empire (the first Persian Empire). Groups that were located closer to the Steppe – the grassy plains separating Europe from Asia that were full of mounted warriors – were also more likely to adopt big dominance hierarchies. If you had lootable resources and felt under threat, you were more likely to band

together and adopt kings and generals or be conquered by someone who did.³¹

Unsurprisingly, war-making has until recently been the biggest item on a state's budget. The Roman Empire spent half to three-quarters of its budget on the military, the Song Dynasty in 1065 directed around five-sixths of its resources towards the military, in early modern Britain between 1685 and 1813 it was about 75 per cent, and in early modern France in 1630–59 it was around 89–93 per cent. Countries spending more on healthcare and welfare than the military is a recent phenomenon limited to the last century or so. Even then there are exceptions: a plethora of countries spent half or more of their budget on bullets and bombs during the First World War.³²

The fact that kingdoms were often created through conquest explains why kings often appeared to be foreigners to those they were ruling. The ruler claims to be different from the masses, whether that be due to a connection with God, or a different, special ethnicity. Many early kings were 'stranger-kings'. We see this in Mesopotamia; the names of early city-states such as Ur, Uruk, and Eridu are all not from the native tongue of Sumerian. These Mesopotamian cities were more likely agricultural communities conquered by outsiders.³³

Enslavement, especially of war captives, was a by-product of conquest. Goliaths refined slavery into an industrial-style project. Some records from ancient Mesopotamia claim that up to 36,000 prisoners were taken as captives of war. These are almost certainly an exaggeration but do show the value placed on captured humans. Even in 1800 up to three-quarters of the global population was in some form of bondage.³⁴ Some became slaves not because they were captured during war, but rather to work off a debt.³⁵ Debt slavery is, in many ways, a perversion of the ancient instinct of reciprocity: to repay the actions of others. In this case, payment is by any means necessary, including by owning another person and having them serve for every moment of every day.³⁶ Slavery wasn't the only form of dominance that accompanied Goliath states. The first states often amplified male dominance and over time led to more patriarchal structures.³⁷

This doesn't mean that participation in early Goliaths was always done at the point of a blade. Opportunities for trade and work, bigger social networks, religious rituals, and awe-inspiring monuments may have also

made people gravitate towards living under a Goliath. Many of the first Goliath-building projects, such as Cahokia and the Pueblos, appear to have begun as ceremonial centres that linked people together. Not everyone living in early states did so in chains; for many it may have been enticing.³⁸

Importantly, most states were relatively authoritarian, but some offered more benefits than others. One study of state formation across thirty cases found significant differences in the number of public goods produced.³⁹ Many states did settle disputes, hand out grain during famines, and coordinate labour to construct roads, bridges, and irrigation canals, although most of this infrastructure was mainly built for moving troops and goods around, not for public welfare. These greater amounts of coordinated action were often better for the people living under a state.⁴⁰ They also tend to correlate with states that had more difficulty in extracting resources from their citizens and had to rely more on negotiating with them. We see this in many cases from the Americas, such as the more egalitarian beginnings of Teotihuacan and Monte Albán. That's probably because they didn't have domesticated animals or bronze weapons. Less Goliath fuel often meant rulers had to resort to more inclusive arrangements.⁴¹

Not all leaders were rapacious pirate-kings. Some pre-modern polities were far more democratic and had far more benevolent leaders than others. The early cities of Jenne-Jeno, Tiwanaku, and Monte Albán, as well as later Teotihuacan, were more egalitarian in how they were governed and how riches were distributed. Similarly, Athens had its famous deliberative democracy, Rome had a republic with some democratic representation (although this was often a thin veil for oligarchy), and the Venetians had a leader (doge) who was chosen by a council of leading families and closely monitored for corruption. Notably, all these examples had representation restricted to free men; and the wealthy and powerful maintained disproportionate influence. Unsurprisingly, it was adult men, especially elites, who were the ones using violence to set up states. In short, the gang members who built the racket ensured that they maintained some control, leading to a more inclusive touch. Well, inclusive as long as you weren't a slave or a woman.⁴²

As with militias in the DRC trying to build a racket around gold mines, a more sophisticated and benevolent approach was usually needed when

subjects had more exit options and less easily lootable resources.⁴³

Tellingly, early statehood is one of the best predictors of how autocratic a state is today. Areas with more Goliath fuel created states sooner and they have left a long dictatorial legacy: just think of China, Egypt, and the Near East.⁴⁴

Even the autocratic rackets could offer numerous benefits to at least a portion of the populace. The free adult males of Athens not only were given a political voice, but also benefited from slavery and patriarchy. Conquests, whether it be in Monte Albán or Rome, meant more booty, plunder, and slaves not just for elites, but also for much of the general populace. Some of the populace was complicit in building Goliaths and putting them back together once they fell apart.⁴⁵

States can and did offer benefits, and some were more inclusive than others, but this shouldn't hide the fact that most were deeply extractive arrangements. Ones that needed to be propped up by armed forces and elaborate ideologies. Gang warfare is still the most apt analogy. Even those that started from more democratic and egalitarian origins tended to drift towards oligarchy, greater concentrations of power, and greater extraction of resources by elites over time.

We've already seen some of the reasons as to why early states were Goliaths. Lootable resources usually resulted in growing wealth inequality over time, which tended eventually to spill over into politics and other forms of power. A second reason is Goliath evolution: societies that were more successful at conquest outcompeted those that were less effective. They grew larger and more persistent over time. And those conquering polities were usually Goliaths. The reason why is rooted in basic human psychology.⁴⁶

THE AUTHORITARIAN IMPULSE

Conflict isn't a completely destructive force. It can spur cooperation, at least within a group. Data from sixteen studies covering forty countries all suggest that exposure to violence from war increases local cooperation within groups.⁴⁷ Groups who have suffered an external attack tend to

develop symbols, stories, and behaviours that bond them together, such as a shared sense of patriotism. We saw that in the wake of the 9/11 terrorist attacks in the US. The day after the attacks, Walmart sold 250,000 US flags; it had only sold 10,000 the same day the previous year. The supermarket chain was almost running out of flags by the end of the week.⁴⁸

Unfortunately, that increased coordination within a group does seem to come at a cost. It requires having an ‘other’ group or enemy to rally against. Across the world the best predictor of levels of prejudice and ‘cultural tightness’ (tighter cultures have stricter norms, harsher punishments, and less tolerance for deviant behaviour) globally is how much a society has been exposed to threats such as war, disease, and scarcity. A people under threat stand together but apart from others.⁴⁹

Conflict, or even just the threat of an external enemy, changes our brains. One study scanned the brains of ninety Chinese students while they read about a potential threat from Japan. Remarkably, their brainwaves started to synchronize. This led to better coordination and completing group tasks more quickly.⁵⁰ It makes sense: if your group is in trouble then the ability to cooperate intensely and survive is critical. It’s a finding backed up by decades of research on terrorist cells, revolutionary groups, and squads of soldiers, all of whom develop strong emotional ties and a willingness to sacrifice themselves for each other (hence phrases like ‘esprit de corps’ and ‘brothers in arms’). It’s a phenomenon that militaries capitalize on, by training small units under duress to break down the individuals and rebuild them into a single team that is both more organized and more willing to kill. An external threat supercharges cooperation within groups and the capacity for violence against outsiders.⁵¹

External threats change our brains in another way. They make us more open to being dominated. Authoritarianism – obedience to high-status authorities and the desire to punish rule-breakers – increases when individuals face a threat to their safety and security. Some percentage of a population appears always to harbour stronger authoritarianism, and studies of twins suggest that it may be largely genetic. Surveys across eight modern high-income countries found that around 10–25 per cent ranked as highly authoritarian (with the US scoring the highest).⁵² These authoritarians tend to become more politically active and aggressive when they are activated by

a social change, most commonly the emergence of a threat.⁵³ Threats triggering authoritarian behaviour don't just happen to those with underlying authoritarian tendencies. They happen to those who generally see the world as a dangerous place and affect even the most devoted peacenik (albeit to a lesser extent).⁵⁴

This *authoritarian impulse* during crisis is so primal that it changes who we prefer as our leaders. Studies show that people want taller leaders with more masculine features during wartime. The old trope that fearful people will turn towards a 'strongman' leader turns out to be true.⁵⁵

There is no shortage of modern-day examples. Vladimir Putin soared into power on the back of a heavy-handed response to a series of unexplained bombings (usually blamed on Chechen rebels). Israeli prime minister Benjamin Netanyahu experienced a resurgence in popularity after invading Gaza. Leaders like Bolsonaro in Brazil and Trump in the US have made a fine art of gaining votes by blaming foreigners for a country's woes. The rise of authoritarian regimes prior to the Second World War was similarly littered with politicians using a dangerous 'other' to activate the authoritarian impulse. Even in democracies, during emergencies, whether it be a war or a pandemic, people are far more likely to (sometimes eagerly) accept less government transparency and more draconian measures. The normal response is to invoke martial law or national emergencies that give sweeping powers to an executive and tend to involve more surveillance and armed enforcement (through the police and military), despite the fact that people usually respond well to disasters and the idea of mass panic is a myth. After 9/11, increased flag sales wasn't the only change: the US government also adopted the 'Patriot Act' that allowed for the indefinite detaining of migrants without charge and permitted the FBI to search email, financial, and telephone records without a warrant.⁵⁶

The threat of violence allowed some groups to become more inwardly cooperative, murderous, and willing to accept authoritarian structures. In other words, simply the shadow of war helped to build Goliaths. More than that, Goliaths then became viral. Once one group becomes a conquering dominance hierarchy, the others who are threatened become more likely to imitate it. We see it in ancient Egypt. Under Narmer's rule Egypt begins to expand and conquer surrounding territories, such as in Canaan; the settlements in these areas begin to erect walls, build fortifications, and arm

themselves.⁵⁷ The Germanic tribes on the frontiers of the Roman Empire, or the nomadic Xiongnu and Xianbei on the edge of China, became more regimented and hierarchically organized in response to conflict with Rome and China.⁵⁸ It can also help explain the patterns we have seen when it comes to war driving state formation: threats triggered people to start to adopt more top-down, authoritarian structures. More aggressive, conquering polities not only had an evolutionary advantage but were also contagious. Violence also selected for certain types of leaders.

THE DARKER ANGELS OF OUR NATURE

Psychologically, not everyone has an equal desire for status or openness to aggression. Those with high rankings in the triad of psychopathy, narcissism, and Machiavellianism are more predisposed to dominating others and are forceful status seekers. Today roughly one in a hundred people or fewer is a psychopath. The most common place to find psychopaths is in prison, where they make up around 25 per cent of the population.⁵⁹ The second most common place is the executive suite. Among CEOs, the rate of psychopaths rises from 1 per cent to 3–21 per cent (the numbers are uncertain, especially as all the dark-triad traits exist on a spectrum).⁶⁰ Psychologists tend to distinguish between ‘successful’ and ‘unsuccessful’ psychopaths. The successful ones end up in boardrooms and parliaments, while the unsuccessful end up in prison.⁶¹

Those ranking highly in the dark triad are the most probable candidates for those who broke the fragile peace of the Palaeolithic and started cycles of violence. The authors of the most recent survey of pre-agricultural violence speculate that violence in the pre-agricultural period may have been driven by a few psychopaths.⁶² Psychopaths are not only strong dominance status seekers, but are also predisposed towards aggression with little empathy holding them back.⁶³ Such individuals, not the average forager, would have been the most likely to break the norms of peace between groups, sparking reprisals and escalation. Once these cycles of conflict began, they were hard to stop.

Psychological phenomena such as the authoritarian impulse and the dark triad are not recent. Like our egalitarianism or love of status, they are ancient, evolved behavioural patterns. Counter-dominance measures and a lack of Goliath fuel both kept a check on dominance-based status-seeking behaviour and the dark triad. That is, until the world warmed, the environment changed, and Goliath fuel became available.

Place yourself back in the body of a forager in the early Holocene. You've started settling down every summer to harvest stands of wild cereal throughout a forest. They are nourishing but leave you stationary and vulnerable. One brash young man begins warning you and your family that he overheard strangers were planning a raid to take the crops. Your group may have good fighters, but humans are fragile, especially when sleeping. There is always an advantage to striking first and using the element of surprise. The usual channels of communication are not open with these new groups and the young upstart continues his warnings and offers to lead a pre-emptive attack. One summer a drought strikes. Your wild stands of grass are low; sharing with your neighbours is becoming increasingly difficult, and those neighbours are growing increasingly anxious. Eventually, you give in and agree to the attack.

The dawn ambush on your new neighbours is hurried and panicked, but successful. Unfortunately, you leave many survivors. A month later, they strike back, launching spears through your camp in the middle of the night. The proud young upstart begins to present himself as a war chief. He is fearless, charismatic, and an excellent tactician. The others listen and begin to shower him with gifts after every victory.

Little do you know that he is a sociopath who had planned this the entire time. The other group never intended to attack you. That doesn't matter now; the slow and bloody spiral towards Goliaths has begun.

Today, many countries have checks and balances on leaders, a media that watches and critiques them, well-established electoral procedures, and, of course, a knowledge of psychological issues such as narcissism and psychopathy. Yet even with that we still end up with world leaders who seem to be steeped in the dark triad. It is difficult to look at some of the most powerful men in recent times and not see signs of narcissism, psychopathy, Machiavellian politicking, or just grandiose ambition. This is a feature of our political systems, not a bug. Politics doesn't select for those who are the most virtuous or most competent, or those who would wield

power best. It selects primarily for those who are the best manipulators and negotiators and are willing to invest the enormous amounts of time and effort (and often relax their morals) necessary to achieve status.

The selection mechanism would have been even worse in the past. The time of the first states lacked many of these accountability mechanisms, and pre-modern leaders tended to be overwhelmingly successful warlords, like Narmar.

Upon taking power, the need to go to battle and the threat of being the victim of a counter-dominance revolution also would have selected for only the strongest status seekers. All of this means that those who were selected – those comfortable with violence, capable of activating the authoritarian impulse in their followers, and ambitious enough to risk death – would have been far more likely to rank high in the dark triad and have stronger status desires. The proportion of psychopaths among the first leaders was probably even higher than it is in boardrooms and parliaments today.

Even those who were more benevolent and ended up in power would probably have become corrupted by it over time. Holding a position high up in a hierarchy leads to marked changes in behaviour, and even in the brain. In one study, college students who had been primed to feel powerful were placed under a brain scanner. The researchers found that a neural process in the students called ‘mirroring’ had been impaired. Mirroring is a kind of neurological mimicry whereby when we see an action performed our brain lights up the same pathways required for that action. Mirroring is a cornerstone of empathy.⁶⁴

Numerous studies, experiments, and laboratory games have repeatedly shown that powerful people tend to have less empathy and less compassion, take more risks, have more affairs, act more impulsively, are more hypocritical, and lie and cheat more frequently.⁶⁵ As summarized by Brian Klaas in his book *Corruptible*: ‘all the available evidence points in one direction. Becoming powerful makes you more selfish, reduces empathy, increases hypocrisy, and makes you more likely to commit abuse.’⁶⁶ Or as Lord Acton once put it: ‘Power tends to corrupt, and absolute power corrupts absolutely.’

Think of this collection of traits – status seeking through dominance, the dark triad, the authoritarian impulse, and power corrupting – as the ‘darker angels of our nature’. They had been kept in check by counter-dominance

measures previously, but were unleashed by Goliath fuel – lootable resources, monopolizable weapons, and caged land – all allowing for the few to dominate the many. It wasn't just about the leaders who were selected for but also about a devoted group of authoritarian followers. The environmental upheaval of the Holocene, growing populations, and the development of Goliaths all provided external threats that made much of the populace more willing to accept authoritarian arrangements.

We'll never fully know what happened during these thousands of years when warfare and Goliaths emerged. We certainly can't run psychology tests on the long-dead leaders and warriors of the time. Yet this explanation – the availability of Goliath fuel meeting the darker angels of our nature and gang warfare ensuing – fits the historical record and what we know of human psychology.

It also helps to explain why early states tended to be so unequal and extractive. A usual explanation for state formation is that big projects (roads, bridges, irrigation, and granaries) and populations needed big rulers to organize them. Yet, irrigation schemes and urban centres often preceded hierarchy. Most importantly, even if a ruler provided decent coordination or military command, why did they need to be worshipped as a god and be given hundreds of times more wealth than the average person? Why didn't we coordinate through democratic coalitions of workers? It's not because we needed big coordinators who demanded even bigger rewards or incentives. It's because of violence, power, the drive for status, and the fear of losing it.⁶⁷

Of course, not *all* early elites were megalomaniacs, psychopaths, or power-drunk. Many could have been well-meaning and well-behaved leaders. Yet even a few rogue rulers would have created a system in which aggression, cheating, and self-aggrandizing were more acceptable (at least among certain classes). And once a few bad precedents were set, it became easier for others to respond and establish similar dynamics. One bellicose city-state could have easily triggered others to devote more resources to walls and spears, and less to growing food and building roads. Those who did not were conquered. It created a new evolutionary pressure, and arms races were born. One egotistical ruler who centralized power and built up a cult of personality provided both a blueprint and justification for others to do the same. Those who didn't had less of a claim to being leader. It became a race to the bottom.

Arms races, races to the bottom, and status races using conspicuous consumption (such as feasts and great monuments) all became competitive traps. Those who chose not to play (pacifists, those who had little desire to rule, or animists who refused to exploit the environment for better weaponry) were either outcompeted or had to change their mind. Think of these as ‘Goliath traps’: destructive competitive dynamics that dominance hierarchies are prone to fall into. They might be good for a few individuals in the short term but are not in the long-term interest of the majority.

The darker angels of our nature – status competition through dominance, the dark triad, the authoritarian impulse, and power corrupting – all combined to help create systems in which one group exploits the rest. The violence of collapse and state formation is just the collateral damage of trying to set up an effective racket. Many were successful in creating vast systems of interconnected states and empires. Like the first Goliaths, these rackets would also eventually break down owing largely to inequality, albeit at a larger scale with far more serious consequences.⁶⁸

Broken Bronze

In Sargon's Footsteps – The Blurry Line Between State and Empire – The Early Bronze Age Collapse – The Beginnings of the West and Capitalism – The Late Bronze Age Systems Collapse – Collapse as a Blessing – The Bright Side of Dark Ages

IN SARGON'S FOOTSTEPS

We do not know the birth name of the world's first emperor. We simply know him as Sargon: 'the true king'.

As with Narmer, we have an engraving that depicts his ascent to power. It's known as the Sargon Victory Stele (from the third millennium BCE) and is held in the Louvre in Paris. It looks unsettlingly similar to the Palette of Narmer. Sargon is shown holding a net of captured enemies in front of the goddess Ishtar while smashing a slave's head with a club ([Figure 21](#)).¹ It is partly a claim to divinity, with the image closely resembling an earlier stele of the god Ninĝirsu (tellingly, a god of farming who later also became a god of war) holding a net of enemies in a similar fashion.²

Sargon's origins are shrouded in myth. In the legend, Sargon is said to have been the illegitimate child of a high priestess. She secretly birthed him and then placed him in a basket to float down the river, a story that would later be echoed by the tale of baby Moses floating in a basket on the Nile. Sargon was adopted by a humble gardener, Akki. In this telling, Sargon is a self-made man who rose to be a cupbearer to the king of Kish, Ur-Zababa.

He overthrew his boss and mobilized Kish for war. He first marched south to conquer the great city of Uruk. He didn't stop there. He conquered cities throughout southern Mesopotamia and then northern Mesopotamia. He founded a new capital called Akkad, the centrepiece of the Akkadian Empire, and changed the official script of the region from Sumerian to Akkadian. The exact date of his reign is unclear, but would have been around 2334–2279 BCE.



[Figure 21](#): Victory Stele SB2 of Sargon of Akkad

Sargon was a ‘marcher king’. He would lead his small army from city to city, conquering, demanding tribute, and installing a puppet leader to rule in his absence. Those leaders often didn’t share Sargon’s culture and weren’t monitored, hence were liable to revolt. It was a tenuous affair, with Sargon frequently needing to revisit cities to quell rebellions and reinstate his authority.

The Akkadian Empire lasted a century after Sargon’s death – an impressive length of time, given how difficult the work of maintaining an empire is.

Sargon faced clear limits to his ambitions, as has every emperor since. There were two key problems that plagued any attempt to create lasting social control. First, people abhorred domination. Cities didn’t want to remain conquered, and the foragers and herders outside them were reluctant to give up their resources or independence. Ideology and external threats

could help people to accept dominance hierarchies, but it wasn't a stable affair. Second was status competition. A ruler could not be everywhere at once, and Sargon's strategy of constantly marching from community to community was not sustainable. Instead, the rulers had to rely on local elites, governors, and eventually bureaucrats to impose rules and extract resources. These elites had their own aspirations for greater status and wealth, and the state had created a hierarchy with a throne and positions of power: the ultimate visible ladder of status. Status competition through dominance was now legitimate, and the stakes were far higher with entire swathes of cities, temples, and armies up for grabs. Elite competition became endemic.

Dominance aversion and status competition made empire-building the equivalent of trying to construct a human pyramid from hostile strangers. Most wanted to be on top, and no one wanted to be shouldering the burden at the bottom. Enormous empires were still eventually built, but only after millennia of struggle and innovation.

We can see it in numbers. For the first 3,600 years after Narmer (3000 BCE to 600 CE) empires varied in size between 0.15 and 1.3 million square kilometres. To put that in perspective, the country of Egypt today is around 1 million square kilometres and 3,600 years is about the same amount of time separating us from the times of Socrates and Classical Greece plus more than another 1,000 years. (Those area estimates are generous, since empires rarely effectively controlled much of their territory.³) Scholars often liken empires to a rimless wheel: an original homeland (the core) conquered lands and cities outside it (the periphery) but kept them separated to ensure they couldn't coordinate against the conqueror. That wheel was usually unstable. Conquerors usually had little control over far-flung territories, with little to no full-time bureaucracy and poor communication technologies (the horse wasn't even introduced into the Near East until 2000 BCE). Conquerors like Narmer and Sargon had to rely on vulgar displays of brute force and punishment – whether that be public monuments showing them smashing people's heads in or hanging flayed skins from city walls – not because they were powerful, but because they were weak. They governed through intimidation.

Over time this changed. Empires found ways to shorten the distance between rulers and the ruled by developing better bureaucracies and quicker transportation and communication technologies, and making the landscape

more easily traversed by chopping down forests, draining swamps, channelling rivers, and laying down roads. It was a long process, but empires evolved and grew in size. As rulers became more powerful and logistically organized they came to rely more and more on bureaucracy, propaganda, and religion to assert their power. It was a move over thousands of years from intimidation to administration.⁴

Throughout history empires have been the default setting for any large-scale state.⁵ China's history is one of empire for over 2,000 years, from the reign of the Qin (221 BCE) until today, punctuated by periods of fragmentation. Empires drove colonization and drew the current borders of the world. Most stories of collapse are stories of empire. Most of what we consider history is just the stories and actions of a few dozen empires.⁶

THE BLURRY LINE BETWEEN STATE AND EMPIRE

While Sargon is usually remembered as the first emperor, he wasn't. Narmer is closer to claiming that title. That's because all states are empires.

Empires are systems of political control in which one group dominates and assimilates others. This control can be either direct conquest and rule (formal) or the effective control of resources (informal).⁷ Sargon's Akkadian Empire is a paradigmatic example of the former: the empire would conquer a city, install a puppet ruler, and then demand tribute (such as gold and slaves). Some claim that the modern US is an example of the latter. It uses unfair 'free-trade' agreements that result in it getting cheaper imports and selling its products at a premium. Empires require violence, or at least the threat of it, to create a system of unequal exchange. The Akkadians ruled through military prowess. The US is a superpower in large part because it has the most advanced military force and arsenal of nuclear weapons in the world. Whether it be through the sword, or negotiations conducted in the shadow of that sword, empires are about domination. In the simplest terms, empire is just the quest of one society to control the resources and people of another.⁸

The borders separating hierarchical hunter-gatherers, states, and empires are blurry. There were frequently enormous differences in the size of the territory and the population they covered, but the way they operated was often similar. The Calusa were one small, concentrated group of fisherfolk violently extracting resources from the communities around them. The earliest city-states were usually a core of palaces and temples dominating a wider hinterland. Archaeologists in general look for a small number of large sites and many small sites as one sign of ‘settlement hierarchy’. Rome was just a city that grew by conquering its neighbours. The British Empire began with the conquest of what we now call England by the Normans. They then slowly subdued the Irish and Welsh. Centuries later, this English empire expanded across the seas, invading India, Australia, the Caribbean, Africa, and North America. The Calusa, Rome, and the British Empire all involved one group dominating and extracting resources from others. Some grew far bigger in size and bureaucracy than others, but the fundamental approach of unequal exchange was the same. All of them were rimless wheels with a core homeland dominating a periphery: the Calusa capital of Calos dominated the surrounding lands and farmers; Rome dominated much of Europe, North Africa, and the Near East; London dominated Britain, then its neighbours, then the world.

Scholars have tied themselves into knots trying to define empire, separate empire from colonization, and decide whether England/Britain was an empire before it began its overseas colonies. It had, after all, already conquered the Irish before it took India.⁹ The reason why empire is so hard to define is because of the unsettling truth that all states are essentially empires. It’s a spectrum, a matter of degree. For simplicity, we can still call states that conquer other states ‘empires’, but the difference is minor.¹⁰

Empires vary in size and rule.¹¹ The Assyrian Empire ruled over 1.4 million square kilometres of land and obsessed over ensuring that its subjects all prioritized the god Assur. The Western Roman Empire covered 5 million square kilometres and cared little about the gods its citizens worshipped, as long as they paid their taxes and didn’t rebel.¹²

Many empires would link together in webs of trade and diplomacy, like the ancient beginnings of the modern hyper-connected, globalized world. The Early and Late Bronze Age of the Mediterranean are both examples of these great Goliath systems of competing and cooperating empires. Both

would undergo a systems collapse: a collapse not just of one city or empire, but of an entire interconnected system of them. If we want to understand how a collapse could unfold in the modern world, then these are some of the more instructive examples.

THE EARLY BRONZE AGE COLLAPSE

Sargon's Akkadian Empire was not alone in the world. To the south was the Old Egyptian Kingdom. Thousands of kilometres to the east was the Indus Valley Civilization: those egalitarian cities full of plumbing and public bathhouses. To the north, in Malta, we have the 'Temple Period'. It was not an empire or state, but a time when thirty gigantic stone buildings were erected across the islands (perhaps by hierarchical early farmers).

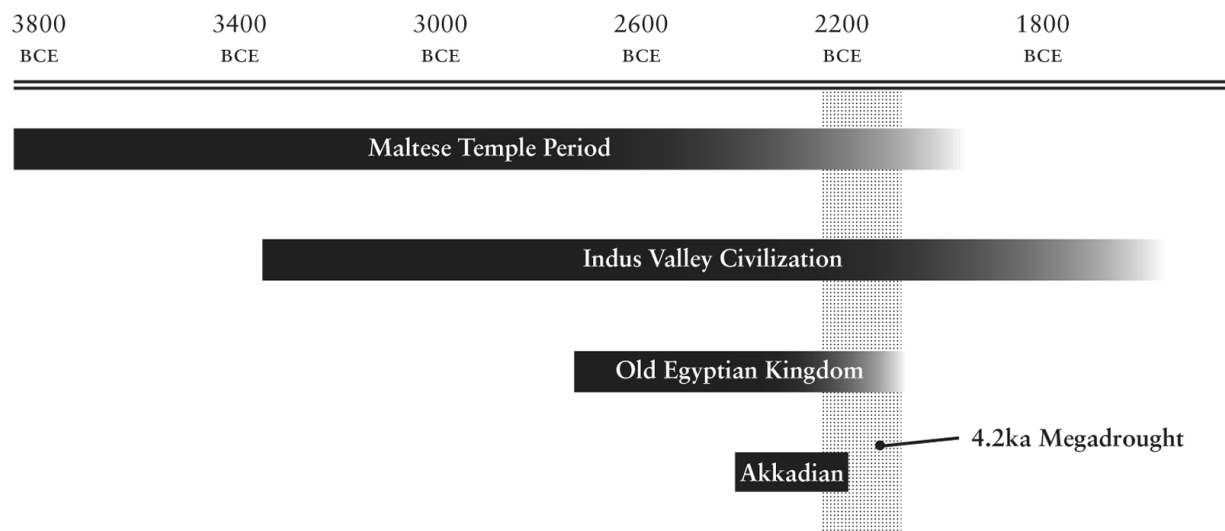
All of these groups eventually failed. The Akkadian Empire that had begun with Sargon in around 2334 BCE had fallen apart by 2150 BCE. The Old Egyptian Kingdom decentralized. There was no longer a single pharaoh, but rather multiple competing claimants to different regions. Numerous settlements throughout Mesopotamia, the Levant, and the Mediterranean were abandoned. The Indus Valley Civilization, which traded with Mesopotamia, reorganized many of its metropolises. The people slowly began to leak from the cities into the surrounding countryside. In Malta, the Temple Period came to an end.

These polities are grouped under what we call the Early Bronze Age, and they all appear to have experienced transformations, even collapses, around 2200–2000 BCE.

It is unclear what caused the collapse of the Early Bronze Age systems. The most widely cited culprit is climate change, in particular a 'megadrought' that appears to have struck the region. Evidence of drought stretches from the Nile to Spain and coincides with the shrinkage of cities and the abandonment of settlements across the area ([Figure 22](#)).¹³

While there is a clear correlation between the drought and the collapses, it is difficult to establish direct causation. We also know that there was significant regional climatic variation. Different areas responded differently and some, such as western Syria, seem to have barely been affected.¹⁴

Some dried abruptly, while for others the reduction in rainfall occurred over centuries and there was little significant change.¹⁵



[Figure 22](#): A timeline of Early Bronze Age polities

None of this signalled a fatal end for these populations. Instead, it meant a decentralization of power. Emperors, pharaohs, and kings lost centralized authority, while local leaders and assemblies gained it. Like the fission-fusion processes of the past, it was a disaggregation. In this case a disaggregation into the building blocks of an empire. In Syria's Euphrates Valley, the Bronze Age had seen a combination of kings ruling alongside popular assemblies and councils of elders. As the region decentralized, the assemblies and councils became more prominent. They even appear to have been a source of resilience, cushioning the fall.¹⁶

Egypt had a similar democratization of both religion and politics. The First Dynasty was, like its monuments, pyramidal in structure. There was a rigid class system, centralized government, and the veneration of leaders, including the ritualistic sacrifice of hundreds at royal funerals. The collapse period is marked by tombs and jewellery becoming more widely available, alongside other signs of greater collective access to religion. While a new pharaoh eventually regained control, the Middle Kingdom was marked by a blurring of class boundaries with new rulers who portrayed themselves as modest shepherds of their people.¹⁷

For Mesopotamia, collapses of empires and kingdoms, such as Sargon's Akkad or the Babylonian Empire (c.1880–1595 BCE), often just meant a

switch back to more autonomous city-states and localized forms of collective governance. Emperors usually had little interest in local matters such as resolving disputes and organizing the day-to-day running of the city, leaving these to local government. This was far from a dreadful outcome. The Akkadian Empire was a predatory, militaristic juggernaut that many would have been delighted to see fall apart.¹⁸

In the Indus Valley, the collapse was more of a slow transformation as people moved from cities into a more village-dominant economy. Over centuries people moved east, began to plant more hardy crops like drought-tolerant millet, and became less reliant on trade between cities and with Mesopotamia.¹⁹

While the drought may have been a catalyst for collapse across the Early Bronze Age, the responses of different regions and polities varied dramatically. None of this constituted an ancient apocalypse. While it was not an overly negative outcome, it was still a systems collapse. It was also not the last time this would happen in the Bronze Age Mediterranean.

THE BEGINNINGS OF THE WEST AND CAPITALISM

A few centuries later, the eastern Mediterranean, south-west Asia, and Aegean was home to a web of interconnected Goliaths (1700 to 1200 BCE). Among them we can count the Mycenaeans of Greece, Minoans, the Hittites of Anatolia (modern Turkey), Cypriots (in Cyprus), the Canaanites of the southern Levant, Egyptians, the Mitanni and Assyrians in Mesopotamia, and the Kassites of Babylon. The Mycenaeans were the original ruthless warrior kings of Greece, residing in multiple palaces across the islands. They received animals and grain from the rural populace in exchange for protection and hosting great feasts. The New Egyptian Kingdom was home to some of the most well-known pharaohs such as Tutankhamun and Ramesses II. It even saw the first possible monotheistic religion in Atenism, the worship of Aten, originally a god who was depicted as a sun disc. The Minoans covered numerous port cities across the Aegean

that had no depictions of male rulers. This Bronze Age network included the fabled city of Troy.

It was a time that also saw the earliest glimpses of capitalism. When capitalism begins depends on how you define it. Most dictionaries give us the rough definition of an economic system where industry has private owners. Academics tend to define capitalism by a list of certain traits: private property, enforceable contracts, wage labour, markets with responsive prices, and supportive governments. We can see most of these traits thousands of years ago in the Bronze Age.^{[20](#)}

In Assyria, around 4,000–3,000 years ago, we have ample evidence of the basic features of capitalism. Over 1,000 recovered cuneiform documents record the use of credit and debt as virtual money, the use of ‘joint-stock’ agreements, interest-bearing debts, and an autonomous merchant class. These were not state-run endeavours overseen by a king. They were privately governed by an assembly of elite merchant families. These families held a monopoly on the trade of certain goods, guaranteed contracts, undertook diplomacy, and regulated the weighing and measurement of precious metals.^{[21](#)}

It is during this time that we see the first signs of multinational corporations as well. In Assyria, different businesses set up shop around the trading city of Kanesh (in modern-day Turkey), a midway point for trading tin from Afghanistan into the Assyrian Empire. These companies had all the hallmarks of what we would call today a multinational corporation: common stock ownership, an organizational hierarchy, and a drive for profit, as well as foreign activities and employees. It’s also in the Bronze Age Assyrian business community of Kanesh that we see the first evidence of businesswomen (and investors), who even penned their own cuneiform letters.^{[22](#)}

States were highly supportive of such arrangements. In one of the first instances of financial deregulation, royal inscriptions from Assur record generous tax breaks given to merchants to encourage commerce in the city. Even the famed ‘Code of Hammurabi’ has rules for granting credit and places taxes and tariffs on commerce. Whether it be capitalism, multinationals, Greek culture, or monotheistic religion, this system was, in short, the root of what many would now call ‘the West’.^{[23](#)}

By 1500 BCE, this Bronze Age network was one of interconnected trade. Each state relied on exchange with others. Most of the gold came from Egypt (Nubia and Sudan), the silver from Greece (the Laurion mines), copper from Cyprus, and tin from modern-day Afghanistan. Tin and copper were vital as they made the alloy bronze.²⁴ There is a reason the Bronze Age was so called, with some comparing the role of bronze to that of oil in the modern world.²⁵ (Well, at least for elites: bronze was primarily a luxury item for weapons, buildings, and decorations – and less used by commoners.) There is an unfortunate penchant for labelling historical periods after metals (such as bronze and iron) that tend to characterize elite life in cities. If we are to name periods after the most used and vital resource, then most of human history should be called the wood age.

Regardless, bronze and trade were central to empire in the Bronze Age.²⁶

If one state wasn't directly trading or bargaining with another then it had indirect contact through one of its neighbours. Tonnes of raw materials and finished goods, as well as dowries, were exchanged across the leaders of Egypt, Mitanni, Assyria, and Babylonia, usually under the guise of gift-giving.²⁷ For instance, one letter detailing gifts from the pharaoh Akhenaten to the Kassite king Burnaburiash II includes gold, copper, silver, bronze, bracelets, necklaces, perfume, mirrors, and ebony boxes.²⁸ The exchange and benefits were limited to the upper echelons of the Bronze Age with little trickling down.

The elite Late Bronze Age trade network happened across cities and was controlled by urban aristocrats who built fortunes off conquest, forced labour, and taxing farmers and client states. Like today, there were tax breaks for the wealthiest. One rich Ugarit trader boasted in a letter of not having to pay import tax when returning from Greece with a ship loaded with beer, grain, and olive oil.²⁹

It wasn't necessarily a pleasant life for the average commoner. Estimates of inequality in the Near East using house sizes suggest that inequality rose with the first cities and states. It continued to increase throughout the Bronze Age in both cities and rural areas.³⁰

The oppression and inequality are reflected in the skeletons of the Bronze Age people. For instance, pharaohs were taller than the rest of the male population. That's because they had access not only to more food but also to

medical services and they lived longer, more leisurely lives. By contrast, Egyptian peasants appear to have suffered regularly from tuberculosis and hernias – the effects of heavy labour. In the Greek Mycenae, kings and elites averaged a height of 172.5cm compared with 166.1cm for the average male population. High-status women were also taller than their poorer peers by around 6cm. This system of bronze Goliaths was deeply unequal.³¹

By 1100 BCE this system had disintegrated. It was a swift unravelling: within two or three generations the small world elite network had collapsed; many of the individual states had ceased to exist, and dozens of cities had been abandoned or burned to the ground. In some cases, even writing was lost, such as the disappearance of the script of Ugarit and the Linear B script from Mycenaean Greece.

THE LATE BRONZE AGE SYSTEMS COLLAPSE

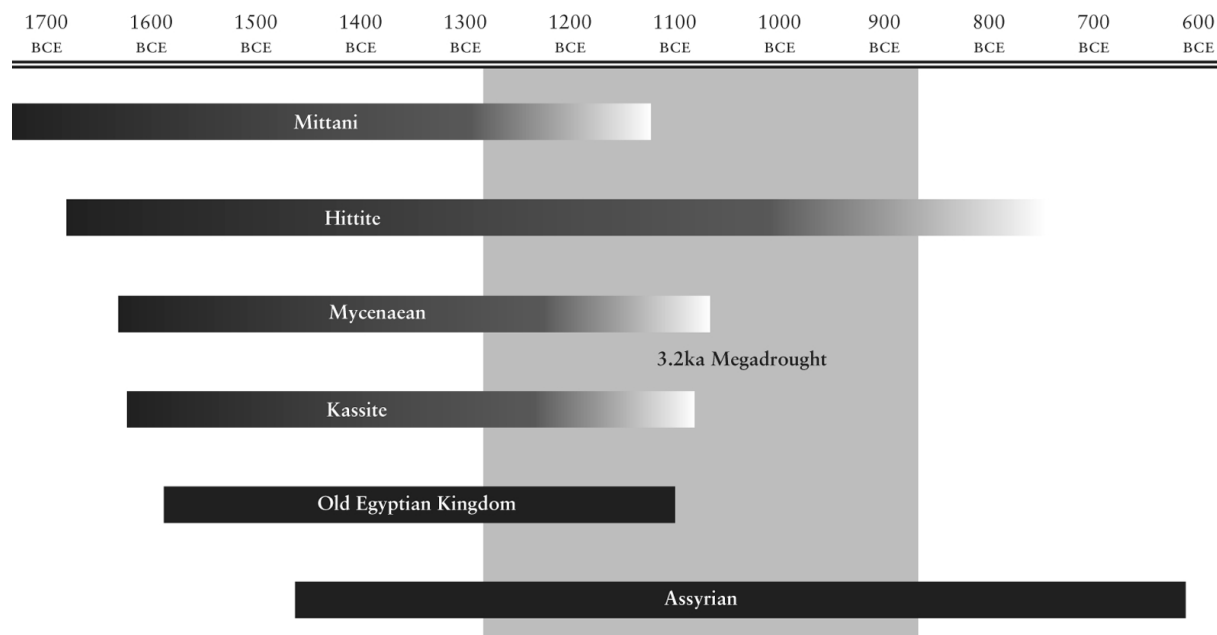
Around 1185 BCE a merchant called Banniya hurriedly penned a letter to his boss. It read, ‘There is famine in your house; we will all die of hunger. If you do not quickly arrive here, we ourselves will die of hunger. You will not see a living soul from your land.’ Banniya was working in the outpost of a commercial company in the city of Emar, located inland in Syria, and messaging his superior in Ugarit, a trading hub on the coast around 250 kilometres away.

Ugarit seems to have been gripped by drought around the same time as Banniya’s letter. In response to a plea from the king of Ugarit the Egyptian pharaoh sent gold, fish, and textiles as aid, yet it was to no avail. Both Emar and Ugarit were destroyed shortly thereafter. The debris left from the razing of Ugarit was two metres high. We do not know who burned the city down.³²

This was the death knell of the Late Bronze Age ([Figure 23](#)).

The Late Bronze Age was hit by a variety of shocks. Perhaps the most decisive was a drought that struck the region around 3,200 years ago and lasted 150 to 300 years.³³ The drought extended from northern Italy down through Greece and Turkey to Cyprus, the Levant, and Egypt, and east to Iran and Iraq. The arid conditions appear to have caused crop yield

shortages and famine. In Egypt, grain prices rose. Some areas appear to have been particularly hard hit as we can see in the letter of Banniya.³⁴



[Figure 23](#): A timeline of Late Bronze Age polities

As with the earlier Bronze Age, the drought was not universal. Some areas do not seem to have experienced a drought at all. Climatic data from a stalagmite near the Pylos palace in Greece suggests that there were wetter conditions before the collapse of the palace, and drier conditions after. The Bronze Age people were also not placid targets. They adapted. During the drought the Egyptians bred more tolerant cattle in Canaan and extended dry farming in the southern and eastern fringes of the Levant to make up for the lower grain production in the north. The drought contributed to the collapse, but its effects differed greatly across regions.³⁵

Alongside the drought came a series of earthquakes. The entire region is riven with fault lines, including the Anatolian Faultline across northern Turkey. Maps of the areas most affected by earthquakes in the twentieth century show an overlap with many sites that were destroyed during the Late Bronze Age. Earthquakes in the region often come in groups, or earthquake ‘storms’, and can be of 6.5 or greater magnitudes, enough to destroy even modern buildings.³⁶ There is evidence of a storm of earthquakes across Mycenaean Greece, Troy, and settlements across Canaan (modern Israel and Jordan). Earthquakes alone, however, seem insufficient

to account for such a broad systems collapse. In both Ugarit and Mycenaean Greece there is evidence of sites being rebuilt after earthquakes around 1250 BCE.^{[37](#)}

These natural disasters may have also been accompanied by pestilence. We now know that the pharaoh Ramesses V died of smallpox, while the entire royal Hittite family was eliminated by an epidemic c.1350 BCE. Yet, there is otherwise little evidence: we've yet to discover any mass graves, or texts that mention an outbreak.^{[38](#)}

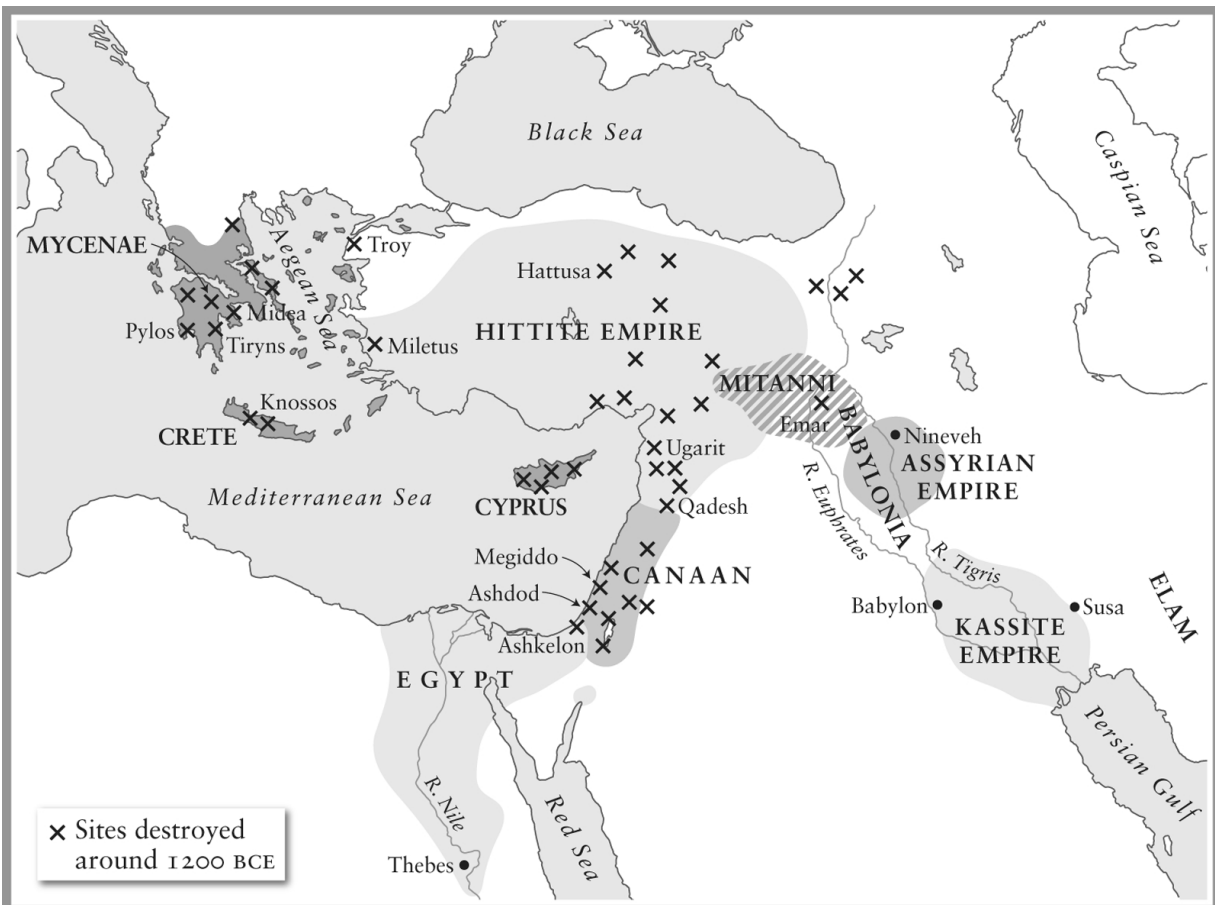
These environmental hazards were accompanied by threats with a human face. Perhaps the most well-known and maligned are the 'Sea Peoples'. These were purportedly invaders who swept through the Mediterranean razing cities to the ground. The idea of marauding Sea Peoples comes from inscriptions by the pharaohs Merneptah and Ramesses III recounting battles in 1207 and 1177 BCE. Ramesses casts himself as the last heroic bastion of civilization holding out against a tide of barbarians: 'They laid their hands upon the lands, as far as the circuit of the earth, their hearts were trusting and confident as they thought "our plans will succeed".' According to Ramesses, their schemes came to nought, for once they met him they were 'slain and made into heaps from head to tail'.

It's a rousing tale but there are a few reasons to not trust Ramesses III. His records aggrandize him as a military genius, but there is essentially no evidence to support his account. Some of the cities Ramesses claimed were destroyed were not, and for other cases much of the destruction occurred before the arrival of the Sea Peoples. In most cases we know it wasn't Sea Peoples who invaded. The Elamites conquered the Hittites, and the Hittites appear to have conquered Cyprus. Some areas, such as Mycenaean Greece, show no signs of external attackers. The more likely culprit is rebellion or conflict between the different palaces within Greece rather than foreign invaders.

There was also no single invading force. Instead, this was more likely to have been a mixed group of refugees, migrants, pirates, and just savvy mobile people from the Aegean and eastern Mediterranean who were moving because of drought, warfare, oppression, and economic hardship. In many cases they would have settled with little conflict, bringing new culture with them (such as Philistine migrants settling in what is modern-day Israel and bringing Aegean-style pottery and architecture). There is, if

anything, an odd double standard in how the migrants of the Late Bronze Age have often been portrayed. When the elites of the Bronze Age traded or warred it was a sign of a thriving culture; but when commoners moved, they are seen as mysterious, apocalyptic invaders.³⁹

A clearer contributor to the systems collapse is civil strife. Rising inequality and conquest empires were not the most stable basis for a social system. Conquered subjects actively resisted and undermined their imperial overlords. There is evidence of rebellion – such as burned settlements and arrowheads from battles but without signs of foreign invasions or a marching army – in Greece, Hazor in Canaan, and many other sites throughout the levant.⁴⁰ Whether it be rebels or invaders, a swathe of cities across the Bronze Age system were razed, as seen in [Figure 24](#).⁴¹



[Figure 24](#): A map of the Late Bronze Age collapse

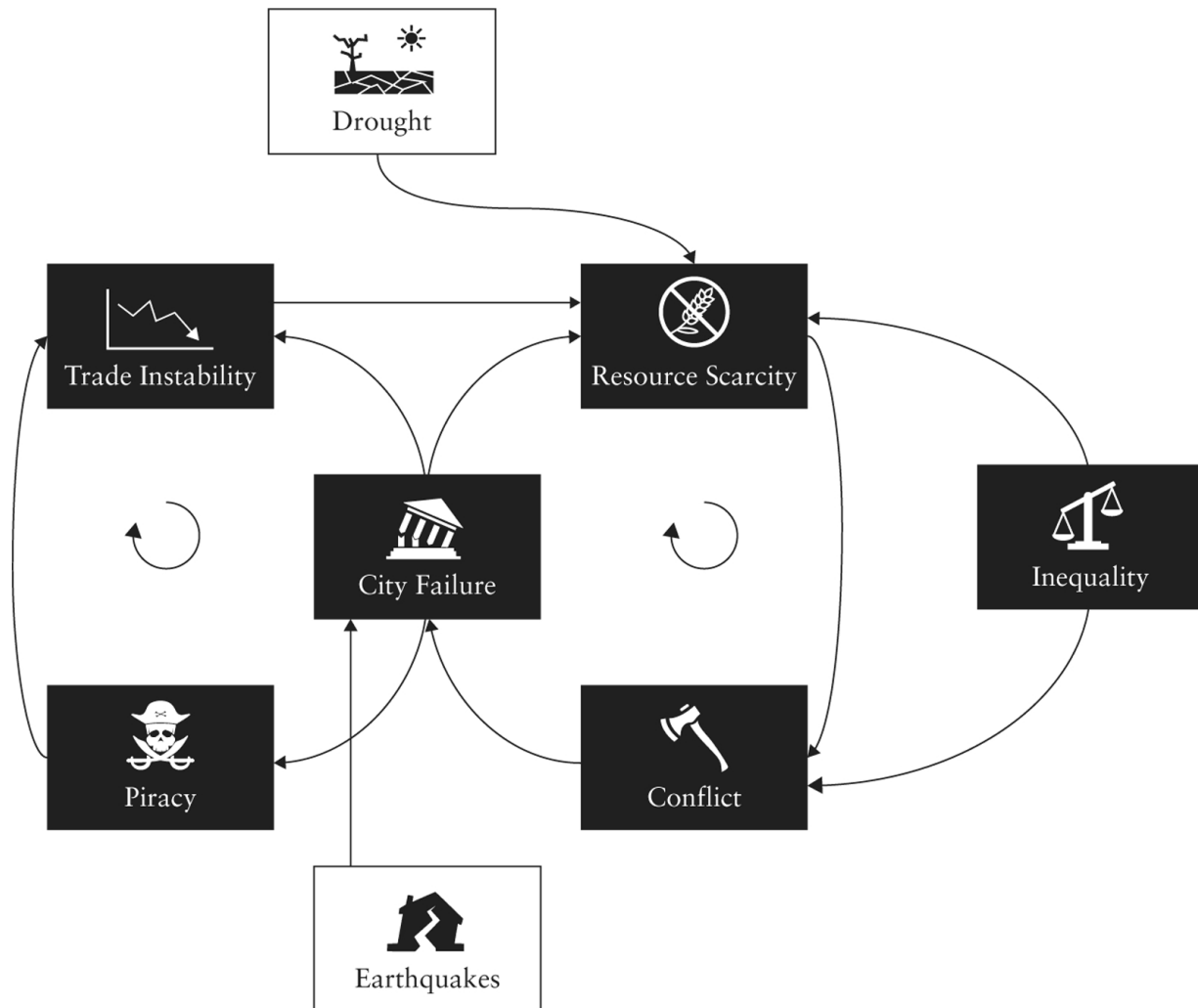
This map shows the rough extent of key empires and cultures preceding the Late Bronze Age collapse, as well as some of the major sites destroyed during the collapse.

My colleague Eric Cline, one of the leading archaeologists of the Late Bronze Age collapse, has suggested that it was caused by no one factor, but by the interaction of many. It was a perfect storm of drought, famine, earthquakes, disease, invasions, and rebellions. More than that, each of these factors created ripple effects that worsened the situation. Revolts spread as commoners stormed palaces and grain stores. The supply of gold from Egypt, silver from Greece, and tin and copper from Afghanistan would have been disrupted. Bronze came in increasingly short supply.⁴²

The very way that the Bronze Age system was structured could have contributed to its swift fall.⁴³ Multiple different stressors cause one system to enter a crisis, which triggers another one to do the same. This can worsen the initial crisis, creating a ‘systems death spiral’.⁴⁴

Two key potential death spirals drove the disintegration of the Bronze Age system. First, drought and rising inequality created more rebellions and interstate wars. Those revolts and invasions destroyed cities, including key trading hubs, destabilizing the trade network. As the trade network frayed and fewer goods were moved around, there was even more resource scarcity, fuelling further conflict. The sheer drop in trade made trading cities less profitable and more vulnerable, making abandonment or external invasion more likely (see [Figure 25](#)). Compounding matters was the impact of the earthquake storm. Pirates emerging from the broken cities, as well as driven by climate change into the area, could have made trade even more dangerous, further unsettling the Bronze Age network. The key behind it all is that the interdependence of the Bronze Age Goliaths was a weakness. Once enough cities were lost it destabilized the network, encouraging more cities to fall apart, slowly unknitting the Late Bronze Age system.⁴⁵

The interdependence of the Bronze Age was a blessing for Goliaths in the good times, but a liability during a time of multiple shocks. One study that mapped the trade and diplomatic networks of the Late Bronze Age found that while it was usually robust, it may have taken just the loss of an empire or two to throw it into disarray. Their model suggests that just the loss of Ugarit and the Hittite Empire could have been enough to upend the Late Bronze Age system. It’s a plausible scenario given that the two seem to have collapsed at roughly the same time.⁴⁶



[Figure 25](#): Death spirals in the Bronze Age collapse

COLLAPSE AS A BLESSING

The Late Bronze Age collapse was no apocalypse. Each polity changed and recovered in different ways. The Hittite Empire fell apart, and its capital of Hattusa was abandoned and then razed. Just a few cities and smaller kingdoms across south-eastern Anatolia and northern Syria were left behind. Egypt broke into rival factions and competing pharaohs, yet its culture and population remained relatively stable. The Assyrians were resilient, with one expert perhaps over-optimistically describing it as a ‘recession’. The palaces of Mycenaean Greece were utterly levelled and used by squatters and herders. Cities and trading hubs such as Ugarit were

burned to the ground. From a wide lens the Late Bronze Age collapse looks like a single systems collapse, but a closer look shows it to be a kaleidoscope of different processes of collapse, resilience, and recovery occurring differently across the region.⁴⁷

We have little idea how this systems collapse impacted people living in rural areas that would have made up 95 per cent of the population. The end of the Hittite Empire and the palace economies of Greece would have meant relief from tax and corvée labour (unpaid, forced labour for a certain period of time, usually for state projects) for most subjects.

The few bits of evidence we do have suggest that the collapse may have been a boon for most people. Inequality estimates based on housing sizes suggest that rural areas became more equal. In contrast, urban settlements continued to become more unequal from the Late Bronze Age into the Iron Age. This is because empires could no longer effectively extract wealth from the countryside.⁴⁸

Tellingly, those who lived under the empires that experienced little change during the collapse did poorly. The Assyrian Empire had a relatively (compared to the Hittites or Mycenaeans) smooth transition into the Neo-Assyrian military juggernaut. It's also one of the few areas where we have data on how people's wealth varied over time, thanks to buried grave goods. In Assyria we see inequality grow from the Late Bronze Age to the Iron Age. Commoners in the Iron Age get buried with fewer and lower-quality grave goods such as pieces of pottery and jewellery. That decrease is probably due to a greater reliance on migrant labour and the Assyrian crown demanding more tax and corvée labour. We don't know if the rich were getting richer, but the poor were clearly getting poorer.⁴⁹

In Greece, more egalitarian regions without palaces didn't even appear to experience a collapse. Areas such as Corinthia, Achaea, and the Euboean Gulf didn't undergo a collapse, and seem to have prospered after the palaces elsewhere fell apart. Unlike the situation with the Assyrians, the poor didn't get poorer. The more egalitarian communities were both more resilient to external shocks and better for those living within them. And the collapse was largely an elite phenomenon driven by inequality. Without palaces and elites there was no conflict and no collapse.⁵⁰

We don't have a sharp, high-resolution image of life during the Late Bronze Age collapse for peasant farmers. Did they really worry much about

their previous Hittite overlords no longer receiving gold and ostrich eggs from overseas? Or did they simply shrug and continue ploughing and enjoy the less frequent visits from the tax collector? We can easily see Mycenaean Greece as a case of ‘escape’ from an unsustainable system of competing palaces extracting wealth from the lands and people around them. Their end was brought about by war and revolt, and the ensuing drought helped to ensure they were never rebuilt. You were better off as a pastoralist grazing your sheep in the ruined palaces of Greece than as a peasant under the extractive dictatorship of the Assyrians.⁵¹

One Bronze Age expert suggested that, rather than think of it as a civilizational collapse, we should view it as a time ‘during which a few large imperial polities succumbed to their structural weaknesses and corruption’, which allowed repressed groups and communities to reorganize and flourish.⁵² While we shouldn’t turn away from the lives lost or the smoke of the burned cities, it is an apt interpretation of the collapse. Perhaps the very idea of civilizational catastrophe followed by a dark age in the Late Bronze Age is ‘imperial rhetoric from beyond the grave’.⁵³ Perhaps we shouldn’t mourn the Egyptians, Hittites, and Mycenaeans.

There may even be grounds for celebrating the ‘dark age’ that followed.

THE BRIGHT SIDE OF DARK AGES

What followed the Late Bronze Age collapse is what was once called a ‘dark age’. It is a common trope in the rise-and-fall vision of history that collapses were followed by dark ages.

It’s a difficult position to hold once you look at what happened in the wake of the Late Bronze Age. Homer, author of the *Odyssey* and the *Iliad*, was a poet of these Greek dark ages.⁵⁴ As was Hesiod, the poet of *Theogony* and *Work and Days*. They are almost certainly part of a wider set of heroic poetry developed during the dark age. Without collapse the world would be bereft of Homer and reams of epic literature.⁵⁵

Collapse could also provide space for more democratic arrangements. The collapse of Mycenae levelled wealth inequality. In the long run, this left emerging aristocrats even after the early iron age (750–480 BCE) weak

and divided. The initial law codes (from around 620 BCE) the Greeks crafted aimed to limit the ability of an individual to dominate the rest. By 525 BCE, many adult males had decided to make collective decisions via mass assemblies. These practices and norms were the foundations for later Athenian democracy.⁵⁶

Athens was a limited democracy that permitted slavery and had political representation for only 10–15 per cent of the population (just free, male, adult citizens). It was nonetheless a significant step beyond the previous autocracy of Mycenaean Greece. With their random selection of representatives and emphasis on deliberation, it may well have been a more advanced and intense form of democracy than the practice of modern, rich countries of simply voting once every four or so years.

Many other benefits accompanied the collapse. New groups formed and prospered. In Canaan, the late-thirteenth- to eleventh-century highlands saw the emergence of village dwellers who were the precursors of the Israelites. Other independent cities such as Tyre and Byblos rose up and took over the trading routes previously serviced by Ugarit. They became the pioneering Phoenicians, who would eventually form a Carthage-based empire. The Phoenicians introduced a new alphabet to the world. Alphabetic scripts had existed before, with Ugarit possessing two of them, but these were elitist scripts with hundreds of symbols that were difficult to learn. In contrast, the Phoenician alphabet was short, simple, and easy to master. Because of that, it spread widely, becoming the basis of the Greek alphabet as well as the Latin alphabet that underpins many languages used across the world today, including English, French, and Spanish.

We see a similar pattern with technology. Most of the key technological breakthroughs of the Bronze Age, such as the alphabet and ironworking, occurred on the peripheries of empire, and/or after the Bronze Age collapse.⁵⁷ Other technologies and practices such as animal husbandry, the plough, irrigation through canals and ditches, the sailing boat, and pottery all preceded the first states of the Bronze Age and were often developed by stateless hunter-gatherers. The idea of golden ages of empire leading to technological and cultural progress, and the fall of empires leading to regression, is at best mistaken.

The term ‘dark age’ has fallen out of favour among historians and archaeologists for good reason. Far from being the beginning of a period of

gloom, the end of the Late Bronze Age led to healthier, wealthier people in many regions, gave us the Latin alphabet and new technology, and made room for the democratic model and culture of Classical Greece.⁵⁸

The systems collapses of the Early and Late Bronze Age are not the only ones to have occurred in world history. Around two millennia later and 11,000 kilometres away a similar phenomenon unfolded across the Atlantic Ocean. There, another Goliath system of numerous states faced drought, war, rebellion, and their own interdependence turning against them.

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Shattered Obsidian

*Collapse in the Yucatán – From Drought and Warfare to Alien Invaders –
When Does the Collapse End? – When Writing Becomes Obsolete –
Collapse in East and West Africa – Systems Collapses Across the World*

COLLAPSE IN THE YUCATÁN

In 1696 the Spanish friar Andrés de Avendaño was one of the first Europeans to encounter the Mayan city-state of Tikal – or what was left of it. His surprise was clear, as he and his companions emerged from thick jungle and stumbled upon a ‘great variety of buildings ... it seemed to us that these buildings must be near a settlement, but we found ourselves, as we saw afterwards, very far from a settlement’. Cities resplendent with great stone pyramids jutted above the canopy, yet they were completely empty. Tikal had largely been abandoned five hundred years earlier. It was one of many Mayan cities that littered the Yucatán peninsula before the arrival of the Spanish.

The Classic Lowland Maya (150–900 CE) followed the same blueprint that we see in the earliest cities of North and Central America, from Monte Albán up to Cahokia. Stone ballcourts and great plazas were enclosed by towering flat-topped pyramid-temples. The deep black volcanic glass of obsidian was an analogue to bronze: it was traded over long distances and used in weapons, jewellery, and tools, mainly by elites (at least initially). The Maya also developed a complicated cosmic calendar replete with

rituals to celebrate royal ancestors, and a glyphic writing system that was regularly used to memorialize the deeds of dead kings.

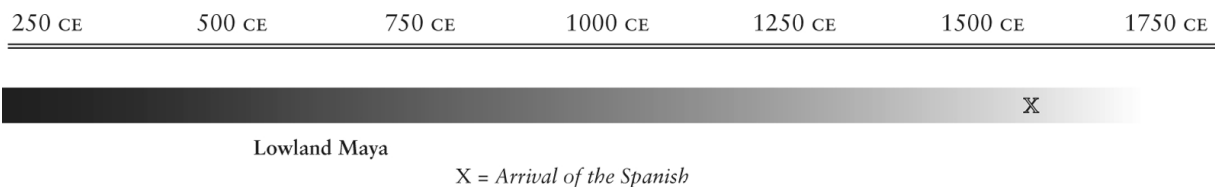
The Classic Lowland Maya did not comprise a single empire; rather it was a collection of around sixty to seventy different city-states spread across a territory encompassing Guatemala, Belize, and parts of Mexico, Honduras, and El Salvador. Each city varied in size and power, but all shared a culture, much like the Greek or Sumerian city-states. Some of the cities were vast, holding up to 120,000 people, not including the network of farms and villages that surrounded them. They dwarfed most of the cities of Europe at the time. The Lowlands were a dense metropolis of 7–11 million people. One airborne laser scan of a 2,144-square-kilometre swathe of Guatemala found 61,480 ancient structures (or twenty-nine for every square kilometre) connected by 106 kilometres of canals.¹ It's difficult to tell how extensive this tropical urban maze was. As recently as October 2024 another previously unexcavated city that would have once housed tens of thousands – nicknamed Valeriana – was found.² The Lowlands were a teeming, sophisticated web of jungle metropolises.

It was also a warring political ecosystem, with cities frequently raiding and conquering each other. This led to a series of cycles in which one polity would expand, often conquering others nearby and becoming an empire, before the conquered would break away and become their own states. For instance, the city-state of Copán expanded to dominate a territory of 10,000 square kilometres from eastern Guatemala to Belize around 500–600 CE. Within 140 years, Quiriguá on its eastern border broke away, celebrating their independence in hieroglyphic texts and expanding. This cycle of rise and fall, expansion and contraction, continued for decades. These cycles are seen periodically, each lasting around 300 years, with the centre of power shifting between different kingdoms and their capitals. In the Classic period it was the Kaanul kingdom centred at Dzibanche, then Calakmul and its rival Tikal. It's hard not to see similarities with the boom-bust/fission-fusion cycles in the Puebloans, first farmers of Europe, and others. As with them, such regular cycles cannot be easily explained by fluctuations in the climate (which tend to lack such regularity).³

The collapse of the southern Lowland Maya (known as the 'Terminal Classical Period', approximately 800–1100 CE) saw multiple cities emptied out. Some cities, such as Tikal, lost nine out of ten residents. The

production of monuments was choked off. The creation of luxury goods and the use of the calendar and glyphic writing system all plummeted. The long-count calendar (an astronomical calendar used to keep track of longer periods of time) was lost altogether. In many cases it's unclear what exactly happened to cause such a collapse.

In others, what happened is all too clear. In about 800 CE the city of Cancuén was attacked suddenly. Unfinished stone and wooden palisades surround the site; spearheads are scattered within it, as are numerous skeletons displaying wounds from axes and spears. The elite were slaughtered and discarded into water storage tanks, along with their sceptres and regalia. The ruling family was dumped into shallow graves outside the city. Whether it was an invasion, peasant rebellion, or coup by noble families is unknown. In other cases, the picture is murkier. But during this period, the Lowlands were a site of war, strife, and flight. By the end of the period, the Goliath of the southern Lowlands – the systems of religious and economic domination we previously would have called Mayan ‘civilization’ – had vanished. The ruling class had, in the words of one expert, ‘gone with the wind’ ([Figure 26](#)).⁴



[Figure 26](#): A timeline of the Lowland Maya collapse

FROM DROUGHT AND WARFARE TO ALIEN INVADERS

As in the Late Bronze Age, the Maya faced a storm of challenges during this terminal period. Among the different failing states, we frequently see massacres of rulers and aristocratic families and targeted destruction of elite monuments and insignia. People, whether they be nobles or peasants, appear to be rejecting the Mayan holy lords, the *k'uhul ajaw*, who previously ruled. What triggered this conflict?

Most of the southern Lowlands appear to have faced an unprecedented drought during this period. The Maya had extensively reshaped the land to produce large amounts of corn, beans, and squash. These were cultivated on terraced hills, forming a cascade of farms down the slope. Unfortunately, corns, beans, and squash are all particularly vulnerable to drought. Even minor reductions in rainfall can have severe effects on both crop harvests and how many people could be fed. Worse, this wasn't just a period of drought. During 700–800 CE rainfall in the area became far less predictable and far more volatile.

Most Mayan cities had experienced periods of drought before, even growing and expanding during them. Most people also farmed far more drought-tolerant crops such as manioc and chaya (tree spinach), drew on forest gardens rich with fruit, and kept extensive home gardens. Even corn could be stored for years, so they weren't entirely vulnerable to climate change. The depopulation of the cities may have been due not only to war and famine, but also to people departing to regional areas to depend on more reliable crops during a period of drought.

Drought was also not uniform for the southern Lowland Maya. The timing and severity varied across the region. For instance, the city-state of Peten appears to have collapsed before any drought occurred. While some cities like Tikal and Cancuén were abandoned in a bloody fashion, these events are spaced out over centuries, and some settlements were never abandoned at all. The western part of the southern Maya Lowlands shows little to no environmental stress, but high levels of warfare. The emerging picture is one of drought playing a role in triggering breakdown, but it was a role that was heavy in some cities and light or even absent in others. When a failure in rain and crops met already fragile states, the result was usually chaotic, but others that were more robust could weather the change.⁵

A key factor here appears to be environmental degradation. In many of the areas of the Maya Lowlands we see increased soil deposited in lakes before the collapse period. It suggests that deforestation was increasing, and soils were being washed away. Similarly, there are often few animal bones discovered in many archaeological sites during this period, suggesting a depletion of forests and animals. This may be one more reason why the

Lowland Maya more easily navigated earlier droughts but suffered during later ones.⁶

We can think of environmental degradation as one way in which societies create energy problems. Societies are in some ways a big marketplace, with energy as the universal currency. Every building, every calorie, is captured solar energy. Taking energy from the environment requires expending energy. That could be the labour required to till, sow, and harvest a field. It could be the rigs, mines, and pipes needed to pull up coal and oil from underground and shuttle it across the world. The ratio of the energy used to extract a resource relative to how much it provides is called the 'Energy Return on Investment' (or, EROI). For simplicity, let's just call it the energy ratio. If I get ten barrels of oil from the ground by using just one barrel to power the machinery used to dig it up, then the energy ratio is 10:1. The energy ratio can, of course, drop, whether that be due to exhausting mines that provide precious metals, or exhausting soils until their yields decline. The energy ratio for the Maya was dropping as they faced drought.

We've seen this in other cases too. The Indus Valley Civilization switched to drought-tolerant crops as an adaptation to climate change.⁷ This unfortunately meant lower yields and a lower energy ratio. The first farmers of Europe had to contend with poorer harvests and declining soil quality. Again, their energy ratio suffered, making droughts worse.

While environmental degradation is a favoured explanation for the Lowland Maya collapse and other cases, it is rarely the sole or primary culprit, as the archaeologist Guy Middleton has shown extensively.⁸ The few cases where it plays a larger role involved bungled responses from rulers in fragile environments.⁹ For example, the Abbasid Caliphate (750–1258 CE) inherited a vast irrigation network across the dry lands of Mesopotamia. Yet the fall of the caliphate was marked by wasteful expenditure, elite competition, and exploitative tax-farming, all of which led to the slow deterioration of the irrigation system. Not only did the caliphate fall apart, but so did the irrigation system. By 1258, as little as only 5 per cent of the population that had resided there during Sassanid times remained. The land had become the wasteland that nineteenth-century European travellers chronicled. Lower Mesopotamia was a unique case: the landscape was especially dry, fragile, and dependent on human infrastructure. The Yucatán too is a fragile ecosystem. In contrast, mainland

Europe, with its varied and temperate climate, never experienced such a catastrophic environmental disruption of a similar scale. Ecocide can happen, but it is thankfully rare.¹⁰

Even taking the fragile environment of the Yucatán into account, collapse there was clearly not just about drought or a declining energy ratio. The environments of the southern Lowlands had bounced back within a century after the collapse, with archaeological digs showing an abundance of animals, including high-canopy animals such as the pig-like tapir. In some sites archaeologists report that a four-square-metre patch of land 10–20cm deep will fill a gallon bag with the bones of animals from this period. This was a time of abundance, yet people didn't move back. If it was purely for environmental reasons we would expect a return to such productive land.¹¹

The most overlooked and perhaps critical source of the collapse was a seemingly innocent one: a movement in trade. Southern Lowland kingdoms were dependent on trade, and their competitors on the coast were booming. People moved from the interior Lowlands to the north, as well as the east and west coasts. It's during this time that states in the north, among them the famed Chichén Itzá, reached their peak in size and prestige. The rise of these areas and the lack of resettlement after the drought ended both support the idea that people were moving to avoid war and seek better economic prospects rather than to escape complete environmental breakdown, although all these factors would probably have contributed: Mayan kingdoms relied most on trade during times of drought, and with less trade available were more likely to turn to war instead.

A final factor is a now familiar one: inequality. Housing sizes suggest that wealth inequality rose steadily throughout the Classical Period and worsened going into the Terminal Period. The great leveller of collapse struck, destroying elite fortunes, and wealth inequality finally declined.¹² The most autocratic kingdoms had the highest levels of inequality, a sign of the link between inequality and extractive institutions.¹³ The cities to the north and the coastal ports that people migrated to were also more equal than the Lowlands they escaped.

The southern Lowland Maya were an increasingly unequal set of kingdoms with a degraded environment and a dependency on trade. As drought struck and trade dried up, the kingdoms turned to war instead. People, predictably, died or moved. Like the Bronze Age collapse, this one

was multicausal, with many threats and vulnerabilities, but the underlying driver was once again a group of polities becoming increasingly top-heavy and extractive.¹⁴

WHEN DOES THE COLLAPSE END?

The Maya did not disappear. While the southern Lowland Maya collapsed, glyphic writing and trade continued. There were hundreds of towns and city-states in the north that grew. The story of the Maya appears to be partly one of people in the south giving up more oppressive religious leaders for more commercial, seafaring societies, particularly in the east, west, and north. Chichén Itzá in the north became a new dominant centre as the imperial cycle restarted. Although these northern settlements also eventually fell apart during a period of drought.¹⁵

Yet even after this period, new cities sprang up and grew. There were still numerous interconnected towns as well as cities resplendent with pyramids and populations of 10,000–20,000. The rebound of the population and regeneration of the Goliath was impressive. The imperial cycle turned once more as power shifted from Chichén Itzá to the city-state of Mayapan (1200–1450) in the Lowlands. It grew to 15,000–20,000 people during this ‘post-Classic’ period. The Mayapan state then collapsed decades before the arrival of the Spanish.¹⁶

The end of the Mayapan polity seems to be a familiar cocktail of drought, inequality, and elite infighting. The period recorded in the Maya K’atun calendar (and evident in a range of archaeological data) includes a time of ‘terror and war’ (1302–23 CE), an inquisition across the noble Xiu family (1361–81 CE), population decline (1381–1401 CE), a massacre of the Cocom noble family (1440–61 CE), and then the eventual abandonment of the city after 1450 CE. These events left behind a series of mass graves. One example from 1302–62 (a likely period of terror and war) contained twenty-five skeletons of people who appear to have been killed by blows to the head and then dismembered after death. Shortly thereafter is another massacre of around twenty individuals who were butchered and burned. This was during a time when one elite faction began to persecute and torture their rivals. The final massacre of the ruling Cocom noble family

and abandonment of the city coincides with a drought. Mayapan had a strong reliance on rain-fed agriculture and lacked centralized grain storage. Rulers who were supposed to be a conduit to the gods could easily be blamed for failed harvests, hence drought could have been used as a way for competing elites to sow discontent.¹⁷

The collapses, including the dramatic southern Lowland collapse, were not wholly negative affairs. Chichén Itzá and Mayapan had greater wealth equality and more open hierarchies than previous polities such as Tikal. While more inclusive, they were still conquest states that venerated priests and rulers in similar ways to those that had occurred in the southern Lowlands.¹⁸

The Mayan story is one not just of collapse but also of resilience and resistance. When the Spanish arrived, they found a prosperous people trading across a vast network of smaller towns and settlements, with different currencies such as cacao beans, shells, stone beads, copper, and cotton. They still wrote in their glyphic script, lived in cities, and warred with each other. Around fifteen states remained. Some had kings while others had ranked noble families. It was a world with fewer people, smaller cities and smaller, less powerful kings. That doesn't mean a worse world. Its size and decentralization might have even been advantageous when they faced European invaders.¹⁹

The Spanish struggled to defeat the Maya. In some battles the Spanish were even defeated by them. In the 1533 Battle of Chichén Itzá the Mayan cities joined together in a force of tens of thousands to repel a Spanish siege. They killed many of the invaders, forcing them to retreat. This was one among several Mayan victories. It is little wonder that it took the colonizers around 170 years to conquer their lands. The dispersed regional population would also have blunted the effects of disease imported by the invading Europeans. Cities like Tikal would have been a demographic time bomb for smallpox, while scattered townships were less vulnerable.

The Mayan people continue to this day, with 6 million speaking different Mayan dialects across countries such as Belize, Mexico, and Guatemala. While the Mayan language and much of their culture persist, other traditions including their glyphic writing system fell into disuse during colonization. Mayan became a dead script.

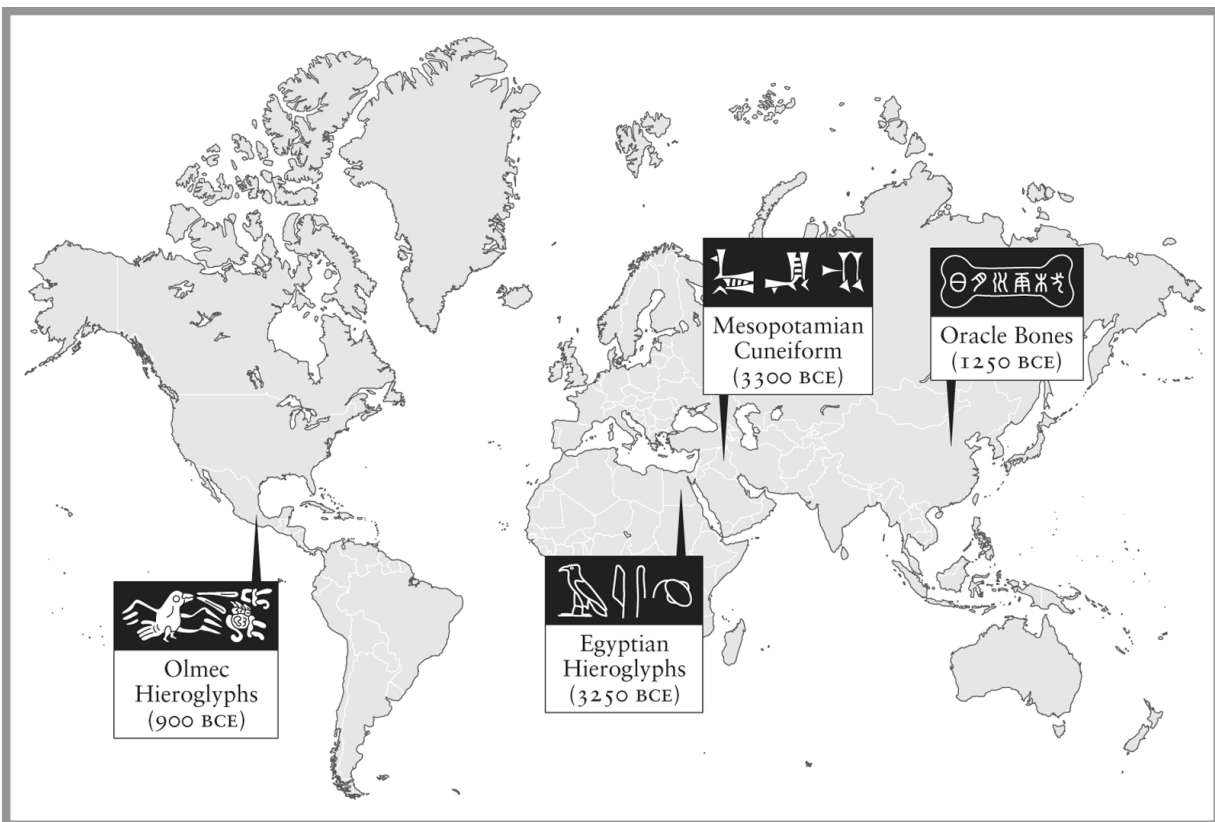
WHEN WRITING BECOMES OBSOLETE

Both the Maya and Mycenaean Greece are (in)famous for losing their scripts. The Indus Valley Civilization script and the Linear A script of the Minoans on Crete also expired during collapses. The loss of writing is central to the story of collapse. What is true about the 'dark ages' is that they often leave historians and archaeologists in the dark about what happened during them. That is because writing dwindles or is lost entirely.

Writing has historically been about power, and writing systems have often been linked to a specific state. Mayan glyphs may have been an original script (or it may have been inspired by the Olmecs), and all the original forms of script such as Mesopotamian cuneiform, Egyptian hieroglyphs, Shang Dynasty oracle bones, and the Olmec script (and the later Zapotec script) were associated with states.²⁰ Similarly, Brāhmī relates to the Mauryan Empire in India, and Kharoṣṭhī to the Kushan Empire that stretched across southern and south-eastern Asia.²¹ Their initial purposes tended to be limited to propaganda, accounting, and administration: tools to stabilize a Goliath. Writing is a recent invention used by very few for most of history for specific elite purposes. This was its weakness.

When power changes, so too do scripts and writing. Use of the Mayan script plummeted during the collapse since the elite class who used it fell apart. Yet it survived. It didn't die until it was intentionally and systematically exterminated by the Spanish in order to impose their own regime. They banned the use of glyphs, burned hundreds of books and scrolls, enforced the use of Spanish, and prosecuted, tortured, and murdered Mayan writers. Nonetheless, the Maya defiantly continued to write in Yucatec (using Spanish script) and record their traditional texts (such as the Chilam Balam books). We see similar cases elsewhere. Egyptian hieroglyphs and other indigenous writing systems were connected to the state. Once Egypt became a Roman province, the use of these writing systems was restricted to temples, which were reliant on state support. This support failed during the early first century CE and Egyptian writing systems were slowly lost in favour of Greek ones ([Figure 27](#)).²²

Writing was often easier to kill because it was practised by a small, elite class. The exclusivity of writing and scripts made them fragile. Mayan was used mainly by a small section of the populace whom the Spanish could effectively target. Similarly, Linear B of the Mycenaeans (mainly used for documenting commercial transactions and the palace production of wool and linen) appears to have been used only in palaces and only by a few dozen writers. In Pylos (southern Greece) there were only some forty-five writers of Linear B and sixty-six in Knossos (Crete). It should be little surprise that Linear B was lost once the palaces were abandoned. Few if any outside of the citadels knew it and the scribes had little use for it (or employment) once the kings were gone.²³



[Figure 27](#): The original scripts of the world

These four scripts are the most widely accepted original scripts in the world. Each was connected to the emergence of a Goliath and an original state. Each eventually led to more developed writing systems.

Languages also tend to reflect power. Mayan still exists, but the lands it once spread over now all have Spanish or, in Belize, English as their official

tongue. Latin, the official language of the Western Roman Empire, also reflected power in the way it changed. It became restricted to powerful networks such as the Church, while local peoples transformed it into dialects, resulting in the Romance languages of today: French, Spanish, Italian, Catalan, and Portuguese.

Collapse rearranges power structures, and the obsolescence of scripts, writing, and language is often just collateral damage.

COLLAPSE IN EAST AND WEST AFRICA

Latin wasn't the only language to become obsolete and restricted to a tiny elite class after a collapse. Another example is Ge'ez, a script created in the Horn of Africa. It is now largely a liturgical language used primarily in Orthodox and Catholic churches throughout Ethiopia, with no one using it as a first language. Once, it was far more widely used, for Ge'ez rose and fell alongside a powerful African empire.

Ge'ez was once the main language and script of the Aksumite Empire (c.150–900 CE). The Aksumite Empire was a trading empire centred on the east coast, stretching across the Red Sea as well as into the modern-day Sudans, Djibouti, Eritrea, and Ethiopia. Its two key cities were Adulis and Axum on the Eritrean coast, both perfectly placed to control the bottleneck of the Red Sea where it meets the Indian Ocean. The Aksumite Empire grew in size and stature as it used its strategic location to dominate the trade network through the Red Sea, establishing a near-monopoly (and, in the process, creating an abundant source of lootable resources, much like a chokepoint of salmon running through a river). It grew rich through controlling the trade in luxury goods including gold, cattle, ivory, and incense. Its power was also built on good environmental luck: from around 100–350 CE there was a shift in the climate towards wetter conditions. The rainy season may have almost doubled in length from about three and a half months to six months.^{[24](#)}

The good fortune was not to last. The rains must have seemed like a blessing, but every drop had a cost. The dry soil became increasingly dislocated. Soil erosion was a particular problem since farms were often built on hills. Environmental degradation was causing an energy ratio

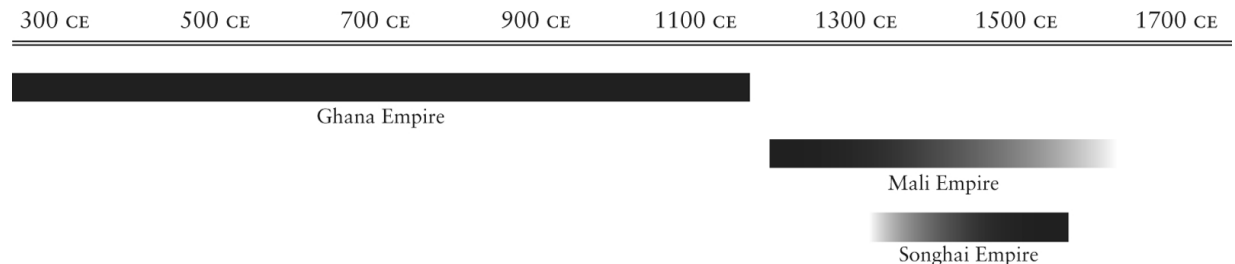
problem for the empire that would soon be exacerbated by another even bigger one.²⁵ Regional politics began to drain the Aksumite Empire of its lifeblood: trade. Wars between the Byzantine and Persian empires (540–561 and 602–632 CE) both slowly sapped the demand for luxury goods. Then things went from bad to worse. The Umayyad Caliphate expanded south and destroyed both Adulis and the Aksumite fleet (702–15 CE). With the loss of its trade networks the Aksumite needed to farm even more intensely, leading to increasingly disastrous soil erosion after 650 CE. The energy ratio of the empire plummeted owing to a combination of soil erosion and the breakdown of trade. The empire stopped minting its own coins and the capital was abandoned within a few generations, becoming a loose cluster of villages. In the tenth century the remnants of the empire were conquered by Queen Gudit, a marauding Jewish-Ethiopian monarch. The language of Ge'ez fell alongside it. The future empires of Ethiopia emerged further south, away from the now barren soils of the land where the Aksumite had once dwelled.²⁶

The Aksumite Empire was not alone in its fate. Africa was the birthplace of numerous indigenous empires, many of which declined or broke down even prior to European invasion.

Five and a half thousand kilometres away on the other side of the continent a different empire rose: the Ghana Empire (around 300–1203 CE). Both Ghana and the Aksumite Empire have a similar origin story: they were both creations of the Sahara Desert. From around 12,500–3000 BCE the Sahara was not an endless expanse of rolling dunes. It was full of lush forests, woodlands, and grassland savannah, home to antelopes, giraffes, gazelles, hyenas, hippos, rhinos, and zebras. We even have cave art in the Sahara depicting cattle and boats. This has been called the African Humid Period, a climatic age that birthed a 'Green Sahara'. When the climate changed and the area began to desiccate into the sandy waves we know today, the inhabitants were largely pushed to its fringes; the eastern edge was where the Aksumite and Egyptian empires emerged while the western frontier was where the Ghana Empire would eventually take root ([Figure 28](#)). (A third exodus was to the north-east, where Egypt emerged.)

The Ghana Empire rose thanks to the introduction of loatable resources and monopolizable weapons. Camels were introduced in the third century, allowing for more effective trade networks across the Sahara, especially the

export of gold, salt, and slaves. The widespread production of iron weapons allowed for an army that was up to 200,000 strong. And there was ongoing conflict with Berber raiders (a group of indigenous pastoralists in north-western Africa).



[Figure 28](#): A timeline of key west African empires

Like Sargon’s Akkadian Empire, Ghana’s was built on one group dominating many others, in this case several other city-states and kingdoms such as the Sosso, Jolof, and Mali. While the exact reasons are unclear, the Ghana Empire eventually fragmented, giving way to the Mali Empire (c.1230–1650 CE).^{[27](#)}

The Mali Empire proved to be even larger and more powerful than its predecessor, growing to cover over 1 million square kilometres. The empire reached its zenith during the reign of Mansa Musa. Mansa Musa was renowned for his great wealth (he may have been the richest individual ever to have lived), his face even appearing on a medieval map.

After Musa’s reign the Mali Empire began to slowly deteriorate. Musa’s successor was deposed by his uncle, Sulayman, after four years. After Sulayman’s lengthy reign of nineteen years, things got worse. There was a series of succession crises and short-lived leaders, with the reigns of the next seven kings lasting four, nineteen, less than one, fourteen, eleven, two, and one year respectively. In the late fourteenth century two leaders were assassinated in just three years. With the empire riven by infighting, Berber raiders to the north began their plunder, cutting off its economic umbilical cord: trans-Saharan trade. Other cities began to break away, including Gao. The city of Gao would slowly take over the empire from Mali, re-establish the cross-desert trade route and entrench in place the Songhai Empire (c. 1376–1591).^{[28](#)}

The Songhai suffered much the same fate as Mali. They grew even larger than Mali but were plagued with succession disputes and civil wars. The

sheer amount of elite infighting meant that the hold of the empire on its subjects was minimal. It mainly claimed tribute and did little else to contribute to the kingdoms it ruled.²⁹ Like the Aksumite and Ghana empires, it also suffered from a decline in trade. In this case it was due to Portuguese ships using the west African coast as a new supply line for gold from Africa to Europe, diverting goods away from the trans-Saharan trade.³⁰

Eventually, Morocco, a new regional power armed with Portuguese guns and cannons, defeated the Songhai. The raiding Berbers redoubled their efforts; the empire fractured once again. The feuding kingdoms and cities began to raid each other for slaves as the Songhai Empire fell into decay. It was finally conquered by European colonizers. Its capital, Gao, was slowly abandoned and had become just a cluster of a few hundred huts when the German explorer Heinrich Barth visited it in 1854.³¹

The rise and fall of west African kingdoms in many ways resembles the problems that Sargon encountered: the difficulty of one kingdom trying to control many. It was a constant vulnerability, an original wound that threatened to burst open whenever there was a large enough shock such as a loss of trade or a succession crisis. It also looks like the Maya, with a cycle of one city or kingdom growing strong before being eclipsed by another. It was a cycle of capture and then fracture.

Yet, there was also an additional problem. The kings of west Africa practised polygyny, taking multiple wives. Under medieval laws kings could take up to four wives, a rule they seemingly made ample use of. One king, Askiya Muhammad, had somewhere between thirty and five hundred children (depending on which source you trust). This was compounded by a second problem: the rules for succession were often unclear, leading to frequent disputes over the throne. These plagued the west African kingdoms, often leaving the regimes weak and distracted.

The Goliaths of the Yucatán and west Africa may have been separated by over 9,000 kilometres and the Atlantic Ocean, yet both learned that interdependence and status competition was deadly.

SYSTEMS COLLAPSES ACROSS THE WORLD

Groups of closely located cities and empires tended to become connected into larger systems. Some, like the Lowland Maya, were highly decentralized and no city or kingdom could ever develop a lasting large-scale empire. Others, like the Early and Late Bronze Age systems, were still more decentralized but had a few larger empires knitted together into one extractive quilt. West Africa had a sequence of three lasting empires (Ghana, Mali, and Songhai), but each struggled to keep hold of its constituent kingdoms. Like the Lowland Maya it was an imperial cycle of one power capturing the others before fracturing and the cycle restarting (much like the fission-fusion processes of early Goliaths). The Aksumite Empire was unified but dependent on the larger system of trade into Egypt and Eurasia. Each represents a different step on the spectrum from more or less integrated Goliath systems. And each had similarities in the way they collapsed.

System collapses across the world varied across time and space. The Hittite Empire collapsed during the Late Bronze Age, but the Phoenicians expanded; most cities in the Mayan Lowlands were abandoned, but others in the north grew. The end of the Mali Empire was a boon for the city of Gao, just as the end of the Ghana Empire had been for Mali.

We see the same variation in the modern world. Talk of the ‘failed state’ of Somalia hides the fact that Somalia broke down into four different regions with four different arrangements and outcomes: there is Ogaden (ruled by an authoritarian single party with little power to tax); Somaliland (an independent republic built on multiparty clan democracy); Puntland (an autonomous region full of warlords); and the southern part of Somalia (an internationally recognized and supported government that has essentially no ability to tax or deliver services and is riven by conflict). Collapse is not a uniform phenomenon.^{[32](#)}

There were also differences in how quickly individual cities and empires collapsed, and how long it took for the overarching system to fall apart. The former could happen in months, while the latter took a century or more. When a change takes centuries, it is almost imperceptible to the people living through it. Individuals rarely recognize that they are living through a collapse. Instead, they sensibly perceive the discrete disasters unfolding in

front of them. They feel the pang of hunger when a grain shipment doesn't arrive. They feel the fear coursing through a city under siege. They smell the smoke and feel the broken rubble under foot as they inspect a house burned in a rebellion. The historian sees the collapse across generations, but those generations experience it very differently. Many would have felt it more as an unfortunate series of wars, crises, and famines, or perhaps not at all.

Systems collapses in Africa, the Americas, and the Mediterranean can be seen as another kind of failed experiment. Large-scale empires in the Mediterranean were a relatively recent invention that had been cobbled together from previous city-states. Think of it as a reversal of Sargon's conquests: an empire was decomposed into its building blocks of independent cities and territories. What followed was often a reversion to more egalitarian city-states, such as those of Classical Greece and the Phoenicians. These proved to be more innovative and resilient in the face of climatic change. Among the Maya, shifts towards more democratic, seafaring polities in the north, and more rural living, can also be seen as a reasonable response to drought and rampant oppression in the Lowlands.

It was also an example of failed interdependence. The Aksumite Empire and other African kingdoms were partly founded on favourable trade monopolies. Both the Late Bronze Age and the Mayan Lowlands were exceptionally interdependent. Those links, especially in the Late Bronze Age, allowed for the intense exchange of letters, goods, and people, including for diplomacy and alliances through intermarriage. These elite networks helped foster thriving empires and city-states, but eventually became a liability. It meant that shocks – whether in the form of disease, aggressive neighbours, or the loss of trading partners – were more easily spread during a period of tumult. Breaking those elite links and dispersing, including to new areas whether it be commoners migrating to northern Mayan cities or the Philistines arriving into Palestine, may have once again been a sensible adjustment.³³

We can see ancient roots behind these systems collapses. Early leaders would use long-distance trade to acquire scarce, distant goods (primarily for status). Similarly, even among hierarchical hunter-gatherers, feasts, alliances, and marriages were all used as tools to bolster power. This had been intensified and scaled up into an entire network of trade in the Bronze Age, as well as among the Maya and the west African kingdoms. Inequality,

status competition, numerous shocks, and the very structure of elite networks proved to be the downfall of these systems. The capture and fracture of the Maya or west African kingdoms is not entirely different from the fission and fusion of early Goliaths, but these Goliath systems had managed to grow far larger and more unequal owing to better systems of control. That growth in scale and interdependence ended up being another vulnerability. Collapse could now occur over a much larger area.

In eastern Asia, collapse looked different. Rather than there being a norm of multiple competing cities and states, China was usually covered by one unified empire. While the Mycenaeans and the Maya lost their scripts, China never did. Yet this empire of eastern Asia was not always unified or stable: it would fall apart again and again.

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The Rise and Fall of the Chinese Goliath

Bronze, Bones, and Betrayal: The Beginnings of Chinese Empire – Cycling in China – Elite Competition, Cooperation, and Corruption – Elite Flight and Eating the Rich – Why Collapse Never Seems to Last

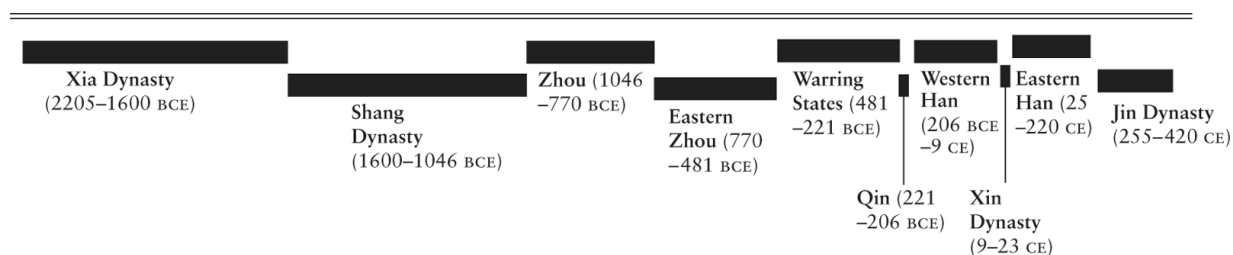
BRONZE, BONES, AND BETRAYAL: THE BEGINNINGS OF CHINESE EMPIRE

China has had over 2,000 years of near-continuous imperial reign. Yet this masks more tentative beginnings.

Sometime between 2200 and 1850 BCE, the Xia Dynasty was established in the north of China, near the Yellow River. It appears at the same time as bronze weapons (another case of monopolizable weapons at work). We know little about the end of the Xia aside from myth. Legend has it that the ruling family grew tyrannical and were eventually overthrown by new rulers: the Shang Dynasty ([Figure 29](#)).

The Shang ruled from 1600 to 1046 BCE, their reign marked by mass sacrifices and Asia's first form of writing. The king, royal family, and special diviners would heat bones (usually from an ox shoulder or tortoise shell) and interpret the heat-induced cracks to predict the future. Despite the purported prophetic power of the 'oracle bones', the Shang did not foresee their own end. The dynasty fell in the same way it had risen, being overthrown by the Zhou.

The Zhou were the first to introduce the Mandate of Heaven: the idea that virtuous rulers were divinely sanctioned by the heavens to oversee the populace. Conversely, a ruler who reigned over a time of drought, flood, famine, and conflict was seen to have lost favour with the heavens. They were no longer fit to rule and their ousting would be entirely legitimate. The Zhou clearly used this to justify their overthrow of the Shang. One of China's oldest, most central ideas is a relic of rebel propaganda. It is also an ideological time bomb: when things go wrong any regime can be accused of having lost the Mandate of Heaven.



[Figure 29](#): A timeline of early imperial China (Xia to Jin)

After around three centuries the Zhou began to decline. We don't know exactly why but hymns from the time hint at factional infighting, and feudal lords appear to have become increasingly autonomous over time.¹ By 481 BCE, these feudal lords had fallen into a state of perpetual war. China's first collapse, known as the Warring States period, had begun.²

It was one of the bloodiest periods of Chinese history. The eventual victors, the Qin Dynasty, slaughtered an estimated 1.5 million soldiers and civilians in their quest to unify China. That included enormous mass murders, such as in 293 BCE when the Qin killed around 240,000 troops. After occupying a territory, the Qin would systematically execute any adult males they found. It was a terror tactic used to minimize the chance of future rebellions.³ One of the Qin's most prolific generals, Bo Qi, would later be branded by some as the 'super butcher'. That butchery laid the gruesome foundations for a continuous period of empire over China.⁴

The Warring States period was also a period of new thinking. Dozens of new ideas and philosophies arose during this time, including two that we associate with China today: legalism and Confucianism. These were among

many doctrines that flourished during this period, an era that would later become known as the time of a ‘Hundred Schools of Thought’.⁵

The Qin (221–206 BCE) brought that time of philosophical innovation to an end. They systematically tried to purge China of these other philosophies, burning countless books. Their aim was to leave no alternative to their preferred philosophy of legalism. This was just one part of a strategy to centralize Chinese society.

The Qin created the first unified large-scale empire across northern China and ushered in a new era. The first emperor, Qin Shihuang, enshrined himself not as a king or an oracle, but as a truly divine being: a

manifestation of heaven on earth.⁶ This new godly leader imposed a simplified script, enforced standardized weight, length, and volume measurements, standardized the currency, and ordered the construction of a 6,840-kilometre imperial highway system (mainly for shuttling troops). He moved elite families to the capital so they could be closely monitored. The empire also began construction of a vast wall of packed earth to the north, to help keep invaders out, taxpayers in, and to better control trade. This was the precursor to the Great Wall of China.⁷

Creating such a vast, new bureaucratic and religious system of authority would usually have taken centuries; the Qin accomplished it within two decades. However, that accomplishment came at a cost. Elites and commoners alike rebelled, labour gangs mutinied, and ‘even many of the Qin state’s own officers and people turned against their rulers’.⁸ Once again, counter-dominance, and conflict between elites, had torn apart a fledgling new Goliath. China’s first large, unified empire had been hoisted up and then burned down in less than fifteen years.

While the Qin’s rule was brief, their project of a divine empire was enduring. Four years later, after a brief civil war, the Han Dynasty was established. The Han is usually portrayed as a golden age for China. They oversaw the invention of paper and the sundial. People in modern China still refer to themselves as ‘Han Chinese’ and to written Chinese as ‘Han characters’. These achievements were built on crushing inequality, which grew over time.

The empire supported a small number of exorbitantly wealthy families of officials, landlords, and merchants. Rich merchants bought land and joined the ranks of government despite bans against them doing so, often amassing

riches that were as great as those held by the highest-ranked imperial officials.⁹ Twenty sub-kings of the Han had fiefs containing 1.35 million households in 2 CE. Just over a century later, in 140 CE, this had swollen to 2 million (a growth from 11 to 20 per cent of the population). During that time the registered households under the Eastern Han fell from 12 million to 10 million, despite the empire expanding. That is because these sub-kings were dodging taxes by hiding their peasants away from government tax collectors. At least a few million peasants were being taxed by their private overlords rather than by the government. The growing elite persuaded the government to abolish conscription and began to grow their own private armies.¹⁰ This was the beginning of the end for the Han. Two million rebelled against a combination of high taxes, corruption, disasters, and disease in a revolution led by a faith ruler (the Yellow Turban Rebellion: 184–205 CE). Eventually, a warlord captured the throne (establishing the Jin Dynasty) before corruption and bickering between elites led to its end.¹¹

Unlike in systems collapses across the world, imperial China always regenerated, and periods of fracture were ultimately brief. China is a story of both how a Goliath can persist despite collapse, as well as of the role of corruption and elite competition in collapse.

CYCLING IN CHINA

The fall of the Han and Jin set in motion a series of imperial cycles in China. After a civil war China saw five periods of unified empire: the Tang (618–907 CE), the Song (960–1276 CE), the Yuan (a foreign occupation by raiders from the Steppe that lasted from 1276 to 1368 CE), the Ming (1368–1644 CE), and the Qing (1644–1912 CE).¹² The interruption during the first millennium from 907 to 960 CE was due to a series of battles among kingdoms called the Five Dynasties and Ten Kingdoms Period; the second millennium was a time of stable empire. There were still rebellions and disasters that could cost millions of lives, and these often occurred at the end of a dynasty, but the imperial architecture of China never truly lapsed. Historians of China may talk about the collapse of a particular dynasty or regime, but rarely of a societal collapse.¹³ This is because while a regime

might be overthrown and thousands might die, the bureaucracy, writing, and overall system of government were never lost.

While imperial rule was stable, being a ruler was a perilous affair. From the first empire of the Qin Dynasty (221–206 BCE) to the Qing Dynasty (1644–1912 CE) China had almost 300 emperors, each reigning on average between six and seven years. That average hides great variation: Emperor Mo of Jin (1234) was on the throne for less than a day, while Emperor Kangxi, the third head of the Qing Dynasty, lasted sixty-two years. Sixty-two was a ripe old age for that time, let alone for an imperial reign. Perhaps the most peculiar trend is that the longest-lasting emperors tended to rule the weakest empires and vice versa. For instance, the Tang Dynasty is renowned as a high point for China, yet the late Tang emperors had a one in two chance of being killed in office. Conversely, Kangxi's extended reign occurred during the Qing Dynasty, an exceptionally weak empire that taxed less than 1 per cent of its economy. The early Song a millennium earlier had taxed around 15 per cent of economic output.¹⁴

Why did weak Chinese empires have long-lived rulers? Yuhua Wang, a professor of government at Harvard University, has used a complex dataset of 282 emperors and 50,000 genealogies to craft a simple answer: elite competition. China faced 'the sovereign's dilemma': a strong, coherent elite that can bolster the state by providing more taxes and resources can also overthrow it. And it did, regularly. The Tang Dynasty shared power more equally between the emperor and other leading aristocratic families and officials. It was a golden age for trade across the silk roads into Europe and the Near East, and a time when China heavily influenced the culture of Japan. Yet five out of a dozen emperors during the late Tang period were ousted through coups.

China was also under consistent threat from horseback warriors from the eastern Steppe (modern Mongolia), including the Xianbei and Xiongnu, and needed a unified elite to defend itself.¹⁵ In one telling example, Emperor Shenzong (1067–1085 CE) of the Northern Song kept his elites so divided to avoid a coup that it weakened the military until the Steppe raiders swept in, conquered the Song, and established the Jin Dynasty. A unified elite meant that the imperial government tended to be eroded by corruption, tax evasion, and coups, while a divided elite meant that the state often lacked the resources and unity to defend against invaders.¹⁶

Status competition between aristocrats and government officials was accompanied by regular popular uprisings, whether it be the Yellow Turban Rebellion (around 184 CE), the Sun En Rebellion (399–403 CE), the Red Turban Rebellions (1351–68 CE), the Nian Rebellion (1851–68 CE), the Taiping Rebellion (1850–64 CE), or the 1899–1901 Boxer Rebellion. Climate change, when it met elite conflict and rebellion, could trigger collapse. The Qing Dynasty (1644–1912 CE) is one instructive case. Droughts and famines triggered rebellions throughout the Qing Dynasty, including the Nian Rebellion and the Taiping Rebellion. The Taiping Rebellion was the bloodiest civil war in human history, potentially carrying off 20 million human souls. It was also an example of elite infighting, having been instigated by budding young bureaucrats who passed their exams but couldn't achieve prestigious positions in the Chinese government. Droughts and floods in the Yellow River region triggered the Boxer Rebellion, which led to the loss of 100,000 lives and temporarily split the Qing government. All these uprisings took a toll on the Qing, until this last (formally) imperial dynasty of China came to an end. Fittingly it was through another rebellion, in 1911–12.^{[17](#)}

The Qing were not alone in being plagued by droughts and floods catalysing slaughter. One study comparing a 1,062-year reconstruction of Chinese warfare against documented periods of cooling caused by volcanic eruptions found a strong correlation between the two. Another drew on millennia of climate data and compared it to a catalogue of 789 rebellions and 2,214 external wars, again finding a significant correlation. While correlation isn't causation, the pattern is difficult to ignore. There are also some compelling explanations. The Mandate of Heaven meant that the moral authority of the emperor could be called into question when natural disasters struck, the weather turned, and harvests shrank. With the Mandate of Heaven lost, rebellion and elite infighting often followed.^{[18](#)}

As with the first farmers of Europe or the Maya, climate alone cannot explain the cycles of dynastic collapse that China experienced. There are numerous cases of crisis where climate played no role. For instance, the Tang Dynasty collapsed without any major environmental stress. Throughout Chinese history there were also many droughts and floods that were handled with little issue. The severity of the drought or flood mattered, and the vulnerability of the dynasty mattered even more. Environmental

shocks triggered a collapse usually only when accompanied by inequality and elite competition. Corruption also seems to have played a recurring starring role.^{[19](#)}

ELITE COMPETITION, COOPERATION, AND CORRUPTION

Dynastic cycles in China demonstrate how elite infighting tears societies apart. The fate of different dynasties depended not just on the intensity of elite infighting, but also on the very structure of elite networks.^{[20](#)}

The dynamic is simple: as elite numbers grow, so too do status-seeking aspirants. The powerful are rarely happy to have just one form of power. As we see throughout Chinese history, wealthy merchants try to buy their way into the political system (as with the Han); learned bureaucrats and princes use their positions to amass fortunes; generals regularly use their swords and axes to take a throne; and religious leaders aren't content to sit in God's good grace, but also need to sit in a palace (the Taiping Rebellion was led by Hong Xiuquan, a disgruntled school teacher who claimed to be the younger brother of Jesus Christ). A more unequal society with a larger number of elites tended to mean more infighting and more potential demagogues looking to capitalize on peasant uprisings. Power politics and status competition are ever-present but tend to escalate as the numbers of those with both the means and desire to seek thrones climb, and as they begin to outpace the number of high-status positions that are available.^{[21](#)}

Such elite competition can be seen in cases of crisis and collapse well beyond China. Lenin was a lawyer, Fidel Castro was a lawyer and a son of wealthy sugarcane planters, and the men who warred over control of Rome were sons of senators, magistrates, and generals from elite families. Increased elite competition tends to happen alongside rising popular immiseration: as inequality rises elite competition intensifies and the rest of society is left poorer and sicker (creating ample opportunity for warring elites to foment rebellion). Researchers, led by Seshat database curators Peter Turchin and Dan Hoyer, have shown that increasing inequality, popular immiseration, and elite competition appear to have played a key

role in a range of collapses throughout history, from the Western Roman Empire in the fifth century CE to the Qing Dynasty in the twentieth (this is commonly referred to as Structural-Demographic Theory).²² It may have even played a role in the collapse of many of the earliest Goliaths, although more evidence is needed.

The driver behind this common story of feuding nobles is, of course, status. We even see it today. Donald Trump's quest to become president intensified after being humiliated by Barack Obama at the White House Correspondents' Dinner in 2011. Just five years later he won the US presidential election. The right-wing firebrand Tucker Carlson refashioned himself as an anti-elite populist after being publicly savaged by comedian Jon Stewart and subsequently fired from CNN. The public loss of status is a great motivator. While both use anti-elite rhetoric, they clearly are elites. They were born into wealth: Trump was a millionaire by age eight thanks to his father, while Carlson's step-mother was an heiress to the Swanson frozen food giant. Both are famous figures and have no material need to work tirelessly to amass even more influence. Yet they do. Status is inexhaustible, especially when you are surrounded by other ambitious competitors.²³

There are two caveats to be aware of when it comes to elite infighting. First, this tends to be a competition between the powerful, seen as much in the billionaires of today as in the princes, wealthy families, and generals of Chinese history. We see that in the largest collapses throughout Chinese history, whether it be the leading general Zhu Wen overthrowing the Tang or the Zhou being carved up by feudal lords. The most common pattern is that it is elites who struggle to transform their wealth into another form of power (usually the power form which is most valued in that society). In China, information power was particularly prized, and hence a surprising number of rebels were individuals who failed to pass their civil service examination, such as Hong Xiuquan, the instigator of the Taiping Rebellion. We see something similar today: it's billionaires such as Donald Trump and Elon Musk trying to weaponize popular discontent and transform their economic power into political power by capturing the US government.²⁴

Second, it's not just a matter of elite competition. It is also about cooperation. Remember the *razones* gangs of Colombia: they would have shoot-outs in the streets over territory but came together and forged pacts

once they jointly faced a larger threat from the police. Similarly, Chinese princes would regularly work together to oust an unwanted emperor such as Wang Mang before turning back to competing with each other. We see the same pattern today with big tech and fossil-fuel companies commercially in competition while jointly lobbying in politics to block legislation that would regulate them. A common part of racketeering is not just that powerful actors will compete to extract resources from those below, but also that they will regularly band together if it's in their own interests, whether that be to quell an uprising or combat an external foe. This is a kind of shifting *elite factionalism*.

Alongside this competition and cooperation also tends to come corruption. Corruption (the misuse of positions of authority for private gain) has been rife throughout history – from Bronze Age traders and Chinese princes dodging taxes to politicians lowering taxes and regulations to benefit their rich donors (and potential future employers). While beneficial for the individual or industry, such gain comes at the cost of trust and shrinking revenues for the state. In the worst cases, it could lead directly to a crisis. The Yellow Turban Rebellion brought about the deaths of hundreds of thousands and left the Han Dynasty in disarray. It was a rebellion fed by the belief that the ruling dynasty had become too corrupt, losing its Mandate of Heaven and causing floods and plagues. It is a repeated grievance in Chinese history: a regime had lost the Mandate of Heaven owing to fraud and mismanagement. (We also see it in the Old and New Egyptian Kingdoms, along with multiple regimes in Mesopotamia.) Across these cases a common, fundamental cause of societal breakdown was institutional failure, especially through incompetence and corruption.

Corruption robs the state of its revenue and legitimacy.^{[25](#)}

Corruption tends to increase over time, alongside wealth inequality. Today, we have ample evidence that inequality and corruption accelerate one another. The greater the discrepancy in power, the more opportunity there is to abuse it. Likewise, corruption can help the rich get richer. In the pre-modern world, with less-advanced bureaucracies and weaker accountability mechanisms, there would have been even more opportunities to exploit the system.^{[26](#)}

China is revealing not just of the role of elite factionalism and corruption, but also of how collapse can affect inequality and elites.

ELITE FLIGHT AND EATING THE RICH

In China, rebellions and coups often meant not only a new regime or breakdown of government but also the intentional culling of the old. We see this during the end of the Tang Dynasty in China. This period witnessed a series of atrocities that ‘effectively wiped out the metropolitan elite’. One of the best-known occurred in 900 CE, when the court eunuchs systematically murdered elites and government officials who were close to the emperor. It was a pre-emptive strike against those who were plotting to eliminate them. It didn’t work, and they were promptly killed in response. A few years later seven of the most powerful remaining ministers were thrown into the Yellow River.^{[27](#)}

In older case studies, such as Tiwanaku and Taosi, we see some evidence of targeting the upper classes, mainly through the defacement of elite monuments and residences. Even without focused assault, elites could often lose more. Their wealth and power were dependent on the status quo; and as the world fractured, much of their fortunes could slip away into the cracks of the social order. Their wealth was either burned, plundered, or meant little in the new, post-collapse world.

Being rich can be a liability in a crisis. It’s a trend not restricted to China or the Americas. In the Crisis Database around a third of the cases involve assassinations. That includes seven in China alone (including the Qin and Han) as well as a host of others, such as the end of the Mali Empire in Western Africa and the fall of the Western Roman Empire. In about 10 per cent of the cases there was an attempt to systematically target and kill off elites. China alone has five of these elite ‘extermination’ events, including the ends of the Tang and Qing, as do many others we’ve encountered, including the Neo-Assyrian Empire that made it through the Late Bronze Age collapse.^{[28](#)}

Being an elite in the time of a crisis is risky. It can be incredibly dangerous, yet when elites are not directly targeted during a collapse, and the collapse isn’t too deep, the rich can often escape the worst effects of collapse and even benefit from them. Bureaucrats and wealthy merchants in China often just changed their loyalties when a new regime came into power, which is a key reason why many of the basic structures of Chinese bureaucracy and writing showed such continuity across dozens of centuries

and regimes.²⁹ Middling elites of the Western Roman Empire that were less affected by the collapse often continued to hold a great degree of property and power. The same happened after the fall of their competitors in the Persian Sasanian Empire. Many Byzantine (the Eastern Roman Empire) elites simply adopted Muslim names and emigrated with much of their wealth after the Ottoman conquest of Constantinople (1453 CE). Powerful individuals usually had more means to escape a crisis and were of great use to lazy conquerors, who could use their cultural knowledge and connections to easily administer and tax their newly acquired territory. Elites frequently escaped the worst effects of collapse by moving or by blending in with the new ruling class. This is often referred to as ‘elite migration’: the powerful and wealthy either changing location or simply shifting their allegiances to stay in high-status positions despite a crisis or collapse.³⁰

Whether elites successfully escape collapse or are swallowed up in the tumult depends on a few factors. Rebellions and revolutions with an anti-elite fervour will usually result in landlords, masters, and royals losing their wealth and their lives via confiscation and executions. It’s a dynamic we see in crises that didn’t even reach the level of collapse, such as the 1793–4 ‘Reign of Terror’ in the wake of the French Revolution and the effects of the 1917 Bolshevik Revolution in Russia. Collapses that are significantly deep will also cost elites more. The ruling priests of Cahokia and the Huhugam probably had nowhere to run and their previous power meant little in the new world born from the collapse.

Yet elites, especially in China, almost always had another empire or kingdom to migrate to. That is because collapse rarely lasts for long. Despite the talk and fear of dark ages, Goliaths, especially in China, are persistent and are almost always reborn.

WHY COLLAPSE NEVER SEEMS TO LAST

We are often enamoured with the question of what causes collapse, yet we ignore the perhaps more interesting question: Why are Goliath societies almost always resurrected?

That question is most pertinent for the case of China. Ever since the Qin, China has tended to be unified by a large-scale empire. Historians don’t

speak of ‘dark ages’ in China since writing, bureaucracy, and empire were never lost.³¹

There are surprisingly few cases of a Goliath-building project failing in a region and not being emulated or reanimated within a century or two. In all five regions of the world where Goliaths first emerged – China, Egypt, Mesopotamia, Mesoamerica, and the Andes – empires would eventually take over. Collapses are rarely permanent or even long-lasting in these regions. Even the deepest collapses, which entirely wipe away the elite class, their culture, and writing – such as in Mycenaean Greece – still led, after a few centuries of more democratic rule, to more kings and militant emperors. The consistent trend is that wherever states reigned and fell, they returned and expanded. The Shang Dynasty was replaced by the Zhou in China, the Wari by the Inca in the Andes, and Egypt had pharaoh after pharaoh for almost three thousand years, interspersed with three periods of more decentralized rule. While early experiments with Goliaths were often abandoned, those that progressed far enough seemed to be surprisingly easy to reboot.

There are a few exceptions where Goliaths either never emerged, or fell and never rose again. People in both North America and Zomia dispersed into smaller groups, sowed less grain and did more foraging (or relied on escape crops), and told tales that warned of the perils of corrupt rulers. Yet, these are rare and peculiar cases: Zomia is an incredibly mountainous and inhospitable terrain; while Cahokia and the Huhugam were novel experiments in areas with low population density and ample exit options, which had entered a prolonged cold spell that depressed crop growth. This geography and the novelty of unequal kingdoms made it easier to avoid Goliath.

Goliath fuel – lootable resources, monopolizable weapons, and caged land – can explain why Goliaths persisted in places like China and the Near East but could be lost elsewhere. Wherever Goliath fuel remained, there was always a decent probability that someone could start and grow another centralized polity. Lootable resources tend not to disappear even if an individual state or Goliath does. There’s a reason why the Kwakwaka’wakw (north-west coast Native Americans) remained as hierarchical slave-owners for centuries: the salmon kept coming. Similarly, once a landscape is tilled into fixed fields and the people are dependent on farming, then there is an ongoing supply of lootable resources. A landscape with few exit options is

also not going to change spontaneously. Population growth will, if anything, reduce exit options as the land becomes more crowded. Similarly, if there were new weapons that allowed one group to dominate another, then they had to be entirely forgotten or intentionally destroyed to avoid being reused. Goliath fuel tends to be sticky, making Goliaths persistent.

The recurring trend is that the longer Goliaths had been around, the more likely it was that one would be reincarnated. A land that has been transformed into tilled fields of wheat or rice paddies cannot easily or quickly be made suitable for hunting and gathering. Nor can farmers quickly learn and adopt an entirely different lifestyle. After a few generations of intensive agriculture, the land is tilted towards a dominance hierarchy.

This is also true of weapons. The longer weapons circulate among a population, and the more people who know about how to build them, the harder it is to dislodge them from the collective memory. If most families have iron weapons and most blacksmiths know how to smelt iron, then it will be difficult to lose iron axes and swords completely from the arsenal of different groups.^{[32](#)}

One of the hardest things to permanently kill is not swords or crops but ideas. Once a legitimized hierarchy is deeply rooted in a culture, it becomes hard to remove. Entire systems for justifying inequality are carried forward for thousands of years. The Mandate of Heaven began with the Zhou around 3,000 years ago yet was still invoked during rebellions against the Qing in the nineteenth and twentieth centuries. Genetic evidence suggests that the caste system of India began around 1,500 years ago with little mixing and births across castes. Once stories of distinction and hierarchy become prevalent, they can easily be picked up again and reused by future leaders after a collapse. An appeal to old symbols and stories is often powerful.^{[33](#)}

Culture can also plant Goliath fuel deeply, making lootable resources seem indispensable. Sierra Leone lies on the Upper Guinea Coast, an area of West Africa known as the 'Grain Coast' or 'Rice Coast'. Declining soil fertility has made rice farming more and more difficult, forcing many farmers to take rice loans (loans of rice seeds to plant). These usually lead to the farmers getting trapped in debt owing to their inability to grow enough rice to pay back the loan. They continue to grow rice despite the

land being better suited for growing alternatives such as potatoes and cassava. Why? School fees are traditionally paid in rice. Many now simply identify as rice farmers and refuse to change or claim that they can't feel 'full' without rice. As one local exclaimed, the problem isn't the land but that 'Sierra Leoneans can only dream of rice!' And of course, grain is central to many modern cultures today whether it be throwing rice at weddings, reaping what you sow, or praying to God to 'give us our daily bread' (a line from the Lord's Prayer). Lootable resources tend to be deeply culturally ingrained.³⁴

Culture played a critical role in the persistence of empire in China. The Qin were successful in establishing a larger-scale bureaucracy as well as standardizing and spreading a single language and writing system across China. Widespread literacy, lengthy historical records, and a tradition of bureaucracy all made it easy to regenerate empire after it fell. Especially after the even more organized Tang Dynasty, rulers always had an easy template to emulate and a workforce to administrate it. Goliath was enshrined into the culture.³⁵

That stronger foundation of bureaucracy and writing was not just a random coincidence. Geography influenced it. China was closely placed to the Steppe and its horseback raiders. The constant threat of invasion helped hasten the construction of empires as farmers had their authoritarian impulse kick in and turned towards hierarchy. All unifications of China, except for one, started in the north or north-east of the country, an area particularly exposed to Steppe raiders. Some have referred to these as 'shadow' or 'mirror' empires: China organized itself more hierarchically in response to mounted warriors from the Steppe, who also adopted more organized hierarchies and formed their own empires (including over China during the Yuan Dynasty). China has few high mountain ranges east of Tibet. Most of the 'core' area of imperial China is relatively flat, lying mainly below 300m and with few areas above 1,500m. It's criss-crossed by rivers that encouraged large-scale farming, supported dense populations, and helped with the shuttling of goods and troops via ships. The geography allowed for an easier expansion of Goliaths with few refuges where people could practise the art of not being governed.³⁶

Modifying the landscape to grow loutable resources, creating stockpiles of monopolizable weapons, spreading ideologies that justify subjugation,

and using bureaucracy and writing to create templates for creating Goliaths made it easy to resurrect them.

Every Goliath leaves its bootprints on the land and people, and the Chinese Goliath left remarkably deep ones. There is perhaps only one other Goliath that rivals the legacy of the Chinese: the Roman Goliath.

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The Falls of Rome

*Born from War – The Three Falls of Rome – Imperial Overstretch –
Diminishing Returns on Extraction – Learning from the Oldest Empire –
Shrinking Houses and Fractured Trade – Why Crises Can Be Good for You
– Eat or Be Eaten*

BORN FROM WAR

In the second century CE, as the Han Dynasty reigned over eastern Asia, the Roman Empire had become the pinnacle of power in the west. It had aqueducts collectively spanning more than 800 kilometres, public baths supplied with plumbing and insulated with double-glazing, and imported goods from over 6,000 kilometres away. Rome, its epicentre, was the first city in human history to reach a population of a million. The influence of Rome stretched north to Germany, east to Mesopotamia, north-west to Britain, and south across parts of northern Africa, including Egypt. In 390 CE the empire covered 4.4 million square kilometres, an area greater than the European Union today.^{[1](#)}

Within a century the reach of the western half of the Empire was effectively nil. It had splintered into several competing kingdoms. And around another century later the city of Rome had dwindled to a mere 30,000 people. It would not reach a million citizens again for more than a thousand years. Europe has never had a lasting empire stretch across it since Rome. Napoleon came closest, establishing a French Empire that spanned

from Spain to Poland in 1812. But the French Empire lasted just ten years. The Western Roman Empire lasted over five hundred. Rome retains an almost mythical status among people of the West today.²

Despite its fabled reputation, Rome began as just one among hundreds of city-states dotting the Mediterranean. The legend is that it was founded in 753 BCE by the twin brothers Romulus and Remus. The two had a human mother and a god for a father: Mars, the god of war.³

It's unsurprising that the Romans imagined their founders to be sons of the god of war, since warriors were the creators of Rome. During the sixth and fifth centuries BCE Italy was a land of budding settlements and roving aristocratic raiders who left behind rich burials replete with weapons. Coalitions of mobile warriors began to slowly settle down in these early city-states. Ruling over a people and continuously taking their resources, rather than occasionally attacking them for plunder, was a more sustainable strategy.⁴

Rome slowly began to scale up and dominate the region from the sixth century BCE onwards. The neighbouring city-states that it did not conquer it controlled through bilateral treaties (150 of them). These 'allies' were forbidden from speaking to one another, lest they band together against Rome. Allies provided military levies while those who were conquered became citizens but had to pay tax to fund military operations. Rome adopted and adapted the culture of those it conquered. The Greek pantheon became the Roman pantheon, with Ares, the god of war, becoming Mars. Yet while the Greeks disliked Ares, Rome celebrated Mars, a god who was literally the legendary father of Rome.⁵

Rome mobilized enormous numbers of young men for battle. From the fourth to the first centuries BCE, 10–25 per cent of men above the age of seventeen were deployed for combat in the Roman Republic. At its peak Rome had an army of around 400,000–500,000. In contrast, most of the armies in Europe during the Middle Ages that followed Rome tended to have just 10,000–20,000 troops.

This enormous military juggernaut was the centrepiece of the Roman strategy of expanding quickly, governing lightly, and using conquest to subsidize even more military campaigns. Taxes were kept low for everyone, especially elites, and Rome interfered little with the governance of its subjects. It thoroughly integrated conquered elites, having them adopt Latin

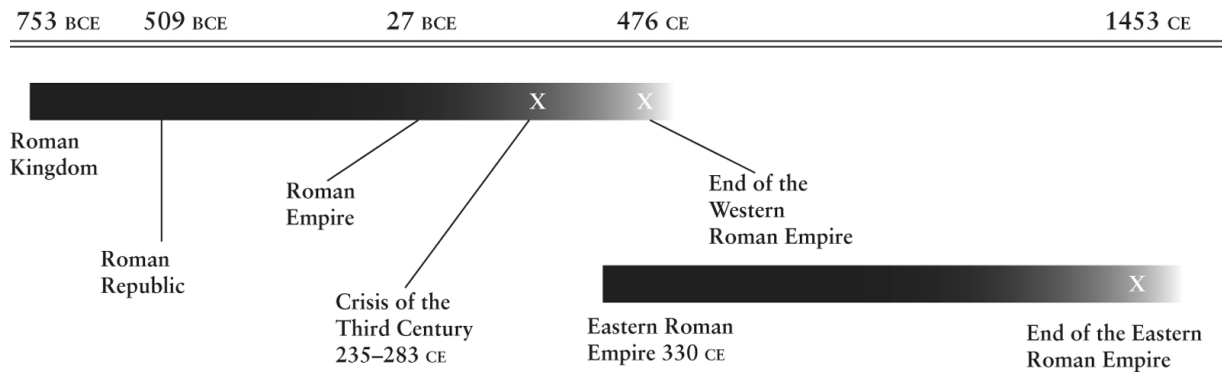
and Roman customs. It was essentially a pyramid scheme: conquering neighbours and then enlisting them to conquer lands even further abroad.⁶

Rome depended on the fruits of conquest. Gold, silver, slaves (used as both house and farm workers, as well as gladiators for entertainment), booty, and even exotic animals from newly conquered territories were paraded in the city of Rome, enriched its citizens, and placated their allies. These invasions could often be highly lucrative. The capture of the treasury of Macedonia in 167 BCE allowed the republic to eliminate taxes back in Rome. The conquered province of Egypt became the breadbasket of the empire, with Rome receiving over half of its subsidized grain (around 200,000 tonnes per annum) from harvests watered by the Nile. Most of the benefits of empire were channelled to the capital, which remained as its 'core'. By the end of the Republic, those living in Rome received free grain, a provision called the *cura annonae*. The Roman poet Juvenal suggested that the citizens were distracted by the superficial attraction of 'bread and circuses'.⁷

Rome thrived owing to the swords, shields, and spears of its legionaries.

THE THREE FALLS OF ROME

During this expansion Rome was a Republic (beginning in 509 BCE, [Figure 30](#)) with some semblance of democracy, including assemblies and voting held in the city of Rome and open to all citizens. This was a thin façade for oligarchy. A few dozen elite families dominated the higher offices of the Senate and the magistrates.⁸ As inequality and the power of the rich and the military grew, the Republic famously fell into empire with a coup by Julius Caesar. This was, in many ways, the first fall of Rome. Rome had always been historically exceptional as a large-scale republic. Now it was just another military dictatorship ruled by a monarch: the Roman Empire (although Rome had effectively been an empire for centuries, having conquered dozens of city-states).



[Figure 30](#): A timeline of Rome

Despite its success in war, Rome was constantly threatened. To the north were the Germanic tribes that it frequently warred with; to the east was its rival the Parthian Empire (and the last Sassanid Empire), while others such as Carthage recurred throughout Roman history. Nonetheless, the Roman Empire persisted and grew, bringing great wealth, especially to its highest classes.

Rome tends to be associated with its famous and infamous emperors, such as Julius Caesar, Augustus, Nero, and Marcus Aurelius. This overlooks the fact that Rome was as much about a land-owning aristocracy as it was about decorated commanders and mad kings. Two of the most important classes were the Senate (around 300–600 individuals who were elected from the patrician aristocracy and acted as public servants) and the governors who oversaw the different provinces. The bureaucracy too was staffed with leading families and local elites. The emperor could not survive without the support of these elites, as Julius Caesar discovered with every blade that was pushed into him. This elite class distinguished themselves from the common folk, whom they branded as *vulgus*, from which we now derive the word ‘vulgar’. Even the expansion of the empire was a matter of distinction: a display of the superiority of Rome over its neighbours.

The power of the elite over the empire meant that they could act with a great degree of autonomy. Since taxes were low and oversight was minimal, governors, senators, and the rich were prone to corruption, rule-breaking, and tax evasion. Despite the occasional exception, they were rarely held responsible for crimes and corruption, with one historian observing that ‘as a general rule a rapacious governor might never expect to be held to account’. This led to gratuitous exploitation of the populace. One governor of southern Spain boasted in private letters of extorting 4 million sesterces

(bronze Roman coins; to put that into perspective, to qualify for the Senate at the end of the Roman Republic you needed a fortune worth 1 million sesterces) from his subjects and then selling some into slavery. Local elites would also shield peasants under their control from taxes and conscription. This was a form of tax evasion, hiding peasants from the imperial tax collector and privately taxing them instead.

Such practices made Rome more economically and politically lopsided. The inequality of Rome had always caused problems. Three Jewish revolts took place between 66 and 135 CE, while uprisings in Gaul and Spain attracted participation from both peasants and the middle class. Slave revolts occurred periodically, including the famous doomed rebellion of ex-gladiator Spartacus. Approximately 6,000 of the surviving rebels were crucified.

The inequality worsened over time. By the peak of the empire one conservative estimate suggests that around a third of economic output was owned by just the top 1.5 per cent of households. Despite the striking villas, bathhouses, and aqueducts, only about a tenth of Rome's people beyond the wealthiest elite were living significantly above subsistence. Most of the population didn't have much beyond what was needed to survive and reproduce.⁹ At least early in the Empire, what was lost in low taxes and corruption was made up for through revenue from conquest which was also used to placate the masses.¹⁰

This arrangement was not sustainable. Rome's invasions further afield became more and more expensive. Ever-longer supply chains of grain and equipment to support armies and fortresses hundreds of kilometres away meant higher costs. At the same time, the most lucrative silver mines of the empire located in Spain began to dry up. Corruption, a costly military, less plunder and booty, and lower silver yields created a fiscal crisis. During times of crisis and war, the denarius – the silver coin of the Romans – was debased to increase the supply of precious metals. This became a common tactic for emperors, who increasingly depleted the amount of gold and silver in the currency. The amount of silver in the denarius fell from 3.5 grams per coin in 50 CE to approximately zero by 269 CE.¹¹

This devaluation of the currency meant that the empire struggled to pay its soldiers. A large, underpaid group of young, trained and armed men is usually a liability – as Rome soon discovered. The third century saw almost

fifty years of civil war from 235 to 284 CE, a period that became known as the 'Third Century Crisis'. Competing generals turned on the empire with the support of disgruntled Roman armies, whom they promised to pay in coin and plunder. Eventually, the crisis was brought under control by Diocletian (284–305 CE) who vanquished the other claimants to the throne. Yet, it came at a cost.

Diocletian introduced a set of reforms, including a significant expansion of the bureaucracy and a doubling of both the empire's military and land taxes (between 324 and 364 CE).¹² Hereditary occupations, conscription, and assessment of all lands for tax purposes were introduced. The reforms shifted power away from local elites towards the bureaucracy and transformed the empire. But it was only a temporary solution, and it never resolved the underlying problem of the wealth pump. Inequality and oligarchy continued to grow across the empire.¹³

The combination of crippling inequality, low returns from conquests, and over-taxation led to the abandonment of land by farmers and an inability to raise large families. This meant that the empire had to rely increasingly on Germanic mercenaries rather than the traditional approach of using conscripted farmer-soldiers. In 476 CE the Germanic troops enacted a coup on the last emperor, Romulus Augustus, when they could no longer be paid. It wasn't just the lack of pay. Germanic migrants were chronically marginalized and mistreated. Military officials who were supposed to dispense grain to them hoarded it for themselves, leading to starvation among the migrants and some purportedly selling their children into slavery to pay for dog meat.

This was the second fall of Rome: the end of the Western Roman Empire. Its woes did not end there. Volcanic eruptions in 536 CE cooled and dried out much of the Mediterranean, while the Plague of Justinian took even more lives. While the effect of this 'one-two punch' of climate and disease is contested, it undoubtedly acted as an additional stressor to an already fragile situation. The Byzantines sought to reclaim Rome and reunite the western and eastern halves of the empire. They succeeded, but not for long, and the capital was lost once again. Their military escapades left the city of Rome and much of Italy depopulated and devastated.¹⁴

Politically the fall was swift. At the beginning of 395 CE, the Western Roman Empire was ruled by a single emperor. Within seventy years,

Roman Britain had been abandoned, the emperor had been deposed, and the city of Rome had been sacked twice. During the fifth century many urban centres were abandoned and much of the empire was depopulated, by almost half in some regions. Other changes were far slower and subtler. Latin slowly over centuries became a secondary language, restricted to elites and monks, and literacy dropped. The eastern half of the empire continued on for centuries until it was defeated by the Ottomans in 1453, marking the third and final fall of Rome. For the Western Roman Empire, many elites simply merged with new Germanic rulers. The Germanic rulers themselves were Roman: they continued on with Roman customs and culture, combining them with their own. With the fall of Rome over the fifth century CE, the Christian Church grew in power, becoming the connective tissue across the new fractured political body of Europe.¹⁵

IMPERIAL OVERSTRETCH

There is ... one nation that at its own cost, through its own exertions, at its own risk has gone to war on behalf of the liberty of others. It renders this service not to those across its frontiers, or to the peoples of neighbouring States or to those who dwell on the same mainland, but it crosses the seas in order that nowhere in the wide world may injustice and tyranny exist, but that right and equity and law may be everywhere supreme.

This statement was penned by the historian Livy, referring to the Roman Republic. Rome never saw itself as the imperial aggressor. Its wars were either (sometimes pre-emptive) self-defence, or in defence of its allies. In the view of some of its members, it created an empire almost accidentally by pursuing noble goals.

This is difficult to believe, given the scale of Rome's growth. In 500 BCE Rome was just one city-state with around 60,000 people and 1,000 square kilometres in territory. It would eventually increase 5,000-fold in size, reaching a peak of about 5 million square kilometres. That is the rough equivalent of the people of a medium-sized American city conquering the entire continental USA. Rome and many of the post Qin-dynasties of China seemed to push the boundaries of size for Iron Age empires. Others, such as

the Mongol empires from the Steppe and different Islamic caliphates, would sometimes surpass this size, but not for long. The Mongol Empire led by Chinggis Khan reached a peak area of 24 million square kilometres but had fractured into four competing khanates within two decades or so.¹⁶

All empires, whether it be Rome, the Qin, or the Mongols, followed expansionist policies. They often depicted their reign not just as legitimate but also as a divine mission to rule all humanity. The British and other colonizing powers believed that they were bringing civilization to savages. Islamic rulers were required to expand their own moral community of believers (the *umma*). *Jihad* translates to ‘struggle’: the struggle to expand the *umma*. The Islamic caliphate was meant to be (eventually) universal. Similarly, Chinggis Khan and the empires that directly followed from the Mongol Empire were all supposed to conquer the entire world.¹⁷

Conquest didn’t just provide growth, status, and legitimacy; it could also offer many practical benefits. It could neutralize hostile neighbours, provide a new source of revenue, offer new lands that could be gifted to competing elites, gift fresh plunder and slaves to placate the masses at home, and allow for low taxes.

As we’ve seen with Rome, the problem is that universal conquest has never been achieved and the benefits of conquest tend to dry up over time. New territories require maintenance. Invasions further away from home cost more and more. New borders and interventions create new enemies. The Germanic tribes on Rome’s borders began to organize themselves more hierarchically in response to Roman expansion. Eventually the costs of maintaining, let alone growing, an empire begin to outstrip the benefits. This is the idea of ‘imperial overstretch’.

Modern empires, such as the Spanish, Ottoman, and British, all overextended the capacity of their militaries. Their vast navies stretched across numerous oceans required more and more resources to run, slowly draining their domestic budgets. Their debts grew and they struggled to procure more money for further wars and military spending, let alone to finance the rest of the economy. Eventually a new competitor less encumbered by an overstretched military rose up and took their spot, much as the US did with the British.¹⁸

While Cahokia and Tiwanaku had to deal more with a lack of mechanisms for controlling a populace, this wasn’t as large a problem for

later Goliaths. Rome introduced the first census (counting of citizens) in Europe, while the Han were the first to do so in China. At its height, Rome had a civil service of around 34,000 people. To put that into perspective, the current European Commission employs only 32,000. Large administrations with better tools acted as watchdogs over elites and unwieldy groups (like the rebelling city-states conquered by Sargon) that were far away from the capital. Rome and the Han were far better equipped than Cahokia or Tiwanaku but had to deal with new problems like corruption and imperial overexpansion. As Goliaths grew bigger and more organized, the Curse transformed accordingly.^{[19](#)}

DIMINISHING RETURNS ON EXTRACTION

We see a pattern re-emerging across case studies. Societies grow more fragile over time and more prone to collapse. Threats that they had always faced such as invaders, disease, and drought seem to take a heavier toll. In the first iteration of Goliath's Curse – the Primal Curse – the primary issue was incompetence and resisted hierarchy. An aversion to dominance, the negative effects of inequality, and different threats collided with states that had little capacity to control their citizens. Because there were so few other Goliaths, collapse could sometimes mean permanent abandonment, as was the case with Cahokia.

Over millennia societies grew bigger – whether it be into systems of interconnected states (such as the Maya and Late Bronze Age) or enormous overarching empires (like Rome or the Han Dynasty) – and more organized. They used administration, innovation, and new sources of Goliath fuel to overcome the Primal Curse and control far larger groups for longer. They no longer faced the same constraints as the earliest Goliaths. As Goliaths transformed, so too did their curse.

They still faced the underlying (and ongoing) problem of rising inequality creating societies where power was more concentrated and institutions more extractive. This had several knock-on effects, some of which were novel for these larger imperial systems. Inequality tends to exacerbate social problems, whether it be violent crime or the mental health of the populace.^{[20](#)} A swelling number of elites increases inequality by

finding ways to transfer wealth from the poorest to the richest, impoverishing the masses. Elites with increasing influence find more ways to siphon money and power away from the state, including through corruption and tax evasion. It's a relationship that multiple studies confirm in the modern world and a pattern we see, whether it be through the princes of China, the governors of Rome, or tax-dodging traders in the Late Bronze Age.²¹

A growing pool of elites competing for status, power, and a limited number of high-status positions turn towards more extreme measures, whether it be starting coups, civil wars, or rebellions (or simply hijacking existing ones). We see it most clearly through feuding nobles and emperors in China conducting coups and rebellions, elite-led coups and civil wars in Rome, a constant 'game of thrones' in West Africa, as well as the city-states of the Maya and the Mycenaean palaces turning on each other. These are just a few examples among a wealth of evidence that shows the importance of elite factionalism to crisis and collapse across the millennia.²²

To appease a disenfranchised populace, and to calm growing elite strife, the state will often push for increasingly expensive conquests or intensifying agricultural and mining production. This degrades the environment, making a society more vulnerable to drought, as we've seen most clearly in the cases of Rome and the Maya. A society that is increasingly top-heavy with a larger class of powerful elites will be less likely to make effective decisions, especially when they involve changing the status quo that they benefit from.²³ One review of sustainability throughout history that I was part of found that responses to environmental disasters – whether it be floods, droughts, or soil erosion – tended to be worse when the decisions were co-opted by elites and prioritized their short-term interests.²⁴ Many have lamented that past societies seem to have suffered paralysis when asked to address obvious problems.²⁵ Oligarchy is why.

The result is more extractive institutions creating growing instability, internal conflict, a drain of resources away from government, state capture by private elites, and worse decision-making. Society – especially the state – becomes more fragile. Private elites tend to take a larger share of extractive benefits. The state, and many of the power structures it helps prop up, then usually falls apart once a shock hits: for Rome it was climate

change, disease, and rebelling Germanic mercenaries; for China it was often floods, droughts, disease, and horseback raiders; for the west African kingdoms it was invaders and a loss of trade; for the Maya it was drought and a loss of trade; and for the Bronze Age it was drought, a disruption of trade, and an earthquake storm.

I call this ‘diminishing returns on extraction’. States seek to extract more resources by becoming more bureaucratic to closely control citizens, intensifying production (whether it be from mines or fields), and expanding through conquest. Elites seek to extract more resources through tax evasion, corruption, private armies, and other means to compete with each other and capture the state. Whether wealth is extracted from the populace, the land, or neighbours, there are eventually diminishing returns. Conquests become more costly over time until they provide more trouble and problems than benefits. More intensive agriculture and mining can mean more productivity, until deforestation, soil erosion, and lowered mine yields set in. An expanding bureaucracy can result in more control at first, but can eventually drain the government coffers. Elites extracting wealth from a populace – and even the state through corruption and oligarchy – can be good for them initially, but eventually they impoverish and deplete the people and government they are dependent on. Similarly, large trade networks primarily used for elite trade can bring prosperity and cultural exchange, but eventually become a liability as they transfer shocks or as a society becomes overly reliant upon them (as was the case for the Aksumite Empire and west African kingdoms). This is why societies appear to become more fragile over time and why there is this repeated cycle of fission and fusion among early Goliaths or capture and fracture among larger empires. It is due to diminishing returns on extraction.²⁶

[Table 4](#): Diminishing returns on extraction

Contributor	Benefits	Diminishing returns	Consequences
Conquest	Benefits from legitimacy,	Benefits dwindle over time as conquering and	Large-scale debt, creation of new enemies,

Contributor	Benefits	Diminishing returns	Consequences
	plunder, and slaves.	maintaining territories further from home cost more and new enemies are created by expansion.	overstretched state capacity, diversion of resources from productive sectors.
Corruption	Initial benefits for elites.	Tends to increase as both inequality and the number of elites increase.	Loss of revenue, loss of legitimacy, and loss of trust in the state.
Declining energy ratio	More intensive production can aid economic growth initially.	Eventually depletion and environmental degradation can weaken the resource base of a society.	Loss of revenue, and lower agricultural and mining productivity.
Elite factionalism	Beneficial for the swelling class of elites initially.	As the number of elites and their power grow (the society becomes top-heavy) they intensify competition for scarce high-status positions.	Polarization, coups, civil war, and rebellion.
Inequality and popular immiseration	Benefits initially for elites who reap the reward of unequal exchange	The benefits eventually fall away as the populace begins to rebel.	Lower government revenue, populace more susceptible to disease, state

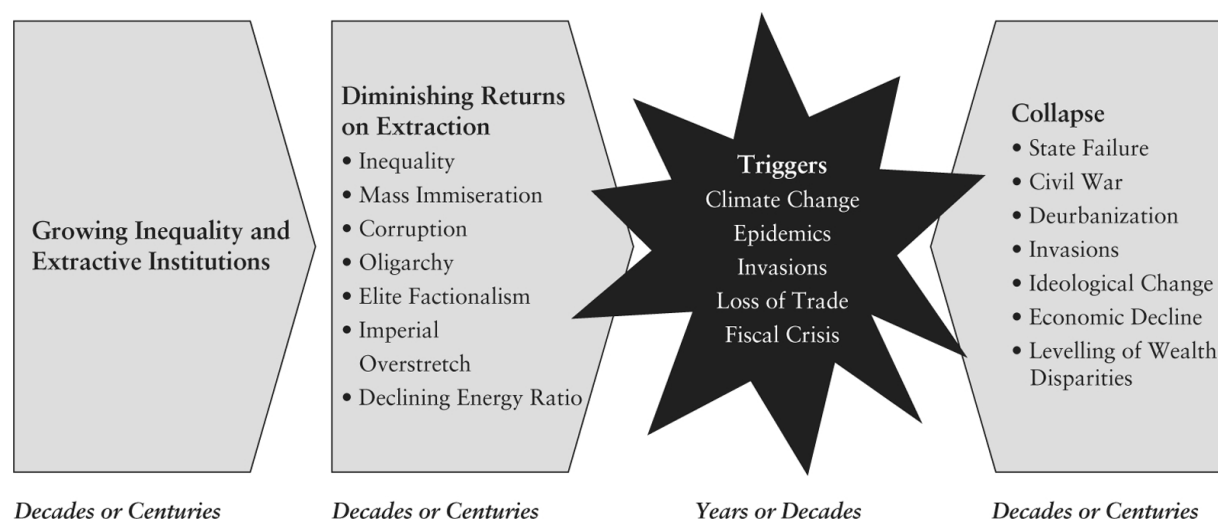
Contributor	Benefits	Diminishing returns	Consequences
	between the rich and poor.		capture, and rebellion.
Oligarchy	Hierarchy can initially be quicker, more efficient, and helpful in unifying a populace through an imposed identity.	A growing concentration of decision-making among elites leads to worse decisions and an inability to reform the status quo.	Poor decision-making and maladaptive responses to disasters.

We see this process embodied in the case of Rome. To begin with, Rome was a republic with at least some limited political enfranchisement. Inequality was lower, and power was more balanced. Over time, it became more unequal, overexpanded, and fell into autocracy. Rome developed a surplus of disenfranchised citizens and ambitious elites. Then there was ample corruption, with Roman governors who would swindle their subjects and sell them into slavery. It also appeared to be suffering from declining yields from both mining and agriculture. The supply of silver from mines in Spain began to dwindle, while deforestation across Italy may have led to eroding soils and lower crop yields in different areas. Like the Maya, they had a dwindling energy ratio.²⁷ Corruption, inequality, oligarchy, and elite factionalism were the key culprits that hollowed out the Roman Empire. These created a predicament in that the populace had become impoverished and hostile, the military was largely privatized, the government had racked up an enormous debt, it was crippled internally owing to elite capture of the highest offices, and had a bloated military and bureaucracy. It had become a brittle shell that was sundered by invaders, civil war, droughts, and disease.

There is a rough consensus on what factors led to Rome's fall, such as corruption, decaying institutions, invaders, climate change, plague, and power struggles over the surplus loatable resources and high-status

positions. Historians now mainly argue about which ones were the most important.²⁸ Diminishing returns on extraction combines all of them into one process. The pre-eminent historian of empire Walter Scheidel has already labelled the collapse of both the Han (and Jin) and the Western Roman Empire as cases of diminishing returns on extraction. Growing inequality along with corruption, elite factionalism, and oligarchic decision-making drained the world's two largest ancient empires.²⁹

Across all the cases of collapse, from the systems collapses in the Mediterranean and Yucatán, to China and Rome, we see similarities. In every case prior to the collapse there is a slow growth in inequality, oligarchy, corruption, and factionalism between elites over the surplus of lootable resources. Diminishing returns on extraction set in. This eventually collides with a set of shocks, usually drought, disease, invaders, floods, or earthquakes. Then things fall apart ([Figure 31](#)).



[Figure 31](#): Diminishing returns on extraction

The first Goliaths and their Primal Curse had some similar elements. For instance, we see some evidence of elite factionalism in cases such as Teotihuacan. They in all likelihood also suffered from poorer decision-making due to oligarchy. Some, such as the first farmers of Europe and the Puebloans, were also facing a declining energy ratio. Yet the bigger problem for the earliest Goliaths was stronger counter-dominance intuitions meeting weak control mechanisms. Other factors – such as imperial overstretch and corruption – began only with large imperial systems. The stronger control

mechanisms of empires like Rome and the Han also meant there was more time for diminishing returns to set in. The commonality across the Primal Curse and diminishing returns on extraction is that more inequality and more extractive institutions made societies more fragile.³⁰

These trends transcend such examples. One review I led looked at ten case studies including the Late Bronze Age collapse, the fall of Rome and several others, such as the Tang Dynasty, the Achaemenid Empire, and the first Arab-Islamic Empire. In almost all of them there was growing inequality and fiercer battles between elites over the surplus wealth and positions of status. Collapse in each case coincided with shocks such as disease, drought, and warfare. The result was that the state fractured, elites and soldiers became more regional, the beliefs justifying hierarchy transformed, and the population dispersed or died.³¹

Behind each of these contributors to imperial collapse we can glimpse cold, ice-age roots. Inequality clashed with our ingrained aversion to dominance. That aversion could be overcome by shared fictions, particularly those that justify hierarchy, such as the Mandate of Heaven. Yet it was a delicate affair, and our allergic reaction to oppression never fully left us. Elite factionalism is principally a battle for higher status, especially given that these elites typically had more than enough riches and security compared with the rest of the population.³²

Yet, some empires seemed to defy the curse for a surprisingly long period of time.

LEARNING FROM THE OLDEST EMPIRE

In the Mortality of States (MOROS) dataset (a compilation of states throughout history) one state stands out: the Eastern Roman Empire, also known as the Byzantine. It is exceptionally long-lived, having lasted over a thousand years. It was, in the words of the historian John Haldon, ‘the empire that would not die’.³³

The Byzantine period began in 330 CE when Rome was split into Western and Eastern empires. Despite the western half falling, the eastern half persisted, even expanding after the fall of Rome to take back most of Italy

and northern Africa by 541 CE. It was then hit by both bubonic plague and the Late Antique Little Ice Age. It struggled with Avars (nomadic raiders) and Slavs invading its eastern front across the Balkans, and a war with the Sasanian Empire across northern Africa and the Near East. Then things got even worse. The Rashidun Caliphate exploded out of the Near East, taking the wealthiest provinces of the Byzantine Empire: Syria, Egypt, and Palestine. The Byzantine capital of Constantinople was besieged every year from 674 to 678. The empire lost around two-thirds of its territory and three-quarters of its income. All seemed lost. Yet somehow it survived and even recovered. Within two centuries the empire had pushed back the Arab forces and regained much of its territory, nearly doubling in size.

What explains the resilience and longevity of the Byzantine Empire? First, it had a strong shared religion in its Orthodox Christian beliefs. These bonds were strengthened by the looming religious enemy of Islam on the borders. This helped to unite the elites as well as the broader populace during the times of crisis. Second, elites at the provincial level became dependent on the empire as the civil administration and military were combined and they were given military titles. Third, it was flexible in its response and willingness to decentralize. After losing so much territory, the empire could barely pay its troops and had to cut military pay by half in 616 and again in 659. As a substitute, soldiers were given land in exchange for military service. The bureaucracy was simplified and the troops, now defending their own land and families, fought more fiercely and came to identify with the empire more strongly. Fourth, the Late Antique Little Ice Age created cooler, wetter conditions that favoured keeping cattle and growing cereals. This meant more food, a healthier population, and less chance of famine. Fifth, the empire was lucky enough to have an easily defended geography, with the Taurus mountain range in the south acting as a natural barrier to the expansion of the caliphate (this became part of the border between the two).^{[34](#)}

Geography and climate were largely matters of luck, but the other three factors were not. They were all responses that helped to unify the populace and quell elite competition. The shared ideology and interests of the empire and local elites reduced status competition, while the more decentralized arrangements bolstered local belief in the empire. The shared religion of Orthodox Christianity across elites and local peasant-soldiers made for a more effective fighting machine while reducing the threats of rebellion and

elite infighting. While the Byzantines were eventually defeated by the Ottoman Empire they managed to reach an unparalleled longevity. Despite their longevity, the fall of the western half of the empire had enormous consequences for Europe.

SHRINKING HOUSES AND FRACTURED TRADE

Collapse destroys wealth and flattens its distribution. As a state falls, state-protected merchants and trade also decline. Luxury goods fall away, and trade becomes more localized.

For much of the former territory of the Roman Empire, especially Roman Britain, the end of the empire meant smaller houses and less wealth. In Britain the median house size went from approximately 150 square metres, prior to Roman occupation, to a high point of around 200 square metres during the empire, before shrinking to under 100 square metres after the Romans left.^{[35](#)}

Losses in wealth can also empty cities and may be responsible for much of the displacement we see during collapse. Rome lost over 95 per cent of its population but this was unlikely to have been due primarily to war or disease. Rome had weathered sieges and outbreaks for centuries. What it could not withstand was the cutting of the *cura annonae*, the shipment of subsidized grain from northern Africa into the city. Free grain had artificially inflated the city's population; once its life-support system was severed, it shrank to a few tens of thousands. There was little point staying in a city that was a target for invaders and pathogens if you no longer had free grain, free entertainment, and reliable protection.^{[36](#)} Settlements and cities were often dependent on such imperial support from far-flung territories. Once those fell away, these places lost their lustre and purpose. We see this from the earliest settlements across Mesopotamia through to the movement of the Maya to coastal areas, and even in the modern 'rust-belt' cities of the Midwest and north-eastern US. People naturally move to follow opportunity.^{[37](#)}

The fall of a Goliath was not always as economically devastating as with Rome, and less overall wealth does not mean that everyone was worse off. It meant greater wealth equality and sometimes even a rise in living standards. Remarkably, collapse, even of the economy, is often beneficial for the health of the populace.

WHY CRISES CAN BE GOOD FOR YOU

The skeletons of Europeans tell a very different story of the Roman Empire from the one most of us have heard. The rise of Rome in the first century CE saw a deterioration in height: Romans were 8cm shorter than they should have been according to projections based on trends prior to its rise. While Roman heights stagnated over the following centuries, areas that were not occupied by the empire, such as eastern Europe, continued to grow in stature. There is a reason why the Germanic ‘barbarians’ were remembered as hulking giants. People in the Roman Empire (with the odd exception of Roman Britain), particularly around the Mediterranean, didn’t start to get taller again until the fifth and sixth centuries, after the empire fell apart. Dental caries and bone lesions decreased, also indicating improved health after the fall. From the point of view of human health, the Roman Empire was the dark age, and the medieval period was the Golden Age.^{[38](#)}

We see a similar pattern in the people of the eastern Mediterranean during the Bronze Age. Height decreases over centuries as people turn increasingly towards urban living. The height of men increases slightly during the early Bronze Age and then stagnates for centuries, while the height of women falls and then slowly starts to rise. The Late Bronze Age collapse reinvigorates health. The height of men begins to increase again, and the growth in women’s height accelerates. The fall of empires often meant a rise in human stature.^{[39](#)}

There are a few contributors to this surprising trend. First, people shifted towards increasingly grain-heavy diets. Most Goliaths relied on grain as a tax crop and feeding larger amounts of people crammed together in cities usually meant eating more wheat, barley, corn, or rice. Crowding also often meant more disease, which would have taken its toll on human health and height. Diets heavier in animal protein tend to result in taller, healthier

populations. Most people in a ‘dark age’ took on more diverse diets with more herding and hunting, less agriculture and less exposure to infectious disease.⁴⁰

Second, the destruction of overarching economic systems can often lead to more free time and healthier habits. Economic recessions and depressions over the last century have often unexpectedly increased life expectancy owing to less air pollution (from less traffic), less drinking and smoking (because people have less money), fewer fatal accidents in the workplace or on the roads, and more leisure and social time (one of the few benefits of losing a job). We see this in the 2008 recession in Europe: the regions with the biggest economic slowdown also experienced the biggest increase in life expectancy. While economic downturns can worsen other health outcomes – such as mental health and rates of suicide – they can also save lives.⁴¹

Third, collapse often resulted in far greater equality. Collapse is a great leveller. If you were to look for one consistent, unwavering commonality across case studies – the Lowland Maya, Rome, the Han, the Late Bronze Age, and the first Goliaths – it would be the simple rule of the deeper the collapse, the greater the wealth equality it produced. This is for a few simple reasons. The destruction of capital – temples, palaces, manors, monuments, bathhouses, and wealthy properties – all disproportionately hurt the rich. The powerful also lost their usual ways of accumulating more riches as centralized institutions fragmented. The dispersal of people, and in some cases loss of life, meant that there were fewer labourers. Those who remained had more bargaining power with bosses and landlords, resulting in higher wages and taller people.

EAT OR BE EATEN

Both antiquity and medieval Europe had the idea of the *translatio imperii*: the handing down (or, often, brutal grabbing) of authority from one empire to another. Empires may fall, but they usually pass the baton, creating a running lineage. The idea dates back to at least the ancient Greek historian Herodotus detailing how the Achaemenid Persian Empire was the direct successor of the Assyrian and Median empires.

Rome is the ultimate example of *translatio imperii*. Many European kingdoms have claimed to be carrying the mantle of Rome ever since its fall. As Voltaire quipped, the Holy Roman Empire of the European Middle Ages was famously not holy, Roman, or an empire. Instead, it was trying to make itself legitimate by invoking the Roman legacy, as did most European empires. Many within the British Empire saw themselves as the inheritors of a Roman tradition. And there is no shortage of comparisons between the US and Rome.

Rome has left a particularly deep legacy beyond the *translatio imperii*. The emergency powers we use during crises (such as the introduction of martial law and states of emergency) were inspired by the Roman practice of selecting a dictator to run things temporarily when facing an immediate threat. Modern notions of property are also rooted in Roman law. We still conceive of property in terms of *usus* (the right to use), *fructus* (the right to enjoy the products of a property, for instance the fruit of a tree), and *abusus* (the right to damage or destroy).⁴²

We have been indelibly moulded by Rome, yet the Etruscans and the Celts whom they conquered left little imprint on the modern world. It's not because they were inferior to Rome. Rather it's because Rome was an unusually large and persistent empire. Evolution is often framed as 'survival of the fittest': those who are best adapted to a particular environment survive and reproduce. Selection of the fittest is caused by competition. In the case of Goliath evolution, it's not the most benevolent groups that leave a lasting mark, but those that are the most successful at conquering others and controlling larger swathes of territory and masses of people. The empires that were the best at doing so tended to have more of their ideas, technology, and culture carried into the future, even through collapses. The world of today is built on that evolutionary process.

Yet when we look back with a more sober gaze, it's clear that evolution does not always select for the wisest or fairest. Rome wasn't just deeply unbalanced and oligarchic, and it wasn't just bad for the health and height of most of its populace. It was also spectacularly savage. Armed gangs of young men selling women into slavery and the public execution of thousands by nailing them to crosses were hallmarks of Roman rule. And we can still see such things in the modern world. Just look at the actions of Islamic State in Iraq and Syria.⁴³ Yet, across the fog of history and through

the lens of the 1 per cent we view Rome far differently from how we see the Islamic State.

Try peering beyond that foggy lens and feeling what the victims of empire must have felt. Try to imagine what it must have been like to be crucified. Crucifixion was designed to be the most public and torturous form of capital punishment.⁴⁴ Every full inhalation would have been excruciating as you dragged your flayed back up the splintered wood and increased the pressure of the ropes and nails on your flesh. Death, when it came, was usually not from dehydration or starvation but from

asphyxiation.⁴⁵ You slowly suffocate as you become too exhausted to take another breath. That was, for many, the experience of a golden age.

Our fascination with the power of Rome also tends to make us enraptured by its collapse and forget all those that it absorbed or destroyed on its way to glory. Take the case of ancient Carthage, once a prosperous trading empire centred in modern Tunisia. In 146 BCE, at the end of the third and final Punic War, Rome razed the city of Carthage to the ground. Roman soldiers went house to house, systematically killing the inhabitants before setting fire to the buildings behind them. The surviving 50,000 citizens were sold into slavery.⁴⁶

History has often been a case of eat or be eaten. Either organize into a conquering dominance hierarchy that will suffer from Goliath's Curse or be consumed by someone else who does. We are the inheritors of that process, not of a linear stepladder of progress.

Underlying this evolutionary race of eat or be eaten are, once again, our old, Palaeolithic roots. Warfare between states and empires is an extension of status competition, no longer between individuals or groups but between entire polities. People commonly think that countries go to war for resources or security. That is a mistaken view, or at least overly narrow. It's increasingly clear that states often go to war for reputation and status. Vladimir Putin has been outspoken in his desire to regain Ukraine in order to restore Russia's status as a great imperial power (and undoubtedly to add to his own legacy). For Rome too, war was often not just about booty or revenge, but also for the glory and prestige of rulers and their empire.

Scholars have now amassed a wealth of data showing how concerns for status, honour, and reputation have shaped both war and peace. The general and military strategist Carl von Clausewitz remarked that 'war is nothing

but the continuation of policy by other means'. War is also nothing but the continuation of status competition by other means.⁴⁷

Our basic human drivers of dominance aversion and status competition didn't change, but the way they were expressed did. They were scaled up as empires grew and technology developed. Collapse too would eventually change as empires devised new weapons and began to cross oceans. Exploration by colonial ships and the beginning of the modern world would not have occurred without the falls of Rome. Our world is built on collapse.

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Fear the Rise

Petrarch's Mistakes – Forgetting Technology – Progress and Collapse – Blessed by the Black Death – How the Fall of Rome Led to the Rise of Colonial Europe – The Origins of Colonial Capitalism – Global Decimation

PETRARCH'S MISTAKES

The term 'dark age' was coined by the fourteenth-century Italian scholar Petrarch. Travelling across Europe, he collected manuscripts, particularly the literature and philosophy of antiquity. Petrarch was such a lover of the Greco-Roman world that he even penned letters to the long-deceased Roman writers Virgil and Cicero.

As much as he was impressed by the writers and art of Classical Greece and Rome, Petrarch was appalled by what he saw as the dearth of good literature and the death of good culture that followed the fall of the Western Roman Empire. He disdained the 'dark' ignorance of his post-Roman time and admired the 'light' of Greece and Rome. In his eyes, the eight centuries from 500 to 1300 were a time of stunted cultural achievement and outright barbarism.

The irony is that the Classical Greece Petrarch longed for was the product of a 'dark age'. The very alphabet that Petrarch wrote in emerged during the Late Bronze Age collapse.

Petrarch made another critical mistake: he saw the end of Rome as a purely catastrophic outcome. When we look at Europe in the wake of Rome

it becomes clear that Petrarch and this ‘rise and fall’ vision of history is spectacularly wrong.

FORGETTING TECHNOLOGY

The history of collapse is tied to the story of progress. The ‘rise and fall’ usually refers not just to the heights of a society’s walls and monuments, but also to the heights and well-being of its people, the dynamism of its culture, and the sophistication of its technology. History is often painted as a long arc of improvement, punctuated by the occasional collapse-induced dark age (such as the medieval ages after Rome).

This isn’t just baked into the outdated notion of dark ages. Even the official chronologies by historians and archaeologists centre and praise state power. The time before the rise of the pharaohs in Egypt is labelled the ‘*Pre-Dynastic*’ period. In Cambodia, the adoption of indigenous writing and the expansion of the first kingdom is the ‘*Pre-Angkorian*’, and the height of empire is the ‘*Mature Angkorian*’ period. For the Maya, their peak of urbanism and elite power is designated as the ‘*Classic*’ period, bookended by the ‘*Pre-Classic*’ and ‘*Post-Classic*’. It is as if people were waiting for empire, fleetingly enjoying its rich fruits, and then lapsing back into obscurity.¹

This progress narrative is often attached to the belief that progress is driven by technological change. It’s best summarized by Microsoft founder Bill Gates, who declared that ‘Innovation is the real driver of progress.’ OpenAI’s Sam Altman put it less elegantly when saying ‘the default state of the world is that things decay and get worse and worse, and the only way that not only do we make life better, but even push against that, like, decay, and all like kill each, destroy each other, is with progress and increasing quality of life. And the only way we get that is through technology.’ Gates and Altman represent a common view: that progress is a train moving forward across history and its engine is technological improvement.

For many, this is why Rome and other peaks in empire are seen to be golden ages: their roads, aqueducts, bathhouses, amphitheatres, and towering monuments all reflect a richer, more technologically advanced society. That technological ability, it is assumed, spilled over into the

welfare and livelihoods of the populace. Neither of these assumptions is true.

Technological change was slow under the Western Roman Empire. Most key productive technologies such as glassblowing and the use of watermills and water-lifting devices were developed in the first three centuries BCE in the Hellenistic east. Rome simply dispersed and further modified these innovations.² There is even a famous story of an inventor who proposed a labour-saving device that could transport heavy columns cheaply. It was rejected by Emperor Vespasian, who preferred to keep his people busy and employed. As we've seen, the virtue of Rome lay in war-making and administration, not in innovation.³

Just as we tend to overestimate the ingenuity of Roman technology, we also underestimate the inventiveness of the period that followed its fall. Medieval England, after the fall of Rome, was not mired in a technological dark age but was genuinely innovative and productive. This period saw the introduction of mechanical clocks, better mirrors, and tidal mills, as well as improved windmills and watermills and the heavy plough, which were deeply helpful for agriculture. In England in the three centuries between 1000 and 1300, agricultural productivity per person looks to have increased by roughly 15 per cent and yields per hectare doubled.

Technologies were lost during collapses. One of the best-known losses is Roman concrete – a unique blend that had been used across Roman architecture since the time of the Republic. It was low-emissions and unusually durable. It can still be seen in some of the empire's surviving bridges, aqueducts, and even the Pantheon. None of the preserved texts after Rome's fall describe how it was made. Only today are scientists discovering its secrets. Another prominent example is the plumbing and underground drainage systems of the Indus Valley Civilization. These were ahead of their time, but they were abandoned alongside the cities they lay under.⁴

Yet we need to bear in mind that technological development is not a straight, rising line across history. Nor is the loss of innovation innately tied to collapse. Even the most basic tools have sometimes been forgotten. When Europeans arrived in Australia the indigenous people didn't have a single bow or arrow, despite both having been invented up to 70,000 years ago in Africa. Instead, they had invented and relied on the boomerang rather

than the bow (alongside spears and throwing clubs). Somehow during the migration from Africa to Australia the technology had been lost. Something similar happened across Oceania, not just for bows and arrows, but also pottery and ocean-going canoes. Some forager groups even lost the ability to make fire.⁵

This collective forgetting was often caused by a catastrophe. In the 1820s the Polar Inuit were struck by an unknown epidemic. Many of the oldest, most learned individuals perished, and their knowledge passed away with them. The group lost basic, critical technologies such as kayaks and bows and arrows. It had taken generations to craft and refine these instruments and, once those who knew how to make them had gone without passing on those skills, they were lost. Technology is a product of cultural learning, and if an event is sufficiently catastrophic it can disrupt the intergenerational chain of learning.⁶

Loss of technology is generally rarer among large societies. That's simply because a larger number of more interconnected people means that knowledge tends to be more widely shared. Writing too helped with this, acting as a source of information storage outside the human brain.

A number of factors could lead to the loss of technology during a collapse. The knowledge behind the technology may have been limited to a small class. As we saw with the Inuit, relying on a small group to carry collective knowledge is a shaky strategy. If the class died, became unemployed, or simply fell out of favour, the techniques were forgotten. This was especially true of technologies that were kept secret, such as Greek Fire: a petroleum-based chemical weapon used by the Byzantines to burn enemy ships. It was tossed in grenades or even sprayed through a pressure-pump like a kind of flamethrower. It was so potent that it could burn on water and may have even led to entire Byzantine naval victories. Yet it was lost, probably in the late twelfth century as the Byzantines' oil supplies were cut off by invaders. Thereafter the fire-projector used on ships disappears and its use in catapults slowly dwindles. The likely reason is that the recipe for Greek Fire was a tightly guarded state secret that only a handful of imperial agents knew. Once the empire lost resources and power, it was no longer made and eventually passed into history. If knowledge is closely tied to a power structure, then it can fall alongside it.⁷

Another cause is that the technology loses its utility. For instance, there was less need for Indus or Roman plumbing after people had moved from cities to rural communities, where they could rely on simpler, cheaper cesspits. Roman concrete was of little use if grand monuments to celebrate the empire were no longer being built. A further reason is deliberate rejection and abandonment. Native Americans in the aftermath of Cahokia stopped building pyramids throughout much of the Mississippi basin.

Rather than being the result of social regression or barbarism, lost technology is – like lost scripts and writing – a matter of technologies becoming obsolescent or rejected during a reshuffling of power. It was, if anything, an adaptation: people dropped and adopted technologies that better suited their circumstances. As we've seen in both post-Roman Europe and the Mediterranean after the Late Bronze Age collapse, technological regression due to collapse is also often an exaggeration. Large empires did not necessarily mean more innovation and their fall was not necessarily accompanied by a loss of inventive ingenuity. And even when technology was lost it did not automatically mean that people led worse lives.

PROGRESS AND COLLAPSE

Across history, neither improved technology nor bigger, wealthier empires have resulted in better lives for most of the populace. Let's look at three things that many would broadly agree are vital to a flourishing life: welfare, health, and happiness (or, as the ancient Egyptians would have said, 'life, prosperity, and happiness').⁸

Welfare and standards of living can be roughly measured and compared across ancient studies using real income (often expressed as kilograms of grain received per year). One of the few studies to do so found fairly small differences in the welfare of most labourers between sixth- and second-century BCE Babylon, the Han Dynasty, and the Roman Empire. There was not a big difference even when comparing them to thirteenth-century England, almost 2,000 years later. Another more systematic study found the same trend. There may have been minor fluctuations and some exceptions such as Athenian Greece. Yet even here, the wealth of Athens was built on the backs of slaves and on the exacting of tribute.

In other words, the average citizen living under a state would have seen little improvement in welfare in the 6,000 years between around 3600 BCE and 1700 CE. An eighteenth-century European peasant was about as well off as an Egyptian peasant 7,000 years earlier. GDP per capita (in real, price-adjusted terms) essentially plateaued during the first millennium CE. These years saw enormous technological change, particularly in agriculture.⁹

The overall trend is that living standards for most people did not improve dramatically over the long run of history. In terms of quality of life, history largely stood still between the transition to agriculture and the end of the Industrial Revolution. This is astounding, given the thousands of years of technological improvements that occurred.

A similar pattern occurs with human height. We've already seen that average height dropped with the move from hunting and gathering to intensified agriculture. Heights also became more variable. In the modern world, differences in wealth both within and between countries are echoed in differences in height. The poor are shorter and the rich taller. This is a historically consistent trend over thousands of years. It wasn't just pharaohs and Mycenaean kings who were taller than their peasants: in the antebellum American South, enslaved women were 6.5cm shorter than southern white women, while the gap for men was 4cm. In contrast, Native American hunter-gatherers were fairly equal in height for both men and women. This is simply a reflection of economic inequality's impact on health.¹⁰

Across the world, height has hovered around 170cm for the past 2,000 years. There were fluctuations, but no lasting increases until the nineteenth century. Peculiarly, as we've seen with both Rome and the Late Bronze Age, height tended to fall under empire and increase during periods of collapse. For human health, the idea of progress rising and falling with empire isn't supported by the evidence.¹¹

Is the same true for happiness? Happiness is perhaps the most important and most difficult to measure variable of progress. We unfortunately have little idea how happy the average citizen was in the Roman Empire or Han Dynasty. What we do have are studies comparing the happiness of modern peoples and hunter-gatherers. One of the few studies to do so focused on the modern Hadza people of northern Tanzania. They were found to be, on average, happier than the inhabitants of over a dozen industrialized countries for which there was comparable data, including the US and Spain.

Other studies have reached similar conclusions. Perhaps most tellingly, ethnographies of hunter-gatherers are replete with reflections on how carefree and happy they are, with close-knit communities and a propensity to spontaneously burst into song and dance.¹²

The historical data is patchy and often hard to measure, but it points towards a surprising and grim conclusion. Despite 11,000 years since the transition towards domesticated crops and animals, most people are sadder and smaller than their Palaeolithic ancestors.¹³

Why was progress painfully slow and new technology not a benefit to most people? Most of the surplus created by technology and economic growth was taken by elites. One estimate from medieval France (1100–1250) suggests that 20 per cent of output went to the construction of cathedrals and churches. Essentially, everything that was not being used to feed people went to building churches and expanding monasteries.¹⁴ As we've seen already, most pre-modern states were staggeringly unequal.

In general, across history, the poorest half of the population would hold less than 5 per cent of the wealth while the top 1 to 10 per cent, comprising merchants, soldiers, priests, and rulers, owned 60–90 per cent.¹⁵ This minority elite extracted essentially all the surplus from the masses of impoverished farmers working the land.¹⁶ That's why thousands of years of technological advances didn't improve the lives of most and why kings stood taller than workers. Technology and wealth can, of course, help immensely (as we see in the modern world), but only when the benefits are widely dispersed.

A more reliable way to boost heights and welfare has been to redistribute power. Sometimes, that has been via collapse, and sometimes in more sinister ways. We can glimpse both in Latin Europe in the wake of Rome.

BLESSED BY THE BLACK DEATH

Alas, this mortality devoured such a multitude of both sexes that no one could be found to carry the bodies of the dead to burial, but men and women carried the bodies of their own little ones to church on their shoulders and threw them into mass graves, from which arose

such a stink that it was barely possible for anyone to go past a churchyard.

This is an extract from the chronicles of one observer of the Black Death in the cathedral priory of Rochester in Kent. The repeated bouts of the Black Death ravaged Europe between 1346 and 1353 CE, killing between a third and over half of the population.^{[17](#)}

It was incredibly traumatic, yet a boon for the lucky survivors. Halving the population meant that, on average, everyone now had twice as much of everything, from ships to land to grain. Because labour had become scarce and land was now abundant, peasants had more bargaining power. And they used it. Uprisings and rebellions occurred across Europe in the fourteenth and fifteenth centuries. The 1381 Peasants' Revolt in England, led by Wat Tyler, was inspired by the preacher John Ball's famous rallying cry of 'Now the time is come in which ye may (if ye will) cast off the yoke of bondage and recover liberty.' Ball was executed, but the push for equality he called for was successful. Peasants were now valuable enough to force concessions out of the landlords and barons. Wages doubled or tripled in many regions, and the pay gap between men and women decreased. People grew taller and healthier. Life expectancy increased and heights grew from a low of 168cm during the Black Death to a high of around 174 cm.^{[18](#)}

While the Black Death was a blessing for the survivors and their descendants, these positive effects were not evenly distributed.^{[19](#)}

Regions that already had inclusive institutions benefited in the long term from the Black Death, while those without them did not. North-western Europe had the strongest presence of guilds, village communities, fraternities, and groups for managing the commons. In these regions the plague acted as a leveller, slashing wealth inequality and potentially helping to hasten the overthrow of feudalism. In contrast, eastern Europe, Russia, the Baltic, Castile, and north and central Italy lacked these inclusive networks and ended up more unequal after the Black Death. Eastern Europe and Russia were also far slower to exit feudalism. In north-western Europe inclusive networks mobilized to ensure elites didn't use the crisis to further their power.^{[20](#)}

The blessings of the Black Death arose not only from the loss of life but also because of the way it shifted power. Mexico in the sixteenth century

provides a useful case study of how power stifled widespread prosperity. Recurrent waves of disease killed off large numbers of workers. This in theory should have given them greater negotiating power and wages, just as the Black Death did in Europe, but the Spanish colonizers suppressed the bargaining power of labourers and wages stagnated. A century later, when these controls were relaxed, real wages shot up.²¹

The fall of Rome and the Black Death not only give us an insight into the surprising relationship between collapse and prosperity, but they can also help us to understand how petty European kingdoms became colonial conquerors.

HOW THE FALL OF ROME LED TO THE RISE OF COLONIAL EUROPE

In the 1480s, Christopher Columbus dreamed of finding a western route to Asia from Europe. He pitched his idea to rulers in England, France, and Portugal, and was promptly rejected. They correctly identified that Columbus massively underestimated the circumference of the earth and size of the oceans. Then, in 1492, he finally received support for his expedition from Spain's Queen Isabella and King Ferdinand.

Columbus got lucky and wandered into the unknown continent of the Americas. He was also fortunate in that he did not live in a unified empire in which a single rejection would have ruined his prospects. He lived in a competitive marketplace of states, and was able to pitch his proposition more widely.

This decentralized landscape of competing states was due to the fall of Rome, following which Europe became a chessboard of warring kingdoms with a common Christian culture. These conditions created a fertile environment for new thinkers to develop and sell their ideas, however crazy they may have been. The relentless competition between European elites also drove a need to innovate, especially in warfare. The advance of science accelerated, resulting in a host of new technologies, including the compass, oceangoing ships, muskets, and cannons.²² Together, these technologies,

and competition to find new lands and trade routes, fuelled the rise of European colonial powers such as Spain and England.

Competing European powers were incessantly at war and strapped for cash. This drove even further innovations. For instance, England's empire created elaborate financial mechanisms to finance bigger overseas conquests, including central banks and joint-stock exchanges in the late seventeenth century. The private sector was combined with the state in new ways to help aid conquest. Private trading companies (such as the Dutch East India Company) were given monopoly rights over newly conquered territories and then jealously defended with vast navies. They became company-states with the ability to raise taxes, punish crimes, and enact treaties with local rulers. Even the conquistadors who invaded the Americas – such as Hernán Cortés and Francisco Pizarro – were not soldiers of Spain. They were armed entrepreneurs who led private expeditions.

Scholars have argued for decades over why the West took over the world. The answers have ranged from geography and better weapons to a better culture for science and invention. Yet, as Walter Scheidel has shown, underpinning all these was an intense competition that wouldn't have existed if Rome hadn't fallen. What did the Romans ever do for us? Well, for future European power, they got out of the way.^{[23](#)}

THE ORIGINS OF COLONIAL CAPITALISM

Colonialism is neither recent nor uniquely European. Colonization, the act of planting settlements of one's own citizens into foreign lands, pre-dates the colonial empires of Spain, Britain, and Portugal. Invaders displacing locals was done by the first farmers of Europe and the Romans.

What was new in post-Roman Europe was that colonization was combined with capitalism, and both accelerated. Capitalism, as we've seen in Assyria, is also ancient. Classical Greece, Rome, Song China, and the Venetian Empire all had the ingredients of capitalism such as wage labour, private property, and profit-orientated entrepreneurs. Yet something changed in the sixteenth century. In this case, a key spur was not the collapse of Rome but the demographic collapse caused by the Black Death.^{[24](#)}

The economic boon for the survivors of the plague meant that they had more disposable income, and a newfound ability to move up the social ladder. Because of this, the demand for luxury items such as gold, furs, silk, sugar, and spices increased. These were goods that needed to be imported from faraway lands, stimulating the demand for colonial ventures to the Caribbean and the Americas.²⁵ A lack of labour meant a higher demand for machines, including a boom in the construction of watermills and windmills. Similarly, the deaths of so many scribes during the plague hastened the mass production of books.²⁶

It wasn't just about the impacts of the plague, but also the eventual response of the rulers. Peasants were getting taller and wealthier – until the rich reacted. As one writer put it in the early 1500s, 'The peasants are too rich ... and do not know what obedience means; they don't take law into any account, they wish there were no nobles ... and they would like to decide what rent we should get for our lands.' The means for putting the peasants in their place was enclosure: the wealthy kicked peasants off their collectively managed lands and commons. Words such as 'pauper' and 'poverty' came into common usage, despite having barely been used in earlier periods. The eighteenth-century protest song 'The Goose and the Common' summarized enclosure in one verse:

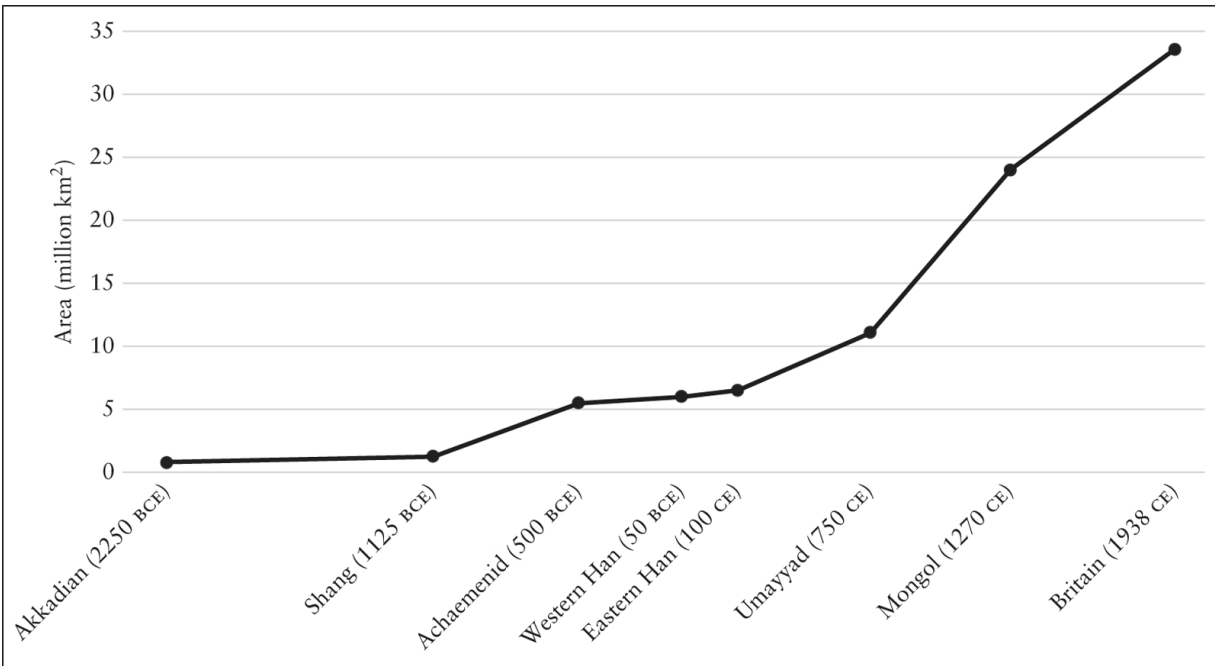
The law locks up the man or woman
Who steals the goose from off the common,
But turns the bigger robber loose
Who steals the common from off the goose.

Much of the land that had been unowned was seized by nobles, who then demanded rent. It was the artificial manufacture of poverty through theft. The peasants became a dispossessed, cheap source of labour for factories in urban areas in the north of England.²⁷ There they were combined with a cheap source of energy in coal. Coal-fired steam engines were less reliable than wind and water power at this time, but had some unique benefits. First, watermills required cooperation between competing owners of mills that drew from the same watercourse, while coal-powered factories could be operated by an independent capitalist. Second, watermills were located in upland river valleys, while coal-powered factories could be placed in cities

to make use of the cheap, excess labour that had been created by enclosure. Factories became the new engines of this European Goliath.²⁸

These factories were supplied by overseas colonies built on stolen land. In 1492 Columbus made landfall in the Bahamas (he was searching for Asia, and died in 1499, convinced he had found it). A year later Spanish settlers began to arrive. Within half a century they had invaded much of the Caribbean and the Americas. A century later, the British had joined. This wasn't just a typical case of conquest. They systematically killed or enslaved the indigenous population, transforming the lands into plantations full of cash crops such as cotton, sugar, and tobacco, or stripped for gold and silver. Lutable resources bloomed. The violent fruits of these plantations were then shipped back to the imperial homelands at low prices, or as gifts to the monarchs. These were new empires built on cash crops, slave-manned plantations, and unequal trade rather than the simpler approach of conquer and demand tribute (or conquer and tax).²⁹

It led to unprecedented imperial expansion. The British Empire became the largest empire to have existed. In 1925 the British Empire controlled 35 million square kilometres, or around a quarter of the world's total land area (approximately seven times the size of the Roman Empire). Sargon's Akkad was stuck in one time zone, while the British just over 4,000 years later had become an empire upon which the sun never set.³⁰ This was an extension of the long-running trend of empires getting bigger and bigger over time (see [Figure 32](#)).³¹



[Figure 32](#): The maximum territorial extent of the largest empires throughout history

These unfathomably large empires required new technology and governance to remain intact. Colonial empires delegated power to formal bureaucracies, local indigenous elites, and state-owned companies to help manage the overseas colonies, while overseas administrators would often settle on informal measures such as taking a local mistress to learn the country's language, customs, and culture. Technology was perhaps even more pivotal. The British used the telegraph and railways to oversee their colonies, rather than runners and horses. In India alone, the British laid down 84,000 kilometres of telegraph lines between 1853 and 1900.^{[32](#)} These proved decisive in allowing the British to react rapidly to rebellions. After the British successfully quelled the 1857 Indian rebellion, one colonial official exclaimed 'The Electric Telegraph has saved India' (saved it for British occupation, that is).^{[33](#)} Goliath continued to evolve and began to cross oceans.

This new form of Goliath was all built on the fall of Rome. Not only that, but the rise of the new empires would prove to be far worse than any imperial collapse had been.

GLOBAL DECIMATION

The emergence of these colonial Goliaths was one of the most violent processes of human history. The colonization of the Americas itself may have claimed up to 10–11 per cent of the world’s people at the time. While the numbers are highly uncertain, only one other event seems to come close to this: the waves of the Black Death in the fourteenth century (see [Table 5](#)).³⁴ We can think of these as ‘global decimations’. Decimation means ‘removal of one tenth’, derived from the Roman military punishment for insubordination in which a cohort (around 480 soldiers) was forced to murder a randomly selected tenth of their comrades.³⁵

[Table 5](#): Historical global decimation

Event	Timeframe	% of global population lost
Black Death	1345–1353	~9.7–10
Invasion of the Americas	1492–1600	~10–11

Global decimation is rare and isn’t about collapse. Instead, it is about disease and the rise of empires. The invasion of the Americas didn’t cause such an enormous body count by collapsing the Inca and Aztec empires. It was due to the actions of European invaders, as well as the diseases they introduced.

Collapse is also rarely the culprit for the worst atrocities that fail to meet global decimation criteria. Mass killings tend to be Goliath-organized affairs. Take, for instance, the case of the Züngharian Empire (1634–1755 CE), the last nomadic, Mongolian empire to threaten China. A combination of elite fracture and conquest by the nearby Qing Dynasty led to its devastation. The Qianlong emperor ordered, ‘Show no mercy at all to these rebels.’ The result was not just the destruction of the state but the genocide of a people. One estimate suggests that 30 per cent of Zünghars were killed, 40 per cent died of disease, 20 per cent fled to Russia, and just 10 per cent

remained as a small minority in Xinjiang province. The Han Chinese then resettled the emptied territory with forced relocations, primarily by ethnic Han Chinese.^{[36](#)}

The transatlantic slave trade was also a matter of imperial ‘success’ rather than collapse. And it wasn’t just Europe. The Ottoman Empire took in (including from African slavers) around 2 million slaves from sub-Saharan Africa over the course of its existence. Europeans tapped into pre-existing slave markets in Africa, which were usually run by local Goliaths.^{[37](#)}

The driver behind these horrors is not a lack of authority structures, but rather organized brutality. As professor of sociology Siniša Malešević has demonstrated, large-scale violence needs highly organized authority structures such as an empire like the British, Spanish, or Qing. Systematically killing, enslaving, or oppressing millions relies on ideology, organization, and creating tight groups unified against an external enemy. Most people do not have a natural proclivity for mutilation, murder, torture, and subjugating others. We do have the capacity for it, but that usually requires a strict chain of command (even the most egregious crimes are often defended as ‘just following orders’), extensive training, and rewiring the brains of soldiers. You need highly organized dominance hierarchies to do this.^{[38](#)}

In some cases, state collapse could even prevent such atrocities. One large data analysis found that failed and failing states are far less likely or able to conduct genocide than a functioning government. That is for the obvious reason that systematic identification and murder require a strong centralized institution.^{[39](#)}

Societal collapse can still carry a heavy toll. Yet its costs are usually due to the quest to create a new racket, such as in the gory rise of the Qin. Historically, collapse has not been the source of the greatest mass killings or atrocities. That we owe, instead, to the creation and expansion of Goliaths. And the creation of the Global Goliath carried the highest price in human life.

Colonization and Collapse

Why Did Cortés Win and Córdoba Fail? – Guns, Germs, Steel, Revolt, and Elite Conflict – Indigenous Allies and Rebellion – Local Apocalypses – The Shatterzone – When Indigenous Elites Helped Europe Eat the World – Babies, Bacteria, Bombs, and Barbarism – Is Colonization Collapse?

WHY DID CORTÉS WIN AND CÓRDOBA FAIL?

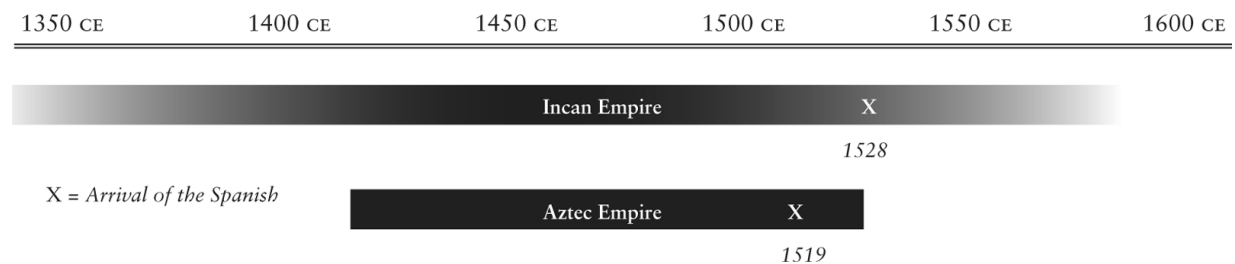
The Spanish slaver Francisco Hernández de Córdoba led the first European expedition to the mainland of the Americas in 1517.¹ He headed a group of 110 Spaniards from Cuba to Isla Mujeres, off the coast of the modern resort city of Cancún. There they spotted the pyramids of the Mayan city El Meco, which they nicknamed ‘El Gran Cairo’ (Great Cairo) because of its resemblance to the Egyptian pyramids. On 5 March 1517, the crew were beckoned ashore by a group of Maya and given an unwelcome surprise.

They were met with a hail of arrows, spears, and stones. Over a dozen Spaniards were injured and the group quickly retreated. They continued to circle the Yucatán coast, going further north and then east towards what is now Mexico City. They grew dehydrated and made the mistake of stopping at the Mayan land of Chakanputún, which they named Champoton, for water. They soon found themselves surrounded by around a thousand Mayan soldiers. The next day, the Mayans closed in on them. Spanish steel blades clashed with wood and obsidian swords. Despite their technological advantage, the Spaniards were soundly defeated. Two were captured, over

fifty were killed, and more succumbed to their wounds while retreating. They eventually made it back to their base in Cuba, where Córdoba died of his injuries. This initial Spanish invasion attempt of Mesoamerica was a disaster.

Two years later, in 1519, the Spanish conquistador Hernán Cortés arrived on the Yucatán coast with an unauthorized expedition of a hundred sailors, eleven ships, 508 soldiers, and sixteen horses. Within two years the Aztec capital of Tenochtitlán was captured and the Aztec Empire had collapsed ([Figure 33](#)).

Colonization resulted in a new wave of collapses across the world. As we'll see, many cases of colonization were collapses, while others were not. Understanding how colonization could cause collapse – and why sometimes it did not – can in part be unravelled by exploring why the Maya soundly defeated Córdoba, while Cortés seemed to swiftly conquer the larger and better organized Aztecs two years later.



[Figure 33](#): A timeline of the Incan and Aztec empires

GUNS, GERMS, STEEL, REVOLT, AND ELITE CONFLICT

Jared Diamond famously summarized the reasons why Cortés won in three words: guns, germs, and steel. Crossbows and muskets could cut through cotton armour at a distance, while metal armour was largely impervious to arrows and obsidian blades. Indigenous Americans could shoot arrows at six times the rate of a Spanish blunderbuss, but to no avail. The Europeans also carried a more silent but deadly weapon: disease. The immune systems of the indigenous Americans were completely unprepared for the diseases the Europeans brought with them: smallpox, bubonic plague, measles,

mumps, chickenpox, scarlet fever, typhus, cholera, and pneumonic plague. It became part of what's called the Columbian Exchange: the widespread transfer of plants, people, pathogens, and culture across the Atlantic Ocean in both directions (with the Europeans taking such crops as corn, tomatoes, and potatoes back with them).²

These reasons alone can't account for why empires such as the Aztec and Incan collapsed. Córdoba had the same military advantages yet was easily crushed by the Maya. There is another critical reason why colonization efforts succeeded: indigenous allies.

The Aztec Empire had inherent vulnerabilities that the Spanish exploited. After the fall of Teotihuacan (c.550 CE), numerous city-states competed, rose, and fell across the Basin of Mexico. Then, an alliance of three Nahua cities – Tenochtitlán, Tetzaco, and Tlacopan – managed to subdue the others and craft an enduring empire (1428–1521). They became known as the 'Triple Alliance' or the Aztec Empire. This was the largest pre-Hispanic empire in Central America, covering over 200,000 square kilometres. They had their own novel number system, a calendar, aqueducts, a taxation system (with taxes paid by provinces in maize, amaranth, textiles, and even warrior costumes), and their capital of Tenochtitlán was built on an island surrounded by floating gardens. Despite this sophistication, it was still a loose empire based on the recent subjugation of many cities by just three.

Like Rome, the Aztecs dominated through their military might. They arose after the introduction of new weapons such as stone swords into the region. The Aztecs believed that humans had a debt to the gods, one that could be repaid through sacrifice and blood offerings. Newly conquered territories didn't just provide sacrificial victims; they also gave tribute to the Aztecs before they were eventually incorporated and graduated to paying tax. It was highly unequal, with the top 1 per cent taking around 41.8 per cent of total income (their share of wealth would have been even higher). Those who resisted the imperial expansion of the Aztecs faced higher taxes and harsher working conditions.³ Many of the conquered territories resented such domination, while unconquered neighbours such as the Tlaxcalans (a confederation of four oligarchic republics) were in a state of near perpetual war with the Aztecs.⁴

Rebellion or total war, however, was not advisable. The Aztecs had mass conscription and could reportedly field an army of 200,000. Imagine a small

city of soldiers sporting obsidian swords, looking to gain status by capturing as many enemies as possible to sacrifice. It was obvious why many, despite despising Aztec rule, acquiesced.

The arrival of the Spanish changed this calculus. At first, the conquistadors fought against the Tlaxcalans. Thanks to their guns, steel, and horses, the Spanish won these battles against the far larger Tlaxcalan forces. It was not all good news for the Spanish, as their numbers dwindled after each fight. Technological advantages could not overcome being massively outnumbered. Cortés and his conquistadors were on the verge of being defeated. Then, the Tlaxcalans changed their minds. They decided that while they could destroy the Spanish, it would cost them too many of their own. Instead, impressed by the Spaniards' prowess in battle, they offered an alliance. The Aztecs would spare Cortés and his men if they would join the Tlaxcalans in attacking the Aztec Triple Alliance. The Spanish wisely agreed.⁵

The Spaniards became one part of a far larger army. A group of 10,000 marched on the Aztec stronghold of Tenochtitlán. Indigenous troops from the Totonac of the Gulf Coast and the Tlaxcalans of the highlands outnumbered their Spanish allies fifty to one and Cortés was by no means the sole head of the army. When they arrived, they were unexpectedly invited into the capital. The emperor, Motecuhzoma II (sometimes known as Montezuma), was a ruthless conqueror, but he was also curious. He wanted to observe these strange Spaniards up close. Motecuhzoma had made a fatal mistake.⁶

Cortés took Motecuhzoma hostage on 14 November 1519 and held him captive inside his own city for over six months. During the months without a leader, the Aztec Empire was largely paralysed and tensions mounted between the occupying Spaniards and the Aztecs. Later, following a massacre by the Spanish of dozens of Aztec nobles and warriors after misinterpreting a ritual as preparation for war, fighting broke out and on 29 June 1520 Motecuhzoma was killed. Spanish accounts blame the Aztec people, claiming they stoned Motecuhzoma to death. The more likely scenario is that the Spanish killed him. As soon as he was dead, the Aztecs elected a new king and ejected the Spanish from Tenochtitlán within days. The retreat of the alliance from Tenochtitlán proved costly for the Spaniards and their allies. While estimates vary, it seems that hundreds of Spaniards

and thousands of indigenous allies were lost. They called it *La Noche Triste*: The Night of Sorrows.⁷

The Spanish–indigenous alliance returned just under a year later, now with a force of 100,000–200,000. Again, the vast majority were indigenous rebels, not conquistadors. The city-states of Tetzaco and Tlacopan, two of the original members of the Aztec Triple Alliance, had also defected, swelling the numbers of the anti-Aztec force. To make things worse for the Aztecs, there was an outbreak of smallpox (introduced by the Spanish) in the city of Tenochtitlán.

The second siege of Tenochtitlán was successful. The superior numbers and devastating effects of smallpox were too much for the Aztec capital. Even so, the small Spanish force is unlikely to have been capable of taking Tenochtitlán (or simply to have survived up till that point). They needed the assistance of indigenous allies.⁸

INDIGENOUS ALLIES AND REBELLION

The key role of indigenous allies can also be seen in the exploits of a relative of Cortés: Francisco Pizarro. Pizarro arrived on the borders of the Incan Empire with 168 soldiers in 1528. Within a decade the Incan Empire had collapsed.⁹

As with Cortés and the Aztecs, Pizarro was not the first European to encounter the Incans. Three years earlier, the Portuguese explorer Aleixo Garcia had led an expedition (including a small army of native allies) against the Incans and was easily defeated. It was such a disaster that the natives turned on Garcia and killed him during the retreat from the Incans. Garcia's struggles should be unsurprising, given the power of the Incan Empire.

The Incan Empire was the largest empire the Americas had ever seen. It encompassed Peru, Bolivia, Chile, Ecuador, Colombia, and parts of Argentina with 40,000 kilometres of roads. An ethnic group (the Inca) of just 100,000 people ruled over between 9 and 11.7 million subjects in a centralized-command economy. It had no traders and the necessities of life were distributed to subjects through state warehouses. With no horses to speed up communication and transport, they had to rely on less orthodox

approaches to administer the empire. Relay runners (*chasquis*) could use the road network to cover an astonishing 200 kilometres per day (four times faster than Spaniards on horseback).¹⁰ They would run with rolled messages, held like a baton, or deliver them by word of mouth, repeating them while running so as not to forget.¹¹

The Incan Empire was also unstable: it had doubled in size in less than a century and maintained a tenuous grip over dozens of different ethnic groups and communities with hundreds of languages.¹² Like most empires, it also suffered from disputes over who should rule. Aleixo Garcia had mistimed his attack; just two years later the Incan Empire would descend into civil war.¹³

In 1527 the Incan emperor Huayna Capac died from a mysterious disease (most likely smallpox introduced by the Spanish). The empire was quickly engulfed by infighting over which of Capac's sons would inherit the throne: Huáscar, who was favoured by the nobility, or Atahualpa, who was backed by the military. The two battled for five years until 1532, when Atahualpa emerged victorious and Huáscar was defeated. Pizarro and his conquistadors walked into the middle of a civil war and made the most of it. They cobbled together an army of disaffected indigenous groups previously oppressed by the Inca, including the Huancas, Chankas, Cañaris, and Chachapoya.

Pizarro confronted the victorious Atahualpa on his return. As with the Aztecs, the element of surprise was key: Atahualpa went with an unarmed contingent to greet the Spanish and was captured. In 1533, Atahualpa was baptized into the Roman Catholic faith. Then, in the main square of Cuzco, he was strangled to death. It was not just a victory due to superior weapons or deadlier bugs; it was, like the defeat of the Aztecs, centrally about conflict between elites and resistance from local elites and commoners against empire.¹⁴

The collapse of the Aztecs and the Incas were not clear-cut cases of 'conquest'. Both can easily be seen as successful rebellions rather than signs of Spanish superiority. Indeed, that's how the Aztec and Incan locals saw them. The allies of Cortés viewed it as a battle against Aztec imperialism with foreign assistance, rather than a foreign invasion. They

saw themselves as successful revolutionaries who overthrew the oppressive Triple Alliance, not as losers against the Spanish.¹⁵

It was only afterwards that the self-aggrandizing accounts penned by conquistadors reframed the conflicts. Suddenly they became legendary stories of a few exceptional Spaniards defeating empires of millions through guile and superior arms. During that time, other more pernicious myths crept in. One is that the Aztecs were overawed by the arrival of Cortés and his conquistadors, mistaking them for gods, particularly the feathered-serpent god Quetzalcoatl. The historian Camilla Townsend has thoroughly debunked this as a myth created in 1552 by Francisco López de Gómara, a chaplain-biographer of Cortés who had never visited Mexico. The Aztecs instead fought bitterly against the invaders, saw them as vile and bizarre beings, and after the conquest frequently refused to adopt their customs, such as Spanish schools and Catholic marriage. Not exactly how one would treat gods. Quetzalcoatl was not even normally depicted as having white skin: he was called the ‘white god’ because it was the colour associated with him, not because it was his ethnicity.¹⁶

Guns, germs (especially), steel, and horses played a role, but elite conflict and local resistance were key. Cortés and his conquistadors were almost defeated by the Tlaxcalans and certainly wouldn’t have managed to beat them and the Aztecs without help. Pizarro would have met the same fate as Aleixo Garcia if he had walked into a unified Incan Empire. And the oligarchic structure of both the Aztecs and the Incans hindered their response: the Aztecs were paralysed when Motecuhzoma was captured, and the Incans were distracted by petty squabbles over the throne.

The critical role of indigenous allies is evident when we look further north. After the conquest of Central America, the Spanish invasion strangely halted. They didn’t move much further north of the present US–Mexico border. But it wasn’t for lack of trying. There were numerous attempts by conquistadors to invade North America, but they failed. Juan Ponce de León attempted to plant a Spanish colony in Florida in 1521. The Calusa (the kingdom of slaving fishers) vehemently disagreed and repelled the Spanish. De León died of wounds received in battle against the Calusa. Several years later Pánfilo de Narváez attempted to lead a conquering Spanish force of 400 inland from Florida. They were constantly attacked by Mississippians and struggled in the terrain. Only four of the original group

survived. Another conquistador, Hernando de Soto, led a group of around 600 Spaniards from Florida to the Mississippi River during 1539–42. The Mississippians constantly harried and ambushed the Spanish, while burning crops to ensure that de Soto's force had nothing to eat. De Soto died of disease on the banks of the Mississippi. Only around half of his force limped back to Mexico.

The English didn't do much better. There were over a dozen attempts to settle the North Atlantic seaboard during the sixteenth century. They all were abandoned, either because of an inability to grow crops and live in the foreign environment or because they were destroyed by Native Americans. The few that did survive, such as the famous first settlement of Jamestown, did so because of support from Native Americans. The colonization of North America took centuries to play out and the colonizers once again had to rely on exploiting pre-existing divisions. The conquest of North America is replete with tales of French and English forces facing off against each other while allied with rival tribes.

The importance of weapons, novel diseases, and native allies varies case by case. Australia was more clearly a matter of guns, germs, and steel. The Aboriginal people had far smaller, dispersed populations that were not well organized for war. This was more a story of newly introduced disease and superior technology leading to a devastated population. In some cases, such as Tasmania, it was simple, brutal colonial genocide.

In contrast, the collapse of the Incan and Aztec empires doesn't look all that different from the cases of Rome or the Han: status competition among elites and counter-dominance across the masses were underlying drivers. All of them faced invaders: the Spanish for the Aztecs and Incans, the Xiongnu for the Chinese, and Germanic tribes and the Sassanids for the Romans. The Spanish had a greater advantage in technology, disease, and surprise, but it was not an entirely different situation. These collapses all had similar root causes.

LOCAL APOCALYPSES

The story of so many kingdoms and empires ends abruptly with these simple words: 'and then the Spanish arrived'. Or the British. Or the Portuguese. Or the Dutch.

There is a long list of states subjugated by European powers. It's easy to forget that Africa was already covered with a litany of different indigenous empires prior to colonization, including the Songhai Empire, the Bamana Kingdom, the Kingdom of Burundi, the Kingdom of Mutapa, the Kingdom of Kongo, the Luba Empire, and the Zulu Kingdom. This map of indigenous African empires would soon be wiped away and redrawn by colonial powers.

Apart from empires and kingdoms, countless indigenous communities, cities, and peoples were lost to colonial expansion. It would take a book thousands of pages long to do justice to the myriad cultures and groups across Australia, Southeast Asia, and the Americas who were hunted, enslaved, and dispossessed. Millions of lives, thousands of cultures, and hundreds of languages were sacrificed for millions of tonnes of gold, silver, sugar, cotton, cardamon, and slaves.

Some of the most tragic examples are the most misunderstood. One such is the Rapa Nui (the natives of what we know as Easter Island). The island off the coast of Chile is renowned for its mysterious *moai* – vast carved stone statues that each weigh up to 270 tons. These dot the coasts and landscapes like eerie sentinels. The Rapa Nui were victims of colonization. A community of 3,000–4,000 had sustainably inhabited the islands for centuries. Then the Europeans arrived. They unintentionally introduced deadly disease, such as smallpox, which ravaged the island. Once discovered, the island became a target. Slavers from Peru carried off hundreds, perhaps as much as half the population, as domestic servants or to work in guano mines. By 1877, only 111 islanders remained.^{[17](#)}

The Rapa Nui are often mistakenly portrayed as a legendary case of self-inflicted environmental collapse. The common story is that they (or imported rats) deforested the island,^{[18](#)} causing massive soil erosion, leading to a decline in agriculture and eventually famine, warfare, and cannibalism before Europeans arrived. This account doesn't withstand scrutiny. There's little evidence of population collapse or warfare prior to the arrival of Europeans. The spears that the Rapa Nui used against each other during their supposed civil war do not appear to be spears, but rather general-purpose tools. While the Rapa Nui did significantly deforest the island, there is little evidence of a mass dying until contact with the Europeans. One DNA analysis of the remains of fifteen Rapa Nui

individuals found no evidence of a large drop in population in the 1600s. Satellite imaging of the island to create estimates of food production produced the same results: there was no collapse prior to European contact. Instead, it appears that their population consistently and gradually increased until the Europeans arrived in the 1860s. The best-known case of collapse is an example of genocide, not ecocide.¹⁹

The mountains of bodies produced by colonization make most other historical collapses pale by comparison. That is not just because the population had grown since the Bronze Age. The proportion of the population killed was also astounding. In the Basin of Mexico, in the century after the conquest of the Aztec Empire, around 55–95 per cent of the population perished. It's easy to brush over those numbers, but pause and ponder it. Imagine your closest friend or partner dying from an unknown disease. They lie writhing in pain, as small blisters form and burst across their body. There is nothing you can do as they take their final rattling breath. Within your lifetime, you lose half or more of the people you are closest to. This was the reality for many communities. In terms of sheer human suffering, these were some of the worst events in history.²⁰

And it wasn't due only to disease. After the Black Death, which killed up to a third or even half of Europe's population in the fourteenth century, Europe recovered in less than a century. In the Basin of Mexico, and many more areas of the Americas, it took two, three, or more centuries to recover. Partly that was because there were multiple waves of different diseases. Yet calculations suggest that if this was due only to disease, the population should have bounced back more quickly, within a few generations. What explains the difference between the Black Death and colonization is oppression. Indigenous groups were frequently starved, brutally overworked, and forced into crowded camps and settlements with poor sanitation and unsafe drinking water. Weakened people died more easily and the camps were superspreaders for imported germs.²¹

This happened around the world, and not just in the Americas. Hawaii and Australia seem to have suffered similar mass death rates of up to 90 per cent over a century or two. Europeans arriving on previously isolated continents would turn 'virgin soils' into bloody fields. In Australia, one official report in the state of Victoria read: 'the most extensive ravages ... will render them extinct within a few years'. Again, diseases were not the

sole culprit. In Australia, the land was treated as *terra nullius* (an empty land, or ‘a territory without a master’) and the indigenous population was forcibly displaced and either detained or killed when they resisted.²²

THE SHATTERZONE

The pustules of smallpox and chickenpox, and the black swellings of the plague, even extended into lands that had not been conquered. The disastrous expeditions of de Soto and other Spaniards into the Mississippi might have failed to claim the land for Spain, but they did succeed in causing numerous societies to collapse. When they had arrived in 1539–40, there were dozens of mound-based chiefdoms and kingdoms with towns of up to 5,000 people. When the French paddled up the Mississippi roughly 135 years later, there were only around three (the Quapaw, the Tunica, and the Natchez). De Soto and other Spaniards had left an unwelcome and fatal legacy. Disease, war, and incessant slave raids were the hammers that shattered these regions, which scholars refer to as the ‘shatterzone’: a realm of instability in which tribes and communities were broken down and new ones were created from the pieces.²³

The shatterzone is essentially a systems collapse of non-state societies. A large group of interconnected, mainly foraging-based groups all fell apart, creating waves of movement and social change. The effects were often hard to predict. Some communities went entirely extinct, their numbers destroyed by outbreaks of disease, slave raids, or attacks from Europeans or other native groups. Some dispersed into smaller egalitarian towns that were more mobile and more easily hidden. In Zomia, huge movements of people fleeing disease and states created a changing mosaic of new groups, identities, and cultures. We see the same occur in the shatterzone of the colonial US. Many of the nations we are familiar with today, such as the Cherokee, were created by people coalescing in the shatterzone.

Others chose a darker path.²⁴ Some groups, such as the Westo, Savannah (Shawnee), and Haudenosaunee (Iroquois), became slavers, incessantly kidnapping their neighbours to sell to the colonists. For the Westo and Savannah, it was good business: they traded slaves for guns, ammunition,

and metal tools, which they could use to catch more people.²⁵ Slave raids by both colonizers and indigenous slavers over the space of a century or two would have chained and dragged away tens of thousands of Native Americans, perhaps over 100,000. In one particularly intense period in Florida (1704–7), native slavers snatched up to 12,000 people from their homes.²⁶

The indigenous groups within the shatterzone became increasingly aggressive, which was an outcome that the colonizers welcomed. In 1680 Governor of South Carolina John Archdale remarked that ‘the Hand of God was eminently seen in thinning the Indians, to make room for the English’ as he watched a civil war unfold between the Westo and the Savannah. The Savannah wiped out the Westo and became the main trading partner of the Carolinian colonists. The Creek (also known as the Muscogee Creek Confederacy), another key slave raider and colonist trader, were later compensated by being recognized as the first Native American group to be ‘civilized’ by the American government. For the Creek and other groups, slaving and aggression was handsomely rewarded.²⁷

The shatterzone was a zone of intermittent collapse. Of the mini-kingdoms or chiefdoms then in existence, few survived. New groups emerged, but into a different world. War and slavery had already existed for centuries, but the shatterzone put these on steroids. Suddenly slavery became a large-scale, commercial endeavour; armed groups would rove over long distances to capture women and children. An arms race developed as new communities competed to sell furs and people in exchange for new iron tools and weapons. Some practices continued: people still hunted, gathered, grew corn, cooperated through reciprocity, had matrilineal kin systems, and organized themselves through clans. Despite these threads connecting them to the past, they suffered a kind of local apocalypse, the end of a world.²⁸

WHEN INDIGENOUS ELITES HELPED EUROPE EAT THE WORLD

Two strange statues flank the entrance of the Royal Palace of Madrid. Both are noticeably different from the other statues of the palace. They don't look European, and are wearing a radically different garb, replete with leg bands and feathered crowns. They are the vanquished Aztec and Incan rulers Motecuhzoma and Atahualpa ([Figure 34](#)). They're not just there to commemorate the conquest of the two empires. Instead, Spanish kings saw themselves as the legal inheritors of the Aztec and Incan crowns. By taking on the crown of the King of the West Indies they were also making themselves the new emperor of the Aztecs and Sapa Inca (the ruler, Sapa Inca meaning 'the one true Inca'). This also meant that the new Incan nobility would become Spanish nobility too under law. It is just another example of *translatio imperii*; even for colonial powers, empire was hereditary.



[Figure 34](#): Statues of Atahualpa (left) and Motecuhzoma (right) at the Royal Palace, Madrid

The Spanish didn't see the Incan and Aztec empires as having collapsed per se. Instead, they had transformed and come under colonial rule. There is a truth to this. There was, at least initially, often little change in the hierarchies of colonized areas. When the Spanish took over the Aztec Empire, they largely adopted the existing taxation system. For the Incans, they simply repurposed the *mita* system of mandatory labour that was already in place. It became the *repartimiento* system, in which Spanish colonists could use indigenous people for forced labour (essentially what was temporary slavery). Similarly, the British initially used the Mughal Empire's taxation system in India and reinforced the caste system.

This was a common approach. European rulers lacked the means, language, or knowledge to govern vast tracts of land and negotiate across myriad groups and dialects. They had to bargain with and rely on indigenous elites. Colonization was not just a matter of subjugation and opposition, but also collusion. In exchange, native elites would get preferential treatment or riches. As noted, Incan and Aztec nobles legally became Spanish nobles, this in return for their assistance in both administering the empire and helping avoid rebellion. Local rulers kept their status by funnelling tax and tribute to the invaders while keeping the peace. It is yet another example of elite migration during a collapse.

European colonialism had non-European assistance.^{[29](#)}

Like the slavers of the shatterzone, many local elites and groups became complicit in the subjugation of their own people. Their cooperation was often not out of a fear of death: indigenous slavers such as the Westo had other options and the local elites of the Incan and Aztec empires still significantly outnumbered their European invaders. Indigenous peoples often allied with invaders out of the allure of benefits, such as guns and tools in exchange for slaves, or the continuation of their high-status noble positions.

That is how remarkably a small group of men managed to take over and run these empires: they relied on striking deals with local elites until more Spaniards arrived. This collaboration meant that the existing culture and systems were often slow to change. It took well over a century for the Spanish to recraft indigenous institutions and impose their language and writing.

Politically and culturally, farmers and peasants would have noticed little change. These were collapses of the pre-existing states – the Aztec and

Incan – and an often slower transformation of much of the rest of the society.³⁰

The economy of colonized lands was melted down and reforged. The Spanish demanded tribute mainly in the form of gold and silver, much of which was then shipped back to Spain. Over the period 1500–1650 around 180 tons of gold, and a staggering 16,000 tons of silver crossed the Atlantic from the Americas to Europe. In agriculture, the exact commodity that was grown varied across colonized lands: in the Caribbean the main exports were ‘cash crops’ such as sugar and tobacco, in the US it was cotton, and in Congo it was ivory and rubber. What this meant was often a simplification of the economy as it became focused on frenetically growing and shipping off luxury commodities to Europe. Energy capture (and often overall economic activity) decreased initially with the large-scale loss of life caused by conquest, but tended to increase over time as more sophisticated tools were introduced to aid the production and distribution of exports.³¹

Some colonized countries did not collapse,³² but nonetheless experienced terrible hardship. In India, GDP growth (a rough proxy for energy capture) increased up to nine-fold during the reign of the British Raj, as did the population and urbanization. It was still a horrifying process. The gutting of India’s manufacturing industry, the introduction of exploitative labour conditions, and a forced shift towards cotton and opium production led to a decline in real wages and an increase in extreme poverty from 23 per cent in 1810 (under the Mughal Empire) to more than 50 per cent in the mid-twentieth century. Life expectancy declined from 26.7 years to 21.9 years. Famines became more frequent and severe, and India experienced somewhere between 50 million and 165 million premature deaths during 1891–1920. Comparing India to Tiwanaku, or many of the other ancient cases we have discussed, is a reminder that there are fates worse than collapse.³³

Perhaps because of cases like India, colonization is often strangely absent from most discussions of societal collapse. Apart from the few cases where an empire quickly unravelled – such as the Aztec and Incan – it is conspicuously missing in most coverage. That is because colonization perversely boosted economic growth and urbanization, even while it ate away the culture, simplified the economy, and murdered millions.

Colonization wasn't just a matter of collapse and genocide; it also marked the global triumph of Goliath.

BABIES, BACTERIA, BOMBS, AND BARBARISM

Five thousand years ago, Uruk and Egypt were fledgling polities covering just a pixel or two of the world map. Despite systems collapses and setbacks, Goliath came to rule the world. While European colonization was the most decisive step in the march of Goliath, empires were already spreading across South and Central America well before the conquistadors arrived. The dominance of Goliath is due to a few basic reasons: bigger populations, more disease, and stronger militaries.

The strongest factor is simple: groups with lootable resources tended to have far more people to stock their armies and economies with. Dense, lootable resources, especially agriculture, allow for far larger, denser populations. It also encourages greater fertility. A carbohydrate-rich, grain-based diet means that children can be weaned onto softer foods earlier, ovulation is encouraged, and a woman's reproductive period is lengthened. Farmers also needed more children to help work the fields.³⁴ Modern studies suggest that, despite significant variation, farmers do seem to be more fertile than foragers.³⁵ Foragers also need far more land. Depending on the environment, a hunter-gatherer may need somewhere between 0.4 and 140 square kilometres. In contrast, Italy has a population density of around 200 people for every square kilometre, while India has 475.³⁶ Agriculture tended to lead to Goliaths with big populations.

Grain states didn't just have higher fertility rates. They also had an obsession with encouraging reproduction. In 18 BCE Emperor Augustus passed a series of laws to increase marriage and children. The Qin of China gave tax breaks to women and their families if they had more kids. And there's no shortage of modern examples of pro-natalist policies that try to boost reproduction. Japan and South Korea both pay people to have babies (about \$2,000 per child). Australia also provides such a subsidy, aptly named the 'baby bonus', and France has had a similar subsidy for years.

Hungary spends 5 per cent of its GDP on trying to stimulate population growth with a slew of loans, grants, and tax-breaks for parents and would-be parents.³⁷ The world's most prominent religions have also promoted patriarchal norms and encouraged fertility. One study of 141 countries found that religiosity was the factor with the strongest correlation to fertility.³⁸

More people living closely together gifted Goliaths with advantages in avoiding disease and winning wars. The barrage of zoonotic infections that developed alongside the first states was initially a curse. Epidemics controlled population levels and tens of thousands died. But the survivors had natural resistances that they passed on. Subsequent generations, especially those living in densely populated urban areas, inherited this biological resilience. Stateless foragers remained vulnerable to these density-dependent diseases; hence first contact, and the first transmissions of tuberculosis, smallpox, syphilis, gonorrhoea, and influenza, proved deadly.³⁹ We see the effects of novel diseases through the mass graves across the Americas and Australia. It may have also been true of earlier forager groups: their mass deaths are just invisible in the archaeological record.

Weapons too made an enormous difference. Steppe-based horseback groups, such as those from the Mongol Empire, periodically invaded and conquered Europe and China because of their mastery of the horse as a tool of war. Their skill as mounted archers and their shrewd tactics made them difficult to defeat. (That is, until military technology outpaced them. Their recurve bows had a longer range than most early guns, but were no match for cannons and rifled breech-loaders.)⁴⁰

Even when non-hierarchical groups did successfully fight back, they usually ended up imitating their enemies. We've seen this repeatedly, such as the Germanic tribes to the north of Rome becoming more state-like. The authoritarian impulse made people more likely to accept a new hierarchy and invading states provided a model to emulate. Even traders introducing new military technology could lead to dominance hierarchies emerging, as with Native American groups turning into militaristic slavers after the introduction of guns. If military innovations didn't lead to the direct conquest of a non-state society, that society would typically convert into a Goliath.

If they didn't change willingly, they were often forced to. The British, when they invaded Burma, preferred to work with autocratic tribes. They had a leader they could directly negotiate with (and bribe) and local elites who could work on their behalf. In contrast, egalitarian groups were a headache to control. To manage the more democratic Chin area of Burma, the British tried to create a Goliath from scratch. They started funding opulent community feasts to help create a high-status chief. In this case it didn't work: the Chins' culture and history proved too powerful and they rejected the chiefly feasts. But for every such failure there are multiple examples where the colonizers got their way and managed at least partly to enforce a new hierarchical structure.⁴¹

The final and most overlooked cause for a Goliath's domination is its willingness to exploit land, animals, and people. Bronze requires tin and copper; iron needs iron ore and smelting; muskets require gunpowder; and the nuclear weapons of today require uranium. All of these involve difficult and often undesirable work such as logging, mining, and smelting. They are usually unscrupulous arrangements involving low pay, gruelling hours, and conditions that will steal years from a worker's life.⁴² As with sugar and cotton plantations, the answer was often to rely on slaves or exploitatively underpaid labourers. That also required an extractive attitude towards the environment and animals. Animist groups who tried to tread lightly on the earth and forbade slavery (like most nomadic egalitarian foragers) were at an evolutionary disadvantage. Without being able to make bronze or steel weapons, or innovate towards guns and bombs, they were likely to be conquered in the long term.

Four key factors led to the global conquest of the world by Goliaths: large populations, better military technology, pandemics, and an eagerness to exploit people and places. In other words: babies, bombs, bacteria, and barbarism. High fertility and large populations protected against disease, allowed for more economic exploitation, and helped maintain stronger militaries. The selection pressure was never for those societies that were the best at maximizing the welfare of their citizens. Rather, it was for who was the best equipped for war. In Mesoamerica that was the Aztecs, until the Spanish arrived. In India it was the Mughal Empire, until the British arrived. European colonizers were just one particularly barbaric and successful example of a long evolutionary process. An evolutionary process

that has always been uniquely tied to collapse. As with Rome, it was eat or be eaten.^{[43](#)}

IS COLONIZATION COLLAPSE?

Colonization meant a new age of collapse. While it paints a complicated picture, there are some general patterns in the relationship between collapse and colonization. When it involved the colonizers meeting a previously separate population, there was almost invariably a demographic collapse owing to the lethal combination of disease and oppression. Where there was already an empire in place, the colonizers would simply hijack the existing arrangements while more slowly destroying and transforming the culture and the economy. When there was no empire, colonization tended to create shatterzones of smashed non-state peoples owing to disease, warfare, or divide-and-conquer tactics. Whether colonization led to collapse, and what the impacts were, depended on geography and the pre-existing polities.

Was colonization collapse? In most cases yes, but colonization was not a single homogeneous force. Instead, colonization was responsible for a mosaic of different collapses across the world. In the Americas entire empires fell apart in years, but the underlying culture took far longer to change. In Australia and elsewhere in the Americas there were shatterzones of transforming indigenous groups that unfolded over centuries. These were far more brutal and horrific than most of the other cases of collapse but tended to involve both hierarchy and energy capture swelling in these newly conquered territories. In places such as India there was extreme hardship yet no collapse. Across the world not all cases of colonization led to collapse, but many did, and, worse, some even resulted in the annihilation of entire peoples and ways of life.

The societies of colonizers themselves never seemed to face a similarly debilitating societal collapse, but there were nonetheless more subdued declines. In the sixteenth century Spain was the most powerful state on the planet with a worldwide empire. Two centuries later it was a second-rate power that had been eclipsed by England and the Netherlands. That decline was largely due to more extractive institutions. The royal fifth – 20 per cent of commodities, including gold and silver, from colonial conquests – was

funnelled straight to the royal treasury. As power was concentrated in the Habsburg monarchy, arrangements became more extractive in general. The monarchs failed to tax the nobility, granted monopolies to preferred guilds, and confiscated property from marginalized groups. The inflow of precious metals from the Americas triggered inflation and led to the Spanish failing to properly allocate resources to productive sectors. They became overstretched and began to fall behind more inclusive and innovative competitors like England.⁴⁴

We see a similar trend take place among European powers even before colonization. In the twelfth to thirteenth centuries, prior to the Black Death, Venice, Florence, and some other Italian city-states became autonomous and began growing rapidly. They still eventually declined, and none became a globe-spanning colonial power. Extractive intuitions were again the cause. In Venice, elites used their power to restrict access to the ruling council and long-distance trade. Eventually, the previously open council became a set of hereditary positions filled by the nobility, who took control of the Venetian navy and used it to monopolize trade. In Florence, elites began to tax the poor heavily and control who could access key markets. The banking sector became monopolized until a few families owned most of it; then one – the Medicis – took political control of the city. Both city-states declined owing to extractive institutions.⁴⁵

While diminishing returns on extraction led to the decline of Spain, Venice, and Florence among others, there were no full-blown societal collapses in the core of Europe. Collapse is a systems-level phenomenon, whether that be of a single unified imperial system like Rome or of a system of states like the Maya or Late Bronze Age. Even if one power (Spain, for example) began to decline, another (England, for example) simply took its place, and the entire system continued to grow. There were crises, but the overall Goliath system ensured that these didn't spiral into collapse. Europe had found a way to subdue Goliath's Curse through technological change, economic development, and more inclusive institutions (in general), and by extracting cheap materials (and later fossil fuels) from abroad. Collapse was something colonizers inflicted on faraway others, not something they suffered themselves.⁴⁶

Some fates were even worse than societal collapse. Working on colonial plantations involved long, intense hours in hot climates in abysmal

conditions rife with tropical disease. For the colonizers, the cost of paying free people a fair wage would have been too high and the huge loss of life in the Americas had made labour scarce and expensive. The easiest alternative was to use slaves. Millions of Africans were captured and transported across the Atlantic to work on sugar, cotton, and other commodity plantations across the Caribbean and the Americas. By the 1780s, almost a million souls were being shipped from Africa every year, accounting for 90 per cent of the trade between Europe and Africa. Europeans also imported malaria into many plantation areas, especially across the southern US, making plantation work even more dangerous. This fed the demand for more malaria-resistant African slaves. There evolved a worldwide military-industrial-slavery complex for producing cheap commodities.⁴⁷

It is hard to overstate the cruelty involved in the transatlantic slave trade. At times the price of slaves was so low that it was simply cheaper for their profit-obsessed owners to work them to death. The conditions were so horrific that the average field hand working on a Caribbean sugar slave plantation would usually only survive for about seven years. The Caribbean became, in the words of one pre-eminent expert on slavery, ‘a slaughterhouse’.⁴⁸ While living, slaves were also subject to humiliation. Slaves under British control were sometimes branded with ‘DY’ (Duke of York) to designate ownership by the British monarchy.

The influx of cheap labour from disposed farms and cheap, often slave-produced raw materials from overseas colonies meant windfall profits for factory owners in Europe, especially England. These enormous sums were reinvested into more equipment and factories, or into the British Navy to help procure more colonies, and hence more slaves and goods. Britain became a machine for accumulating wealth. Capitalism and colonization were combined in slave plantations and crowded factories, beginning around half a millennium ago. This grisly steamroller passing through the colonies also wreaked destruction closer to home, causing the deterioration of the masses of farmers, peasants, and workers back in the European heartland.

Height, wages, health, and life expectancy all fell in England and much of Europe as colonial capitalism rose. Real incomes plummeted as capitalism became ascendant during the ‘long sixteenth century’ (1450–

1640) and enclosure pushed people from the land into towns. In England, life expectancy declined from a high of forty-three in the 1500s to the low twenties during the 1700s. Men shrank from their peak of around 174cm in the wake of the Black Death to around 168cm in 1800. Industrialization and colonial capitalism didn't just usher in a new age of collapse; they also impoverished the ordinary subjects at the core of empire.⁴⁹

Collapse and catastrophe looked different in each region of the world, whether it be imperial collapse in the Americas, shatterzones in Mississippi and Australia, or plummeting health in England. Yet when we zoom out and look at the world across centuries there is one coherent picture. All these collapses and transformation were the labour pains of birthing a new Goliath.

Violence had accompanied the creation of the very first Goliaths across the world. In this case, however, it was not a single state like Uruk or the Xia Dynasty being born; it was a new world system – a global Goliath.

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This long and bloody genesis is carved into the city of Madrid. Just 900 metres from the statues of Motecuhzoma and Atahualpa is the Kilómetro Zero, the point from which distances from the capital are measured and from which six great highways snake out across Spain.

To stand on the Kilómetro Zero is to stand on the physical representation of the rimless wheel of empire, six roads radiating out from one core, one capital. Many capitals have such a monument. Rome's Milliarium Aureum (Golden Milestone) is believed to be the origin of the phrase 'All roads lead to Rome'. The rail networks of the UK and France look alike: rimless wheels leading to London and Paris, respectively. Even Cahokia was structured similarly, with paths across the Mississippi converging on the now desolate collection of earthen pyramids.

The rimless wheel no longer stops at the oceans. In Ghana the railway lines from the interior to key ports on the coast continue as shipping lanes, along which goods such as cocoa from the plantations are sent to Britain. Colonization created a global set of such rimless wheels whose spokes led back to European capitals.

Colonization brought about a new age of collapse, but underlying it are the same old root causes of status competition and dominance aversion.

Colonization was underpinned by a search for luxury commodities such as tea, sugar, tobacco, and coffee – all of which were initially status goods. The empires of the Aztecs and the Incas, as well as many not covered here, were weakened by elite infighting and popular rebellion, which the colonizers then exploited. The Spanish monarchs who approved expeditions into the Americas were searching not just for gold and silver, but ultimately for power and status. Conquistadors such as Cortés and Pizarro were also searching for riches and fame. Their pursuits were successful.

Just over a kilometre away from Kilómetro Zero are streets named Calle De Hernán Cortés and Calle De Pizarro. The two conquistadors received the fame and status they craved. Back at the Royal Palace, beyond the statues of Motecuhzoma and Atahualpa, are vast rooms full of golden chandeliers, ornaments coated in gold and silver from the Americas, walls plastered in ceramics from Paris, the finest fabrics from weavers from Brussels, feasting tables, and marble statues of Spanish kings posing as Caesar. The Royal Palace is not unique; you can see similar opulence in the castles and palaces of Paris and Vienna. They are great acts of conspicuous consumption, physical embodiments of status competition.

But if we dig beneath the palaces and plantations we find ice-age roots. It took just a few psychological features to reshape the world into a system ruled by a few empires and a single, global Goliath.

Collapse in the Modern World

The End of Empire? – Goliath Goes Global – How to Kill a Modern State – A Failed Order – Becoming Anti-Fragile – Earned Through Struggle – A World Without Collapse?

THE END OF EMPIRE?

At the start of the First World War, European empires controlled around 84 per cent of the world's land surface. The Austro-Hungarian Empire covered over 600,000 square kilometres of central Europe, the Ottoman Empire stretched over most of the Near East, the French ruled most of western and northern Africa, while Britain still possessed Australia, Canada, and India. The US was also a colonial power. It had acquired what presidents Woodrow Wilson and Theodore Roosevelt unashamedly called colonies: Hawaii, Guam, the Wake Islands, Puerto Rico, American Samoa, and the Philippines. Together these accounted for around a fifth of the land area of US territory and 19 million people.^{[1](#)}

At least officially, there are no empires left. Over the course of the twentieth century the last land empires fell apart, the maritime empires fragmented, the colonies became independent, and everyone started calling themselves nation-states instead of empires.

As we'll see, collapse today appears to be slowly becoming less common. Yet it would be unwise to assume that Goliath's Curse is disappearing.

The two world wars were the primary spurs to imperial disintegration. The first marked the loss of the Ottoman, Austro-Hungarian, and German empires. While Germany contracted into a smaller nation-state, the Austro-Hungarians lost their empire and became the smaller states of Austria and Hungary (alongside a host of smaller states), and the Ottoman Empire was abolished and eventually divided into countries including Iraq, Syria, Egypt, Turkey, and Palestine. The second war saw the end of the short-lived German Third Reich and the Japanese Empire. The next few decades saw the break-up of the former colonial empires of the British, Spanish, and Portuguese.²

These failures and the contraction of empires were not, however, *caused* by the external shock of the world wars. Instead, the world wars were battles *for* empire. While the Second World War is often framed as an epic clash between democracies and fascism, it was primarily a conflict over land and people (and thus an extension of its predecessor). In 1939, Britain, the Netherlands, France, Belgium, China, the USSR, and the US all had empires. Together they held about a quarter of the global population at the time (some 600 million people) in colonial shackles. They controlled around fifteen times as much land as the Axis powers of Italy, Japan, and Germany combined.

The Axis powers were all growing economies that had largely missed out on the colonial scramble for Africa, the Pacific, and South America. They had increasing power, but few colonies. This was at a time when shipping raw materials from distant colonies was a key driver of economic growth. Hitler's obsession with *Lebensraum* (which roughly translates to living room or space) was forged in the colonial spirit of the day. The Allies were happy with the world order, while the Axis powers wanted to redistribute the spoils. As Hitler himself said, 'What India was for England the spaces of the East will be for us,' adding that the German conquerors would 'look upon the natives as Redskins'. He was mainly referring to eastern Europe. The Axis empires were unsuccessful imperial upstarts thirsting for colonies.³

It clearly wasn't mainly a war of ideals for the Allies either. The US, despite its apparent hatred of fascism, only reluctantly joined the war in December 1941 once it was attacked by Japan.⁴ The allies of the US, Britain, France and the Netherlands – despite their claim to be the defenders

of freedom – still didn't allow their colonies to operate democratically or independently. Just over a year before the beginning of the war, British troops at Vidurashwatha massacred thirty Indians and wounded another hundred for the crime of trying to hoist the flag of the Indian National Congress. The British also showed little concern for their colonies once the war started. During 1943 in Bengal, they seized 40,000 tons of rice and 65,000 boats to deny the advancing Japanese army food and resources; up to 3 million people died during the ensuing famine. Many in India still refer to it as the 'Bengal Holocaust'. Churchill attributed the disaster to the Indians 'breeding like rabbits' and asked why the shortages hadn't killed Gandhi. Such actions are those of empires exploiting their colonial assets to defend their power, not democratic ideals.⁵

The world wars laid the basis for decolonization in the decades that followed. At home, governments struggled to convince citizens who were still recovering from a war to retain possession of their overseas assets. Colonies became a luxury too expensive to keep. Internationally, it seemed hypocritical to keep hold of foreign lands after ostensibly fighting a war in the name of democracy and basic human rights. In the colonies, the wars left populations that were armed and not eager to return to European powers. The indigenous elites who had once collaborated in colonial rule began to turn on their former masters. They joined anti-colonial nationalists and overturned European rule. Importantly, the US and USSR actively supported different independence movements to win them over to their side. The colonial administrators exited in a panicked hurry, often leaving broken and divided countries in their wake.⁶

These were not societal collapses but imperial contractions. There were no large-scale or enduring losses of energy capture or population, and often little change in the rulers, elites, culture, government, or tax system. What did happen is that the empires shrank back to their core homelands.

They shrank, but they didn't disappear. Most countries are the direct progeny of empires. Even in 1948, inhabitants of the UK were still referred to as 'subjects' of the Crown, not citizens. Despite now using the term 'citizens', the country is still built on the past subjugation of many nations. Similarly, France may have lost its overseas colonies, but it was created by the kings of the Île-de-France (where modern-day Paris sits), expanding their rule over surrounding principalities such as Burgundy and Brittany.

The borders of Russia, Germany, and Italy represent previous Russian, Prussian, and Piedmontese aggression. China is the direct descendant of the Qin and in its annexation of Tibet still acts like an empire. The US has 750 military bases scattered across eighty countries around the world and spends \$849 billion annually on defence: around 40 per cent of all government spending on defence globally, and more than the next nine countries combined.⁷ Between 1776 and 2023 it conducted roughly 400 military interventions, with about half of those since 1950.⁸ That long list includes Korea (1950–53), Guatemala (1954), the Congo (1960), Laos (1962), Ecuador (1963), Panama (1964), Brazil (1964), Vietnam (1965–73), Ghana (1966), Cambodia (1969), Chile (1973), and Grenada (1983). The US uses its economic clout, subtly backed by its military might, to shape international policy and acquire favourable trade arrangements. It also still has its overseas territories such as Puerto Rico. Many other former colonial empires still maintain island possessions, with the UK alone having fourteen overseas territories.⁹

Occasionally the mask slips and empire is spoken of clearly. As a senior advisor to President George W. Bush said, ‘We’re an empire now, and when we act, we create our own reality. And while you’re studying that reality ... we’ll act again, creating other new realities ... We’re history’s actors ... and you, all of you, will be left to just study what we do.’¹⁰

We tend to think of the world as being made up of ‘nation-states’. A nation is a group with a claim to a common ancestry – derived from the Latin *nascor*, meaning to be born – yet those rarely exist. Most nations are recent, artificial creations while others with a shared language and history, such as the Kurds (15 million), the Catalans (over 7 million), and the Scots (over 5 million), don’t have their own state. We are closer to living in a world of united empires than united nations.¹¹

Empire in the twenty-first century has also taken a new, subtler form. Demanding tribute as did the Assyrians or tax like the Romans has been phased out in favour of unequal trade. After the end of the Second World War, both of the new superpowers – the US and the Soviet Union – flexed their newfound status. The US would have the more prominent legacy of the two. It helped to shape the global institutions that order the world’s finances today: the World Bank and the International Monetary Fund (IMF). Since then, the decolonized countries have been saddled with onerous debt

and an unbalanced trade system. For instance, the EU heavily subsidizes its agricultural products, which are then imported into Africa at low costs, undermining local producers who don't have the benefit of government support. It's easier, cheaper, and now more politically acceptable to take resources through loans, debt, unfair agreements, and infrastructure projects than to send out an army to a distant land.¹²

The result is a new global system of extraction. Industrialized countries, including all the former colonial empires, extract cheap labour and materials from poorer countries. Researchers in 2022 tallied up the costs and benefits of this unequal international trade. They found that over the period 1990–2015 \$10 trillion was transferred every year from the global south to the global north. That is thirty times what poorer countries receive in aid each year. Overall, \$242 trillion was moved during that twenty-five-year period, enough to eliminate extreme poverty seventy times in a row.¹³ A more recent study just looking at labour over the period found that there was a net flow of \$17.6 trillion. This is largely due to differences in wages: labourers in the global south are paid 87–95 per cent less than their northern counterparts even when equally skilled.¹⁴

Unequal trade isn't the only reason these countries are rich, but it is one key contributor. The more autocratic institutions left by some colonizers in former colonial countries also contribute to large differences in economic growth today.¹⁵ Empire never truly came to an end. The world is still covered with empires operating under a single Goliath system.

GOLIATH GOES GLOBAL

We can use the different sources of power to understand the Goliath we live under today. Economically, every country is now capitalist.¹⁶ They all abide by similar rules and agreements, with many enforced by a few international bureaucracies such as the IMF, World Bank, and World Trade Organization. The flow of capital is dictated by shared blueprints of collateral law, property law, trust law, and corporate law. It's why money and capital can flow so easily across borders. Most of these legal systems are modelled on English common law and New York state law. Information

power is increasingly concentrated. A small network of surveillance capitalist firms – such as Google, Amazon, and Meta – gather, direct and monetize data. This network of surveillance capitalists is accompanied by emerging mass surveillance states – especially in the UK, the US and China – that unblinkingly monitor these streams of data.

Many of the world's elite, tyrants and progressives alike, have had a similar cultural upbringing and education. Monarchs, oligarchs, and the megarich tend to be educated at a handful of elite schools and universities in the UK, US, and Switzerland. North Korean leader Kim Jong Un was schooled in Bern, Switzerland. The former kings and princes of Belgium, Burundi, Iran, and Yugoslavia all attended the Institut Le Rosey in Switzerland, sharing the playground with members of dynastic families such as the Rockefellers and Rothschilds.

Even the practice of shaping the world through law is clustered. All the world's top hundred law firms are headquartered in just two cities: London and New York. Politically, every polity is now a nation-state. To be part of the UN (headquartered in New York) you need to have a state and be recognized by other major powers.

Violence is still mainly (although not always successfully) controlled by a few key powers, namely the US and, to a lesser extent, China and Russia. Yet violence is no longer the main source of control. Why launch a missile or a battleship when you can rely on the law to enforce extractive arrangements? We now all live in a single capitalist system pioneered by the British and then moulded by the US.¹⁷

The creation of that single Goliath has also changed the nature of collapse itself.

HOW TO KILL A MODERN STATE

Modern states die regularly. Since 1816, 202 states have existed. Fifty of those no longer exist, including imperial powerhouses such as the Ottoman Empire as well as the lesser-known Kingdom of Hawaii.¹⁸ Thirty-five of them came to violent ends.

Academics usually call these examples 'state death'. State death can be something of a misnomer. It is not that the government entirely disappears,

or everyone dies. Rather it is simply the loss of sovereignty. The state formally loses its ability to make foreign policy and is no longer recognized as independent. It usually means one state conquering another or breaking up into multiple smaller ones. A violent (albeit temporary) example would be that of Poland being invaded by Germany at the outset of the Second World War. A less bloody case would be Bavaria joining the Prussian Kingdom in 1871.¹⁹

The two most common causes of sovereignty loss are being a 'buffer state' trapped between two great powers and getting caught in a territorial dispute. One review of hundreds of cases during the period 1816 to 2004 found that being a buffer state, like Poland caught between Germany and the Soviet Union, was the leading cause of sovereignty loss.²⁰ An updated analysis found that conflict over territory is 40 per cent more likely to contribute to a loss of sovereignty than being a buffer state.²¹

Small states with little power over the international system have been the biggest victims. While great powers like the US have proved robust, small states tend to lose their sovereignty frequently. We currently have around 150 small states in the world, but there were fewer than forty in 1900, and almost 200 in 1648.²² That is a lot of state births and many losses in sovereignty. The key reason seems to be the actions of great powers. For instance, the explosion in state numbers during the twentieth century was largely down to the break-up of imperial powers and competitive decolonization between the US and the Soviet Union.

Today sovereignty loss appears to be disappearing from the world. Only nine states have lost their sovereignty since the end of the Second World War. States today are 94 per cent more likely to survive in any given year than states prior to 1945.

Again, the actions of the most powerful are the reason for the slow vanishing of sovereignty loss. Since 1945 a strong norm of 'territorial integrity' has reigned over the world order. International outrage follows any intervention or invasion into a sovereign country.²³ This regard for sovereignty has been largely enforced by the nuclear-armed US to protect the world order it built. In the twenty-first century we still have the hypocritical invasions of Iraq and Afghanistan, the invasion of Ukraine by Russia, and the brutal campaign of Israel in Gaza. While this is still far less

than what was experienced in the twentieth century or earlier, they are signs that the protection of sovereignty may be eroding.^{[24](#)}

Most cases of sovereignty loss were transfers of power, not societal collapses. Even the few examples where a state did entirely fall apart follow a peculiar pattern.

A FAILED ORDER

The modern world has seen many cases not only of sovereignty loss but also of ‘state failure’. These are what most of us think of as modern collapses: Somalia, Sudan, Afghanistan, the Democratic Republic of Congo, and many others.

There is no single universal definition of state failure, which has made analysis of modern collapse difficult. One of the few systematic investigations of state failure tried to address this by providing a clear, operational definition.^{[25](#)} Failed states were defined as those that had lost their capacity to make laws, extract taxes, and control violence within their borders. Indicators were made for each of these and then searched for case studies, and then applied to seventeen cases of potential state collapse over the period 1946 to 2014. The study found five cases that met all these criteria, five that met two, and four that met only one. Yet not one of these cases lasted longer than six months. Within half a year, each one had a government that was able to collect taxes, make laws, or enforce some control over violence.^{[26](#)}

Fragile and failed states too are largely a legacy of colonial empires. Both colonization and the hurried, muddled process of decolonization left a patchwork of scarred countries with nonsensical borders. About 40 per cent of the world’s borders today were drawn up by France or Britain. These borders were usually imposed for the convenience of colonial administrators and rarely reflect the different political and ethnic groups they encompass. It’s entirely unsurprising that artificial borders drawn by foreigners with a history of stripping the land for slaves and resources left many countries fragile.^{[27](#)}

Not all colonizers were equally brutal. The British and the Spanish, despite their atrocities, were more light-handed than the French and Portuguese. We see this in rates of modern state failure today. One study of 118 states found that those with more European intervention, especially the more extractive French or Portuguese empires, had a higher risk of state failure. The actions of colonial administrators, including the judicial systems they put in place, the ethnic divides they deepened, and the groups they empowered, all left weak, divided, unaccountable governments that were more prone to falling apart.²⁸ As the English writer and correspondent Robert Fisk reflected,

After the Allied victory of 1918, at the end of my father's war, the victors divided up the lands of their former enemies. In the space of just seventeen months, they created the borders of Northern Ireland, Yugoslavia and most of the Middle East. And I have spent my entire career – in Belfast and Sarajevo, in Beirut and Baghdad – watching the people within those borders burn.²⁹

We can also see the legacy of colonialism when we contrast different kinds of colonized countries. Compare the settler colony of Australia to the non-settler colony of Sudan. Both were colonized by the British, but while settler Australians were given stable property rights and relatively inclusive governmental institutions modelled on Britain's, Sudan got neither. Those differences in institutions made for large changes in their long-term trajectories. (Notably, even in the case of Australia, Britain actively demolished the property rights and inclusive institutions of the indigenous Australians.)³⁰

The corrosive effect of extractive institutions isn't just an economic or political phenomenon. It is also gendered. The more strongly states subjugate women, the more likely they are to be both autocratic and prone to failure.³¹

Given their colonial origins as extractive constructs, it is difficult to call most of these 'failed states'. Afghanistan, Congo, and Somalia were never 'successful' and robust modern welfare states. They were the haphazard constructs built from the last dying gasp of colonialism. The failure here is more the international order, rather than the states that were born from it.

And there is the genuine problem that thinking of them as ‘failed’ can justify military interventions rather than humane policy.³²

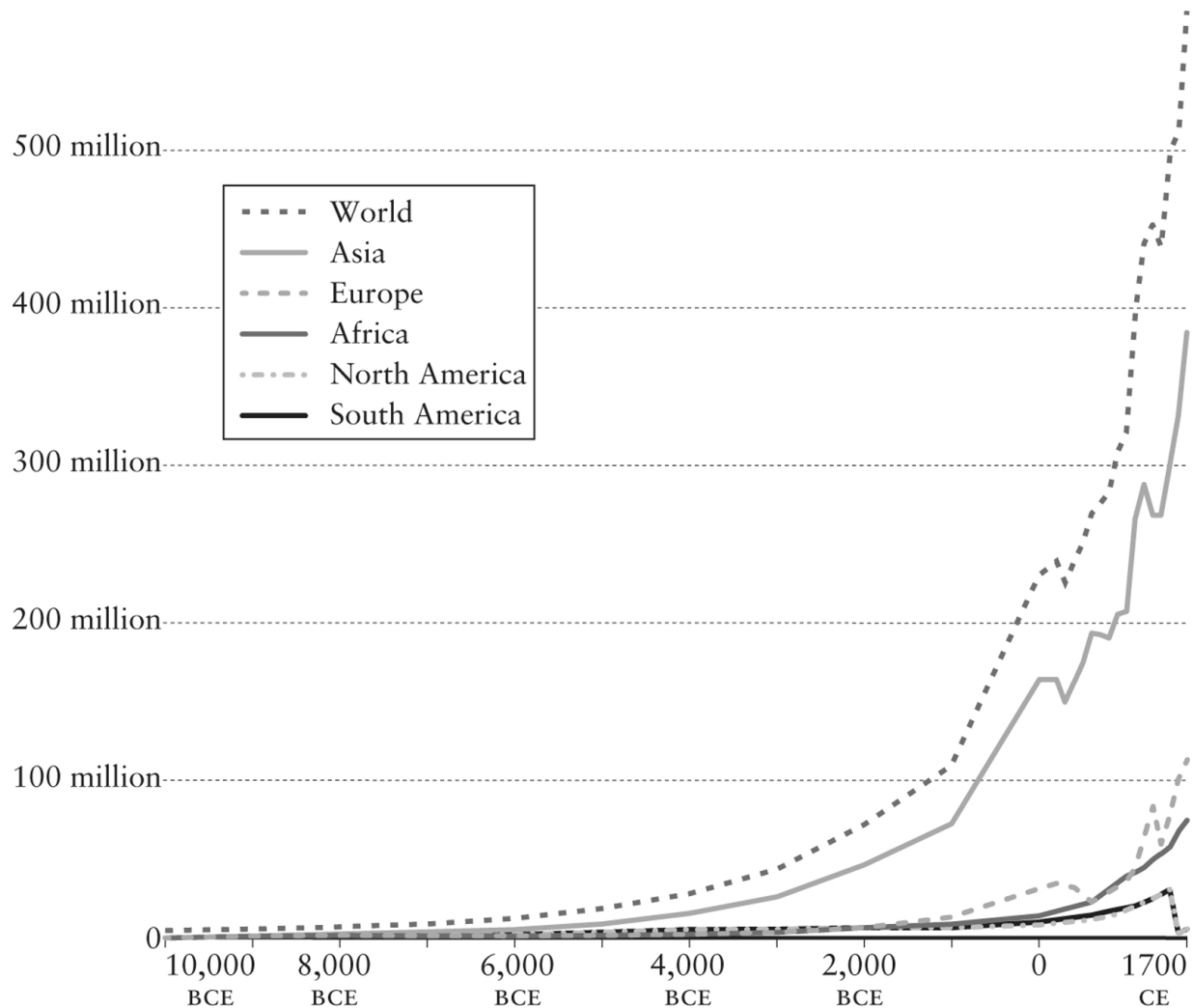
There was more to it than just colonization. Colonial institutions were a key vulnerability, but other factors such as environmental degradation and population pressure may have played a role. These varied across cases and don’t appear to be as common or strong a contributor as colonial institutions.³³

State failure, when it happens in our modern world, seems to be briefer (usually less than six months) and more quickly recovered from. Look at the fall of the Soviet Union in 1989–91. GDP was halved within a few years but it had tripled above pre-fall levels within two decades.³⁴ The ruling elites changed, but not dramatically. Putin was, after all, already a high-ranking lieutenant colonel in the KGB before the Soviet Union fell apart and he switched to politics. We can talk about the collapse of the Communist regime, but not of societal collapse.

There’s a clear trend here: societal collapse, or anything remotely close to it, like state collapse, is happening more quickly, as is recovery. This is part of a longer-term process of the world accelerating, and both people and Goliaths getting better at dealing with collapse.

BECOMING ANTI-FRAGILE

The history of humanity is one long story of resilience. Global population has grown steadily over the past 12,000 years. Despite plague, earthquakes, societal collapses, world wars, and volcanic eruptions, our numbers climbed from around 4 million at the beginning of the Holocene 11,700 years ago to around 600 million by 1700 ([Figure 35](#)). From there it exploded to the 8 billion souls living today.³⁵



[Figure 35](#): World population 10,000 BCE to 1700 CE

We see that same resilience at a more zoomed-in level in the most extreme scenarios – even a nuclear attack. In August 1945, Hiroshima was struck by a sixteen-kiloton atomic bomb, reducing 40 per cent of the city to ash, destroying 90 per cent of the buildings, and extinguishing 140,000 souls. Yet, within a day 30 per cent of homes had electricity restored, within two days the local Bank of Japan was running again, and within a week some telephone lines were working again. Today Hiroshima is a thriving city with over 1 million inhabitants.^{[36](#)}

Resilience is the flipside of collapse. It is the ability of a system to experience disturbances without changing its fundamental functions or identity. Imagine a tree being buffeted by a hurricane but never breaking, or

a blade of grass being stepped on but springing back up. Both are displaying resilience.^{[37](#)}

Societies can be something even stronger than resilient: they can be anti-fragile. That is, they gain and improve from disturbances and disorder. Think of the human body. Lifting a heavy object will temporarily shred your muscle fibres, but they will grow back stronger. A healthy amount of stress makes us stronger. It appears that this may hold true not just for human bodies, but for entire human populations.^{[38](#)}

Populations that have experienced more downturns in the past tend to be more resilient and more likely to recover from later downturns. One study covering sixteen world regions (including China, Europe, Australia, North America, South America, South Africa, and the Near East) over 30,000 years found that populations that had experienced more frequent downturns became more resilient to later ones. The effect appeared to be more pronounced in herders and farmers than in hunter-gatherers. Farmers and herders faced far more demographic busts but became more resilient in the long term. In other words, their populations are anti-fragile.^{[39](#)}

The reason for this anti-fragility is cultural evolution. Over time, people develop technologies and cultural practices that allow them to survive and prosper in hostile environments. Indigenous Australians have developed cooking methods to detoxify local foods and the Inuit built kayaks and specialized harpoons to prosper in the Arctic.^{[40](#)} This is not a matter of individual intelligence, but rather methods to deal with adversity that are refined and passed on down numerous generations.^{[41](#)}

When societies encounter frequent hardships, such as climate change or disease, they tend to learn from experience and develop better ways of coping and even benefiting from shocks. Learning from disasters has been the trigger for a range of innovations, including quarantine for managing the spread of disease, buildings that can sway and buckle without breaking during earthquakes, and the use of back-burning to prevent bush-fires. Communities that experienced these shocks more frequently were more likely to develop the technologies and practices to survive and even thrive.^{[42](#)}

This effect is even stronger among farming societies across the world because their reliance on agriculture meant they faced more shocks. While

hunter-gatherers could move and switch to other food sources, farmers were often tied to their lands and particular crops. They were forced to experiment more. Technologies such as writing then enabled the larger-scale storage of information. This helped ensure that more successful adaptations were identified and passed down to the next generation.⁴³

Goliath evolution is an extension of this. Bigger, longer-living empires tended to pass on their ideas, not only to their own citizens, but also to those they conquered. Those that were bigger tended to have technology and practices that allowed them to control more people, more tightly, and field larger, better-equipped armies. The growing size of empires over time is a sign of their anti-fragility. Individual empires like Rome may have fallen apart, but the project of empire became more effective over time.

We see this in the increasing maximum size of empires over the course of history. Between 3000 and 600 BCE empires varied in size between 0.15 and 1.3 million square kilometres. Then there was a burst in growth between 600 BCE and 1600 CE, to between 2.3 and 24 million square kilometres. While the Mongol Empire reached 24 million, it couldn't hold it for long; instead, most lasting empires, such as Rome or the longest Chinese dynasties, tended to be 5–6 million square kilometres. The maximum size advanced again after 1600 with the ascent of maritime colonial powers. Then, empires became enmeshed to create the Global Goliath we have today.⁴⁴

This slow change in the size (in both territory and population) of empires and Goliaths represents a slow accumulation of new forms of Goliath fuel. The most obvious was monopolizable weapons. Hand-held bronze weapons provided the bloodied bricks for building the original states in Egypt, China, and Mesopotamia. Later, larger empires like the Achaemenid, Roman, and Han all came to rely on horses and iron weapons. The spread of colonial maritime empires was similarly in part due to their advanced guns, ships, cannons, and steel weaponry.

Lootable resources have also changed and scaled up over time. Grain has been a perennial example of a loutable resource. Not only is it highly stealable, but it also favours fertility and bigger populations. For most of history, this didn't change much.⁴⁵ The biggest change came with colonization, when slavery-based mines and plantations allowed for the far greater production of loutable resources.

Ways to cage land also transformed over time. While many of the earliest states erected walls, others had to settle for smaller palisades (as was the case for Cahokia). Later empires made a fine art of wall building. One only needs to look at the Great Wall of China or the Romans' Hadrian's Wall. Modern empires established far clearer borders with customs, passports, and hawkishly guarded points of entry (and exit).

Goliath fuel didn't just change; Goliaths also developed better ways of governing. The Iron Age didn't just see the formation of larger empires like Rome and the Han; it also saw the creation of far bigger bureaucracies and the first censuses.⁴⁶ New transportation and communication tools enabled rulers to send orders and troops more quickly across vast distances. For thousands of years Goliaths could not cross the ocean. Then, European caravels allowed for it, while the telegraph and rail allowed for the control of distant colonies.

That process of Goliath evolution – developing new fuel, as well as better governance and technology – was driven by competition. The collapse, or at least decline, of some states, was a requirement for a Goliath to evolve. Some states have even proved to be anti-fragile during a systems collapse. This can be seen during the Late Bronze Age collapse. The Phoenicians profited immensely from taking over the trade routes that had previously been run by the now destroyed city of Ugarit as well as by the Egyptians and the Hittites. They pushed their trade south and west, becoming a key regional player in the wake of the collapse. By contrast, the Neo-Assyrians (911–609 BCE) opted for a more sinister approach. They began to conquer the weakened states that surrounded them in the post-collapse region. They slowly annexed the remains of the Hittite Empire (the 'Neo-Hittite' states), defeated the Armenian Kingdom of Urartu (in modern Turkey), and marched all the way to the Egyptian border. They became the largest empire the world had seen until then. Both the Phoenicians and the Neo-Assyrians were anti-fragile, but in dramatically different ways. Yet the most persistent and anti-fragile project is the 5,000-year-long process of Goliath, not any individual empire.⁴⁷

The quicker recovery from state failures in the modern world may partly reflect this long history of anti-fragility. States have developed police forces, full-time militaries, and administrations that make it easier to regain control after a disaster – or even a civil war – has struck. Similarly,

emergency-response and risk-management professionals can draw on decades of studies and accumulated practices to deal better with new crises. It doesn't mean they always get it right, but it may help explain why state failures today are usually so brief.

The brevity may also simply be due to technology. Military, police, peacekeepers, diplomats, and aid workers can now be quickly deployed by air and closely coordinated using the internet. Whatever the reasons, collapse is certainly shortening and disappearing while many areas of life are improving.

EARNED THROUGH STRUGGLE

The modern world has undoubtedly experienced enormous improvements that should be celebrated. It is a lengthy list that includes eliminating smallpox, increasing literacy from 12 per cent in 1820 to 87 per cent in 2022, and, over the past 120 years, cutting infant mortality from 43 per cent to 4.5 per cent, and doubling average life expectancy. As a testament to our burgeoning health, we are also now taller. Since around the nineteenth century, global average height has increased by around 8cm for women and 9cm for men, although men today are still shorter than their Palaeolithic forefathers. These are momentous achievements that we should appreciate.^{[48](#)}

The catch is that all these trends are recent, usually within the last century or two. The long-run history of progress is something more akin to a U-curve. Happiness, health, peace, and prosperity start relatively high, fall during the adoption of Goliaths, stagnate for about 5,000 years, and then begin to rise again (at least in some areas). That rise is primarily for two reasons: massive increases in energy capture and collective action.

In the last two centuries, humankind has tapped into a geological gift. Fossil fuels have bequeathed us a previously unimaginable source of energy. Measured in calories, the average modern European captures over fifty times more than the average European ice-age hunter-gatherer. This enormous increase is largely recent: global energy consumption increased over thirty-five-fold from 1800 to 2022.

We tend to forget that fossil fuels come primarily from long-dead plants and animals. These organisms died between 360 and 286 million years ago during the Carboniferous period, after capturing sunlight through photosynthesis or other means. It is that fossilized energy that we are consuming. According to one estimate, it would take 400 years of global photosynthesis to power the modern world for one year. It takes ninety-eight tons of organic matter buried during the Carboniferous to become just five litres of petrol. We are now a high-energy Goliath, powered by dead matter.^{[49](#)}

With such gargantuan increases in energy, it would be surprising if we did not see significant bursts of human progress. It's easy to assume that the rise of capitalism around 1600 CE, and then industrialization about 150 years later, led to improvements in the human condition. But this isn't the case. In fact, the opposite occurred.

As we've seen, even the vast releases of power and energy during industrialization in the mid-eighteenth to nineteenth centuries did not automatically lead to better living conditions for most. Early on it was a period of exploitation and deterioration, with both heights and real wages falling. In England, the pioneer of industrialization, reports by Parliament in 1833 detailed children working twelve- to fifteen-hour days in overheated and cramped coal mines, who were beaten near daily, and had little in the way of compensation or rights. Reliable figures are hard to come by, but fatalities must have numbered in the thousands per year and injuries in the tens of thousands. Many were caused by unguarded machinery and children fitted with loose clothing, leading to truly grisly reports of children being pulverized in machines. One 1892 memoir recounts the death of Mary Richards, who was caught in a malfunctioning drawing frame before being dashed to the ground and when 'she was extricated, every bone was found broken – her head dreadfully crushed'. The average height of men enlisting in the army in the 1890s through to the First World War was 168cm, still below their peak before capitalism. There is a reason why the poet William Blake referred to factories at the time as 'dark Satanic Mills'.^{[50](#)}

What eventually caused industrialization to translate into shared prosperity was shared struggle. Improvements in health and welfare occurred during the 1880s in northern Europe and the mid-twentieth century in Latin America, south Asia, and sub-Saharan Africa. In Europe

this coincided with the rise of unions, public health movements, and the expansion of the vote. Elsewhere it was anti-colonial movements and public pressure for improved services that turned the wheels of progress. The biggest single measure to lengthen life expectancy – the gift of sanitation – was rolled out following public protests, especially from unions. Innovation and invention are important: vaccines and sanitation needed technological breakthroughs. Energy and technology are vital: they are necessary conditions for the improvements we've seen in the past two centuries. But they weren't enough in and of themselves. Improvements needed to be demanded and fought for.⁵¹ It needs the right social structures and sharing of power. As Frederick Douglass once said, 'If there is no struggle, there is no progress.'

If power structures remain exploitative, human welfare can even worsen while technology improves. It is visible across history and even today. The US has the most well-funded and advanced biotechnology sector in the world. The National Institutes of Health (NIH) funds around \$9 billion in biotech research every year, while tens of billions are invested in the private sector. Despite this, the US has the highest rate of maternal and infant mortality, the highest death rates for avoidable conditions, and the lowest life expectancy at birth among rich countries. The US may have far more start-ups and investment, but what it lacks is universal healthcare. Having the government negotiate with big pharma on behalf of people would make a bigger difference to people's lives (and premature deaths) than the next innovation breakthrough.⁵²

Progress is also always a matter of perspective. We forever need to ask: progress of what? The advances we usually talk about and plot onto graphs are improvements in health and material comfort. But if we begin to talk about improvement in our inner lives – happiness and social connection – or treatment of the environment, suddenly it appears that most societies have regressed since egalitarian hunter-gatherers. As we've already seen, happiness studies of nomadic egalitarian groups suggest that they are happier than their industrial counterparts in over a dozen countries examined. The average life today is, though, not that of a middle-class Dane, but closer to a far less happy Chinese factory worker or Indian peasant farmer.⁵³

We also need to ask: progress for whom? The rise and expansion of the Global Goliath has been responsible for the extinction of tens of thousands of species. Countless unique pieces of life trampled under an industrial boot. In the eyes of the dodo or the passenger pigeon, our recent advances don't seem particularly admirable. The over 100 billion animals that are tortured and killed every year in factory farms might not find our increasing heights or health to be a sufficient excuse for their unimaginable suffering.

The same can be said for the ways of life that have been devastated and sidelined at the barrel of a gun. Today there are a maximum of only 5 million hunter-gatherers across the globe and their numbers dwindle every decade, alongside hundreds of languages and cultures.⁵⁴

A WORLD WITHOUT COLLAPSE?

Since at least 1945, enduring societal collapse has been absent from the world. The decades have been scarred by civil wars, genocide, losses of sovereignty, and state failure, but none has resulted in a lasting societal collapse.⁵⁵

This suppression of collapse is caused by a globalized, state-based, capitalist system underpinned by US supremacy. Whenever a state is even close to falling apart, the international community tends to mobilize. The US, desiring stability and seeking to prevent any potential wellspring for terrorism, has deterred most territorial conquests and subtly interfered to prevent (and sometimes induce) state failure. It even had its own government-sponsored and CIA-commissioned taskforce on political instability and state failure.⁵⁶

Fossil-fuelled capitalist growth has similarly ensured that losses in energy capture are not long-lasting. As Naomi Klein documents in *The Shock Doctrine*, natural disasters, coups, and state failures were instead often seen as a commercial opportunity. These ruptures were capitalized on by outside forces to open new markets and force in 'free-market' policies. These were often a thin veil for economic exploitation and unequal exchange.⁵⁷ For instance, US economists and international institutions pushed for privatization in the wake of the fall of the Soviet Union. The result was the

rise of Russian oligarchs who picked up former government enterprises at bargain prices, often promising favourable deals with the US and others.⁵⁸

It may seem that Goliath is on an arc towards defeating the curse. Goliath first faced the Primal Curse of a rebellious populace and weak ways to control them. Larger empires overcame these challenges but faced diminishing returns on extraction. More advanced colonial empires appeared to find a way of avoiding the curse: rapid technological change and overseas extraction allowed them to spread globally without a societal collapse in Europe. States still eventually fell apart during the world wars and decolonization, but it was more of a transformation than a collapse.

True collapse appears to be slowly disappearing from the modern world. That 'disappearance' is, however, an illusion. Over the long run, collapse appears to have, if anything, accelerated from hunter-gatherer cultures which endured for tens or hundreds of thousands of years, through the long-lived earliest settlements and cities, fragile empires, to often short-lived and quick-collapsing modern regimes. While states have taken over the world, they are not particularly long-lived.⁵⁹ Collapse just seems to have temporarily slowed, but has been accelerating ever since the Holocene and the creation of Goliaths. Yet the threat of collapse still hangs overhead and, if it comes, it will be far worse than anything that has gone before. The curse is now global and more dangerous than ever.

Part Three

ENDGAME

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Mors ex Machina

Digging Up Death – From Human Exploitation to Human Extinction – Goliath Becomes Death – When Our Luck Runs Out – Reviving (and Upgrading) our Greatest Killers – Automated Cognition – The AI Accelerant

DIGGING UP DEATH

Your plastic flip-flops slip in the dirt as you swing the pickaxe. The earth is red, but down here deep in the bowels of Congo, it looks a shadowed, bloody brown. It's dark and the walls seem to press around you. Those walls have always seemed closer, more ominous, ever since your oldest brother, Akile, died when a tunnel collapsed on him. Your other brother, Kijani, was pulled out but hasn't walked since. You remember that day, the way the tears felt on your dusty face, that moment you realized that Akile's smile wasn't coming up from the ground ever again. The crying didn't last long. You have to work, especially now that Kijani can't.

Your stomach rumbles as you swing again. You skipped lunch and now your head aches with exhaustion as you pass your tenth hour of work. It seemed like a good idea at the time: you'll make an extra dollar without the lunch break and by working overtime. It wasn't a hard choice since lunch is just two tasteless bread rolls and a carton of sugary water that barely passes as juice. The air is heavy with black dust which sticks to your clothes and coats your throat. If you're lucky, you might make a bit over \$4 today.^{[1](#)}

The cobalt – that bright, blue-green-grey rubble that you smash out from the rocks a hundred metres below the surface – will be worth far more than your tears. After being carried by wheelbarrow to the truck it'll be shipped to China for transformation into those tiny, patterned silver and gold squares that we call microchips.

Some of the cobalt you have mined eventually ends up in a chip that is soldered to a computer now in San Francisco, powering a new Large Language Model for an AI company. The owner of that company has over 1,000,000 times more wealth than you do. That won't change, no matter how long you spend toiling underground.^{[fn1](#)}

Other grains of cobalt line the chips inside a drone that cuts through the dry desert air of Libya at 72 kph. Packed snugly next to those microchips are explosives. When it spots its target, the multicopter drone will dive down at 120 kph until it smashes into them. It won't even need a human to approve of that kamikaze murder. The microchips are all the authorization it needs. It's killing autonomously. Blood and bone will blend with the cobalt you once dug out of the ground.

This is just one story of the unseen faces that lurk behind the production of global catastrophic risk today. The specific names are fictional, but the story is not. And that one tale doesn't do justice to the scale of the exploitation. Around three-quarters of the global cobalt supply – used in the manufacture of batteries for phones, computers, and electric vehicles – today comes from the Democratic Republic of Congo. Some 15 to 30 per cent of that is mined by what are referred to as 'artisanal' miners. It's a euphemism for people working in unregulated and dangerous conditions. Up to 40,000 of them are children who spend their days in squalid tunnels that are prone to collapse. Around the world there could be up to 100 million informal miners of all ages digging up not just cobalt, but many other rare earth metals such as the lithium used in your phone, or the tantalum used in semiconductors. The costs of quarrying cobalt are carried across bloodlines. Workers have such high rates of toxic metals in their blood that they are five times more likely than average to have children with birth defects.^{[2](#)}

Over eight decades ago, in what was then the Belgian Congo, a different mine was birthing a different global threat. The Shinkolobwe mine was worked by Congolese miners day and night in appalling conditions under

the draconian supervision of its colonizing authorities. Medical and birth records were seldom kept, but there are reports of many children born with congenital defects (as with cobalt). The mine is named after a kind of apple that, if squeezed, leaves a skin burn. It's a telling name, since many of the people working in the mine suffered from the hallmarks of radiation sickness. That's because Shinkolobwe is the world's purest uranium mine.

Shinkolobwe was the mine that provided around two-thirds of the uranium for the Manhattan Project: the 1940s US research and development programme to build the first atomic weapons. Uranium from Shinkolobwe was present in the first atomic test bomb – the Trinity bomb – that was detonated in the deserts of New Mexico in 1945. Shinkolobwe uranium was also in the two bombs (named 'Little Boy' and 'Fat Man') dropped on Hiroshima and Nagasaki later that year. Some of the 1,000 tons of silvery-white rock that was carved out of the Congo would eventually be vaporized and its particles burned into the bodies and buildings of Japan.³

There are many links in this chain of production from exploited miners to global risk. Another is the coal miners in the Shanxi province of western China during the rise of that country's economy two decades ago. There, workers lived in shared dormitories with unclean sheets and beds as hard as stale bread. They were paid \$7.50 a day to risk their lives. One man whose injuries when a mine caved in resulted in him becoming a paraplegic received no compensation from his employers, nor even a car ride to the hospital. Another, already without front teeth owing to mining accidents, continued to work despite having a broken shoulder.⁴

The coal the Shanxi miners excavate is shipped both across China and around the world to be burned at around 400–500°C. Coal is around 45–86 per cent carbon, and that carbon reacts with oxygen during the burning process to create carbon dioxide (CO₂), a greenhouse gas that floats into the atmosphere and acts like a blanket, trapping solar radiation and heating the earth. The earth's atmosphere now includes 427 parts per million of CO₂, and that figure is growing every year. The coal dug up in Shanxi will eventually become the gases that supercharge hurricanes and melt the polar ice caps.

FROM HUMAN EXPLOITATION TO HUMAN EXTINCTION

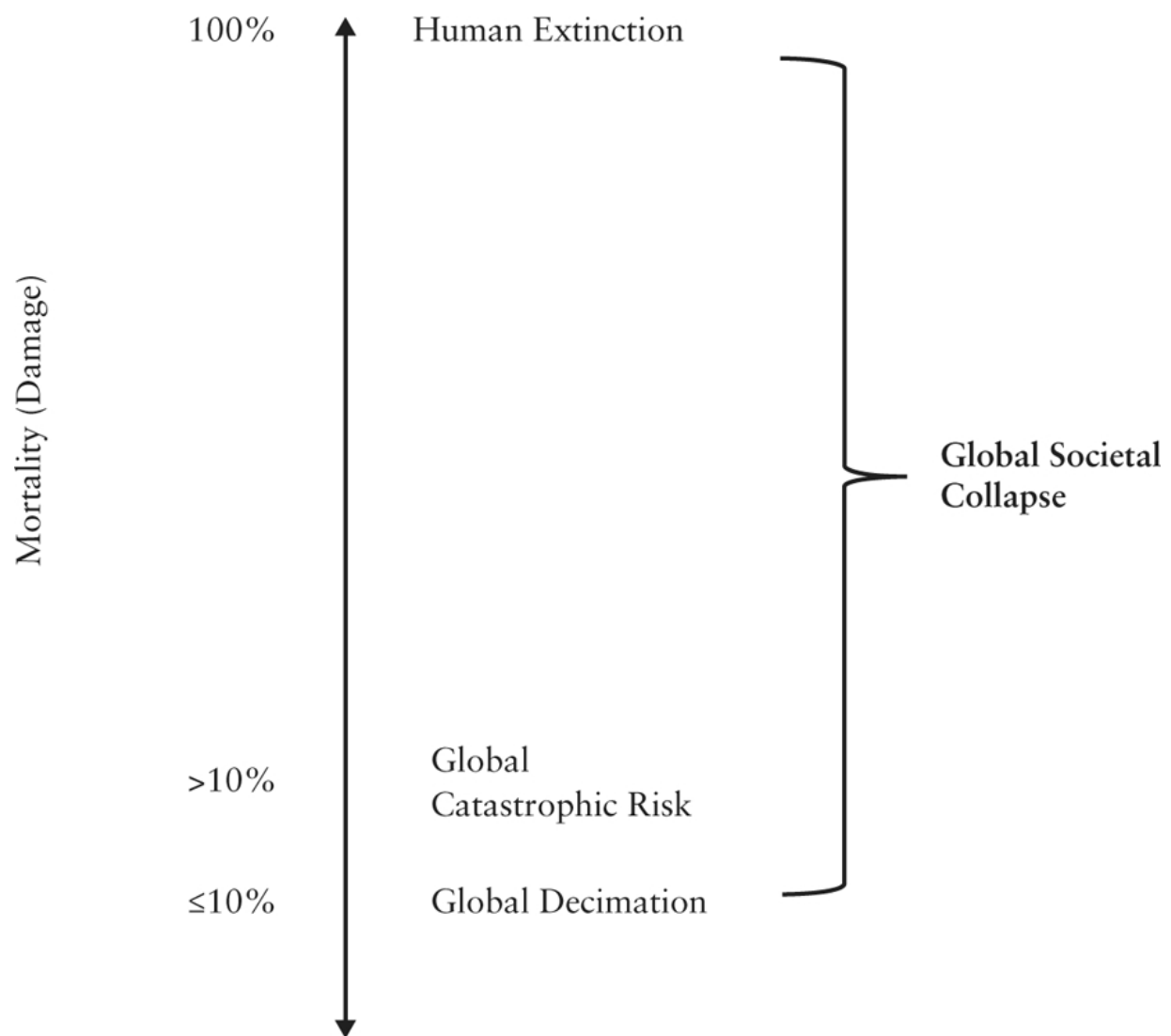
It is deceptively easy to think about killer drones, AI, nuclear weapons, and climate change as being separate, isolatable threats. Scholars and politicians alike have tended to regard these as a list of distinct global problems – unfortunate side-effects spewed out by innovative technology, whether that be the machinery of the Industrial Revolution or the algorithms of today.⁵ They are not. Each is a link in an economic chain from toiling miners to the cleaners sweeping the floors of a military base or a fossil-fuel-company headquarters. These threats are not unavoidable; they are consciously created by powerful groups who profit handsomely from the endeavour. Global catastrophic risk is the product of a worldwide system of extraction: the Global Goliath.

Global catastrophic risk means the likelihood of a severe, unprecedented, and irreversible catastrophe that could happen in the space of decades. Past generations have experienced the global decimation of the Black Death and the invasion of the Americas. A worldwide disaster that kills more than 10 per cent of the global population while disrupting the critical systems the world depends on, such as food and telecommunications, would be an unprecedented global catastrophe. If that were to happen, we would enter into dark, uncertain territory and a potential spiral into global societal collapse.⁶

Scholars today aren't worried just about global catastrophes or even a global societal collapse; some also fear even worse fates, such as human extinction. Think of this as a spectrum of catastrophe, from precededented global decimation risk through to unprecedented global catastrophic risk, including global societal collapse and human extinction. The upper end of this spectrum, from a deep collapse to human extinction, is often called 'existential risk'. This would be a catastrophe that threatens the existence of either global society or our species, *Homo sapiens* ([Figure 36](#)).⁷

This is not about individual threats like AI or nuclear weapons. This is about the overall level of risk produced by the Global Goliath. The world is not just facing one individual challenge like climate change, thermonuclear war, autonomous armed drones, enormous volcanic eruptions, or dangerous

new technologies. It is facing all of them at the same time. And alongside these threats are new vulnerabilities, such as interconnected global financial markets where a failure in one country can quickly spread across the world. Then there are responses that could make a collapse worse, such as our tendency to turn to autocrats during times of crisis. We call issues that can significantly contribute to catastrophic risk across these risk areas (threats, vulnerabilities, exposure, and responses) ‘drivers’ (for instance, climate change creates certain hazards such as lethal heatwaves, spawns new vulnerabilities like droughts, deepening debt, and may worsen other threats like conflict). All these drivers are interrelated.



[Figure 36](#): The spectrum of global catastrophe

Think about climate change. We should not ask whether climate change is an existential risk but rather try to discover how much climate change contributes to the level of existential risk. Whether the world can handle 3°C of warming above pre-industrial temperatures depends on how the future looks. A world characterized by strong international cooperation, high trust in public institutions, and few other problems may be able to handle that level of global heating. A world that is riven with conflict, distrust, autocracy, rampant inequality, and out-of-control technology might well collapse.⁸

These drivers share another feature: many are driven by modern technology. The Latin phrase *deus ex machina*, a god from the machine, was used in the theatres of antiquity to describe a situation in which a character required divine intervention to escape from an impossible situation. A god figure was then lowered onto the stage from above. Gods don't just bring salvation, of course; they also mete out punishment and death. The world today faces a set of problems that could be termed *mors ex machina*: death from the machine. Advanced technology can help improve our standard of living and manage global problems, but it has also introduced unprecedentedly destructive threats. Collapse in the future is likely to be global and far worse than in the past. That is in large part because of the technological threats being spawned by the Global Goliath. Threats such as nuclear weapons, engineered pandemics, AI, and killer robots.

GOLIATH BECOMES DEATH

On 16 July 1945 Robert J. Oppenheimer uttered the words 'now I am become death, the destroyer of worlds', quoting the Bhagavad Gita. He was watching a fireball erupt in the deserts of New Mexico, the first explosion of a nuclear weapon. This was the Trinity Test. Within a month two atomic bombs had been dropped on the cities of Hiroshima and Nagasaki, killing between 110,000 and 210,000 people.⁹ The blast in Hiroshima was so powerful that it etched the shadows of some of the victims into stone.

Less than a decade later, in 1954, the US conducted the Castle Bravo test, which ignited a thermonuclear bomb around 1,000 times more powerful

than the one dropped on Hiroshima. Since then, over 2,000 nuclear weapons have exploded across a dozen or more sites around the world. Tests have been conducted by at least eight countries, although two have dominated: the US and Russia. There was a distinct colonial flavour to the test sites. France picked Algeria (the Algerians won their fight for independence within two years of the first test); the UK used its former colony Australia, and the US used the Marshall Islands in the central Pacific as a test ground for Castle Bravo.

The constant pursuit of bigger arsenals that could be more easily concealed from the enemy, and deliver missiles more quickly, continued throughout the Cold War. By 1986 – just over four decades since the Manhattan Project – over 70,000 nuclear weapons had been built.^{[10](#)}

Imagine just a single 800-kiloton weapon dropped in the middle of New York City. Everything within a five-kilometre radius would be vaporized. Everyone within twelve kilometres would be killed almost instantly. Even concrete buildings would be demolished. All those within eighteen kilometres who survived would die within a month or two of radiation poisoning. Within the first twenty-four hours a minimum of around 1.5 million people would be dead. Another 1.5 million would be injured and the Manhattan skyline would be essentially empty: a flat line of smouldering rubble.^{[11](#)}

Nuclear war is unlikely to end after the first explosion. Imagine in this case that the US is struck by Russia. Russia is unlikely just to bomb New York City. They are more likely to send a swarm of missiles with the aim of crippling the US economy and military, ideally minimizing the ability of the US to retaliate. However, the US has nuclear submarines scattered around the world, hidden silos, and more than sixty nuclear bombers (the forces of land, sea, and air are referred to as the ‘nuclear triad’), all of which could be quickly mobilized. Nuclear war will not be easily contained: even a single nuclear missile is more likely to spiral into thousands of explosions than remain unanswered. A plausible simulation of a nuclear war between the US and Russia predicted that within the first few hours 34.1 million people would die and another 57.4 million would be left injured.^{[12](#)}

Covering the earth with mushroom clouds brings a new, even darker threat into play: nuclear winter. Nuclear blasts turn cities into infernos and skyscrapers into billowing chimneys of burned materials and soot. These

particles float up high into the atmosphere and act as heat sponges, absorbing incoming solar radiation and preventing it from reaching the surface below. This cools the planet. A war between India and Pakistan (both countries with nuclear weapons) resulting in around 47 million tons of soot in the skies would cause global temperatures to drop by around 4°C. A war between the US and Russia would throw up about 150 million tons of soot, dropping global temperatures by 9–10°C. The latest modelling suggests that the ensuing famines would kill up to 5 billion people, well over half the world's current population.¹³

It could be worse, as there are many impacts not included in such models.¹⁴ When a nuclear bomb explodes it produces a pulse of electromagnetic radiation (commonly called an EMP). This EMP explodes outwards, a bubble that expands at the speed of light. It acts like a powerful lightning strike, overcharging and frying any electrical infrastructure it touches. Depending on the power of the bomb and the height of the detonation, one EMP could affect the entirety of the contiguous US. Think of hundreds of these across the world. What regions would be engulfed by these EMPs depends on who is involved in the war. One involving the widespread participation of nuclear powers would see most of the northern hemisphere fried. Much of the electrical infrastructure in these areas would be rendered useless. Electricity and the internet would be lost. For how long, we don't know.

It wouldn't be easy to replace all the now impotent electrical infrastructure. Large power transformers are highly customized and take up to two years to manufacture. It would take far longer to produce one in a world without the internet and with most of its factories and supply chains destroyed. It's also difficult to tell how long it would take to recover the internet. The internet relies on Border Gateway Protocol (a standardized set of rules that determines how connections between computers are routed), which assumes that any servers that go offline can reconnect to a functioning network. If most servers have been destroyed, the protocol may fail.¹⁵

In the aftermath, people would, of course, move, seeking shelter, food and loved ones, relying on our old power of mobility, as always. Unfortunately, unlike in the Palaeolithic, this could have a severe downside. A large movement of people with compromised immune systems (due to

both malnutrition and radiation sickness), alongside a rapid change in the climate and animal movement, would turn the world into a breeding ground for new zoonotic infections. Medicines and vaccines would dry up and healthcare would be limited in a world where the internet, electricity grids, and supply chains are severed.

It's hard to imagine what the knock-on effects would be on war and politics. Fewer than 3,000 people died in the 9/11 terrorist attacks and that led to \$4–6 trillion dollars' worth of wars in Iraq and Afghanistan, as well as the approximately 777,000 killed and the 13.4 million displaced. Imagine what millions dead and a levelling of the richest city in the world might do. Or what would happen if thousands of cities were annihilated, with the sky becoming a web of criss-crossing missiles and then a veil of ash.

The existential risk and collapse expert Florian Jehn has explored how a nuclear winter could impact the natural world. The release of nitrogen oxide and smoke from the nuclear firestorms would degrade the ozone layer that shields the earth from ultraviolet radiation. We could lose up to three-quarters of ozone, resulting in UV levels of 35–45. (The current advice is to stay indoors if UV levels are above 11.) The oceans would turn more acidic (by around 0.05 on the fourteen-point pH scale). The acidified oceans, winter conditions, and radiation from both sun and bombs would strangle an already struggling biosphere. How much extinction rates would further increase is unknown (and perhaps unknowable), but it would be devastating.^{[16](#)}

WHEN OUR LUCK RUNS OUT

A nuclear war would be the worst event in human history. It is difficult to see how such a conflict between the US and Russia would not result in a global collapse. That day could still occur at any time.

Currently nine countries have stockpiled around 10,000 nuclear warheads. That's a welcome relief and a sevenfold reduction from the high of over 70,000 weapons some four decades ago. Mass protest movements and successful arms-reduction agreements (such as the 1979 Strategic Arms Limitation Treaty and the 1988 Intermediate-Range Nuclear Forces Treaty) drove these reductions. The spread of nuclear weapons has also been

restricted. One country (South Africa) has willingly dismantled its stockpile, Ukraine gave up 1,900 warheads it had inherited from the Soviet Union, and several countries with the capability to produce those weapons chose not to.

Despite these achievements, the threat remains. The nuclear war scenarios outlined above are all possible with existing nuclear arsenals. Unfortunately, all signs now point towards these arsenals growing in both size and strength. China has expanded its nuclear arsenal from around 200 in 2020 to 500 in 2023. The US and Russia are both developing new technologies that could make the use of nuclear weapons more likely and the global situation more precarious. Russia's hypersonic missiles are harder to detect and intercept than older, slower-moving weapons, and they reduce the time available for any target to react to an incoming barrage. The US is planning to spend \$1.7 trillion over three decades to upgrade and modernize its own nuclear arsenal.

All this creates unnecessary risk. There is in any case no strong evidence that nuclear weapons are effective in preventing wars or ensuring peace. Some have argued that the deterrent effect has ushered in a 'long peace', and at first glance the second half of the twentieth century does seem relatively peaceful. But this is an illusion that has been debunked on numerous occasions. The number of wars increased for three decades after nuclear weapons were built, and didn't decline even at the point when nuclear arsenals were largest. Yes, international conflict has decreased slightly since the fall of the Berlin Wall, but that might be due to random chance. (If it was attributable to nuclear weapons, we should have expected that fall to coincide with their invention, not the end of the Soviet Union. Even if the world has seen a decrease in conflict between countries since the world wars, that would need to last another 100–140 years before it became statistically significant.)¹⁷ What is true is that we haven't seen any great-power wars – such as a direct conflict between the US and Russia – since the invention of nuclear weapons. Yet even here, we'd still need a longer time frame to know that this was due to nuclear deterrence and not just luck.¹⁸

We do have evidence that the absence of nuclear war is largely down to sheer dumb luck. On 26 September 1983, the Soviet duty officer Stanislav Petrov saw on his screen a wave of nuclear missiles coming towards Russia.

He faced a choice: escalate the decision to his superior officer (who might have launched a massive retaliation, given tensions at the time), or not. Luckily for all of us, Petrov decided not to. He reasoned that the US would have sent a far larger fleet of warheads if this was a first strike intended to wipe out the USSR. He was right: it was a false alarm caused by a malfunction in the satellite warning system. Later Petrov reflected that if someone else had been on shift, it could easily have resulted in a global nuclear holocaust. Even at the time, he said it was a 50/50 decision for himself.^{[19](#)}

Such nuclear ‘near misses’ have occurred at least a dozen times throughout history. As my colleague Professor Benoît Pelopidas (the leading expert on luck and nuclear weapons) suggests, this may be an underestimate since nuclear-armed states have been notoriously secretive and many details are still classified. States also show few signs of learning from these near misses.^{[20](#)} And each time one of these near misses occurs we should consider the existing peace to have been shattered.^{[21](#)}

Today, our luck looks increasingly fragile. Russian President Vladimir Putin has already warned that a deployment of NATO troops into Ukraine would result in ‘tragic consequences’ – one of several not-so-subtle allusions to a nuclear strike. Meanwhile China has more than doubled its nuclear arsenal. A senior US official announced that the US may expand its own stockpile of nuclear weapons. That expansion would be in addition to the \$1.5-trillion overhaul and upgrade of the US nuclear arsenal. The fact that we have yet to face a thermonuclear war is not because of strategic brilliance or a sign of a functioning balance of power. It’s mainly a matter of chance. If tensions continue to escalate between nuclear-armed states, then it becomes ever more likely that our luck will finally run out.^{[22](#)}

REVIVING (AND UPGRADING) OUR GREATEST KILLERS

Manipulating subatomic particles for weapons is a dangerous affair that threatens the entire world, and so is manipulating viruses. Avian influenza, also known as the H5N1 virus, has a mortality rate of over 50 per cent. It is

over fifty times deadlier than Covid-19. Luckily, we know of only a handful of cases making a leap from birds to humans, in none of which it has been contagious. Had it been, then potentially the worst pandemic in history would have begun.

It is the sheer lethality of H5N1 that made a series of experiments in 2011 so concerning. A group of researchers across the US, Japan, and the Netherlands found a way to mutate the virus and make it transmissible between ferrets. In one study they spliced together genes from the H5N1 virus with swine flu (H1N1); in another they modified H5N1 and then ran it through numerous generations of ferrets until it had mutated. The scientists made a virus that could now spread through the air between mammals. While they didn't create a virus that could be passed between humans, and the lethality of what they had made is debated, for many it was one step too close. This is called 'gain-of-function' research: experiments that add to or enhance the functions of an organism, such as the transmissibility, lethality, or types of hosts a virus can infect compared with the original versions found in the wild. Advocates claim that such research can help us identify and prepare for the next pandemic; critics say it isn't worth the risk of starting the next pandemic.

Gain-of-function research is just one example of 'dual-use research of concern': research in the life sciences that can be misused to do great harm. There are others. In 2017 a small group of Canadian scientists spent less than \$100,000 to recreate the then-extinct horsepox virus in six months using mail-order DNA. Synthesizing a virus from scratch ('de-novo synthesis') has now been done with poliovirus and influenza virus. There are fears that this could be done with variola virus, the virus that causes smallpox. That is a disease that kills roughly a third of its victims and leaves most of the survivors with deep pitted scars. It has through decades of global effort now been eradicated, but its genome is publicly available and there are no substantial technical barriers to reviving it like the small group did with horsepox.^{[23](#)}

It's plausible that such a high-mortality pandemic could trigger a global collapse. If we were hit by a plague such as the Black Death, we would be likely to fall apart in a way that medieval Europe did not. The people of medieval Europe were not dependent on collective infrastructure such as electricity grids and mass transit systems, and their governments didn't base their legitimacy on providing public goods like healthcare. The modern

world is different. Even a virus with a lower fatality rate, like smallpox, would almost certainly break the world's healthcare system. Existing systems are already under strain and increasingly overburdened by climate-change-related chronic illnesses and infectious diseases. They are also just not equipped to handle a huge influx of patients. Bear in mind that Covid-19 had a fatality rate of less than 1 per cent. Even then, during the height of the pandemic Italian hospitals could not always provide ventilators for those who were struggling to breathe. A high-mortality pandemic would also hit front-line health-service workers incredibly hard. If a third of them were to die, while others were left incapacitated for weeks, how would an already overwhelmed healthcare system cope?

Without the guarantee of a functioning system to take care of the ill, the legitimacy of states would begin to dissolve, and those employed in many other essential sectors would probably stop working: why expose yourself to a one-in-three chance of dying, especially if you won't even be able to get professional treatment? Suddenly, rubbish piles up in the streets. Power lines are no longer repaired. Deliveries fail to arrive. Water and sewage systems begin to falter. Undoubtedly, our old altruistic ways would kick in: many would still go outside and brave the risk. Yet even courage and compassion have limits. Our entire economic system would begin to fray. The peasants of medieval Europe didn't expect hospital care from their states as part of a 'social contract', nor did they rely on large, complicated, collective infrastructure like we do. Global society appears to have become better-equipped to produce vaccines, yet we are also more vulnerable to a large enough disease shock.^{[24](#)}

That situation is likely to get worse. In 2020 I led a WHO process to identify the most significant issues in dual-use research of concern. The process didn't just highlight the reconstruction of the variola virus as one frightening possibility; the convened group of experts also foresaw several others, including the use of machine-learning algorithms that trawl through vast amounts of data to identify new dangerous pathogens that don't yet exist, and platforms that allow for the quick reconstruction and engineering of different diseases and bacteria.^{[25](#)}

Our ability to create new, dangerous pathogens is increasing rapidly alongside our technology. Sequencing the human genome, a project that began in 1990, took thirteen years and cost \$2.7 billion. By 2020

sequencing a human genome cost less than \$300 and could be performed in days. We can now also modify and edit genomes in real time thanks to CRISPR-Cas9 (a technology that essentially allows us to cut, copy, and paste genes). Both the new dual-use problems on the horizon and the rapid development of biotechnology have led many to fear that a small, untrained group may soon be able to create something far worse than smallpox. Evolution tends to constrain naturally occurring viruses such as smallpox: those that are too lethal kill off their hosts, while those that kill too quickly don't spread far enough or infect enough people. An engineered virus would be designed rather than evolved and, at least in theory, would not face such limits. Rather than reviving a natural virus, a group could create an unnaturally lethal one that kills more of its victims than the plague but spreads quicker than Covid-19. In other words, an engineered doomsday disease.

Luckily, carrying out a large-scale bioterror attack is exceedingly difficult. It requires researchers with extensive training, tacit knowledge of how to work in a lab with pathogens, access to advanced labs and equipment, substantial experience with clandestine operations to avoid being caught (probably someone who is a former military or intelligence operative), and the fortunately rare desire to murder all of humanity. Finding a highly trained virologist who is capable of recreating a virus, let alone designing and creating a new one, is already exceptionally difficult. Finding all these skills and experience in one individual is nigh impossible. And more people means more coordination and an even greater chance that someone will defect and tell the authorities.^{[26](#)}

Even with current advanced technology, bioterror is surprisingly rare and ineffective. Currently we have a biocrime or bioterrorism incident around every 2.9 to 3.5 years. Most are hoaxes, and even those that do kill usually only claim a few casualties.^{[27](#)} The largest one to date was carried out by followers of the cult leader Rajneesh (also known as Osho) who poisoned ten salad bars in Oregon with salmonella, resulting in 751 cases of food poisoning and luckily no deaths. This had led some researchers to conclude that even if you have the right amount of money and resources, killing large numbers of people with germs is just incredibly hard.^{[28](#)}

We should still guard against the possibility of catastrophic bioterrorism, but be aware that this is a highly unlikely possibility. For now, the risk

comes from large, powerful organizations such as militaries and giant bioengineering firms. These are the institutions that will have the resources to create and release an engineered virus. That's true historically as well. While the greatest bioterror incident injured fewer than a thousand people, Japan killed around 300,000 Chinese during the Second World War by experimenting with typhus, cholera, the plague, and a host of other bioweapons.^{[29](#)}

Even if there was a collapse-inducing bioterror attack because of cheap, dangerous technology, the bigger cause would be the governments who fail to regulate these technologies appropriately, and their failure to invest in public healthcare and pandemic preparedness. Our most robust line of defence against pandemics – whether they be natural or lab-made – is still the simple, often overlooked and underfunded work of having well-resourced functioning healthcare systems, a healthy populace with strong immune systems, strong biosecurity practices in labs, ample vaccine capacity, effective contact-tracing processes, and other pandemic preparedness measures. These remain largely absent, despite the fresh scars of Covid-19.

The risk of collapse is likely to rise over the coming decades, driven in part by advances in another potentially dangerous technology: AI.

AUTOMATED COGNITION

In 2023 a long list of AI luminaries signed a letter on the risks of AI. These included Geoffrey Hinton (known as the 'Godfather of AI'), Stuart Russell (who co-wrote the textbook on AI: *Artificial Intelligence: A Modern Approach*), Demis Hassabis (head of Google DeepMind), Sam Altman (head of OpenAI), and Dario Amodei (leader of Anthropic). The letter stated that 'Mitigating the risk of extinction from AI should be a global priority alongside other societal-scale risks such as pandemics and nuclear war.'^{[30](#)}

That concern has been expressed in more modern language by the rapper Snoop Dogg: 'And I heard the dude, the old dude that created AI [referring to Hinton] saying, "This is not safe, 'cause the AIs got their own minds, and

these mother----ers gonna start doing their own s---.” I’m like, are we in a f---ing movie right now, or what? The f--- man?’

One problem is that it’s unclear what Hinton or Snoop are talking about. In general, AI tends to be used as a vague term to cover the technologies that make machines do things that, if a human did them, we would consider intelligent. That’s a capacious and ambiguous definition.³¹ Intelligence alone has dozens of potential definitions, from achieving goals in a complex environment through to acting appropriately with foresight.³² AI is, of course, also a marketing term that can overstate the capabilities of many algorithms and make them seem almost ethereal.³³

Let’s not get distracted by abstract, marketing terms and stick to what these machines actually do. They are material systems that vastly speed and scale up specific cognitive tasks. That could be moving a robotic arm (spatial navigation), recognizing faces (image recognition), or stringing together words to maximize the probability of providing the correct answer to a question (text processing). While intelligence is a broad, fuzzy term, cognition refers to specific, concrete abilities that different algorithms can excel at. As the Oxford cognitive scientist Carla Zoe Cremer observes, we are better off thinking of this as ‘automated cognition’.

What Hinton and others truly fear is that one algorithm will be able to automate and combine a large number of different cognitive abilities. It will start to possess the kind of broad, flexible abilities that a human has. It may even be able to learn and develop new cognitive abilities. Think of this as ‘automated general cognition’. Then there is the much more speculative chance that an algorithm might not just become general, but also orders of magnitude quicker and better performing than a human. From now on, when I use ‘AI’ or ‘AGI’, I’ll be referring to automated cognition or automated general cognition.³⁴

The idea of machines annihilating or enslaving humans is by no means new. In 1872 the novel *Erewhon* by Samuel Butler depicted a society that had destroyed most of its mechanical inventions after the machines began to gain consciousness, self-replicate, and rise up owing to natural selection among machines.³⁵ In 1965 the Oxford-based AI researcher I. J. Good published an essay on ‘Speculations Concerning the First Ultraintelligent Machine’. He hypothesized that if a machine were to become more intelligent than its human programmers then it could design even smarter

algorithms, triggering an explosion in AI. Humans would be quickly left behind. Good warned, ‘Thus the first ultraintelligent machine is the last invention that man need ever make provided that the machine is docile enough to tell us how to keep it under control.’³⁶

In the past few decades, experts, including leading computer scientists, have joined Good in taking this risk seriously. The perceived threat is the creation of an ‘unaligned’ AGI. That is, an AGI with a mission that is either not under human control or does not reflect human values (as difficult as they may be to define across 8 billion people). The concern is not that the machine will be malevolent or murderous; rather, we’ll just be collateral damage for them to achieve their programmed goals.

This wouldn’t be a mere collapse, but potentially complete human extinction, or perhaps an even worse fate, like perpetual enslavement.³⁷

Unlike nuclear weapons, this is a hypothetical concern. We don’t know exactly what such a machine would look like. We don’t know how it would control or destroy all humanity. The ways that scholars have suggested all tend to be sci-fi-like, such as the algorithm manufacturing and releasing billions of microscopic drones to simultaneously assassinate all 8 billion of us.³⁸ The ultraintelligent machine quickly begins to sound like a malevolent technological god. And we should always be sceptical of appeals to gods, whether they reside in real clouds or the digital one.

We don’t even know if such a machine can be built, or how long it would take. Numerous attempts have been made to find out. A 2022 survey asked 352 AI experts when unaided machines would be able to complete every task better and cheaper than a human: 90 per cent said it would occur within a century and 1.1 per cent said it would never be built. The most recent survey of 2,778 researchers found a 5 per cent chance that superhuman AI would result in extremely bad outcomes by the end of the century, such as human extinction. Superforecasters put the risk of AI causing human extinction lower, at around 0.38 per cent. Few of us would hop on a plane with a 5 per cent (or even 0.38 per cent) chance of crashing. In this case, a crash would potentially kill not just a few hundred, but billions.³⁹

There are reasons to be sceptical about these results: surveys are one of the least accurate forms of forecasting. We have no evidence that they work on long time frames for such a complex question as the creation of an unprecedented technology. They don’t allow experts to deliberate, and

individual experts don't have a great track record of prediction, even in their own fields.⁴⁰ Importantly, these surveys show an enormous range of opinions, all the way from 'it will be built within years' to 'it will never be built'.⁴¹

AI could also slow or stagnate for a range of reasons, including increasing scarcity of the materials needed for labs and microchips, hard physical limits (such as how small one can make a microchip), or innovations simply becoming too difficult and costly. The polymath Vaclav Smil, in his magisterial book *Growth*, puts forward the case that the growth of all systems eventually stops. Diminishing returns doesn't just apply to extraction, but also to innovation: every dollar of investment results in fewer and fewer benefits. That halting point for new technologies, whether it be in bioengineering or AI, may be far sooner than anticipated.⁴²

AI may already be facing limits. Projections suggest that large text models like ChatGPT will be using datasets equivalent to all the available online, public human-generated text by 2026 to 2032. Within a few years we could run out of data to develop bigger, better algorithms. This limit to data could be overcome by relying more on 'synthetic data' created by AI, but whether it's possible and what problems this would create is uncertain.⁴³ There could also be limits to energy use. One report estimated that energy usage by data centres could more than double between 2023 and 2027, causing 40 per cent of data centres to face energy constraints.⁴⁴ The approach of building bigger algorithms to get better results could also plateau. Some researchers in the field of machine learning think that the approach of simply scaling up existing algorithms (with more data and more parameters) may already be hitting a wall.⁴⁵

AI itself is an industry product that is already prone to hyperbole and exaggeration. The heads of AI companies, such as OpenAI's Sam Altman, are happy to feed the concerns about AI safety because it makes their products seem far more powerful than they are. They are artificial hype-men.⁴⁶ Today we have forecasts of AI significantly boosting the economy and speculative visions of a world run by virtual AI workers. Yet we haven't seen AI have much impact on growth or productivity. Even going back to 1987 the Nobel Prize-winning economist Robert Solow quipped, 'You can see the computer age everywhere but in the productivity statistics.' That

fundamentally hasn't changed: we still haven't seen any significant economic boost from AI.

We also don't know what is necessary to build intelligence. One hypothesis is that intelligence only evolves in a body. It is an outcome of an embodied interaction with an environment using different senses. This would make the proposition of building machine intelligence far more difficult. Robotics lag far behind the development of large text models, while computer simulations would need to be run at an incredibly high resolution (requiring an enormous amount of data and raw materials like semiconductors) to produce an embodied intelligence.^{[47](#)}

Despite all these reasons for caution, we still can't rely on the assumption that an AGI is impossible to build or will face limits to growth. History shows how quickly previously unthinkable technology can be built. On 12 September 1933 the physicist Leo Szilard read a newspaper that would shape history. An article in it quoted Lord Rutherford, the man known as the father of nuclear physics, who dismissed the use of nuclear fission as 'moonshine'. Szilard took a walk afterwards. During that stroll he came up with the idea of a nuclear chain reaction. That became central to the creation of the atomic bomb, which was built just a dozen years later. AGI may seem unlikely now, but in an uncertain world we can't blithely dismiss fears of powerful new machines. We need to be cautious. The stakes are so grave that even an incredibly low likelihood of building a catastrophic AGI should give us concern. More importantly it should give us a reason to pause efforts to construct such a machine. We need to stop efforts to build something that could potentially kill everyone until we can make an informed, genuinely democratic decision about whether it is worth the risk.^{[48](#)}

Stopping AI is possible. All it may require is simply addressing the economic exploitation that underpins increasingly powerful machines. Even the most impressive algorithms are still built on cobalt miners in the DRC, unfathomable amounts of data provided by artists, writers, and ordinary people, and hundreds of hours of training and reviewing by flesh-and-blood humans. Some of that work is done by humans known as 'content moderators'. These poor souls need to trawl through thousands of sentences and images, telling the algorithm whether or not material is appropriate. Every day they are exposed to the worst of humanity, ranging from torture

to sexual abuse. OpenAI hired many of its content moderators from Kenya, paying them between \$1.46 and \$3.74 per hour and offering them no support. Many of the workers have complained of psychological trauma.⁴⁹ Data annotators in Uganda are paid similarly low wages, despite the fact that building an algorithm requires far more hours of work in annotating, organizing and checking data than coding an algorithm.

The data that is annotated in Uganda is then exported along a fibreoptic cable that runs close to the old East African railway. That was a railway financed by the British to take resources from Uganda's interior to the port of Mombasa, from where it was shipped back to the imperial core. Lurking even under AI there is the global, rimless wheel of empire. Now, data, finance, cobalt, and silicon are moving along these spokes, not just colonial commodities.⁵⁰

All this data is, of course, stolen.⁵¹ Data is fast becoming the latest lootable resource that can be easily seen, captured, transported, and stored. While we celebrate the CEOs of leading AI companies, the algorithms themselves are models of collective intelligence founded on our collective data. All this data is extracted from the web without our consent, while none of us receives a single dollar in compensation for our valuable data. AI is more extractive than artificial: it's dependent on cheap workers, cheap materials, and stolen information. It wouldn't be possible (or at least inexpensive and highly profitable) without the Global Goliath. If we started paying workers fairly and requiring consent and compensation for the use of public data, the development of AI would both slow and become far more controllable.⁵²

THE AI ACCELERANT

AI doesn't even need to reach such heights to be dangerous. It already poses enough problems. That is because AI accelerates many existing concerns, such as allowing for sophisticated automation of hacking, amplifying the threat of cyberwarfare between countries, and allowing for the construction of deepfake videos, causing problems for both our political and personal lives.⁵³ That is just the beginning of a far longer list of potential woes.

One particularly worrisome possibility is that the responsibility of whether to fire nuclear missiles is placed in the digital hands of an algorithm. There is already a precedent for this. Russia has since 1985 possessed the Perimetr system, more charmingly known as ‘the Dead Hand’. When activated it has contact with Russian command. If that connection were to be lost, the computer system would assume that they had been destroyed and automatically launch a radio-tipped rocket that would shower down radio signals instructing all nuclear silos below to launch. The intention is to provide foolproof deterrence: even if the Soviet commanders are killed in a first strike, the rest of their arsenal will still launch on the US. It’s a pre-programmed doomsday device. Fortunately, it’s active only during crises, though we don’t know whether it is active at the moment.

In 2024 Adam Lowther, former professor of the US Army’s School of Advanced Military Studies, and former US Air Force Colonel Curtis McGiffin advocated for the US to adopt its own dead-hand system through the integration of AI. The US military already has a worrying fondness for automation and integrating AI into its forces. This has led to a growing chorus of scholars warning the government not to hand over the nuclear launch codes to an algorithm. It would be the automation of nuclear holocaust.^{[54](#)}

The military has already shown a willingness to use AI to scale and speed up murder. In 2016 the National Security Agency (NSA) of the US hooked up a new machine-learning algorithm to its drones hunting across the skies of Pakistan. For over a decade the US had been using drones to launch missile strikes against people and compounds across the country. This new programme was different. It used the metadata (information about data, for instance where and when a call took place) of 55 million Pakistanis to determine how likely they were to be a messenger for terrorist cells. It analysed how they moved, who they called, and who they met, to decide whether they should die. The programme would then create a ‘final kill list’ of the most suspicious suspects for the drones to target. The program’s name was SKYNET. We don’t know how many were killed because of SKYNET, or even whether they were guilty of aiding terrorist organizations.^{[55](#)}

Using AI to accelerate bombing campaigns isn't just a US pastime. In 2023, in the opening weeks of Israel's bombing campaign of Gaza, a new set of AI systems was deployed. It had been developed by Unit 8200, Israel's version of the NSA. The 'Lavender' system would use swathes of data to predict who was an appropriate target connected to Hamas or the Palestinian Islamic Jihad. At its height Lavender identified as many as 37,000 Palestinian men. These had to be vetted by a human operator, but the procedure was inappropriately brisk. One Israeli using the Lavender system admitted they would spend around twenty seconds checking a target before giving approval for them to be bombed. Another AI system called 'The Gospel' would identify buildings and facilities that were most likely used by targets. The targets were then tracked and a third algorithm called 'Where's Daddy?' would alert the Israeli military when the targets had arrived back at their family homes and were ready to be assassinated. Dropping a bomb on an unguarded house was seen to be more reliable than attacking military facilities. According to internal sources, the Israeli military used cheap 'dumb bombs' that wiped out entire houses and were permitted to kill up to twenty civilians for every target. This was largely done as a cost-cutting measure. We don't know how many innocent lives were lost due to these AI systems, but it could easily be in the thousands.⁵⁶

Lavender at least required some oversight. There are some AI military systems that require no human approval at all to decide who lives and who dies. These are usually called 'lethal autonomous weapons', but for clarity let's just call them 'killer robots'. The first confirmed attack from a killer robot came in 2020, when a Turkish-made Kargu-2 drone (the same kamikaze drone mentioned earlier) hunted down and assaulted people in Libya. A year later in 2021, Israel launched a swarm of AI-coordinated autonomous drones during an attack on Gaza. AI-enhanced drones packed with explosives are, of course, already being used in the war between Russia and Ukraine.

This is just a small taste of things to come. The US is developing fleets of unmanned ships and swarms of up to 250 AI-managed drones. This is likely to be boosted by the \$1 billion Replicator Initiative that plans to build even larger AI-managed drone swarms that will hunt down targets. Not to be outdone, the US Air Force is also looking to construct a fleet of 1,000 unmanned fighter jets to be flown by AI systems. Russia is also working on

AI-run drones, vehicles, and even missiles. Perhaps the most concerning is an autonomous, nuclear-armed underwater vehicle called Poseidon.⁵⁷

Proponents of AI warfare suggest that the use of killer robots and integration of AI into military operations could make war more restricted and humane. Algorithms are not expected to be clouded by emotion or bloodlust and could be hypothetically programmed to abide by strict rules. After looking at operations like Lavender and SKYNET, it is hard to take such a position seriously. The goal of using these systems is speed, scale, and cost, not compassion, restraint, or a desire to abide by the rules of warfare.

The debate over whether killer robots will be humane also overlooks bigger problems. Killer robots won't stay on the battlefield. It is only a matter of time before one is used to assassinate a head of state, or a swarm is outfitted with facial recognition to exterminate a rival political group (a scenario chillingly depicted in a short video titled 'slaughterbots'). Even that may not be as bad as how a government could use killer robots to intimidate and even kill its own citizens. Historically, larger, non-violent movements are more likely to succeed.⁵⁸ Part of that is because violent action is less likely to be carried out by police and the military when a gathering has such a broad cross-section of society. Will you still fire into a crowd when you might shoot a cousin or a friend? Killer robots, however, would have no such qualms. They would, at least in theory, be obedient to whoever controls the code.⁵⁹ The scenario of killer robots becoming widespread across a population is terrifying. The scenario in which they become the ultimate monopolizable weapons – a new, modern Goliath fuel – is perhaps worse.⁶⁰

Machine learning could also supercharge our global ecological crisis. One search on Google uses as much electricity as is required to power a lightbulb for two minutes. One query to ChatGPT is twenty-five times more costly: you could light up a bulb for fifty minutes. That is an enormous increase, and one that can accelerate the already substantial emissions of the digital world. If all our digital technology was a single country, it would be the third highest consumer of electricity, behind the US and China.⁶¹

Major fossil-fuel companies such as Shell intend to use machine learning to improve their ability to find and tap new sources of oil and gas.⁶² Any

new fossil-fuel developments are completely incompatible with limiting global heating to 1–2°C. To have just a one in two chance of keeping the planet below a safer 1.5°C level of warming, around 60 per cent of oil and gas, and 90 per cent of known coal reserves, need to remain in the ground.⁶³ Using AI to search for more at this stage is a public commitment to heating the planet.

An automated nuclear response, killer robots, mass surveillance, digging up and burning more coal: these are all outcomes of AI. Then there are the myriad other problems that just the first wave of chatbots and generative algorithms have created: supercharging plagiarism in school, recommendation algorithms that deepen political divisions, bots that speed up cyberwar and cybercrime, the theft of intellectual property from artists and writers, and the spectre of mass unemployment due to automation. These systems offer numerous benefits as well. AI-enhanced detection of cancers and identifying new drugs is important, AI-generated art is cool, chatbots can be nifty, and AI could help progress areas of science, but it's hard to see these outbalancing the shadow of a crowd of protesters being cut down by a buzzing cloud of drones.

This is the biggest threat of AI: it accelerates the creation of global catastrophic risk. Arms races and military operations – whether it be the race to a bigger swarm of autonomous armed drones or deciding which families to drop bombs on – get speeded up. The hunger of the global economy for data, energy, and resources will deepen. The need for more and more data for better algorithms will drive mass surveillance. While thermonuclear war or engineered pandemics (or, more speculatively, AGI) could directly cause collapse, this AI accelerant is closer to a collapse by a thousand cuts. It is the equivalent of pouring a tank of petrol onto an inferno.

AI will also slowly and irrevocably take our fate out of our own hands. As the world speeds up, humans can no longer keep pace. That slow theft of decision-making power is already happening on battlefields and in courtrooms across the world. Speed comes at a price. Algorithms may be quicker and more efficient than humans, but we just need to think of Stanislav Petrov to remember the costs. If an algorithm had automatically responded to that nuclear alert forty years ago, it would have made a different decision from Petrov's. You wouldn't be reading this book.

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Gaia vs Goliath

*Forever Corrupted – Playing With the Planetary Life-Support System –
How Bad Could Climate Change Get? – Tipping into Catastrophe –
Breaking Boundaries – The Roots of the Anthropocene*

FOREVER CORRUPTED

If you've been out in the rain recently, or drunk a glass of water, then you have probably been exposed to poly- and perfluoroalkyl substances, also known as PFAS. They have been called 'forever chemicals' owing to their persistence. They can last for years, as the environment has no clear way to break them down. There are around 4,500 of these fluorine-based chemicals that were originally intended to be used in heat-, water-, and stain-resistant products. Since then, they have spread from these materials into the earth and our bodies. PFAS are now found in the bodies of 97 per cent of Americans. They have also been found in soils and rainwater around the world. Often in levels that exceed European and US safety guidelines. Even if you are in remote regions of Antarctica, you will be exposed to PFAS.

There is simply no way to avoid them.^{[1](#)}

Substances such as PFAS have been linked to a range of health outcomes globally, including higher cancer rates as well as declining fertility and sperm count levels. They have also been associated with an increase in birth defects, less-effective vaccination responses, raised levels of liver enzymes and cholesterol, and kidney and testicular cancer. What is more worrying is

what we don't know. We lack long-term safety assessments for most of these long-lasting pollutants. We know even less about how they interact with the environment and other compounds. Such assessments may be impossible, given the quantities in which these artificial compounds are produced: over 100,000 types and 2 billion tonnes every year.² Also, they may not be reversible, since PFAS cycle continuously through rivers, rain, and other water sources. Concerningly, the costs of trying to clean them up may simply be prohibitive. It is currently estimated that trying to remove just one type of PFAS, perfluoroalkyl acids (PFAAs), at the same rate they are produced every year would cost between \$20 trillion and \$7,000 trillion. In other words, the cost of just cleaning up one type of PFAS probably exceeds the global GDP (which is around \$106 trillion). And that wouldn't even get rid of what has already been released. These truly are forever chemicals.³

PFAS are part of a wider basket of synthetic compounds and chemicals that are called 'novel entities'. They are geologically novel, as they are synthetic and have never previously existed on earth. Novel entities include PFAS alongside different types of plastics (including microplastics), bisphenol A (BPA), and persistent organic pollutants such as aldrin (a pesticide), highly carcinogenic polychlorinated biphenyls (PCBs), and DDT. DDT was banned after decades of use since it was found to be highly persistent and toxic to many forms of life, as well as carcinogenic to humans.⁴

The amount of novel entities like PFAS that the earth system can handle is what researchers now consider to be a 'planetary boundary'. Planetary boundaries are zones of human impact on the earth system. Once you pass a boundary, the risk of abrupt and irreversible change increases dramatically. Change that could have disastrous consequences for humanity and the planet. We are now thought to be beyond the boundary for PFAS: we have already entered the uncertain danger zone for novel entities. This is not the only planetary boundary we have crossed. We are already feeling the effects of a more visible trespassed boundary: climate change.⁵

PLAYING WITH THE PLANETARY LIFE-SUPPORT SYSTEM

The history of life on earth is a story about the climate. We've had four mass extinction events in the last 541 million years (the Phanerozoic period of earth with larger-scale multicellular lifeforms).⁶ Each of these is associated with a loss of around 75 per cent or more of species. Climate change, namely global warming, played a key role in all of them.⁷

The human story is also entwined with the climate. For the first 95–99 per cent of our species' history, we lived in climatic conditions in which sustained, intensive agriculture was almost impossible. The dry, cold conditions with abrupt, volatile climatic shifts weighed against intensive food production, or even just settling down in one place.⁸ Then, 12,000 years ago, the entry into the warmer, more stable Holocene allowed for Goliath fuel, especially lootable resources, leading to the incredibly slow but fundamental shift from hunting and gathering to farming, from civilization to Goliath.

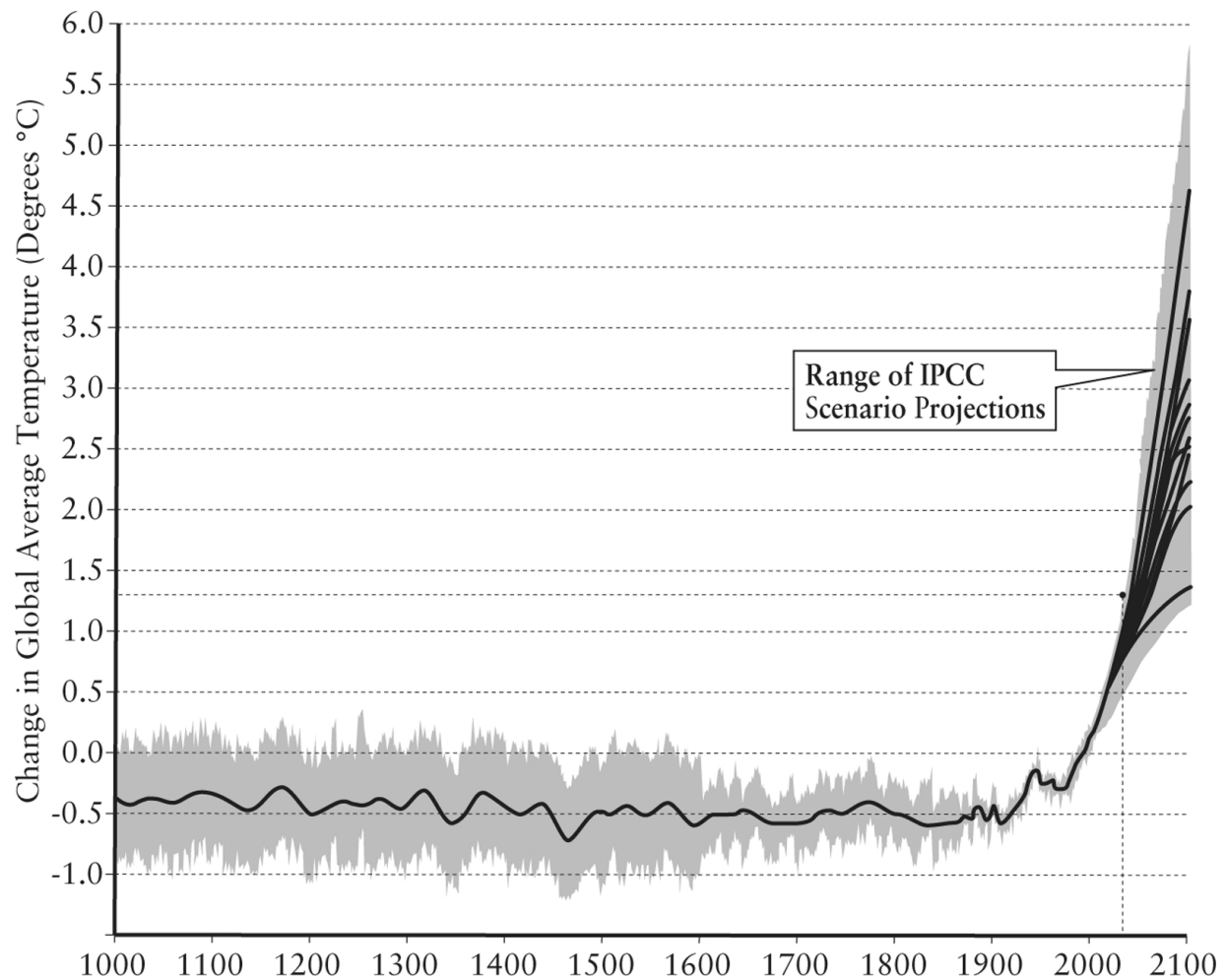
Ever since then we have been fundamentally constrained by the climate. For the past 6,000 years the vast majority of humans have lived in a narrow climatic envelope of ~11–15°C (along with a second smaller envelope of ~20–25°C, for India's monsoon region).

This is the 'climate niche' of Goliath. The first five original political states of the world all arose in this peculiar climatic bandwidth. Since then, most cities, crops, people, and livestock have stayed clustered in these areas. Even today, this climatic zone contains the greatest amount of economic activity. While we can live in numerous other environments, such as the Arctic and the Sahara, those areas don't tend to support cities or industrial infrastructure.⁹

Shifts in temperature have played a major part in shaping human history. We've repeatedly seen the fingerprints of climate change, from the fall of the Late Bronze Age during a megadrought to the synchronized collapse of Goliath systems across the north of America during the Little Ice Age. This was never a straightforward relationship of climate causing collapse, but we shouldn't ignore the role that climate plays. Even the rise of great empires often coincided with a change in the climate. Rome's expansion coincides

with the ‘Roman climatic optimum’: conditions that were favourable for growing crops and supporting a larger army.¹⁰ The formation of the Mongol Empire was also given a helping push by the climate. The rise of Chinggis Khan’s empire, the largest contiguous empire in history, occurred during a period of persistent wet and warm conditions.¹¹ These conditions meant more productive grasslands that Mongol horses grazed. More and healthier horses meant a stronger army. The contraction of the Mongol Empire and its sudden retreat from Hungary in 1242 occurred during a period of colder conditions that undermined the pastureland and mobile cavalry.¹² Climate is one important factor that moved the needle of history.

It would be a mistake to feel as though our industrial Global Goliath is now outside the reach of the effect of climatic shifts. That overlooks how dramatic the amount of warming we are facing is. The changes that rocked and directed the world in the past were regional, short term and usually less than 1°C. We are now sitting at the precipice of global heating of perhaps 2–6°C above pre-industrial levels within the next hundred years. By the end of this century we could experience temperatures that have not been felt since the Early Eocene, around 55–50 million years ago.¹³ Even a rise of just 2°C would lead to temperatures that have not been experienced for over 2.6 million years, since the Pleistocene epoch.¹⁴ We are facing an atmosphere and a climate that will be radically different from the one in which our species and Goliath evolved ([Figure 37](#)).¹⁵



[Figure 37](#): Temperature change from 1000 to 2100 CE

The greatest mass extinction event in the earth's history, the Great Permian Dying (around 252 million years ago), was sparked by an initial temperature rise of 6–8°C over tens of thousands of years; 80–90 per cent of all life on earth was eventually lost. We are releasing carbon today ten times faster than the carbon pulse that caused the Great Dying. What we are doing to the climate is geologically unprecedented. We are playing with the atmospheric life-support system of the planet.

HOW BAD COULD CLIMATE CHANGE GET?

The declaration that followed the 1988 Toronto Conference, one of the first climate summits, described the consequences of climate change as

potentially ‘second only to a global nuclear war’. We’ve already seen how cataclysmic nuclear war could be. Could climate change approach that level of global destruction?

In truth, we don’t know. Both higher-temperature scenarios and climate risk are dangerously understudied. Analysis of reports by the Intergovernmental Panel on Climate Change (IPCC) and the wider scientific literature shows that they disproportionately focus on warming of 2°C or less. We are betting on the best case.¹⁶

What we do know is that the warming we are facing is severe and unprecedented. Our current emissions trajectory puts us on track for a temperature rise of around 2.1–3.9°C above pre-industrial levels by 2100.¹⁷ These numbers are highly uncertain. It all depends on the tremendously difficult task of predicting economic growth, energy technologies, and geopolitics decades ahead. It also depends on how the earth reacts. The earth system could be more sensitive to our emissions than we are expecting, meaning we get more warming than anticipated for every additional tonne of carbon put into the atmosphere. Our best estimates forecast that a doubling of CO₂ will see temperatures rise by around 2.5–4°C. Yet that leaves an 18 per cent chance of warming of more than 4.5°C.¹⁸ This is not exactly what anyone would call ‘low probability’. One recent (and controversial) paper led by NASA scientist James Hansen suggests that a doubling is highly likely to cause warming of 4.5°C. In the long run, over more than a century, this could result in heating of 10°C.¹⁹

It could be even worse than that if the earth system springs any hidden surprises. One poorly understood factor is how clouds respond. One recent simulation suggests that at very high levels of greenhouse gas concentrations in the atmosphere (roughly a tripling of current levels), the stratocumulus cloud decks (fluffy, low-lying clouds) around the tropics will begin to break up. Clouds usually reflect sunlight, effectively cooling the earth. Their loss will result in even more warming, perhaps up to 8°C of it.²⁰ This would be in addition to the warming of around 6°C already caused by high concentrations, resulting in a spike of heating of about 14°C. This is, for now, a speculative simulation. Let’s hope we don’t get the chance to test its accuracy.

So far, it appears that we are on track for higher levels of warming and may have been underestimating how sensitive the climate system is. As of 2025 we have already temporarily exceeded 1.5°C. Over the period June 2023 to May 2024 global temperatures were 1.63°C above pre-industrial levels. This is a significant acceleration of previous warming trends, but whether it is temporary is unclear.

Even a rise of 1.5°C would have dramatic consequences. It would significantly increase the likelihood of ‘multiple breadbasket failures’: simultaneous failed harvests in the major crop-production centres of the world, such as the US Midwest.²¹ This would shock the price of staple crops like maize and wheat, with repercussions for the world. One small example of such a ripple effect comes from over a decade ago. In 2010 a climate-change-driven heatwave scorched Russia. Wheat yields dropped, causing Russia to institute a cereal export ban. This caused a spike in global cereal prices, especially wheat. In the UK, food-bank usage surged as people struggled to pay for increasingly expensive staples. In Egypt, where 40 per cent of people lived below the poverty line, the effect was even more noticeable. Bread is crucial in Egypt, where it is referred to as *aish*, meaning ‘life’. The increasing price of bread contributed significantly to the 2011 Arab Spring revolts there, which were initially dismissed as yet another bread riot. Now imagine that rather than a single heatwave in Russia, a far worse surge in temperatures occurred simultaneously in Russia, Brazil, and the US Midwest. We don’t know what the knock-on effects would be, and our models don’t even try to capture such possibilities, but it’s safe to assume they wouldn’t be good.²²

That is just for 1.5°C. The impacts of climate change do not scale linearly with temperature. That means a temperature rise of 3°C would be more than twice as damaging as 1.5°C. Such levels of warming would violently shift the climatic niche that our societies are adapted to, with over a third of people pushed beyond the niche by the end of the century.²³ Some areas would be especially hard hit. Currently only about 30 million people live in areas that have average yearly temperatures above 29°C. These are largely non-urban peoples living in the Sahara Desert or on the Gulf Coast. Under a medium-high emission scenario, by 2070 around 2 billion people are expected to live in regions where the average yearly temperature exceeds 29°C.²⁴ Much will depend on whether they can successfully adapt.

This extreme heat will make for a far more volatile world and could even trigger other catastrophic threats. These domes of extreme heat will also encompass two nuclear states (Pakistan and India) and seven Biosafety Level 4 labs (labs that handle the most dangerous pathogens in the world). They would also disproportionately fry the countries that are already politically fragile and likely to fall into state failure (see [Figure 38](#)).²⁵ What would happen next is necessarily speculative. Pakistan and India have always had a tense relationship, with several border skirmishes over the past decades. They also share a water resource in the Indus River. Around 70 per cent of Pakistan's economy relies on the Indus, which is set to dry up under future climate change.²⁶ The chances of a nuclear conflict, accidental or intentional, increase alongside geopolitical tensions. Without remedying action, climate change in the Indus Basin is likely to inflame these tensions. (This doesn't mean that climate change will directly cause a nuclear war between the two, simply that it heightens the conditions for conflict that could escalate into one.)²⁷

Then there is, of course, the heat. Humans have a physiological limit to how much heat they can withstand. Prolonged exposure to heat can be lethal, even for young healthy adults. Such lethal effects are usually reached once the wet-bulb temperature – the temperature at 100 per cent humidity – of 31°C is reached.²⁸

Under global warming of 4°C vast swathes of the Middle East, South and East Asia, and North and South America would experience extreme heat conditions that reach lethal levels. Every year 2.7 billion people would have to endure weeks of such heat during the daytime; 1.5 billion would experience a month of it and 363 million would face an entire season of such intolerable outdoor temperatures.²⁹ Global supply chains would be strained as people are limited to working outside for a few hours during darkness for weeks or months on end. Air conditioning isn't a sustainable solution as it would simply increase energy demand, emissions, and blackouts.³⁰ (It is also an option that the poor of the world rarely have access to.) The ripple effects of such a rise would make the 2010 Russian heatwave look mild.

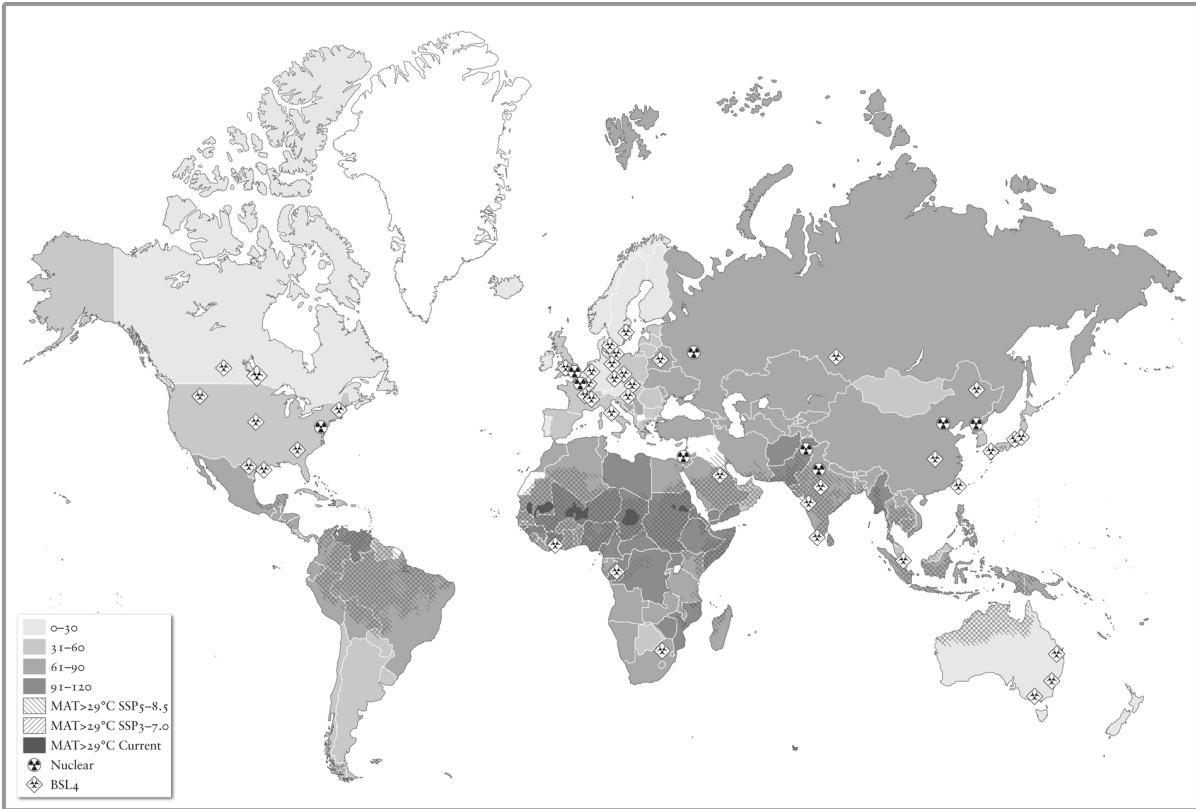


Figure 38: Extreme heat, fragile states, and the production of global catastrophe

This map depicts the overlap between fragile states (rated on a scale of 0–120, with higher numbers meaning greater fragility), nuclear weapons producers (in this case, the capitals of countries that possess nuclear weapons), Biosafety Level 4 labs, and areas of extreme heat. The shaded areas show regions of extreme heat exceeding 29°C Mean Annual Temperature (MAT) by 2070. SSP5–8.5 is the highest emissions scenario used by the Intergovernmental Panel on Climate Change (IPCC). SSP3–7.0 is a medium-high emissions scenario.

A rise of 6°C or more would lead to mass extinction. Lethal heat would engulf even more than 3 billion every year. The Arctic would be ice-free year-round while Antarctica would lose around 70 per cent of its ice, leading to sea levels twelve metres higher than today.³¹ Billions would be displaced from the climate niche. As the world warms, animals and infectious diseases will also move with the changing climate niche. This will all heighten the risk of new zoonotic infections (like Covid-19) as we get novel mixes of people and animals in hot conditions.

This picture of the future is bleak enough, but it gets worse once we consider tipping points in the climate system.

TIPPING INTO CATASTROPHE

Tipping points are thresholds where just a small push can propel a system into a totally different state, sometimes with no possibility of return, just as when you rock too far back on a chair you commit yourself to toppling over. Tipping points are one of the key reasons that 1.5°C is considered the planetary threshold for dangerous climate change. Even at this temperature, six global tipping elements are at a high risk of being passed. These include the collapse of the Greenland and West Antarctic ice sheets as well as the loss of low-latitude coral reefs.³² In other words, we are already careening into a world that will lose its coral reefs and two of its largest ice sheets.

These tipping points don't just threaten to create significant disruption. They could also propel further warming. They are what we call 'positive feedbacks': self-amplifying changes. Think of the loss of an ice sheet. The bright white of snow reflects sunlight back out into space. Dark seawater absorbs sunlight, heating the earth. As ice sheets shrink and more seawater is exposed, the earth warms up even more, melting even more sea ice. It becomes a self-propelling, sometimes runaway, change.

We already have numerous reasons to worry about such self-perpetuating cycles. Currently, the Arctic is heating nearly four times faster than the global average rate and twice as fast as previously expected. The permafrost across the Arctic hides a ticking time bomb: it holds nearly twice as much carbon as the atmosphere. As it thaws, increasing amounts of methane are released. It's a process that is irreversible over the space of centuries (it would take far longer to reverse). According to one model, we may have already reached the point of no return: methane will continue to leak from the Arctic for hundreds of years even if we don't release one more tonne of carbon. These are just the tipping points we know of. A world that is 3–4°C warmer than pre-industrial temperatures would be radically unfamiliar. If it happens, it will be so different that it is immensely difficult to know when the warming will stop.³³

Even tipping points that could occur this century with lower levels of warming are alarming. One of the most concerning is the shutdown of the Atlantic meridional overturning circulation, known as the AMOC. The AMOC acts as a huge, liquid conveyor belt, taking warm salty water from the tropics and transporting it to the North Atlantic. This warmer water releases heat before cooling and becoming denser, sinking down before moving south back towards the tropics. Melting ice from the polar regions and higher temperatures in the ocean are combining to slow the AMOC.

A complete shutdown of the AMOC was once considered a speculative possibility. That is no longer the case. The latest evidence using sea surface temperatures over the past two decades suggests that the AMOC could pass the threshold towards collapse during 2025–95.³⁴ A shutdown of the AMOC would reshape the world. It would plunge Europe into an ice age while warming the tropics. London could be cooled by around 10°C, while areas further north could be worse. Bergen in Norway could see temperatures drop by 15°C.³⁵ Much of the arable farming land in the UK would become unsuitable for growing crops. Globally, land that can be used for growing corn and wheat would be halved compared to a world without climate change.³⁶ This is not a far-fetched or distant scenario. Once the AMOC passes a tipping point the collapse can unfold within decades. While the numbers are highly uncertain, Stefan Rahmstorf, one expert studying the AMOC, put the odds of crossing the tipping point by the end of this century at 50/50, the flip of a coin. As with AI, we shouldn't rely on the

estimates of a single individual, but all the evidence points towards an AMOC collapse within the next century as being plausible and not even low probability. While the AMOC would lead to a regional ice age in western Europe, unfortunately the rest of the globe would continue to heat.^{[37](#)}

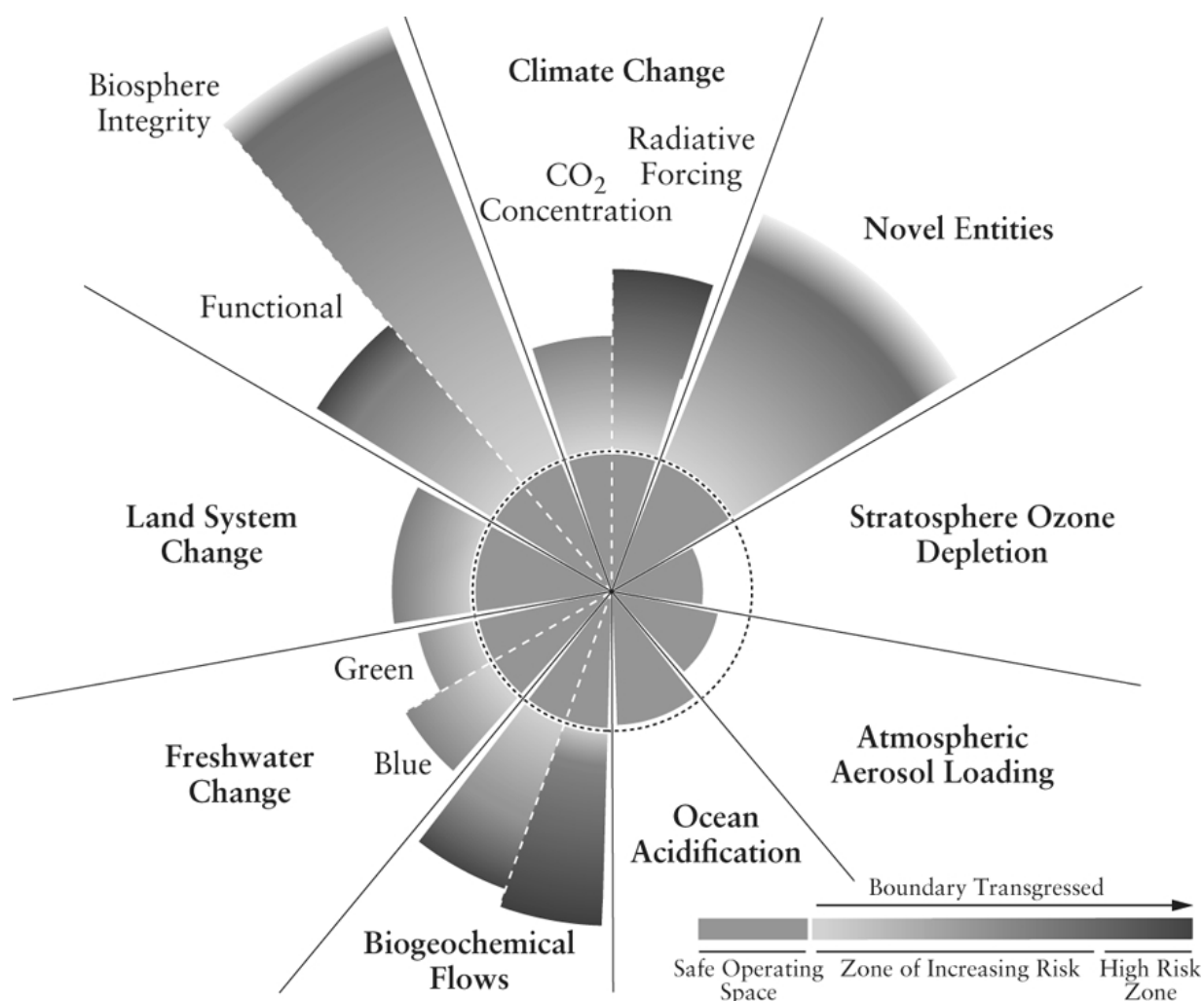
The danger of the AMOC is that, once we cross a certain threshold, we are committed to its collapse, even if we stop global carbon emissions entirely. This is the unsettling dynamic of how most tipping points work. The AMOC is only one among several interlinked tipping points that are now coming to light, including the Antarctic Circumpolar Current (a current surrounding Antarctica that could be 20 per cent slower by 2050 owing to climate change). Tipping points don't just characterize the climate system, but also much of the earth system that we are degrading. We aren't just risking pushing the climate system over the edge, or pushing into hazardous unknown territory with novel entities; we are wading into the danger zone for multiple planetary boundaries.

BREAKING BOUNDARIES

There are nine planetary boundaries. These are estimated thresholds past which we will leave the safety of Holocene-like conditions for the planet and enter a state in which the earth can no longer effectively self-regulate. We've transgressed six of these, four of which are in the high-risk zone ([Figure 39](#)).^{[38](#)} None of these show any signs of abating. The high-risk zone for these boundaries means two things. First, we've moved beyond the conditions in which our societies developed (as with our shift of the human climate niche). Second, the potential for uncertain, abrupt, and irreversible change and damage increases significantly. We've already seen this for novel entities and climate change: tipping points and uncertainties are in many ways the scariest part of what we are doing to the earth system.^{[39](#)}

Regardless of tipping points and uncertainties, we are already heading towards the next mass extinction event. Current rates of species extinction are around 10,000 times what they would be without human intervention.^{[40](#)} In other words, species are going extinct 10,000 times quicker than they

otherwise would because of human activity. Thirty per cent of all mammals are now at risk of extinction. Some species are suffering far more than others. Insects are dying out at rates eight times those of birds, reptiles, and mammals. Total insect populations are decreasing by around 2.5 per cent per year. Based on current trends, there is the real possibility of most insect species disappearing from the earth by the end of the century.⁴¹ Again, there are potential tipping points. The fear here is that the loss of enough critical species will start a cascade across the webs of life. As one prey species dies, its predator goes extinct, and the losses of both ripple across ecosystems.



[Figure 39](#): Transgressing the planetary boundaries

All planetary boundaries are interconnected; transgressing any one of them could plausibly be a key contributor to collapse. We are transgressing

six so far. One study of extinction rates among marine fossils found that a temperature rise of around $>5.2^{\circ}\text{C}$ tends to be a trigger for mass extinction.⁴² Similarly, other studies have found that large changes in the earth's carbon cycle are associated with mass extinction events. The current burst of carbon we are releasing may have already passed the threshold for previous mass extinction events.⁴³ That is just the interaction between biodiversity loss and climate change; there are many others.

These are just the boundaries we have studied and know about. There may be others that we are already beginning to transgress. These include deoxygenation of the oceans and the cycling of different elements in the earth such as boron, arsenic, and mercury. One could even be in space. Competing companies are racing to fill the atmosphere with fleets of satellites. Projects such as SpaceX's Starlink will lead to up to a million new spacecraft being launched in the coming decades. This wave of floating metal has foreseeable dangers. As the defunct satellites drop from orbit, they burn up in the lower atmosphere, leaving conductive, electrically charged particles. The earth is surrounded by a magnetic sphere that holds in place two belts of charged particles (known as the Van Allen Radiation Belts). The debris from burned satellites will leave a trail of conductive particles that is billions of times greater than the current Van Allen Belts.

Adding billions of tonnes of unknown chemicals to the atmosphere hasn't worked out well so far. Doing the same to the earth's magnetic field is also unlikely to be a wise move. There is already evidence that this could weaken the earth's magnetic field, and the re-entry of spacecraft could deplete the earth's repairing ozone layer. The weakening of both the magnetic field and the ozone layer will leave humanity even more vulnerable to solar storms (which could destroy electrical infrastructure) and cancer-causing solar radiation. It could eventually push us into the high-risk zone for a fifth planetary boundary: stratospheric ozone depletion.⁴⁴ Then there are the unknown effects, for instance whether billions of floating metal particles littering the atmosphere could warm or cool the planet.⁴⁵

From the oceans and soils to the skies and beyond into space, the earth system is being pushed into a perilous state.

Infringing on some of these planetary boundaries, such as climate change, could directly lead to collapse. Others act as indirect drivers of

collapse by slowly weakening societies. For instance, novel entities could lift the rate of chronic illnesses and disease, overburdening already struggling healthcare systems and creating further distrust in public institutions. This doesn't make such drivers less relevant to collapse. Remember that the fall of Rome and Chinese dynasties were occasioned by both numerous vulnerabilities (such as corruption and elite competition) and bigger shocks (like disease and invaders). The important lesson is that these interacting contributors to collapse all emerge from an extractive Goliath system.

THE ROOTS OF THE ANTHROPOCENE

We no longer live in the Holocene but in the Anthropocene. Ours is a geological epoch in which human activities have become a key shaper of the earth system. Today you can find not only PFAS in the Arctic but also plastic 10,975 metres below sea level in the deep-ocean Mariana trench.

Our gouging environmental fingerprints are now inescapable.^{[46](#)}

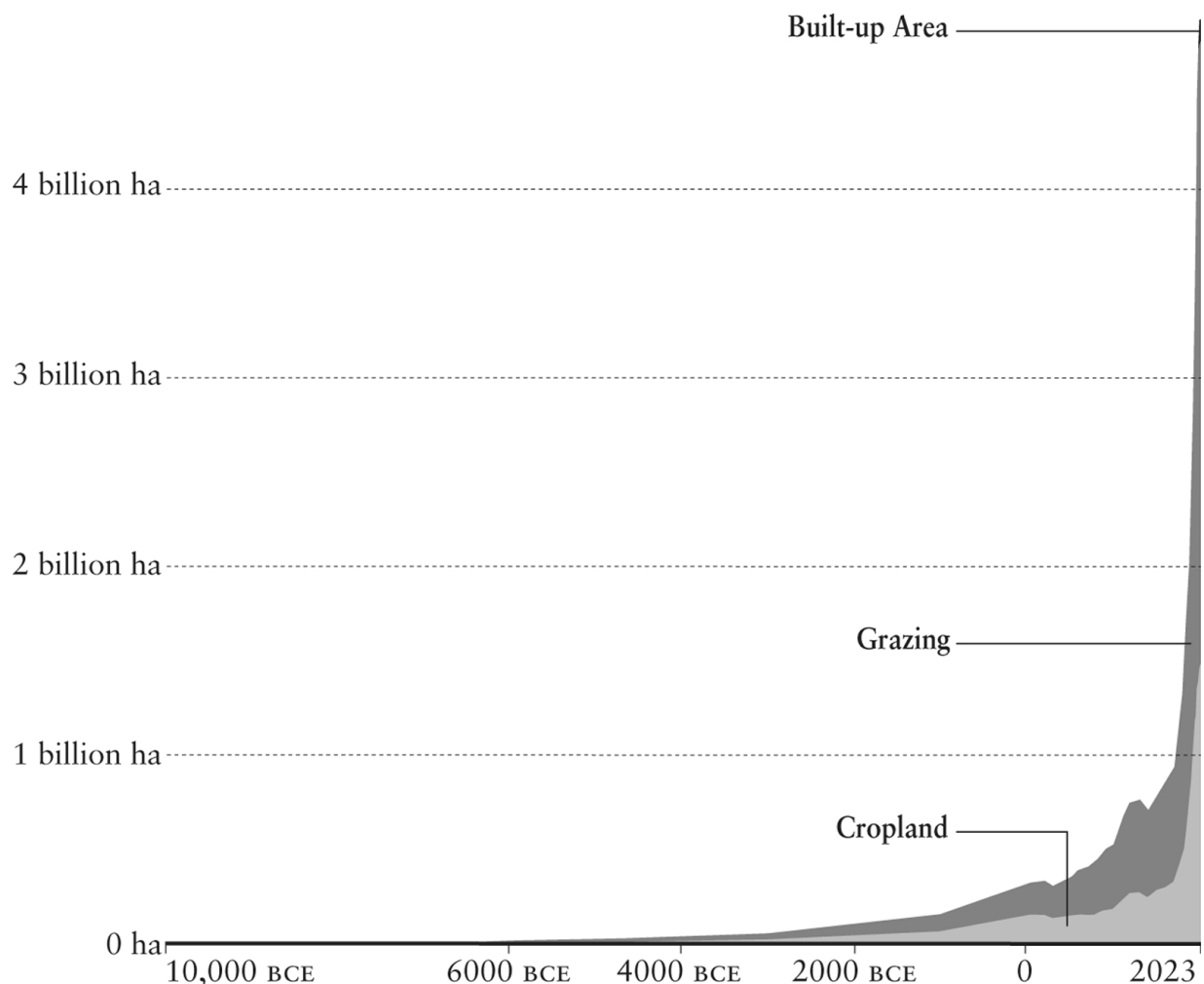
While our actions now will determine both the climate and the fate of most species (including our own), the Anthropocene is not a recent phenomenon. It is a process with deep historical roots. Early hominids were habitually using fire at least 400,000 years ago, and perhaps as early as 800,000 years ago. Fire remade the landscape. It not only favoured species that could withstand burning (pyrophytes), or benefited from it (pyrophiles), but also selected for certain types of animals. As we spread from Africa, our footsteps on the land became heavier. Hunting animals, burning forests, and harvesting wild crops started to transform the earth. Before the first plough had been picked up, foragers had already modified over three-quarters of the world's regions.^{[47](#)}

As we raised sickles and scythes our prints on the land deepened. The turn towards domesticated crops and animals meant a new kind of landscaping. The wetlands of Mesopotamia were slowly replaced with man-made irrigation canals and pastures for goats, sheep, and cattle. These new lands looked wild and diverse compared with the cash-crop plantations of spices, sugar, tea, and cotton that would follow 4,500 years later under colonization.

The biggest acceleration of our environmental impact came with the end of the Second World War. If a future archaeologist were to examine this point in the fossil record, they would notice some peculiar trends. From 1952 to 1962 there would be a thin layer of radioactive materials such as plutonium spread across the world due to nuclear tests. Plastics and previously unseen novel entities would begin to appear. The fossils of animals become increasingly scarce. As the Global Goliath grew the entire world was re-engineered.^{[48](#)}

One leading candidate for the date that should mark the beginning of the Anthropocene is 1945. Others think it should be the invasion of the Americas, when the sheer scale of death cooled the climate. Some believe it began when our ancestors begin to pick up ploughs to farm the land. Yet, any single date would be misleading. The Anthropocene isn't an event; it's a long, accelerating process. We've now entered a thick-and-fast Anthropocene that is rapidly pushing beyond our planetary boundaries.^{[49](#)}

This quickening, deepening stamp on the land is not mainly due to human numbers or the general greed of humans. When we look at human activity across the tropics, whether it be the Yucatán, or Africa, there are clear thresholds past which our environmental impact intensifies. The introduction of domesticated crops such as rice, colonization, and the post-1945 acceleration are the three clearest turning points. Each of these is marked by the creation and evolution of Goliath: the first states, colonial empires, and the forging of the Global Goliath. At each point there was a clear acceleration in the acres of forests cleared, the number of species wiped out, and the amount of pollution released into the soil, rivers, and oceans. The Anthropocene is the environmental footprint of Goliath.^{[50](#)}



[Figure 40](#): Land use over the past 12,000 years

Land use is a measurement of how humans have modified the global environment. It is one key indicator of the Anthropocene. The story of land use is simple. The conversion of land into fields for crops and pasture for grazing animals began to pick up around 8,000 years ago as lootable resources were adopted and Goliaths began to emerge. It has grown and accelerated over time with two big upticks in the eighteenth century (the Industrial Revolution and spread of Goliath) and the 1950s (the entrenchment of the Global Goliath) (see [Figure 40](#)).

Not everyone bears equal culpability for this planetary conundrum. The rich are disproportionately responsible for our current ecological predicament. The richest 1 per cent worldwide account for more emissions than the poorest 66 per cent.^{[51](#)} Those living in high-income countries use six times as many resources and produce ten times more emissions than

those living in low-income countries.⁵² One study across five planetary boundaries found that the top 10 per cent of income earners globally were responsible for between 31 and 67 per cent of planetary degradation.⁵³

The richest 1–10 per cent not only exert the biggest environmental impact but also tend to set status and consumption norms for everyone else to strive for. There is a reason why the most expensive perfumes and designer bags are marketed by the richest and most glamorous celebrities. It is also the wealthiest who hire lobbyists, make hefty political donations, and even run for office (Trump being one recent example). Lurking beneath the crisis of planetary boundaries is inequality and status competition.

The Anthropocene is largely driven by the long historical process of turning the principle of least effort of our ancestors on its head and instead using the environment and the labour of others for conspicuous consumption. The principle of least effort was all about expending as little energy as possible to meet your needs so you can maximize leisure. Conspicuous consumption is about wastefully using as much energy as possible to signal power and status to others. Chiefs and kings of hierarchical hunter-gatherers regularly cajoled, coerced, and encouraged their wider communities to intensify the production of food for feasts. This may have even been a historical contributor to the uptake of domesticated crops and animals. It continued from there with growing a greater surplus (usually through conquest or intensifying agriculture) to fuel more material displays of status. Whether it be throwing a feast in Mycenaean Greece (or essentially anywhere else for that matter), a maize-beer party in Monte Albán, human sacrifices by the Shang Dynasty, a parade of war captives through Rome, the colonial gold and silver that lines the churches and palaces of Spain, mountains of plantation-grown sugar and tea in Britain, or showing off a lavish set of private jets and yachts in New York, material growth is needed when the underlying goal is status. From feasts and pyramids to rockets and skyscrapers (especially those emblazoned with one man's name), such conspicuous consumption is largely status competition that has become manifest in grain, stone, metal, and concrete.

Today, this growth fetish underpins our environmental problems. Economic growth is one of the few factors that pushes against each of the planetary boundaries. To date, economic growth and the material footprint of societies are closely correlated. Growth across countries is set to increase

material extraction and is expected to rise by 60 per cent by 2060, endangering our ability to meet climate, biodiversity, and pollution targets.⁵⁴ Economic growth de facto makes us less likely to keep the climate sufficiently cool. Higher growth means we need to cut emissions more rapidly, a task we are already failing at abysmally. Plausibly high economic growth rates could make the worst-case emissions trajectory (which puts us on track for warming of around 4–6°C) 35 per cent likely.⁵⁵

Although it is possible to grow the economy while reducing emissions (usually called ‘decoupling’, a phenomenon we’ve seen in countries such as the UK) the rates required go far beyond anything we’ve seen. Equitably limiting warming to 1.5°C means that high-income countries would need to decouple emissions from growth an average of ten times faster than current rates.⁵⁶ There is just the raw, inescapable reality that more growth means sharper, more difficult emissions cuts. Staying within 1.5°C of temperature rise means annual reductions of global emissions of 7.6 per cent, a rate faster than that achieved by Covid-19 (6.3 per cent), for decades.⁵⁷ Doing that while rapidly expanding the economy – which would increase the amount of reductions required – seems at best unwise and at worst impossible. Moreover, even if we did miraculously decouple emissions sufficiently, we’d still have the problem of such material consumption pushing against the rest of the (too frequently ignored) planetary boundaries. The more we remain enslaved to Goliath’s impulse to grow, the less likely we are to keep the planet stable.

This is not to say that we should stop all economic growth: the poorest countries – from Afghanistan to Zambia – still desperately need to grow economically to eliminate poverty and increase prosperity. Instead, to stay within planetary boundaries we’ll need to eventually overcome the growth fetish and rein in conspicuous consumption. We’ll also need to realize that, past a certain point (which rich countries have already passed), growth doesn’t mean progress. Simon Kuznets, the inventor of Gross Domestic Product (GDP), our modern measure of economic growth, remarked in 1934 that ‘the welfare of a nation can scarcely be inferred from a measure of national income’. GDP was never intended to be a yardstick of progress, and it is a mistake to consider it as such: Portugal, for example, has 65 per cent less income per capita than the US but outranks it on almost every social indicator.⁵⁸

Rather than focusing on raw numbers of GDP, we should focus on what we are growing. Renewable energy, vaccine capacity, affordable housing, and leisure time are all things we should all be striving to increase and innovate in. Not bullets, bombs, cigarettes, SUVs, private aeroplanes, fossil-fuel infrastructure, or algorithms to keep teenagers glued to their screens. We need to start focusing on progress rather than growth. That means moving beyond growth and putting our production back under democratic control.

Growth simply measures economic activity, which includes the production of risk. The construction of every nuclear warhead, the manufacturing of every killer robot, and the chemical synthesis of every tonne of novel entities contributes to GDP. The biggest contributors to global catastrophic risk, whether it be in emissions, dangerous new technologies, powerful new AI models, nuclear arsenals, carbon emissions, or novel entities, are also the biggest economies: the US and China.

This isn't a problem of humans in general eating the earth. It's Gaia vs Goliath.⁵⁹

The Death-Star Syndrome

When the World Stood Still – The Hyperconnected World – The Accelerating World – The Concentrated World – Falling from the Zenith

WHEN THE WORLD STOOD STILL

As a species, we are more powerful than ever. In the last 10,000 years the yields of our food supply have grown by an order of magnitude. We can move 50 to 150 times faster than walking. Our life expectancy has tripled, thanks largely to reducing infant mortality and combating bacteria through better sanitation and antibiotics. Economic activity per person in wealthy nations is around a hundred times higher than at the peak of the Roman Empire. Availability of knowledge for the average individual has increased by an unfathomable amount. Today, a simple internet connection gives you access to 64 trillion gigabytes of data, a complete compendium from astronomy to zoology.^{[1](#)}

We are more powerful than ever, yet in some ways we are also more vulnerable than ever, as we discovered in 2020. New Year's Day that year was much like any other, but within four months over 3 million people had died, entire countries had been turned into fortresses and hundreds of millions had been locked in their homes.

This wasn't just a moment where we realized our own collective biological vulnerability. It was also a triumph of sorts. Vaccines were developed within weeks and it ended up being the fastest vaccine

development and rollout in history. It could have been far quicker if countries had invested more in vaccine production and companies had not hoarded the intellectual property of life-saving drugs.²

In March 2021, while the pandemic was still unfolding, another shock hit the globe when the *Ever Given*, a 200,000-tonne, 400-metre-long container ship, became stuck in the Suez Canal. Around 369 ships, representing approximately 30 per cent of global container traffic, backed up at either end of the waterway, costing the world \$6–10 billion in lost trade.³

Both these events could have been much worse: a virus that had killed 10 per cent of its victims rather than <1 per cent would have had a very different outcome, and the canal could have taken far longer to clear.

The world appears to be growing both increasingly robust and more fragile. I call this the ‘Death-Star Syndrome’, after the space-weapon in the film *Star Wars: A New Hope* that can annihilate entire planets but can itself be destroyed by a single well-placed blow. Our world is incredibly powerful and robust, yet surprisingly fragile if hit hard enough in the right place.

THE HYPERCONNECTED WORLD

Both the spread of Covid-19 and the economic fallout from the blockage of the Suez Canal were worsened by the interconnectedness of the world. During the twenty years from 1950, commercial air traffic grew in volume by over 10 per cent per year. By 2014 around 3.3 billion passengers were being carried every year. In 2024 that number was around 5 billion, and it could double in the 2030s. We can see a similar trend in shipping. Over the past 150 years, global shipping capacity has increased around sixty-fold.⁴ Today, exported goods account for around 25 per cent of global economic activity by value compared with 8 per cent or less before the Industrial Revolution.⁵

The world is economically more connected than ever. This can mean more connections between people, cheaper goods, and more countries dependent on one another and so less likely to go to war. Hyperconnectivity is also a key reason why collapse is suppressed in the modern world. When a state begins to fall, the international community can and usually does

mobilize resources to prop it back up (although, as we can see in Palestine, this depends on geopolitics). Yet this is not a complete good-news story.

The same links that can aid in recovery can also transmit a crisis around the entire system. One of the most striking examples was the WannaCry ransomware attack. WannaCry was a computer virus that could replicate itself through a network. The initial attack occurred at 07:44 UTC on 12 May 2017. In less than eight hours it had corrupted 300,000 computers across 150 countries, including networks in FedEx and the UK's National Health Service. Again, we got lucky: the virus just demanded a ransom payment from victims rather than destroying data, and it was able to be quickly defused. At 15:03 UTC the hacker turned cybersecurity expert Marcus Hutchins found a way to prevent the virus from spreading further. The problem with connectivity can be seen not just in computer viruses, but also in biological ones: the hyperconnected world enabled Covid-19 to move across entire continents and leapfrog oceans in just a few days.

Connectivity can often make a system resilient to small shocks but vulnerable to large ones. It is a pattern we see across systems, be they financial, digital, or ecological.⁶ Think of how quickly the global financial crisis of 2008 spread from a single housing bubble in the US across most of the world in the space of months.⁷

In the Late Bronze Age, interdependence was a blessing for elites in the best of times but became an Achilles heel in the worst. Their close trade in tin and copper – the prerequisites for making bronze – and their close diplomatic links, led to a state of interdependence. This meant that just the loss of a few key hubs, such as Ugarit and the Hittite Empire, was potentially enough to destabilize the system. The countries of today are, of course, far more technologically advanced and resilient than the Hittite and Assyrian empires of the Late Bronze Age, but the world is also far more interconnected and fast-moving. The Late Bronze Age also reveals that the problem is not connectivity per se but rather deep interdependence: when one part falls, so too do all those that are overly reliant on it.

THE ACCELERATING WORLD

The Black Death of the fourteenth century spread at the speed of medieval sailing ships; the Covid-19 virus, carried on a commercial airliner, travelled one hundred times faster; our online financial systems communicate around a hundred times faster than the speed of sound; most stock-market trading is outsourced to algorithms that can react far faster to minuscule microsecond-to-microsecond changes in the market and maximize profit.

A market ruled by lightning-quick machines can lead to lightning-quick disaster. In 2010 the US stock market lost over \$1 trillion on stock indices within thirty-six minutes. That flash-crash was caused by algorithmic trading, which accounts for about three-quarters of stock-market trades. The system operates far too quickly for any human to intervene and prevent a crash.⁸

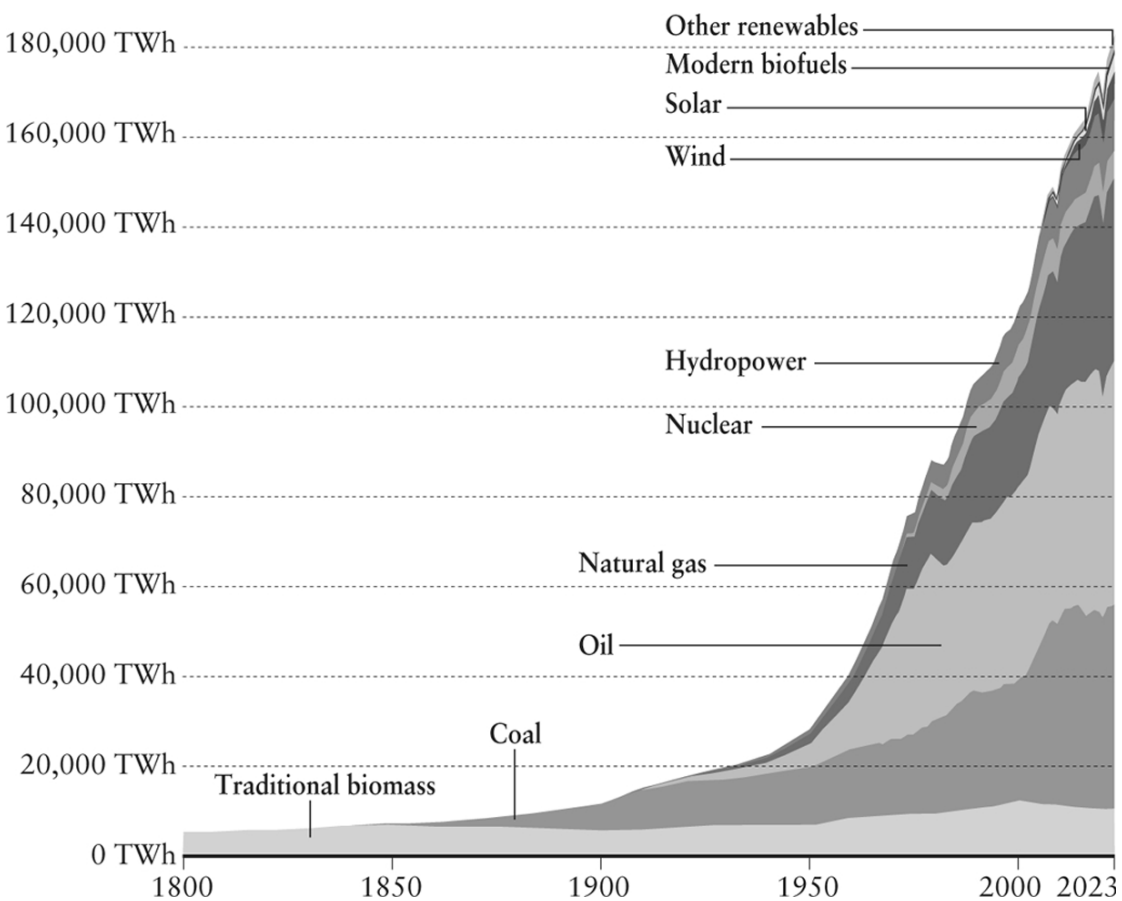
Algorithmic trading is one example of ‘tight-coupling’: when two components in a system are closely interdependent and respond quickly to each other’s moves. Another common example is ‘just-in-time’ supply chains, whereby manufacturers avoid keeping expensive inventories by having parts arrive when needed. As with algorithmic trading, it makes for an efficient but vulnerable system. In 2011, floods in Thailand engulfed a central hub for producing car parts and hard-disk semiconductors that most of their customers relied on for just-in-time delivery. There were factory slowdowns and closures across Japan, Malaysia, and Indonesia. The price of hard drives rose by up to 190 per cent, exports from Thailand to the European Union decreased by 35 per cent (and over 20 per cent for the US), and car manufacturers in Japan saw a decrease of up to 50 per cent in profits as well as 174 days of disrupted production.⁹

Having more back-up stockpiles and slower trading with more human involvement would lessen the potential for factory shutdowns and flash-crashes, yet countries and companies have little incentive to do so. Efficiency, low costs, and speed are essential in a deregulated, high-competition, profit-driven economy, and building resilience costs money. Under the tyranny of the quarterly earnings report, long-term planning takes a back seat.

That quickening pace is also evident in the environmental footprint of the Global Goliath. After 1945 most socioeconomic trends speeded up rapidly,

whether it be population growth, the production of paper, the consumption of fertilizers, or the constructions of dams. Between 1950 and 2022 the global economy increased fourteen times over, going from around \$10 trillion to approximately \$140 trillion.¹⁰ All of those rapid increases required huge material extraction from the environment. Whether it be the amount of water dammed, greenhouse gases released into the atmosphere, or the number of shrimps farmed, human impacts on the environment all soared dramatically. This is sometimes called ‘the Great Acceleration’ ([Figure 41](#)).¹¹

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[Figure 41](#): The Great Acceleration

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The upper graph shows global GDP (international \$ at 2017 prices) over the past 2,000 years, while the lower shows energy consumption over roughly the last two centuries. Economic growth was incredibly slow over most of history: global GDP was around \$213 billion in 1 CE and took around 1,500 years to double to \$500 billion. Both global GDP and energy consumption have a similar trend, accelerating with the Industrial Revolution and again in the 1950s. Many other indicators, whether it be plastic production, water consumption, global greenhouse-gas emissions, or the number of fish caught globally, all show a similar spike around the 1950s.

But it wasn't just a great acceleration; it was a great coupling. The fate of the earth system – whether it be through the proliferation of novel entities into water streams, the number of species going extinct, or changes in the temperature of the planet – is now tightly intertwined with the Global Goliath. And whether we like it or not, the Global Goliath is still dependent on the earth system (remember where all that cobalt, silicon, uranium, steel, and coal comes from).

THE CONCENTRATED WORLD

The world is becoming increasingly monopolized. Economic power, the goods and services we use, and even our infrastructure are all becoming more monopolized and concentrated. You can taste it in every meal you eat. Around 60 per cent of calories consumed globally come from just four crops: soy, corn, rice, and wheat (the last three were the original lootable resources and tax crops), and around three-quarters of global production of these four staples comes from just five countries.¹² In the private sector, two companies own about 40 per cent of the global seed market, four control 44 per cent of the global agricultural machinery markets, and four dominate 62 per cent of the market for agrochemicals.¹³ That concentration of market power and control shows no signs of reversing: mergers and acquisitions between companies are making the global food system more and more concentrated. It's a process driven by dynamics we've already seen throughout history: those who have power (or in this case, market share) find it easy to acquire more power.¹⁴

The concentration and monopolization of market shares can be seen everywhere one looks. In the US in 1993 there were 193 defence contractors. Now it is essentially just five colossal firms. In the US, over half of printed book sales and 80 per cent of eBook sales happen through

Amazon (the numbers globally are about the same). In 2015, 64 per cent of internet advertisement revenue went to either Google or Facebook (although Google's share triples that of Facebook). One city (Hsinchu) and two companies (UMC and TSMC) in Taiwan satisfy 60 per cent of the world's appetite for semiconductors. And finance is so concentrated that some institutions are now deemed 'too big to fail'. This is a problem in itself: banks and airlines can often indulge in risky and irresponsible behaviours because they know they can rely on government bail-outs (a phenomenon known as 'moral hazard').¹⁵

These concentrated, homogeneous systems tend to be the most vulnerable to shocks.¹⁶ If diets are highly dependent on one crop, then one disease or shock can create a famine, as in 1845–52, when Ireland lost about a quarter of its population due to potato blight in combination with the policies of the British government.

There is also some evidence to support the old truism 'the bigger they are, the harder they fall'. A 2020 study found that larger ecosystems tend to shift more sharply than smaller ones. Think of the change from a coral reef to an algae-dominated ecosystem after bleaching. After analysing twenty-five marine ecosystems, fifteen freshwater ones, and five terrestrial environments, one study found that that any increase in ecosystem size meant quicker change. The team concluded: 'The collapse of large vulnerable ecosystems, such as the Amazon rainforest and Caribbean coral reefs, may take only a few decades once triggered.' Whether this applies to societies is unknown.¹⁷

A more concentrated world poses vulnerabilities that could be catastrophic. Some scholars think that a volcano would need to rank as an 8 on the volcanic explosivity index (VEI) (a phenomenon colloquially called a 'supervolcano') to trigger a global catastrophe in the modern world.¹⁸ Yet a smaller volcano could also have an equally high impact owing to the structure of our world.

There are spots critical to the global economy where shipping lines, underwater internet cables, telecommunications cables, and other international infrastructure are clustered. These are what the University of Cambridge volcanologist Lara Mani has called 'pinch points'.¹⁹ A category 7 volcano hitting one of these vulnerable pinch points could be calamitous. The chance of one of these erupting in the next century is around one in

five.²⁰ Vulnerable points, interdependence, and rapidly reacting systems are all key ingredients that allow for a big shock or crisis to spiral into collapse.

FALLING FROM THE ZENITH

‘From morn to noon he fell, from noon to dewy eve/ a summer’s day; and with the setting sun/ dropped from the zenith like a falling star.’ John Milton penned those lines in 1667 for the epic poem *Paradise Lost*, yet they are surprisingly relevant to our modern world. While the current Global Goliath may seem unassailable, it contains several chinks in its silicon armour.

More hyperconnected, hyperconcentrated, hyperhomogeneous, and hyperaccelerated systems make sufficiently large shocks more likely to occur and spread. This is systemic risk: the potential for individual disruptions or failures to cascade into a system-wide failure.²¹ It’s something that scholars have been warning about for over a decade. Such systems create ‘synchronous failures’ in which crises spill across systems.²² We see that in the global, industrial world in the close association of oil crises with financial crises: financial crises tend to impact fossil fuels heavily, and often vice versa (such as when the 1973 oil crisis helped trigger the 1973–5 financial crisis).²³ Today, crises across global systems – like food, energy, and finance – are entangled in causes and effects, creating what some call a ‘global polycrisis’.²⁴ Underneath all this are just some simple structural features that are fundamental to the Global Goliath: speed, connectivity, and concentration.²⁵

None of this is about bad luck; it’s about the structure of the system. The Global Goliath is like a stack of ladders. Imagine a hundred ladders lined up against a wall. Individually, it’s easy to make any of them topple, but you must hit them one by one. Now, imagine you tie them together into a single giant ladder. It takes an act of great strength to push this one over, but everything falls together in a single gargantuan crash.²⁶ That is the situation we find ourselves in today.

The Rungless Ladder

Swords of Damocles – Fattening Tails – Termination Shock – The Problem With Techno-Fixes – The Somalia-Denmark Rule – Not as Improbable as You Might Think – Falling Fossil Fuels – The Rungless Ladder

SWORDS OF DAMOCLES

In Greek mythology, Damocles, a courtier of the ancient King Dionysius I of Syracuse, Sicily, is fawning over the monarch and telling him how lucky he is to be king. Dionysius offers to let him take his place on a throne at the head of a table overflowing with the finest food and surrounded by flatterers. The catch is that he hoists a sword directly above Damocles' head. Suspended by a single horse hair, it could fall at any time. It is the king's way of showing that, while he may have wealth and power, he lives every moment under constant threat of assassination. Power sits in the dark shadow of precariousness.

This is a fable made for our modern world. Dan Zimmer, a catastrophic-risk researcher at Stanford University, has compared the creation of nuclear weapons and surging economic growth in the post-war period to a contemporary Sword of Damocles. Distracted by growing material comfort, we ignore the atomic sword hanging overhead. We live under not one sword, but many. As riches multiply, so too do the existential threats delicately suspended above us.

Violence and deaths from infectious disease are at historical lows and life expectancy is at a historical high. Yet we also live in the most dangerous

time in human history. This is because of catastrophic technological threats, planetary boundaries, systemic risk, and our responses to the prospect of catastrophe. Together these are all building up ‘latent risk’: risk that does not cause harm today but would do so in a post-collapse world. Our Global Goliath is continuously inflating the costs of collapse, meaning that a future societal breakdown will be far worse than any we have seen in the past.

FATTENING TAILS

You are about twice as likely to take your own life than to be murdered. You are more likely to die from cancer than from any infectious disease. Unfortunately, this does not mean that we live in the safest of times or are on the cusp of defeating disease. Far from it.

Deaths from disease and violence follow a ‘fat-tailed distribution’.^{[1](#)} They are marked by a few outsized events, such as the Spanish flu, the Black Death, and the Columbian Exchange. This makes observing an average from one year, or even ten, essentially meaningless. The average will be brutally skewed by these ‘tail events’. The extremes, not the average, are the main source of risk. Importantly, these tails are becoming fatter and longer over time. Even if the world on average seems safer and healthier, the extremes are growing in likelihood and severity.

When it comes to violence, you may have a lower probability of being bludgeoned to death than a citizen of Uruk in 4000 BCE. But in 4000 BCE there was no weapon that could kill half of the people on the planet. The risk of violence hasn’t decreased per se; we’ve simply lowered the number of observed killings per year while enormously inflating the risk of global mass murder. All it takes is one bad day, one mistake, and suddenly the violence statistics begin to look very different.

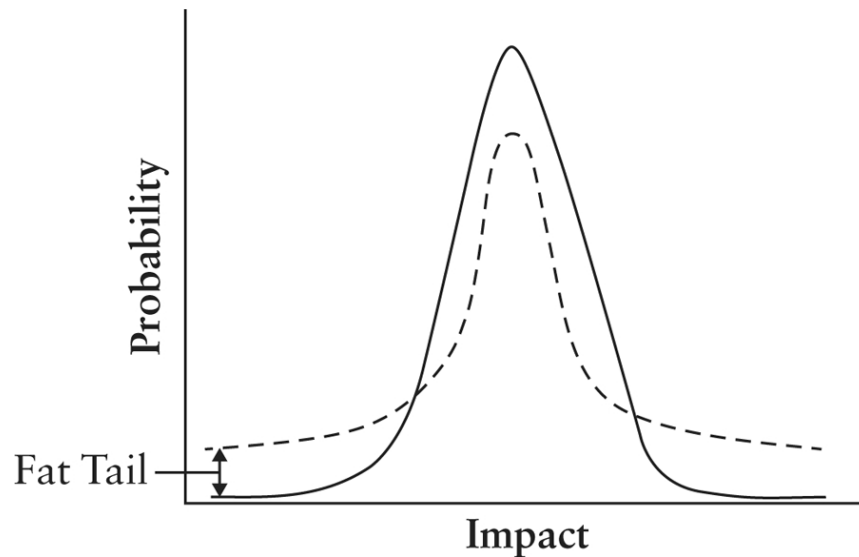
Every action or scenario has risks that vary according to their likelihood and severity. The combination of those will form a ‘risk distribution’. Imagine you are forced to play one of two games of chance. In one, you are flipping a coin for a prize and penalty of \$10,000. There’s a 50 per cent chance you lose \$10,000 and a 50 per cent chance you win \$10,000. Imagine a second game: Russian roulette. You sit around a table with five others, a revolver loaded with a single bullet in front of you. If you survive,

you win \$1 million. There is a five in six chance you walk out a millionaire, and a one in six chance that you die. Technically, your odds of victory are far higher, and the monetary payoff is far larger, yet most of us would opt to flip the coin instead. That is because the risk distribution of the two games is overwhelmingly different, especially the ‘tail risk’ of a low-probability, high-impact bad outcome. The two games have different risk distributions: Russian roulette has a better average outcome, but a huge tail risk.²

What we have done with violence is to shift the risk distribution much more towards a game of Russian roulette. We have inflated the tail risk, while making the average outcome better.

It is illogical to say that the world is becoming safer while tail risk is growing. It’s like saying that your neighbour is a pacifist, except for that one stabbing spree when they killed fifty-five people.³ It is a fallacy to believe that we live in a peaceful age with the nuclear Sword of Damocles hanging overhead. This is true for disease as well. Infectious disease has followed a fat-tail distribution throughout history.⁴ A mass-fatality engineered virus would, like nuclear weapons, be growing the tail risk ([Figure 42](#)).

The greatest threats we face today, whether it be from AI, climate change, or nuclear weapons, all began after the Industrial Revolution. They all require combustion engines, electricity, and international supply chains fuelled by oil, coal, and gas. Coal, cobalt, and uranium are all built on cheap, exploited labour arrangements that began with colonization. These threats are the tail risk of the Global Goliath.



[Figure 42](#): Fat Tails

The taller curve is roughly a ‘normal’ distribution, while the dashed line is a ‘fat tail’ distribution. The key difference between the two is that the fat tail distribution has higher probabilities of big losses or big gains. This can be seen in the gap between the tails of the two distributions.

This is pivotal for collapse. These tail risks are either key triggers for a collapse or could be set off domino-like during a societal breakdown. A nuclear war could trigger a global breakdown, or, conversely, a period of international instability and conflict due to climate change and runaway technology could trigger a nuclear war. These planetary Swords of Damocles make collapse a dark prospect. Latent risk is growing not only due to fatter tail risk, but also because of our responses to managing catastrophe.

TERMINATION SHOCK

In the future, we may live in a world where a fleet of planes patrol the skies 24/7, pumping out chemicals that reflect sunlight, such as sulphur dioxide, to cool the earth – a process known as stratospheric aerosol injection. It is the cheapest and most likely form of solar geoengineering: large-scale efforts to change the reflectiveness of the earth and so lower global temperatures and combat climate change. (For simplicity I’ll be using ‘solar geoengineering’ in place of ‘stratospheric aerosol injection’). It involves

spraying particles that reflect sunlight, such as sulphur dioxide, to reduce temperatures. These aerosols act as minuscule space mirrors, reflecting incoming solar radiation back out to space and cooling the planet. Solar geoengineering would reduce global temperatures.

Solar geoengineering will be experimented with, and perhaps even globally deployed, in the coming decades. It's relatively cheap: solar geoengineering would cost only around \$2–10 billion per year.⁵ To put that into perspective, the US Department of Defense requested a budget of \$849.8 billion for 2025. It's also likely to work. Solar geoengineering replicates volcanic eruptions, which also spew reflective particles into the sky with similar effect, as in the 1991 Mount Pinatubo eruption that cooled the earth by around 0.5°C during the following year. Oil-producing countries (and their rich customer states) would support such a move, of course, since it would allow more breathing space to use fossil fuels. The most vulnerable might eventually see it as the lesser of two evils.

Ironically, human-emitted pollutants are already cooling the planet by about 0.5°C. If we were to ban them, we would experience a surge in warming that would push up temperature increases close to 2°C. In 2020 the International Maritime Organization (IMO) introduced new regulations that lowered the sulphur content of shipping fuels from a maximum of 3.5 per cent to 0.5 per cent. It effectively cut sulphur emissions from shipping by 70 per cent.⁶ Since then, the world has experienced an acceleration of global warming, with temperature increases now exceeding 1.5°C. This speeding-up of warming has probably been due to the reduction of shipping pollution that has lowered the amount of coolants in the atmosphere. Whether we like it or not, we are already solar geoengineering the climate by emitting both aerosols and greenhouse gases.

Is solar geoengineering safe? Or could the cure be worse than the disease? In 2021 Dr Aaron Tang and I examined what its catastrophic downsides could be. We discovered that, unlike the potential of global warming, surprisingly little is known about the impacts of cooling by solar geoengineering. It could slow the monsoon and change rainfall patterns across the world, but much remains uncertain. The risk will also depend on how it is deployed, the amount of cooling, and the kind of world it is used in. Most models bet on the best-case scenario by assuming solar geoengineering will be deployed with perfect international cooperation for

decades on end. This isn't a safe assumption. It is more likely that a small cadre of countries will resort to a rushed deployment because the world is warming even more rapidly than expected.⁷

Solar geoengineering will also pose other unique problems such as political instability. After a solar geoengineering project is initiated, every extreme weather event may be attributed to it. Conspiracy theories and accusations of wrongdoing will abound, especially if there are no reliable scientific methods to determine just how much a solar geoengineering experiment contributed to a drought, flood, or hurricane. If the project were to be undertaken unilaterally, the country responsible would be blamed by the rest of the world.

Finally, solar geoengineering creates the problem of termination shock.⁸ Imagine that the world is facing a temperature rise of 4°C by 2100. In one future, we didn't use solar geoengineering. It took 250 years to go from pre-industrial temperatures to this hothouse earth state. That is because temperature rises alongside increasing greenhouse-gas emissions. Now think of a second future where the world implemented a solar geoengineering system in 2040. We've had six decades of cooler, more stable temperatures. Then, in 2100, the system is knocked out by a major event such as a geomagnetic storm, a nuclear war, or a severe pandemic. The solar-reflective particles unfortunately wash out of the atmosphere with a half-life of around eight months. Within a year and a half, over 75 per cent of the chemical mirrors that have been cooling the earth are gone, yet we still have high levels of greenhouse gases blanketing the atmosphere.⁹ The world begins to heat up fast. Rather than taking 250 years to reach 4°C of heating, it only takes about two decades. This failure of solar geoengineering would result in a warming rate a hundred times faster than the greatest mass extinction event in history. This is termination shock: it is the latent risk of solar geoengineering.

Termination shock is not inevitable. Perhaps we won't be hit by a debilitating catastrophe; perhaps we'll wisely use the time that solar geoengineering provides to reduce emissions until we can safely phase out the system. Yet it is nonetheless a gamble. Solar geoengineering does not solve climate change per se: it reduces the average risk but fattens the tail risk. The average outcome of a solar geoengineering effort might be far less

damaging than the climate change it is preventing, but in the worst-case scenario termination shock could make matters far worse.

Solar geoengineering would also only be a partial fix. It wouldn't stop the oceans from turning more acidic because ocean acidification is primarily caused by a chemical reaction between ocean water and carbon in the atmosphere. Solar geoengineering would also be helpless to reverse the effects of passing certain tipping points. Lowering global temperatures can't stop the AMOC from shutting down once a threshold has been breached or reverse it once it has shut down because the tipping-point-induced shutdown is driven by dynamics in the ocean, not by global temperature. Similarly, if we lose ice sheets in the future, we won't regain them by lowering the temperature back to present levels.^{[10](#)}

Even so, solar geoengineering could be one way to significantly reduce climate risk. We could also try to design a system to safeguard against different potential catastrophes, such as strong insulation against electromagnetic pulses and multiple back-ups. A well-constructed solar geoengineering system that is implemented with global cooperation, that doesn't act as an incentive to keep burning fossil fuels with impunity, that is robust to most global catastrophes, and that is used temporarily to shield the world while we rapidly decrease emissions, would be better than facing the direct impacts of climate change. We need to question how likely such a scenario is and how much of an increased tail risk we are willing to tolerate. There is no right or wrong answer as to whether that trade-off is worth it. We simply need to carefully and democratically pick our poison.

THE PROBLEM WITH TECHNO-FIXES

Solar geoengineering is representative of a wider set of techno-fixes to environmental problems. Almost every environmental threat has a matching engineering solution that mitigates the hazard without addressing the causes of the risk.

This holds true for even regional-scale disasters such as cyclones.^{[11](#)} There are several engineering interventions to prevent the formation or decrease the intensity and duration of cyclones. These include cloud seeding (injecting silver iodide or other chemicals that help form clouds and

precipitation), and aerosol injection (firing up sunlight-reflecting or other particles to de-energize the cyclone or influence clouds). This has yet to be done at scale, but it would probably work, and a group of Australian scientists have already floated the idea.¹²

Similarly, there are many ways to prevent the threat of volcanoes, whether at a regional or global level. These include drilling, fracturing volcanic rocks, and cooling off the magma that surrounds the central reservoir. These volcanic interventions could help release gas that is pent up, lower the intensity of an eruption, or even completely prevent it.¹³ For now, all these remain understudied and none has been implemented at scale.

There are even ways to use engineering to combat the biodiversity crisis. Imagine a world in which you go to smell a flower and a robot bee flies off the stamens. This is not sci-fi. The world is losing pollinators such as honey bees at an alarming rate and potential replacements include mechanical ones.¹⁴ It has also been suggested that we could artificially resurrect or ‘de-extinct’ lost species. There are multiple ways of doing so, but the one that has received the most attention involves using preserved genetic material to create a clone embryo that is then carried to birth and raised by a near relative (an elephant raising a woolly mammoth, for example).

These are all part of a wholesale drift towards a more artificial world.¹⁵ In this case, deeper institutional, or behavioural policies are overlooked in favour of a techno-fix. This doesn’t mean that all these interventions are undesirable. Volcanic engineering is something we’ll have to turn to eventually if we want to avoid the devastation that would accompany a large-scale eruption. It is something that we should be putting far more money into researching and exploring. Similarly, given the rate of warming and consequences of climate change, we should be exploring and ramping up technical efforts to either delay or reverse warming. This could include attempts to restabilize melting ice sheets or to draw down large amounts of carbon by growing trees and using mechanical filters.

But we also need to be aware that an increasingly artificial world comes with a cost. Technological solutions to environmental problems are more likely to fall apart during a collapse. If we rely on using aerosols to prevent cyclones, then imagine what the loss of such methods during collapse would mean: an unprepared and unprotected world would be facing cyclones intensified by global heating. Becoming dependent on mechanical

pollinators means exactly that: relying on a pollinator that can be lost owing to fracturing of the energy supply or a breakdown of industrial infrastructure. A nuclear war, or even an explosion of solar energy from the sun, could knock out the global electricity supply and all our techno-fixes with them.

Growing latent risk can also be seen in disease. Conditions such as HIV are treatable and manageable with modern medicine, but in a post-collapse world without antiretrovirals, HIV could become an even more dangerous prospect than in the 1980s. The same applies to the many devastating diseases that today have been neutered by improved nutrition, health, and treatments.

As with ecological risk, there are tempting techno-fixes for the next pandemic. For instance, some biosecurity experts have pushed for fitting Far UVC lighting across as many buildings as possible.¹⁶ These are essentially lights that sterilize a room without negatively impacting human health. Yet, if overused, a decreased exposure to bacteria could leave our immune systems weak and vulnerable. This could still be an intervention worth exploring, but we do need to think about the downsides and unintended consequences.

New technologies, including techno-fixes, tend to be built on the pre-existing stack of technology. AI requires reams of data sourced from the internet, all of which depend on computers, and computers in turn rely on electricity, which is built on an expanse of industrial infrastructure from tankers moving coal to the mass smelting of steel. The constant layering of technologies on top of each other makes the overall technological system more complicated, interconnected, and potentially fragile. If the electricity grid goes down, it would take most technology with it. Technological complexity and dependency carry a cost.¹⁷

THE SOMALIA-DENMARK RULE

Not all states are alike. Many states today are far more accountable and spend far more on welfare (and far less on the military) than in the past. The US government (in 2025), which has a much larger military than any other country, still spends only 13 per cent of its budget on its military while

allocating 21 per cent for social security. If you are a citizen of Denmark, you'll get twenty-four weeks of paid leave after having a child (as will your partner), whom you can then send to heavily subsidized childcare. When they leave school they can study for free at a university, before eventually retiring with a decent pension. The benefits offered by these democratic welfare states, and the amount of control citizens have over them, is almost unimaginably high when compared with the Aztec or Akkadian empires.

States of today are also far more robust than those of the past. Sargon and Narmer had to rely on exaggerated intimidation, and most states throughout history didn't even have a full-time police force. In contrast, rich states today have sophisticated bureaucracies (including specialized emergency services), full-time policing and defence forces, mass surveillance systems, and sophisticated intelligence agencies. There has been an enormous shift from intimidation to administration. In short, most citizens in rich countries have less reason to revolt, and far less likelihood of doing so successfully, than those in pre-modern states. After five millennia of Goliath evolution, states have reached an unprecedented level of power and stability. More democratic states tend to provide more public goods, maintain a social safety net, and integrate their citizens more thoroughly.

As always, however, the benefits come at a cost. A welfare state is a good thing, but it becomes a liability if the state disappears. One survey of thirty historical regimes found that good governments (more democratic states providing public goods) were more likely than autocracies to experience severe collapses marked by severe, system-wide decline and a significant population loss.^{[18](#)}

Today's inclusive, highly centralized institutions handle crises well, but the strong reliance they generate means that if they do go down, the fall can be harder.

We can think of this as the Somalia-Denmark rule: the more predatory a state, the less harmful (or even more beneficial) its collapse; the more inclusive and benevolent a state, the more costly its collapse. If you are a citizen of Denmark, you depend on a relatively benevolent state. You have much to lose from its collapse. A Somali living under the Barre regime was better off being governed by a local warlord.

Not only are more of us dependent on states, but we are also simply more dependent on globalized infrastructure. You might be reading this while sipping coffee grown in Guatemala, made using a machine powered by a

national electricity grid reliant on coal from China or gas from the US, with a phone next to you that is partly composed of rare earth metals from across the horn of Africa. Not only that, but most of us outsource vast amounts of knowledge to the systems around us. Individually, we know infinitesimally little about how the world works. Most of us lack the knowledge of how even basic items such as zips work. Scholars have compared us to ants: as we have come together in larger, more hierarchical groups, individuals have become more specialized.¹⁹ Individual hunter-gatherers might have had the knowledge to live from the land. Most of us now are hyperspecialized while lacking basic survival skills. Knowledge is increasingly held not by individuals but rather by groups and institutions. Our hyper-specialization and dependence on global supply chains can give us great power, but that dependency is also a vulnerability.

This might not be an issue if the risk of a global catastrophe or of these systems failing were sufficiently low. Unfortunately, that is not the case.

NOT AS IMPROBABLE AS YOU MIGHT THINK

Global catastrophe is often written off as being speculative, long term, and of such a low probability that we don't need to worry. For instance, some analysts of solar geoengineering have downplayed termination shock, since a calamity large enough to knock the system offline is too rare and humanity would simply recover and reinstall it.²⁰ Most estimates suggest that running a solar geoengineering system would cost less than 1 per cent of global GDP. Even an unprecedented economic catastrophe, something that wipes out 75 per cent of economic activity, would still leave plenty of money to reactivate the system.

Neither of these is a convincing objection. Assuming states will swiftly reinstate solar geoengineering, or another artificial support system, in the wake of calamity means ignoring recent experience. In 2021, creating 3 billion doses of Covid-19 vaccines would have cost \$18–120 billion and provided benefits of \$17.4 trillion.²¹ Despite this, vaccine production remained stubbornly low. States and leaders are not rational actors who always act in the wisest, most cost-effective ways during a catastrophe.

Relying on them to coolly and quickly reactivate a solar geoengineering system is naïve.

Moreover, global crises are not as improbable as commonly assumed. The chance of a high-impact volcanic eruption in a century is one in five. Extreme space weather events are also unnervingly likely. These include solar flares (mass ejections of electromagnetism from the sun) that cause geomagnetic storms in the earth's magnetic field, disrupting and frying electrical infrastructure. The 1859 Carrington Event is usually used as the threshold for a space weather disaster. This was an intense geomagnetic storm that luckily occurred before most of the world was dependent on electricity. If it occurred today, a similar event would cause widespread electrical failure and disruptions for months. It would probably be enough to knock out a solar geoengineering system. Estimates for a future event that is as bad as or worse than the Carrington Event, range up to 20.3 per cent per decade. While there is no consensus, and we should place little faith in these numbers, they are not negligible either.^{[22](#)}

The same can be said of nuclear weapons. We don't – and can't – know the exact likelihood of a nuclear war, but all the estimates agree that it isn't low. One study mathematically modelled the likelihood of an inadvertent nuclear war between the US and Russia based on past near misses and false alarms. They calculated an annual probability of between 0.001 per cent and 7 per cent.^{[23](#)} One expert suggested it was closer to 1 per cent per year.^{[24](#)} In 2022, Harvard Professor Graham Allison put the odds of Russia nuclear bombing the US at between 0.1 and 1 per cent.^{[25](#)} It's unclear whether that likelihood has since increased. Since Russia invaded Ukraine, superforecasters have calculated the likelihood of Russia using nuclear weapons outside its territories at 2 per cent.^{[26](#)}

Pandemics too are not a very low-probability occurrence. Pandemic outbreaks are a matter of when, not if. Using a well-documented track record of outbreaks since at least 1600, scientists have calculated that one occurs roughly every 129 years. That means there is a 38 per cent chance of experiencing a pandemic similar to Covid-19 within an average lifetime. This risk is based on past experience, but the world is changing. As deforestation, environmental destruction, and climate change worsen, more animals are pushed out of their usual habitats and the chances of another

zoonotic infection increase. The probability of a pandemic could double in the coming decade.^{[27](#)}

Then there are other, more speculative threats that we can't place clear numbers on, such as the creation of a powerful AGI or an engineered virus. All these threats add up. Even just one, like nuclear war, might seem low at 1 per cent per year, but becomes unnervingly high over the space of a century. We are not just facing the threat of nuclear war, but of all these agents of catastrophe together. These drivers also interact. For instance, extreme climate change both heightens the risk of a mass extinction event and inflames geopolitical tensions.

If such a catastrophe does occur it may be far more difficult to recover from than is commonly assumed. That is because the energy ratio of fossil fuels is drying up.

FALLING FOSSIL FUELS

On 10 January 1901, the Texas Oil boom began. An oil drill tapped into the Spindletop Hill reservoir, sending a forty-five-metre geyser of oil into the sky.

These 'gushers' can now be seen only in black and white photos. The fossil-fuel industry is instead having to resort to increasingly technical, complicated, and costly procedures, such as fracking for shale oil and gas. They have also had to resort to extracting oil from dirtier, 'unconventional' sources like tar sands. Tar sands are seas of loose sand that contain bitumen (a kind of crude oil) mixed with clay, sand, and water. Tar sands are hard to extract oil from and produce around 31 per cent more emissions than conventional oil wells.^{[28](#)} And fossil-fuel companies are also turning to machine learning to locate harder-to-find sources of coal, oil, and gas. While these techniques have expanded the supply of fossil fuels, they're also more costly, in both money and energy.

Over the years, it is taking more oil to extract a barrel of oil from the ground. In the US, by 2007 the energy ratio of oil and gas (that is, the number of barrels that can be extracted from the ground using one barrel of oil) had plummeted from 1200:1 to 5:1. That is a 240-fold decrease. Globally, there is a race between a decline in the energy ratio due to the

depletion of fossil fuels and technological efforts to increase it. Right now, depletion is winning.²⁹ The energy ratio of coal is expected to fall similarly by 2040 at the latest.³⁰ One study looked at the ratio at the final stage, once energy reaches the economy, rather than when it is first extracted from the earth. The researchers suggests that we are overestimating the current energy ratio of fossil fuels. By their calculations it is currently just 6:1 worldwide and declining.³¹

Future projections are no less comforting. By 2050 the energy ratio for oil could fall to 1.5:1.³² The long-term energy ratio of gas is also set to fall dramatically.³³

The trajectory of the energy ratio has led to fears of an ‘energy cliff’ or ‘energy cannibalism’. If the energy ratio drops to such a level then the world can no longer maintain its current production or affluence. One solution would be to move to renewable energy. The downside is that the energy ratio of renewables is not particularly high compared to the fossil fuels extracted during the nineteenth and twentieth centuries. The upside is that they are still reasonable (approximately 10:1 for wind and solar) and increasing owing to technical improvements.³⁴ If we look at the energy ratio for when energy reaches the economy, then renewables may already offer better returns than fossil fuels. All technologies tend to reach a point of diminishing returns and stagnate, but hopefully the improving energy ratio of renewables continues for decades to come.³⁵

It is difficult to overstate how important this trend – and the race between the energy ratio of renewables and the falling energy ratio of fossil fuels – may be. Progress over the past two centuries has been dependent on vast infusions of low-cost fossil fuels. We often don’t see it since we are largely ‘energy blind’ to how much coal, oil, and gas goes into the food we eat and the systems that warm, cool, and illuminate our homes.³⁶

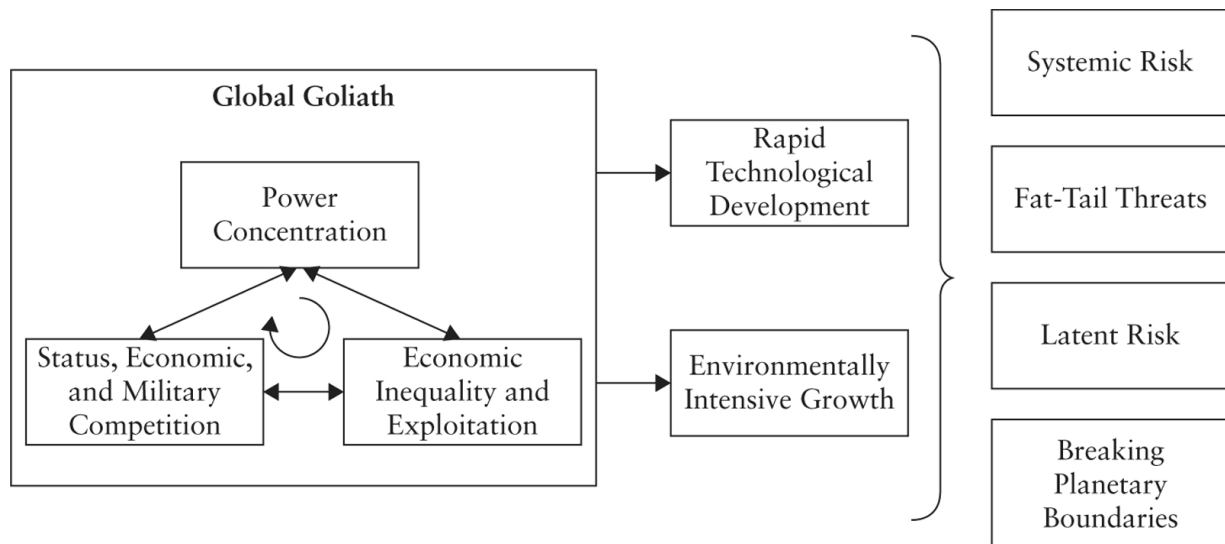
The first wave of industrialization was fuelled by geysers of oil and seemingly bottomless pits of easily accessed coal. These low-cost sources are no more. If the world collapses, it may be impossible to re-industrialize.

THE RUNGLESS LADDER

Think of our global society as a poorly built ladder. As we climb, the rung we have just used falls away. The higher you climb, the more dangerous the fall. Eventually, any drop from the ladder becomes fatal. The missing rungs represent the heightened severity of collapse due to more powerful weapons, more disease burden, a more deteriorated global environment, and a reliance on industrial agriculture, states, increasingly complex technological systems, and global systems suffering from the Death-Star Syndrome. The first farmers of Europe had only axes and arrows to fight with, had room to disperse, and could slowly transition towards doing more herding and foraging and less farming. Their fall, when it came, was piecemeal, gradual, and from just a few rungs high. It was often even beneficial. Now the fall is likely to be far more bruising.

The global technological and environmental threats we face today along with our hyperconnected and hypercoupled system breed new vulnerabilities, making it easier for big shocks to travel across both borders and sectors, gaining in power as they do. And some of our attempts to ward off disaster build latent risk into the system. The Global Goliath is creating new, unprecedentedly destructive threats, new vulnerabilities through systemic risk, increasing the ability for regional shocks to amplify into global ones, and inflating latent risk. Collapse today looks very different from the liberation of Cahokia or the decay of Rome. We are facing a genuine global catastrophe that could end our industrial world and much of life with it.

In chess, the endgame refers to the stage where only a few last moves are left. The world of Goliath that started around five millennia ago also appears to be reaching its endgame ([Figure 43](#)). The endgame is when we face the genuine possibility of existential risk: deep, potentially permanent, global collapse or even human extinction. In the next few decades or centuries, it is highly likely that the system will self-terminate unless it is fundamentally reformed. For the first time, there is the alarmingly high risk that Goliath will destroy itself permanently and may even take our species with it. Such high stakes naturally raise the question of who or what is to blame for this precarious predicament.



[Figure 43](#): The endgame

All the contributors to global catastrophic risk, whether it be fat-tail threats like nuclear weapons, planetary boundaries like climate change, systemic risk or latent risk, are driven by the Global Goliath. The Global Goliath doesn't only grow or drive technological development. Since the social and environmental costs (like the costs of coal in terms of air pollution and climate change) aren't included in the price, it selects for particularly environmentally and socially destructive types of technology and economic growth.

The Roots of Our Endgame

*Are We All to Blame? – The Agents of Doom – Crafting Threats in the Dark
– Make Your Own Nightmare – Hate the Game and the Players – Root
Causes – The Tree of Doom*

ARE WE ALL TO BLAME?

It's easy to think that we are all to blame for any future collapse. That we find ourselves in the precipitous position of facing unprecedented climate change, thermonuclear war, systemic risk, and dangerously rapid technological change due to human nature or each of our everyday choices.

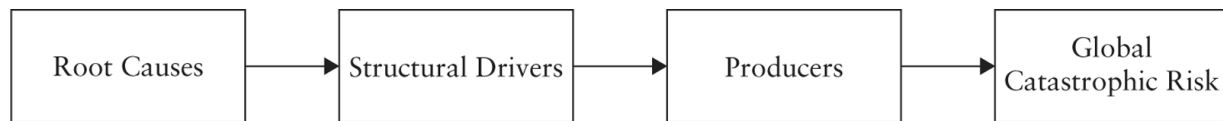
This idea of collective blame did not spontaneously appear; it was planted into the public conversation by a few powerful actors. This can be clearly seen in the case of climate change. In 2004 the fossil-fuel giant British Petroleum (BP) hired the public relations firm Ogilvy & Mather to help deflect responsibility away from itself and onto consumers. As part of this 'beyond petroleum' media campaign (which cost over \$100 million) they popularized the term 'carbon footprint', a measure of how much carbon individuals emit going about their everyday activities, whether it be eating breakfast or taking the bus to work. They even developed their own carbon footprint calculator to help everyone estimate just how ashamed they should be of their lifestyle. In 2019 they revamped their efforts through a 'Know Your Carbon Footprint' campaign. While individual action is, of

course, important, the aim was always to shift responsibility onto the public.¹

It was a strategy that was more broadly shared. One analysis of 212 documents from Exxon, another leading fossil-fuel producer, found that they had systematically framed climate change as a problem of individual culpability. For instance, in 1998 Exxon asked the public to combat growing emissions by exercising ‘a little voluntary “can do”’, before merging with Mobil to become the largest oil company in the world, selling 5.4 million barrels of oil per day and reaping an annual revenue of \$137.2 billion. For decades, oil producers have framed rising fossil-fuel demand as inevitable and placed the burden of cutting emissions squarely on the shoulders of the masses. It was only in internal memos and articles in little-read scientific journals that they admitted their own culpability.²

The argument that we are all to blame occurs again and again. Sometimes we are all to blame because of our desires: people endorse nuclear weapons because they want security or are content with dangerous algorithms because they want convenience, entertainment, and economic growth. Sometimes we are all to blame because the root of our problems is human nature itself: nuclear arms races and a lack of cooperation on climate change are because of our innately parochial, suspicious, and self-interested nature.³ But blaming the public is a manufactured distraction – a PR myth that dissolves as soon as the hard figures on who produces catastrophic risk are examined. The reality is that a mere handful of giant corporations, countries, and militaries are responsible for creating the great majority of catastrophic risk, shrouding their activities in secrecy to avoid scrutiny, and undermining efforts to regulate them. Think of this small number of companies and institutions – big tech, the fossil-fuel industry, and military-industrial complexes (a collection of militaries and the arms manufacturers they tend to contract to) – who disproportionately contribute to the apocalypse as the ‘Agents of Doom’.

Agents of Doom are in turn shaped and driven by larger incentives. At root, there lie just a few causes that are embedded in both the earth and our skulls. We can trace the causal chain of catastrophic risk from these roots via the primary producers of risk to the endgame we face today ([Figure 44](#)). Most catastrophic risk doesn’t emerge randomly but is produced by just a few actors for profit and power.



[Figure 44](#): The causal chain of global catastrophic risk

We can think of global catastrophic risk as being part of a chain of cause-and-effect. For human-made threats, this causal chain goes back from catastrophic risk today to the actors producing it, through the structures which shape their behaviour, back to the deepest root causes which have created those structural drivers.

THE AGENTS OF DOOM

There's a good reason that Exxon and BP were so eager to shift the blame onto individuals. Both are part of a cohort of just thirty companies that between them are responsible for a third of global energy emissions. No matter how we slice up responsibility, these fossil-fuel giants are some of the key drivers of climate change. Over half of the greenhouse gases pumped into the atmosphere from 1854 to 2022 stem from just seventy-eight corporate and state-owned companies. Since 2016, with the legal adoption of the Paris Climate Agreement, 80 per cent of the rise in emissions has come from just fifty-seven fossil-fuel and cement companies.^{[4](#)}

It's not due to a lack of knowledge. The greenhouse effect was discovered in 1856 by US scientist Eunice Foote and we've had a high-resolution understanding of climate change since the publication of the first IPCC report in 1989.^{[5](#)} These companies are well aware of the damage that each tonne of carbon is causing, yet they are choosing to ignore it.

It's a similar picture regarding countries. Just ten states (counting the EU as one) are responsible for around 80 per cent of cumulative global emissions since the Industrial Revolution. The US alone accounts for about a quarter of greenhouse gases presently in the atmosphere.^{[6](#)} It's a trend that is consistent over time. Even today the top ten emitters account for roughly three-quarters of global emissions.^{[7](#)}

Two countries, the US and Russia, hold 85 per cent of stockpiled nuclear warheads and continue to develop their technological advancement in

delivery.⁸ The production and maintenance of those weapons is carried out mainly by a small group of twenty-four companies who operate largely out of the public eye but are financed by 287 financial institutions (such as banks and, ironically, pension funds and insurance companies).⁹

Biological weapons, now officially outlawed, followed a similar historical pattern from the First World War onwards. As with nuclear weapons, it was the most powerful and technologically advanced states – eleven countries such as Japan, Russia, Germany, the US, and the UK – that created biological weapons. Since then, there have been around eighteen incidents from biological weapons, the worst being Japan's attacks on China during the Second World War. It remains states and large biotech companies who have the necessary resources, equipment, and expertise to create new dangerous pathogens.

This tendency for a few powerful actors to produce new world-damaging technologies holds true for AI as well. The frontier of large text models is largely concentrated in three key companies: Microsoft's OpenAI, Google DeepMind, and Anthropic. Microsoft and Google rank third and fifth as the richest companies in the world. They are all headquartered on the US west coast (although Google's DeepMind itself is in London). That means they all fall under the jurisdiction of one country: the US. Even if we cast a wider net beyond large text models, we still see just a small number of actors trying to build AGI. In 2020 over seventy projects were explicitly trying to build an AGI, around half of them in the US, as were eight of the nine projects with military connections.¹⁰

There are just a handful of powerful actors that are producing the world's greatest threats. These Agents of Doom are large fossil-fuel companies, big-tech titans, and the military-industrial complexes of a few empires such as the US, China, and Russia. One could, of course, argue that, while the creation of these global threats stems from just a few agents, they are responding to the demands of the masses. It's another myth that doesn't hold up to scrutiny.

CRAFTING THREATS IN THE DARK

Democracy dies in the dark. It is exceedingly difficult to democratically regulate any activity that is kept out of the public eye. Yet, that is exactly what is done when it comes to the production of the world's greatest threats.

It is an approach that is, despite their efforts to hide it, all too visible in fossil-fuel companies, who have known about climate change and its potential dangers for decades. The largest trade association for the oil and gas industry has known since the 1950s, the coal industry since the 1960s, and fossil-fuel giants such as Total and Exxon since the 1970s. During the 1970s Exxon made remarkably accurate internal forecasts of future global warming, as did many of their employees in scientific journals. In 1979 an internal Exxon study even admitted fossil fuels were causing climate change and could be responsible for 'dramatic environmental effects before the year 2050 ... the potential problem is great and urgent'. Rather than alert the public, they buried the evidence. Then they collaborated to cast doubt on the science of climate change.^{[11](#)}

Fossil-fuel companies have since at least the 1970s run well-funded disinformation campaigns to undermine action on climate change. They funded conservative think-tanks with millions of dollars, which in turn financed and gave a prominent platform to scientists infamous for falsely challenging the scientific consensus on issues such as the link between smoking and lung cancer, the effects of acid rain, and the existence of the hole in the ozone layer. They included the tobacco lobbyist Frederick Seitz and William Nierenberg, who was both a scientist in the Manhattan Project and co-founded the conservative think-tank the George C. Marshall Institute. Oil, coal, and gas companies didn't just fund and platform professional contrarians who were sympathetic to their cause; they even helped set up and fund fake grass-roots organizations that pushed against the science of climate change.^{[12](#)} Much of this strategy was coordinated by the International Petroleum Industry Environmental Conservation Association (IPIECA) lobbying group, which has over forty corporate members including BP, Shell, Chevron, and ExxonMobil.^{[13](#)} This deliberate campaign of misinformation successfully set back the public discussion (and action) on climate change by decades.

The fossil-fuel companies are not alone in their deception. While microplastics and PFAS may be novel, the dangers were not unforeseen by the companies producing them. The largest industrial producers of PFAS such as 3M and DuPont were running safety studies on these compounds for decades. By 1970 they had established that the chemical C8 (a type of PFAS known as PFOA) was ‘highly toxic when inhaled and moderately toxic when ingested’. They didn’t publish the findings. Instead, a decade later 3M announced to the public that C8 was ‘about as toxic as table salt’. Their internal findings included reports on the possible links between PFAS and prostate cancer that were shared between 3M and DuPont, and records of children with eye and facial defects born to workers exposed to PFAS. While the manufacturers shared this information among themselves, they did not share it with the public. Instead, these studies on the negative health impacts were intentionally buried and suppressed.^{[14](#)}

They and others continued to insist that these products were safe for decades, sought approval for them from the Environmental Protection Agency (EPA), and denied the adverse health impacts of PFOA. The producers knew these chemicals were potentially toxic forty years before the public health community yet chose to endanger the public in pursuit of profit. Their disinformation campaigns and lobbying efforts would never have been possible if it had not been for their own internal secrecy.^{[15](#)}

Secrecy has also been central to the production of nuclear weapons. The public has been largely kept in the dark since their very invention. Both the Manhattan and Castle Bravo projects of the US are now almost synonymous with ‘large, secretive government project’. The initial classification system was so extensive that even basic physics published in textbooks were considered ‘born secret’ (i.e. classified as soon as they were written). The obsession with secrecy even led to the Atomic Energy Commission burning hundreds of copies of an edition of the *Scientific American* in 1950. Since then, nuclear weapons have remained staunchly overclassified and outside the domain of public oversight.^{[16](#)}

A key point of US secrecy was to prevent proliferation. It failed. Eight other countries subsequently developed atomic weapons after the US did, while numerous others could have (and had the knowledge to do so) but chose not to. Similarly, secrecy played no role in having Ukraine give up their inherited stockpile or South Africa dismantling theirs. What secrecy

has done instead is handicap public discussion of nuclear weapons. The clandestine approach of the US has been replicated elsewhere. Israel has not admitted to having nuclear weapons, although it is an open secret that they do. Some believe that powerful states still have illegal biological weapons capabilities that have successfully been kept secret.

It's not just the threats themselves that are being hidden; it's often also crude corruption, lobbying, and profiteering in the companies and organizations. One of the most common approaches is the creation of 'revolving doors' in which individuals move between lucrative positions in industry (or lobbying for industry) and powerful political positions which are supposed to regulate that industry. One illuminating example comes from the US nuclear-defence industry. The former senator for Arizona Jon Kyl was one of the key politicians who pushed for more money to be spent on the multi-trillion-dollar US nuclear modernization programme in 2010. In 2013 he left office and joined the lobbying group Covington & Burling, working on behalf of the defence contractor Northrop Grumman, the same company that had received the contract to modernize the US arsenal. Northrop Grumman also spent \$1.55 million on political donations in 2020, while its Political Action Committees (PACs) doled out \$3.77 million to both Democrats and Republicans. Together, Boeing, Raytheon, Northrop Grumman, and other aeronautical defence companies spent \$46.9 million on lobbying.¹⁷

Kyl is just one example of dozens of generals and politicians who push for increased military spending while in office and then cycle into handsomely paid positions with defence contractors (who benefited from their actions) afterwards. It is an endemic problem that amounts to legalized bribery: making the right moves while in office can guarantee a career with a six-figure salary in industry later.

It also doesn't just happen in the US military. One report in the EU found that six fossil-fuel companies (Shell, BP, ENI, Total, Galp, and Equinor) and five of their lobby groups were involved in seventy-one 'revolving door' cases over the period 2015–21. During that time, they also averaged six meetings a month with the European Commission and spent €170 million on lobbying.¹⁸ In Australia, there is the well-documented case of how oil, coal, and aluminium lobby groups – a coalition who call themselves 'the Greenhouse Mafia' – have shaped the country's energy and

environment policies.¹⁹ There is also another revolving door between politics and industry. Every resources minister in the Australian government since 2001 who has retired from parliament has taken a position in the mining, consultancy, or fossil-fuel industries after finishing their time in office. Since 1990 more than a quarter of federal or assistant federal ministers have taken jobs with special interest groups after their political careers ended. It's why Australia, the world's sunniest and second-windiest continent, has no price on carbon emissions and also has some of the highest emissions per person in the world. Action on nuclear weapons, climate change, and other global threats are the products of intentional efforts to undermine democracy and capture state power.

We see the same revolving-door problem even when it comes to responses to pandemics. In 2021 the Moderna vaccine was approved by Stephen M. Hahn, the head of the US Food and Drug Administration (FDA). Moderna was created by Flagship Pioneering, a bioengineering venture-capital firm that spits out enormous numbers of start-up firms (with the strategy of having the vast majority fail but winning big on one or two). The vaccine wasn't even a Moderna invention: it relied on innovations from the US government-owned National Institutes of Health (NIH) as well as Scripps and Dartmouth universities. Moderna was then given \$10 billion in taxpayer-funded money before making \$36 billion in profit. Hahn left the FDA a year after authorizing the vaccine and joined Flagship as its chief medical officer. He remains there as CEO-Partner. This happens not just with Flagship. One of Hahn's predecessors, Scott Gottlieb, moved from being head of the FDA to being on the board of the pharmaceutical company Pfizer. Between 2001 and 2010 over a quarter of FDA employees who approved cancer and haematology drugs went on to work in or consult for biopharmaceutical companies.²⁰ It is little wonder that the US public has such a deep distrust of public institutions and why there has been such a fertile ground for vaccine conspiracy theories. The revolving door also makes proper regulation of biotechnology far less likely: if you are going to go and work in the industry afterwards then you have far less incentive to burden them with legislation to improve risk management. Those revolving doors and legalized corruption undermine efforts to reduce and respond to global risk.

Survey after survey has shown strong public support for reducing catastrophic risk. One survey across six countries found that a strong majority in each one, including Israel and the US, supported the creation of an enforceable international agreement that would eliminate all nuclear weapons. Europeans in particular heavily endorse a ban on nuclear weapons. For years, opinion polls across the US, EU, China, Australia; and others have shown substantial support for greater action on climate change. One poll covering twenty-six countries found that a majority in each country opposed the development and use of lethal autonomous weapons. If anything, these surveys underestimate public support, since they are just gathering the opinion of people who often aren't fully informed (especially given government and corporate secrecy). Open democratic experiments where citizens are convened in a jury (of 9–12 people) or assembly (of a few hundred), are briefed by experts, and then deliberate, tend to show even stronger support for aggressively tackling global risks such as climate change.²¹

Whatever the threat, we see the same pattern of a few Agents of Doom profiting from creating global catastrophic risk while cloaking their actions in secrecy and extensively lobbying to prevent regulation.

MAKE YOUR OWN NIGHTMARE

In 2023 when a long list of academics and corporate leaders signed a letter warning of the risk that machine intelligence could wipe out the human race, a few names stood out. Sam Altman, Demis Hassabis, and Dario Amodei are the leaders of the three largest, most advanced AI companies in the world, constructing the most dangerous algorithms.

They continue to fret publicly about the machines they are building. Amodei, CEO of Anthropic, thinks there is a 10–25 per cent chance of AI destroying humanity; Altman, CEO of Open AI, has worried that the ‘development of superhuman machine intelligence is probably the greatest threat to the continued existence of humanity’. There are dozens of other remarks from AI pioneers and producers warning that it could cause human extinction. Yet these tech giants continue to pursue an invention they believe could destroy all humanity, perhaps all life on earth. Even though

Amodei thinks there is around a one in four chance that the algorithms he is helping to construct could murder not only all his closest family and friends, but also you and everyone you know, every single one of us, he continues to go to work every day.²²

Not only are these leaders ostensibly making their own nightmare while warning the world about their actions, but they are also actively thwarting attempts to better oversee and regulate the creation of dangerous algorithms. Whistleblowers at OpenAI have warned that employees must gain permission from the company first before they contact the media. That's the equivalent of asking a local mafia for permission to tell the police about their drug deals. Employees leaving the firm also need to sign highly secretive non-disclosure agreements or else they will lose their shares in the company. They are also bound by 'non-disparagement agreements', which essentially mean that employees who have left the company cannot say anything critical of OpenAI, again at risk of losing their shares. It seems to be a widespread practice, with past and present employees at both OpenAI and DeepMind having penned an open letter calling for greater whistleblower protection and safety measures. 'Open' AI appears to be ironic at best.²³

While coders get most of the attention, the role of lobbyists should not be underestimated. From 2022 to 2023 the number of groups lobbying the US government about AI almost tripled. These lobbyists are, perhaps unsurprisingly, not pushing for tighter regulation to prevent human extinction. On the contrary, Google, Meta, OpenAI, and TechNET (an industry-wide lobbying group representing Google, OpenAI, Anthropic, Amazon, and others) have all openly opposed SB 1047, a regulation under the California State Legislature. The bill would require risk assessment of any models that cost more than \$100 million to train and are bigger than 10^{26} FLOPS (a measurement for the number of calculations a computer can perform in a second) before they are deployed to ensure that these programs can't cause catastrophic harms by, for example, carrying out cyberattacks or helping to build nuclear weapons. It would also provide whistleblower protections. In September 2024, Gavin Newsome, the governor of California, vetoed the bill. He did so largely because of immense industry pressure from the likes of Meta, OpenAI, and Google, all of whom cautioned that the bill would lead to an exodus of AI firms from California.

The same companies warning that their machines could threaten human extinction successfully helped veto policy that would better control these machines. They are crying out for regulation in the abstract while killing specific policies which could regulate them.²⁴

Perhaps that's because people like Altman don't truly fear their own creations. Warning of an apocalypse is instead a way to build hype about their products while gaining more leverage over regulators. Their plea to governments is that if you want to regulate a doomsday machine then you need the advice of those building it. Their motive is profit and power, not safety.

There are two alternative, more charitable explanations, ones that the developers themselves have offered. One is that the technology is so powerful that there is a moral imperative to build it. AI has been declared a potential solution to extreme poverty, energy shortages, climate change, and essentially every other global challenge faced by humanity. This belief that a technology will create utopia is by no means new. Proponents of weapons of mass destruction have repeatedly claimed that more advanced weaponry could end all wars by simply making conflict too dangerous and inhumane. That was said for artillery, dynamite, machine guns, submarines, smokeless powder, and nuclear weapons. Every time, that claim of using more powerful weapons to end war has been proved wrong.²⁵ It's also an argument that is more about a messiah complex (essentially a declaration that 'I'll decide whether the potential benefits justify the risk to humanity') than altruism.

The other excuse offered by developers is that they need to build the potentially world-ending machine first because if they don't, someone else will. And they are the (self-proclaimed) good actors who can build it safely and benevolently. It's a view summarized in a 2024 op-ed by Sam Altman in which he framed AI as a race between democracies and

authoritarianism.²⁶ Often the developers use both utopian aspirations and fears of a race simultaneously as excuses. On another occasion Altman said 'the upside of AGI is so great, we do not believe it is possible or desirable for society to stop its development forever'.²⁷ Similarly, the machine-learning expert Geoffrey Hinton once responded to a question of whether the development of AI could be stopped with, 'I don't think that's possible,

and I think we should continue to develop it because it could do wonderful things.’²⁸

The arms-race defence is interesting because it is so common. CEOs have used it to explain why they need to build AI, and to defend environmentally destructive practices; generals have used the same excuse to justify building nuclear weapons and killer robots; and presidents have used it to explain why they can’t cut emissions. The excuse is that if we don’t do it, someone else will, and it’s better if we ‘good guys’ do it first.

We shouldn’t allow this arms-race defence let the Agents of Doom off the hook. Even if you need to compete, no one is forcing chemical companies to bury the evidence on the harms of PFAS, or fossil-fuel companies to run scientific disinformation campaigns, or defence contractors to lobby and use revolving doors to push for bigger defence budgets and nuclear arsenals. That is clearly being done for the benefit of a small number of firms and individuals. Yet there are also deeper incentives at play here. Unless we address those structural incentives, then regulating or penalizing the Agents of Doom will not be a lasting solution.

HATE THE GAME AND THE PLAYERS

In 1928 the polymath John von Neumann wrote a paper called ‘On the Theory of Games of Strategy’ and started the now popular field of game theory. Its aim was to identify what would be the mathematically optimum course of action for a rational agent to take when trying to win a game, whether it be chess or war. Game theory is one of the commonly used explainers of arms races.

Arms races have often been explained in game theory as an example of a ‘security dilemma’. Imagine two neighbours, both self-interested, rational (they act logically to achieve their self-interested goals) individuals who prioritize their own survival.²⁹ They would prefer a situation in which both remain unarmed. Guns are expensive and fatal accidents can occur. However, they can’t trust their neighbour’s benevolence. The worst-case scenario is you have no weapons when your neighbour attacks. The better alternative would be if you get weapons and your neighbour doesn’t. If your neighbour has also chosen to purchase weapons, then you are of course

relieved that you decided to stockpile armaments as well. Hence, both are incentivized to begin stockpiling weapons just in case. A move which is good for each actor individually is bad for the collective.

Globally, it's often assumed that this security dilemma and subsequent arms races are the result of 'international anarchy'. In the absence of one overarching arbiter or world government, states can't help but compete with one another.

A similar game-theoretic explanation is often used to justify environmental destruction. Different competing groups overusing a common resource has often been referred to as a 'tragedy of the commons'. The tragedy is usually posed as a thought experiment: imagine an open-access good (a common) such as a sea shared between fishers. If all of them overfish, it will lead to the destruction of the resource. However, if one of them does so, they can get away with it while getting a larger catch (at least in the short term). A few will always act selfishly, and those who practise restraint will be outcompeted. As with an arms race, individual actors pursue rational short-term concerns, and the result is collective tragedy. This idea was first proposed by the American ecologist Garrett Hardin in a 1968 essay. Hardin wasn't interested in environmental commons but was instead trying to argue for population control. Nonetheless, his essay has become a classic explanation for our environmental woes, from regional overfishing to global climate change.³⁰

We can think of the tragedy of the commons as emblematic of a wider problem: a race to the bottom. Imagine a confectionery company that wants to create sweets without the involvement of child labour. Paying adults a fair wage will put altruistic firms at a disadvantage compared with those who opt for the cheapest but most exploitative labour conditions. The company with no scruples and a cheaper product will win out. Both arms races and races to the bottom are underpinned by game-theoretic logic: self-interested, rational actors pursuing short-term, self-interested goals results in collective harm. It's easy to see how such dynamics can be used to explain the actions of the Agents of Doom, increasing the likelihood of nuclear war or extreme climate change.

We have already seen the problem of a third type of race that underlies the others: status races. Whether it be between chiefs competing with feasts, rulers through monuments, or billionaires through high-emissions stocks, superyachts, and private jets, status races tend to drive conspicuous

consumption and expansion. In the long run, it is driving us into an unpleasant collision with planetary boundaries. As we've seen, that race for status also partly underpins arms races and races to the bottom.

While appealing, there are a few issues with viewing the problem through this game-theoretic lens. While it's mathematically neat, we now have years of studies showing that people aren't as rational as game theory assumes they are. In laboratory experiments, irrational humans frequently fail to match the predictions of game theory. Rather than being rational and self-interested, people often show strong signs of reciprocity, a concern for self-image (strongly related to status), and an aversion to inequity (which often overlaps with an aversion to dominance).³¹ They often don't end up in arms races. Similarly, if you put people in a tragedy-of-the-commons-type situation in a laboratory, they rarely end up enacting a tragedy. Simply allowing them to talk to each other prevents the tragedy of the commons from emerging.³²

The tragedy of the commons also frequently doesn't occur in the world outside of lab experiments. For instance, Hardin was quickly disproved by the economists Elinor and Vincent Ostrom, who showed that many communities (including non-state groups) regularly solve the tragedy of the commons. Groups such as the Khoisan and central African hunter-gatherers seem to have managed their environment just fine for hundreds of thousands of years.³³

Our deep past also undermines the idea that security dilemmas plague all humans and human groups. Our Palaeolithic fluid civilizations lived in a state of 'anarchy', yet appear to have been peaceful. The first signs of organized war and arms races begin only with the rise of dominance hierarchies. There is also a wide array of different 'peace systems' throughout world history (including hunter-gatherers) that don't fall into arms-race-type dynamics. They manage to create peace not just through governments but also through norms and culture.³⁴

Yet, there is a reason that game theory seems convincing. It does seem to explain how many states and corporations today behave, as well as the struggle between elites and empires in the past. Clearly some groups fall prey to this dynamic far more than others. Why?

ROOT CAUSES

There is a reason why arms races often happen across empires of the past and the Agents of Doom today but rarely occur among most individuals. Empires and the Agents of Doom are not human beings; they are dominance hierarchies. They are artificial agents who act very differently from the average person. Hobbes, when he first spoke of the Leviathan state, defined it as ‘an artificial man, though of greater stature and strength than the natural, for whose protection and defence it was intended; and in which the sovereignty is an artificial soul, as giving life and motion to the whole body’. The courts were its artificial joints, the administration its memory, its economic riches were its strength, and collapse was its death. Similarly, the corporation is a legal entity imbued with the rights of an individual. Legally, it is an artificial person. Alongside states and corporations, large-scale algorithms are fast becoming a third type of artificial agent, although one largely controlled by states and companies.³⁵

Both states and companies are artificial creations: inventions that both amplify decision-making power and allow for a being that could outlive its creators. A state can enter agreements and take on loans, which it is expected to fulfil, even if the signing head of state has died. ‘Corporation’ is derived from the Latin *corpus* meaning body (which, ironically, is one thing they lack). The ‘state’ is derived from the Latin *status*, for a manner of standing or ranking. In perhaps the ultimate Freudian slip, the roots of the state are in status.³⁶

States and companies (and perhaps now algorithms) are not just amplifiers of power and decision-making; they also inflate the worst in us: the darker angels of our nature. Think of a modern state, such as the US or Russia, embodied as a single person stalking the streets. Imagine what they are like. Overwhelmingly self-obsessed and self-interested, constantly boasting of their achievements, continuously suspicious of their neighbours, persistently posturing and hoarding guns and knives, and greedy to the extent of never giving even just 1 per cent of their earnings to those in need.³⁷ That state person, if it took a psychiatry test, would rate disturbingly high on the dark triad (psychopathy, narcissism, and Machiavellianism). Bear in mind that Machiavellianism comes from the

Florentine philosopher Niccolò Machiavelli (1469–1527), famous for his work *The Prince*, which advises leaders to be amoral to achieve political glory. A corporation as a person is much the same: status-seeking, profit-focused, self-interested, Machiavellian, and overly competitive. When Hobbes thought about the state of nature, he appears to have been thinking of a world populated by these artificial agents, not with actual flesh-and-blood humans.³⁸

In both game theory and economics, the model used for humans (often called *Homo economicus*) resembles these artificial persons much more than the average human. They are self-interested, attempt to act rationally, and pursue material wealth. Races for arms, resources, or status aren't innate to human groups. They are more likely to occur only among groups that exhibit some peculiar characteristics: dominance hierarchies marked by status competition, corruption by power, higher levels of the dark triad, and bureaucracies that aim for rationality (although they rarely completely achieve it).³⁹ While game theory often does a poor job of predicting the behaviour of people as a whole, it does work better for smaller segments of the population. In laboratory experiments, individuals who rank higher on the dark triad (specifically psychopathy and Machiavellianism) are more likely to defect and not cooperate.⁴⁰ Dominance hierarchies are not just less likely to cooperate and fall into harmful race dynamics; they also select for and amplify the individuals and characteristics that hinder cooperation.

This is not just about races and competition. Militaries and defence companies regularly trade, collaborate, and cooperate. In 2016 the Israeli NSO Group developed a powerful spyware weapon called Pegasus that could covertly hack into phones and laptops. It didn't share this only with allies but also with ostensible enemies such as Saudi Arabia and the United Arab Emirates. These sales needed formal approval by the Israeli Defense Forces. Today the US accounts for around 42 per cent of global arms exports, including to its rivals. Profit was, and often is, deemed more important than national defence. Countries and companies don't just compete and race; they collude and cooperate.⁴¹

The same is true of the global ecological crisis. Fossil-fuel companies and PFAS manufacturers are technically competitors. They are fighting each other for a bigger market share. Yet, when necessary, they will lobby and work together to suppress science and undermine policy.

This witches' brew of competition and cooperation is a common feature of racketeering. We saw earlier how the *razones* gangs of Columbia competed for territory and drug sales, but still governed among themselves and could combine forces against a common enemy, such as the police. They weren't just racing against each other to control turf and the drug market; they were part of a wider racket that was trying to maximize extraction. Similarly, emperors and competitors for the throne regularly cooperate to tax the peasants or fight off invaders, all while still feuding. It is equally accurate to describe this as a racket rather than a race. We can see the modern world in a similar light. As states mainly developed as protection rackets to begin with, so too do Agents of Doom both produce the risk and then benefit by offering the solution. Even OpenAI began with the ostensible goal of creating 'safe' and 'aligned' AI. They are rackets made up not by communities or individuals, but by very strange artificial agents.

THE TREE OF DOOM

The root causes of our endgame were put down when the darker angels of our nature (status competition, power corrupting, the dark triad, and the authoritarian impulse) met the new environmental conditions of the Holocene (such as lootable resources). Goliaths began to emerge. At the individual level, those who were the strongest status seekers (and often the most adept at violence and manipulation) were selected for. At the group level, communities that were more effective at conquest tended to outcompete their neighbours and grow. It was multi-level selection. These new groups even invented myths to valorize growth and conquest. From the civilizing mission of European colonizers to the *jihad* of the Islamic caliphate, dominance hierarchies have had ideologies which justify universal expansion.

These new Goliaths amplified our darker angels. They fell into competitive 'Goliath traps': arms races, races to the bottom, and status races using conspicuous consumption. Yet these weren't just races; they were rackets. Groups who were competing to extract resources from a given population would regularly trade (as with the Late Bronze Age or today) and band together when necessary. Those races/rackets, whether it be for

protection or the extraction of data or fossil fuels, are the main drivers of our endgame today (Figures 45 and 46).⁴²

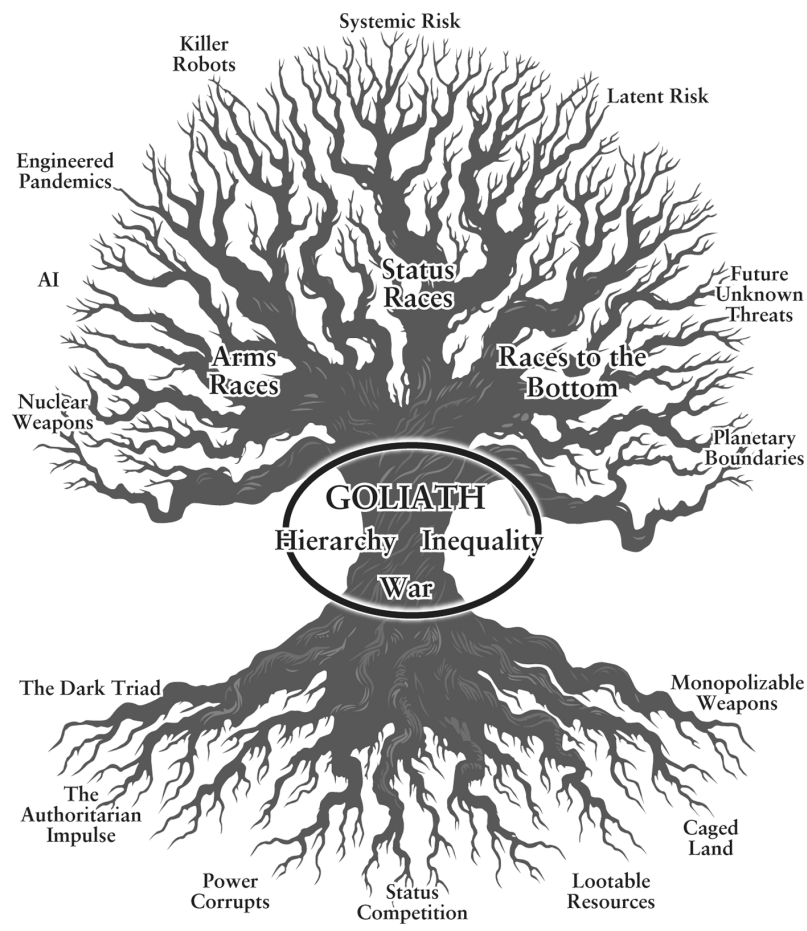


Figure 45: The Tree of Doom

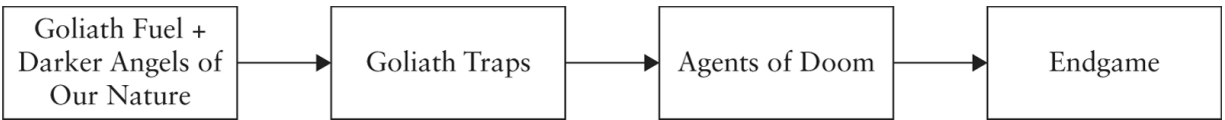


Figure 46: From root causes to our modern endgame, or the Tree of Doom

The Tree of Doom is at heart a causal chain from root causes (the combination of Goliath fuel and the darker angels of our nature) through to the endgame we face today.

Goliath fuel and our darker angels are the root causes of our endgame. They spurred the creation of Goliath systems which became stuck in different, interlinked races whether it be for status, security, or a competitive advantage. Today these Goliath traps have branched out into all the contributors to global catastrophic risk: nuclear weapons, AI, killer

robots, climate change, the infringement of planetary boundaries, and others. This tree, from the root causes to our unfolding endgame, is still growing.⁴³ Unfortunately, it's difficult to see how this thousand-year-old process is going to end well.

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The Fates of Goliath

Putting a Date on Doomsday – Our Curse – Evolutionary Suicide – When Ideas Dry Up – Silicon Goliath – Killing Goliath – Choose Your Curse

PUTTING A DATE ON DOOMSDAY

[Table 6](#): Apocalyptic probabilities

Probability	Timeline	Conditions
49 per cent	2050	Collapse
50 per cent	2100	Collapse or extinction
>25 per cent	2100	Existential risk
19 per cent	2100	Extinction
16 per cent (1 in 6)	2100	Existential risk

The most common way of trying to calculate when doomsday might arrive has been for experts to guess what our odds are of making it past 2100 ([Table 6](#)). The Astronomer Royal Lord Martin Rees says it's a coin flip between survival and collapse or extinction. Jared Diamond is even more pessimistic, giving us a 49 per cent chance of collapsing by 2050.

University of Oxford researcher Toby Ord believes it is more like Russian roulette, our chances of being wiped out within the next eighty years being one in six. His colleague Nick Bostrom believes it is at least one in four. Averaging these experts' opinions gives us a one in three chance of facing a global collapse, or something far worse, by the end of the century.¹

These numbers are frighteningly high. They also tend towards suspiciously round figures and clichés, such as 50/50, a quarter, or Russian roulette. This is just one sign that they aren't to be trusted. These are all examples of unreliable forecasts. Superforecasters (the gold standard of forecasters) draw on an exceptional group of experts with diverse information to make predictions about specific, concrete questions. Even superforecasters have little predictive ability beyond a year or two and there is little evidence that they have any accuracy after a decade.² These apocalyptic predictions are being made about a vague question, looking decades ahead, usually by individuals who aren't superforecasters. Don't trust any particular date on doomsday.³

While we can't put clear numbers on the likelihood of collapse, what we can do is understand the most likely future scenarios based on the characteristics of Goliath and how they have changed over time.

OUR CURSE

The Global Goliath we inhabit has its own unique fuels. The most powerful monopolizable weapons are now nuclear: the US and Russia achieved superpower status by having bigger, more advanced nuclear arsenals (and militaries) than anybody else. More than that, the decision to commit the world to a war that could cost billions of lives lies in the hands of one individual: the US president. It is the most consequential and important decision any leader could make, yet no democratic deliberation or approval is required. It has led some to brand the US a 'thermonuclear monarchy'.⁴ The dead-hand in Russia is, of course, no better. These monopolizable weapons have tilted the world away from democracy.

The lootable resources that enable and power the Global Goliath are fossil fuels. Fossil fuels, like grain, are concentrated, defensible, and easily

monitored, moved, and stored. Unsurprisingly, they have regularly been appropriated at massive scale by states and corporations. Most of the world's flows of oil, gas, and coal are controlled by a few companies that are or were often state-owned. Saudi Aramco is Saudi Arabia's state-owned oil and gas company, the gas titan Gazprom was previously owned by Russia (which maintains a majority share), and Royal Dutch Shell is the merger of Royal Dutch Petroleum and Shell Transport and Trading Company (the former was strongly connected with the Dutch government, and the latter to the UK).

Also, like grain, fossil fuels are energy-dense and have permitted a surge in population. Global population has ballooned from 1 billion prior to the Industrial Revolution to 8 billion today, fed in large part thanks to synthetic fertilizers derived from fossil fuels: grain-based agriculture on petrochemical steroids. Hence, loutable resources tend to build on each other rather than replace each other: fossil fuels enabled larger harvests of grain rather than replacing them. Today, every calorie we eat was produced using around ten calories from fossil fuels.

Land is also now more caged than ever. The world is carved into precise borders that are often hawkishly monitored. Some are physical: Israel, Egypt, India, and Saudi Arabia have all erected border walls and fences covering thousands of kilometres and surpassing the efforts of Rome, China, and Persia.⁵ We now also have invisible cages: identity cards, passports, and mass surveillance systems have all become new ways to control borders and people.

Most people alive today have been lucky enough to inhabit a world in which many of the most powerful countries at the core of the Goliath were, by historical standards, inclusive and democratic. This was in large part due to the two twentieth-century world wars, which destroyed enormous amounts of capital that was primarily held by the wealthy. Governments began to tax the rich more aggressively to bankroll the war effort and put in place measures to appease workers (who were now armed, and upon which countries were entirely dependent) such as rent, wage, and price controls. Unionization rates peaked shortly after the Second World War. Scholars have come to call this great levelling of wealth the 'Great Compression'. The world wars didn't just level wealth. Wealth equality spilled into the other forms of power and made countries less extractive in general. Key welfare-state provisions were passed directly afterwards, and democracy

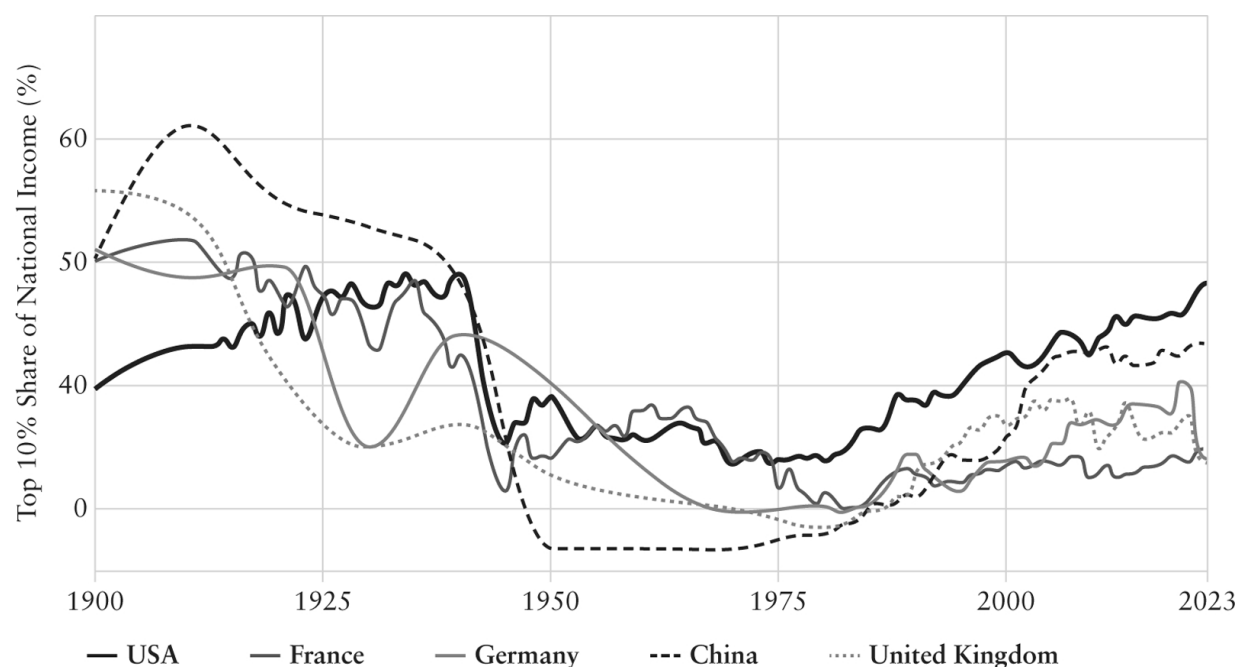
was strengthened. Conscription during the world wars was frequently followed by extending the vote in many countries, including to women in the US and UK directly after the First World War.⁶

It would be a mistake to think that this was a lasting and global transformation. The global political and economic system is still grossly unequal. Wealth inequality began to creep back up across many countries in the 1970s. Now the richest 1 per cent of the world own almost half its wealth, while just eighty-one billionaires possess more wealth than the poorest half of humanity. The situation appears to be worsening. The share of global wealth for the top 1 per cent swelled from 25–30 per cent in the 1980s to approximately 40 per cent in 2022. For the US, China, and Europe, the share of the top 1 per cent rose from 28 per cent in 1980 to approximately 33 per cent in 2022. In contrast, the share of the bottom three-quarters has continued to hover around 10 per cent. These numbers are underestimates. They don't capture the money and capital siphoned into overseas tax havens, for instance ([Figure 47](#)).⁷

Wealth inequality has spilled over into our politics globally. Democracy is backsliding. The number of people living in democracies has been decreasing and the number living in autocracies has been increasing over the last fifteen years. Today 5.7 billion (around 71 per cent of the global population) live in an autocracy. Forty-two countries are currently becoming more autocratic while just eighteen are becoming more democratic by a measurement that includes sixty indicators such as the fairness of elections, independence of the media, and equality of access to resources. Such democracy measurements don't include workplace democracy or even whether the most pivotal decisions – like whether to launch a nuclear weapon – are made democratically.

The democracy and public services we often take as being the natural condition of states are an historical aberration. One that is slowly being undone as the world drifts back towards the historical default of greater inequality, autocracy, and extraction of wealth and power by elites. There may be a lag (as there was between wealth inequality rising and democratic backsliding), and some reversals, but in the long run we can expect wealth inequality to spill into the other forms of power. It is happening today. Elon Musk is not only one of the world's richest men but also owns X, wields enormous cultural clout, played a starring role in getting Donald Trump re-

elected, and will be instrumental in making policy during Trump's second term. He is melding together all the different forms of power. Similarly, Amazon founder Jeff Bezos is not just one of the world's richest men; he also owns the *Washington Post* (the third largest newspaper by readership in the US), audiobook giant Audible, the streaming service Twitch, and the film company MGM. Similar patterns have been repeated throughout history, including the gilded age of the late nineteenth century and the rise of autocracies in the 1930s. The record is simply replaying today.



[Figure 47](#): Inequality in key empires across the twentieth century

All major countries share a rough trend of inequality peaking before the world wars, plummeting during them (especially during the Second World War), and then rising again from the 1970s onwards. Wealth inequality follows a similar pattern but at a higher rate. Today, the top 10 per cent capture around half of global income, but three-quarters of global wealth.

The curse of our Global Goliath is not necessarily diminishing returns on extraction, though the US does show many signs of this. There remains intense elite capture of politics and factionalism. In 1983 the number of US households worth over \$10 million was 66,000. It increased (in real terms) within a few decades to 693,000 in 2019. Since the late 1970s real wages for most of the population have remained largely stagnant (or even declining) despite GDP per capita more than doubling. As Peter Turchin has shown, the role of popular immiseration and inequality in driving populism

and political division is far too apparent.⁸ The enormous military overstretch of the US has helped fuel a growing public debt of \$35 trillion. Yet, this is not the most worrisome curse. The US may decline: a civil war could even break out, although this seems unlikely, given its large full-time, well-equipped police force, military, and intelligence agencies. Even if it were to happen, the Global Goliath might be shocked for a few years or even a decade, but it is difficult to imagine how it would trigger a global societal collapse. And the US staying as a stable, coherent society is also not an antidote to our global woes. Remember that it was during the 1950s, 60s, and 70s, when elite competition was low and wealth equality higher, that the US built a nuclear stockpile that could destroy the world and had the largest acceleration in its environmental impact. Diminishing returns on extraction remains important and relevant. Yet there is now a new and more dangerous curse: global existential risk. The global system of extraction now creates threats and vulnerabilities which could destroy the whole world.

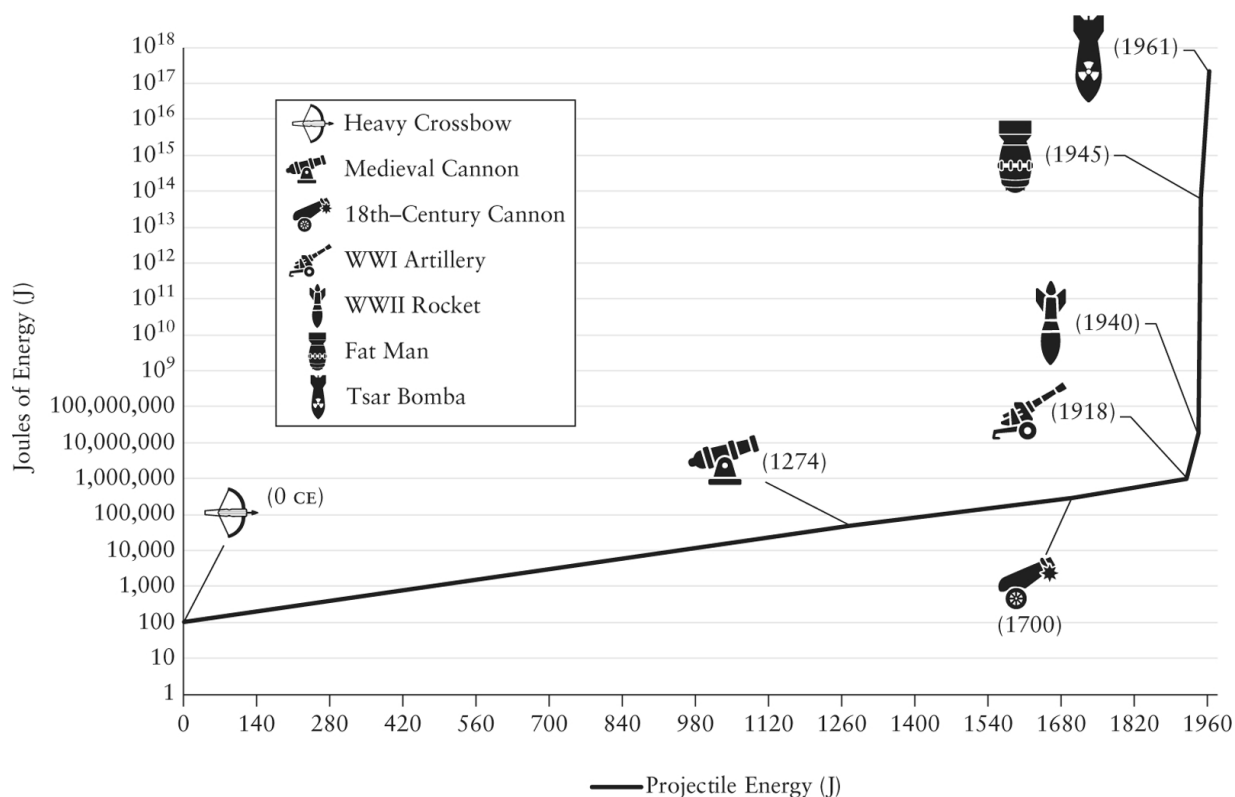
EVOLUTIONARY SUICIDE

The most likely fate of the Global Goliath is self-termination. The pressures that have been selected for over time – larger populations, more disease, greater military capacity, and economic exploitation (categorized earlier as ‘babies, bombs, bacteria, and barbarism’) – may have been evolutionarily advantageous for individual empires, but now underpin the catastrophic risk we face today. Bigger, more interconnected populations mean worse disease outbreaks; stronger militaries mean more damaging arms races; and barbarism is the economic backbone of the systems that produce carbon, weapons, and dangerous technology.

It is a persistent and long-running trend that is difficult to stop. Goliaths have reliably turned economic power and larger populations into worse weapons. Human history is one of growing tail risk. Take the history of projectile weapons. The power and speed of projectile weaponry have grown consistently over time. An average modern-day nuclear weapon is fifty trillion times more powerful than a bow and arrow (as measured in joules of energy released). When we plot the most effective projectile

weapons of different ages, we see a steadily rising line, with a sharp uptick in the twentieth century (see [Figure 48](#)).⁹ GDP shows a similar trend over that same period. Increases in economic power have reliably been translated into more potent weapons.

There is no sign of this trend abating. Even today we see the mass development of killer robots and other even more disturbing weapons, including the weaponization of space. States are now competing to create and unleash space-based directed-energy weapons (DEWs), such as ultra-short-pulse lasers. The end goal is to have these space-weapons establish an orbital ‘kill web’ that can identify and eliminate targets anywhere on the planet.¹⁰



[Figure 48](#): Energy released by projectiles over time

The shape of this graph will be familiar by now. It is similar to what we’ve seen with global GDP, land use change, and energy consumption. More energy and money have been reliably transformed into more destructive weapons.

We see the same trend when we look at environmental impacts over time. The problem is that the selection pressure for Goliaths has created entities

more effective at extraction, whether it be from people or the environment. Goliaths have evolved new forms of Goliath fuel – such as fossil fuels, data, and nuclear weapons – to persist and grow larger over time. Yet now those new sources of fuel threaten to destroy the entire planet.

Think of this as evolutionary suicide: short-term evolutionary pressures lock the population into a path of self-destruction. Short-term adaptation leads to long-term extinction. We see such a pattern with cancer (which destroys its host and itself in the process) and in laboratory experiments where profligate bacteria change the pH of their environment, slowly rendering it inhospitable and committing themselves to extinction.

Given how deeply locked in are Goliath traps (status races, arms races, and races to the bottom) and other problems, it is hard to see how we can escape without a global collapse unfolding.

Technology is usually mooted as the potential saviour from evolutionary suicide. Bioterrorism can be prevented through mass surveillance and nuclear weapons can be deterred by better nuclear weapons. Many powerful figures now hold up AI as a solution to all our woes. Demis Hassabis explained his motivation for developing AI as to ‘solve the problem of intelligence and then use intelligence to solve everything else’. Some are even willing to take the bet that technological progress will outpace and solve any problems it creates along the way. Ex-Google CEO Eric Schmidt said in 2024, responding to a question about the emissions and energy demand of AI, ‘we’re not going to hit the climate goals anyway because we are not organized to do it and yes the needs in this area [AI] will be a problem. But I’d rather bet on AI solving the problem than constraining it.’ That rhetoric from the producers of AI has proved influential, with AI now almost always framed as not only a positive development, but one that is necessary for economic growth and meeting the challenges of today.¹¹ For many, technology is a *deus ex machina*. Unfortunately, there is little reason to believe that betting on technology indefinitely is going to work out.

WHEN IDEAS DRY UP

The belief that we can engineer our way out of Goliath’s Curse is widespread. Cleaning up PFAS and other novel entities may be exceedingly

expensive, but we can perhaps engineer a new bacterium to eat it away at a lower cost. AI systems might be dangerous, but we just need more technical research to make sure they are faithful to human values and commands.

This approach is what most of the world's economic system is relying on. Yet it is a huge gamble that hinges on our ability to endlessly find new ideas to address new problems before they become irreversible.

Most economists seem to think this is a safe bet; that human ingenuity is a bottomless resource. They place particular faith in 'recombinant growth': new inventions in one area spilling over into others, sparking more innovation. Think of how computers could be used to increase our knowledge in areas ranging from finance to climate modelling, so enabling even more developments. Innovation should create constant or even accelerating returns.¹²

Yet innovation appears to be slowing down. In the US, the average number of patents-per-inventor has decreased since the 1970s. This does not seem to be compensated for by higher-quality inventions. On the contrary, early innovations in a field tend to be the ones that produce the largest improvement.¹³ The first combustion engines were revelatory compared with the incremental improvement in engines today. Research productivity is also declining across almost all sectors at the rate of around 50 per cent every thirteen years.¹⁴ Ideas appear to be getting harder to find.

Why is this? One explanation is that we have picked all the lowest-hanging fruit. New inventions require bigger research teams, more money, and more effort. Another is that materials used in technology are becoming more costly (true for some, but certainly not all industries). There is also a political component: industries often suffocate new competitors, including through 'defensive' research and development. This means investing to protect market share and using tactics such as lobbying, secrecy, buying up potential competitors, and using legal teams to tie up competitors with patent-infringement lawsuits.¹⁵ For instance, fossil-fuel companies such as ExxonMobil had a significant hand in the demise of early electrical cars. Oil companies filed for over twenty patents for electric vehicles since the 1940s as a way to block their production. They simultaneously lobbied against policies that would encourage consumers to adopt electric cars.¹⁶ Companies don't want innovation per se; they want profit.

It's not impossible that there could be a technological revolution that resets the playing field and allows for a new wave of inventions and growth, as happened during the Industrial Revolution. Many see AI as filling this role, while others suggest more bizarre (and disturbing) workarounds such as cloning particularly gifted researchers to boost research productivity.^{[17](#)}

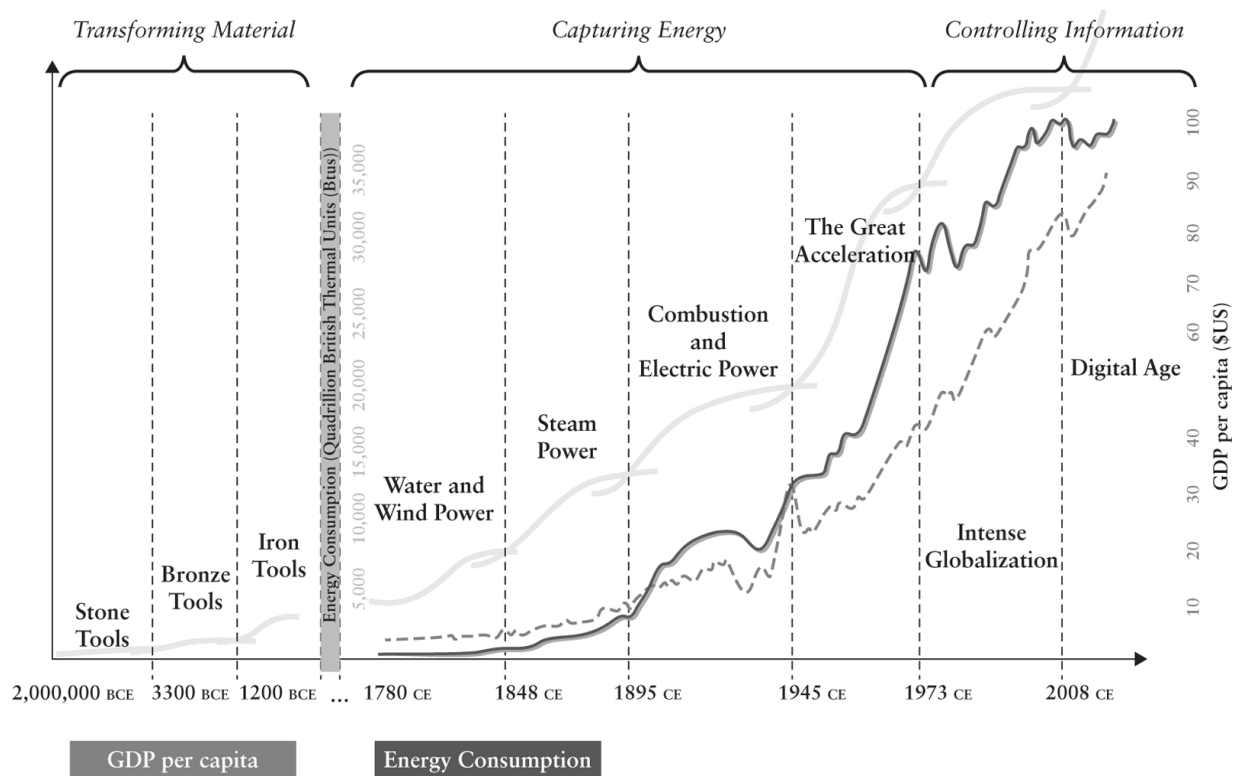
Even if such an AI revolution were to occur, it's not guaranteed that it would be an effective or sustainable cure. A new age of innovation would come with a host of new problems. Indeed, most of the core problems we have covered thus far have been enabled by advances in technology: splitting the atom created nuclear weapons; the combustion engine spews greenhouse gases into the atmosphere. Outrunning technological problems by developing more technology doesn't seem like a sensible long-term strategy. Especially if technological development is still being driven largely by Goliath traps: the pursuit of profit, status, and a military edge.

Even overcoming the current hurdle of diminishing returns on research productivity might simply worsen our situation. The acceleration of economic growth and innovation may be the sickness rather than the remedy. As we've already covered, growth makes staying within planetary boundaries more challenging. Given that this is already a Herculean task, growing for growth's sake (or because of status competition) is not a prudent course of action.

Economic and technological growth has another downside: legal lag. Every new technology introduces new problems. Legal and regulatory systems are already struggling to produce new norms, laws, and legislation to keep up with issues such as copyright infringement in large text models, whether armed drones can kill without human supervision, and the regulation of novel entities such as PFAS. Further accelerating the Global Goliath will exponentially increase the number of issues our already slow and overburdened policy systems need to manage. And there is little evidence that we will solve these international problems. A review of 224 studies found that the vast majority of international treaties don't achieve their intended outcomes. This is especially true of treaties that deal with areas such as environmental protection and human rights. The few exceptions relate to trade and finance, domains that tend to serve the

wealthiest.¹⁸ We should have little confidence that our global governance can keep up.

Exponential growth could eventually simply outstrip the ability of our societies to adapt. Using simple yet sound mathematics, the complexity scientist Geoffrey West has shown that exponential economic growth requires increasingly quick paradigm shifts to keep pace. It took around two millennia to go from the Iron Age to the age of navigation, two more centuries to leap to the Industrial Revolution, just over a century to move into the computer age, and just forty years to accelerate into the digital age. It's like having to jump from one treadmill to another, each one running faster than the last ([Figure 49](#)).¹⁹



[Figure 49](#): Accelerating waves of innovation and economic growth across human history

The lines tracing energy consumption and GDP per capita are for the US.

If the next treadmill is fast enough, we may be thrown off. Our societies might simply not be equipped to adapt to a new internet-level innovation every ten, five, or even two years. Human societies simply cannot keep up with the changes and problems generated by unregulated innovations and

the speeding up of life. This is to say nothing of whether the earth system can withstand even further demands created from ballooning growth. A technological singularity could lead to collapse, not utopia.²⁰

Apart from technological stagnation and over-acceleration there is still the slim chance of a *deus ex machina* that will allow us to innovate our way out of catastrophic risk while growing exponentially, but even this does not necessarily offer a bright future.

SILICON GOLIATH

In 2023, 230,000 tonnes of cobalt and 9 million tonnes of silicon were exhumed from the earth. That is 10 million tonnes less than the amount of plastic that is leaked into the environment every year. That exchange of materials – the extraction of silicon and cobalt and the dumping of synthetic materials into the waters of the world – in many ways summarizes our global system of extraction. Cheap materials, fuel, data, and labour go into the Global Goliath; profit, plastic, carbon, novel entities, and new technologies and weapons (as well as many good things, like medical devices) come out.²¹

It is the unequal exchange of a new type of emerging system: the Silicon Goliath. Silicon is still not the most important or bountiful of our resources: cement, steel, and ammonia are far more vital.²² Silicon is, however, representative of a new Goliath fuel that is reshaping and speeding up the world system. Data is a crucial new lootable resource. The world's most powerful intelligence agencies capture vast streams of it to enable mass surveillance. The world's most powerful companies strip-trawl the internet to gather the data to train new generative AI systems and to better target advertisements. That data is being used to automate cognition, to speed up and scale up all the existing global systems, whether it be the search for fossil fuels, the production of novel entities, mass surveillance, or the organization of robust-yet-fragile global supply chains.

Data and AI are also critical to new monopolizable weapons such as coordinated swarms of killer robots. Imagine a leader who can not only fire a nuclear weapon without democratic oversight (which is already a reality), but also enforce martial law through fleets of armed and loyal killer robots.

Then there is the silicon-laced cage of surveillance. Mass surveillance systems are of little use without vast streams of data and sophisticated AI systems to process them. Disturbingly, unless you are protecting your privacy, almost everything you've searched on the internet, every email you've sent, and every phone call you've made or received is likely to be stored in a server in the US where an agent of the National Security Agency (NSA) can access it. This is due to the 'PRISM' program, a mass surveillance system that was exposed by the former NSA analyst Edward Snowden in 2013. PRISM taps into the data flowing through big network providers, such as AT&T and Verizon in the US, as well as companies such as Meta, Google, and Microsoft.

It doesn't just capture the data of US citizens, either, but also that of most people across the world. Around 25 per cent of internet traffic is routed through (and 30–40 per cent of data centres are contained in) the US (where it is tapped by intelligence agencies), while a lot more is captured by tapping big-tech conglomerates such as Google, and sharing data from mass surveillance systems with allies such as the UK.²³ PRISM is just one example of a government-led mass surveillance system; Australia, the UK, China, Russia, and many others all have their own. That is in addition to the extensive 'surveillance capitalism' programs from which Google and Meta profit handsomely.²⁴ Searching and organizing all that data wouldn't be possible without AI.

Surveillance capitalists are accompanied by intelligence communities to create a 'stalker complex' that constantly watches the world. The breadth and intrusiveness of this mass surveillance apparatus is the work of a small number of intelligence agencies such as the NSA and its UK counterpart the Government Communications Headquarters (GCHQ). That stalker complex is constantly seeking to use technology to more closely watch and understand people, whether it be to better target ads or to identify terrorists and political dissidents.²⁵

Such AI-based massive surveillance programs are increasingly being paired with robotics and other technologies. In 1993 the US government's Defense Advanced Research Project Agency (DARPA) awarded grants for the development of insect-sized drones that could carry out surveillance.²⁶ The US today has a range of sci-fi-like surveillance projects, including the SMART program, which looks to create a planet-wide monitoring system

by harvesting data from different sources such as satellites, drones, and planes. China, during the Covid-19 pandemic, used drones employing facial-recognition technology to enforce lockdowns.²⁷ Then there is the sheer size of the rollout of surveillance cameras. The UK alone has somewhere between 4 and 6 million CCTV cameras, or roughly one for every ten citizens. That is just a small taste of things to come.

Goliaths looking to improve ways to control and coordinate their citizens is a long-running trend. As we've seen, societies across history have steadily marched from relying on intimidation to increasingly relying on more effective forms of administration and control. It's a trend reflected in the growing size and effectiveness of bureaucracies across history, as well as an inexorable increase in the amount of information processed by states. While Narmer and Sargon knew vanishingly little about their subjects, the CEOs and executives in charge of the stalker complex have a wealth of information about citizens of the modern world, including their names, income, addresses, immigration histories, political beliefs, purchases, and even their movements and whereabouts. Rather than relying on a few part-time bureaucrats, they have enormous highly skilled administrations aided by algorithms.²⁸

The modern world is no exception to this trend. Governments everywhere are investing hundreds of millions of dollars to improve a range of surveillance measures, including facial recognition. As new disasters emerge, surveillance will only deepen. The PRISM program was built off the Patriot Act, a piece of legislation that was rushed through Congress just forty-five days after the 9/11 terrorist attacks. It is a classic ploy to use crises as an opportunity to expand surveillance powers, as also occurred during Covid-19, a savvy way in which the powerful capitalize on our authoritarian impulse during a disaster.

A world of growing catastrophic threats will provide ample opportunities to expand the unblinking eyes of the stalker complex.²⁹ Some scholars have already suggested installing a system of ubiquitous surveillance and pre-emptive policing to stop any (for now, imaginary) decentralized weapons of mass destruction like engineered pathogens. Imagine what kind of security and surveillance measures the US or China might respond with if they suffered a case of nuclear terrorism? Simply the spectre of

catastrophic terrorism may be enough to scare the world into greater autocracy.³⁰

Such systems don't seem to be effective at catching terrorists at the moment, let alone creating a comprehensive, real-time watchful eye over the entire world. Similarly, the production and abilities of killer robots are limited and crude. We should be sceptical of techno-hype, yet the levels of power and precision they could reach as the decades pass is significant.³¹ For some, this may not be worrisome. Being watched is barely noticeable for the luckiest of us, except for the occasional creepy indication like seeing more tailored ads. But relying on these systems to remain in relatively safe, accountable hands in a world of democratic backsliding is unwise. We are constructing systems of extensive control in a world that is becoming more unequal and extractive.³²

The transition between one form of Goliath and another is rarely a clear-cut matter. The same is true of the current transition from the Global Goliath to Silicon Goliath ([Table 7](#)). As our systems become more dependent on advanced algorithms, and as mass surveillance (and new weapons) curtails the ability to resist through protest, we cross increasingly into the realm of Silicon Goliath. We aren't quite there yet: many systems around the world can still effectively function without AI, mass surveillance often lacks effectiveness, and new monopolizable weapons such as drone swarms are still limited in their use. Yet we are transforming into a Silicon Goliath at a rapid pace.

[Table 7](#): The evolution of Goliath across history

Goliath	Main lootable resources	Main form of caging	Main monopolizable weapons	Curse
Early Goliaths	Aquatic resources, grain	Geographic, palisades	Stone, obsidian, copper	Primal Curse
Empires	Grain	Walls	Bronze, iron, horses	Diminishing returns on

Goliath	Main lootable resources	Main form of caging	Main monopolizable weapons	Curse
				extraction
Colonial empires	Grain, commodities	Walls	Gunpowder (cannons and muskets)	Shatterzones, genocide, and collapse caused by colonial expansion
Global Goliath	Fossil fuels, grain, and commodities	Passports, identity cards, walls, surveillance	Nuclear weapons	State failure and global catastrophic risk
Silicon Goliath	Data, fossil fuels, grain, and commodities	Advanced mass surveillance systems	Nuclear weapons, killer robots, space weapons, and others	Global catastrophic risk and locked-in mass surveillance and autocracy

Even if the best-case scenario of techno-optimists occurs and we manage to technofix our way out of all the catastrophic risk we face, if we reach some kind of technological singularity, the outcome is unlikely to be appealing to many. Yes, we may have far greater power over nature and technological prowess – perhaps even space flights and robotic slaves – but probably far less freedom, democracy, equality, and human connection. The same rule throughout all history applies to the future too: technology alone won't give us a civilization. Yet there are actions that will.

KILLING GOLIATH

The last path forward is the least likely: that we collectively kill (or perhaps tame) the Global Goliath. Across history Goliaths have collapsed, but wherever there has been Goliath fuel, they have tended to rise again. While some societies have been more democratic, they have tended to be either outcompeted by others or become more extractive over time. There are a few factors in the modern world that could help us to kill Goliath and create a more equal and less exploitative global system. That is, a global civilization.

The first is making resources less lootable. Imagine if the data of today could be gathered only with the full and informed consent of citizens and was fairly remunerated. The world of AI would change dramatically as the development of algorithms slowed and the risk was alleviated.

The second is increasing exit options. Creating more open borders and, importantly, more ways for people to escape mass surveillance, would mean that states would have to negotiate more with their citizens. It's a trend we've seen throughout history: geographies with more exit options favoured democracy. While this would need to be done carefully to address security and cultural concerns, it would be a step towards the fluid civilizations of the past.

A third is using technology to enable both larger-scale democracy and greater oversight of the powerful. The Agents of Doom use secrecy because they know that transparency would lead to greater regulation. While Rome and other early republics and democracies were limited in how many citizens could attend assemblies, today we can use online platforms and deliberative assemblies to run democracy at a far larger scale more intensely. Technology is expanding the realm of what is democratically possible.

The fourth and final possibility is knowledge. The survivors of the first Goliaths of northern America knew the oppression that accompanied a dominance hierarchy and could organize to avoid it. Yet no society through history understood the darker angels of our nature or Goliath fuel. Understanding why Goliaths emerge and how they fall into Goliath traps can help us to create interventions and institutions able to stop history from repeating (I'll highlight several promising interventions in the Epilogue).

The final two factors – technology and knowledge – have changed dramatically in the past two decades. They are also the two that give the modern world the best chance of escape from Goliath. But it is a path with few precedents, and there is little sign that we are currently on it.

CHOOSE YOUR CURSE

A new Silicon Goliath is emerging. The existential risk we face is in many ways the same risk that always accompanies the rise of a new Goliath. The first states were littered with blood-covered bronze swords and pits of sacrificed slaves. The creation of empires like the Qin and Rome created seas of fractured skulls. The establishment of a new colonial, capitalist world system was enabled by the sacrifice of the tens of millions made to work under the whip, or die alone drowned in the Atlantic, crushed in mines, or slowly perishing from the pitted skin and scarred lungs of smallpox. The legacy of the Silicon Goliath could be the destruction of the entire planet, already disrupted by new weapons that can annihilate cities in the blink of an eye, and the slow removal of decision-making and autonomy from our hands into the cold and fast machinations of AI. Ironically, the rise of our Silicon Goliath could trigger its fall.

Looking at both the long-run trajectory of Goliath and current trends, our Global Goliath appears destined to hurtle down one of three paths. The first, and most likely, is self-termination. The second is a world in chains. The third is a world in which we somehow manage to shackle and control it. Reversing inequality and establishing deep democracy in a lasting fashion has unfortunately been a rarity throughout history. It is a highly unlikely outcome, which makes it worth fighting for even more desperately. While throughout history the collapse of a Goliath was usually temporary and a liberation, in the future it threatens to be permanent and take us and much of the planet with it. We are passengers on a journey that looks likely to end in chains or evolutionary suicide. Our Global Goliath will die explosively unless we kill it first.

The prognosis here may seem dire and speculative. Yet, ask yourself, where do races to the bottom and arms races lead to in the long term but large-scale destruction? We can't rely on our luck never running out or somehow finding a plucky technical fix to every new technological

problem. Similarly, are there any signs of the drive towards mass surveillance slowing or reversing? Even the world's most ardent democracies are spending at least an order of magnitude more on improving facial recognition and surveillance, and at least a hundred times more on their militaries, than they are on improving democracy (whether that be through initiatives to improve voter turnout or experiments in deliberative democracy). This is what economists call 'revealed preferences': follow actions and money to understand the underlying motivations. The preferences of our emerging Silicon Goliath are clear.

This is not to say that these scenarios will come to fruition in a few years, or even a decade. There will be delays and reversals. But over the course of decades and centuries, the deeper trends we've seen across history will show through.

If self-termination, some kind of global societal collapse, is the most probable trajectory for our Silicon Goliath, we have to ask: just how bad could it get?

After the Fall

Hart's Mystery – From Homo solus to Homo finalis – A Second Industrial Revolution? – A Hellish Earth – Survival of the Richest? – Goliath Rebooted? – A Kaleidoscope of Futures

HART'S MYSTERY

Are we alone in the universe? Some think that, despite there being no signs of extraterrestrial life, the universe should be teeming with technologically advanced species. This riddle began with an equation formulated by the astrophysicist Frank Drake in 1961. Drake calculated that the number of alien societies we could contact is determined by the number of planets that could carry life, the number of times that life begins and evolves into an intelligent, advanced society, and the length of time for which it has persisted. Given that the universe is around 13.7 billion years old, and contains 100 billion planets, it seems highly unlikely that there are no spacefaring aliens out there. But where are they?

This is often mistakenly termed the 'Fermi Paradox'. The legend is that during a lunch with colleagues in the 1950s the physicist Enrico Fermi (one of the architects of the first atomic bomb) asked, 'Where is everybody? Where are all the aliens if there should be so many of them out there?' The truth is that Fermi was actually complaining about the impossibility of intergalactic space travel. The question of 'Where is everybody?' was, in fact, first asked by the astrophysicist Frank Hart in 1975.^{[1](#)} It is also not a

paradox. You can end up with an infinitesimal number of potential extraterrestrial cultures if the number of habitable planets or the likelihood of life beginning is tiny. Once you factor in the uncertainty about how likely life is to start, suddenly the paradox dissolves and it becomes mathematically unsurprising that we might be alone in the universe.²

Rather than dealing with Fermi's Paradox we are dealing with Hart's Mystery: the mystery of our seemingly solitary existence. When we gaze out to the stars, we realize that for now we are more than likely the only flickering light of sentience in a vast sea of empty blackness. That raises the stakes of continuing life on this planet: the choices we make now could determine whether the universe is potentially left as a collection of rock and burning stars whirling in a lifeless vacuum for millennia to come. If a global societal collapse does happen, then will it be our species' final curtain call or just another rebirth?

FROM *HOMO SOLUS* TO *HOMO FINALIS*

Of all the species that have ever existed, 99.9 per cent are now gone.

It's almost certain that *Homo sapiens* will eventually go the same way. Unless other realities exist that we can access with god-like technology (an enormous if), then when this universe ends – whether it is ripped apart owing to its own expansion or collapses in on itself – we will too. The more pressing question is whether a collapse in the coming decades or centuries could spiral into human extinction. It's a difficult question to answer since collapse can take so many forms, from a relatively non-violent decline of our global economy and power structures over decades to an apocalyptic spasm of nuclear-fuelled destruction. Still, it's important to trace how many people would need to perish for it to result in probable longer-term extinction and what impediments the survivors would face to surviving.³

Humans have survived for at least 200,000 years. If one looks at the extinction rates of other mammals, then statistically that makes our odds of being wiped out in any given year (naïvely not assuming any other catastrophe) one in 14,000. That figure balloons out to one in 870,000 if we consider all our hominid ancestors.⁴ Yet it is questionable how relevant such calculations are for a species that is spread across the entirety of earth

and which can piece cities back together after they have been split apart by atom bombs. There are other, more useful ways of trying to calculate our odds of survival.

Imagine you are one of a group arriving on a lush, deserted island. How big would the group need to be in order to survive and create a self-sustaining society? You need enough to avoid inbreeding and other deleterious effects of a too-small gene pool. You also need a few extra as a buffer to guard against accident, drought, or disease. This is known as the 'minimum viable population'.

We have some historical examples which can give us a glimpse of what this number is likely to be. The original peopling of both the Americas (at least 23,000 years ago) and Polynesia (around 1000 BCE to 700 CE) may have occurred with a group of around only a hundred. The initial raft of people landing in Polynesia who eventually led to the indigenous cultures of the Māori and the Samoans may have just been a few dozen. We get other surprisingly low numbers when we look further back in time. The first successful migrations out of Africa into Europe and east Asia may have totalled as few as 1,000–3,000 people. The human groups there still thrived.⁵

Others have looked to the stars to estimate the minimum viable population. How many people would it take to start an intergalactic colony? Australian geneticist Ian Franklin and American biologist Michael Soulé in 1980 proposed that fifty individuals would be needed to combat inbreeding, and 500 to prevent genetic drift (small random fluctuations in genes can lead to certain alleles disappearing in a small population, leading to a loss of genetic variation). More recent, refined guesses have turned up larger figures of 14,000–44,000. The big spread in numbers is due to varying estimates of the required buffer against calamity. Disasters happen, and even if a minimum viable number is enough in theory, it would be precarious in practice. As the Neanderthals found out, sparse dispersed populations can dwindle into extinction with just a bit of bad luck. That larger estimate of 14,000–44,000 would also need to be in close proximity so they could reproduce, protect each other, and form a community. Even a community of 44,000 might be vulnerable if the conditions were dangerous enough.⁶ While few have explored it, these numbers also depend on the genetic diversity of the group.

Even a minimum viable population number of 44,000 is, of course, very low compared with the current global population of 8.2 billion. This makes the possibility of extinction appear far-fetched. The only attempt to examine this issue suggested that even an extreme collapse with extensive infrastructure damage, temporary climate change, and a population loss of 99.99 per cent would have only a 0.1–0.3 per cent chance of causing extinction. There would still be 810,000 survivors, who would have a ‘grace period’ of easy access to remaining supplies (canned goods and produce) and equipment. That might buy them enough time to learn the necessary survival skills such as basic foraging and agriculture. Killing off every single human is an incredibly high bar.⁷

That said, there are also reasons to believe that it will be trickier in the future for a population to bounce back post-collapse. The survivors would be living in an environment that is more hostile than settling the warm, bountiful islands of Polynesia. That is due to the latent risk we face: the survivors would not be trying to scavenge and rebuild in the stable climate we’ve had throughout the Holocene. They would be living in a far warmer world, one with fewer survivable habitats and where foraging or farming would be far more difficult. They would also potentially have to cope with more diseases than our ancestors did, and far fewer means of control or cure once the supply of leftovers available in hospitals and health centres ran out.⁸ Survivors would be affected by the long-term effects of novel entities,⁹ including declining fertility and sperm counts.¹⁰ All this means that the viable population is more likely to be at the 44,000 end of the range, or perhaps even higher.

How we would fare in a global collapse is also dependent on how prepared we are. In the north of Norway in the Svalbard archipelago is the Svalbard Global Seed Vault, also known as the ‘Doomsday Vault’. There are over 1,700 gene banks around the world which house different seed types (for agricultural research and development) and the Doomsday Vault aims to have a copy of each unique seed. It was intended to be an icy, impregnable fortress sunk into the permafrost, but in a warming world, that permafrost is now melting. In 2017 the vault was flooded, although luckily the meltwater did not reach the seeds. Nonetheless, it’s hard to predict how well the vault will fare in a 4°C warmer world.¹¹

Whether we survive a catastrophe depends in large part on what food is available afterwards. Most scenarios of global catastrophe involve a large-scale climatic disruption, whether that be a post-war nuclear winter, a high-impact volcanic eruption, or extreme climate change. Scientists at the organization ALLFED (the Alliance to Feed Earth in Disasters) have already started to cook up schemes of how to feed the entirety of the world's population if traditional agriculture fails. Such plans include mass food factories producing mushrooms, insects, algae, and seaweed, or even the straight chemical synthesis of food. However, the research is nascent and more needs to be done at scale. Certainly, we are nowhere close to being able to feed the survivors of a nuclear war. Even if we do scale up these systems, they don't solve the political problem of distribution. (Today, the world produces more than enough food to provide 2,750 calories per person every day, yet 9 million people starve every year because food isn't distributed appropriately.) More support is needed for the vital work of organizations like ALLFED if we are to be prepared for the worst-case scenario.^{[12](#)}

There are a few lessons we can draw from all this. First, extinction is implausible unless a truly cataclysmic situation involving a combination of catastrophic threats and vulnerabilities occurs. Second, even under these conditions humanity is likely to survive in the short term, at least. Third, we are dramatically underprepared. Fourth, the longer the status quo persists, the worse a future collapse will be. We need to start worrying not just about the fall, but also where we are falling from.

Extinction in the longer term, over a span of thousands of years, might also be more likely if we aren't able to rebuild a sufficient level of cultural and technological power.^{[13](#)}

A SECOND INDUSTRIAL REVOLUTION?

Almost every collapse has been followed by a 'technical recovery': regaining the previously established levels of energy capture and population density. This doesn't mean recovery in the sense of improving people's lives. If a future technical recovery is anything like the first bout of

industrialization, it will carve deep scars into the body of humanity all over again.

Whether the survivors of a deep global collapse would be able to recover is questionable. We've only ever had one industrial revolution, and that was an exceptional one-off that occurred in a stable climate with plentiful, easily accessed coal and oil. If we had a deep collapse and the entire industrial skeleton of the modern world fell to pieces, would we be able to repeat it?

It is a topic that few have dared broach. There have been some pieces of analysis, ranging from presentations and blog posts through to Lewis

Dartnell's *The Knowledge: How to Rebuild Our World from Scratch*.¹⁴ It is an absurdly underexplored topic, given its importance, partly because it is necessarily speculative. It would depend on how deep the collapse is, and how great is the energy ratio of the fossil fuels that remain in the ground (as well as the renewable energy we have at our disposal). These are largely unknowable factors. Dartnell is the only one to have seriously tried to answer the question.

He took a related but different approach: could the Industrial Revolution have been achieved without fossil fuels? His answer is a tepid maybe. Charcoal from sustainably harvested trees could provide one source of dense energy, although lack of land would keep the supply limited. Solar PV is durable but difficult to make. It relies on extremely thin slices of pure silicon components. This, in turn, needs technology similar to that required to make semiconductor electronics components. Renewable electricity from wind and water are more promising. We could start with windmills and watermills before progressing to wind turbines and hydroelectric dams. This would be possible only in a few favourable locations. The second industrial revolution would not take place in the UK but more likely in Scandinavia. Dartnell concluded ambivalently: 'An industrial revolution without coal would be, at a minimum, very difficult ... a combination of renewable electricity and sustainably grown biomass, might be possible after all. Then again, it might not.'¹⁵

Dartnell is right: an industrial revolution without coal would be a momentous challenge at best and physically impossible at worst. We don't know, but there are reasons to think that the challenge may be easier in a post-collapse world. Even a (near-term) collapse that incurred large-scale infrastructure loss, such as a nuclear war, would still leave plenty of

working coal mines, oil rigs, and stored gas. Knowledge of high-efficiency engines and power plants would smooth the transition. If it happened within the coming decades, there would still be many fossil fuels to draw on. There are currently around 1.7 trillion tonnes of coal that could be profitably recovered using existing technology, and 188 trillion cubic metres of gas. If we continue at the current rate of production, we have around 134 years of coal reserves left, 51 of oil, and 44 of gas.

Burning coal, oil, and gas is, of course, eating the future in more ways than one. Using those reserves would be more than enough to cause climate chaos. That is a problem that a second industrial revolution would face: achieving a technical recovery while worsening climate change. This in itself may make the task insurmountable. There is then a second problem. The main impediment is not the amount of fossil fuels left in the ground but rather their energy ratio.¹⁶ We would be approaching industrialization with hard-to-reach and hard-to-use resources. The declining energy ratio of fossil fuels is likely to make a second industrial revolution impractical.

Would post-collapse communities even want to reindustrialize? Studies almost always assume that *of course* they would want to.¹⁷ Yet it's not a given that a post-collapse population would have the same pro-technology sentiments that many have today. If collapse was triggered by a nuclear war, extreme climate change, or another technological calamity, then the survivors might have little appetite to repeat past mistakes. They might choose to reject the old system, like the people of Cahokia and Tiwanaku, who destroyed elite symbols and monuments as they walked away. In a world ruined by runaway technology, that might be an entirely sensible decision.

The downside is that, although in the short term an unregenerated industrial collapse would almost certainly lower the threat of human extinction, in the very long term we'll require advanced technology to survive.

A HELLISH EARTH

Over tens or hundreds of millennia we are likely to face many world-shattering natural hazards, such as gargantuan volcanic eruptions. Over

millions of years, the chance of a collision with a large asteroid or comet suddenly becomes non-negligible. Survivors living in a non-industrialized world would not have the resources to deflect an asteroid or prevent an eruption. However, the potentially biggest threat is the changes in the earth we are triggering today.

Climate change and temperature rise will not magically stop in 2100 (where the models frequently end). The earth will continue to transform. There were once fears that enough global warming could trigger a runaway greenhouse effect which would boil the water on the surface and sterilize the earth, turning it into a planet essentially like Venus. To everyone's relief, the IPCC suggested in 2009 that there was virtually no chance of human activities creating a runaway greenhouse effect (although there are a few theoretical models and scenarios that disagree).¹⁸ It would also be a process more likely to unfold over hundreds of thousands of years.

A runaway greenhouse effect is not needed to reach human extinction, though. Over tens of thousands of years, the world would be utterly unrecognizable with 6°C of warming. The most deadly and disturbing change would come from the oceans, carbon in the atmosphere making them more acidic while melting ice caps dump freshwater into them, blocking deep-sea circulation and cutting off the circulation of oxygen. This would create 'anoxic' oceans with little oxygen and essentially no marine life. The sea would become a dead zone colonized by purple and green sulphur bacteria. The surface would become a vast expanse of purple with a slick oil-like surface. Hydrogen sulphide gas produced by bacteria would belch from its surface, destroying the ozone layer. Along with lethal heat, mass extinction, and dead oceans, we would also have to contend with more ultraviolet radiation. This would skyrocket skin cancer rates among the already ragged, vulnerable survivors.¹⁹

These anoxic oceans and their effects seem to be key to explaining how climate change has triggered mass extinction events in the past.²⁰ Add in the destruction we have wreaked on other planetary boundaries and long-term extinction doesn't seem implausible. Certain parts of the earth may still be technically habitable, and we are a remarkably ingenious and resilient species that has weathered ice ages and dramatic climate swings before. Yet every species has its limits. That's true of even our species' most powerful members.

SURVIVAL OF THE RICHEST?

In the past, elites have used their wealth to migrate or change their identities when trouble looms. Power is helpful even during a crisis (unless the holders of it are directly targeted and killed off during the tumult). For now, the mega-rich seem convinced that they will repeat the feats of past elite survivors and escape the apocalypse.

The main customers of the most luxurious doomsday bunkers are billionaires. Peter Thiel, the billionaire co-founder of PayPal and AI (or rather, AI for military, security, and policing) company Palantir is a prime example. In 2011, Thiel bought a 477-acre former sheep station in the South Island of New Zealand. He also purchased New Zealand citizenship, secretly lobbying the government and pledging to invest in New Zealand start-ups as part of his application. Citizenship was granted, even though he had spent less than twelve days in the country up until then (the usual criterion is to have lived in New Zealand for 1,350 days to be granted citizenship). OpenAI CEO Sam Altman revealed in 2016 that he and Thiel have a deal by which if a global collapse occurs they will board a private jet and fly to Thiel's property.^{[21](#)}

Thiel and Altman's preference for New Zealand is widely shared, since New Zealand will be one of the most resilient countries in the case of a global catastrophe. Its location deep in the southern hemisphere means it will be one of the least affected by the fallout from a nuclear war, and its isolation makes it well protected from pandemics. It is a doomsday safe haven. This has led to it being recommended as a kind of national Armageddon bunker in the libertarian manifesto *The Sovereign Individual: How to Survive and Thrive During the Collapse of the Welfare State*, a book that has been endorsed by Thiel and other billionaires such as Marc Andreessen.

Then again, why buy a doomsday bunker on a small piece of land when you could buy an entire country instead? That is exactly what the former cryptocurrency mogul Sam Bankman-Fried was considering doing. Bankman-Fried was head of the cryptocurrency company FTX before being jailed for fraud. He and his brother Gabe (who led the philanthropic FTX Foundation) discussed buying Nauru, the third-smallest state in the world, located in Micronesia. Their plan was to 'purchase the sovereign nation of

Nauru in order to construct a “bunker/shelter” that would be used for ‘some event where 50–99.99 per cent of people die [to] ensure that most EAs survive’. (In this case ‘EAs’ refers to the Effective Altruism movement, which Bankman-Fried was part of.)²²

Self-preservation is a widespread instinct among the richest and has spawned an entire industry of doomsday bunkers across the world. Reinforced, luxury bunker-manors with pools, wine vaults, artificial gardens for sunbathing in simulated sunlight, and underground hydroponic farms, can be found from Texas to Czechia. One group of super-rich preppers asked the media theorist and writer Douglas Rushkoff for advice on how best to prepare for a world-ending tragedy of whatever stripe, which they described as ‘the event’. Some have purchased not just bunkers but also their own security forces (in one case a dozen Navy SEALs) to defend their mini-fortresses from mobs and raiders.

Their questions to Rushkoff were tellingly self-interested: ‘How do I maintain authority over my security force after the event?’ The options they had considered included replacing their guards with more loyal robots or ensuring obedience with disciplinary collars (the kind that zap you if don’t follow orders). When Rushkoff suggested that they instead befriend their employees and treat them well now to build a connection that might last through a crisis, they rolled their eyes.²³

This bunker mentality is a flawed strategy. It’s unlikely that these bunkers would stand the test of a true worst-case scenario. Even the Svalbard Doomsday Vault was almost compromised by a single flood. Underground farms might work in laboratory settings but are a different matter in reality. They would need to function and avoid contamination for years while under duress. All it would take is one rogue bug and the entire food supply could be ruined. It’s also unlikely to work because it does exactly the opposite of what has allowed communities to persist and prosper through disasters of the past and present.

To prepare for a societal breakdown, you are better off stockpiling friends, favours, and skills rather than guns and canned beans. That’s the central message we have from the Palaeolithic and studies of past societal collapses and modern disasters. Communities that respond the best tend to have broad, strong social ties. They quickly cooperate when the wider political structure falls apart, with neighbours and acquaintances providing

the role that the state once did, whether that be by sharing resources, organizing medical aid, or protecting each other. The billionaires of today are pursuing a strategy of investing in places rather than people. Not only is that likely to backfire, it's also exactly the kind of thinking that is fuelling collapse today.²⁴

There are even some scenarios in which the poor, not the mega-rich, may be the most resilient. Countries that are most reliant on global industrial infrastructure will be most affected when it is suddenly stripped away. Imagine a ravenous computer virus shuts down critical infrastructure and we are left without functioning machinery, or the ability to create new pesticides and fertilizers. Europe, the Americas, and parts of India and China could experience losses of up to 75 per cent of their rice, corn, wheat, and soybean crops. Africa, in contrast, would see drops of only around 5 per cent for soybeans and 25 per cent for rice (although almost half of wheat crops could be lost).²⁵ Many African farmers and city-dwellers are also already accustomed to living with unreliable utilities.²⁶

Much will depend on how collapse unfolds. (In the case of a pandemic, New Zealand would be the safer option.) It's difficult to see through this fog of uncertainty which groups would fare best post-collapse. For now, being poor in wealth but rich in friends is a better strategy than being a suspicious billionaire under siege in a bunker. Not only that, but cultivating bonds of solidarity and mutual aid rather than ruthless self-interest means that the survivors are more likely to help recreate a world that is worth preserving. This is perhaps the most important and least appreciated question about a post-collapse world: whether it will be a world like today's – of patriarchy, rich and poor, slaves and masters – or a world of civilization.

GOLIATH REBOOTED?

Ancient Greece looked very different from the rest of the Mediterranean after the Late Bronze Age collapse. The Assyrian Empire continued to have an empire, one that expanded and immiserated its subjects more with each passing decade. Egypt continued to have pharaohs. Greece remained a land without large cities or kings for several centuries. When Classical Greece did emerge, their societies had little connection to the Mycenaean palaces

that came before them. Even their writing was entirely different. Linear A and B were lost and the Greeks instead adopted and modified the Phoenician alphabet. The collapse of the Mycenaean kingdoms was unusually deep. And the Greek polities that emerged afterwards were unusually egalitarian.²⁷

City-states such as Athens began to pop up from the ruins. While many sought to restore old hierarchies, other communities remained egalitarian. Greece became an ecosystem of hundreds of city-states in which much of the organizing was done by citizens, rather than by religious leaders or bureaucrats. They maintained a surprising amount of wealth equality (at least for free men) and their economic growth and well-being far outstripped that of their neighbours. Compared with the empires that surrounded them, Greece was a hotbed of democracy and equality. Why did post-collapse Greece look so different?

The severity of the collapse, Iron Age weapons, and the very geography of Greece all played a role in its democratic recovery. The collapse was exceptionally deep. Wealth was dramatically levelled. Essentially, everyone in Greece became equally poor with few differences in power. The terrain of Greece also tilted away from autocracy. It is a land of islands, and around 80 per cent of it is mountainous. In other words, there were plenty of exit options. If you didn't like the edicts of a budding king, you could more easily hop to another island or retreat to the mountains away from the grasp of armies and tax collectors.

New iron weapons also played a role. They were far cheaper and easier to make than bronze, meaning they became more widely dispersed. Communities that could field the largest armies of iron spear- and shield-wielding infantry were the ones that survived. Politically inclusive groups were usually the ones that could mobilize more citizens for war and give them a greater sense of camaraderie and solidarity. This was vital for success in battle. Greeks began to use the phalanx formation: a tightly packed rectangle of troops fighting shoulder to shoulder behind an interlocked wall of shields. Such shared struggle also gave the troops a greater sense of connection beyond the battlefield. Conquered slaves aside, post-collapse Greece selected for democracy and equality.²⁸

Greece is one example of how deep collapse can lead to a more democratic recovery. The society that emerges from the ashes is freer, more

equal, and more prosperous than what came before. It is a far cry from the militant raiders and despotic patriarchs we see in dystopian films like *Mad Max* or *The Road*. It's not only movies that envisage a more autocratic post-apocalyptic world. Some scholars have suggested that post-collapse societies will revert to the historical average of predatory autocracies. The current prevalence of modern democracies and republics is not representative of how most of history has looked: it's an aberration. These scholars argue that in the absence of other evidence, we should assume that we'll end up looking more like a normal, pre-modern kingdom than a Norwegian welfare state. This might make sense if collapse were a random reset, but it's not.^{[29](#)}

As we've seen in the case of Greece or the collapse of the first Goliaths of the Americas, there are some factors that reliably lead to more democratic recoveries. These include more wealth equality, more exit options, fewer lootable resources, and either a rejection of previous hierarchies or a cultural memory of inclusive practices. Less Goliath fuel, a flat playing field, and a cultural distaste for domination all favour democracy. All these should increase after a deep collapse. The more wealth that is levelled, the flatter the playing field, and the more responsible great powers are for the fall, the more Goliath structures will be blamed and discarded. A deep collapse could provide fertile soil for democracy to take root.

However, numerous modern factors could gain momentum, pushing us towards autocracy. There would be cultural memories and technologies of how to watch and control others. Earlier despots didn't have surveillance equipment or centuries of development on how to bureaucratically track and understand citizens. Parts of the world would still be likely to have an abundance of lootable resources, including fuel and machinery. Then again, the sheer abundance of leftover materials might work against a Goliath re-emerging. As in the case of the Black Death, if there is enough population loss then labour may become the limiting factor. This would give people far greater bargaining power over any would-be despots.^{[30](#)}

The biggest wildcard determining what kind of society emerges from the ashes is culture. Whether it be in Zomia, the Mississippi, or the south-west US, people passed down stories of evil wizards, overbearing monarchs, and tyrannical trickster gods to warn of the dangers of vesting too much power

in leaders. Those were for comparatively less traumatic disasters. Imagine what kinds of tales might be crafted in the wake of the floods and fires of a 4°C world or the sickly glow of mushroom clouds and irradiated rainfall. Would we speak of mad machine-obsessed priests, myopic cults of an earth-eating stock market, and delusional dictators with the power of gods and the restraint of toddlers? It's easy to see why the survivors of a true cataclysm would not look fondly on the Global Goliath of today and why they might enshrine the myths and practices that would help them avoid its resurrection.

As in Greece, the deeper the collapse, the greater likelihood of a democratic recovery. This is, unfortunately, not a good news story, at least not in the long term. Classical Greece was still covered by slave-owning Goliaths, even if there were democratic practices between free adult men. Eventually it was covered by larger, less democratic empires. North America may have had stories and collective memories to avoid Goliaths, but an indigenous empire would still probably have emerged. Despite the initial conditions favouring democracy, history is always likely to repeat if Goliath fuel remains. That's because the darker angels of our nature also remain, and there will be a selection pressure for more militaristic societies over the long run. We can't naïvely rely on collapse killing Goliath.

A KALEIDOSCOPE OF FUTURES

Collapse is not uniform. From a wide view the world system may have collapsed, but once you zoom in, you'll notice different recovery processes in different regions. Some regions will have better odds of recovering in a politically inclusive manner or kickstarting a low-carbon industrial revolution.

This would mimic the collapses of the past. We see a similar kaleidoscope of outcomes during the Late Bronze Age collapse. The Mycenaeans experienced a near-total collapse followed by a democratic and cultural efflorescence after centuries. The survivors of the Hittite Empire persisted in a few minor, rump states in northern Syria. In contrast, the Assyrians and Egyptians kept their kings and centralized economies.³¹ The Neo-Assyrians went on a warpath while the Phoenicians focused on trade

and cooperation. We would see a similar panoply of outcomes in a post-collapse world, which could be populated by aggressive Australian pirates or democratic sub-Saharan trading cities.

One of the few trends that seems to hold across regions is that the longer current trends persist, the deeper the collapse will be. Latent risk, whether it be antibacterial-resistant microbes or global heating, will continue to build. The energy ratio of fossil fuels will continue to fall. Unless things change radically, we'll continue to climb the rungless ladder. The one bright spot is that the deeper the fall, the more likely it is that we'll rise again as democratic equals. Yet waiting for a cataclysm to rebalance power is an unnecessary, grotesque gamble. We don't need to roll the dice with evolutionary suicide; we can stop this five-millennia-long trajectory and slay Goliath.

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Fiat Justitia, Ne Pereat Mundus

*The Trinity Jury – From Darker Angels to Evolutionary Suicide – Justice
Lest the World Perish – Memory Over Myth*

THE TRINITY JURY

In 1945, on the eve of the Trinity atomic bomb test, the scientists behind the Manhattan Project began to place morbid bets. During the construction of the bomb, the physicist Edward Teller had calculated that the fission chain-reaction of the bomb they were about to detonate could potentially ignite the atmosphere, extinguishing all life on earth. The probability was incredibly low, but Teller could not rule out the possibility of incinerating the entire biosphere.

The scientists were gambling over not just the end of the world, but potentially the end of all life across the cosmos. Earth contains the only observable signs of life in all the universe. This blue marble floating in the vast ink of space may be the only refuge for the miracle we call life. The bomb could have left the entirety of existence empty and silent.

It was a huge risk, and also an unnecessary one. The Americans knew that the Nazis were no longer seriously pursuing the creation of their own atomic weapon, and that even the design they had once tried to pursue would never be successful. These were facts that the US government chose not to pass on to the scientists dutifully creating the world's first atom bomb at Los Alamos. The Trinity Test was no longer about preventing the Nazis from creating a nuclear Third Reich.

I want you to imagine that the decision to detonate the bomb – to risk igniting the atmosphere – had rested not with betting scientists or the machinations of wartime generals or politicians but with a jury of randomly selected US citizens fully briefed both on the impacts and risks of the bomb and on the geopolitical and military situation. Now, ask yourself this: would that jury – with their friends, sons, daughters, parents, husbands, and wives waiting at home – have taken the same gamble?

What would be a similar jury's verdict on the other global risks we face today? Would they decide to speed up the development of AI? Would they be happy to continue to increase emissions or search for new sources of oil despite the risk of catastrophic climate change? Would a citizens' jury sitting in on the board meetings of 3M, DuPont, BP, and Shell have hidden evidence of the global harm caused by their products, or run disinformation campaigns?

The early evidence says they wouldn't: democracy is risk-averse. In 2020, President Emmanuel Macron of France approved a French Climate Citizens' Convention. It was a response to the Yellow Vest protests in which citizens protested against rising energy prices, inequality, and perceived corruption. The Convention was designed to address all these concerns by having a group of ordinary members of the French public determine climate policy while informed with the latest science. Macron agreed to adopt all the recommendations advanced by the assembly. They spent the next two years learning and debating. At last, they agreed on a set of 148 recommendations: a progressive policy package to make all products sold in France disclose their carbon content, impose a carbon tax on all imports into the EU, and even make 'ecocide' (the destruction of entire ecosystems) a criminal offence.

Macron then faltered and broke his word. Many of the key provisions proposed by the citizens were watered down and weakened. In the end, less than 40 per cent of the proposals were properly adopted and implemented, and the most ambitious were scrapped. It seems unlikely that Macron reneged on his deal because there were fundamental flaws in the citizens' proposals (remember that the entire process was overseen by large groups of experts). He opposed them because they were too ambitious.¹

Imagine that Macron had asked the citizens' assembly whether France should continue to have nuclear weapons and they had said no. It's difficult to see him accepting their decision. When citizens are given the reins, they

tend not to want to risk the world or the future of their children. So much so that leaders need to seize power back. Currently companies and countries can privately profit from producing public risks. We need to put these actors back under democratic control.²

The use of deliberative assemblies and juries is just one example of ‘open democracy’: accountable, inclusive governance led by the masses with ordinary citizens at its decision-making heart. Open democracy aims to continuously improve and innovate how we make collective decisions. It’s about building more resilient societies, and also wiser, fairer ones. It is a method built on an appreciation of our deep past and evolutionary inclinations.

FROM DARKER ANGELS TO EVOLUTIONARY SUICIDE

The biologist E. O. Wilson once identified the fundamental problem of humanity as the fact that we have ‘Palaeolithic emotions, medieval institutions, and godlike technology’. There is some truth to that. Yet it’s not as simple as our outdated emotions meeting superpowered technology. The psychology of most people is not the problem.

On the contrary, we can be immensely sociable, intelligent, altruistic, wonderful at self-organizing, and quick to learn from each other. We see that throughout the ages. For most of human history we lived in fluid civilizations. These were webs of exchange, mutual aid, and reciprocity that were full of mobile, egalitarian, social individuals with a strong aversion to being dominated. Status competition lurked underneath these arrangements but was kept in check through counter-dominance strategies and sheer environmental circumstances. Collapse occurred only when environmental shocks disconnected populations or severed them from the wider civilization.

The fluid civilizations of the Palaeolithic allowed us to survive the ice age and cover regions and continents with shared tools, instruments, and culture. It’s little wonder we make better decisions when we come together, deliberate, and debate. That shared heritage is why communities regularly persevere and even flourish through collapse, and why earthquakes and

hurricanes often end up providing tales about the triumph of the human spirit, with even the injured quickly providing mutual aid, food, shelter, and love to those in need. Those same better angels of our nature can explain why we would trust a group of ordinary citizens not to detonate the Trinity bomb.

Open democracy and inclusive institutions are essentially harnessing these better angels of our Palaeolithic psychology. They are using the best of us to make better decisions and closer communities: to harness our collective intelligence and impulses to help each other. They are moving closer to our past as equals in a web of sharing, constantly moving communities: a fluid civilization. Our Palaeolithic emotions are not a curse. The curse is the institutions and arrangements which bring out the worst in us.

These institutions began to emerge 12,000 years ago, when the climate warmed. People became increasingly dependent on lootable resources, such as grain and fish, and the world got smaller. When weapons became available, and if the landscape was caged, small groups began to seize control of these new lootable resources. We began to slowly, hesitantly adopt more primal forms of organization: dominance hierarchies. It was a case of evolutionary backsliding: we began to give up the arrangements that had made us uniquely human and instead more closely resemble our more hierarchical chimpanzee and gorilla cousins.

As inequalities and hierarchies rose, so too did war. The threat of mass violence caused many to reach for strong, aggressive leaders, an authoritarian impulse that is buried deep in us. They became protection rackets of sorts, and the state was born.

All it took was one or two societies turning into a Goliath to set off a chain reaction. The authoritarian impulse was triggered in neighbours who now had a model to emulate. These were not thoughtfully designed institutions. Those who sought power were the least fit to wield it. Frequently the most violent and overconfident were selected for and the sheer concentration of power in masters, kings, pharaohs, and emperors led to rulers who were far too often corrupt.

These Goliath racketeers behaved rather differently from most people: they were more intent on growth, status, and aggression than the average flesh-and-blood human. Goliaths competed, and elites fought over the surplus of lootable resources within them. Dominance hierarchies

metastasized like a cancer: growing, spreading, and co-opting or conquering the groups and civilizations around them.

Yet they were also self-terminating. They frequently contained the seeds of their own demise: Goliath's Curse. The driving engine was that wealth inequality tended to increase. That then spilled over into inequalities of power, whether that be in autocracy (inequality in the control of decision-making), theocracy (inequality in the control of information), or rule by the military (inequality in the control of violence). Societies became more extractive as time passed. The social landscape was increasingly tilted towards the interests of the most powerful, be they plutocrats, priest-kings, military dictators, or entrenched bureaucrats. The benefits that Goliaths could provide began to dry up while the costs increased. Early on, this frequently engendered a strong reaction from people with fresh memories of egalitarianism. In weak Goliaths with few mechanisms of control, people often easily rebelled or fled. This was the Primal Curse of Goliath and could lead to completely abandoned cities or a boom-and-bust cycle among early farmers and kingdoms.

Larger empires had a different curse. They could control their citizens more effectively, but inequality and societal top-heaviness drove corruption, elite factionalism, poor decision-making, overexpansion, environmental degradation, and immiseration of the masses. The state was slowly hollowed out to the stage where shocks – drought, floods, war, disease – could split it apart. This diminishing return on extraction was the next iteration of Goliath's Curse.

Eventually, far larger colonial empires that could cross the oceans developed. They seem to have largely escaped the curse. Their technological and economic growth was so rapid that even declines rarely spiralled into collapse. Instead, the curse was the damage they inflicted on other societies across the world. They constructed a global system of extraction built on slavery, dispossession, genocide, collapse, and shatterzones. It created a rimless wheel of global empire where resources were extracted from a periphery which often experienced societal collapse, while the core received resources and grew increasingly resilient. We see a similar pattern in the world today. State failure is largely limited to states that have a particularly harsh history of colonization, which left behind broken, extractive institutions.

This Global Goliath now has a different curse. The rimless wheel of empire now produces catastrophic risk which is capable of destroying the entire world system. The production of catastrophic risk is largely limited to a few powerful, secretive actors. These are the Agents of Doom: the fossil-fuel industry, big tech, and military-industrial complexes. They are rackets that both cooperate and compete with each other to produce global risk for private profit and power. The Global Goliath has also incentivized the creation of ever faster and more interconnected systems. Whether you are worried about climate change, nuclear weapons, or overextended just-in-time supply chains, the answer is the same. It is the Goliaths and Goliath traps that began evolving millennia ago.

Throughout history, collapse has usually been a good thing for most citizens. There could be costs, especially when it came to increased violence. This was due less to collapse than to status seekers battling to create a new state. The bloodiest and most disturbing processes throughout history have been due to the birth and expansion of a Goliath, not its fall. This was especially true of the creation of today's Global Goliath through the marriage of colonization and capitalism.

We now live in a world system that is transforming into a Silicon Goliath. We have entered the endgame. Either we kill Goliath or we potentially face a final collapse that might succeed where the ice age failed and turn us from *Homo solus* into the final footnote of our species' history.

To rephrase E. O. Wilson, the problem of humanity is that our Bronze Age institutions suppress the best and bring out the worst of our Palaeolithic psychology to produce humanity-killing technology. It's less catchy, but more accurate. Or we can drop Wilson's formulation and just say that the darker angels of our nature are flying us towards evolutionary suicide.

JUSTICE LEST THE WORLD PERISH

There is an old Latin maxim *Fiat justitia, et pereat mundus*, which roughly translates to 'Let justice be done, though the world perish'. It hints at what many see as an underlying tension between justice and resilience.³

It's common to hear similar sentiments today. We often hear that democracy must be sacrificed to cut emissions, unleash technology, or prevent terrorism. That we need to sacrifice privacy on the altar of national security. That we need to restrict freedom and democracy to make it out of the twenty-first century alive.⁴

Yet all the evidence we do have suggests that more inclusive institutions tend to make better, more future-focused decisions in the long run. We've already come across evidence from fields ranging from computer science to forecasting suggesting that bigger, more diverse groups that share information reach better decisions.⁵ Similarly, one of the few studies to look at what types of societies make the best long-term policies found that democracies significantly outperformed autocracies.⁶ Giving in to the impulse to embrace despotism is not wise. It is simply repeating a mistake from thousands of years ago, on a global scale. It is a decision built on following a darker angel of our nature rather than evidence. It is turning to the authoritarian impulse when we should be turning towards each other.

We live in a remarkably lucky world. Preventing collapse and escaping the endgame requires building a better one: lower carbon emissions and fewer deaths from air pollution; fewer weapons and more money for schools and hospitals; more involvement of regular citizens in governance; less corruption; fewer plutocrats and more democracy. These are frequently what populists from both the left and right are calling for. Supporters of both Donald Trump and Bernie Sanders would agree on the basic idea that politics and the economic system no longer represent or fairly support most people and that democracy is fraying and needs to be resuscitated.

The challenges of populism, avoiding collapse, and building a better world all have the same solution: the redistribution of power. We must use technology to help us run open democracy at scale (with widespread use of deliberative assemblies and juries) and keep inequality in check. Those who are worried about the rise of populism need to start listening to the anger of the people around them rather than vilifying them. They need to offer a path that is more appealing than an authoritarian impulse fed by scapegoating minorities, protestors, and immigrants. That is not one based on bland pleas for hope or growth. It requires real, concrete change: introduce wealth taxes and progressive taxation; crack down on corruption (including banning legalized bribery like the revolving door between politics and industry); and

give people a genuine direct say in government and workplaces through open democracy.

Such changes are eventually likely to be embraced and celebrated. When people speak of ‘making America great again’ they are thinking of the 1950s, a time when the tax on the highest income bracket was over 90 per cent, unionization was high, and wealth inequality was low. When people think of the glory of Classical Greece, they are usually thinking of Athenian democracy and its egalitarian culture. Now we have the opportunity to build societies without the patriarchy and racism of the 1950s US, or the slavery of Athens, and with far greater material comfort and well-being.

It is not a matter of trying to achieve justice even if it destroys the world. It is *fiat justitia, ne pereat mundus*: let justice be done lest the world perish.

MEMORY OVER MYTH

Escaping Goliath’s Curse will be a Herculean task. We need to reroute the path of human history. Making it out of the endgame will require deep-rooted and systemic changes.

Such reforms will clash with myths that have evolved over thousands of years. The most entrenched is that subjugation is a necessity. We now live in a world that would have been unthinkable to our ancestors: one filled with bosses, managers, masters, and dictators. Each has some kind of dominating power over those they control: bosses can fire their employees, landlords can evict their tenants, dictators can jail or execute their citizens, and gods can eternally punish and torture their creations. Many would claim that we intrinsically need elites and hierarchies. That the rich and rulers (and even domineering gods), despite the problems they create, are necessary to any well-functioning large-scale society.

The problem is that the term ‘elites’ has always been a misnomer. What we need is real elites: people who are exceptionally skilled at particular tasks and jobs, be they plumbers, cleaners, caretakers, researchers, doctors, or teachers. We want and need people to train the next generation, take care of the most vulnerable, and use human knowledge to improve our freedom and well-being, whether that be creating new antibiotics or helping forge new forms of democracy.

We have little evidence that what we traditionally call elites – the richest and most powerful throughout history – have the same proven skill as athletes, cooks, artists, and writers. We have no reason to believe that the pharaohs and kings of the past were particularly good decision-makers. On the contrary, societies that overthrew their initial dominators remembered them very differently: Native Americans recalled the rulers of Cahokia, the old Puebloan societies, and others as evil wizards and gamblers, not great leaders of decorated civilizations. Most of our history is viewed through the lens of the 1 per cent. The Native Americans and the people of Zomia glimpsed history through the eyes of wary survivors.

Today we have similarly little evidence that CEOs or investment bankers are particularly skilful at their jobs. Companies may be successful, but it is difficult to say whether they owe their success to the individual brilliance of a leader or founder (usually the preferred option), the grit and skill of their workers, or simply pure dumb luck. We are all fooled by randomness. Let a random group of gamblers haggle long enough on the stock market and eventually a billionaire will emerge. We'll attribute their success to their personal philosophy, diet, or morning routine, but in reality it was just fortuitous. It is far more difficult to disentangle individual skill from sheer luck in finance or commerce than it is with a firefighter, an athlete, or a chef.⁷

We rarely have good evidence that those in elite positions with enormous salaries are giving an elite performance. (For instance, financial advisors' stock picks appear to perform no better than random chance.⁸) In general, the higher up the ladder of a corporation or a hierarchy someone ascends, the less evidence there is of personal skill contributing to performance.

Statistician Nassim Nicholas Taleb calls this 'the inverse rule'.⁹

That inverse rule is built upon the 1 per cent view of history and the countless myths of elite superiority: from the idea that kings have a divine right to rule to the notion that the world's biggest companies are dependent on the genius of one or two individuals rather than the tireless labour and courage of underpaid workers in schools, hospitals, mines, and factories. The algorithms and skyscrapers of today, like the pyramids and great walls of yesteryear, are built on the sweat of tens of thousands of ordinary labourers. The large text models of today are not founded on the intellect of Sam Altman but on the work of thousands of academics who pioneered

machine learning, the countless workers who dig up and process the materials, and the myriad underpaid and unpaid individuals who provide the data and train the machines. Our greatest achievements are not because of the 1 per cent; they are collective endeavours. As George Eliot once wrote, ‘The growing good of the world is partly dependent on unhistoric acts ... half owing to the number who lived faithfully a hidden life, and rest in unvisited tombs.’¹⁰ It’s time to drop the 1 per cent view and start paying attention to the unmarked tombs and unknown heroes upon whose shoulders we’ve built our world.

Another myth we’ll need to overcome is that we can’t reshape society and upend dominance hierarchies. We live in a peculiar world in which many believe that we can split atoms and even recreate intelligence on silicon, yet it is impossible to do democracy at scale or resolve international arms races. This bafflingly selective failure of imagination is not because of evidence. We’ve controlled arms races before and have the technology and know-how to have large-scale, intense democracy. It is instead just another useful story of subjugation: we can hope for technological singularities but not social emancipation. The truth is that we *can* change our social structures and escape Goliath traps. And we’ll need to. Reshaping the world to avoid races (or rackets) for status, arms, and advantages is our great challenge. It won’t be easy, but it is possible. And it could determine whether this is our finest century or our final century.

Dispelling these myths will be difficult. It’s difficult to find a utopian idea among the Greeks or Romans that did not involve a hierarchical world. The thinkers of antiquity were far more capable of imagining the abolition of marriage than war, patriarchy, or class domination. Similarly, today, the feverish techno-dreams of billionaires involve reproducing human intelligence on silicon, colonizing far-flung icy planets, and harvesting stars, but tell us little of whether we have equality between the sexes. Our world and our myths are the product of thousands of years of Goliath evolution.

A central myth we’ll need to unravel is the myth of progress. History is a not a story of improvement from the stirring of civilization and agriculture through to the rise of empires to the skyscrapers and satellites of the modern world. Agriculture and sedentism took a wrecking ball to our physical and mental health. The rise and fall of empires did result in bloodshed and hardship, but it also often meant the fall and rise of our heights and

freedoms. Technology did not automatically improve the overall human condition; the benefits were usually eaten up by a select elite class. Technology can still be pivotal and beneficial: vaccines, plumbing, anaesthetics, antibiotics, central heating, the printing press, and so much more have helped to improve the standards of living of billions. Yet we need to realize that shared progress in health and well-being came only when technological developments were paired with collective struggle; that to have the benefits of technology without the catastrophic risk of climate change, nuclear weapons, killer robots, and novel entities, we'll also need to struggle for democratic control of how technology is produced and used. We'll need to fight to escape Goliath traps. And that fight will need to be waged every day.

Without that fight, our Silicon Goliath is likely to take us on a path to either an unfree world or a global collapse. A potential, final curse.

It won't be easy, but anything worth fighting for rarely is. Even if there seems to be little hope, take comfort in defiance. It is worth doing the right thing even if the odds are stacked against us.

The world is not a Greek tragedy.¹¹ We can't rely on a *deus ex machina* like AI solving our problems. Real progress and an escape from Goliath's Curse will come only from the battle to rebalance power. We can write a better story together. A story not about collapse or rise and fall. Rather, it would be a story of redemption. A story about rebuilding civilization.

Epilogue: Slaying Goliath

Escaping the Endgame – Recreating Civilization – Becoming a David

ESCAPING THE ENDGAME

Collapse may seem inescapable. This is an illusion. Most of the challenges we face are entirely solvable. If our world falls apart in the cold of a nuclear winter or the unending blaze of climate change, it won't be because there was nothing we could have done. There are many ways to reduce the risk of global collapse, to defeat Goliath, and even actions that individuals can take.

Staying Within Planetary Boundaries

Every large country now has a 'pathway to decarbonization': a detailed road map made by large groups of experts detailing how to decarbonize their society and reach net zero emissions. Most of these plans depend on a few pillars: switching to renewable energy for electricity production, electrifying transport, adopting aggressive energy efficiency measures, and reducing any remaining emissions through reforestation and direct air capture (machines which directly suck greenhouse gases out of the atmosphere and sequester them).¹ Reducing our consumption of meat and dairy would also help cut emissions, as well as enormously reducing the amount of suffering in the world.

One key way to slash emissions is to stop supporting fossil fuels. Currently governments around the world reduce the price of fossil fuels

through measures such as tax breaks. More than that, the price of fossil fuels doesn't include their costs to society, such as the roughly 5–10 million deaths caused annually by fossil-fuel air pollution.² These are costs that we all pay. All of these are 'subsidies' on the price of fossil fuels. Currently fossil fuels are subsidized by around \$7 trillion per year, or around 7 per cent of global GDP.³ Scrapping subsidies and properly pricing carbon could both dramatically cut emissions and save lives. The money saved by no longer handing out tax breaks for fossil fuels, as well as the revenue from a carbon tax, could generate a stream of funding worth around 3 per cent of global GDP. That is enough money, according to some estimates, to eradicate extreme poverty, provide universal access to sanitation and clean water, end hunger, and still have over \$1 trillion to spare.⁴

Decarbonizing the world will be a momentous effort, but it won't necessarily be an expensive one (unless you are a fossil-fuel company). The IPCC in its latest review suggests that limiting global warming to 2°C will mean global GDP losses by 2050 of 1.3–2.7 per cent, or 2.6–4.2 per cent to restrict warming to 1.5°C.⁵ Understanding the world purely through GDP misses many things, including less volatile energy prices and millions of lives saved from air-pollution-related deaths. Once you factor these in, decarbonization could well be a net economic benefit.⁶ (Although quibbling over whether saving the world will be a net economic cost is perhaps the height of stupidity.)

Climate change is just one of many planetary boundaries we face. Staying within planetary boundaries will mean giving up the elite dream of endless economic growth. There are many ways we can progress as a society while remaining within planetary boundaries. These include reducing the working week to three or four days, dropping GDP as an economic target in favour of more holistic measures like the Genuine Progress Indicator (GPI) – which factors in poverty, inequality, pollution, and resource depletion – and scaling down destructive forms of production such as fossil fuels, advertising, and factory farming.⁷ An even more sweeping action is to ensure that the effects of products on the environment and human health – whether it be cancer and lost fertility from PFAS, or lost biodiversity – are reflected in their market price. This would be a seismic shift in our economic system built on just one small common-sense

change: companies should have to pay for the economic and social damage they do. It is simply making our economies more truthful and realistic. This would not leave us in shared poverty. We can provide decent living standards for 8.5 billion people using just 30 per cent of existing global energy and resource use.⁸

Controlling AI

Controlling AI is just as doable as cutting emissions. We can – and should – ban the development of killer robots and automated general cognition (AGI). Technology bans may seem unrealistic, but there are hundreds of examples of technologies throughout history that have been developed and rescinded or intentionally never developed.⁹

There are leverage points in every part of the supply chain for building advanced algorithms. One way to both create a fairer economic system and indirectly slow AI is by paying workers properly along the entire supply chain, from the miners in the Congo, to the factory workers in Taiwan and China, through to the cleaners in San Francisco.

AI is dependent not just on cheap labour but also on stolen data. Giving citizens control of their data and properly compensating them for it would not only be the fair thing to do, but would also substantially slow the development of AI by forcing AI companies to actually pay for the resources they currently use for free. Rather than it being stolen through the digital trawl-nets of OpenAI and others or having it silently (and often unknowingly) taken as part of archaic user-service agreements that no one reads, people could instead sell their data to companies who want it through ‘data unions’ that would negotiate a fair price (rather than every individual having to bargain with Google or Apple).¹⁰

Three key ingredients for AI are computing power (in the form of microchips, also frequently called ‘compute’), energy, and coders. Compute is concentrated in a few supply chains that flow primarily through Taiwan. There are multiple ways that a coalition of countries could cooperate to restrict and slow the supply of compute. One is to put a cap on the size of a compute cluster, or how much can be used in a single model.¹¹ Pricing

carbon to reflect the true cost of electricity or setting a cap on the maximum energy usage of data centres or models would also act in a similar way to slow AI. Whistleblower protections and stronger unions could also help budding tech workers organize against companies and programs that lack scruples.^{[12](#)}

Once models are released, legislation could hold companies responsible for the harms they cause (as the vetoed SB 1047 bill in California attempted to do) and limit their use in dangerous domains, whether that be banning facial-recognition algorithms or the incorporation of AI into the military.^{[13](#)}

A World Without Nuclear Weapons

Dismantling nuclear weapons stockpiles is also eminently possible. The US was typically dismantling 1,000 or more nuclear weapons every year during the 1990s. With about 10,000 weapons in the world today, a global effort could easily see stockpiles eliminated within a decade. Studies have shown that many countries could plausibly rid themselves of nuclear weapons within ninety days.^{[14](#)}

That could be done through an international treaty, and indeed we already have one: the 2017 Treaty on the Prohibition of Nuclear Weapons (which binds countries to not use, test, possess, acquire, or develop nuclear weapons), a treaty that no nuclear power has joined but seventy-three others have.

A ban is needed because no level of nuclear weapons is safe. As a minimum initial step, countries could reduce their stockpiles to a low enough level that it would provide a ‘deterrent’ against war. That could be around 300 low-yield weapons per country.^{[15](#)} Yet even a regional war between Pakistan (170 weapons) and India (172 weapons) would still have global impacts, dropping temperatures by 4°C.^{[16](#)} That is to say nothing of the millions of lives lost and the knock-on effects. Weapons that can kill millions indiscriminately will never be safe and any society that has them can never call itself civilized. Our species can never truly be worthy of the mantle ‘sapiens’ (‘wise’ in Latin) until this scourge is lifted from the planet.

The Problem Is Power

The natural rebuttal to each of these measures, such as banning AGI, killer robots, and nuclear weapons, is the defence of political feasibility. It is seemingly a golden rule of the politics of collapse: what is effective is unfeasible and what is feasible won't be effective.

Political feasibility may seem like an immovable force, but it can change rapidly. During the spread of Covid-19 governments enacted far-reaching measures that previously seemed unthinkable. When countries do act decisively, miraculous change is possible.^{[17](#)}

For many it's somehow easier to imagine a nuclear winter or even an AGI apocalypse than to imagine a world without abhorrent weapons of mass destruction. It's easier to imagine the end of the world than the end of Goliath. That is because we are still stuck in Goliath traps: status races, arms races, and races to the bottom (where firms or states try to outcompete each other by exploiting people and environments more intensely). Decreasing nuclear stockpiles, slashing carbon emissions, and making our societies more democratic are all completely feasible, and doing so in the long term will mean escaping these traps and digging up the root causes of existential risk. We'll need to do what few societies have successfully done: kill Goliath.^{[18](#)}

RECREATING CIVILIZATION

Today we have measures that can level each source of power, reduce Goliath fuel, and tame the darker angels of our nature.

Democratizing Political Power

We need to introduce open democracy, with deliberative juries and assemblies creating national policies for governments and providing oversight of corporations. If decision-makers are randomly selected from society, we will no longer be selecting for those who crave status and power

or who rank higher in the dark triad. The constant cycling through new citizens to make decisions will also help ensure no one is in authority long enough (or holds enough power) to be corrupted. But we don't just need to rely on physical assemblies of randomly selected citizens as in ancient Athens. New technologies allow for online platforms to aggregate the opinions of enormous numbers of people when making critical decisions about how we live.¹⁹ We're often told that open democracy is not practically possible, but that has always been an argument for hoarding political power.

This is just the tip of the iceberg. There are many other options, such as banning the revolving door between regulators and industry. If we start to properly fund experiments in democracy, and reward them like we reward start-ups in tech, then who knows in what profound new ways we'll be able to harness the wisdom of crowds.²⁰

Democracy is not just for governments. If you believe that democracies work because they result in better decision-making, and that all of us deserve a say in collective endeavours, then there is no reason to accept workplace autocracy. The problem throughout history has been not states (some states have been, and many are, important providers of social support, public services, and public goods) but concentrations of unaccountable power. Corporations such as Exxon or the East India Company have been just as destructive as empires, and firms today are less accountable and less democratic than most states. Corporations could similarly be democratized by reforming them into large-scale worker cooperatives governed by workers alongside deliberative juries, and with an overriding legal goal of providing social and environmental benefits, not short-term returns to investors.

Make the World Equal Again

Levelling political power won't be enough. Even a deep democracy that uses citizens' assemblies and juries, as well as digital technology to carry out regular direct votes, will eventually be undermined if some people have billions of dollars with which to rig any system we come up with. One could argue that we have democracy and massive inequality in the world

today, and that the two are not in tension. It's a mistaken idea. What we ordinarily call democracy – systems in which a subset of people (who are aggressively propagandized by political marketers and billionaire-owned media empires) vote every four to five years for a tiny number of (usually rich) representatives who are funded and lobbied by corporations (for whom they frequently work afterwards), who then enact policies which usually better represent elite interests than popular opinion – is better described as an oligarchy with democratic furnishings. It is far more inclusive than most governments throughout history, but that is a low bar compared to what is possible. Even this threadbare democracy is being frayed by increasing wealth inequality. It is no coincidence that (with a slight time lag) democracy started backsliding after inequality began rising across the world in the 1970s. Today, the grinning face of many ailing democracies is that of tech and property billionaires. Inequality in one form of power or another will eventually spill into others: the rich buy elections, overly powerful generals launch coups, and autocrats amass fortunes. The original and most powerful inequality was found in wealth: lootable resources provided a form of power that could be easily translated into other forms of power and passed down through generations. Political democracy will die without economic equality.

Wealth becoming more unequal may be close to an iron law throughout history, but reversing it is surprisingly easy. The simplest way is through taxation. The US had an income tax of above 90 per cent on the highest earners from 1944 through to 1963 (the highest rate in the US is now 37 per cent). It didn't lead to an economic bust. It actually helped to usher in an economic golden age for the US. A highly progressive taxation system should be combined with an even higher tax on wealth such as land and stocks.

Other measures include placing a cap on wealth, \$10 million for example (a level that is well beyond what any individual needs), or capping the income of the highest earners within companies at five times that of the lowest-paid worker. While tax dodging and tax evasion may seem like the inescapable consequence of such proposals, there are a raft of ways to prevent this, including establishing a coalition of countries that set a minimum tax rate on large multinational corporations and penalizing tax havens that charge less.^{[21](#)}

There is also an abundance of ways to reduce global inequality. One option is a global minimum wage, which could be half the median wage of a country to take account of local conditions. Another is a comprehensive debt jubilee – in other words, writing off all debts – a policy that has been surprisingly common (and successful) throughout history.^{[22](#)}

Democratic Control of Information and the Military

One of the best ways to distribute the control of information is to break up existing monopolies, specifically in big tech and the media. This could mean creating new legislation as well as rigorously implementing existing anti-trust tools, such as the Sherman Act and Clayton Act in the US, which have already been used to block big mergers.

Another is to make power structures more transparent. Governments' classification and secrecy practices, particularly for military projects, are excessive, lack structure, and are often used to hide incompetence and corruption rather than important state secrets.^{[23](#)} Corporate non-disclosure agreements are little better, as we've seen with OpenAI and others cracking down on safety-concerned whistleblowers. The foundation of any democracy is a well-informed electorate, and thus transparency should be the rule, not the exception. Secrecy should be permitted only if it has been signed off by an independent arbiter or a citizens' jury. Another key step will be empowering and protecting whistleblowers, whether it be in the military or in big tech. That means providing stronger legal protection for those who have the courage to report misconduct and repealing archaic laws that allow governments and corporations to hunt, silence, and imprison whistleblowers, such as the Espionage Act (1917) in the US.

Rebalancing power will inevitably require increased protest and activism in the coming decades. Yet governments have made a concerted recent effort to curtail the right to protest, such as through the UK's Police, Crime, Sentencing, and Courts Act 2022, or the 323 different bills that have been introduced in the US since 2017 that restrict the right to protest, and even *encourage* violence against peaceful protesters.^{[24](#)} Finding ways to roll these back and unshackle concerned citizens will be critical to levelling power and holding the Agents of Doom to account.

Democratically Controlling Technology and Production

We can also slow – or even halt – the production of dangerous new technologies and weapons by placing them back under democratic control. Monopolizable weapons and technologies under the control of the few have been the norm for just the last 1 per cent of human history. This is not a call to halt technological progress; it is a call to democratically control technology to ensure it *is* progress. AI can be used to better spot cancers, vaccines have prevented the suffering of billions, space exploration ignites the enthusiasm and imagination of millions, and we'll need new technologies to capture carbon from the atmosphere and stabilize melting ice sheets. It is a question of better directing technological development and production.

Researchers have already suggested ways in which companies and countries could use experts to map out the most likely development pathways of different technologies (such as AGI) and then use deliberative juries or assemblies to decide whether to pursue them and what regulation to put in place. This would help us democratically and pre-emptively regulate technology, as opposed to the current losing strategy of rushing haphazardly to make legislation *after* a new technology is released.^{[25](#)}

We could use a similar democratic process to navigate difficult trade-offs on the path ahead, such as whether we should use emergency options (these might include negative emissions, short-term geoengineering, or interventions to stabilize ice sheets) to avoid the worst effects of climate change: examine the potential worst-case effects of both the warming we are facing and the potential emergency solutions, and then let a group of expert-informed citizens decide.^{[26](#)}

This is not a complete blueprint for solving our problems. Trying to foist a vision of utopia on the world has never ended well. Instead, we should democratize the world, level power, reduce Goliath fuel, reduce existential risk, and see what world blossoms from this new civilization.

BECOMING A DAVID

In the biblical tale, Goliath is slain by a single slingshot-toting future king by the name of David. Defeating Goliaths, in reality, is not done alone. Instead, it requires actions that leverage the people and communities around us.

Don't Be a Dick

First, I propose a simple pledge to not be a dick. This is a pledge to not work for, invest in, or support any firm, institute, or individual that significantly contributes to global catastrophic risk. Don't work for an Agent of Doom, whether it is an AGI lab, a fossil-fuel company, or an arms manufacturer. The idea that you are going to change an Agent of Doom *from the inside* is nonsense, and the excuse of 'if I don't do it someone else will' is the sort of bullshit used by guards at concentration camps. Hold yourself to a higher standard. Stopping arms races, status races, and races to the bottom begins when each of us refuses to get trapped in them.

More than that, it is a pledge to not become a Goliath yourself. All of us rank somewhere on the dark triad. All of us are capable of being corrupted by power. All of us crave status to some degree. Don't let the darker angels of your nature win. If you find yourself atop a hierarchy, democratize it and distribute your power among your peers. Every year, try to make sure that your actions leave the world a little freer than it otherwise would have been.

This same pledge can be adopted not just by individuals but also by institutions. While Google did a poor job of fulfilling its credo of 'don't be evil', all is not lost. Corporations can and should reorganize themselves to be more internally democratic and minimize the pay discrepancies between the CEO and interns (who should be paid). Firms can also use their money and portfolios to fight against doomsday. Large institutional investors like pension funds and sovereign wealth funds can and should pledge to use their investments to minimize global catastrophic risk (not just moving their investments away from fossil-fuel companies, big tech, and arms producers, but moving them *into* sectors that help make the world more resilient).^{[27](#)}

No institution should be financing our global descent, and no one should support one which does.

Be a Democrat

Second, practise democracy. Democracy is not just a form of government; it is a culture and a way of life. One that all of us need to recapture. Be a democrat: don't just vote every few years, but join a union, join an activist group (while there is no group fighting against all the sources of catastrophic risk, there hopefully will be one soon), advocate for workplace democracy, and discuss political matters in a productive way – whether it be nuclear weapons or mass surveillance or climate collapse – with friends and family members.

Vote Against the Apocalypse

Third is voting. You are first and foremost a citizen, not a consumer. While democracy is more than voting, voting is a fundamental part of democratic practice in the modern world. If you know about climate change, a nuclear winter, and the other real risks to the survival of our species but base your vote on who promises to cut your taxes, then you are culpable for our current path towards self-destruction. It is a travesty that no election to date has been decided by the candidates' positions on nuclear weapons and climate change. There is no reason to contribute to that trend in the future. Choose the party and candidate that offer genuine, practical ways to empower citizens through deliberative juries and assemblies, not those who use empty rhetoric about 'draining the swamp', 'protecting the nation', or providing 'change we can believe in'. Choose those who have concrete plans to adopt a high carbon price, to price environmental impacts in general, to ban fossil-fuel subsidies, to join the 2017 Treaty on the Prohibition of Nuclear Weapons, and to place a ban on the pursuit of AGI and killer robots. If there are none, then contact your local representative and lobby for it, or join a grassroots movement that is fighting for real change.

Voting may seem inconsequential, but it can change things extremely quickly. If everyone in Australia (or the US, or the UK) began switching off their lights prudently and chose the lowest-carbon travel options, then it would barely make a dent in national emissions. If they changed their vote to those offering the strongest decarbonization plans, then suddenly emissions could be on track to be eliminated within decades. You of course only have one vote, but talking to family and friends can multiply your impact. Debate and discussion are the lifeblood of democracy.

Don't Be Dominated

Fourth is to re-cultivate the counter-dominance intuitions that guided us through the Palaeolithic. Oppose domination in all your relationships, whether they be personal, family, or workplace. Whenever you come across a hierarchy, whether it is based on wealth, gender, or age, ask whether it is legitimate and whether it justifies domination. If it doesn't, then try to overturn it.

One of the first and most pernicious stories justifying subjugation was that of the saviour leader. High priests, god-kings, and budding rulers claimed to predict the future by reading bones, ensure the harvest through ritual sacrifice, and provide protection from unseen spirits and neighbouring groups. Now, we have billionaires and oligarchs offering salvation by building walls between us, or inventing a superintelligent AI god to solve our problems, or building colonies on Mars as an escape plan. These have always been – and still are – glorified parlour tricks to keep people distracted while the bosses pick their pockets.

It is time to say 'enough'. It is time to realize the bright and terrifying truth: no gods, kings, heroes, or masters are going to save us. Slaying Goliath and avoiding evolutionary suicide is, like all great achievements, going to be a collective action. It is on us. It is going to take the thankless, unglamorous work of having difficult conversations with your friends, giving up power, trusting your fellow citizens rather than a strongman leader when you're scared, going into the streets even when you are comfortable at home, and taking an ethical stand even if it might cost you your job.

Each of those little acts is another stone flung at Goliath, and each crack in its skull is a doorway to freedom. The stand will be worth it. A world free of nuclear weapons, carbon emissions, killer robots, and the threat of collapse, one full of genuine democracy and economic equality, is no utopia. It is within reach.

Revolutions begin where myths end; Goliath ends when civilization begins.

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Glossary

Authoritarian impulse

The psychological tendency for individuals to be more accepting and supporting of authoritarian structures and leaders when faced with a threat to their safety and security.

Caged land

Territory with few exit options. This is a form of Goliath fuel that tends to lead towards the establishment of dominance hierarchies. Caged land can include geography (such as Egypt, which is cordoned off by desert and sea); a population tied to certain resources (such as a stream of salmon or a field of wheat); or technological changes (such as the construction of walls). All of these can – and do – interact and change over time.

Chaos

A social system in which everyone is trying relentlessly and violently to dominate, but no one ever succeeds.

Civilization

A social structure characterized mainly by non-hierarchical, democratic cooperation.

Collapse

The relatively rapid contraction and fragmentation of different power systems. This results in large-scale losses in hierarchy, population density, and energy capture.

(Cumulative) cultural evolution

Knowledge, practices, and technologies that are selected for, passed down across generations, and accumulated over time.

Dark triad

Three antisocial personality traits: psychopathy (callousness, and a lack of empathy and remorse), narcissism (an inflated sense of entitlement and self-importance), and Machiavellianism (manipulating others for personal gain).

Darker angels of our nature

A combination of psychological traits that have led to Goliaths: status competition through domination; the dark triad (psychopathy, narcissism, and Machiavellianism); embracing strong authority structures when threatened (the authoritarian impulse); and becoming corrupted by power.

Death-Star Syndrome

Large, robust systems that are surprisingly fragile if hit in the right spot with a sufficient amount of force.

Democracy

A spectrum of (more or less) equal and inclusive political practices.

Democracies can be hierarchical (ancient Athens for adult males) or non-hierarchical (egalitarian foragers such as the Hadza).

Diminishing returns on extraction

As states and elites try to extract more from their citizens, resources, and neighbours, this causes increased inequality, corruption, oligarchy (poor decision-making), overstretch, popular immiseration, elite factionalism, and a declining energy ratio. These different effects hollow out the state until it falls apart or is broken apart by different shocks. The second iteration of Goliath's Curse.

Dominance hierarchy

A social-ranking system in which one group or individual is placed above others owing to their ability to impose penalties, including violence.

Empire

Essentially synonymous with a state. For ease, we should think of these as particularly big states, or states that have conquered other states.

Endgame

A situation in which there is a significant risk of global catastrophic or existential risk. In short, a time when Goliath is likely to self-terminate permanently or be fundamentally reformed.

Evolutionary backsliding

The movement from egalitarian fluid civilizations into dominance hierarchies (Goliaths) which more closely resemble the social structure of gorillas and chimpanzees.

Evolutionary suicide

Short-term evolutionary pressures lock the population into a path of self-destruction. Short-term adaptation leads to long-term extinction.

Existential risk

The likelihood of a deep, lasting, global collapse or human extinction.

Extractive institutions

Great concentrations of power leading to institutions (whether it be an authoritarian government, an oppressive theocracy, or a bloated bureaucracy controlled by the rich) in which elites extract more resources, labour, and energy through unequal exchange.

Global catastrophic risk

The likelihood of an unprecedented loss of more than 10 per cent of global population and disruption of the critical systems on which the world depends.

Global decimation risk

The likelihood of an event or process that leads to a loss of around 10 per cent of global population within a year.

Global Goliath

A collection of interconnected hierarchies across the entire world in which some individuals dominate others to control energy and labour. In our modern world this includes hierarchies such as the state, patriarchy, rich and poor (based on a capitalist global economy), and myths about the need for hierarchy.

Goliath

A collection of interconnected hierarchies in which some individuals dominate others to control energy and labour.

Goliath evolution

The evolution of dominance hierarchies. Violence selects for Goliaths that are better at warfare, and usually larger and more centralized.

Goliath fuel

Ecological and technological conditions that allow for the establishment of a Goliath. These include lootable resources, monopolizable weapons, and caged land.

Goliath traps

The destructive competitive dynamics that dominance hierarchies are liable to fall into. The pursuit of short-term gain by status seekers leads to everyone losing out in the long term. There are three main Goliath traps: races for status, arms races, and races to the bottom.

Lootable resources

Resources that can easily be seen, stolen, and stored. A form of Goliath fuel.

Monopolizable weapons

Weapons or tactics that can be concentrated (or even monopolized) and give one group an advantage over their competitors.

Primal Curse

The first iteration of Goliath's Curse. This involves early Goliaths falling apart owing to weak institutions meeting strong counter-dominance intuitions, as well as rising inequality (and different shocks).

Rungless ladder

A metaphor for how more powerful Goliaths tend to have deeper and more widespread collapses. For the modern world this heightened severity of collapse is due to more powerful weapons, more deteriorated environments, and a reliance on states and complex technological systems.

Silicon Goliath

A Goliath that covers the entire world where the populace is controlled through mass surveillance and dominance hierarchies, which are in turn overwhelmingly dependent on advanced algorithmic systems. The main forms of Goliath fuel for this form of Goliath are data (lootable resource),

killer robots and nuclear weapons (monopolizable weapons), and mass surveillance systems (caged land).

Somalia-Denmark rule

The more predatory a state, the less harmful (or even the more beneficial) its collapse; and the more inclusive and benevolent a state, the more costly its collapse.

State(s)

A set of centralized institutions that impose rules on and extract resources from a population in a territory. States can also provide public goods, whether that be roads, bridges, or (in later cases) education and healthcare.

Stories of subjugation

Myths, ideologies, and religions that justify the domination of the many by the few.

Systemic risk

The potential for individual disruptions or failures to cascade into a system-wide failure.

Tree of Doom

The causal chain from root causes through Goliaths and Goliath traps to existential risk.

Unequal exchange

The unfair exchange of value between individuals or groups owing to power imbalances.

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Notes

INTRODUCTION: A PEOPLE'S HISTORY OF COLLAPSE

1. The forms of political power are an extension and refinement of Michael Mann's sources of social power. See Mann, *The Sources of Social Power I: A History of Power from the Beginning to AD 1760*, 2nd edn (Cambridge University Press, 2012). I've added in 'demographic power' and expanded 'ideological power' to 'information power', and 'military power' into 'violent power'. The overall framework is elaborated on in Luke Kemp et al., 'Societal Collapse as a Powershift' (forthcoming, 2025). A note about the endnotes in this book. In general, I'll be aiming to both provide scholarly references where necessary and supply broad overviews of the archaeological case studies through more accessible, popular (yet reliable) books and pieces. I've often opted for the latter since many readers won't have the university affiliations or money to buy access to expensive journal articles or academic books (which is a shame). I've aimed to provide page numbers for references except when either these are taken from an e-reader (in which case I've provided a chapter), or it is a key point made repeatedly throughout the source. [↩](#)
2. My definition of collapse is a refinement of the many that have come before. One of the first suggested was by the famed Cambridge archaeologist Colin Renfrew in his 'Systems Collapse as Social Transformation: Catastrophe and Anastrophe in Early State Societies', in *Transformations* (Academic Press, 1979), 482–4, <https://doi.org/10.1016/B978-0-12-586050-5.50035-X>. Renfrew focuses on four general features of collapse: the collapse of the centralized administrative state, the loss of an elite, the collapse of a centralized economy, and a population decline and settlement shift. In other words, he's referring to the large-scale fragmentation and

contraction of political, economic, information, and demographic power networks. My definition is also an evolution of Tainter's canonical definition of collapse as a rapid and enduring reduction in a pre-established level of 'complexity' (a term later reused by Jared Diamond). See Joseph A. Tainter, *The Collapse of Complex Societies* (Cambridge University Press, 1990); Jared Diamond, *Collapse: How Societies Choose to Succeed or Fail* (Penguin, 2011). As we'll see later, the term 'complexity' refers mainly to hierarchy, centralization, and inequality. 'Complexity' is a misnomer as both complexity scientists and common sense usually see it as the diversity and interconnectivity of parts. Hierarchy and centralization often act to reduce diversity and reduce the interconnectivity of certain groups, such as between peasants (or, as we see later, Rome preventing city-states from communicating). Similarly, administration is an art of simplification. The point is most eloquently expressed in James C. Scott, *Seeing Like a State: How Certain Schemes to Improve the Human Condition Have Failed* (Yale University Press, 2008); Norman Yoffee, *The Evolution of Simplicity* (University of Chicago Press, 2001). States and hierarchies simplify as much as they increase complexity. Hence, their collapse can often increase certain kinds of 'complexity'. The diversity of political forms, economic relations, languages, and cultural practices can often increase after a collapse. While long-distance trade and communication are often reduced, connectivity at a local and regional level can also increase during the breakdown of a state. The only genuine commonalities across all case studies are losses in hierarchy, energy capture, and population density. What Tainter and others are really talking about here are centralized power structures, hence the emphasis. The other aspects of more diverse and stratified roles and goods are by and large a by-product of the few trying to organize and control the labour and resources of the many. Even the most basic definitions of societal complexity frame it in this way. For instance, Brian Hayden, 'Social Complexity', *The Oxford Handbook of the Archaeology and Anthropology of Hunter-Gatherers*, ed. Vicki Cummings (Oxford University Press, 2014), 644, <https://doi.org/10.1093/oxfordhb/9780199551224.013.047>. There have been numerous other definitions. Scholars in the field of existential risk have defined collapse as the loss of the ability to create industrial

and post-industrial technologies, or the complete loss of agriculture and the other clichéd trappings of civilization, such as writing and the rule of law. See Will MacAskill, *What We Owe the Future* (Basic Books, 2022); Toby Ord, *The Precipice: Existential Risk and the Future of Humanity* (Hachette, 2020). The fathers of the popular French collapsology movement, Pablo Servigne and Raphaël Stevens, see it as a loss of the services provided by the state (*How Everything Can Collapse*, Polity, 2020). In these views collapse is the sudden theft of any virtues that the modern, globalized world offers. They are more reminiscent of post-apocalyptic films like *The Road* and *Mad Max* and not applicable to pre-modern cases, or even precise enough to be useful. Perhaps the best recent attempt to define collapse comes from the ecologists Garry Peterson and Graeme Cumming in 2017. They frame collapse as a quick, enduring loss of a system's identity and a large amount of 'socio-ecological capital' ('Unifying Research on Social–Ecological Resilience and Collapse', *Trends in Ecology & Evolution* 32, no. 9 (2017): 698–9). The problem is that identity is difficult to define, and many cases of collapse see the recuperation and expansion of ecosystems. The vast reforestation of central Mexico and the Andes after the collapse of the Aztec and Incan empires are just two examples. The definition presented here tries to cut to the heart of collapse, focus on only the most common consequences, and avoid arbitrary and value-laden terms like 'complexity'. It still has some issues, most notably that 'rapid' and 'enduring' will always be subjective terms. It's not a fatal flaw as even the best definitions of complex phenomena rely on some expert judgement. As noted later, we should see collapse as a spectrum, ranging from the shallow (less severe, rapid, and enduring) to the deep (more severe, rapid, and enduring). Thank you for reading this short essay of an endnote.↵

3. Timothy R. Pauketat, ed., *Cahokia: Ancient America's Great City on the Mississippi* (Penguin, 2010), chap. 1.↵
4. The image is from Flickr and the original artist has the excellent name of 'Thank You (25 Millions) views'. See www.flickr.com/photos/prayitnophotography/32434956237/. The image is published under the Creative Commons Licence. For further information on the Creative Commons Licence, see <https://creativecommons.org/licenses/by/2.0/>.↵

5. I'll be using the abbreviations of BCE (Before the Common Era) and CE (Common Era), derived from the old Gregorian calendar of BC (before the estimated birth of Christ) and AD (after the birth of Christ, or Anno Domini, medieval Latin for 'in the year of Our Lord').↵
6. Pauketat, *Cahokia*, chap. 6. The burials occurred over the space of a century, including numerous elite burials. This was likely one lineage of rulers accompanied by retinues of sacrificial victims.↵
7. On Cahokia's collapse, see Larry V. Benson, Timothy R. Pauketat, and Edward R. Cook, 'Cahokia's Boom and Bust in the Context of Climate Change', *American Antiquity* 74, no. 3 (2009): 467–83, <https://doi.org/10.1017/S000273160004871X>; Joseph A. Tainter, 'Cahokia: Urbanization, Metabolism, and Collapse', *Frontiers in Sustainable Cities* 1 (2019): 11–14, <https://doi.org/10.3389/frsc.2019.00006>. Interestingly, and surprisingly, the initial emigration occurred while Cahokia was still at its zenith in terms of construction and power.↵
8. This definition covers the most commonly accepted features of statehood. See Walter Scheidel, 'Studying the State', in *The Oxford Handbook of the State in the Ancient Near East and Mediterranean*, ed. Peter Fibiger Bang and Walter Scheidel (Oxford University Press, 2013), 5–9, <https://doi.org/10.1093/oxfordhb/9780195188318.013.0002>. I have not used the popular definition from Weber of the state as a legitimate monopoly on lethal force. That's because it was always a bit of a silly and circular definition. Most pre-modern states never had a monopoly on the means of violence and who decides whether it is legitimate other than the state itself. Relying on government records and writing is a little mistaken given that rulers had a vested interest in making themselves seem more powerful and their control over violence legitimate.↵
9. Calculations were made using the Mortality of States (MOROS) database.↵
10. Eric Cline, *1177 B.C.: The Year Civilization Collapsed* (Princeton University Press, 2015; 2nd revised and updated edn, 2021); Eric Cline, "Mind the Gap": The 1177 BCE Late Bronze Age Collapse and Some Preliminary Thoughts on Its Immediate Aftermath', in *How*

Worlds Collapse: What History, Systems, and Complexity Can Teach Us About Our Modern World and Fragile Future, ed. Miguel A. Centeno et al. (Routledge, 2023), 98–107, provides a useful overview of the Late Bronze Age collapse. For the estimates of ‘peak empire’, see Walter Scheidel, ‘The End of “Peak Empire”: The Collapse of the Roman, Han and Jin Empires’, in the same volume, and ‘The Scale of Empire: Territory, Population, Distribution’, in *The Oxford World History of Empire*, vol. I: *The Imperial Experience*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), 91–110,
<https://doi.org/10.1093/oso/9780199772360.003.0003>.↵

11. Patrick Wyman, ‘How Do You Know If You’re Living Through the Death of an Empire?’, *Mother Jones*, 2020,
www.motherjones.com/media/2020/03/how-do-you-know-if-youre-living-through-the-death-of-an-empire/.↵
12. Even the Vacant Quarter was eventually reoccupied by Native Americans. See A. J. White et al., ‘After Cahokia: Indigenous Repopulation and Depopulation of the Horseshoe Lake Watershed AD 1400–1900’, *American Antiquity* 85, no. 2 (2020): 263–78,
<https://doi.org/10.1017/aaq.2019.103>.↵
13. For an accessible timeline of nuclear near misses, see Future of Life Institute, ‘Accidental Nuclear War: A Timeline of Close Calls’, Future of Life Institute, 23 February 2016,
<https://futureoflife.org/resource/nuclear-close-calls-a-timeline/>. The current carbon pulse is an order of magnitude faster than the release during the Great Permian Dying, although it remains to be seen whether we will reach the 6–8°C of temperature rise that it saw. For an easy overview, see Mark Lynas, *Our Final Warning: Six Degrees of Climate Emergency* (HarperCollins, 2020).↵
14. Ord, *The Precipice*; Michael Lawrence et al., ‘Global Polycrisis: The Causal Mechanisms of Crisis Entanglement’, *Global Sustainability* 7 (2024): e6, <https://doi.org/10.1017/sus.2024.1>.↵
15. Tyler Ausin Harper, ‘The 100-Year Extinction Panic is Back, Right on Schedule’, *New York Times*, 26 January 2024,
www.nytimes.com/2024/01/26/opinion/polycrisis-doom-extinction-humanity.html.↵

16. Luke Kemp, 'Agents of Doom: Who Is Creating the Apocalypse and Why', *BBC Future*, 2021, www.bbc.com/future/article/20211014-agents-of-doom-who-is-hastening-the-apocalypse-and-why.↵
17. For prominent examples of this hazard-centric framing, see Ord, *The Precipice*; MacAskill, *What We Owe the Future*; Martin J. Rees, *Our Final Century: Will Civilisation Survive the Twenty-First Century?* (Arrow, 2004).↵
18. Peter T. Leeson, 'Better Off Stateless: Somalia Before and After Government Collapse', *Journal of Comparative Economics* 35, no. 4 (2007): 699–701, <https://doi.org/10.1016/j.jce.2007.10.001>; Benjamin Powell, Ryan Ford, and Alex Nowrasteh, 'Somalia After State Collapse: Chaos or Improvement?', *Journal of Economic Behavior & Organization* 67, nos 3–4 (2008): 657–70, <https://doi.org/10.1016/j.jebo.2008.04.008>.↵
19. There is also the case of the 1975 Lebanese civil war. It was a tragic event which led to the deaths of over 100,000 and the displacement of at least a million people. Yet, the welfare of the survivors appears to have improved post-collapse relative to their pre-civil-war condition. Ersun N. Kurtuluş, 'Exploring the Paradoxical Consequences of State Collapse: The Cases of Somalia 1991–2006 and Lebanon 1975–82', *Third World Quarterly* 33, no. 7 (2012): 1285–1303, <https://doi.org/10.1080/01436597.2012.691831>. Whether this is due to remittances from abroad, improved governance, or other factors, is debated. It is also not to say that this was a net benefit. The civil war also led to the emergence of armed groups which persist through to today, such as Hezbollah.↵
20. Walter Scheidel, *Escape from Rome: The Failure of Empire and the Road to Prosperity* (Princeton University Press, 2019), 89.↵
21. See footnote 1 of Walter Scheidel, 'Slavery in the Roman Economy', Princeton/Stanford Working Papers in Classics (Stanford University, 2010): 2, <https://ssrn.com/abstract=1663556>.↵
22. These massive relocations appear to have been done to enhance political control: rebellious groups were moved to more fortified areas while loyal ones were shifted to more hostile, peripheral terrains. In Quechua the rulers were usually referred to as 'Inka', a term which is

still used today. However, I've opted to use the more commonly known 'Inca' spelling to avoid confusion.[↵](#)

23. Owen Lattimore, 'The Frontier in History', in *Studies in Frontier History: Collected Papers 1929–58* (Oxford University Press, 1962), 476–81.[↵](#)
24. These kinds of periodizations, which celebrate empire and treat times of decentralization as temporary intervals of chaos or nothingness, are common throughout historiography. Even attempts by archaeologists to push against common misconceptions and the language of imperial periodization have failed and backfired. In 1978, the British Egyptologist Kenneth Kitchen railed against using the term 'Third Intermediate Period'. He stated that the period was not chaotic and that we should use an alternative: the 'Post-Imperial Epoch'. Unfortunately, he chose to call his book on the subject *The Third Intermediate Period in Egypt (1100–650 BCE)*. His poor choice of title may have helped with sales, but simply added to the popularity of imperial periodization.[↵](#)
25. Ellen Morris, "'Lo, Nobles Lament, the Poor Rejoice": State Formation in the Wake of Social Flux', in *After Collapse: The Regeneration of Complex Societies*, ed. Glenn M. Schwartz and John Jackson Nichols (University of Arizona Press, 2006).[↵](#)
26. Ibid.[↵](#)
27. In academia, such groups, which are excluded or marginalized from power structures, are usually referred to as 'the subaltern'. [↵](#)
28. The 90–95 per cent estimate comes from Rebecca Storey and Glenn Reed Storey, *Rome and the Classic Maya: Comparing the Slow Collapse of Civilizations* (Routledge, 2017).[↵](#)
29. On the difficulties of distinguishing population dispersal from depopulation, see Storey and Storey, *Rome and the Classic Maya*. For the case of the Bronze Age collapse, see Luke Kemp and Eric Cline, 'Systemic Risk and Resilience: The Bronze Age Collapse and Recovery', in *Perspectives on Public Policy in Societal-Environmental Crises*, ed. Adam Izdebski, John Haldon, and Piotr Filipkowski (Springer, 2022).[↵](#)

30. Most early languages, including Classical Greek and Latin, were a *scriptio continua*, where there is no punctuation or spaces between words or sentences, making it particularly difficult to parse without extensive training. The slow abandonment of the *scriptio continua* was one factor that favoured education and literacy becoming more widespread in the pre-modern era. Some modern languages such as Thai maintain the *scriptio continua*. ↵
31. Documentary evidence has several other biases. It favours the temperate west. Paper in tropical climates tends to become dust in one or two centuries, unless it is stored in protective conditions. And western countries tend to be excavated by archaeologists more frequently. See Peter Crooks and Timothy H. Parsons, 'Empires, Bureaucracy and the Paradox of Power', in *Empires and Bureaucracy in World History*, ed. Peter Crooks and Timothy Parsons (Cambridge University Press, 2016), 3–28, <https://doi.org/10.1017/CBO9781316694312.002>. ↵
32. This is a necessarily crude simplification. However, the general rule of elites being those with disproportionate power in a society holds true across different definitions. The problem is that what forms of power were most ascendant varied across societies. Some valued warriors, some preferred traders, and others prioritized wisdom. For an entertaining discussion and overview, see David Priestland, *Merchant, Soldier, Sage: A New History of Power* (Penguin, 2013). ↵
33. In 2023 global energy consumption was 183,230 terawatt hours (TWh = 1 trillion watts per hour). For perspective, the average LED lightbulb uses 10 watts per hour. Energy consumption has risen by more than an order of magnitude (ten-fold) since 1800, when it was just 5,653 TWh. For energy consumption, see Hannah Ritchie, 'Energy Production and Consumption', Our World in Data, 2024, <https://ourworldindata.org/energy-production-consumption>. ↵
34. Estimates of long-term per capita energy capture come from Ian Morris, *The Measure of Civilization: How Social Development Decides the Fate of Nations* (Princeton University Press, 2013). ↵
35. The FAO provides the estimate of two-thirds of global food energy intake coming from three staples: FAO, 'Staple Foods: What Do People Eat?' (Food and Agricultural Organization), accessed 30 May

- 2023, www.fao.org/3/u8480e/u8480e07.htm. The contribution of wheat to global caloric intake was calculated by Bekele Shiferaw et al., ‘Crops That Feed the World 10. Past Successes and Future Challenges to the Role Played by Wheat in Global Food Security’, *Food Security* 5, no. 3 (2013): 291–317, <https://doi.org/10.1007/s12571-013-0263-y>.↵
36. Urbanization figures come from Hannah Ritchie and Max Roser, ‘Urbanization’, *Our World in Data*, February 2024, <https://ourworldindata.org/urbanization>.↵
37. Unsurprisingly, achieving this ambition in under 500 pages will mean a sacrifice of depth for breadth.↵

CHAPTER 1: HOBBS’S DELUSION

1. Thomas Hobbes, *Leviathan*, ed. Christopher Brooke (Penguin, 2017).↵
2. Frans De Waal, ‘Part I: Morally Evolved: Primate Social Instincts, Human Morality, and the Rise and Fall of “Veneer Theory”’, in *Primates and Philosophers: How Morality Evolved*, ed. Stephen Macedo and Josiah Ober (Princeton University Press, 2016), 1–58.↵
3. For evidence on the Mahabharata and Dīgha Nikāya, see Upinder Singh, *Political Violence in Ancient India* (Harvard University Press, 2017), 34, 60. A broader overview is provided in Monica L. Smith, ‘The Fundamentals of the State’, *Annual Review of Anthropology* 51, no. 1 (2022): 496, <https://doi.org/10.1146/annurev-anthro-041320-013018>.↵
4. Charles Edward Merriam, ‘Hobbes’s Doctrine of the State of Nature’, *Proceedings of the American Political Science Association* 3 (1906): 152–3, <https://doi.org/10.2307/3038543>.↵
5. Lee Clarke, ‘Panic: Myth or Reality?’, *Contexts* 1, no. 3 (2002): 21–6, <https://doi.org/10.1525/ctx.2002.1.3.21>.↵
6. John Drury, David Novelli, and Clifford Stott, ‘Representing Crowd Behaviour in Emergency Planning Guidance: “Mass Panic” or Collective Resilience?’, *Resilience* 1, no. 1 (2013): 18–37, <https://doi.org/10.1080/21693293.2013.765740>. This survey of 448

UK citizens, including 120 sport events stewards, 115 police officers, 89 members of the public, 78 students, and 46 civilian safety professionals, found that most had fears around mass panic and civil disorder.[↵](#)

7. Technically, the Palaeolithic included multiple glacial (ice age) and interglacial (significantly warmer) periods. However, these interglacial periods were brief during the Palaeolithic and for the ease of the lay reader I'll be sticking to the casual term of the 'ice age'.[↵](#)
8. Eleanor M. L. Scerri, 'The North African Middle Stone Age and Its Place in Recent Human Evolution', *Evolutionary Anthropology: Issues, News, and Reviews* 26, no. 3 (2017): 119–35, <https://doi.org/10.1002/evan.21527>; Doron Shultziner et al., 'The Causes and Scope of Political Egalitarianism During the Last Glacial: A Multi-Disciplinary Perspective', *Biology & Philosophy* 25, no. 3 (2010): 319–46, <https://doi.org/10.1007/s10539-010-9196-4>.[↵](#)
9. The classic texts here are: Peter J. Richerson, Robert Boyd, and Robert L. Bettinger, 'Was Agriculture Impossible During the Pleistocene But Mandatory During the Holocene? A Climate Change Hypothesis', *American Antiquity* 66, no. 3 (2001): 387–411, <https://doi.org/10.2307/2694241>; Robert Bettinger, Peter Richerson, and Robert Boyd, 'Constraints on the Development of Agriculture', *Current Anthropology* 50, no. 5 (2009): 627–31, <https://doi.org/10.1086/605359>.[↵](#)
10. Smithsonian Institute, 'Pleistocene Volcano List', Smithsonian Institute, 2024, https://volcano.si.edu/volcanolist_pleistocene.cfm.[↵](#)
11. Shultziner et al., 'The Causes and Scope of Political Egalitarianism during the Last Glacial'.[↵](#)
12. Kim R. Hill et al., 'Co-Residence Patterns in Hunter-Gatherer Societies Show Unique Human Social Structure', *Science* 331, no. 6022 (2011): 1286–9, <https://doi.org/10.1126/science.1199071>.[↵](#)
13. While often called 'immediate return' hunter-gatherers, this is something of a misnomer. They planned game hunts months in advance and spent lifetimes cultivating relationships that they could rely on during times of need.[↵](#)

14. A. B. Migliano et al., 'Characterization of Hunter-Gatherer Networks and Implications for Cumulative Culture', *Nature Human Behaviour* 1 (2017): 0043, <https://doi.org/10.1038/s41562-016-0043>. Many groups begin to refer to friends who become close enough as family, such as brother, sister, and cousin. It's a tradition that many cultures continue today. Anthropologists call it 'fictive kinship'.↵
15. R. I. M. Dunbar, 'Do Online Social Media Cut Through the Constraints that Limit the Size of Offline Social Networks?', *Royal Society Open Science* 3, no. 1 (2016): 1, <https://doi.org/10.1098/rsos.150292>; R. Bretherton and R. I. M. Dunbar, 'Dunbar's Number Goes to Church: The Social Brain Hypothesis as a Third Strand in the Study of Church Growth', *Archive for the Psychology of Religion* 42, no. 1 (2020): 63, <https://doi.org/10.1177/0084672420906215>.↵
16. Numerous studies of group size, as well as neocortex size relative to group size, have found a far wider variety of numbers. See Patrik Lindenfors, Andreas Wartel, and Johan Lind, "'Dunbar's Number' Deconstructed", *Biology Letters* 17, no. 5 (2021): 20210158, <https://doi.org/10.1098/rsbl.2021.0158>.↵
17. Douglas W. Bird et al., 'Variability in the Organization and Size of Hunter-Gatherer Groups: Foragers Do Not Live in Small-Scale Societies', *Journal of Human Evolution* 131 (2019): 96–108, <https://doi.org/10.1016/j.jhevol.2019.03.005>.↵
18. The ostrich egg social network is discussed in Jennifer M. Miller and Yiming V. Wang, 'Ostrich Eggshell Beads Reveal 50,000-Year-Old Social Network in Africa', *Nature* 601, no. 7892 (2022): 234–9, <https://doi.org/10.1038/s41586-021-04227-2>. The obsidian network is covered in Nick Blegen, 'The Earliest Long-Distance Obsidian Transport: Evidence from the ~200ka Middle Stone Age Sibilo School Road Site, Baringo, Kenya', *Journal of Human Evolution* 103 (2017): 1–19, <https://doi.org/10.1016/j.jhevol.2016.11.002>.↵
19. These are hunter-gatherer groups who now reside in central Africa, but previously had a larger range. See Cecilia Padilla-Iglesias et al., 'Cultural Evolution of Central African Hunter-Gatherers Reflects a Deep History of Interconnectivity', preprint, 26 October 2022, <https://doi.org/10.21203/rs.3.rs-2205369/v1>; Cecilia Padilla-Iglesias et

- al., 'Deep History of Cultural and Linguistic Evolution among Central African Hunter-Gatherers', *Nature Human Behaviour* 8 (2024): 1263–75, <https://doi.org/10.1038/s41562-024-01891-y>.↵
20. Cecilia Padilla-Iglesias, 'Did Humanity Really All Arise in One Place?', *Sapiens*, 2 January 2023.↵
 21. See Cecilia Padilla-Iglesias, 'Societies of Perpetual Movement', *Aeon*, 5 March 2024, <https://aeon.co/essays/the-hunter-gatherers-of-the-21st-century-who-live-on-the-move>.↵
 22. The map of the Clovis Culture is adapted from Michael R. Waters, Thomas W. Stafford, and David L. Carlson, 'The Age of Clovis – 13,050 to 12,750 Cal Yr B.P.', *Science Advances* 6, no. 43 (2020): fig. 1, <https://doi.org/10.1126/sciadv.aaz0455>. The image of the Aurignacian is adapted from Carolyn C. Szmidt, Laurent Brou, and Luc Jaccotey, 'Direct Radiocarbon (AMS) Dating of Split-Based Points from the (Proto)Aurignacian of Trou de La Mère Clochette, Northeastern France. Implications for the Characterization of the Aurignacian and the Timing of Technical Innovations in Europe', *Journal of Archaeological Science* 37, no. 12 (2010): fig. 1, <https://doi.org/10.1016/j.jas.2010.08.001>.↵
 23. J. Colette Berbesque et al., 'Hunter-Gatherers Have Less Famine Than Agriculturalists', *Biology Letters* 10, no. 1 (2014): 20130853, <https://doi.org/10.1098/rsbl.2013.0853>.↵
 24. Andrei Irimia et al., 'The Indigenous South American Tsimané Exhibit Relatively Modest Decrease in Brain Volume With Age Despite High Systemic Inflammation', *Journals of Gerontology: Series A* 76, no. 12 (2021): 2147–55, <https://doi.org/10.1093/gerona/glab138>.↵
 25. H. Pontzer, B. M. Wood, and D. A. Raichlen, 'Hunter-Gatherers as Models in Public Health: Hunter-Gatherer Health and Lifestyle', *Obesity Reviews* 19 (2018): 24–35, <https://doi.org/10.1111/obr.12785>. A wider review of the literature can be found in Herman Pontzer and Brian M. Wood, 'Effects of Evolution, Ecology, and Economy on Human Diet: Insights from Hunter-Gatherers and Other Small-Scale Societies', *Annual Review of Nutrition* 41, no. 1 (2021): 363–85, <https://doi.org/10.1146/annurev-nutr-111120-105520>. Full details on atherosclerosis and the Tsimané can be found in Hillard Kaplan et al., 'Coronary Atherosclerosis in Indigenous South American Tsimané: A

Cross-Sectional Cohort Study’, *The Lancet* 389, no. 10080 (2017): 1730–39, [https://doi.org/10.1016/S0140-6736\(17\)30752-3](https://doi.org/10.1016/S0140-6736(17)30752-3). The diets of our ancestors would of course have varied enormously, depending on what environments they lived in. While there is the modern fad of the ‘paleo’ diet, there would have been no single diet for the Palaeolithic. Today, the Inuit subsist on a largely carnivorous diet of fish, seal and caribou. In contrast, 70–95 per cent of the calories for foragers in the Andes of South America from thousands of years ago came from plants. Despite this diversity, they all have ample exercise, social connections and varied diets with no highly processed foods.

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26. Their guts, not just their brains, also seem to be healthier. Studies of the foraging Hadza in northern Tanzania have found that they have far more diverse gut microbiomes (collections of bacteria, fungi and genes found in the gut) than do people from Nepal, Italy, or California. The average Hadza has 730 species of gut microbe while the average Californian has just 277, many of which showed signs of chronic stress and inflammation. See Hannah C. Wastyk et al., ‘Gut-Microbiota-Targeted Diets Modulate Human Immune Status’, *Cell* 184, no. 16 (2021): 4137–53.e14, <https://doi.org/10.1016/j.cell.2021.06.019>; Stephanie L. Schnorr et al., ‘Gut Microbiome of the Hadza Hunter-Gatherers’, *Nature Communications* 5, no. 1 (2014): 3654, <https://doi.org/10.1038/ncomms4654>; Matthew M. Carter et al., ‘Ultra-Deep Sequencing of Hadza Hunter-Gatherers Recovers Vanishing Gut Microbes’, *Cell* 186, no. 14 (2023): 3111–24.e13, <https://doi.org/10.1016/j.cell.2023.05.046>. The Hadza microbiome also contained numerous, previously undocumented microbes that appear to have vanished from other populations. This is no insignificant finding. Microbiome health affects not only gastrointestinal health, but also mental health, with poor biomes linked to higher rates of anxiety and depression. Notably, modern studies on mental and physical health are awash with advice that simply reflects a hunter-gatherer lifestyle: frequent exercise, copious social contact, diets low in processed foods, lots of time outdoors in nature, and a regular sleep schedule of 7–8 hours.↵
27. For an overview, see Karl Widerquist and Grant S. McCall, *Prehistoric Myths in Modern Political Philosophy* (Edinburgh University Press,

- 2018). For the Agta case study, see Mark Dyble et al., ‘Engagement in Agricultural Work Is Associated With Reduced Leisure Time Among Agta Hunter-Gatherers’, *Nature Human Behaviour* 3 (2019): 792–6, <https://doi.org/10.1038/s41562-019-0614-6>. The estimate of seventy-seven hours is derived from US Bureau of Labor Statistics, ‘American Time Use Survey – 2023 Results’, www.bls.gov/news.release/pdf/atus.pdf. They estimate that the average American parent who travels to work spends forty hours working, as well as around 2.4 hours per day on housework, and 1–2.3 hours per day on childcare. I’ve added these together, alongside a conservative estimate of one hour per weekday travelling to and from work.↵
28. There are far more nuanced definitions of violence that can incorporate indirect effects. For a summary, see Siniša Malešević, *The Rise of Organised Brutality: A Historical Sociology of Violence* (Cambridge University Press, 2017), chap. 2. Since this is a book for a popular audience, I’ve constrained the discussion to the most simple and common-sense definition. ↵
 29. Philip Thomson and John Halstead, ‘How Violent Was the Pre-Agricultural World?’, *SSRN Electronic Journal*, 2023, 46–53, <https://doi.org/10.2139/ssrn.4466809>.↵
 30. Douglas P. Fry and Patrik Söderberg, ‘Myths About Hunter-Gatherers Redux: Nomadic Forager War and Peace’, ed. Kirk Endicott, *Journal of Aggression, Conflict and Peace Research* 6, no. 4 (2014): 259–60, <https://doi.org/10.1108/JACPR-06-2014-0127>.↵
 31. Widerquist and McCall, *Prehistoric Myths in Modern Political Philosophy*.↵
 32. Jonathan Haas and Matthew Piscitelli, ‘The Prehistory of Warfare: Misled by Ethnography’, in *War, Peace, and Human Nature*, ed. Douglas P. Fry (Oxford University Press, 2013), 168–90, <https://doi.org/10.1093/acprof:oso/9780199858996.003.0010>. On Jebel Sahaba, see R. Brian Ferguson, ‘Archaeology, Cultural Anthropology, and the Origins and Intensifications of War’, in *The Archaeology of Warfare: Prehistories of Raiding and Conquest*, ed. Elizabeth Arkush and Mark W. Allen (University Press of Florida, 2006), 482–3.↵
 33. Thomson and Halstead, ‘How Violent Was the Pre-Agricultural World?’ Thomson and Halstead do adjust their figures upwards to 3.2

per cent for the Palaeolithic and 9.7 per cent for the End-Palaeolithic. Their adjustment is based on both some such deaths not leaving skeletal traces and others possibly not being preserved. Hence, this suggests that we could be undercounting, a problem they have tried to correct for. However, they also note that the skeletal remains are a small percentage of the overall population, meaning we could be overcounting or undercounting substantially (p. 76). There is the far larger issue that any adjustments rely on and reflect the pre-existing biases of the authors. This is critical, given how sensitive and political the issue of violence is. Despite the shortcomings, it is safer and more reliable to rely on the unadjusted figures. Statistics for the modern world come from Max Roser, ‘Causes of Death Globally: What Do People Die From?’, Our World in Data, 2021, <https://ourworldindata.org/causes-of-death-treemap>. One could say that the Palaeolithic appears to be more peaceful than the modern world. That wouldn’t be warranted. The data is too sparse and Palaeolithic deaths from suicide may have involved poison or other means that didn’t leave marks on bones.↵

34. For pessimistic estimates of prehistoric violence, see Steven Pinker, *The Better Angels of Our Nature: Why Violence Has Declined* (Viking, 2011); Azar Gat, *War in Human Civilization* (Oxford University Press, 2008); Ian Morris, *War! What Is It Good For? Conflict and the Progress of Civilization from Primates to Robots* (Farrar, Straus and Giroux, 2014). A critique of Pinker’s analysis of prehistoric warfare is provided in R. Brian Ferguson, ‘Pinker’s List: Exaggerating Prehistoric War Mortality’, in *War, Peace, and Human Nature*, ed. Douglas P. Fry (Oxford University Press, 2013), 112–31, <https://doi.org/10.1093/acprof:oso/9780199858996.003.0007>. There have also been several detailed critiques of Pinker’s contention that war has been declining in the modern world. The political scientist Bear Braumoeller has published a scathing statistical critique showing that the rate of conflict increased in the two centuries prior to the Cold War, and the rate of civil wars has increased thereafter. Full details can be found in Bear F. Braumoeller, *Only the Dead: The Persistence of War in the Modern Age* (Oxford University Press, 2019). See also Tanisha M. Fazal, ‘Dead Wrong? Battle Deaths, Military Medicine, and Exaggerated Reports of War’s Demise’, *International Security* 39,

- no. 1 (2014): 95–125. Last but not least, an entire cadre of scholars of violence covering areas from Native America to Soviet Russia have put forward an extensive critique of *The Better Angels of Our Nature* in the appropriately titled Philip G. Dwyer and Mark S. Micale, eds., *The Darker Angels of Our Nature: Refuting the Pinker Theory of History & Violence* (Bloomsbury Academic, 2021).↵
35. Haas and Piscitelli, ‘The Prehistory of Warfare:’, figs 10.2 and 10.3. Credit to Jill Seagard, The Field Museum. Redrawn from Jean Clottes and Jean Courtin, *The Cave Beneath the Sea: Paleolithic Images at Cosquer* (Abrams, 1996).↵
 36. José Maria Gómez et al., ‘The Phylogenetic Roots of Human Lethal Violence’, *Nature* 538, no. 7624 (2016): 233–7, <https://doi.org/10.1038/nature19758>. Note that this is a limited study and essentially just a best guess based on genetic correlations.↵
 37. Kevin D. Hunt, *Chimpanzee: Lessons from Our Sister Species* (Cambridge University Press, 2020), 233–8.↵
 38. R. Brian Ferguson, *Chimpanzees, War, and History: Are Men Born to Kill?* (Oxford University Press, 2023).↵
 39. Dave Grossman, *On Killing: The Psychological Cost of Learning to Kill in War and Society* (Open Road Media, 2014): 23–6. For useful and accessible overviews, see Randall Collins, *Violence: A Micro-Sociological Theory* (Princeton University Press, 2009), chap. 2; Rutger Bregman, *Humankind: A Hopeful History* (Bloomsbury, 2021), chap. 4. While the initial work of S. L. A. Marshall (who estimated in the Second World War that 25 per cent of soldiers never fired their guns, but was found to have shoddy methods afterwards) has been debunked, most of the subsequent work in the field of ‘killology’ has reaffirmed the basic insight that soldiers are often deeply reluctant to engage in violence.↵
 40. Grossman, *On Killing*, 35.↵
 41. For the most comprehensive and up-to-date overview of the literature covering the Palaeolithic, read Tibor Rutar, ‘The Prehistory of Violence and War: Moving beyond the Hobbes–Rousseau Quagmire’, *Journal of Peace Research*, 18 October 2022, 002234332210901, <https://doi.org/10.1177/00223433221090112>. In it, Rutar provides a

more detailed critique of the methods of different studies, an approach that I don't have the space for here (and the average reader likely doesn't have the appetite for). He comes to the same conclusion: violence was low in the Palaeolithic, war was not likely to have occurred, and violence follows an inverted U-pattern rising from hunter-gatherers to the medieval ages before decreasing in the modern world.↵

42. Douglas P. Fry and Patrik Söderberg, 'Lethal Aggression in Mobile Forager Bands and Implications for the Origins of War', *Science* 341, no. 6143 (2013): 270–73, <https://doi.org/10.1126/science.1235675>. For a useful overview of the wider literature, see Thomson and Halstead, 'How Violent Was the Pre-Agricultural World?', 31–3.↵
43. Clarke, 'Panic'.↵
44. For an overview of the relevant literature, see John Drury, David Novelli, and Clifford Stott, 'Psychological Disaster Myths in the Perception and Management of Mass Emergencies', *Journal of Applied Social Psychology* 43, no. 11 (2013): 2259–70, <https://doi.org/10.1111/jasp.12176>. Bregman, *Humankind*, and Rebecca Solnit, *A Paradise Built in Hell* (Viking, 2009), provide entertaining accounts of the best-known case studies.↵
45. Yes, people do occasionally respond poorly. There are cases of price-gouging, panic-buying, and unruly behaviour during disasters. But the reason that Australian shoppers fighting over toilet rolls during Covid-19 became a newsworthy meme is because it is exceptional (as well as admittedly funny). ↵
46. Variants of the myth of mass panic also prove to be wrong. The conventional wisdom was once that pandemics ignited division, hatred, and blaming of 'the other'. Yet this isn't true either, historically. For instance, bouts of the Black Death in the late medieval and Renaissance periods did not lead to violence against or persecution of Jews, despite evidence linking its spread to Jewish traders who transmitted the plague by breaking quarantine to trade infected goods. In general, most outbursts of disease, from the Plague of Athens to influenza in the modern world, have brought people together rather than driving them apart. See Samuel K. Cohn, 'Pandemics: Waves of Disease, Waves of Hate from the Plague of Athens to A.I.D.S.',

Historical Research 85, no. 230 (2012): 535–55,
<https://doi.org/10.1111/j.1468-2281.2012.00603.x>.↵

47. For an overview, see Bas van Bavel et al., *Disasters and History: The Vulnerability and Resilience of Past Societies* (Cambridge University Press, 2020), <https://doi.org/10.1017/9781108569743>.↵
48. On food sharing in modern egalitarian hunter-gatherers, see James Woodburn, ‘Egalitarian Societies’, *Man* 17, no. 3 (1982): 437–42, <https://doi.org/10.2307/2801707>. Deer remains from 400,000–100,000 years ago suggest that butchering was initially done collectively. Over time this evolved into a more systematic practice with a few skilled individuals butchering and allocating the meat, as we see in modern hunter-gatherers. See Mary C. Stiner, Ran Barkai, and Avi Gopher, ‘Cooperative Hunting and Meat Sharing 400–200 Kya at Qesem Cave, Israel’, *Proceedings of the National Academy of Sciences* 106, no. 32 (2009): 13207–12, <https://doi.org/10.1073/pnas.0900564106>. Others have used mathematical and conceptual models to help explain how food sharing between pair-bonded couples helped us spread into a wider range of environments: Ingela Alger et al., ‘The Evolution of Early Hominin Food Production and Sharing’, *Proceedings of the National Academy of Sciences* 120, no. 25 (2023): e2218096120, <https://doi.org/10.1073/pnas.2218096120>. This also resulted in cultures of extreme charity. Early European explorers were stunned by the generosity of indigenous Americans, even during times of famine. See David Graeber, *Debt: The First 5,000 Years* (Melville House, 2011), 100–101.↵
49. Graeber, *Debt*, 99.↵
50. For a fuller discussion, see Hélène Landemore, *Democratic Reason: Politics, Collective Intelligence, and the Rule of the Many* (Princeton University Press, 2017); Hélène Landemore, *Open Democracy: Reinventing Popular Rule for the Twenty-First Century* (Princeton University Press, 2020).↵
51. On the origins of parliamentary democracy, see John Keane, *The Shortest History of Democracy: 4,000 Years of Self-Government – A Retelling for Our Times* (The Experiment, 2022), 87–9. On examples of early democracy across the world, read David Stasavage, *The*

Decline and Rise of Democracy: A Global History from Antiquity to Today (Princeton University Press, 2021), chaps. 1 and 2.↵

52. Our ancestors would still have had a dizzying array of cultures. We see that in foragers today. The Hiwi of Venezuela dance almost daily and rarely divorce while the Ache of Paraguay rarely dance and divorce frequently, and there are many differences in their humour, food, beliefs, and language. Despite this variety in lifestyles, they all seemed to keep a level playing field. One can have diversity while maintaining a common democratic thread.↵
53. Walter Scheidel, *The Great Leveler: Violence and the History of Inequality from the Stone Age to the Twenty-First Century* (Princeton University Press, 2017).↵
54. Sarah Blaffer Hrdy, *Mothers and Others: The Evolutionary Origins of Mutual Understanding* (Belknap Press of Harvard University Press, 2011). For an accessible overview, see Camilla Power, ‘Gender Egalitarianism Made Us Human: Patriarchy Was Too Little, Too Late’, *Open Democracy*, 2018, www.opendemocracy.net/en/gender-egalitarianism-made-us-human-patriarchy-was-too-little-too-late/. It’s also why our grandmothers so often help with childrearing and passing on generations of accumulated wisdom.↵
55. Ethan G. Harrod, Christopher L. Coe, and Paula M. Niedenthal, ‘Social Structure Predicts Eye Contact Tolerance in Nonhuman Primates: Evidence from a Crowd-Sourcing Approach’, *Scientific Reports* 10, no. 1 (2020): 6971, <https://doi.org/10.1038/s41598-020-63884-x>.↵
56. For the evolutionary case, see Power, ‘Gender Egalitarianism Made Us Human’. On settlement patterns and genetic diversity, consult M. Dyble et al., ‘Sex Equality Can Explain the Unique Social Structure of Hunter-Gatherer Bands’, *Science* 348, no. 6236 (2015): 796–8, <https://doi.org/10.1126/science.aaa5139>.↵
57. On the division of labour, see Robert L. Kelly, *The Lifeways of Hunter-Gatherers: The Foraging Spectrum*, 2nd edn (Cambridge University Press, 2013), chap. 9. On the evidence for women hunting: Abigail Anderson et al., ‘The Myth of Man the Hunter: Women’s Contribution to the Hunt across Ethnographic Contexts’, ed. Raven Garvey, *PLOS ONE* 18, no. 6 (2023): e0287101,

<https://doi.org/10.1371/journal.pone.0287101>; Randall Haas et al., 'Female Hunters of the Early Americas', *Science Advances* 6, no. 45 (2020): eabd0310, <https://doi.org/10.1126/sciadv.abd0310>; Sarah Lacy and Cara Ocobock, 'Woman the Hunter: The Archaeological Evidence', *American Anthropologist* 126, no. 1 (2024): 19–31, <https://doi.org/10.1111/aman.13914>. The evidence for women hunting has been used to suggest that there was no division of labour between the sexes. That is an overstated position which has been heavily criticized for a lack of data (Vivek V. Venkataraman et al., 'Female Foragers Sometimes Hunt, Yet Gendered Divisions of Labor Are Real: A Comment on Anderson et al. (2023) *The Myth of Man the Hunter*', *Evolution and Human Behavior* 45, no. 4 (2024), S1090513824000497, <https://doi.org/10.1016/j.evolhumbehav.2024.04.014>).↵

58. Tatiana Zerjal et al., 'The Genetic Legacy of the Mongols', *American Journal of Human Genetics* 72, no. 3 (2003): 717–21. Others have found similar results for Chinese emperors: see Patricia Balaresque et al., 'Y-Chromosome Descent Clusters and Male Differential Reproductive Success: Young Lineage Expansions Dominate Asian Pastoral Nomadic Populations', *European Journal of Human Genetics* 23, no. 10 (2015): 1413–22, <https://doi.org/10.1038/ejhg.2014.285>. One recent study suggests that the widely spread genes today may not come from Chinggis Khan himself, but rather the Niru'un clan of the Mongols. See Lan-Hai Wei et al., 'Whole-Sequence Analysis Indicates that the Y Chromosome C2*-Star Cluster Traces Back to Ordinary Mongols, Rather Than Genghis Khan', *European Journal of Human Genetics* 26, no. 2 (2018): 230–37, <https://doi.org/10.1038/s41431-017-0012-3>.↵
59. Walter Scheidel, 'Fitness and Power: The Contribution of Genetics to the History of Differential Reproduction', *Evolutionary Psychology* 19, no. 4 (2021): 14747049211066599, <https://doi.org/10.1177/14747049211066599> provides a nuanced overview of how conquest and status differences seem to have impacted reproductive differences across history. His conclusion is that often conquest resulted in the male lineage of the conquerors replacing the genetic lineage of the vanquished, but this varies across cases.↵

60. Monika Karmin et al., 'A Recent Bottleneck of Y Chromosome Diversity Coincides With a Global Change in Culture', *Genome Research* 25, no. 4 (2015): 461–2, <https://doi.org/10.1101/gr.186684.114>.↵
61. Michael Westaway, David Lambert, and Monika Karmin, 'There Was a Decline of Male Diversity When Humans Took to Agriculture', *The Conversation*, 20 April 2015, <https://theconversation.com/there-was-a-decline-of-male-diversity-when-humans-took-to-agriculture-38725>.↵
62. Barbara F. Walter, *How Civil Wars Start: And How to Stop Them* (Crown, 2022).↵
63. Brian Gallagher, 'How Can We Discourage Mass Shootings?', *Nautilus*, 15 May 2023, <https://nautil.us/how-can-we-discourage-mass-shootings-304104/>. For an overview between the frustrated longing for status and mass shooters, read Will Storr, *The Status Game* (William Collins, 2021), chap. 8.↵
64. Jason Manning, *Suicide: The Social Causes of Self-Destruction* (University of Virginia Press, 2020), <https://doi.org/10.2307/j.ctv103xf27>.↵
65. Most of his kids did well for themselves, with two (Nelson and William) becoming billionaires who tried and failed to corner the world market in silver. ↵
66. Joey T. Cheng, Jessica L. Tracy, and Cameron D. Anderson, eds., *The Psychology of Social Status* (Springer, 2016), 19.↵
67. Francesco Margoni, Renée Baillargeon, and Luca Surian, 'Infants Distinguish Between Leaders and Bullies', *Proceedings of the National Academy of Sciences* 115, no. 38 (2018), <https://doi.org/10.1073/pnas.1801677115>.↵
68. Justin Jennings, *Finding Fairness: From Pleistocene Foragers to Contemporary Capitalists* (University Press of Florida, 2021), 3; Christopher Boehm, *Moral Origins: The Evolution of Virtue, Altruism, and Shame* (Basic Books, 2012).↵
69. For an overview, see Tian Chen Zeng, Joey T. Cheng, and Joseph Henrich, 'Dominance in Humans', *Philosophical Transactions of the Royal Society B: Biological Sciences* 377, no. 1845 (2022): 20200451, <https://doi.org/10.1098/rstb.2020.0451>; Joey T. Cheng, 'Dominance,

Prestige, and the Role of Leveling in Human Social Hierarchy and Equality', *Current Opinion in Psychology* 33 (2020): 238–44, <https://doi.org/10.1016/j.copsyc.2019.10.004>.↵

70. Cameron Anderson, John Angus D. Hildreth, and Laura Howland, 'Is the Desire for Status a Fundamental Human Motive? A Review of the Empirical Literature', *Psychological Bulletin* 141, no. 3 (2015): 574–601, <https://doi.org/10.1037/a0038781>.↵
71. Christopher Boehm et al., 'Egalitarian Behavior and Reverse Dominance Hierarchy [and Comments and Reply]', *Current Anthropology* 34, no. 3 (1993): 227–54. Also see Boehm's personal correspondence with Brian Klaas in Dr Brian Klaas, *Corruptible: Who Gets Power and How It Changes Us* (John Murray, 2022), 29. There is also evidence that testosterone tends to increase the desire for status, but only in unstable hierarchies. See A. B. Losecaat Vermeer et al., 'Exogenous Testosterone Increases Status-Seeking Motivation in Men With Unstable Low Social Status', *Psychoneuroendocrinology* 113 (2020): 104552, <https://doi.org/10.1016/j.psyneuen.2019.104552>. Women still engage in status competition as well, especially with other women. This trait can be especially intense in some female primates. However, when it comes to dominance-based status-seeking in humans, men are far more likely to do so. ↵
72. For the most recent definitions of the dark triad, read Stephane A. De Brito et al., 'Psychopathy', *Nature Reviews Disease Primers* 7, no. 1 (2021): 1, <https://doi.org/10.1038/s41572-021-00282-1>; Zlatan Krizan and Anne D. Herlache, 'The Narcissism Spectrum Model: A Synthetic View of Narcissistic Personality', *Personality and Social Psychology Review* 22, no. 1 (2018): 6, <https://doi.org/10.1177/1088868316685018>; David Sloan Wilson, David Near, and Ralph R. Miller, 'Machiavellianism: A Synthesis of the Evolutionary and Psychological Literatures', *Psychological Bulletin* 119, no. 2 (1996): 285. On the link between the dark triad and dominance-based status seeking, read Adam C. Davis and Tracy Vaillancourt, 'Predicting Dominance and Prestige Status-Striving from the Dark Tetrad: The Mediating Role of Indirect Aggression', *Current Psychology* 42, no. 16 (2023): 13680–92, <https://doi.org/10.1007/s12144-021-02492-y>; Peter K. Jonason and

Virgil Zeigler-Hill, ‘The Fundamental Social Motives that Characterize Dark Personality Traits’, *Personality and Individual Differences* 132 (2018): 98–107, <https://doi.org/10.1016/j.paid.2018.05.031>. Note that I’m referring here to people’s scores on the dark triad, not necessarily full-blown personality disorders, such as narcissist personality disorder (NPD).↵

73. Although chimps also engage in social behaviour such as grooming babies to gain prestige, much like campaigning politicians kissing babies for a photo.↵
74. Christopher R. von Rueden and Adrian V. Jaeggi, ‘Men’s Status and Reproductive Success in 33 Nonindustrial Societies: Effects of Subsistence, Marriage System, and Reproductive Strategy’, *Proceedings of the National Academy of Sciences* 113, no. 39 (2016): 10824–9, <https://doi.org/10.1073/pnas.1606800113>; Dan P. McAdams, ‘The Appeal of the Primal Leader: Human Evolution and Donald J. Trump’, *Evolutionary Studies in Imaginative Culture* 1, no. 2 (2017): 5, <https://doi.org/10.26613/esic.1.2.45>.↵
75. Cecilia Padilla-Iglesias et al., ‘Population Interconnectivity Over the Past 120,000 Years Explains Distribution and Diversity of Central African Hunter-Gatherers’, *Proceedings of the National Academy of Sciences* 119, no. 21 (2022): e2113936119, <https://doi.org/10.1073/pnas.2113936119>.↵
76. Carina M. Schlebusch et al., ‘Khoe-San Genomes Reveal Unique Variation and Confirm the Deepest Population Divergence in *Homo sapiens*’, *Molecular Biology and Evolution* 37, no. 10 (2020): 2944–54, <https://doi.org/10.1093/molbev/msaa140>.↵

CHAPTER 2: COLLAPSE FOR 99 PER CENT OF HUMAN HISTORY

1. The most common estimate is that by the beginning of the Common Era three-quarters of the human population was held in just four empires: the Roman Empire, the Kushan Empire, the Parthian Empire, and the Han Dynasty of China. See Walter Scheidel, ‘From Plains to Chains: How the State Was Born’, *Financial Times*, 5 October 2017, www.ft.com/content/aa39bc10-a836-11e7-ab66-21cc87a2edde. However, this is based on a fairly unreliable source: the 1978 *Atlas of*

World Population History. The estimates are largely based on old guesswork, especially for non-state peoples. Today, light detection and ranging (LIDAR) scans of non-state areas such as in the Amazonian basin, regularly suggest larger population estimates than were once assumed. For a full discussion, see David Wengrow, 'Beyond Kingdoms and Empires', *Aeon*, 5 July 2024, <https://aeon.co/essays/an-archeological-revolution-transforms-our-image-of-human-freedoms>.↵

2. The 1–3 per cent figure is from Benjamin Vernot and Joshua M. Akey, 'Resurrecting Surviving Neandertal Lineages from Modern Human Genomes', *Science* 343, no. 6174 (2014): 1017–21, <https://doi.org/10.1126/science.1245938>.↵
3. It's worth stating a few caveats and clarifications here. Collapse is something that affects people and power structures, namely Goliaths. In contrast, extinction occurs in other biological life forms while collapse does not. That doesn't change the basic fact that extinction of a human population will result in a complete permanent collapse: a destruction of all its power structures. Nonetheless, the more sceptical reader can simply see this discussion as a precursor to collapse (like the coverage of early fission-fusion cycles) if they prefer.↵
4. Benedict R. O'G. Anderson, *Imagined Communities: Reflections on the Origin and Spread of Nationalism* (Verso, 1990); Yuval Noah Harari, *Sapiens: A Brief History of Humankind*, trans. John Purcell and Haim Watzman (Vintage, 2015). On the importance of storytelling, see Daniel Smith et al., 'Cooperation and the Evolution of Hunter-Gatherer Storytelling', *Nature Communications* 8, no. 1 (2017): 1, <https://doi.org/10.1038/s41467-017-02036-8>. Note that their study is only for the Agta; however, there are powerful reasons to believe that this holds true for other groups, especially given its evolutionary advantages in promoting cooperation. While stories and symbols are vital, we must also be careful not to overemphasize the power of imagined communities. We are incredibly cooperative even without them. Infants of 12–24 months can start cooperating, giving help to adults (such as through pointing), and playing social games (and even reversing their roles in those games), all before they can even speak. See Michael Tomasello, *Why We Cooperate* (MIT Press, 2009).↵

5. Joseph Patrick Henrich, *The Secret of Our Success: How Culture Is Driving Human Evolution, Domesticating Our Species, and Making Us Smarter* (Princeton University Press, 2016).↵
6. Andrea Bamberg Migliano and Lucio Vinicius, ‘The Origins of Human Cumulative Culture: From the Foraging Niche to Collective Intelligence’, *Philosophical Transactions of the Royal Society B: Biological Sciences* 377, no. 1843 (2022): 5–6, <https://doi.org/10.1098/rstb.2020.0317>.↵
7. The estimate for cumulative cultural learning beginning around 600,000 years ago was calculated in Jonathan Paige and Charles Perreault, ‘3.3 Million Years of Stone Tool Complexity Suggests That Cumulative Culture Began During the Middle Pleistocene’, *Proceedings of the National Academy of Sciences* 121, no. 26 (2024): e2319175121, <https://doi.org/10.1073/pnas.2319175121>. On ochre, see Rimtautas Dapschauskas et al., ‘The Emergence of Habitual Ochre Use in Africa and Its Significance for the Development of Ritual Behavior During the Middle Stone Age’, *Journal of World Prehistory* 35, nos 3–4 (2022): 233–319, <https://doi.org/10.1007/s10963-022-09170-2>. The oldest forms of artwork could be patterns and sculptures carved on rocks at South Africa’s beaches: Charles Helm, ‘Ancient Humans May Have Made Patterns and Sculptures on Africa’s Beaches’, *The Conversation*, 22 September 2019, <https://theconversation.com/ancient-humans-may-have-made-patterns-and-sculptures-on-south-africas-beaches-123546>. There is still some dispute as to whether this is cumulative, but the evidence is pointing that way.↵
8. For an overview of recent evidence, see Rebecca Wragg, *Kindred: Neanderthal Life, Love, Death and Art* (Bloomsbury Sigma, 2020); Emma Pomeroy et al., ‘New Neanderthal Remains With the “Flower Burial” at Shanidar Cave’, *Antiquity* 94, no. 373 (2020): 11–26, <https://doi.org/10.15184/aqy.2019.207>; Julia Galway-Witham, James Cole, and Chris Stringer, ‘Aspects of Human Physical and Behavioural Evolution During the Last 1 Million Years’, *Journal of Quaternary Science* 34, no. 6 (2019): 355–78, <https://doi.org/10.1002/jqs.3137>.↵
9. D. L. Hoffmann et al., ‘U-Th Dating of Carbonate Crusts Reveals Neandertal Origin of Iberian Cave Art’, *Science* 359, no. 6378 (2018):

- 912–15, <https://doi.org/10.1126/science.aap7778>; Adhi Agus Oktaviana et al., ‘Narrative Cave Art in Indonesia by 51,200 Years Ago’, *Nature* 631, no. 8022 (2024): 814–18, <https://doi.org/10.1038/s41586-024-07541-7>.↵
10. The image is taken from the Neanderthal Museum in Mettmann, Germany.↵
 11. It left them vulnerable, with problems in finding genetically viable mates and a predicament where small changes in fertility or the ratio of births to deaths led to large population swings, a phenomenon called ‘stochastic fluctuations’.↵
 12. Krist Vaesen, Gerrit L. Dusseldorp, and Mark J. Brandt, ‘An Emerging Consensus in Palaeoanthropology: Demography Was the Main Factor Responsible for the Disappearance of Neanderthals’, *Scientific Reports* 11, no. 1 (2021): 4925, <https://doi.org/10.1038/s41598-021-84410-7>. Experts included only those who had published on Neanderthal extinction over the period 2014–20. The outcome is surprising since demographic explanations are recent, with most relevant articles having been published only five years or so before 2020.↵
 13. Laurits Skov et al., ‘Genetic Insights into the Social Organization of Neanderthals’, *Nature* 610, no. 7932 (2022): 519, 523–4, <https://doi.org/10.1038/s41586-022-05283-y>.↵
 14. Calculated by the palaeoanthropologist April Nowell using data from Erik Trinkaus, ‘Neanderthal Mortality Patterns’, *Journal of Archaeological Science* 22, no. 1 (1995): 121–42, [https://doi.org/10.1016/S0305-4403\(95\)80170-7](https://doi.org/10.1016/S0305-4403(95)80170-7). See Kristina Killgrove, ‘Did We Kill the Neanderthals? New Research May Finally Answer an Age-Old Question’, *Live Science*, 10 April 2024, www.livescience.com/archaeology/did-we-kill-the-neanderthals-new-research-may-finally-answer-an-age-old-question.↵
 15. Ali R. Vahdati et al., ‘Exploring Late Pleistocene Hominin Dispersals, Coexistence and Extinction With Agent-Based Multi-Factor Models’, *Quaternary Science Reviews* 279 (2022): 107391, <https://doi.org/10.1016/j.quascirev.2022.107391>.↵
 16. See Nicholas P. Simpson et al., ‘A Framework for Complex Climate Change Risk Assessment’, *One Earth* 4, no. 4 (2021): 489–501,

<https://doi.org/10.1016/j.oneear.2021.03.005>; Luke Kemp et al., ‘Climate Endgame: Exploring Catastrophic Climate Change Scenarios’, *Proceedings of the National Academy of Sciences* 119, no. 34 (2022): e2108146119, <https://doi.org/10.1073/pnas.2108146119>. This is also echoed in the way that the Intergovernmental Panel on Climate Change (IPCC) uses risk in its Sixth Assessment Report. See IPCC, ‘The Concept of Risk in the IPCC Sixth Assessment Report: A Summary of Cross-Working Group Discussions’ (IPCC, 2020), www.ipcc.ch/site/assets/uploads/2021/02/Risk-guidance-FINAL_15Feb2021.pdf. The one difference is that the IPCC separates risk and its three determinants (threat, exposure, and vulnerability) from response risks.↵

17. Steven E. Churchill et al., ‘Shanidar 3 Neandertal Rib Puncture Wound and Paleolithic Weaponry’, *Journal of Human Evolution* 57, no. 2 (2009): 163–78, <https://doi.org/10.1016/j.jhevol.2009.05.010>; Christoph P. E. Zollikofer et al., ‘Evidence for Interpersonal Violence in the St. Césaire Neanderthal’, *Proceedings of the National Academy of Sciences* 99, no. 9 (2002): 6444–8, <https://doi.org/10.1073/pnas.082111899>.↵
18. Katerina Harvati et al., ‘Apidima Cave Fossils Provide Earliest Evidence of *Homo sapiens* in Eurasia’, *Nature* 571, no. 7766 (2019): 500–504, <https://doi.org/10.1038/s41586-019-1376-z>.↵
19. Ryan J. Rabett, ‘The Success of Failed *Homo sapiens* Dispersals out of Africa and into Asia’, *Nature Ecology & Evolution* 2, no. 2 (2018): 212–19, <https://doi.org/10.1038/s41559-017-0436-8>.↵
20. Andrew Bevan et al., ‘Holocene Fluctuations in Human Population Demonstrate Repeated Links to Food Production and Climate’, *Proceedings of the National Academy of Sciences* 114, no. 49 (2017), <https://doi.org/10.1073/pnas.1709190114>.↵
21. Martin A. J. Williams et al., ‘Environmental Impact of the 73ka Toba Super-Eruption in South Asia’, *Palaeogeography, Palaeoclimatology, Palaeoecology* 284, nos 3–4 (2009): 295–314, <https://doi.org/10.1016/j.palaeo.2009.10.009>.↵
22. Chris Clarkson et al., ‘Human Occupation of Northern India Spans the Toba Super-Eruption ~74,000 Years Ago’, *Nature Communications* 11, no. 1 (2020): 961, <https://doi.org/10.1038/s41467-020-14668-4>;

- Michael Haslam et al., 'The 74ka Toba Super-Eruption and Southern Indian Hominins: Archaeology, Lithic Technology and Environments at Jwalapuram Locality 3', *Journal of Archaeological Science* 37, no. 12 (2010): 3370–84, <https://doi.org/10.1016/j.jas.2010.07.034>.↵
23. Benjamin A. Black et al., 'Global Climate Disruption and Regional Climate Shelters after the Toba Supereruption', *Proceedings of the National Academy of Sciences* 118, no. 29 (2021): e2013046118, <https://doi.org/10.1073/pnas.2013046118>.↵
24. Felix Riede, Gerald Oetelaar, and Richard VanderHoek, 'From Crisis to Collapse in Hunter-Gatherer Societies. A Comparative Investigation of the Cultural Impacts of Three Large Volcanic Eruptions on Past Hunter-Gatherers', in *Crisis to Collapse: The Archaeology of Social Breakdown*, ed. Tim Cunningham and Jan Driessen (UCL Presses Universitaires de Louvain, 2017). All of these examples are technically during the Holocene, but since they occurred to hyper-egalitarian foragers, they can provide some insight into how prehistoric hunter-gatherers may have responded to volcanic disruptions.↵
25. For a lively discussion of the role of mobility in human evolution, read Padilla-Iglesias, 'Societies of Perpetual Movement'.↵
26. Jennings, *Finding Fairness*, 46–7; Matt Grove, Eiluned Pearce, and R. I. M. Dunbar, 'Fission-Fusion and the Evolution of Hominin Social Systems', *Journal of Human Evolution* 62, no. 2 (2012): 191–200, <https://doi.org/10.1016/j.jhevol.2011.10.012>; Robert Foley and Clive Gamble, 'The Ecology of Social Transitions in Human Evolution', *Philosophical Transactions of the Royal Society B: Biological Sciences* 364, no. 1533 (2009): 3267–79, <https://doi.org/10.1098/rstb.2009.0136>.↵
27. Iain D. Couzin and Mark E. Laidre, 'Fission-Fusion Populations', *Current Biology* 19, no. 15 (2009): R633–5, <https://doi.org/10.1016/j.cub.2009.05.034>. The two-thirds estimate comes from analyses of human conversations, in which around two-thirds is devoted to social topics (grouped under the term 'gossip'). See R. I. M. Dunbar, 'Gossip in Evolutionary Perspective', *Review of General Psychology* 8, no. 2 (2004): 100–110, <https://doi.org/10.1037/1089-2680.8.2.100>.↵

28. I. Crevecoeur et al., 'Late Stone Age Human Remains from Ishango (Democratic Republic of Congo): New Insights on Late Pleistocene Modern Human Diversity in Africa', *Journal of Human Evolution* 96 (2016): 35–57, <https://doi.org/10.1016/j.jhevol.2016.04.003>.↵
29. For an entertaining overview, see Padilla-Iglesias, 'Did Humanity Really All Arise in One Place?'. Some even see these adaptations in bipedalism and bigger brains enabling fission-fusion practices as two of five major transitions in hominid evolution. See Foley and Gamble, 'The Ecology of Social Transitions in Human Evolution', 3276.↵
30. This extended beyond the buffalo hunt into organizing migrations, preventing warriors from going on unauthorized raids, and overseeing crowds at festivals. The composition of the police force was regularly cycled between seasons. This was likely an effective accountability mechanism: you'd have far less incentive to abuse your coercive power if you knew you'd be on the receiving end next summer. See David Wengrow and David Graeber, 'Farewell to the "Childhood of Man": Ritual, Seasonality, and the Origins of Inequality', *Journal of the Royal Anthropological Institute* 21, no. 3 (2015): 597–619, <https://doi.org/10.1111/1467-9655.12247>.↵
31. On the Shoshone, see Robert H. Lowie, 'Some Aspects of Political Organization Among the American Aborigines', *Journal of the Royal Anthropological Institute of Great Britain and Ireland* 78, no. 1/2 (1948): 11, <https://doi.org/10.2307/2844522>. On the Wik-Mungkan, see Donald F. Thomson, 'The Seasonal Factor in Human Culture Illustrated from the Life of a Contemporary Nomadic Group', *Proceedings of the Prehistoric Society* 5, no. 2 (1939): 209–21, <https://doi.org/10.1017/S0079497X00020545>.↵
32. Marcel Mauss, *Seasonal Variations of the Eskimo: A Study in Social Morphology* (Routledge, 2004). The anthropologist Marcel Mauss called this behaviour a 'twofold morphology'.↵
33. Chris J. Stevens and Dorian Q. Fuller, 'Did Neolithic Farming Fail? The Case for a Bronze Age Agricultural Revolution in the British Isles', *Antiquity* 86, no. 333 (2012): 707–22, <https://doi.org/10.1017/S0003598X00047864>; Bruce Bower, 'Herders, Not Farmers, Built Stonehenge', *Science News*, 9 June 2012, www.sciencenews.org/article/herders-not-farmers-built-stonehenge;

Caroline Lang et al., 'Gazelle Behaviour and Human Presence at Early Neolithic Göbekli Tepe, South-East Anatolia', *World Archaeology* 45, no. 3 (2013): 410–29,

<https://doi.org/10.1080/00438243.2013.820648>.↵

34. Rather than houses, these mammoth-bone structures seem to have been celebrations of successful hunts and ancient trading hubs for the exchange of amber, animal pelts, and marine shells. There are other sites of grand burials. In Dolní Věstonice, in the south of the Czech Republic, a triple burial dating back 27,000 years involved two males with elaborate headdresses placed alongside a female on ochre-stained soil. For summaries of the relevant literature, see Hayden, 'Social Complexity'; Wengrow and Graeber, 'Farewell to the "Childhood of Man"'.↵
35. Hayden, 'Social Complexity'; David Graeber and David Wengrow, *The Dawn of Everything: A New History of Humanity* (Allen Lane, 2021).↵
36. Bennett Bacon et al., 'An Upper Palaeolithic Proto-Writing System and Phenological Calendar', *Cambridge Archaeological Journal* 33, no. 3 (2023): 1–19, <https://doi.org/10.1017/S0959774322000415>.↵
37. Graeber and Wengrow, *The Dawn of Everything*, chap. 3.↵
38. Some cultures periodically shift their societies with the moon rather than with the seasons. They do so not because they worship the moon, but rather to align with menstruation. This is based on Chris Knight's 'sex-strike' theory of human origins, and Camilla Power's 'Female Cosmetic Coalitions'. They propose that gender and political equality was maintained through coalitions of women banding together with males to prohibit and punish any philandering or domineering men. They did so by doing monthly rituals where they would paint themselves in red ochre to disguise women who were menstruating. Menstruation is one of the few visible signs that women give for their impending fertility. The ochre ritual was intended to scramble that information and thwart the efforts of any man or group to monopolize reproduction. See Chris Knight, *Blood Relations: Menstruation and the Origins of Culture* (Yale University Press, 1991); Power, 'Gender Egalitarianism Made Us Human'. Modelling and genetics suggest that ice-age hunters did have some kinds of fission-fusion dynamics, but

these can take many forms, and there is no evidence that this involved any changes in social structure.↵

39. Mauss, *Seasonal Variations of the Eskimo*, 33. Note that this is in direct contradiction to Graeber and Wengrow's *The Dawn of Everything*. In it, they claim that Mauss recorded that 40 per cent or less of the variance in the seasonal structure of the Inuit was due to environmental changes, even though that number appears nowhere in Mauss's original text.↵
40. That's possibly why we see signs of seasonal cycles during the Upper Palaeolithic near choke points of migrating herds.↵
41. Graeber and Wengrow, *The Dawn of Everything*; Wengrow and Graeber, 'Farewell to the "Childhood of Man"'.↵

CHAPTER 3: FROM HUNTING AND GATHERING TO BEING HUNTED AND GATHERED

1. For an overview of the Calusa, see Fernando Santos-Granero, *Vital Enemies: Slavery, Predation, and the Amerindian Political Economy of Life* (University of Texas Press, 2009), chap. 4.↵
2. For an accessible overview of the north-west coast foragers, see David Wengrow and David Graeber, "'Many Seasons Ago': Slavery and Its Rejection Among Foragers on the Pacific Coast of North America', *American Anthropologist* 120, no. 2 (2018): 237–49, <https://doi.org/10.1111/aman.12969>. Notably, Wengrow and Graeber differ from other experts in seeing the origins of hierarchy across the north-west coast peoples as being due to conscious political choice, rather than to material circumstances. As we'll see, while this is a liberating perspective, it's not convincing. For a summary of the Calusa and the latest evidence on their watercourts, see Victor D. Thompson et al., 'Ancient Engineering of Fish Capture and Storage in Southwest Florida', *Proceedings of the National Academy of Sciences* 117, no. 15 (2020): 8374–81, <https://doi.org/10.1073/pnas.1921708117>.↵
3. Brian Hayden, *The Power of Ritual in Prehistory: Secret Societies and Origins of Social Complexity* (Cambridge University Press, 2018), chap. 2; Wengrow and Graeber, "'Many Seasons Ago'". This is also a

recurring theme in Flannery and Marcus's theory of how inequality and states first arose (Kent V. Flannery and Joyce Marcus, *The Creation of Inequality: How Our Prehistoric Ancestors Set the Stage for Monarchy, Slavery, and Empire* (Harvard University Press, 2012)).↵

4. Manvir Singh and Luke Glowacki, 'Human Social Organization During the Late Pleistocene: Beyond the Nomadic-Egalitarian Model', *Evolution and Human Behavior* 43 (2022): 418–31, <https://doi.org/10.1016/j.evolhumbehav.2022.07.003>. There may have been many more whose traces are buried beneath the concrete and roads of current agricultural societies.↵
5. For the classic overview of hunter-gatherer diversity, see Kelly, *The Lifeways of Hunter-Gatherers*.↵
6. T. Douglas Price and Ofer Bar-Yosef, 'Traces of Inequality at the Origins of Agriculture in the Ancient Near East', in *Pathways to Power*, ed. T. Douglas Price and Gary M. Feinman (Springer, 2010), 147–68, https://doi.org/10.1007/978-1-4419-6300-0_6; Heather Pringle, 'The Ancient Roots of the 1%', *Science* 344, no. 6186 (2014): 823, <https://doi.org/10.1126/science.344.6186.822>. These signs of inequality, as well as earlier examples during the Palaeolithic, do appear to be temporary and ephemeral, as explored in Samuel Bowles and Mattia Fochesato, 'The Origins of Enduring Economic Inequality', Santa Fe Institute, 31 April 2024, <https://sites.santafe.edu/~bowles/wp-content/uploads/Text-JEL-31-Jan-sent-.pdf>. There is an intense debate on whether we can accurately measure and compare inequality across societies. Many of the most common criticisms are summarized in David Wengrow, 'The Mismeasure of Human History?', *Stone Econ*, 29 June 2024, www.stone-econ.org/news-and-blogs/the-mismeasure-of-human-history. Many of these are also addressed in Timothy A. Kohler and Michael Ernest Smith, eds., *Ten Thousand Years of Inequality: The Archaeology of Wealth Differences* (University of Arizona Press, 2018), chap. 1. The critics largely fall into the camp of 'post-processual' archaeologists who emphasize the subjectivity of interpreting any historical artefacts. The processual camp, while noting and accounting for cultural subjectivities, tries to find material, scientific ways to account for these and still do comparative analysis.

Processualists are comfortable in trying to quantify inequalities, while post-processual archaeologists tend to lament such attempts. Throughout the book, I'll be firmly sitting in the processual camp and using, where valid, quantified estimates of things like inequality. The problem is that most of the critiques raised by post-processual archaeologists were already known and being addressed by processual experts (Earle et al., 'Processual Archaeology and the Radical Critique [and Comments and Reply]'). Importantly, post-processual critics rarely have any explicit methodology or effective way of trying to compare cases. This tends to result in the far worse vices of wild speculation and a total inability to answer questions about the bigger patterns seen across history (*The Dawn of Everything* by Wengrow and Graeber is the most well-known, entertaining, and bewildering example).↵

7. For Figure 7, war is defined as large-scale, organized lethal violence between different politically autonomous groups. Estimates for the beginning of war are taken from Douglas P. Fry, Charles A. Keith, and Patrik Söderberg, 'Social Complexity, Inequality and War Before Farming: Congruence of Comparative Forager and Archaeological Data', in *Social Inequality Before Farming? Multidisciplinary Approaches to the Study of Social Organization in Prehistoric and Ethnographic Hunter-Gatherer-Fisher Societies*, ed. L. Moreau (McDonald Institute for Archaeological Research, 2020), 314–15, <https://doi.org/10.17863/CAM.60639>; Claudio Cioffi-Revilla and David Lai, 'War and Politics in Ancient China, 2700 B.C. to 722 B.C.: Measurement and Comparative Analysis', *Journal of Conflict Resolution* 39, no. 3 (1995): 467–94; Elizabeth Arkush and Tiffany A. Tung, 'Patterns of War in the Andes from the Archaic to the Late Horizon: Insights from Settlement Patterns and Cranial Trauma', *Journal of Archaeological Research* 21, no. 4 (2013): 307–69, <https://doi.org/10.1007/s10814-013-9065-1>; R. Brian Ferguson, 'The Prehistory of War and Peace in Europe and the Near East', in *War, Peace, and Human Nature*, ed. Douglas P. Fry (Oxford University Press, 2013), 210–22, <https://doi.org/10.1093/acprof:oso/9780199858996.003.0011>; Kent V. Flannery and Joyce Marcus, 'The Origin of War: New ¹⁴C Dates from Ancient Mexico', *Proceedings of the National Academy of Sciences*

- 100, no. 20 (2003): 11801–5,
<https://doi.org/10.1073/pnas.1934526100>. Estimates of state formation are taken from MOROS.↵
8. Also known as the ‘new stone age’. Note that this is an archaeological or cultural period, while the Holocene is a geological one. Technically, at least in Mesopotamia, the time before the first states is usually divided into the Neolithic and the ‘Chalcolithic’ (a time marked by the increased use of smelted copper). My simplification here is done for clarity.↵
9. Peter J. Richerson, Robert Boyd, and Robert L. Bettinger, ‘Was Agriculture Impossible During the Pleistocene But Mandatory During the Holocene? A Climate Change Hypothesis’, *American Antiquity* 66, no. 3 (2001): 387–411, <https://doi.org/10.2307/2694241>. Some have argued that we also saw hierarchical hunter-gatherers and Goliaths during the Palaeolithic since some lootable resources were available. There are a few signs that people harvested whale meat and shellfish as early as 160,000–130,000 years ago. For an overview of the earliest signs of lootable resources, see Singh and Glowacki, ‘Human Social Organization During the Late Pleistocene’, 424–6. This is unlikely to have resulted in hierarchical hunter-gatherers for a few reasons. First, we have no signs of such communities. There are the seasonal burials and monuments in Upper Palaeolithic Europe we have already dealt with: these appear to be more outbursts of social activity rather than any prolonged dominance hierarchy. Second, exit options were also abundant, meaning if people felt threatened then they could easily move to new food sources. Even if a Palaeolithic upstart had tried to lay claim to a coastal source of shellfish, it would have been exceedingly easy to move away. Third, military technology was underdeveloped: there were no monopolizable weapons. Once we think in terms of Goliath fuel it is abundantly clear why the Palaeolithic was marked by nomadic-egalitarian foragers and not hierarchical ones.↵
10. Eric Alden Smith and Brian F. Coddling, ‘Ecological Variation and Institutionalized Inequality in Hunter-Gatherer Societies’, *Proceedings of the National Academy of Sciences* 118, no. 13 (2021): e2016134118, <https://doi.org/10.1073/pnas.2016134118>. See also Singh and

Glowacki, 'Human Social Organization During the Late Pleistocene'. This is the 'behavioural ecology' perspective on hunter-gatherer variation which is dominant in the field. The key idea is that dense, reliable resources tend to lead to more hierarchical social arrangements. There are dissenters, including those who believe that a reliance on salmon was not the main contributor to the hierarchical structure of many of the north-west coast Native Americans; see Kenneth E. Sassaman, 'Complex Hunter-Gatherers in Evolution and History: A North American Perspective', *Journal of Archaeological Research* 12, no. 3 (2004): 227–80, <https://doi.org/10.1023/B:JARE.0000040231.67149.a8>; Graeber and Wengrow, *The Dawn of Everything*; Wengrow and Graeber, "Many Seasons Ago". However, these other accounts which believe that cultural changes are the more important driver suffer from one key problem: they can't explain the wider patterns in the variation of hierarchical hunter-gatherers, while the behavioural ecology view can. Similarly, the behavioural ecology view can help explain why farming societies ended up being brutally unequal and stratified in the long term. The idea of grain as a lootable resource, and that some crops are more lootable than others, was best explored by James C. Scott in a series of influential books, including *The Art of Not Being Governed: An Anarchist History of Upland Southeast Asia* (Yale University Press, 2009), esp. chap. 6, and *Against the Grain: A Deep History of the Earliest States* (Yale University Press, 2017), esp. chap. 4.↵

11. James C. Scott, 'Response', *Cambridge Archaeological Journal* 29, no. 4 (2019): 718, <https://doi.org/10.1017/S0959774319000441>. More speculatively, the flourishes of what could be inequality during the Upper Palaeolithic could also be due to foragers capitalizing on choke points of slightly lootable resources, since large surpluses of meat can be salted or smoked and stored.↵
12. Monique Borgerhoff Mulder et al., 'Intergenerational Wealth Transmission and the Dynamics of Inequality in Small-Scale Societies', *Science* 326, no. 5953 (2009): 682–8, <https://doi.org/10.1126/science.1178336>. Notably, hunter-gatherers do have some degree of transmissible wealth, but it is comparatively tiny.↵

13. The power framework here is a modified version of Mann's sources of social power.↵
14. Lynn H. Gamble, *The Chumash World at European Contact: Power, Trade, and Feasting Among Complex Hunter-Gatherers* (University of California Press, 2011).↵
15. This is a recent overturning of the old idea that horses were directly introduced by Europeans to North America. See William Timothy Treal Taylor et al., 'Early Dispersal of Domestic Horses into the Great Plains and Northern Rockies', *Science* 379, no. 6639 (2023): 1316–23, <https://doi.org/10.1126/science.adc9691>.↵
16. Some have even called them the 'Comanche Empire'. Pekka Hämäläinen, *The Comanche Empire* (Yale University Press, 2008). This is part of a larger literature on 'kinetic' or 'horseback' empires.↵
17. Although most modern horses are derived from a later equine strain originating around 2000 BCE; see Pablo Librado et al., 'The Origins and Spread of Domestic Horses from the Western Eurasian Steppes', *Nature* 598, no. 7882 (2021): 634–40, <https://doi.org/10.1038/s41586-021-04018-9>.↵
18. Sandra L. Vehrencamp, 'A Model for the Evolution of Despotic Versus Egalitarian Societies', *Animal Behaviour* 31, no. 3 (1983): 667–82, [https://doi.org/10.1016/S0003-3472\(83\)80222-X](https://doi.org/10.1016/S0003-3472(83)80222-X).↵
19. This is a slight amendment to Robert Carneiro's original 'circumscription theory'. See Robert L. Carneiro, 'The Circumscription Theory: Challenge and Response', *American Behavioral Scientist* 31, no. 4 (1988): 497–511. Carneiro's original theory has come under significant critique (Terry Stocker and Jianyi Xi'ao, 'Early State Formation: A Complete Rejection of the Circumscription Theory', *Social Evolution & History* 18, no. 2 (2019), <https://doi.org/10.30884/seh/2019.02.09>; Julia Zinkina, Andrey Korotayev, and Alexey Andreev, 'Circumscription Theory of the Origins of the State: A Cross-Cultural Re-Analysis', *Cliodynamics: The Journal of Quantitative History and Cultural Evolution* 7, no. 2 (2017), <https://doi.org/10.21237/C7CLIO7232817>.) My theory of state formation avoids most of the problems by seeing circumscription as just one variable that edged a community towards hierarchy, alongside lootable resources, and military technology. One important criticism is

that circumscription theory can't account for the fact that many early autonomous communities were not conquest orientated. My theory accounts for this by noting that just one warring polity would have likely turned its neighbours more authoritarian (see the 'authoritarian impulse') in Chapter 8.↵

20. Partly inspired by Eric Alden Smith, Jennifer E. Smith, and Brian F. Coddington, 'Toward an Evolutionary Ecology of (in)Equality', *Philosophical Transactions of the Royal Society B: Biological Sciences* 378, no. 1883 (2023): fig. 1, <https://doi.org/10.1098/rstb.2022.0287>.↵
21. It is what physicists call 'an attractor state': a state that the system will tend to evolve towards given initial conditions. We can all make choices, but they are bound and influenced by our past and current material circumstances.↵
22. Rupert Neate, 'All Billionaires under 30 Have Inherited Their Wealth, Research Finds', *Guardians*, 3 April 2024, www.theguardian.com/business/2024/apr/03/all-billionaires-under-30-have-inherited-their-wealth-research-finds.↵
23. A deep dive into violence and the Oaxaca Valley is provided by Flannery and Marcus, *The Creation of Inequality*.↵
24. For Japan, see Hisashi Nakao et al., 'Violence in the Prehistoric Period of Japan: The Spatio-Temporal Pattern of Skeletal Evidence for Violence in the Jomon Period', *Biology Letters* 12, no. 3 (2016): 20160028, <https://doi.org/10.1098/rsbl.2016.0028>; Tomomi Nakagawa et al., 'Violence and Warfare in Prehistoric Japan', *Letters on Evolutionary Behavioral Science* 8, no. 1 (2017): 8–11, <https://doi.org/10.5178/lebs.2017.55>.↵
25. The data for conflict in Native north-eastern America can be found in David H. Dye, 'Trends in Cooperation and Conflict in Native Eastern North America', in *War, Peace, and Human Nature*, ed. Douglas P. Fry (Oxford University Press, 2013), 132–50, <https://doi.org/10.1093/acprof:oso/9780199858996.003.0008>.↵
26. Ferguson, 'The Prehistory of War and Peace in Europe and the Near East'. One could even think of this as a kind of evolutionary backsliding.↵

27. Ibid.[↵](#)
28. The overarching argument on the archaeological sequence of violence is in Douglas P. Fry, Charles A. Keith, and Patrik Söderberg, 'Social Complexity, Inequality and War before Farming: Congruence of Comparative Forager and Archaeological Data', in *Social Inequality Before Farming?*, ed. Luc Moreau (McDonald Institute for Archaeological Research, 2020), <https://doi.org/10.17863/CAM.60639>. For Figure 9, see R. Risch and H. Meller, 'The Representation of Violence in the Rock Art of the Sahara and the Spanish Levant', in *Archéologie Européenne: Identités et Migrations –Hommages au Professeur Jean-Paul Demoule* (Sidestone Press, 2017), 374.[↵](#)
29. Fry, Keith, and Söderberg, 'Social Complexity, Inequality and War Before Farming', 305–10.[↵](#)
30. Christopher Knüsel and Martin J. Smith, eds., *The Routledge Handbook of the Bioarchaeology of Human Conflict* (Taylor & Francis, 2019).[↵](#)
31. Linda Fibiger et al., 'Conflict Violence, and Warfare Among Early Farmers in Northwestern Europe', *Proceedings of the National Academy of Sciences* 120, no. 4 (2023): e2209481119, <https://doi.org/10.1073/pnas.2209481119>.[↵](#)
32. The ability to replace one victim with anyone else from the same group is usually referred to as 'social substitutability'.[↵](#)
33. Malešević, *The Rise of Organised Brutality*.[↵](#)
34. The main proponents of the raiding-for-sex theory of war (a tale in which women are only objects to be fought over) are Gat, *War in Human Civilization*, and Pinker, *The Better Angels of Our Nature*.[↵](#)
35. Fry and Söderberg, 'Lethal Aggression in Mobile Forager Bands', 272; Fry and Söderberg, 'Myths about Hunter-Gatherers Redux', table 1.[↵](#)
36. We should exercise some caution here, as there are weaknesses to the evidence. The archaeological record contains only a small number of skeletons relative to the number of people who lived and died in the Palaeolithic. There are also some perplexing cases. For instance, according to observations from first-contact Europeans (an admittedly unreliable source), Australian Aboriginal people appear to have had

high rates of intergroup conflict. This is also reflected in cave art suggestive of armed confrontations. Yet we see no evidence of group violence in the archaeological record. See Christophe Darmangeat, 'Vanished Wars of Australia: The Archeological Invisibility of Aboriginal Collective Conflicts', *Journal of Archaeological Method and Theory* 26, no. 4 (2019): 1556–90, <https://doi.org/10.1007/s10816-019-09418-w>.↵

37. Lewis Henry Morgan, *Ancient Societies*, ed. Leslie A. White, 2nd edn (Belknap Press of Harvard University Press, 2000). Morgan's work was partly inspired by his time among the Haudenosaunee (previously known as the Iroquois), an alliance of six Native American groups across north-eastern America. He was adopted as part of the Hawk Clan of the Seneca nation and given the name Tayadawahkugh, meaning 'the one lying across'. While they saw him as a bridge between them and their colonial oppressors, Morgan's ideas instead dug a deeper cleft between them. They were now seen as the defunct relics of humanity's immature past. In Morgan's schema, both barbarism and savagery were split into the ascending categories of 'lower', 'middle', and 'upper', while civilization was divided into 'ancient', 'medieval', and 'modern'. Morgan was undoubtedly influenced by evolutionary theory, but never actually referenced evolution in any of his major works. Sir John Lubbock (1834–1913), a contemporary and friend of Morgan, as well as a fellow pioneer of social evolutionism, had a similar schema of savages, barbarians, and the civilized. Unlike Morgan, he did draw a direct connection between evolutionary theory and the progress of humanity. This was a convenient misreading of evolutionary theory. In biological evolution, the species best-adapted to its environment reproduces and lives on. In social evolutionism, evolution has a purpose and a direction.↵
38. We lack detailed comparative studies of the energy capture of egalitarian and hierarchical hunter-gatherers. Nonetheless, it seems reasonable that the latter captured significantly more energy than the former. Egalitarian hunter-gatherers live at a subsistence level and capture little to no energy beyond what they need on a given day to survive. By contrast, hierarchical hunter-gatherers capture enough surplus energy to throw extravagant feasts, to decorate and bury elites with luxury items often requiring hundreds of hours of labour and

longer-distance trade, and more frequently construct large-scale monuments.↵

39. Brian Hayden, *The Power of Feasts: From Prehistory to the Present* (Cambridge University Press, 2014), chap. 4. There are numerous problems with the idea of chiefdoms, including when it's seen as an evolutionary stepping stone towards the state. These critiques are summarized in Norman Yoffee, 'Too Many Chiefs? (Or, Safe Texts for the '90s)', in *Archaeological Theory*, ed. Norman Yoffee and Andrew Sherratt (Cambridge University Press, 1993), 60–78, <https://doi.org/10.1017/CBO9780511720277.007>. Nonetheless, it's a helpful term for the diverse space of centralized, hierarchical groups which haven't quite reached statehood, but are also clearly not leaderless. ↵
40. For a good summary of common criticisms of the stepladder approach (although applied to the more advanced approach of 'band', 'tribe', 'chiefdom', 'state' of the 1960s), see Yoffee, 'Too Many Chiefs?'.↵
41. Tainter, *The Collapse of Complex Societies*, 4; Diamond, *Collapse*, 3.↵
42. Bruce G. Trigger, *Understanding Early Civilizations: A Comparative Study* (Cambridge University Press, 2007), 46.↵
43. This definition of complexity largely dates back to the cultural evolution literature of the 1950s and 1960s. It still remains the predominant way of talking about complexity in archaeology. When most archaeologists talk about 'complex societies' they are referring to more unequal, centralized, and hierarchical ones. There are, of course, exceptions. Tim Kohler provides a useful summary of how archaeologists can approach the topic from the lens of complex systems and reviews some relevant books and articles in Timothy A. Kohler, 'Complex Systems and Archaeology', in *Archaeological Theory Today*, ed. Ian Hodder (Polity Press, 2012), 93–123. ↵
44. For similar points on how states and administrations simplify their citizens and landscapes, see Yoffee, *The Evolution of Simplicity*; Scott, *Seeing Like a State*. For a general elaboration of this argument, see Luke Kemp, 'Diminishing Returns on Extraction: How Inequality and Extractive Hierarchy Create Fragility', in *What History, Systems, and Complexity Can Teach Us About Our Modern World and Fragile*

Future, ed. Miguel A. Centeno, Thayer S. Patterson, and Peter W. Callah (Routledge, 2023),

[www.taylorfrancis.com/chapters/edit/10.4324/9781003331384-5/diminishing-returns-extraction-luke-kemp?](http://www.taylorfrancis.com/chapters/edit/10.4324/9781003331384-5/diminishing-returns-extraction-luke-kemp?context=ubx&refId=9e978dc6-db65-4abe-b79a-a6b38531c307)

[context=ubx&refId=9e978dc6-db65-4abe-b79a-a6b38531c307](http://www.taylorfrancis.com/chapters/edit/10.4324/9781003331384-5/diminishing-returns-extraction-luke-kemp?context=ubx&refId=9e978dc6-db65-4abe-b79a-a6b38531c307). Some archaeologists and anthropologists, such as Joseph Tainter, do use a definition of complexity which speaks of the number of different moving parts in a system (although this is still tied up with inequality, centralization, and hierarchy). Even here they seem to be thinking only about complexity in areas such as military, capital, and bureaucracy.↵

45. In contrast, a non-hierarchical arrangement means that everyone is unranked, or at least that there are no permanent, overarching positions of authority. A merchant may have greater bartering skills than a soldier, but neither has direct authority over the other. This is usually referred to as ‘heterarchy’: see Carole L. Crumley, ‘Heterarchy and the Analysis of Complex Societies’, *Archaeological Papers of the American Anthropological Association* 6, no. 1 (1995): 3, <https://doi.org/10.1525/ap3a.1995.6.1.1>. However, the term is a little confusing.↵
46. David Wengrow, ‘A History of True Civilisation Is Not One of Monuments’, *Aeon*, 2 October 2018, <https://aeon.co/ideas/a-history-of-true-civilisation-is-not-one-of-monuments>. There was of course still some coercion: counter-dominance strategies. But such lethal violence was shared across the community, not commonly deployed, and rarely needed to enforce rules. ↵
47. What Is Politics? 10.3 *The Ingredients of Hierarchy: Graeber & Wengrow’s Dawn of Everything*, Chapter 3, 2022, www.youtube.com/watch?v=nsIxMzLjEfs.↵
48. There is a prolific literature on ‘heterarchy’, but this isn’t easily understood (or even very helpful) for a general audience. Similarly, corporate vs despotic power in archaeology is both incredibly vague and confusing to most people who associate ‘corporate’ with private companies. A simple, back-to-basics approach is warranted here.↵
49. Paul Cartledge, ‘Ancient Greeks Would Not Recognise Our “Democracy” – They’d See an “Oligarchy”’, *The Conversation*, 3 June

2016, <https://theconversation.com/ancient-greeks-would-not-recognise-our-democracy-theyd-see-an-oligarchy-60277>; Martin Gilens and Benjamin I. Page, 'Testing Theories of American Politics: Elites, Interest Groups, and Average Citizens', *Perspectives on Politics* 12, no. 3 (2014): 564–81, <https://doi.org/10.1017/S1537592714001595>.↵

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CHAPTER 4: ABANDONED EXPERIMENTS I: FAILED FARMERS

1. Ferguson, 'The Prehistory of War and Peace in Europe and the Near East'; Douglas P. Fry and Geneviève Souillac, 'Prehistoric Warfare', in *Oxford Bibliographies in Military History* (2023), <https://doi.org/10.1093/obo/9780199791279-0224>.↵
2. Fry and Souillac, 'Prehistoric Warfare'; Augusta McMahon, 'From Sedentism to States, 10,000–3000 BCE', in *A Companion to the Ancient Near East*, ed. Daniel C. Snell (Wiley, 2020), 27–43, <https://doi.org/10.1002/9781119362500>, chap. 2.↵
3. Oana Borcan, Ola Olsson, and Louis Putterman, 'Transition to Agriculture and First State Presence: A Global Analysis', *Explorations in Economic History* 82 (2021): 1, <https://doi.org/10.1016/j.eeh.2021.101404>.↵
4. Unintentional domestication of crops dates back at least to 25,000 years ago as hunter-gatherers developed new tools and killed off large mammals. In a lighter sense it began 780,000 years ago with the first use of fire to shape landscapes, selecting for more fire-tolerant plants (especially pyrophytes and pyrophiles) while killing off others. Robin G. Allaby et al., 'Emerging Evidence of Plant Domestication as a Landscape-Level Process', *Trends in Ecology & Evolution* 37, no. 3 (2022): 268–79, <https://doi.org/10.1016/j.tree.2021.11.002>; ↵
5. Amaia Arranz-Otaegui and Joe Roe, 'Revisiting the Concept of the "Neolithic Founder Crops" in Southwest Asia', *Vegetation History and Archaeobotany* 32, no. 5 (2023): 482, <https://doi.org/10.1007/s00334-023-00917-1>.↵
6. Gary M. Feinman and Jill E. Neitzel, 'The Social Dynamics of Settling Down', *Journal of Anthropological Archaeology* 69 (2023): 101468, <https://doi.org/10.1016/j.jaa.2022.101468>.↵
7. Stephen Shennan, *The First Farmers of Europe: An Evolutionary Perspective* (Cambridge University Press, 2018), <https://doi.org/10.1017/9781108386029>. The most recent evidence suggests that around 3.6 per cent of the migrating farmers interbred with the local hunter-gatherers: Joaquim Fort and Joaquim Pérez-

Losada, 'Interbreeding Between Farmers and Hunter-Gatherers Along the Inland and Mediterranean Routes of Neolithic Spread in Europe', *Nature Communications* 15, no. 1 (2024): 7032, <https://doi.org/10.1038/s41467-024-51335-4>.↵

8. Some put it down to a combination of population growth, technological innovation, and climate change: Gregory K. Dow and Clyde G. Reed, 'The Origins of Sedentism: Climate, Population, and Technology', *Journal of Economic Behavior & Organization* 119 (2015): 56–71, <https://doi.org/10.1016/j.jebo.2015.07.007>.↵
9. There are some important caveats: some regions strangely started getting smaller *before* picking up agriculture, while others experienced increases as they became lactose tolerant and more reliant on animal products (what's known as the 'lactase growth hypothesis'). Nonetheless, the general picture between ancient foragers and post-ice-age farmers is one of reducing height. The most recent and nuanced study of 3,507 skeletons from 366 sites can be found in Jay T. Stock et al., 'Long-Term Trends in Human Body Size Track Regional Variation in Subsistence Transitions and Growth Acceleration Linked to Dairying', *Proceedings of the National Academy of Sciences* 120, no. 4 (2023): 1, 3–7, <https://doi.org/10.1073/pnas.2209482119>. For a useful overview of the literature, consult Amanda Mummert et al., 'Stature and Robusticity during the Agricultural Transition: Evidence from the Bioarchaeological Record', *Economics & Human Biology* 9, no. 3 (2011): 284–301, <https://doi.org/10.1016/j.ehb.2011.03.004>. If you're looking for a concrete number, then one of the most robust is that early European farmers were 3.82cm shorter than expected (given their genetics) relative to Palaeolithic hunter-gatherers. See Stephanie Marciniak et al., 'An Integrative Skeletal and Paleogenomic Analysis of Stature Variation Suggests Relatively Reduced Health for Early European Farmers', *Proceedings of the National Academy of Sciences* 119, no. 15 (2022): 1, <https://doi.org/10.1073/pnas.2106743119>.↵
10. Theya Ivitsky Molleson, 'The People of Abu Hureyra', in *Village on the Euphrates: From Foraging to Farming at Abu Hureyra*, ed. M. T. Moore, G. C. Hillman, and A. J. Legge (Oxford University Press, 2000), 301–24. The earliest Neolithic villages display evidence of substantial oral infections and ubiquitous enamel defects – Clark

Spencer Larsen et al., ‘Bioarchaeology of Neolithic Çatalhöyük Reveals Fundamental Transitions in Health, Mobility, and Lifestyle in Early Farmers’, *Proceedings of the National Academy of Sciences* 116, no. 26 (2019): 12621, <https://doi.org/10.1073/pnas.1904345116>. The association between ill health and agriculture was not lost on our ancestors. The meaning of the Greek root for the word parasite is ‘beside the grain’.↵

11. Happiness is, in many ways, ensuring that your desires are met. You can do so by either producing a lot or desiring little. Egalitarian hunter-gatherers take the second path, wanting mainly friendship and leisure time over the endless pursuit of more material luxuries. These tendencies have earned egalitarian foragers the mantle of ‘the original affluent society’. See Marshall Sahlins, *Stone Age Economics* (Routledge Classics, 2017), chap. 1.↵
12. On peasants rarely producing a surplus, see Scott, *Against the Grain*, 152, 270; A. V. Chayanov, *The Theory of Peasant Economy* (University of Wisconsin Press, 1986). When a peasant family has more working than non-working members they will also reduce their activity once subsistence is reached. They aren’t particularly interested in producing surplus food. Richard B. Lee, Irvén DeVore, and Wenner-Gren Foundation for Anthropological Research, eds., *Man the Hunter* (Aldine Transaction, 1968); Ronald M. Berndt and Catherine Berndt, *Man, Land and Myth in North Australia: The Gunwinggu People* (Michigan State University Press, 1970); Hayden, *The Power of Feasts*.↵
13. For a review and nuanced discussion of the principle of least effort, see B. G. Trigger, ‘Monumental Architecture: A Thermodynamic Explanation of Symbolic Behaviour’, *World Archaeology* 22, no. 2 (1990): 122–4, <https://doi.org/10.1080/00438243.1990.9980135>. The principle of least effort doesn’t govern all human activities. Foragers will still spend a fair bit of time acquiring foods that are renowned for their flavour or prestige. But it’s accurate as a rule of thumb when it comes to the daily business of feeding a group.↵
14. For an overview, see Chris J. Stevens, Enrico R. Crema, and Shinya Shoda, ‘The Importance of Wild Resources as a Reflection of the Resilience and Changing Nature of Early Agricultural Systems in East

- Asia and Europe', *Frontiers in Ecology and Evolution* 10 (2022): 10, <https://doi.org/10.3389/fevo.2022.1017909>.↵
15. For the floodplain hypothesis, see Scott, *Against the Grain*, 63–7. For a broader discussion, especially on the importance of wider demography, see Shennan, *The First Farmers of Europe*.↵
 16. At the individual level, modern analyses suggest that higher reliance on farming correlates strongly with higher fertility. However, context and culture matter significantly. See Abigail E. Page et al., 'Women's Subsistence Strategies Predict Fertility Across Cultures, But Context Matters', *Proceedings of the National Academy of Sciences* 121, no. 9 (2024): e2318181121, <https://doi.org/10.1073/pnas.2318181121>.↵
 17. Scott, *Against the Grain*, 47–56; Hayden, *The Power of Feasts*, chap. 5.↵
 18. Hayden, *The Power of Feasts*, chap. 4.↵
 19. Ibid., chap. 5; Brian Hayden and Suzanne Villeneuve, 'A Century of Feasting Studies', *Annual Review of Anthropology* 40, no. 1 (2011): 433–49, <https://doi.org/10.1146/annurev-anthro-081309-145740>. Feasting can also offer other benefits, including creating social alliances: see Brian Hayden, 'The Proof Is in the Pudding: Feasting and the Origins of Domestication', *Current Anthropology* 50, no. 5 (2009): 597–601, <https://doi.org/10.1086/605110>.↵
 20. Trigger, 'Monumental Architecture'. ↵
 21. Zuzana Hofmanová et al., 'Early Farmers from across Europe Directly Descended from Neolithic Aegeans', *Proceedings of the National Academy of Sciences* 113, no. 25 (2016): 6886–91, <https://doi.org/10.1073/pnas.1523951113>.↵
 22. Shennan, *The First Farmers of Europe*.↵
 23. Detlef Gronenborn and Barbara Horejs, 'Expansion of Farming in Western Eurasia, 9600 – 4000 Cal BC (Update Vers. 2023.1)', Zenodo (2023), <https://doi.org/10.5281/ZENODO.10047818>.↵
 24. Stephen Shennan et al., 'Regional Population Collapse Followed Initial Agriculture Booms in Mid-Holocene Europe', *Nature Communications* 4, no. 1 (2013): 2486, <https://doi.org/10.1038/ncomms3486>.↵

25. Shennan, *The First Farmers of Europe*.[↵](#)
26. Fibiger et al., ‘Conflict, Violence, and Warfare Among Early Farmers in Northwestern Europe’, 8.[↵](#)
27. Shennan, *The First Farmers of Europe*; Bevan et al., ‘Holocene Fluctuations in Human Population’.[↵](#)
28. Shennan, *The First Farmers of Europe*.[↵](#)
29. Fibiger et al., ‘Conflict, Violence, and Warfare Among Early Farmers in Northwestern Europe’.[↵](#)
30. The Navajo term ‘Anasazi’, which means ‘ancient enemy’, was previously used and understandably discarded.[↵](#)
31. Timothy A. Kohler, R. Kyle Bocinsky, and Darcy Bird, ‘Fluctuat Nec Mergitur’, in *How Worlds Collapse*, ed. Miguel Centeno et al. (Routledge, 2023), 159–60, <https://doi.org/10.4324/9781003331384-11>.[↵](#)
32. Ibid., 160. On the declining rate of violence, see Timothy A. Kohler et al., ‘The Better Angels of Their Nature: Declining Violence Through Time Among Prehispanic Farmers of the Pueblo Southwest’, *American Antiquity* 79, no. 3 (2014): 444–64, <https://doi.org/10.7183/0002-7316.79.3.444>.[↵](#)
33. Bevan et al., ‘Holocene Fluctuations in Human Population’.[↵](#)
34. Scott, *Against the Grain*, 73, 124. Scott refers to this unprecedented concentration of grain, animals, and humans, as ‘Neolithic resettlement camps’.[↵](#)
35. Ibid., chap. 3.[↵](#)
36. Kathleen DuVal, *Native Nations: A Millennium in North America* (Random House, 2024), chap. 3; Scott, *Against the Grain*, chaps. 3 and 5. This is especially true for those that had yet to develop practices of quarantine and containment (although there are references to separating the sick from the healthy dating back to Leviticus in the Bible).[↵](#)
37. Frederik Valeur Seersholm et al., ‘Repeated Plague Infections Across Six Generations of Neolithic Farmers’, *Nature* 632, no. 8023 (2024): 114–21, <https://doi.org/10.1038/s41586-024-07651-2>.[↵](#)

38. Scott, *Against the Grain*, 97. Most acute infections don't leave a trace on bones, and ancient DNA studies are still relatively new.↵
39. Gregory Johnson, 'Organizational Structure and Scalar Stress', in *Theory and Explanation in Archaeology*, ed. Colin Renfrew, Michael Rowlands, and Barbara A. Segraves-Whallon (Academic Press, 1982), 389–421. On the super-linear scaling of some crimes, see Marcos Oliveira, 'More Crime in Cities? On the Scaling Laws of Crime and the Inadequacy of Per Capita Rankings – A Cross-Country Study', *Crime Science* 10, no. 1 (2021): 27, <https://doi.org/10.1186/s40163-021-00155-8>.↵
40. Dániel Kondor et al., 'Explaining Population Booms and Busts in Mid-Holocene Europe', preprint, SocArXiv, 11 May 2022, <https://doi.org/10.31235/osf.io/c32up>.↵

CHAPTER 5: ABANDONED EXPERIMENTS II: VOTING WITH FEET AND FIRE

1. Louis Wirth, 'Urbanism as a Way of Life', *American Journal of Sociology* 44, no. 1 (1938): 1. This is usually known as the sociological definition of a city and is widely used in archaeology alongside the functional definitions which stress how the activities in a settlement's centre impact the surrounding rural areas. There is no clear quantitative definition for a city; however, under these definitions many of the cases covered here would be cities, including Taosi, Cahokia, Moundville, Tiwanaku, Monte Albán, and San Lorenzo. Others, such as Jenne-Jeno, are more disputable. It's worth noting that cities aren't always permanent, but that is a realm of scholarly contestation which is beyond the scope of this book.↵
2. For an overview of the archaeology of Çatalhöyük, see Ian Hodder, *Çatalhöyük, the Leopard's Tale: Revealing the Mysteries of Turkey's Ancient Town* (Thames & Hudson, 2011); Ian Hodder, 'Çatalhöyük: The Leopard Changes Its Spots. A Summary of Recent Work', *Anatolian Studies* 64 (2014): 1–22, <https://doi.org/10.1017/S0066154614000027>; Justin Jennings, *Killing Civilization: A Reassessment of Early Urbanism and Its Consequences* (University of New Mexico Press, 2016), chap. 4. Note that there were

signs of interpersonal violence. More recent evidence suggests that the population of the city could have been as low as 600–800: Ian Kuijt and Arkadiusz Marciniak, ‘How Many People Lived in the World’s Earliest Villages? Reconsidering Community Size and Population Pressure at Neolithic Çatalhöyük’, *Journal of Anthropological Archaeology* 74 (2024): 101573, <https://doi.org/10.1016/j.jaa.2024.101573>. If this is true, Çatalhöyük suddenly looks much less exceptional. It is just another Neolithic village, albeit one with a fascinating culture. Such a low population would also undermine the idea that the abandonment of Çatalhöyük was due to scalar stress, and lend more weight to the idea of this being a slow, intentional move away from inequality.↵

3. Hodder, *Çatalhöyük, the Leopard’s Tale*, 98.↵
4. Jennings, *Killing Civilization*, chap. 4; Samuel Bowles and Mattia Fochesato, ‘The Origins of Enduring Economic Inequality’, Santa Fe Institute, 31 January 2024, 29–33, <https://sites.santafe.edu/~bowles/wp-content/uploads/Text-JEL-31-Jan-sent-.pdf>.↵
5. Larsen et al., ‘Bioarchaeology of Neolithic Çatalhöyük’.↵
6. Jennings, *Killing Civilization*, chap. 4; Lech Czerniak and Arkadiusz Marciniak, ‘Abandoning Çatalhöyük: Re-Shuffling, Re-Location and Migration as the Means of Mitigating Social Unease in the Late Neolithic’, in *6000 BC*, ed. Peter F. Biehl and Eva Rosenstock (Cambridge University Press, 2022), 136–57, <https://doi.org/10.1017/9781107337640.011>.↵
7. Justin Jennings et al., ‘Cities, Surplus, and the State: A Re-Evaluation’, *Journal of Urban Archaeology*, 2021, 19–21, <https://doi.org/10.1484/J.JUA.5.126592>; Jennings, *Killing Civilization*, chap. 7.↵
8. Jennings et al., ‘Cities, Surplus, and the State’, 19–21; Jennings, *Killing Civilization*, chap. 7.↵
9. Jennings et al., ‘Cities, Surplus, and the State’, 20–21.↵
10. For an overview, see David Wengrow, ‘The Origins of Civic Life – A Global Perspective’, *Origini: Preistoria e Protostoria Delle Civiltà*

Antiche – Prehistory and Protohistory of Ancient Civilizations 42, no. 2 (2018): 25–44.↵

11. He Nu, 'Taosi: An Archaeological Example of Urbanization as a Political Center in Prehistoric China', *Archaeological Research in Asia* 14 (2018): 21, 26, <https://doi.org/10.1016/j.ara.2017.03.001>.↵
12. Ibid., 24; He Nu, 'The Longshan Period Site of Taosi in Southern Shanxi Province', in *A Companion to Chinese Archaeology*, ed. Anne P. Underhill (Wiley, 2013), 269.↵
13. Jennings et al., 'Cities, Surplus, and the State', 22.↵
14. David Cheetham and Jeffrey P. Blomster, 'Materializing the San Lorenzo Olmecs', in *The Early Olmec and Mesoamerica*, ed. Jeffrey P. Blomster and David Cheetham (Cambridge University Press, 2017), 9–36, <https://doi.org/10.1017/9781316228098.002>; Jennings et al., 'Cities, Surplus, and the State', 21–3.↵
15. Some contend it may have allowed the use of forearms and even paddles.↵
16. Linda M. Nicholas and Gary M. Feinman, 'The Foundation of Monte Albán, Intensification, and Growth: Coactive Processes and Joint Production', *Frontiers in Political Science* 4 (2022): 4–6, <https://doi.org/10.3389/fpos.2022.805047>.↵
17. There was also a temporary disruption to its collective rule in 200 CE as Monte Albán expanded. This was reversed within decades, for what exact reason we do not know.↵
18. Gary M. Feinman and Linda M. Nicholas, 'After Monte Albán in the Central Valleys of Oaxaca: A Reassessment', in *Beyond Collapse: Archaeological Perspectives on Resilience, Revitalization, and Transformation in Complex Societies*, ed. Ronald K. Faulseit (Southern Illinois University Press, 2016), 59. For an accessible overview of the Monte Albán site, see Jennings, *Killing Civilization*, chap. 9.↵
19. It may have been the seat of an empire, or just an oversized city. For an overview of the debates about whether Teotihuacan was centralized or decentralized, see Tom Froese, Carlos Gershenson, and Linda R. Manzanilla, 'Can Government Be Self-Organized? A Mathematical Model of the Collective Social Organization of Ancient Teotihuacan,

- Central Mexico', *PLOS ONE* 9, no. 10 (2014): e109966, <https://doi.org/10.1371/journal.pone.0109966>.↵
20. George L. Cowgill, *Ancient Teotihuacan: Early Urbanism in Central Mexico* (Cambridge University Press, 2015); Kenn Hirth, David M. Carballo, and Bárbara Arroyo, eds., *Teotihuacan: The World Beyond the City* (Dumbarton Oaks Research Library and Collection, 2020). Smith et al. estimate the Gini coefficient of Teotihuacan to be 0.12 (Michael E. Smith et al., 'Quantitative Measures of Wealth Inequality in Ancient Central Mexican Communities', *Advances in Archaeological Practice* 2, no. 4 (2014): 311–23, <https://doi.org/10.7183/2326-3768.2.4.XX>, 321). In contrast, the Gini of Denmark is around 0.35.↵
21. For good overviews, see Guy Middleton, *Understanding Collapse: Ancient History and Modern Myths* (Cambridge University Press, 2017), 222–43; Deborah L. Nichols and Ryan H. Collins, 'Episodes of the Feathered Serpent: Aztec Imperialism and Collapse', in *How Worlds Collapse*, ed. Miguel Centeno et al. (Routledge, 2023), <https://doi.org/10.4324/9781003331384-12>, 174–6. The elite revolt theory is best advocated for by Linda R. Manzanilla, 'Cooperation and Tensions in Multiethnic Corporate Societies Using Teotihuacan, Central Mexico, as a Case Study', *Proceedings of the National Academy of Sciences* 112, no. 30 (2015): 9210–15, <https://doi.org/10.1073/pnas.1419881112>. On climate change, see Jungjae Park, Roger Byrne, and Harald Böhnelt, 'Late Holocene Climate Change in Central Mexico and the Decline of Teotihuacan', *Annals of the American Association of Geographers* 109, no. 1 (2019): 104–20, <https://doi.org/10.1080/24694452.2018.1488577>.↵
22. For an overview, see Middleton, *Understanding Collapse*.↵
23. Justin Jennings and Timothy K. Earle, 'Urbanization, State Formation, and Cooperation: A Reappraisal', *Current Anthropology* 57, no. 4 (2016): 474–93, <https://doi.org/10.1086/687510>.↵
24. John Wayne Janusek, *Ancient Tiwanaku* (Cambridge University Press, 2008), 191.↵
25. Kenny Sims, 'After State Collapse: How Tumilaca Communities Developed in the Upper Moquegua Valley, Peru', in *After Collapse: The Regeneration of Complex Societies*, ed. Glenn M. Schwartz and

- John Jackson Nichols (University of Arizona Press, 2006), 114–37; Nicola Sharratt, ‘Collapse and Cohesion: Building Community in the Aftermath of Tiwanaku State Breakdown’, *World Archaeology* 48, no. 1 (2016): 144–63, <https://doi.org/10.1080/00438243.2015.1110043>; Janusek, *Ancient Tiwanaku*. For decent overviews of recent archaeology on collapse in the site, see Middleton, *Understanding Collapse*, 276–98; Guy D. Middleton, ‘Collapse Studies in Archaeology from 2012 to 2023’, *Journal of Archaeological Research*, 9 (2024): 33–5, <https://doi.org/10.1007/s10814-024-09196-4>.↵
26. See Jason Nesbitt’s coverage of Tiwanaku in Kemp et al., ‘Societal Collapse as a Powershift’.↵
 27. For an overview, see Middleton, *Understanding Collapse*, 276–98. Pukura was built around a thirty-two-metre-high terraced platform that was used for rituals. It was a publicly accessible monument. The first signs of inequality emerge with some trying to restrict access to the platform and control the production of obsidian, polychrome pottery, and stone sculptures. The site was rapidly abandoned as people returned to autonomous villages. See Jennings and Earle, ‘Urbanization, State Formation, and Cooperation’, 478; Janusek, *Ancient Tiwanaku*, 88–95.↵
 28. I’ve left Taosi and Çatalhöyük out of this graph, as they are far older and extend the timeline too far.↵
 29. There were differences: Çatalhöyük was not a city and its abandonment took centuries, while Tiwanaku was a state that fell apart in decades.↵
 30. Pauketat, *Cahokia*, 131–4; DuVal, *Native Nations*, chap. 1. It also may have helped decrease conflict. Like most later empires, the borders around Cahokia’s territory became hotter with conflict, while its own territory was cooled with less violence. Some have even compared it to the peace that the Roman Empire provided to conquered peoples: the *Pax Romana*. However, the idea of *Pax Cahokia* is currently disputed.↵
 31. For an overview of the timeline of Cahokia and key contributors to its collapse, see Benson, Pauketat, and Cook, ‘Cahokia’s Boom and Bust in the Context of Climate Change’; Tainter, ‘Cahokia’; DuVal, *Native Nations*, chap. 2.↵

32. Note that, as covered earlier, ‘Goliath’ does not mean ‘state’. Few would argue that Cahokia, the Huhugam, the Puebloans, or the multiple mound-based Mississippi mini-kingdom ever became fully fledged states. That doesn’t change the fact that they were becoming unequal dominance hierarchies in which rules were imposed through force.↵
33. Pauketat, *Cahokia*, chap. 10.↵
34. Moundville looks much like Cahokia but with some important differences. The most noticeable is the absence of any sign of slavery, coerced labour, or human sacrifice. ↵
35. Vernon James Knight Jr and Vincas P. Steponaitis, ‘A New History of Moundville’, in *Archaeology of the Moundville Chiefdom* (University of Alabama Press, 1998), 1–25.↵
36. For an overview, see DuVal, *Native Nations*, chaps. 1, 2.↵
37. In the final stages of the collapse the most populous areas were having to hunt more and different kinds of animals, suggesting they were running low on food.↵
38. On the collapse period, consult Rebecca M. Dean, ‘Hunting Intensification and the Hohokam “Collapse”’, *Journal of Anthropological Archaeology* 26, no. 1 (2007): 109–32, <https://doi.org/10.1016/j.jaa.2006.03.010>; J. Brett Hill et al., ‘The “Collapse” of Cooperative Hohokam Irrigation in the Lower Salt River Valley’, *Journal of the Southwest* 57, no. 4 (2015): 609–74; DuVal, *Native Nations*, chap. 2.↵
39. David J. Hally, ‘The Settlement Patterns of Mississippian Chiefdoms in Northern Georgia’, in *Settlement Pattern Studies in the Americas: Fifty Years Since Virú*, ed. Brian R. Billman and Gary M. Feinman (Smithsonian Institution Scholarly Press, 1999), chap. 8; John H. Blitz, ‘Mississippian Chiefdoms and the Fission-Fusion Process’, *American Antiquity* 64, no. 4 (1999): 577–92, <https://doi.org/10.2307/2694206>. For the model of prestige and status being central to both individual and group warfare in the Mississippian kingdoms, see Karl T. Steinen, ‘Ambushes, Raids, and Palisades: Mississippian Warfare in the Interior Southeast’, *Southeastern Archaeology* 11, no. 2 (1992): 132–7. On a lack of control mechanisms, see Jaeweon Shin et al., ‘Scale and

Information-Processing Thresholds in Holocene Social Evolution’, *Nature Communications* 11, no. 1 (2020): 2394, <https://doi.org/10.1038/s41467-020-16035-9>. Control mechanisms in the original research are framed as ‘information processing’ mechanisms, such as text, money, writing, and an administrative hierarchy.↵

40. Hayden, *The Power of Ritual in Prehistory*; Hayden, *The Power of Feasts*.↵
41. Benson, Pauketat, and Cook, ‘Cahokia’s Boom and Bust in the Context of Climate Change’; David P. Pompeani et al., ‘Severe Little Ice Age Drought in the Midcontinental United States During the Mississippian Abandonment of Cahokia’, *Scientific Reports* 11, no. 1 (2021): 13829, <https://doi.org/10.1038/s41598-021-92900-x>. For an overview of temperature change in the US south-west and the predominance of cooler, more drought-prone conditions, see Andrew Gillreath-Brown, R. Kyle Bocinsky, and Timothy A. Kohler, ‘A Low-Frequency Summer Temperature Reconstruction for the United States Southwest, 3000 BC–AD 2000’, *The Holocene* 34, no. 4 (2024): 451–66, <https://doi.org/10.1177/09596836231219482>.↵
42. Washington Matthews, ‘Noqoilpi, the Gambler: A Navajo Myth’, *Journal of American Folklore* 2, no. 5 (1889): 89, <https://doi.org/10.2307/533316>. Much like modern investors he turned gambling into enormous wealth. See also Stephen H. Lekson and Catherine M. Cameron, ‘The Abandonment of Chaco Canyon, the Mesa Verde Migrations, and the Reorganization of the Pueblo World’, *Journal of Anthropological Archaeology* 14, no. 2 (1995): 184–202, <https://doi.org/10.1006/jaar.1995.1010>. There are many well-known problems with oral traditions, including their lack of precision in chronology (e.g. a lack of precise dates), often wide variation in tales, and problems with non-locals interpreting highly contextual stories. Despite this, there is an emerging acknowledgement that oral traditions can be as reliable and valuable as written texts if used wisely. For some accessible discussions, see Stephen E. Nash, ‘The Underestimated Reliability of Oral Histories’, *The Atlantic*, 22 September 2018, www.theatlantic.com/science/archive/2018/09/dead-sea-scrolls-oral-written-history/571039/; Scott, *The Art of Not Being*

Governed, chap. 6.5. Oral traditions are now regularly used across the historical sciences. [↵](#)

43. Joshua B. Nelson, *Progressive Traditions: Identity in Cherokee Literature and Culture* (University of Oklahoma Press, 2014), 132–3. [↵](#)
44. Lynn S. Teague, ‘Prehistory and the Traditions of the O’Odham and Hopi’, *KIVA* 58, no. 4 (1993): 438–42, <https://doi.org/10.1080/00231940.1993.11758220>. [↵](#)
45. For a summary of the egalitarian power-sharing arrangements post-collapse, see DuVal, *Native Nations*, chap. 2. [↵](#)
46. I’m indebted to Tim Kohler for raising this hypothesis with me during an initial round of reviews for this book. [↵](#)
47. Blitz, ‘Mississippian Chiefdoms and the Fission-Fusion Process’. [↵](#)
48. Jennings et al., ‘Cities, Surplus, and the State’; Jennings, *Killing Civilization*; Jennings and Earle, ‘Urbanization, State Formation, and Cooperation’. [↵](#)
49. Christopher I. Beckwith, *Empires of the Silk Road: A History of Central Eurasia from the Bronze Age to the Present* (Princeton University Press, 2011). [↵](#)
50. Gary M. Feinman and Linda M. Nicholas, ‘Framing Inequality in Ancient Civilizations’, in *Poverty and Inequality in Early Civilizations*, ed. Richard Bussman and Tobias Helms (Verlag Dr. Rudolf Habelt GmbH, 2020), 107–17, 113. [↵](#)
51. Dan Lawrence et al., ‘Long-Term Trends in Settlement Persistence in Southwest Asia: Implications for Sustainable Urbanism, Past, Present and Future’, *Urban Studies* 62, no. 3 (2023): 543–59, <https://doi.org/10.1177/00420980231161245>. One caveat here is that their measurement of centralization and hierarchy (what they call ‘complexity’) is time. That is, in general, settlements tend to become more unequal, hierarchical, stratified, and war-like as centuries pass. Compare Çatalhöyük to the Assyrian Empire. While imperfect, this is a decent proxy, and one of the few we can use, given the available evidence. It could also highlight an additional cause for declining longevity: conquest. As the first Goliaths arose, they razed and sacked many nearby settlements. For many that remained, the incentive to flee

from these targets must have been strong. We can then expect that, over time, unequal settlements would start to become more long-lasting while egalitarian ones would lose their longevity. While this is the best evidence we have for now, we should stay sceptical. The analysis is based on the sparse remains of ceramics, which can easily overlook a hiatus in occupation of one or two centuries (personal communication from Tim Kohler).↵

52. There is an interesting preliminary analysis of hundreds of sites covering 20,000 years (using the GINI Database) which suggests that inequality is slightly correlated with greater longevity. This correlation increases over time as settlements get bigger. Larger settlements in larger systems also tend to last longer. See Dan Lawrence et al., ‘Two Sides of Sustainability: Comparing Trends in Wealth Inequality and Settlement Persistence in Preindustrial Societies’, *Proceedings of the National Academy of Sciences* (forthcoming). At first blush, this may seem to be in tension with the argument. That’s not the case. First, it can’t tell as much about the first Goliaths per se since it is covering a far larger span of time, including after Goliaths become established and spread. Second, the authors themselves highlight that this is not a case of inequality leading to more persistent settlements over time. Instead, to me, the most likely explanation here is that the long-lasting settlements are the centres of larger systems of extraction (Goliaths). As time goes on and Goliaths become more established, these settlements become less likely to be abandoned. They eat and destroy smaller, more egalitarian settlements (hence decreasing their persistence) and become more embedded in the culture and landscape. As we’ll see in Part II, Goliaths both tend to be resurrected (meaning big sites are rarely completely abandoned) and to spread at the expense of other societies.↵
53. Stéphen Rostain et al., ‘Two Thousand Years of Garden Urbanism in the Upper Amazon’, *Science* 383, no. 6679 (2024): 183–9, <https://doi.org/10.1126/science.adi6317>; Heiko Prümers et al., ‘Lidar Reveals Pre-Hispanic Low-Density Urbanism in the Bolivian Amazon’, *Nature* 606, no. 7913 (2022): 325–8, <https://doi.org/10.1038/s41586-022-04780-4>.↵

CHAPTER 6: GOLIATH RISES

1. Pauline Wilson Wiessner, Akii Tumu, and Nitze Pupu, *Historical Vines: Enga Networks of Exchange, Ritual, and Warfare in Papua New Guinea* (Smithsonian Institution Press, 1998).[↵](#)
2. Adapted from Fuller et al., ‘Convergent Evolution and Parallelism in Plant Domestication Revealed by an Expanding Archaeological Record’, fig. 1. Note, this only includes archaeologically documented cases, not biogeographically inferred. These don’t change the underlying conclusions. Eastern Tibet domesticated buckwheat and yak. While it did eventually develop a state – the Tibetan Empire in 618 CE – this was after interaction with the Chinese. Hence it usually isn’t considered an original state. Amazonia cultivated peanuts and manioc, neither of which are effective Goliath fuel. The western Savannah had cowpea, fonio, and African rice. It is in the direct vicinity of the western Sahel, and African rice was a staple of the Ghana Empire. The eastern Savannah domesticated cattle and sorghum. Notably, domesticated cattle did lead to hierarchical pastoralist groups, and eventually confederations of raiders known as the Ṣanhājah.[↵](#)
3. In most of the literature these are usually referred to as ‘pristine states’. I’ve used the term ‘original’ here instead since it is more intuitive and avoids unnecessary positive connotations.[↵](#)
4. The original version of ‘Circumscription Theory’ proposed by Carneiro emphasized that such conditions created a pressure for war. However, this has been recently challenged. See Zinkina, Korotayev, and Andreev, ‘Circumscription Theory of the Origins of the State’. Most initial states did not have neighbours to conquer. Instead, the stronger version of this is that people simply had less ability to flee from a centralized elite. See Scott, *Against the Grain*, 150–52.[↵](#)
5. People could and did still, of course, travel across and on rivers. The point here is that both geography and population density generally constrained (although never entirely prevented) movement.[↵](#)
6. Peter Turchin et al., ‘Disentangling the Evolutionary Drivers of Social Complexity: A Comprehensive Test of Hypotheses’, *Science Advances* 8, no. 25 (2022): 7–8, <https://doi.org/10.1126/sciadv.abn3517>.[↵](#)

7. Domestication refers to a co-evolutionary process in which one species shapes the fitness of another to gain more resources or services. For plants, this meant a human-managed environment selected for plants, especially specific cereals and grains. The domestication process did not start with the Holocene. The use of fire to prepare food dates to around 780,000 years ago, several hundred thousand years before *Homo sapiens*. It is thought that around 25,000 years ago both the kill-off of large mammals and new tools used for foraging changed both the environment and which crops had a competitive advantage. Plant domestication was not an intentional or rapid process, at least not to begin with. The initial changes to cereals, such as larger, non-shattering seeds, would have been largely imperceptible to the eyes of early farmers; see M. D. Purugganan, 'What Is Domestication?', *Trends in Ecology & Evolution* 37, no. 8 (2022): 663–71, <https://doi.org/10.1016/j.tree.2022.04.006>; Allaby et al., 'Genetic Revelations of a New Paradigm of Plant Domestication as a Landscape Level Process', 324–8; Zohar et al., 'Evidence for the Cooking of Fish 780,000 Years Ago at Gesher Benot Ya'aqov, Israel'. ↵
8. Joram Mayshar, Omer Moav, and Luigi Pascali, 'The Origin of the State: Land Productivity or Appropriability?', *Journal of Political Economy* 130, no. 4 (2022): 1091, <https://doi.org/10.1086/718372>. ↵
9. Thilo R. Huning and Fabian Wahl, 'You Reap What You Know: Appropriability and the Origin of European States', *European Journal of Political Economy* 79 (2023): 102432, <https://doi.org/10.1016/j.ejpoleco.2023.102432>. ↵
10. Scott, *Against the Grain*, 128. ↵
11. Timothy A. Kohler et al., 'Greater Post-Neolithic Wealth Disparities in Eurasia Than in North America and Mesoamerica', *Nature* 551, no. 7682 (2017): 619–22, <https://doi.org/10.1038/nature24646>. The domestication of animals appears to have been much more of a revolution. It was speedy and most likely intentional. ↵
12. Jared Diamond, 'Evolution, Consequences and Future of Plant and Animal Domestication', *Nature* 418, no. 6898 (2002): 700–707, <https://doi.org/10.1038/nature01019>; Thomas Cucchi and Benjamin Arbuckle, 'Animal Domestication: From Distant Past to Current Development and Issues', *Animal Frontiers* 11, no. 3 (2021): 6,

<https://doi.org/10.1093/af/vfab013>. The extinction of megafauna – such as the dire wolf, mammoth, and diprotodon (essentially a 3.5-tonne, 1.8-metre-tall wombat) – driven by both climate change and, to a lesser extent, human pressure, also influenced the spread of domesticates.↵

13. Adapted from Scott, *The Art of Not Being Governed*, table 3. I've used Scott's estimates for labour intensity, added in a subjective judgement of visibility, and specified his estimates of storability (refining these to month and year estimates, rather than low/medium/high judgements).↵
14. It may have even been a pristine state, although this is disputed. See H. J. M. Claessen, 'The Emergence of Pristine States', *Social Evolution & History* 15, no. 1 (2016): 25.↵
15. C. Leipe et al., 'Discontinuous Spread of Millet Agriculture in Eastern Asia and Prehistoric Population Dynamics', *Science Advances* 5, no. 9 (2019): 9, <https://doi.org/10.1126/sciadv.aax6225>.↵
16. Adam S. Green makes the strongest case for the egalitarian order of the Indus in 'Killing the Priest-King: Addressing Egalitarianism in the Indus Civilization', *Journal of Archaeological Research* 29, no. 2 (2021): 153–202, <https://doi.org/10.1007/s10814-020-09147-9>.↵
17. Peter Frankopan, *The Earth Transformed: An Untold History* (Alfred A. Knopf, 2023), 93. ↵
18. Leo von Geusau, *Akha Internal History: Marginalization and the Ethnic Alliance System* (Curzon, 2011), 134.↵
19. Scott, *The Art of Not Being Governed*. It's a perspective that isn't without controversy. For critiques, see Victor Lieberman, 'A Zone of Refuge in Southeast Asia? Reconceptualizing Interior Spaces', *Journal of Global History* 5, no. 2 (2010): 333–46, <https://doi.org/10.1017/S1740022810000112>; Jean Michaud, 'Editorial – Zomia and Beyond', *Journal of Global History* 5, no. 2 (2010): 187–214, <https://doi.org/10.1017/S1740022810000057>. However, even these critiques still usually praise the work and simply suggest that it is more complex than Scott often proposes. For instance, people fled to Zomia to escape not just states but also popular conflict. These are important nuances to Scott's argument, not refutations. ↵

20. On the Great Dismal Swamp and maroonage in the antebellum, see Sylviane A. Diouf, *Slavery's Exiles* (New York University Press, 2014). It's worth noting that some even grew loutable resources such as rice. A key point here is that each source of Goliath fuel will push a society towards a Goliath structure, but it doesn't automatically create a Goliath. A group with loutable resources but many exit options, few monopolizable weapons, and strong egalitarian ethics may never develop a Goliath.↵
21. Angela M. Chira, Russell D. Gray, and Carlos A. Botero, 'Geography Is Not Destiny: A Quantitative Test of Diamond's Axis of Orientation Hypothesis', *Evolutionary Human Sciences* 6 (2024): e5, <https://doi.org/10.1017/ehs.2023.34>.↵
22. Mayshar, Moav, and Pascali, 'The Origin of the State', 1097, 1138.↵
23. Scott, *The Art of Not Being Governed*, chap. 6.↵
24. This art of not being governed looks similar to the nomadic egalitarian hunter-gatherers of today. This has led some to claim that most hunter-gatherers, including the egalitarian Ju'/hoansi and central African foragers, are simply like the peoples of Zomia: their egalitarian, nomadic way of life is a way of avoiding state capture, rather than a window onto life in the Palaeolithic. For instance, many have claimed that central African hunter-gatherers live in hostile lands because they were pushed there by Bantu farmers who forcibly took all the arable land. This view is mistaken. Comparisons of genetic and archaeological data against reconstructions of the climate show that central African foragers have inhabited their current areas of residence since 120,000 years ago. Their genes, instruments, and languages all seem to have ancient origins rather than having been influenced by or adopted from Bantu farmers. And, as covered earlier, many of the nomadic-egalitarian foragers of Africa seem to have genetically diverged from most of humanity over a hundred thousand years ago. The people of Zomia are simply adopting an old way of life which prevented rulers and inequalities from emerging within a society. It just so happened to have the added benefit of preventing being captured by rulers from outside of a society. This scepticism is best expressed by James C. Scott in his review of Jared Diamond's *The World Until Yesterday* (Scott, 'Crops, Towns, Government', *London*

Review of Books, 21 November 2013, www.lrb.co.uk/the-paper/v35/n22/james-c.-scott/crops-towns-government). He finishes his critique with the line ‘We have virtually no credible evidence about the world until yesterday and, until we do, the only defensible intellectual position is to shut up.’ While Scott may be my favourite intellectual, he is simply, profoundly wrong on this one. For the evidence on central African hunter-gatherers, see Padilla-Iglesias et al., ‘Population Interconnectivity over the Past 120,000 Years Explains Distribution and Diversity of Central African Hunter-Gatherers’; Padilla-Iglesias et al., ‘Cultural Evolution of Central African Hunter-Gatherers Reflects a Deep History of Interconnectivity’. Essentially, anything from Camilla Padilla-Iglesias on human origins is excellent.↵

25. Echoed by the archaeologist Bruce Trigger in his *Understanding Early Civilizations*. He analysed seven early societies (Egypt, Mesopotamia, North China, the Maya, the Inka/Inca, the Aztecs, and the Yoruba in Benin), noting the close similarities in basic social structure. The evolution I’m talking about here is usually referred to as ‘macrocultural evolution’ or ‘cultural macroevolution’: that is, the evolution of culture not just at a small scale, but at the largest level of states and societies. This is usually driven by competition between groups, or ‘multi-level selection’ (a process whereby natural selection operates on the level not just of genes or individuals, but of groups at the highest level). For a general discussion, see Russell D. Gray and Joseph Watts, ‘Cultural Macroeolution Matters’, *Proceedings of the National Academy of Sciences* 114, no. 30 (2017): 7846–52. To avoid the jargon I’ll be using ‘Goliath evolution’, or occasionally ‘cultural evolution’, rather than cultural macroevolution.↵
26. What counted as a day trip depended on the land you had to traverse and how you could move your troops through it. Roads and horses, or ships upon easily navigable rivers helped; forests, marshes, swamps, hills, and mountains hindered. A lack of technology or unwelcoming terrain could all create further ‘distance’ between rulers and their potential subjects. See Charles S. Spencer, ‘Territorial Expansion and Primary State Formation’, *Proceedings of the National Academy of Sciences* 107, no. 16 (2010): 7119, <https://doi.org/10.1073/pnas.1002470107>. I’ve excluded Spencer’s

cases of the Moche in Peru and the Indus Valley. There is too much debate over whether the Moche were formally a state, and too little evidence that the Indus Valley Civilization was even hierarchical, let alone a state. On the role of geography, see Mann, *The Sources of Social Power I*, chap. 4; Scott, *Against the Grain*, chap. 1; Scott, *The Art of Not Being Governed*, chap. 5.↵

27. On how cultural evolution operates at the level of intergroup competition, see Henrich, *The Secret of Our Success*, chap. 10. The strongest critique of the idea of larger populations driving cultural evolution comes from K. Vaesen et al., 'Population Size Does Not Explain Past Changes in Cultural Complexity', *Proceedings of the National Academy of Sciences* 113, 16 (2016): E2241–7, <https://doi.org/10.1073/pnas.1520288113>. Others have even suggested that population size can inhibit cultural evolution (N. Fay et al., 'Increasing Population Size Can Inhibit Cumulative Cultural Evolution', *Proceedings of the National Academy of Sciences* 116, 14 (2019): 6726–31, <https://doi.org/10.1073/pnas.1811413116>). Yet, that is based on a single, small experiment just using the task of paper-plane construction. ↵
28. Hopefully it's clear by now that these rarely came as one single package and there were exceptions. As we've already covered, the Inca had a large-scale empire but no writing. We're dealing with strong, general trends here.↵
29. Christina Cheung et al., 'Diets, Social Roles, and Geographical Origins of Sacrificial Victims at the Royal Cemetery at Yinxu, Shang China: New Evidence from Stable Carbon, Nitrogen, and Sulfur Isotope Analysis', *Journal of Anthropological Archaeology* 48 (2017): 28–9, <https://doi.org/10.1016/j.jaa.2017.05.006>.↵
30. Michael Winkelman, 'Political and Demographic-Ecological Determinants of Institutionalised Human Sacrifice', *Anthropological Forum* 24, no. 1 (2014): 47, <https://doi.org/10.1080/00664677.2014.860888>. Another factor is political organization through alliances, although this is tricky since many alliances are created through coercion.↵
31. Joseph Watts et al., 'Ritual Human Sacrifice Promoted and Sustained the Evolution of Stratified Societies', *Nature* 532, no. 7598 (2016):

- 228–31, <https://doi.org/10.1038/nature17159>.↵
32. Peter Turchin, *Ultrasociety: How 10,000 Years of War Made Humans the Greatest Cooperators on Earth* (Beresta Books, 2016).↵
33. David Graeber and Marshall Sahlins, *On Kings* (Hau Books, 2017), 181–2.↵
34. M. Fiskesjö, ‘On the “Raw” and the “Cooked” Barbarians of Imperial China’, *Inner Asia*, 1, no. 2 (1999): 139–68, <https://doi.org/10.1163/146481799793648004>; Emma Dench, ‘Barbarian’, in the *Oxford Classical Dictionary*, ed. Thomas Wiedermann (Oxford University Press, 2022), <https://doi.org/10.1093/acrefore/9780199381135.013.1051>. For a broader discussion, see Jennings, *Killing Civilization*. ↵
35. On the importance of ideological integration being used to create lasting states and empires, see Jack A. Goldstone and John F. Haldon, ‘Ancient States, Empires, and Exploitation: Problems and Perspectives’, in *The Dynamics of Ancient Empires*, ed. Ian Morris and Walter Scheidel (Oxford University Press, 2009), 10–15, <https://doi.org/10.1093/oso/9780195371581.003.0001>. We don’t know what beliefs the Natufians adopted to justify their rising inequalities, nor what the priests of Cahokia claimed to be.

Not everyone believed the stories. Across cultures we have regular signs that slaves didn’t think their masters were superior, peasants didn’t think highly of their lords, and citizens didn’t believe their emperor was actually a god. In the antebellum South one curse uttered by slaves was ‘I pray that death and disease shall be forever with them and that their crops shall not multiply’. Nor were states all-powerful. Early kings often went to great lengths to showcase their strength, yet in reality they frequently had little impact on the day-to-day affairs of most people. Their boasts were a sign more of weakness than strength. In Southeast Asia, kingdoms and empires were often just seasonal. During the heavy rains of the monsoon they had little influence beyond the palace walls. Despite this, their influence, and their power over the belief of their citizens, grew over generations and centuries. See James C. Scott, *Domination and the Arts of Resistance* (Yale University Press, 1990), 34. The curse is recounted by the black novelist and anthropologist Zora Neale Hurston.↵

36. Ian Morris et al., *Foragers, Farmers, and Fossil Fuels: How Human Values Evolve* (Princeton University Press, 2015). For an overly detailed exploration of how ideologies throughout history have justified inequality, read Thomas Piketty, *Capital and Ideology*, trans. Arthur Goldhammer (The Belknap Press of Harvard University Press, 2020).↵
37. Jonathan J. B. Mijs, ‘The Paradox of Inequality: Income Inequality and Belief in Meritocracy Go Hand in Hand’, *Socio-Economic Review* 19, no. 1 (2021): 7–35, <https://doi.org/10.1093/ser/mwy051>.↵
38. The transition from animism to monotheistic religion is covered in Hervey C. Peoples, Pavel Duda, and Frank W. Marlowe, ‘Hunter-Gatherers and the Origins of Religion’, *Human Nature* 27, no. 3 (2016): 261–82, <https://doi.org/10.1007/s12110-016-9260-0>. Granted, this is an understudied area, but this is by far the best evidence we have. For the analysis of big kings preceding big gods, see Harvey Whitehouse et al., ‘Testing the Big Gods Hypothesis With Global Historical Data: A Review and “Retake”’, *Religion, Brain & Behavior* 13, no. 2 (2023): 124–66, <https://doi.org/10.1080/2153599X.2022.2074085>.↵
39. A similar point is made by Michael Mann in *The Sources of Social Power I*, chap. 3.↵
40. A. E. J. Miller et al., ‘Gender Differences in Strength and Muscle Fiber Characteristics’, *European Journal of Applied Physiology and Occupational Physiology* 66, no. 3 (1993): 254–62, <https://doi.org/10.1007/BF00235103>. For an indication of what this means, just look at the differences in world records for strength-based sports. The male deadlift and bench-press world records are 501kg and 355kg. For females it is 325kg and 207.5kg. There are similar gaps for Olympic weightlifting as well as javelin, shot-put, and a range of other strength-based sports.↵
41. A similar point is made in Nancy Lindisfarne and Jonathan Neale, *Why Men? A Human History of Violence and Inequality* (Hurst, 2023), chap. 9. Initial physical differences could compound into large social difference in societies where violence was key.↵
42. Valerie M. Hudson, Donna Lee Bowen, and Perpetua Lynne Nielsen, *The First Political Order: How Sex Shapes Governance and National*

- Security Worldwide* (Columbia University Press, 2020), chap. 2. For a similar account of the importance of patrilocality, but with less emphasis on violence, see Anil Ananthaswamy and Kate Douglas, 'The Origins of the Patriarchy', *New Scientist* 238, no. 3174 (2018): 34–5, [https://doi.org/10.1016/S0262-4079\(18\)30707-3](https://doi.org/10.1016/S0262-4079(18)30707-3).↵
43. Angela Saini, *The Patriarchs: The Origins of Inequality* (Beacon Press, 2023), 112.↵
 44. Eric Berkowitz, *Sex and Punishment: Four Thousand Years of Judging Desire* (Westbourne Press, 2013), 19–21.↵
 45. A similar case for the origins of patriarchy is made by Alice Evans, a senior lecturer at King's College London and a leading expert on the global history of patriarchy. She points to inheritable wealth (such as land, domesticated animals, and grain), conquest by male-dominated cultures, and religion as three key factors (among others) that drove the emergence of patriarchy. See Alice Evans, 'Ten Thousand Years of Patriarchy!', 3 June 2022, <https://www.ggd.world/p/ten-thousand-years-of-patriarchy-1>.↵
 46. Monika Karmin et al., 'A Recent Bottleneck of Y Chromosome Diversity Coincides with a Global Change in Culture', *Genome Research* 25, no. 4 (2015): 461–2, <https://doi.org/10.1101/gr.186684.114>.↵
 47. Shin et al., 'Scale and Information-Processing Thresholds in Holocene Social Evolution'.↵
 48. Flannery and Marcus, *The Creation of Inequality*, chap. 10.↵
 49. Timothy Earle, 'The Evolution of Chiefdoms', *Current Anthropology* 30, no. 1 (1989): 87, <https://doi.org/10.1086/203717>. Earle was technically referring to 'chiefdoms'.↵
 50. Shin et al., 'Scale and Information-Processing Thresholds in Holocene Social Evolution'. This echoes the thesis of the Stanford historian and archaeologist Ian Morris. In *Why the West Rules – For Now* he argues that when states grew too large without the right measures they often stagnated and collapsed, and he tracks imperial power over time (by their energy capture, information processing (such as written materials), urbanism, and military strength). Morris spotted a pattern: states in both the east (China) and west (Europe and the Near East)

repeatedly hit a ‘ceiling’ to their power before collapsing. Morris also suggests that Rome and the later Song Dynasty hit a ‘hard ceiling’ of what was possible for an agrarian empire. See Ian Morris, *Why the West Rules – For Now: The Patterns of History and What They Reveal About the Future* (Profile Books, 2011). Goliath evolution and Goliath’s Curse provide one way of explaining this pattern.↵

CHAPTER 7: GOLIATH’S CURSE

1. We used a technique called ‘survival analysis’. For more details, read Marten Scheffer et al., ‘The Vulnerability of Aging States: A Survival Analysis Across Premodern Societies’, *Proceedings of the National Academy of Sciences* 120, no. 48 (2023), <https://doi.org/10.1073/pnas.2218834120>. There are numerous caveats to keep in mind. First, these terminations range from relatively bloodless coups to deep collapses. Second, the numbers are based on broadly accepted start and end dates in historical and archaeological accounts. These are often disputed. For instance, did the Eastern Roman Empire (the Byzantine) end in 1453 with the fall of its capital, Constantinople, with the sack of Constantinople and partitioning of its territories by crusaders in 1204, or with a large-scale loss of territory to the Islamic caliphates during the seventh century? To help address this, we used both upper and lower estimates for both the beginning and end of a state. On the Crisis Database study, see Daniel Hoyer et al., ‘All Crises Are Unhappy in Their Own Way: The Role of Societal Instability in Shaping the Past’, preprint, SocArXiv, 15 February 2024, <https://doi.org/10.31235/osf.io/rk4gd>. One previous study of forty-two empires found a different ‘ageless’ pattern of mortality: that is an empire was as likely to collapse in its first year as its 100th. See S. Arbesman, ‘The Life-Spans of Empires’, *Historical Methods: A Journal of Quantitative and Interdisciplinary History*, 44, no. 3 (2011): 127–9, <https://doi.org/10.1080/01615440.2011.577733>. Notably, that database is almost eight times smaller than the MOROS database and covers far less time, although this ageless pattern of mortality is what we see in businesses and the extinction of past species in the fossil record. On companies, see M. I. G. Daepf et al., ‘The Mortality of Companies’, *Journal of the Royal Society Interface* 12, no. 106 (2015),

<https://doi.org/10.1098/rsif.2015.0120>. The pattern in fossils was first discovered by the biologist Leigh Van Valen and is usually attributed to the ‘Red Queen Effect’: the constant race to adapt to a changing environment and competition means that a species (or company) can die as a result of one mistake. ↵

2. Marten Scheffer et al., ‘Loss of Resilience Preceded Transformations of Pre-Hispanic Pueblo Societies’, *Proceedings of the National Academy of Sciences* 118, no. 18 (2021): e2024397118, <https://doi.org/10.1073/pnas.2024397118>; Sean S. Downey, W. Randall Haas Jr, and Stephen J. Shennan, ‘European Neolithic Societies Showed Early Warning Signals of Population Collapse’, *Proceedings of the National Academy of Sciences* 113, no. 35 (2016): 9751–6, <https://doi.org/10.1073/pnas.1602504113>.↵
3. For an overview, see Scheidel, *The Great Leveler*. Scheidel covers a series of collapses from different periods and regions: the Tang Dynasty, the Western Roman Empire, the Lowland Maya, the Bronze Age Mediterranean, Pre-Columbian Americas, the first post-imperial period in ancient Egypt, and, most recently, Somalia. Analysts of inequality disagree over the peaceful mechanisms of wealth redistribution, but largely agree on the more sinister ones. See Pim de Zwart, ‘The Global History of Inequality’, *International Review of Social History* 64, no. 2 (2019): 309–23. Similarly, former World Bank economist Branko Milanović referred to the ‘malignant mechanisms’ of equalization, such as civil strife, war, and epidemics. See Milanović, *Global Inequality: A New Approach for the Age of Globalization* (Harvard University Press, 2016). For instance, the role of the two world wars in reversing the wealth imbalances in the earlier twentieth century is well documented; see Thomas Piketty, *Capital in the Twenty-First Century*, trans. Arthur Goldhammer (The Belknap Press of Harvard University Press, 2017); Thomas Piketty and Emmanuel Saez, ‘Inequality in the Long Run’, *Science*, 344, no. 6186 (2014): 838–43, <https://doi.org/10.1126/science.1251936>. Others have more recently contended that for natural disasters, such as pandemics, levelling is not an automatic outcome. Many shocks since the medieval period have exacerbated rather than alleviated wealth inequality. The result depends on pre-existing institutions. If economic and political power is skewed, then elites can capitalize on a shock to further their

advantages. See Bas van Bavel and Marten Scheffer, ‘Historical Effects of Shocks on Inequality: The Great Leveler Revisited’, *Humanities and Social Sciences Communications* 8, no. 1 (2021): Article 76, <https://doi.org/10.1057/s41599-021-00763-4>. However, this analysis is for more moderate external blows.↵

4. Stasavage, *The Decline and Rise of Democracy*, chap. 1; Kohler and Smith, *Ten Thousand Years of Inequality*, chap. 11.↵
5. This is often known as the ‘Matthew Principle’, from Matthew 25:29: ‘For to everyone who has will more be given, and he will have abundance; but from him who has not, even what he has will be taken away.’↵
6. Piketty, *Capital in the Twenty-First Century*; Peter Turchin, *End Times: Elites, Counter-Elites, and the Path of Political Disintegration* (Penguin, 2023).↵
7. Branko Milanović, Peter H. Lindert, and Jeffrey G. Williamson, ‘Pre-Industrial Inequality’, *Economic Journal* 121, no. 551 (2011): fig. 2, <https://doi.org/10.1111/j.1468-0297.2010.02403.x>. For an accessible summary, see Max Roser, ‘How Unequal Were Pre-Industrial Societies?’, Our World in Data, 2 December 2013, <https://ourworldindata.org/how-unequal-were-pre-industrial-societies>.↵
8. Branko Milanović, ‘How Rich Were the Rich? An Empirically Based Taxonomy of Pre-Industrial Bases of Wealth’, *Explorations in Economic History* 93 (2024): 101592, <https://doi.org/10.1016/j.eeh.2024.101592>. The estimate that the 1 per cent had an even bigger share of wealth is my own. It’s a robust trend across countries today and historically that wealth inequality is larger than income inequality. That is for the simple reason that wealth, whether it be in land, gold, or stocks, tends to be more concentrated than income. For modern billionaires such as Elon Musk and Jeff Bezos, the majority of their riches come from what they own, not their salaries.↵
9. In the institutional economics literature, this is usually referred to as a difference between ‘inclusive institutions’ and ‘exclusive institutions’, most notably used in Daron Acemoglu and James A. Robinson, *Why Nations Fail: The Origins of Power, Prosperity, and Poverty* (Profile

Books, 2013). However, this is usually reserved for formal structures such as the rule of law, not the broader set of relations that organize how a society runs. I've tried to refine the definition by specifying it as the inequality in each form of power. There are clear metrics for measuring some inequalities in power, such as for wealth (the Gini coefficient, Palma ratio, and others) and decision-making (the V-Dem index, although this notably doesn't include companies). For others, such as the control of information and violence, there are fewer well-refined measurements. Ideally, we would have measurements for each of these. Developing them is outside the scope of this book and I'll be analysing the cases in a broader manner looking at, for instance, how privatized the military is and how autocratic or captured by private interests the government is. Note that the underlying assumption here is that large differences in power tend to result in more extraction (unfair transfers of value) in the long run. It's a fairly safe assumption when we look across the long arc of history: whether it be patriarchy, wealth inequality in the earliest states, or large differences between groups in military technology, the end result was usually an extractive arrangement.↵

10. The exact outcome will depend on culture and particularities, of course.↵
11. Alf Hornborg, 'Imperial Metabolism: Empire as a Process of Ecologically Unequal Exchange', in *The Oxford World History of Empire*, vol. I: *The Imperial Experience*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), <https://doi.org/10.1093/oso/9780199772360.003.0014>.↵
12. Richard Wilkinson and Kate Pickett, *The Spirit Level: Why Equality Is Better for Everyone* (Penguin, 2010), and *The Inner Level: How More Equal Societies Reduce Stress, Restore Sanity and Improve Everyone's Well-being* (Allen Lane, 2018).↵
13. Turchin, *End Times*; Turchin, *Ages of Discord: A Structural-Demographic Analysis of American History* (Beresta Books, 2016). We technically don't know how much of a role popular immiseration played in the case of these early Goliaths. However, we can see it in a plethora of later collapses (see structural-demographic theory in chapter 11 of this book). ↵

14. DuVal, *Native Nations*, chap. 2.↵
15. Daniel Hoyer et al., 'Navigating Polycrisis: Long-Run Socio-Cultural Factors Shape Response to Changing Climate', *Philosophical Transactions of the Royal Society B: Biological Sciences* 378, no. 1889 (2023): 11, <https://doi.org/10.1098/rstb.2022.0402>; Dániel Kondor et al., 'Explaining Population Booms and Busts in Mid-Holocene Europe', preprint, SocArXiv, 11 May 2022, <https://doi.org/10.31235/osf.io/c32up>; Bevan et al., 'Holocene Fluctuations in Human Population'.↵
16. An initial analysis using the Crisis Database suggests that crises unfold in a range of ways and there is no one paradigmatic pattern to crises (Hoyer et al., 'All Crises Are Unhappy in Their Own Way').↵
17. Timothy M. Lenton, 'Climate Change and Tipping Points in Historical Collapse', in *How Worlds Collapse*, ed. Miguel Centeno et al. (Routledge, 2023), 261–81, <https://doi.org/10.4324/9781003331384-17>. ↵
18. This roughly echoes my colleague Tilman Hartley, who frames this as a crisis being more likely to lead to breakdown or collapse when it involves more 'scarcity' (growing vulnerabilities due to diminishing returns, as we'll cover later), more shocks, and more closed institutions. See Tilman Hartley, 'State Crisis Theory: A Systematization of Institutional, Socio-Ecological, Demographic-Structural, World-Systems, and Revolutions Research', preprint, 18 April 2023, <https://doi.org/10.32388/G2GNJR>.↵
19. It's tempting to see collapse as just a matter of bad luck. One review of nineteen cases of collapse found, along with sixty-four different explanations, that 44 per cent were bolts from the blue, such as a changing climate or invaders. But growing vulnerability was just as important as the threats which triggered a collapse. See Joseph A. Tainter, 'How Scholars Explain Collapse', in *How Worlds Collapse*, ed. Miguel Centeno et al. (Routledge, 2023), 32–3, <https://doi.org/10.4324/9781003331384-4/scholars-explain-collapse-joseph-tainter>.↵
20. See U. Büntgen et al., 'Cooling and Societal Change During the Late Antique Little Ice Age from 536 to Around 660 AD', *Nature Geoscience* 9 (2016): 231–6, <https://doi.org/10.1038/ngeo2652>; P. N.

- Peregrine, 'Climate and Social Change at the Start of the Late Antique Little Ice Age', *The Holocene* 30, no. 11 (2020), <https://doi.org/10.1177/0959683620941079>.↵
21. Peregrine, 'Climate and Social Change at the Start of the Late Antique Little Ice Age'.↵
 22. Groups have significant differences in whether there are leaders, what privileges and powers those leaders have, how much authority they have and how widely shared that authority is. The difference could be stark: leaderless polities like early Monte Albán or Teotihuacan had no depictions of elites and numerous public buildings; whereas the Maya had boastful leaders glorified on painted murals and rampant inequality.↵
 23. Peregrine, 'Social Resilience to Climate Change during the Late Antique Little Ice Age'. There are a few caveats to be aware of. First, the proxies for determining how much different regions cooled during this time are uncertain. Second, measuring social change, let alone the inclusivity of institutions, is always going to be a difficult and imperfect exercise. P. N. Peregrine, 'Social Resilience to Climate-Related Disasters in Ancient Societies: A Test of Two Hypotheses', *Weather, Climate, and Society* 10, no. 1 (2018): 145–61, <https://doi.org/10.1175/WCAS-D-17-0052.1>.↵
 24. On social capital, see Robert D. Putnam, 'Social Capital: Measurement and Consequences', *Isis: Canadian Journal of Policy Research* 2 (Spring 2001): 41–51. Inclusive institutions in particular are what Putnam would call 'bridging capital'.↵
 25. D. K. Yoon, Jung Eun Kang, and Samuel D. Brody, 'A Measurement of Community Disaster Resilience in Korea', *Journal of Environmental Planning and Management* 59, no. 3 (2016): 436–60, <https://doi.org/10.1080/09640568.2015.1016142>; Katherine Hore et al., 'People's Participation in Disaster-Risk Reduction: Recentering Power', *Natural Hazards Review* 21, no. 2 (2020), [https://doi.org/10.1061/\(ASCE\)NH.1527-6996.0000353](https://doi.org/10.1061/(ASCE)NH.1527-6996.0000353); D. O. Kasdan, 'Considering Socio-Cultural Factors of Disaster Risk Management', *Disaster Prevention and Management* 25, no. 4 (2016): 464–77, <https://doi.org/10.1108/DPM-03-2016-0055>; Ilan Kelman, Jessica Mercer, and J. C. Gaillard, *The Routledge Handbook of*

Disaster Risk Reduction Including Climate Change Adaptation (Routledge, 2017).↵

26. Hore et al., ‘People’s Participation in Disaster-Risk Reduction’, 1.↵
27. There is also growing evidence that they promote more economic growth. The main proponents of the theory that political inclusion drives economic growth are Daron Acemoglu and James Robinson: see Acemoglu and Robinson, *Why Nations Fail*; Acemoglu et al., ‘Democracy Does Cause Growth’, *Journal of Political Economy* 127, no. 1 (2019): 47–100, <https://doi.org/10.1086/700936>. A recent meta-analysis of 2,000 different studies supports this basic idea, with the caveat that the democracy–growth link varies across context: Marco Colagrossi, Domenico Rossignoli, and Mario A. Maggioni, ‘Does Democracy Cause Growth? A Meta-Analysis (of 2000 Regressions)’, *European Journal of Political Economy* 61 (2020), <https://doi.org/10.1016/j.ejpoleco.2019.101824>.↵
28. On collective intelligence, see Lu Hong and Scott E. Page, ‘Groups of Diverse Problem Solvers Can Outperform Groups of High-Ability Problem Solvers’, *Proceedings of the National Academy of Sciences* 101, no. 46 (2004): 16385–9, <https://doi.org/10.1073/pnas.0403723101>, and ‘Problem Solving by Heterogeneous Agents’ *Journal of Economic Theory* 97, no. 1 (2001): 123–63, <https://doi.org/10.1006/jeth.2000.2709>; Vicky Chuqiao Yang and Anders Sandberg, ‘Collective Intelligence as Infrastructure for Reducing Broad Global Catastrophic Risks’, *Proceedings of the Stanford Existential Risks Conference 2023*: 194–206, <https://doi.org/10.25740/mf606ht6373>. On prediction markets, consult Arrow et al., ‘The Promise of Prediction Markets’, *Science* 320, no. 5878 (2008): 877–8, <https://doi.org/10.1126/science.1157679>. For a general overview of the wisdom of the crowds, see James Surowiecki, *The Wisdom of Crowds* (Abacus, 2005); C. Simoiu et al., ‘Studying the “Wisdom of Crowds” at Scale’, *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing* 7 (2019): 171–9, <https://doi.org/10.1609/hcomp.v7i1.5271>.↵
29. On superforecasters, see Philip Tetlock and Dan Gardner, *Superforecasting* (Random House, 2016). On the benefits of deliberative democracy, consult John S. Dryzek et al., ‘The Crisis of

Democracy and the Science of Deliberation’, *Science* 363, no. 6432 (2019): 1144–6, <https://doi.org/10.1126/science.aaw2694>; Landemore, *Democratic Reason*; Landemore, *Open Democracy*. On post-conflict reconciliation, see E. Ugarriza and D. Caluwaerts, *Democratic Deliberation in Deeply Divided Societies: From Conflict to Common Ground* (Macmillan, 2014). For an overview of recent experiments in deliberative democracy, read OECD, ‘Innovative Citizen Participation and New Democratic Institutions: Catching the Deliberative Wave’, www.oecd.org/en/publications/innovative-citizen-participation-and-new-democratic-institutions_339306da-en.html. On horizon scanning, see Luke Kemp, ‘Foreseeing Extreme Technological Risk’, in *Managing Extreme Technological Risk*, ed. Catherine Rhodes (Centre for the Study of Existential Risk, 2024).↵

30. A. Izdebski, L. Mordechai, and S. White, ‘The Social Burden of Resilience: A Historical Perspective’, *Human Ecology: An Interdisciplinary Journal* 46, no. 3 (2018): 291–303, <https://doi.org/10.1007/s10745-018-0002-2>.↵

CHAPTER 8: GANG WARFARE

1. We don’t know who exactly Narmer was. He may have been the same person as other early pharaohs, such as ‘Menes’ or ‘King Scorpion’. The sketch of the Palette of Narmer is available through Wikimedia and uploaded by Maker166. See <https://commons.wikimedia.org/wiki/File:Narmer-Palette.jpg>. Available through Creative Commons License Version 4.0.↵
2. Pauketat, *Cahokia*, chap. 12.↵
3. Christian T. Miller and Jonathan Jones, ‘Firestone and the Warlord: The Untold Story of Firestone, Charles Taylor and the Tragedy of Liberia’, *ProPublica*, 18 November 2014, www.propublica.org/article/firestone-and-the-warlord-intro.↵
4. Ferguson, ‘The Prehistory of War and Peace in Europe and the Near East’, 223.↵
5. Katherine Hirschfeld, *Gangster States* (Macmillan, 2015), 3, <https://doi.org/10.1057/9781137490292>.↵

6. Christopher Blattman, *Why We Fight: The Roots of War and the Paths to Peace* (Viking, 2022), chap. 9. [↵](#)
7. Raúl Sánchez de la Sierra, 'On the Origins of the State: Stationary Bandits and Taxation in Eastern Congo', *Journal of Political Economy* 128, no. 1 (2020): 32, 34–7, <https://doi.org/10.1086/703989>. [↵](#)
8. Katherine Hirschfeld, 'Failing States as Epidemiologic Risk Zones: Implications for Global Health Security', *Health Security* 15, no. 3 (2017): 291–2, <https://doi.org/10.1089/hs.2016.0077>. [↵](#)
9. Robert I. Rotberg, *State Failure and State Weakness in a Time of Terror* (World Peace Foundation/Brookings Institution Press, 2003). [↵](#)
10. Pun intended. Hirschfeld, *Gangster States*. [↵](#)
11. Ferguson, 'The Prehistory of War and Peace in Europe and the Near East', 223. [↵](#)
12. Hirschfeld, *Gangster States*. [↵](#)
13. Joerg Baten, Giacomo Benati, and Arkadiusz Sołtysiak, 'Violence Trends in the Ancient Middle East between 12,000 and 400 BCE', *Nature Human Behaviour* 7 (2023): 2064–73, <https://doi.org/10.1038/s41562-023-01700-y>. Redesigned from Figure 2 in the article. Data points that were disconnected from trend lines have been deleted and the two trend lines for Turkey have been blended (a previous line was discarded owing to a small sample size of nine). While sparse in parts, this is the best dataset we have access to. Moreover, it supports a far wider body of evidence suggesting that increased violence accompanies state creation and collapse. [↵](#)
14. Thanks to Tim Kohler for raising the idea of 'score-settling' based on his previous work on violence in the Pueblo societies. [↵](#)
15. This is likely to be a historical high-point. Women's political empowerment surged during the twentieth and twenty-first centuries and during this time the US was undertaking the largest historical integration of women into the military. See Joshua S. Goldstein, *War and Gender: How Gender Shapes the War System and Vice Versa* (Cambridge University Press, 2001), 10. [↵](#)

16. Hollie Nyseth Nzitatira, Jared F. Edgerton, and Laura C. Frizzell, 'Analyzing Participation in the 1994 Genocide in Rwanda', *Journal of Peace Research* 60, no. 2 (2023): 291–306, <https://doi.org/10.1177/00223433221075211>.↵
17. Marvin E. Wolfgang, Robert M. Figlio, and Thorsten Sellin, *Delinquency in a Birth Cohort* (University of Chicago Press, 1987); Nyseth Nzitatira, Edgerton, and Frizzell, 'Analyzing Participation in the 1994 Genocide in Rwanda'.↵
18. Lizzie Wade, 'The Wari's Grisly End – the Fall of a South American Empire', *Science*, 17 August 2016, www.science.org/content/article/waris-grisly-end-fall-south-american-empire; Arkush and Tung, 'Patterns of War in the Andes from the Archaic to the Late Horizon'.↵
19. This sample covers four continents and two millennia, and includes islands (the Rapa Nui), large state-systems (the Late Bronze Age (LBA)), vast empires (the Western Roman Empire), and collections of city-states (the Lowland Maya).↵
20. Kemp et al., 'Societal Collapse as a Powershift'. The Maya estimates come from B. L. Turner, 'The Rise and Fall of Maya Population and Agriculture, 1000 BC to Present: The Malthusian Perspective Reconsidered', in *Hunger in History: Food Shortage, Poverty, and Deprivation*, ed. L. Newman (Blackwell, 1990), 178–211; R. L. Wilkinson, 'Yellow Fever: Ecology, Epidemiology, and Role in the Collapse of the Classic Lowland Maya Civilization', *Medical Anthropology* 16, no. 3 (1995): 269–94, <https://doi.org/10.1080/01459740.1994.9966118>, 270–84. The Rapa Nui estimate is compiled using the figures from Chapter 15.↵
21. On the reliability of state census data, see J. Haldon, 'Comparative State Formation: The Later Roman Empire in the Wider World', in *The Oxford Handbook of Late Antiquity*, ed. S. F. Johnson (Oxford University Press, 2012), 1111–47; Pinker, *The Better Angels of Our Nature*, 194; Matthew White, *The Great Big Book of Horrible Things* (W. W. Norton, 2012).↵
22. Turchin, *End Times*, 223.↵

23. Kemp et al., 'Societal Collapse as a Powershift'. While these statistics give a general picture, bear in mind that they are averages, and the effects of a societal breakdown varied dramatically by region. During the fall of Rome's empire, the city itself may have lost up to 95 per cent of its people, yet many other areas had stable populations.↵
24. Flannery and Marcus, 'The Origin of War', 11803–5.↵
25. In Mesopotamia, the very creation of the universe was the result of an act of violence. The god Marduk kills the chaotic, primordial sea goddess Tiamat (often depicted as a sea dragon), allowing for the creation of humankind (and in some versions resulting in Marduk becoming the supreme god). An act of sacred violence tames the chaos and creates order. It's a basic myth that reappears in numerous faiths including the Hittite religion. It has obvious echoes of the Hobbesian myth: there once was chaos and now there is peace due to the shadow of an iron fist. Although tellingly, Tiamat was peaceful until Marduk declared war.↵
26. Steven Garfinkel, 'Violence and State Power in Early Mesopotamia', in *The Cambridge World History of Violence*, ed. Garrett G. Fagan et al. (Cambridge, 2020), 219–37, <https://doi.org/10.1017/9781316341247.012>, 219–21; Davide Nadali et al., 'Representations of Violence in Ancient Mesopotamia and Syria', *ibid.*, 629–53, <https://doi.org/10.1017/9781316341247.033>, 633–4.↵
27. The most disputed is the case of Tiwanaku. Although even here the prolific creation of serrated and barbed arrows (which are poor for hunting but ideal for penetrating flesh and causing an infection) suggests that warfare, or at least coercion, also played a role. See D. Hu, 'War or Peace? Assessing the Rise of the Tiwanaku State through Projectile Point Analysis', *Lithics* 37 (2017): 83–5.↵
28. On the specific case studies, see Flannery and Marcus, *The Creation of Inequality*, chap. 17.↵
29. Charles Tilly, 'Reflections on the History of European State-Making', in *The Formation of National States in Western Europe*, ed. Charles Tilly (Princeton University Press, 1975), 3–83, 42.↵
30. L. E. Cederman et al., 'War Did Make States: Revisiting the Bellicist Paradigm in Early Modern Europe', *International Organization* 77, no.

- 2 (2023): 324–62, <https://doi.org/10.1017/S0020818322000352>, 324.↵
31. Turchin et al., ‘Disentangling the Evolutionary Drivers of Social Complexity’; Thomas E. Currie et al., ‘Duration of Agriculture and Distance from the Steppe Predict the Evolution of Large-Scale Human Societies in Afro-Eurasia’, *Humanities and Social Sciences Communications* 7, no. 1 (2020): 34, <https://doi.org/10.1057/s41599-020-0516-2>; Turchin, *Ultrasociety*. Note that initially some military technologies, particularly iron weapons, resulted in greater wealth equality. In the case of iron, it was simply because it was more widely available. Yet eventually a small elite was still able to monopolize these and use them as the foundation for new Goliaths.↵
32. For a useful summary of statistics and sources, read Yuval Noah Harari, *Nexus: A Brief History of Information Networks from the Stone Age to AI* (Vintage, 2024), chap. 11.↵
33. Scott, *Against the Grain*, 179. In *On Kings*, 5, the anthropologists David Graeber and Marshall Sahlins claim that these ‘stranger-kings’ were the dominant and even original form of polity, although they also claim that these were mainly indigenous inventions and provide surprisingly little evidence to support this. For the original theory of stranger-kings, see Marshall Sahlins, ‘The Stranger-King, or, Elementary Forms of the Politics of Life’, *Indonesia and the Malay World* 36, no. 105 (2008): 177–99, <https://doi.org/10.1080/13639810802267918>.↵
34. Adam Hochschild, *Bury the Chains: The British Struggle to Abolish Slavery* (Pan, 2012), 2.↵
35. Seth Richardson, ‘Mesopotamian Slavery’, in *The Palgrave Handbook of Global Slavery Throughout History*, ed. Damian A. Pargas and Julianne Schiel (Springer International, 2023), 17–39, https://doi.org/10.1007/978-3-031-13260-5_2; J. N. Reid, ‘Slavery in Early Mesopotamia from Late Uruk Until the Fall of Babylon in the Longue Durée’ (2014), <https://ora.ox.ac.uk/objects/uuid:a693cd93-092e-4118-ae02-b9775bc2285e/files/md69d238573b3389b44745f39cc219dac>.↵
36. Scott, *Against the Grain*; Scott, ‘Response’, 719.↵

37. Saini, *The Patriarchs*, 112.↵
38. Robert D. Drennan, 'Pathways from Farmers to States around the Globe', *Cambridge Archaeological Journal* 29, no. 4 (2019): 699–702, <https://doi.org/10.1017/S0959774319000374>.↵
39. This is often framed as the 'cooperative' account of state formation. The most influential work here is Richard E. Blanton and Lane F. Fargher, 'Collective Action in the Evolution of Pre-Modern States', *Social Evolution & History* 8, no. 2 (2009): 133–66. The authors significantly overstate their case by claiming that this shows that many states were cooperative endeavours. Providing collective goods (including the presence of things like a bureaucracy) is an extraordinarily low bar for considering a state to be created for collective purposes. Especially when these primarily tended to benefit elites. The roads of Rome were built by legionaries for legionaries. They were (at least initially) mainly a conduit for information and shuttling troops, not trade and public transport. ↵
40. We should think of this as 'coordinated action' rather than 'collective action', as it is usually called. It's only when this is paired with clear signs of more inclusive, democratic governance that we can talk of more collective action.↵
41. Archaeologists of South and Mesoamerica have often been the biggest proponents of the collective action account of state formation. A lack of Goliath fuel can help explain why early states in the Americas often had more coordinated action and rebellion.↵
42. For an overview of some more inclusive polities which provided public goods but eventually drifted towards oligarchy, see Richard E. Blanton et al., 'Moral Collapse and State Failure: A View From the Past', *Frontiers in Political Science* 2 (2020), <https://doi.org/10.3389/fpos.2020.568704>. Thanks to Dan Hoyer for the 'rapacious pirate-kings' term.↵
43. The phenomenon of there being more democracy when states had less information about their subjects is well articulated in Stasavage, *The Decline and Rise of Democracy*, chap. 1. Similarly, Richard E. Blanton et al., 'The Fiscal Economy of Good Government', *Current Anthropology* 62, no. 1 (2021), <https://doi.org/10.1086/713286>, find that the provision of public goods and greater coordination of labour

- (‘good government’) is usually correlated with (and probably caused by) more reliance on the income of subjects/citizens.↵
44. Jacob G. Hariri, ‘The Autocratic Legacy of Early Statehood’, *American Political Science Review* 106, no. 3 (2012): 471–94, 471, 489.↵
 45. That is the hidden benefit of the dominance hierarchy: it’s not only those at the top who benefit, but those on the lower rungs still have others they can lord it over: such as women in Rome owning slaves or the city dwellers of China being fed by rural peasants.↵
 46. Even the very long-lasting egalitarian settlements we came across earlier – such as Jenne-Jeno – would usually either grow extractive or be conquered by a neighbour.↵
 47. Bauer et al., ‘Can War Foster Cooperation?’, *Journal of Economic Perspectives* 30, no. 3 (2016): 249–74, <https://doi.org/10.1257/jep.30.3.249>, 270–72.↵
 48. Julian E. Barnes, ‘A Nation Challenged: Proud Spirits; As Demand Soars, Flag Makers Help Bolster Nation’s Morale’, *New York Times*, 23 September 2001, www.nytimes.com/2001/09/23/business/nation-challenged-proud-spirits-demand-soars-flag-makers-help-bolster-nation-s.html.↵
 49. J. C. Jackson et al., ‘Ecological and Cultural Factors Underlying the Global Distribution of Prejudice’, *PLOS ONE* 14, no. 9 (2019): e0221953, <https://doi.org/10.1371/journal.pone.0221953>. Note that this isn’t one study per se: it examines fifty US states, forty-seven pre-industrial historical societies, and twenty-five modern countries. ↵
 50. Y. Mu, S. Han, and M. J. Gelfand, ‘The Role of Gamma Interbrain Synchrony in Social Coordination When Humans Face Territorial Threats’, *Social Cognitive and Affective Neuroscience*, 12, no. 10 (2017): 1614–23, <https://doi.org/10.1093/scan/nsx093>.↵
 51. Malešević, *The Rise of Organised Brutality*, chap. 2.↵
 52. Cameron Easley, ‘U.S. Conservatives Are Uniquely Inclined Toward Right-Wing Authoritarianism Compared to Western Peers’, *Morning Consult*, 28 June 2021, <https://pro.morningconsult.com/trend-setters/global-right-wing-authoritarian-test>.↵

53. The seminal works here are Bob Altemeyer, *Right-Wing Authoritarianism* (University of Manitoba Press, 1981), and Karen Stenner, *The Authoritarian Dynamic* (Cambridge University Press, 2005). Stenner stresses that these are latent traits that are activated only by certain conditions, such as external threats. In general, the literature here refers to ‘right-wing authoritarianism’, but there are growing studies on left-wing authoritarianism.↵
54. Danny Osborne et al., ‘The Psychological Causes and Societal Consequences of Authoritarianism’, *Nature Reviews Psychology* 2 (2023): 220–32, <https://doi.org/10.1038/s44159-023-00161-4>, is the most authoritative and up-to-date review of the authoritarianism literature, including on its genetic basis.↵
55. Hannah S. Ferguson et al., ‘Context-Specific Effects of Facial Dominance and Trustworthiness on Hypothetical Leadership Decisions’, *PLOS ONE* 14, no. 7 (2019): e0214261, <https://doi.org/10.1371/journal.pone.0214261>; Daniel E. Re et al., ‘Facial Cues to Perceived Height Influence Leadership Choices in Simulated War and Peace Contexts’, *Evolutionary Psychology* 11, no. 1 (2013): 147470491301100, <https://doi.org/10.1177/147470491301100109>.↵
56. For a broader discussion of how violent groups build these ‘micro-solidarities’, see Malešević, *The Rise of Organised Brutality*, chap. 2.↵
57. Ferguson, ‘The Prehistory of War and Peace in Europe and the Near East’, 224–5.↵
58. For a similar point, see Turchin, *Ultrasociety*, chap. 8. Turchin also highlights how the Germans went from being egalitarian to borrowing state-style hierarchies from the Romans, although he attributes the spread of dominance hierarchies more to their innate success in battle than to an authoritarian impulse. There might be something to this, although military historians are split on what is the ideal level of decentralization in a fighting force. ↵
59. Kent A. Kiehl and Morris B. Hoffman, ‘The Criminal Psychopath: History, Neuroscience, Treatment, and Economics’, *Jurimetrics* 51 (Summer 2011): 355–97, <https://pubmed.ncbi.nlm.nih.gov/24944437/>.↵

60. On the base rate of 1 per cent, see James E. Freeman, Freya Samson, and Gavan R. Palk, 'Identifying the Presence of Psychopathy in the Community', APS Forensic Psychology National Conference, 2011-08-04 to 2011-08-06 (unpublished), <https://eprints.qut.edu.au/37263/>. On the current level among CEOs, see Simon Croom, Katarina Fritzson, and Nathan Brooks, 'Personality Differences and Buyer-Supplier Relationships: Psychopathy in Executives, Gender Differences and Implications for Future Research', *Journal of Purchasing and Supply Management* 27, no. 4 (2021): 100721, <https://doi.org/10.1016/j.pursup.2021.100721>. P. Babiak, C. S. Neumann, and R. D. Hare, 'Corporate Psychopathy: Talking the Walk', *Behavioral Sciences & the Law* 28, no. 2 (2010): 174–93, <https://doi.org/10.1002/bsl.925>, 183, find a prevalence rate of 3.9 per cent among high-profile corporate professionals. Karen Landay, Peter Harms, and Marcus Credé, 'Shall We Serve the Dark Lords? A Meta-Analytic Review of Psychopathy and Leadership', *Journal of Applied Psychology* 104 (2018), <https://doi.org/10.1037/apl0000357>, suggest that this could be lower, finding a positive but weak relationship between psychopathy and leadership positions in a meta-analysis of ninety-two existing studies. An accessible overview of some of the literature can be found in Gardner Selby, '1 in 100 People Are Sociopaths', *Politifact*, 30 March 2018, www.politifact.com/factchecks/2018/mar/30/michael-mccaul/michael-mccaul-says-1-100-people-sociopath/. These figures are ultimately best guesses. Désiré G. C. Palmen, Emile W. Kolthoff, and Jan J. L. Derksen, 'The Need for Domination in Psychopathic Leadership: A Clarification for the Estimated High Prevalence of Psychopathic Leaders', *Aggression and Violent Behavior* 61 (November–December 2021): 101650, <https://doi.org/10.1016/j.avb.2021.101650>, suggest that the connection is largely due to a psychopathic need to dominate, which is best achieved through leadership positions. One of the problems across the literature is that psychopathy – like all the dark triad – is on a spectrum.↵
61. B. N. Persson and S. O. Lilienfeld, 'Social Status as One Key Indicator of Successful Psychopathy: An Initial Empirical Investigation', *Personality and Individual Differences* 141 (2019): 209–17, <https://doi.org/10.1016/j.paid.2019.01.020>. ↵

62. Thomson and Halstead, 'How Violent Was the Pre-Agricultural World?', 107.↵
63. Those high on the dark triad also tend to score significantly higher on 'social dominance orientation': a measurement of how much one craves power over others.↵
64. Jeremy Hogeveen, Michael Inzlicht, and Sukhvinder Obhi, 'Power Changes How the Brain Responds to Others', *Journal of Experimental Psychology: General* 143, no. 2 (2014): 755–62, <https://doi.org/10.1037/a0033477>.↵
65. For an overview, see T. O. Tobore, 'On Power and Its Corrupting Effects: The Effects of Power on Human Behavior and the Limits of Accountability Systems', *Communicative & Integrative Biology* 16, no. 1 (2023): 2246793, <https://doi.org/10.1080/19420889.2023.2246793>.↵
66. Klaas, *Corruptible*, 102.↵
67. This also holds true for scholars who claim that states and hierarchies were a way to deal with 'scalar stress'. Even if that were the case, it does little to explain the inequality and authoritarian arrangements. ↵
68. Others call these Goliath traps 'multipolar traps': situations in which competition forces self-interested, rational agents to resort to actions which are to their short-term benefit but to the long-term detriment of the group. ↵

CHAPTER 9: BROKEN BRONZE

1. Nadali et al., 'Representations of Violence in Ancient Mesopotamia and Syria', 636.↵
2. Known as the Stele of the Vultures. See *ibid.*, 634.↵
3. There are other problems with territorial estimates. Often the land was not arable, the areas exaggerated by written sources to flatter rulers, and many of the borders were often porous and constantly raided. They are still ultimately the best estimates we have. For a full discussion of the problems with territorial estimates, see Scheidel, 'The Scale of Empire: Territory, Population, Distribution'; Peter Fibiger Bang, 'Empire – A World History: Anatomy and Concept, Theory and Synthesis', in *The Oxford World History of Empire* vol. I: *The Imperial Experience*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), <https://doi.org/10.1093/oso/9780199772360.003.0002>.↵
4. This was described by Michael Mann as the move from 'despotic power' to 'infrastructural power'. See Mann, *The Sources of Social Power I*. This doesn't mean that intimidation was phased out in favour of administration. Rather administration was relied on more but was still always underpinned by the threat of violence.↵
5. John Darwin, *After Tamerlane: The Rise and Fall of Global Empires, 1400–2000* (Penguin, 2008).↵
6. Similarly, the historian Niall Ferguson once quipped 'much of what we call history consists of the deeds of 50 to 70 empires'. See Niall Ferguson, 'Empires with Expiration Dates', *Foreign Policy* 156 (2009): 46.↵
7. Michael W. Doyle, *Empires* (Cornell University Press, 1986), 19; Rein Taagepera, 'Size and Duration of Empires Growth-Divide Curves, 3000 to 600 B.C.', *Social Science Research* 7, no. 2 (June 1978): 181, [https://doi.org/10.1016/0049-089X\(78\)90010-8](https://doi.org/10.1016/0049-089X(78)90010-8). Definitions vary but usually focus on the control of 'sovereignty'. Taagepera focuses on 'large sovereign political entities' while Doyle looks at 'political societies', which are never clearly defined. See also Darwin's

definition of imperialism (p. 416) as ‘the attempt to impose one state’s predominance over other societies by assimilating them to its political and cultural and economic system’. Although his definition of empire differs slightly as ‘systems of influence or rule in which ethnic, cultural or ecological boundaries were overlapped or ignored’. I’ve opted for the simpler and broader ‘political groups’ over the use of sovereign polity, societies, or systems. The basic idea of a core exploiting a periphery is also central to the idea of a ‘galactic polity’ (Graeber and Sahlins, *On Kings*, 9.), as well as a world-system (Immanuel Maurice Wallerstein, *World-Systems Analysis: An Introduction* (Duke University Press, 2004). For a useful overview of this idea of core–periphery relationships in world-systems theory, read Christopher Chase-Dunn and Dmytro Khutkyy, ‘The Evolution of Geopolitics and Imperialism in Interpolity Systems’, in *The Oxford World History of Empire* vol. I: *The Imperial Experience*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021),

<https://doi.org/10.1093/oso/9780199772360.003.0004>.↵

8. David Harvey, *The New Imperialism* (Oxford University Press, 2005), 1; Darwin, *After Tamerlane*. The idea of empire simply being one group trying to control another can be found in Bang, ‘Empire – A World History’.↵
9. Some call this process ‘internal colonization’: a nonsensical idea since every empire begins by conquering its neighbours.↵
10. For some of these definitional debates, see Krishan Kumar, *Empires: A Historical and Political Sociology* (Polity Press, 2021), chaps. 1 and 4. There are a few potential exceptions, such as more democratic confederations. ↵
11. John F. Haldon, ‘The Political Economy of Empire: “Imperial Capital” and the Formation of Central and Regional Elites’, in *The Oxford World History of Empire* vol. I: *The Imperial Experience*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), <https://doi.org/10.1093/oso/9780199772360.003.0006>, 179.↵
12. Territory estimates come from Taagepera, ‘Size and Duration of Empires: Growth-Decline Curves, 3000 to 600 B.C.’, 187.↵

13. Harvey Weiss, ed., *Megadrought and Collapse: From Early Agriculture to Angkor* (Oxford University Press, 2017). This megadrought (known as the 4.2ka Megadrought, since it happened around 4.2 thousand years ago) marks the beginning of the current geologic age known as the ‘Meghalayan Age’.↵
14. Dan Lawrence, Alessio Palmisano, and Michelle W. de Gruchy, ‘Collapse and Continuity: A Multi-Proxy Reconstruction of Settlement Organization and Population Trajectories in the Northern Fertile Crescent during the 4.2kya Rapid Climate Change Event’, *PLOS ONE* 16, no. 1 (2021): e0244871, <https://doi.org/10.1371/journal.pone.0244871>.↵
15. Z. B. Ön et al., ‘A Bayesian Test for the 4.2 ka BP Abrupt Climatic Change Event in Southeast Europe and Southwest Asia Using Structural Time Series Analysis of Paleoclimate Data’, *Climatic Change* 165, nos 1–2 (March 2021): 7, <https://doi.org/10.1007/s10584-021-03010-6>.↵
16. Lisa Cooper, ‘The Demise and Regeneration of Bronze Age Urban Centres in the Euphrates Valley of Syria’, in *After Collapse: The Regeneration of Complex Societies*, ed. Glenn M. Schwartz and John Jackson Nichols (University of Arizona Press, 2006).↵
17. Morris, “‘Lo, Nobles Lament’”; Ellen Morris, ‘Ancient Egyptian Exceptionalism: Fragility, Flexibility and the Art of Not Collapsing’, in *The Evolution of Fragility: Setting the Terms*, ed. Norman Yoffee (Cambridge University Press, 2019).↵
18. Norman Yoffee and George L. Cowgill, eds., *The Collapse of Ancient States and Civilizations* (University of Arizona Press, 1991).↵
19. Alena Giesche et al., ‘Recurring Summer and Winter Droughts from 4.2–3.97 Thousand Years Ago in North India’, *Communications Earth & Environment* 4, no. 1 (2023): 103, <https://doi.org/10.1038/s43247-023-00763-z>. This shift from cities and trade to self-reliant villages was not a completely easy one. In Harappa, one of the largest settlements of the Indus Valley, there are signs of heightened violence and more infectious diseases such as tuberculosis and leprosy among those who remained. For an overview, see Cameron A. Petrie, ‘Diversity, Variability, Adaptation and “Fragility” in the Indus

- Civilization’, in *The Evolution of Fragility: Setting the Terms*, ed. Norman Yoffee (Cambridge University Press, 2019), chap. 7, 123.↵
20. Larry Neal, ‘Introduction’, in *The Cambridge History of Capitalism*, ed. Larry Neal and Jeffrey G. Williamson (Cambridge University Press, 2014), 2, <https://doi.org/10.1017/CHO9781139095099.001>; Branko Milanović, *Capitalism, Alone: The Future of the System That Rules the World* (The Belknap Press of Harvard University Press, 2021).↵
 21. Michael Jursa, ‘Babylonia in the First Millennium BCE – Economic Growth in Times of Empire’, in *The Cambridge History of Capitalism*, ed. Larry Neal and Jeffrey G. Williamson (Cambridge University Press, 2014), 28–33, <https://doi.org/10.1017/CHO9781139095099.002>.↵
 22. Karl Moore and David Lewis, ‘The First Multinationals: Assyria circa 2000 BC’, *MIR: Management International Review* 38, no. 2 (1998): 95–107, www.researchgate.net/publication/233925545; Sophie Hardach, ‘The Secret Letters of History’s First-Known Businesswomen’, BBC News, 14 January 2021, www.bbc.com/worklife/article/20210111-the-secret-letters-of-historys-first-businesswomen.↵
 23. Even Babylonian temples and palaces were dependent on the market: selling up to half of their agricultural income for silver.↵
 24. Cline, “‘Mind the Gap’”, 101.↵
 25. Carol Bell, ‘The Merchants of Ugarit: Oligarchs of the Late Bronze Age Trade in Metals?’, in *Eastern Mediterranean Metallurgy and Metalwork in the Second Millennium BC: A Conference in Honour of James D. Muhly; Nicosia, 10th–11th October 2009* (Oxbow Books, 2012), 180–87.↵
 26. For the most compelling case for the centrality of wood, see Roland Ennos, *The Wood Age: How One Material Shaped the Whole of Human History* (William Collins, 2021).↵
 27. Eric H. Cline and Diane Cline, ‘Text Messages, Tablets, and Social Networks: The “Small World” of the Amarna Letters’, in *There and Back Again – the Crossroads II: Proceedings of an International Conference Held in Prague, September 15–18, 2014*, ed. Jana

- Mynářová et al. (Charles University [Prague], Faculty of Arts, 2015), 17–44.↵
28. These gifts were transported by merchants, traders, and sailors who no doubt also exchanged staple goods such as grain. The banal practice of haggling and trading was below rulers, who instead relied on the more regal tradition of gift-giving. We see this pattern of commercial trade occurring under the cover of gift-giving across the world. Trobriand Islander chiefs in the South Pacific would come together to exchange shell armbands and necklaces, while the crew who transported them busied themselves trading practical staples with the locals.↵
29. Cline, 1177, 2nd edn, 75.↵
30. Pertev Basri and Dan Lawrence, 'Wealth Inequality in the Ancient Near East: A Preliminary Assessment Using Gini Coefficients and Household Size', *Cambridge Archaeological Journal* 30, no. 4 (2020): 701, <https://doi.org/10.1017/S0959774320000177>. Kohler and Smith, *Ten Thousand Years of Inequality*, chap. 1, and Christian E. Peterson and Robert D. Drennan, 'Letting the Gini out of the Bottle', in *Ten Thousand Years of Inequality: The Archaeology of Wealth Differences*, ed. Timothy A. Kohler and Michael Ernest Smith (University of Arizona Press, 2018), 39–66, discuss and address the problems with measuring inequality using household sizes.↵
31. As the Oxford archaeologist Josephine Quinn observed, 'The whole Bronze Age system produces a lot of discontent' (quoted in Annalee Newitz, 'The Elites Were Living High. Then Came the Fall', *New York Times*, 5 November 2020). On the height of pharaohs, see Michael E. Habicht et al., 'Body Height of Mummified Pharaohs Supports Historical Suggestions of Sibling Marriages', *American Journal of Physical Anthropology* 157, no. 3 (2015): 519–25, <https://doi.org/10.1002/ajpa.22728>. For evidence on the Mycenaeans, as well as height and inequality more broadly, see Carles Boix and Frances Rosenbluth, 'Bones of Contention: The Political Economy of Height Inequality', *American Political Science Review* 108, no. 1 (2014): 1–22, <https://doi.org/10.1017/S0003055413000555>.↵
32. Cline, 1177, 2nd edn, 155–8.↵
33. David Kaniewski and Elise Van Campo, '3.2 Ka BP Megadrought and the Late Bronze Age Collapse', in *Megadrought and Collapse*, ed.

- Harvey Weiss (Oxford University Press, 2017),
<https://doi.org/10.1093/oso/9780199329199.003.0005>.↵
34. For an overview of the evidence, see Cline, 1177, 2nd edn.↵
35. Martin Finné et al., ‘Late Bronze Age Climate Change and the Destruction of the Mycenaean Palace of Nestor at Pylos’, *PLOS ONE* 12, no. 12 (2017): e0189447,
<https://doi.org/10.1371/journal.pone.0189447>.↵
36. Amos Nur and Eric H. Cline, ‘Poseidon’s Horses: Plate Tectonics and Earthquake Storms in the Late Bronze Age Aegean and Eastern Mediterranean’, *Journal of Archaeological Science* 27, no. 1 (2000): 43–63, <https://doi.org/10.1006/jasc.1999.0431>.↵
37. Ibid. It can be difficult to distinguish an earthquake from another force that destroyed a city, whether it be a revolt or invasion. Archaeologists tended to rely on telltale signs, such as skeletons pinned under fallen walls and pillars knocked askew. See Cline, “Mind the Gap”, 100.↵
38. Kemp and Cline, ‘Systemic Risk and Resilience in the Bronze Age’, 214.↵
39. A. Bernard Knapp, *Migration Myths and the End of the Bronze Age in the Eastern Mediterranean* (Cambridge University Press, 2021). Guy D. Middleton, ‘Should I Stay or Should I Go? Mycenaeans, Migration, and Mobility in the Late Bronze Age and Early Iron Age Eastern Mediterranean’, *Journal of Greek Archaeology* 3 (2018): 115–45, <https://doi.org/10.32028/jga.v3i.525>, provides a useful overview of the problems with the Sea Peoples narrative and puts forward a more realistic interpretation of this as simply a case of adaptive mobility.↵
40. For an overview, see Cline, 1177, 2nd edn. Sometimes this even took the form of linguistic resistance. Ugarit, a multilingual, metropolitan trading hub, adopted a local vernacular language and its own alphabetic cuneiform script. This was seemingly a rejection of the Hittite Empire, which it had come under the control of. See Philip J. Boyes, ‘Negotiating Imperialism and Resistance in Late Bronze Age Ugarit: The Rise of Alphabetic Cuneiform’, *Cambridge Archaeological Journal* 29, no. 2 (2019): 185–99, <https://doi.org/10.1017/S0959774318000471>.↵

41. Inspired by a figure from Austin Crouch, 'Egypt and the Bronze Age Collapse', *Culture Frontier*, 2023, www.culturefrontier.com/bronze-age-collapse/. The original has been significantly revised and expanded, including the addition of estimated territorial extracts of different empires and regions.↵
42. Cline, *1177*, 2nd edn, chap. 6.↵
43. These are similar to 'synchronous failures': Thomas Homer-Dixon et al., 'Synchronous Failure: The Emerging Causal Architecture of Global Crisis', *Ecology and Society* 20, no. 3 (2015): art6, <https://doi.org/10.5751/ES-07681-200306>. Catalytic failures means that the different crises, rather than simply being synchronized, directly catalyse each other. The Late Bronze Age collapse had both.↵
44. This term was originally coined by my colleague Lauren Holt in a paper I co-authored with her and several others: S. J. Beard et al., 'Assessing Climate Change's Contribution to Global Catastrophic Risk', *Futures* 127 (March 2021): 2, <https://doi.org/10.1016/j.futures.2020.102673>.↵
45. Kemp and Cline, 'Systemic Risk and Resilience in the Bronze Age', 214–17. Inspired by Figure 1.↵
46. Igor Linkov et al., 'Are Civilizations Destined to Collapse? Lessons from the Mediterranean Bronze Age', *Global Environmental Change* 84 (2024): 102792, <https://doi.org/10.1016/j.gloenvcha.2023.102792>.↵
47. Eric Cline, *After 1177 B.C.: The Survival of Civilizations* (Princeton University Press, 2024), 15–18.↵
48. Basri and Lawrence, 'Wealth Inequality in the Ancient Near East', 701. Kohler and Smith, *Ten Thousand Years of Inequality*, chap. 1.↵
49. Petra M. Creamer, 'Inequalities in Wealth Distribution Within Imperial Assyrian Graves', *Antiquity* 98, no. 400 (2024): 920–35. There are a few problems with measurements using grave goods. In particular, they are performative and can change over time with culture. Creamer discusses and addresses these potential problems on pages 5–6.↵
50. Guy D. Middleton, 'Mycenaean Collapse(s) c. 1200 BC', in *Collapse and Transformation: The Late Bronze Age to Early Iron Age in the*

Aegean, ed. G. D. Middleton (Oxbow Books, 2020), 12, 18.↵

51. On the ‘escape’ from Mycenaean Greece see Erika Weiberg and Martin Finné, ‘Resilience and Persistence of Ancient Societies in the Face of Climate Change: A Case Study from Late Bronze Age Peloponnese’, *World Archaeology* 50, no. 4 (2018): 595, <https://doi.org/10.1080/00438243.2018.1515035>. Also, Joseph Maran, ‘The Crisis Years? Reflections on Signs of Instability in the Last Decades of the Mycenaean Palaces’, *Scienze Dell’Antichità* 15 (2009): 241–62. Maybe the rural populations were more exposed to bandits and robbers in Dark Age Greece, but for now that is difficult to see.↵
52. Raphael Greenberg, ‘What Collapsed in 1177?’, *Journal of Eastern Mediterranean Archaeology and Heritage Studies* 10, no. 2 (2022): 193, <https://doi.org/10.5325/jeasmedarcherstu.10.2.0191>.↵
53. Claudia Glatz, *The Making of Empire in Bronze Age Anatolia: Hittite Sovereign Practice, Resistance, and Negotiation* (Cambridge University Press, 2020), 299, <https://doi.org/10.1017/9781108867436>.↵
54. Poet or poets? There is a continued scholarly disagreement as to whether Homer was a single, historical figure or a composite of multiple authors.↵
55. Although, funnily enough, Homer was largely glorifying the Late Bronze Age in his myths.↵
56. Ian Morris, ‘The Collapse and Regeneration of Complex Society in Greece, 1500–500 BCE’, in *After Collapse: The Regeneration of Complex Societies*, ed. Glenn M. Schwartz and John Jackson Nichols (University of Arizona Press, 2006), 72–85. We can’t say that the collapse directly led to Athenian democracy (the two were centuries apart), but it seems exceedingly unlikely that the latter would have occurred without the former.↵
57. V. Gordon Childe, *The Prehistory of European Society* (Spokesman, 2009). For a more complete discussion on the relationship between technological change and production during dark ages, see Chris Harman, *A People’s History of the World: From the Stone Age to the New Millenium* (Verso, 2017).↵

58. As archaeologist William Dever has observed, speaking of Canaan in the wake of the Bronze Age collapse, 'Perhaps the most important conclusion to be drawn about the "Dark Age"... is that it was nothing of the sort.' See W. G. Dever, 'The Late Bronze – Early Iron I Horizon in Syria-Palestine: Egyptians, Canaanites, "Sea Peoples," and Proto-Israelites', in *The Crisis Years: The 12th Century B.C.*, ed. W. A. Ward and M. S. Joukowsky (Kendall/Hunt, 1992); Cline, 1177, 2nd edn.↵

CHAPTER 10: SHATTERED OBSIDIAN

1. Marcello A. Canuto et al., 'Ancient Lowland Maya Complexity as Revealed by Airborne Laser Scanning of Northern Guatemala', *Science* 361, no. 6409 (2018): 1, <https://doi.org/10.1126/science.aau0137>.↵
2. Luke Auld-Thomas et al., 'Running out of Empty Space: Environmental Lidar and the Crowded Ancient Landscape of Campeche, Mexico', *Antiquity* 98, no. 401 (2024): 1340–58, <https://doi.org/10.15184/aqy.2024.148>.↵
3. Joyce Marcus, 'Political Fluctuations in Mesoamerica', *National Geographic Research and Exploration* 8, no. 4 (1992): 406–9.↵
4. 'Gone with the wind' comes from Michael D. Coe, *The Maya* (Thames & Hudson, 2005), 162. On the Cancuén case study, see Arthur A. Demarest et al., 'Economy, Exchange, and Power: New Evidence from the Late Classic Maya Port City of Cancuen', *Ancient Mesoamerica* 25, no. 1 (2014): 210, <https://doi.org/10.1017/S0956536114000121>. For an overview of the collapse across the different cities and kingdoms, see Arthur Andrew Demarest, Prudence M. Rice, and Don S. Rice, *The Terminal Classic in the Maya Lowlands: Collapse, Transition, and Transformation* (University Press of Colorado, 2005). Storey and Storey, *Rome and the Classic Maya*, chaps. 4, 5, and 6, provide a view of the collapse across the Maya as a conjoined system.↵
5. Tobias Braun et al., 'Decline in Seasonal Predictability Potentially Destabilized Classic Maya Societies', *Communications Earth & Environment* 4, no. 1 (2023): 82, <https://doi.org/10.1038/s43247-023-00717-5>, provide the most recent evidence on declining predictability.

Shanti Morell-Hart, Lydie Dussol, and Scott L. Fedick, ‘Agriculture in the Ancient Maya Lowlands (Part 1): Paleoethnobotanical Residues and New Perspectives on Plant Management’, *Journal of Archaeological Research* 31, no. 4 (2023): 593–5, cover the surprisingly diverse and resilient nature of Mayan food production. For nuanced accounts on the impact of drought and its interactions with the political drivers of collapse, see Peter M. J. Douglas et al., ‘Impacts of Climate Change on the Collapse of Lowland Maya Civilization’, *Annual Review of Earth and Planetary Sciences* 44, no. 1 (2016): 613–45, <https://doi.org/10.1146/annurev-earth-060115-012512>; B. L. Turner and Jeremy A. Sabloff, ‘Classic Period Collapse of the Central Maya Lowlands: Insights about Human–Environment Relationships for Sustainability’, *Proceedings of the National Academy of Sciences* 109, no. 35 (2012): 13908–14, <https://doi.org/10.1073/pnas.1210106109>. Arthur A. Demarest, ‘The Classic Maya Collapse, Water, and Economic Change in Mesoamerica: Critique and Alternatives from the “Wet Zone”’, in *The Great Maya Droughts in Cultural Context*, ed. G. Iannone (University Press of Colorado, 2014), 177–206, outlines how the Peten collapse appeared to have preceded the drought. For a nuanced overview of the archaeology of the collapse period, and particularly the drought hypothesis, see Middleton, ‘Collapse Studies in Archaeology from 2012 to 2023’, 30–33.↵

6. For a summary, see Douglas et al., ‘Impacts of Climate Change on the Collapse of Lowland Maya Civilization’, 636.↵
7. Petrie, ‘Diversity, Variability, Adaptation and “Fragility” in the Indus Civilization’.↵
8. John Haldon et al., ‘SDG 13 – How Societies Succeeded or Failed to Respond to Environmental Disruption’, in *Before the UN Sustainable Development Goals: A Historical Companion*, ed. M. Gutmann (Oxford Scholarship Online, 2022), 385–424, <https://doi.org/10.1093/oso/9780192848758.003.0014>; Joseph A. Tainter, ‘Archaeology of Overshoot and Collapse’, *Annual Review of Anthropology* 35, no. 1 (2006): 59–74, <https://doi.org/10.1146/annurev.anthro.35.081705.123136>; Guy D. Middleton, ‘Nothing Lasts Forever: Environmental Discourses on the

- Collapse of Past Societies’, *Journal of Archaeological Research* 20, no. 3 (2012): 257–307, <https://doi.org/10.1007/s10814-011-9054-1>;
- John Haldon et al., ‘Demystifying Collapse: Climate, Environment, and Social Agency in Pre-Modern Societies’, *Millennium* 17, no. 1 (2020): 1–33, <https://doi.org/10.1515/mill-2020-0002>.↵
9. Tainter, ‘Archaeology of Overshoot and Collapse’.↵
 10. K. W. Butzer, ‘Collapse, Environment, and Society’, *Proceedings of the National Academy of Sciences* 109, no. 10 (2012): 3632–9, <https://doi.org/10.1073/pnas.1114845109>, 3634–6.↵
 11. Thanks to Marilyn Masson for this point and for her invaluable comments and assistance with this chapter.↵
 12. Urszula Strawinska-Zanko et al., ‘Capital in the First Century: The Evolution of Inequality in Ancient Maya Society’, in *Mathematical Modeling of Social Relationships*, ed. Urszula Strawinska-Zanko and Larry S. Liebovitch (Springer International, 2018), 161–92, https://doi.org/10.1007/978-3-319-76765-9_9.↵
 13. Amy E. Thompson, Gary M. Feinman, and Keith M. Prufer, ‘Assessing Classic Maya Multi-Scalar Household Inequality in Southern Belize’, ed. Jacob Freeman, *PLOS ONE* 16, no. 3 (2021): e0248169, <https://doi.org/10.1371/journal.pone.0248169>.↵
 14. Turner and Sabloff, ‘Classic Period Collapse of the Central Maya Lowlands’.↵
 15. David M. Pendergast, ‘Stability through Change: Lamanai, Belize, from the Ninth to the Seventeenth Century’, in *Late Lowland Maya Civilization: Classic to Postclassic*, ed. Jeremy A. Sabloff and E. Wyllys Andrew (SAR Press, 1986), 223–49.↵
 16. For a full review of the population and political recovery of the Maya, see Marilyn A. Masson et al., ‘Postclassic Maya Population Recovery and Rural Resilience in the Aftermath of Collapse in Northern Yucatan’, *Journal of Anthropological Archaeology* 76 (2024): 101610, <https://doi.org/10.1016/j.jaa.2024.101610>. As with the Late Bronze Age, the collapse and recovery varied substantially depending on the region.↵
 17. Douglas J. Kennett et al., ‘Drought-Induced Civil Conflict Among the Ancient Maya’, *Nature Communications* 13, no. 1 (2022): 3911,

<https://doi.org/10.1038/s41467-022-31522-x>.↵

18. Some have claimed that this was partly a rejection of autocracy in favour of more democratic arrangements. See Diane Z. Chase and Arlen F. Chase, 'Framing the Maya Collapse: Continuity, Discontinuity, Method, and Practice in the Classic to Postclassic Southern Maya Lowlands', in *After Collapse: The Regeneration of Complex Societies*, ed. Glenn M. Schwartz and John Jackson Nichols (University of Arizona Press, 2006), 168–88. That is perhaps an overstatement, given that the new rulers carried on many of the symbols and practices of the southern Lowland holy lords. On continuity of practices in the postclassic period, see Marilyn A. Masson, 'Postclassic and Contact-Period Maya Rulership', in *Faces of Rulership in the Maya Region*, ed. Patricia A. McAnany (Harvard University Press, 2025).↵
19. Marilyn A. Masson, 'The "Othering" of Maya Political Economies', *Archaeological Papers of the American Anthropological Association* 32, no. 1 (2021): 109–27, <https://doi.org/10.1111/apaa.12147>.↵
20. The two potential exceptions are the Indus Valley script and the Rongorongo script of the Rapa Nui, although whether these are original scripts is highly contested.↵
21. Stephen Houston, John Baines, and Jerrold Cooper, 'Last Writing: Script Obsolescence in Egypt, Mesopotamia, and Mesoamerica', *Comparative Studies in Society and History* 45, no. 3 (2003): 432, <https://doi.org/10.1017/S0010417503000227>.↵
22. Ibid. For Figure 27, the icons for Shang Oracle Bones and Mesopotamian Cuneiform are stylized replicas and hence may not be perfect characters. The icons for Olmec Hieroglyphs come from Wikimedia by the author Madman2001. See https://commons.wikimedia.org/wiki/File:San_Andreas_Cylinder_Seal_Print_1.svg. Published under the Creative Commons License Version 4.0. The icons for Egyptian Hieroglyphs come from Wikimedia by Riccardo.mebere. See https://commons.wikimedia.org/wiki/File:Hieroglyph_brain.org. Published under the Creative Commons License Version 4.0.↵
23. James T. Hooker, *Linear B: An Introduction* (Bristol Classical Press, 1980); Houston, Baines, and Cooper, 'Last Writing'.↵

24. Another example of trade bottlenecks acting as a source of lootable resources, much like the Nabataeans mentioned in Chapter 3. Karl W. Butzer, 'Rise and Fall of Axum, Ethiopia: A Geo-Archaeological Interpretation', *American Antiquity* 46, no. 3 (1981): 471–95, <https://doi.org/10.2307/280596>, 489–92.↵
25. Ibid.↵
26. Ibid.; Gianfrancesco Lusini, 'The Decline and Collapse of the Kingdom of Aksum (6th–7th CE): An Environmental Disaster or the End of a Political Process?', in *The End of Empires*, ed. Michael Gehler, Robert Rollinger, and Philipp Strobl (Wiesbaden: Springer Fachmedien Wiesbaden, 2022), 321–36, https://doi.org/10.1007/978-3-658-36876-0_15. The pastoralists on its borders for whom it had previously exacted tribute also turned on it.↵
27. On army size, see David C. Conrad, *Empires of Medieval West Africa: Ghana, Mali, and Songhay*, rev. edn (Chelsea House, 2010), 71. On the lack of clarity regarding the break-up of the Ghana Empire, see Nikolas Gestrinch, 'The Empire of Ghana' (Oxford Research Encyclopedias, 2019), <https://doi.org/10.1093/ACREFORE/9780190277734.013.396>.↵
28. Madina Ly-Tall, 'The Decline of the Mali Empire', in *General History of Africa IV: Africa from the Twelfth to the Sixteenth Century*, ed. D. T. Niane (UNESCO/Heinemann, 1984), 172–86, 173–4.↵
29. Margaret Shinnie, *Ancient African Kingdoms* (Edward Arnold, 1968), 85.↵
30. Ibiang Oden Ewa, 'Pre-Colonial West Africa: The Fall of Songhai Empire Revisited', *Journal of the Historical Society of Nigeria* 26 (2017): 1–24, 20, www.jstor.org/stable/48562076.↵
31. Ibid. For a more entertaining overview of the rise and fall of Ghana, Mali, and Songhai, read Paul Cooper, *Fall of Civilizations: Stories of Greatness and Decline* (Duckworth, 2024), chap. 11.↵
32. Tobias Hagmann and Markus V. Hoehne, 'Failures of the State Failure Debate: Evidence from the Somali Territories', *Journal of International Development* 21, no. 1 (2009): 53, <https://doi.org/10.1002/jid.1482>.↵

33. Some even suggest that we should not think of cases like the Lowland Maya or the Late Bronze Age as singular collapses. Rather they are just a sequence of different state failures and disasters detonating over centuries. The failure of each city – such as Tikal and Ugarit – and state had its own unique background and trajectory. That case is made most forcefully by the archaeologist Guy Middleton: see Guy Middleton, ‘Do Civilisations Collapse?’, *Aeon*, 16 November 2017, <https://aeon.co/essays/what-the-idea-of-civilisational-collapse-says-about-history>. Cases like the Lowland Maya and the Late Bronze Age can, of course, be both. There were chains of numerous catastrophes and state failures over time, but they were also connected. Many of the Mayan city-states were destroyed by competitors and neighbours, but they were also reacting to a shared challenge of drought. The same applies to the Late Bronze Age. The fall of the Ugarit trading hub had ramifications for the greater Mediterranean. We even see this in African cases. The rise of Portuguese shipping undermined the gold trade of the Songhai, just as wars between the Byzantines and Persians, or the expansion of the caliphate, bled the Aksumite Empire dry. For the Maya and Late Bronze Age, the impacts cascaded and amplified to bring down the broader economic and cultural system. There’s no reason to ignore the connections or presume that the wider systems collapse was just a coincidence.↵

CHAPTER 11: THE RISE AND FALL OF THE CHINESE GOLIATH

1. This time of decentralization (770 to 481 BCE) became known as the Spring and Autumn period.↵
2. In 771 BCE Quanrong pastoralists (a Tibeto-Burman group) destroyed the Zhou capital of Haojing (near present-day Xi’an) and killed King You. The Zhou moved to a capital in the east, creating the Eastern Zhou Dynasty (the earlier reign became known as the Western Zhou). Unfortunately for the Zhou, the loss of their capital undermined their power and their feudal lords began to act more and more independently, negotiating their own alliances and declaring war on each other.↵

3. This wasn't just a Qin tactic: the Athenian Empire carried out similar mass executions, as have others.↵
4. Victoria Tin-bor Hui, 'Genocide, Extermination and Mass Killing in Chinese History', in *The Cambridge World History of Genocide*, vol. I: *Genocide in the Ancient, Medieval and Premodern Worlds*, ed. Ben Kiernan, T. M. Lemos, and Tristan S. Taylor (Cambridge University Press, 2023), 289–93, <https://doi.org/10.1017/9781108655989.017>.↵
5. Fernanda Pirie, *The Rule of Laws: A 4,000-Year Quest to Order the World* (Profile Books, 2021). Both legalism and Confucianism were direct reactions to the disorder of their times. Shang Yang (390–338 BCE) and Han Fei (280–233 BCE), the two most prominent philosophers of legalism, advocated a stringent, punishment-based version of law coupled with a strong government and rigid bureaucracy. It was a strategy aimed at restoring imperial control – a strategy that some credit as a strong influence on the centralized approach of modern China today. It was a philosophy that took a dim view of humans, one that looked remarkably similar to the Hobbesian worldview. Confucianism, in contrast, emphasized the obligations of rulers towards the ruled and opposed codified law. Confucius (551–479 BCE) sought to lay a blueprint for the reestablishment of political order.↵
6. Mark Edward Lewis et al., 'The First East Asian Empires: Qin and Han', in *The Oxford World History of Empire*, vol. I: *The Imperial Experience*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), 218–20, <https://doi.org/10.1093/oso/9780197532768.003.0008>. He took on the title of 'Huang Di'. It was a pivotal linguistic shift. Previous leaders had called themselves *wang*, loosely translating to 'king'. 'Di' referred to mythical, heroic sages who first created 'civilization', and had previously been used to designate high gods under the Shang. The emperor also began the Feng and Shen rituals in which he would ascend Mount Tai to converse with the gods and become immortal. He had philosophers craft a new cosmic cycle based on the old calendar, which justified his claim to the throne.↵
7. Ibid., 220–23. The most commonly cited reason for the wall is protection from invaders, but Owen Lattimore suggests that keeping wayward peasants in the grip of the Qin was another major motivation.

See Owen Lattimore, 'The Frontier in History', in his *Studies in Frontier History: Collected Papers, 1929–58* (Oxford University Press, 1962), 476–81.[↵](#)

8. Lewis et al., 'The First East Asian Empires', 223.[↵](#)
9. This process of amassing elite treasures was interrupted briefly by the Wang Mang rebellion (9–23 CE). The revolt appears to have been triggered by flooding of the Yellow River, with the ensuing famine and disaster undermining the emperor's already fraying rule. While typically remembered in Chinese history as a villain, Wang Mang was a Confucian social reformer who abolished slavery and redistributed the land more equally. His forward-looking actions made him enemies. Soon he was overthrown and killed during rebellions fostered by disgruntled elites. Tristram R. Kidder et al., 'New Perspectives on the Collapse and Regeneration of the Han Dynasty', in *Beyond Collapse: Archaeological Perspectives on Resilience, Revitalization, and Transformation in Complex Societies*, ed. Ronald K. Faulseit (Southern Illinois University Press, 2016), 70–88.[↵](#)
10. Scheidel, *The Great Leveler*, chap. 2; Scheidel, 'The End of "Peak Empire"'.[↵](#)
11. The general history in this chapter relies on Lewis et al., 'The First East Asian Empires', in *The Oxford World History of Empire*, vol. II: *The History of Empires*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), <https://doi.org/10.1093/oso/9780197532768.001.0001>. This particular rendition of the collapse of the Han and Jin is best expressed in Scheidel, 'The End of "Peak Empire"'.[↵](#)
12. The period between the Tang and the Song is one without unified empire. It is typically called the Five Dynasties and Ten Kingdoms Period. It ended in 979 CE when the Song defeated their competitors and re-established a coherent empire.[↵](#)
13. The notable exceptions being the Warring States period and the end of the Tang Dynasty, which resulted in the battles of the Five Dynasties and Ten Kingdoms Period.[↵](#)
14. Yuhua Wang, *The Rise and Fall of Imperial China: The Social Origins of State Development* (Princeton University Press, 2022), chaps. 3 and

- 5; Yuhua Wang, 'The Curse of Long-Ruling Autocrats', 18 January 2023, <https://press.princeton.edu/ideas/the-curse-of-long-ruling-autocrats>.↵
15. The marauders constantly threatened and harassed China's northern frontiers, leading to the Han even paying them tribute in an attempt to avoid confrontation. ↵
 16. Wang, *The Rise and Fall of Imperial China*. ↵
 17. Georg Orlandi et al., 'Structural-Demographic Analysis of the Qing Dynasty (1644–1912) Collapse in China', *PLOS ONE* 18, no. 8 (2023): e0289748, <https://doi.org/10.1371/journal.pone.0289748>.↵
 18. Wang, *The Rise and Fall of Imperial China*, chap. 2; Chaochao Gao et al., 'Volcanic Climate Impacts Can Act as Ultimate and Proximate Causes of Chinese Dynastic Collapse', *Communications Earth & Environment* 2, no. 1 (2021): 234, <https://doi.org/10.1038/s43247-021-00284-7>. For some wise warnings about mistaking climate correlation for causation, see Dagomar Degroot et al., 'Towards a Rigorous Understanding of Societal Responses to Climate Change', *Nature* 591, no. 7851 (2021): 539–50, <https://doi.org/10.1038/s41586-021-03190-2>; Dagomar Degroot et al., 'The History of Climate and Society: A Review of the Influence of Climate Change on the Human Past', *Environmental Research Letters* 17, no. 10 (2022): 103001, <https://doi.org/10.1088/1748-9326/ac8faa>.↵
 19. Hoyer et al., 'Navigating Polycrisis'.↵
 20. It's not as clear in cases discussed earlier, including the systems collapses of the Late Bronze Age and the Lowland Maya, owing to less writing and documentation being available, but there is little reason to suspect it was missing.↵
 21. In this account, there is a limited number of elite positions and competition becomes a problem once they are all filled. Like a game of musical chairs (or better put, musical thrones), those left without a seat become frustrated elite aspirants willing to upend the status quo to achieve power. While important, there are some nuances and minor problems with this theory. First, many of the apparent frustrated elites come from highly privileged backgrounds (see the examples of Donald Trump and Tucker Carlson below and many of the instigators of

rebellion throughout Chinese history). They are usually outraged that they can't transfer one form of power into another (such as from billionaire to politician) or because they have been publicly humiliated or denied status (such as the failed bureaucrats of China, or, again, Trump and Carlson). Second, elite positions are not always clearly limited in number. There is no limit on how many people can be wealthy in any society. This was more of a problem in pre-modern states and in certain domains (such as government). Third, many elite aspirants clearly have an insatiable hunger for status. Trump is a billionaire, has his name plastered across skyscrapers, and is perhaps the best-known person in the world. Yet, he still ferociously competes for power. Clearly, achieving high office once was not enough for him. For the most ambitious, status is an infinite resource and getting to sit in one seat in the game of musical thrones will never be enough. Rather than a case of elite overproduction creating 'counter-elites', it is better to see this as status competition heating up as the number of players increases, the number of elite positions becomes saturated, and the size of the prize swells.↵

22. This process of growing elite competition is part of 'structural-demographic theory'. The theory was originally pioneered by the demographer Jack Goldstone. Goldstone helped establish the 'Political Instability Task Force' of the US, which used structural-models focused on elite competition to predict which states would fail next. For an overview, see Peter Turchin, *War and Peace and War: The Rise and Fall of Empires* (Penguin, 2007); Turchin, *End Times*; Peter Turchin and Sergey A. Nefedov, *Secular Cycles* (Princeton University Press, 2009). Peter Turchin, the biggest developer of structural-demographic theory, is now warning that the US is entering a phase of heightened elite competition and crisis: Peter Turchin, 'Political Instability May Be a Contributor in the Coming Decade', *Nature* 463, no. 7281 (2010): 608, <https://doi.org/10.1038/463608a>. ↵
23. On the encounter with Jon Stewart acting as a turning point for Carlson, see Devin Gordon, 'What Happened to Jon Stewart?', *The Atlantic*, 21 April 2022, www.theatlantic.com/culture/archive/2022/04/the-problem-with-jon-stewart-tucker-carlson/629608/. The calculation on Trump being a millionaire at age eight comes from Ross Buettner, Susan Craig, and

David Barstow, '11 Takeaways From The Times's Investigation Into Trump's Wealth', *New York Times*, 2 October 2018, www.nytimes.com/2018/10/02/us/politics/donald-trump-wealth-fred-trump.html.↵

24. Notably, even competition for bureaucratic positions in China and elsewhere was usually among rich or upper-middle-class families. Those who failed to translate their riches into other forms of power or more status rebelled.↵
25. Butzer, 'Collapse, Environment, and Society', 3638. There is the historiographical problem that later regimes often had a good reason to brand their predecessors as corrupt. That way, their coup or rebellion was justified. That said, both the recurrence, the genuine evidence of tax dodging and growing fortunes of public officials, and the modern relationship between corruption and inequality, should all give us confidence that this is a real phenomenon.↵
26. Laura Policardo and Edgar J. Sánchez Carrera, 'Corruption Causes Inequality, or Is It the Other Way Around? An Empirical Investigation for a Panel of Countries', *Economic Analysis and Policy* 59 (2018): 92–102, <https://doi.org/10.1016/j.eap.2018.05.001>; Jong-Sung You and Sanjeev Khagram, 'A Comparative Study of Inequality and Corruption', *American Sociological Review* 70, no. 1 (2005): 136–57, <https://doi.org/10.1177/000312240507000107>.↵
27. Scheidel, *The Great Leveler*, 134.↵
28. From a version of the Crisis Database provided by Dan Hoyer in August 2024.↵
29. There are, of course, exceptions. The Yuan Dynasty, a regime of Mongol invaders, adopted many of the structures of previous Chinese dynasties, but kept much of the previous intelligentsia excluded from critical positions.↵
30. John Haldon et al., 'Lessons from the Past, Policies for the Future: Resilience and Sustainability in Past Crises', *Environment Systems and Decisions* 40, no. 2 (2020): 295, <https://doi.org/10.1007/s10669-020-09778-9>.↵
31. Naturally there is some nuance here. It took three centuries after the collapse of the Han for there to be reunification under the Sui/Tang.↵

32. There are some caveats here. More complex weapons cannot be easily replicated. As we'll see in later chapters, groups such as the Inuit lost technologies such as kayak making. However, this only tends to occur with smaller, isolated groups.↵
33. Analabha Basu, Neeta Sarkar-Roy, and Partha P. Majumder, 'Genomic Reconstruction of the History of Extant Populations of India Reveals Five Distinct Ancestral Components and a Complex Structure', *Proceedings of the National Academy of Sciences* 113, no. 6 (2016): 1594–9, <https://doi.org/10.1073/pnas.1513197113>.↵
34. Catherine E. Bolten and Richard 'Drew' Marcantonio, 'Usurious Strangers and "a Better Tomorrow": Agricultural Loans, Education, and the "Poverty Trap" in Rural Sierra Leone', *Economic Anthropology* 10, no. 1 (2023): 77 and 87, <https://doi.org/10.1002/sea2.12256>.↵
35. Bennet Bronson, 'Patterns of Political Regeneration in Southeast and East Asia', in *After Collapse: The Regeneration of Complex Societies*, ed. Glenn M. Schwartz and John Jackson Nichols (University of Arizona Press, 2006), 144. See also Scheidel, *Escape from Rome*, chap. 9.↵
36. Ibid., chap. 8. For some nuance and criticisms, see Peter Turchin, 'The Great Escape: A Review Essay on *Escape from Rome: The Failure of Empire and the Road to Prosperity* by Walter Scheidel', *Cliodynamics: The Journal of Quantitative History and Cultural Evolution* 11, no. 2 (2021): 78–81, <https://doi.org/10.21237/C7clio11251633>.↵

CHAPTER 12: THE FALL OF ROME

1. For population, see Morris, *The Measure of Civilization*, 147–8, table 4.1, and 155–6, table 4.2. For territorial extent, see Rein Taagepera, 'Size and Duration of Empires: Growth-Decline Curves, 600 B.C. to 600 A.D.', *Social Science History* 3, no. 3 (1979): 115–38, <https://doi.org/10.1017/S014555320002294X>. The EU is closer to 4.2 million. Bear in mind that such territorial estimates are problematic. Often an empire would claim ownership of a territory even if their control was tenuous and only really consisted of a garrison. Nonetheless, Rome's was remarkably large. For summaries of the

general history of Rome that I draw upon, see Scheidel, *Escape from Rome*; Bang, ‘The Roman Empire’; Walter Scheidel, ‘Ancient Mediterranean City-State Empires: Athens, Carthage, Early Rome’, in *The Oxford World History of Empire*, vol. II: *The History of Empires*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), 137–58,

<https://doi.org/10.1093/oso/9780197532768.003.0005>. Rome was also remarkably large in population. While estimates are difficult, at its peak the Roman Empire may have governed between a quarter and a third of the world’s population. No other empire has ruled over a larger fraction of humanity (the British Empire only ever encompassed around a fifth of global population).↵

2. According to one viral meme in 2023, most men apparently still muse about Rome several times a week. ‘The Roman Empire: Why Men Just Can’t Stop Thinking about It’, *Guardian*, 19 September 2023, www.theguardian.com/lifeandstyle/2023/sep/19/the-roman-empire-why-men-just-cant-stop-thinking-about-it. I confess to being in this category, although hopefully with a better reason than most.↵
3. A king ordered them to be killed and, like Sargon and Moses, they were saved by being set in a basket upon a river. They floated down the Tiber until the two were suckled by a female wolf and then adopted by a shepherd. Rome was built upon the site.↵
4. Jeremy Armstrong, *War and Society in Early Rome: From Warlords to Generals* (Cambridge University Press, 2016). ↵
5. Rome began as a kingdom in 753 BCE, before the nobles overthrew the last king (with the memorable name of King Lucius Tarquinius Superbus).↵
6. Scheidel, *Escape from Rome*, 98–104. Young, land-owning male citizens had to serve in the military for six to seven years (usually while they were young, aggressive, and had yet to inherit family farms), and more during times of crisis.↵
7. Grain estimates are based on Paul Erdkamp, ‘Annona (Grain)’, in *Oxford Classical Dictionary*, ed. Thomas Wiedermann (Oxford University Press, 2022), 9, <https://doi.org/10.1093/acrefore/9780199381135.013.8000>. I’ve taken the common estimate of 30 million *modii* of subsidized grain imports

into Rome and converted it into metric tonnes by assuming each *modius* is around seven kilograms (the most widely accepted conversion).↵

8. For those outside the city, participation was limited to those rich enough to easily travel to Rome regularly.↵
9. For a summary of Rome's inequality, see Scheidel, *The Great Leveler*, chap. 2.↵
10. On corruption, see Scheidel, *The Great Leveler*, 39, 42. The rapacious governor is cited in Bang, 'The Roman Empire', 241.↵
11. On silver mine yields, see J. C. Edmondson, 'Mining in the Later Roman Empire and Beyond: Continuity or Disruption?', *Journal of Roman Studies* 79 (1989): 85, <https://doi.org/10.2307/301182>.↵
12. Joseph A. Tainter, 'Collapse and Sustainability: Rome, the Maya, and the Modern World', *Archaeological Papers of the American Anthropological Association* 24, no. 1 (2014): 204, <https://doi.org/10.1111/apaa.12038>.↵
13. Scheidel, 'The End of "Peak Empire"', 113.↵
14. On the effects of the Byzantine invasion (and a contrarian take on the fall of Rome) see Edward Jay Watts, *The Eternal Decline and Fall of Rome: The History of a Dangerous Idea* (Oxford University Press, 2021). The most forceful case for the impact of climate and disease is made by Kyle Harper in *The Fate of Rome: Climate, Disease, and the End of an Empire* (Princeton University Press, 2017). This has been significantly challenged, most notably by John Haldon et al., 'Plagues, Climate Change, and the End of an Empire: A Response to Kyle Harper's *The Fate of Rome* (1): Climate', *History Compass* 16, no. 12 (2018): e12508, <https://doi.org/10.1111/hic3.12508>. The effect of the Plague of Justinian (the first recorded outbreak of bubonic plague in Europe, which began in 541 CE during that emperor's reign) in particular has been called into question, since multiple lines of evidence such as burials and coin deposits don't show the usual telltale signs of a deadly epidemic: Lee Mordechai et al., 'The Justinianic Plague: An Inconsequential Pandemic?', *Proceedings of the National Academy of Sciences* 116, no. 51 (2019): 25546–54, <https://doi.org/10.1073/pnas.1903797116>.↵

15. Kemp et al., 'Societal Collapse as a Powershift'.[↵](#)
16. Walter Scheidel, *What Was the Roman Republic Really Like?*, Humanities West 2014 Roman Republic Program, www.youtube.com/watch?v=U4Stzxi-m4s. Land area estimates are taken from Taagepera, 'Size and Duration of Empires: Growth-Decline Curves, 600 B.C. to 600 A.D.', 125. Scheidel gives the example of a city with the population of Palo Alto and size of Santa Clara County overtaking the contiguous US. Palo Alto is also around 60,000 people while Santa Clara county is approximately three times the size of Rome (3,344 km²). The US (9,372,610 km²) is approximately half the size of the Roman Empire at its height. Hence, if anything the comparison slightly underestimates the scale of Rome's expansion.[↵](#)
17. Ayşe Zarakol, *Before the West: The Rise and Fall of Eastern World Orders* (Cambridge University Press, 2022). For a further discussion of the importance of expansion and the myth of civilization to empire, see Bang, 'Empire – A World History'; Kumar, *Empires*, chap. 2.[↵](#)
18. Paul M. Kennedy, *The Rise and Fall of the Great Powers: Economic Change and Military Conflict from 1500 to 2000* (Vintage Books, 1989); Paul Kennedy, 'Survival of Empires', in *Survival: The Survival of the Human Race*, ed. Emily Shuckburgh (Cambridge University Press, 2008).[↵](#)
19. Crooks and Parsons, 'Empires, Bureaucracy and the Paradox of Power'.[↵](#)
20. Covered primarily in Chapter 7. For summaries of the evidence, see Wilkinson and Pickett, *The Spirit Level*; Wilkinson and Pickett, *The Inner Level*.[↵](#)
21. Covered primarily in Chapter 11. For key recent pieces, see Policardo and Carrera, 'Corruption Causes Inequality'; You and Khagram, 'A Comparative Study of Inequality and Corruption'.[↵](#)
22. Covered primarily in Chapters 7 and 11. For summaries, see Turchin, *End Times*; Turchin and Nefedov, *Secular Cycles*; Hoyer et al., 'Navigating Polycrisis'.[↵](#)
23. See Chapter 7 as well as Kemp, 'Diminishing Returns on Extraction'.[↵](#)
24. Haldon et al., 'SDG 13'.[↵](#)

25. This is particularly evident in Scott A. J. Johnson, *Why Did Ancient Civilizations Fail?* (Routledge, 2017); Diamond, *Collapse*. Technically Scott is an archaeologist and Diamond is more of an anthropologist, but they are both writing in the field of deep history.↵
26. For a full summary and further evidence, see Kemp, 'Diminishing Returns on Extraction'.↵
27. Thomas Homer-Dixon, *The Upside of Down: Catastrophe, Creativity, and the Renewal of Civilization* (Island PR, 2008); Scheidel, 'The End of "Peak Empire"'.↵
28. Scheidel, *Escape from Rome*, 174.↵
29. Scheidel, 'The End of "Peak Empire"', 112. Remarkably, Rome and the Han Dynasty of China collapsed in a similar way. They even fell at roughly the same time. Both had reigns that were punctuated by periods of crisis and disorder. Rome was beset by the Third Century Crisis and the Han by the Wang Mang rebellion. The Han fell apart just fifteen years before Rome's crisis period, and the Jin fell just a few decades before Rome capitulated. One leading theory is that the Antonine Plague originated in China and spread to Rome via the Silk Roads (or originated in central Asia before spreading to China and Rome). Like Rome, the Han also had to contend with external invaders. Rome and the Han were also beset by a changing climate. Crises during the third century CE for both Rome and the Han occurred during a time of cooling across the northern hemisphere. See Harper, *The Fate of Rome*; Kyle Harper and Michael McCormick, 'Reconstructing the Roman Climate', in *The Science of Roman History*, ed. Walter Scheidel (Princeton University Press, 2018), 11–52, <https://doi.org/10.23943/9781400889730-006>. Compare Haldon et al., 'Plagues, Climate Change, and the End of an Empire'.↵
30. This can be seen as a blending of different theories of collapse. Turchin, Hoyer, and Goldstone all stress the role of popular immiseration and elite conflict in their structural-demographic theory. Diamond highlighted the role of climate change, loss of trading partners, war, environmental degradation, and poor responses in his five-part explanation. Joseph Tainter suggested that collapse was due to 'diminishing returns on complexity': societies become more complex to address new problems but face diminishing returns which

eventually overwhelm them. This is usually due to popular discontent (from being overworked and overtaxed) and loss of revenue (due to the sheer complexity of regulatory and social systems). Daron Acemoglu and James Robinson point towards extractive institutions as being the key driver for why some states ‘succeed’ and others ‘fail’ in economic development. Diminishing returns on extraction knits these all together and provides additional clear pathways from inequality to collapse based on evidence. That includes growing studies on the weakness of oligarchy in decision-making through the empirical work on the link between corruption and inequality. Each of these theories has its weaknesses. Structural-demographic theory struggles to explain the role of other factors beyond elite factionalism and popular immiseration. Diamond’s framework is largely dependent on external shocks and lacks a clear rationale for why some societies respond poorly to catastrophe. Tainter’s theory is elegant but fundamentally flawed by the fact that complexity is far too ambiguous (scholars have measured this as the number of monuments built, trees felled, the energy ratio, or the size of an army, and, as covered earlier, complexity is usually just used as a euphemism for inequality and hierarchy) and it’s not at all clear why a society needs to increase ‘complexity’ (whatever that is) to resolve oncoming problems, or how exactly this leads to collapse. For a full critique, see Kemp, ‘Diminishing Returns on Extraction’. Bringing these all together into a single explanation links ancient and modern case studies, addresses the shortcomings of the individual theory, and links them to a wider explanation of how and why societies, states, and empires developed over time. It also has overlap with the frameworks provided by Hartley, ‘State Crisis Theory’, and Butzer, ‘Collapse, Environment, and Society’. I agree with their more nuanced frameworks, which suggest that crises can lead to reform, entrenchment, or full-blown collapse. Collapse is not the only possible outcome. For brevity and clarity I’m focusing purely on collapse in this section.↵

31. Kemp et al., ‘Societal Collapse as a Powershift’.↵
32. As we’ll see later, the key crisis aspect of structural-demographic theory sees elite competition as a battle over limited high-status positions. In other words, elite competition is status competition.↵

33. The age of the Byzantine Empire would be around 1,058–1,145 years. This depends on when exactly you choose to date the formation and end of the empire. Constantinople was conquered by the Ottomans in 1453, but the last outpost of the empire valiantly fought on until 1479. Personally, I'd opt for the loss of the capital as the marker for state termination. The empire had lost its sovereignty and independence even if a few troops continued to hold out. This would put its age at 1,145 years. Notably, there was a minor collapse in 1204 when the crusaders of the Fourth Crusade sacked Constantinople and partitioned the empire. It was recaptured in 1261 and limped on for another two centuries. Even if we take this lower bound of 1204, the empire still lasted for an impressive 874 years. The quote of an empire that would not die comes from John F. Haldon, *The Empire That Would Not Die: The Paradox of Eastern Roman Survival, 640–740* (Harvard University Press, 2016).↵
34. Ibid.↵
35. Robert Perry Stephan, 'House Size and Economic Growth: Regional Trajectories in the Roman World', unpublished PhD dissertation, Stanford University, 2013.↵
36. Wyman, 'How Do You Know If You're Living Through the Death of an Empire?'↵
37. Lawrence et al., 'Long-Term Trends in Settlement Persistence in Southwest Asia'.↵
38. Nikola Koepke, 'The Biological Standard of Living in Europe from the Late Iron Age to the Little Ice Age', in *The Oxford Handbook of Economics and Human Biology*, ed. J. Kolmos and I. Kelly (Oxford University Press, 2016), 78–81, <https://doi.org/10.1093/oxfordhb/9780199389292.013.34>; Nikola Koepke and Joerg Baten, 'The Biological Standard of Living in Europe During the Last Two Millennia', *European Review of Economic History* 9, no. 1 (2005): 76–7, <https://doi.org/10.1017/S1361491604001388>; Irene Barbiera and Gianpiero Dalla-Zuanna, 'Population Dynamics in Italy in the Middle Ages: New Insights from Archaeological Findings', *Population and Development Review* 35, no. 2 (2009): 375, <https://doi.org/10.1111/j.1728-4457.2009.00283.x>; Monica

- Giannecchini and Jacopo Moggi-Cecchi, 'Stature in Archeological Samples from Central Italy: Methodological Issues and Diachronic Changes', *American Journal of Physical Anthropology* 135, no. 3 (2008): 290, <https://doi.org/10.1002/ajpa.20742>.↵
39. Boix and Rosenbluth, 'Bones of Contention', fig. 1.↵
40. Koepke, 'The Biological Standard of Living in Europe from the Late Iron Age to the Little Ice Age', 86–8, 88–90.↵
41. For useful summaries, see Joan Ballester et al., 'Effect of the Great Recession on Regional Mortality Trends in Europe', *Nature Communications* 10, no. 1 (2019): 679, <https://doi.org/10.1038/s41467-019-08539-w>; Lynne Peeples, 'How the Next Recession Could Save Lives', *Nature* 565, no. 7740 (2019): 412–15, <https://doi.org/10.1038/d41586-019-00210-0>.↵
42. On the influence of the Roman model on modern emergency powers, see Marc de Wilde, 'Just Trust Us: A Short History of Emergency Powers and Constitutional Change', *Comparative Legal History* 3, no. 1 (2015): 110–30, <https://doi.org/10.1080/2049677X.2015.1041728>; Daniel D. Slate, 'Crisis Government's Legitimacy Paradox: Foreseeability and Unobservable Success', in *Intersections, Reinforcements, Cascades: Proceedings of the 2023 Stanford Existential Risks Conference*, The Stanford Existential Risks Initiative, 2023, <https://doi.org/10.25740/ZJ321VJ7513>; Luke Kemp, 'The "Stomp Reflex": When Governments Abuse Emergency Powers', *BBC Future*, 28 April 2021, www.bbc.com/future/article/20210427-the-stomp-reflex-when-governments-abuse-emergency-powers.↵
43. Scheidel, *What Was the Roman Republic Really Like?*↵
44. Although it wasn't invented by the Romans; it dates back to at least the Assyrians of the Bronze Age.↵
45. The loved ones of the crucified often mistakenly gave them water, prolonging their suffering.↵
46. Although eventually it was resettled. It is now a suburb of Tunis, the capital of Tunisia.↵
47. States are, as Shakespeare once said, 'jealous in honour, sudden and quick in quarrel, seeking the bubble reputation even in the cannon's mouth'. There is an abundant and growing literature on war and

competition between states as a struggle for status. Seminal works include Jonathan Renshon, *Fighting for Status: Hierarchy and Conflict in World Politics* (Princeton University Press, 2017); William C. Wohlforth, 'Unipolarity, Status Competition, and Great Power War', in *International Relations Theory and the Consequences of Unipolarity*, ed. G. John Ikenberry, Michael Mastanduno, and William C. Wohlforth (Cambridge University Press, 2011), 33–66, <https://doi.org/10.1017/CBO9780511996337.002>. States are usually seen as being most likely to resort to war when their actual capabilities outstrip their reputation or as a way to gain face after being subjugated to humiliating events, such as the wartime reparations imposed on Germany after the First World War under the Treaty of Versailles. It is usually produced by either a leader, or leaders, seeking status themselves through their countries' actions, or domestic groups using national status as a tool to shift foreign policy. For a critical stocktake of the literature on status and outstanding problems and questions, see Paul K. MacDonald and Joseph M. Parent, 'The Status of Status in World Politics', *World Politics* 73, no. 2 (2021): 358–91, <https://doi.org/10.1017/S0043887120000301>; Allan Dafoe, Jonathan Renshon, and Paul Huth, 'Reputation and Status as Motives for War', *Annual Review of Political Science* 17, no. 1 (2014): 371–93, <https://doi.org/10.1146/annurev-polisci-071112-213421>. ↵

CHAPTER 13: FEAR THE RISE

1. Many archaeologists are aware of the problems with these terms and are trying to combat them. ↵
2. Scheidel, *Escape from Rome*, 453.↵
3. There were a few exceptions, such as Roman concrete, but most of the developments we associate with the Roman Empire came earlier, often from the peoples they conquered.↵
4. Marie D. Jackson et al., 'Phillipsite and Al-Tobermorite Mineral Cements Produced through Low-Temperature Water-Rock Reactions in Roman Marine Concrete', *American Mineralogist* 102, no. 7 (2017): 1435–50, <https://doi.org/10.2138/am-2017-5993CCBY>.↵

5. W. H. R. Rivers, *The Disappearance of Useful Arts* (Forgotten Books, 2017); Joseph Henrich, *The Secret of Our Success*, chap. 12.[↵](#)
6. Henrich, *The Secret of Our Success*, chap. 12.[↵](#)
7. On Greek Fire, see John Haldon, “‘Greek Fire’ Revisited: Recent and Current Research’, in *Byzantine Style, Religion, and Civilization: In Honour of Sir Steven Runciman*, ed. Steven Runciman and Elizabeth Jeffreys (Cambridge University Press, 2006).[↵](#)
8. I use welfare and standards of living interchangeably here. This is gauged by real income (income accounting for inflation), as this is the way it is usually measured in the economic history literature. Thanks to Eric Cline for noticing how this echoes the Egyptian saying of ‘life, prosperity, and happiness’.[↵](#)
9. Bas van Leeuwen, Jieli Li, and Reinhard Pirngruber, ‘The Standard of Living in Ancient Societies: A Comparison between the Han Empire, the Roman Empire, and Babylonia’, 2013, https://pure.knaw.nl/ws/files/5691366/CGEHWP45_vanleeuwenetal_9.pdf. Notably, this paper has not been published in peer-review literature, although the authors are each well-published economic historians with expertise in most of the relevant case studies. More importantly, the broader point is supported by calculations (drawing from a range of sources) in Daron Acemoglu and Simon Johnson, *Power and Progress: Our Thousand-Year Struggle over Technology and Prosperity* (PublicAffairs, 2023).[↵](#)
10. On the height of pharaohs, see Habicht et al., ‘Body Height of Mummified Pharaohs’. For evidence on Native Americans and the Mycenaeans, as well as height and inequality more broadly, see Boix and Rosenbluth, ‘Bones of Contention’.[↵](#)
11. For the estimates of agricultural productivity during 1000–1300 CE, see Acemoglu and Johnson, *Power and Progress*. There was, of course, some variation. Being a farmer in Rome almost certainly was an improvement on being a farmer in Uruk, one of the earliest cities in Mesopotamia. Yet many of those improvements came on the back of conquest, appropriation, and widespread slavery. In any case, these differences were not enormous. For the best estimate of human height over the long run (in the UK), see Gregori Galofré-Vilà, Andrew Hinde, and Aravinda Meera Guntupalli, ‘Heights Across the Last 2,000

Years in England', in *Research in Economic History*, ed. Christopher Hanes and Susan Wolcott, vol. XXXIV (Emerald Publishing Limited, 2018), 67–98, <https://doi.org/10.1108/S0363-326820180000034003>.

The general idea is supported by a range of other estimates summarized in Boix and Rosenbluth, 'Bones of Contention', and others.↵

12. Tomasz Frackowiak et al., 'Subjective Happiness Among Polish and Hadza People', *Frontiers in Psychology* 11 (2020): 1173, <https://doi.org/10.3389/fpsyg.2020.01173>; Robert William Martin and Andrew J. Cooper, 'Subjective Well-Being in a Remote Culture: The Himba', *Personality and Individual Differences* 115 (2017): 19–22, <https://doi.org/10.1016/j.paid.2016.10.021>. It's worth noting that people in modern industrialized countries are taller and happier than those living in the Roman Empire, hence there should be an even bigger difference in happiness between nomadic egalitarian foragers and those living during the peak of historical empires.↵
13. See preceding references in this chapter, as well as chapters 1 and 4. The evidence that average human height is only just reaching Palaeolithic levels (for men, at least) is strong. While there are only a few studies comparing egalitarian hunter-gatherer happiness to that of industrialized countries, they do produce the common result that the hunter-gatherer societies are mentally better off. Today, OECD countries account for around 17 per cent of global population. A fairer comparison of average global happiness would be a factory worker in China or a peasant farmer in sub-Saharan Africa against an egalitarian forager. The difference in happiness is likely to be far more striking.↵
14. For an overview, see Acemoglu and Johnson, *Power and Progress*.↵
15. Piketty, *Capital in the Twenty-First Century*, 336.↵
16. Scheidel, *The Great Leveler*, 448.↵
17. It may have been as much as 60 per cent in places such as Greece and the British Isles.↵
18. On height after the Black Death, see Galofré-Vilà, Hinde, and Guntupalli, 'Heights Across the Last 2,000 Years in England'; Emily J. Brennan and Sharon N. DeWitte, 'Sexual Stature Difference

Fluctuations in Pre- and Post-Black Death London as an Indicator of Living Standards', *American Journal of Human Biology* 34, no. 10 (2022): e23783, <https://doi.org/10.1002/ajhb.23783>. Interestingly, women seem to benefit less in terms of height, and in some cases average heights even decrease. This is likely to have been because better nutrition meant that women reached puberty earlier and stopped growing. Sharon N. DeWitte, 'Mortality Risk and Survival in the Aftermath of the Medieval Black Death', ed. Andrew Noymer, *PLOS ONE* 9, no. 5 (2014): e96513, <https://doi.org/10.1371/journal.pone.0096513>, found lower mortality rates in the wake of the Black Death. On wages, see Fernand Braudel, *Capitalism and Material Life: 1400–1800* (Fontana/Collins, 1979), chap. 3. As the historian of epidemics Samuel Cohn Jr remarked, the Black Death did not result in the 'decline, or collapse of civilizations, at least in Europe, but instilled remarkable resilience'. Samuel K. Cohn, 'The Black Death: Collapse, Resilience, and Transformation', in *How Worlds Collapse*, ed. Miguel Centeno et al. (Routledge, 2023), 191, <https://doi.org/10.4324/9781003331384-13>.↵

19. For an overview, see Scheidel, *The Great Leveler*, chap. 10. On the demise of feudalism and wider institutional impacts, see Remi Jedwab, Noel D. Johnson, and Mark Koyama, 'The Economic Impact of the Black Death', *Journal of Economic Literature* 60, no. 1 (2022): 132–78, <https://doi.org/10.1257/jel.20201639>; David D. Haddock and Lynne Kiesling, 'The Black Death and Property Rights', *Journal of Legal Studies* 31, no. S2 (2002): S545–87, <https://doi.org/10.1086/345566>. On the economic benefits of the plague (or, the 'plague bonus'), see James Belich, *The World the Plague Made: The Black Death and the Rise of Europe* (Princeton University Press, 2022).↵
20. Van Bavel and Scheffer, 'Historical Effects of Shocks on Inequality'.↵
21. Leticia Arroyo Abad, Elwyn Davies, and Jan Luiten van Zanden, 'Between Conquest and Independence: Real Wages and Demographic Change in Spanish America, 1530–1820', *Explorations in Economic History* 49, no. 2 (2012): 149–66, <https://doi.org/10.1016/j.eeh.2011.12.001>. A broader analysis of why

power and not just technology matters for real wages can be found in Acemoglu and Johnson, *Power and Progress*. Inclusive institutions and how power is distributed can also help explain why societies fared differently when facing the plague. Both medieval England (1348–50) and Egypt under the rule of the Mamluk Sultanate (1347–9) were struck by the Black Death. Both suffered in a similar way. Mamluk Egypt lost around a third of its population, which was then followed by a flooding of the Nile, grain shortages, famine, and conflict. The Mamluks completely mismanaged the crisis: they gave little aid to the commoners and instead spent lavishly on new construction projects. Rural areas suffered large loss of life, lowering the tax revenue of the empire, and it underspent on its irrigation infrastructure, worsening the lot of the farmers. This mismanagement and recurring bouts of the plague for the next century both contributed to its fall in 1517. In contrast, we've already seen that England in the decades that followed had a large-scale improvement in the health, wages, and well-being of the population, especially peasants. The difference in outcomes results from a difference in how both elites and commoners responded. And part of that difference was due to inclusive institutions: England already had a parliament, while the Mamluks had a supreme sultan who ruled through generals and provincial governors.↵

22. While many others had access to gunpowder and iron (including China and the Mughal Empire in India), Europeans more aggressively refined and developed their arsenals, as well as the military tactics to use them. See Philip T. Hoffman, *Why Did Europe Conquer the World?* (Princeton University Press, 2015).↵
23. Scheidel, *Escape from Rome*; Hoffman, *Why Did Europe Conquer the World?* New financial institutions essentially helped to extend the monetary supply and fuel the military-industrial complex. One pivotal development was the creation of the Bank of England, in 1694. It was not constructed to drive economic growth and stability, but rather to assist a war against the French (the War of the League of Augsburg). It issued mutual shares with a guaranteed 8 per cent return and within a year had raised £1.5 million for the war effort. Within a decade Britain had defeated the French. See Tim Di Muzio and Matt Dow, 'Uneven and Combined Confusion: On the Geopolitical Origins of Capitalism and the Rise of the West', *Cambridge Review of International Affairs*

- 30, no. 1 (2017): 14–18,
<https://doi.org/10.1080/09557571.2016.1256949>; Tim Di Muzio,
Carbon Capitalism: Energy, Social Reproduction and World Order
(Rowman & Littlefield International, 2015); Haldon, ‘The Political
Economy of Empire: “Imperial Capital” and the Formation of Central
and Regional Elites’, 201–2.↵
24. Larry Neal and Jeffrey G. Williamson, eds., *The Cambridge History of
Capitalism* (Cambridge University Press, 2014),
<https://doi.org/10.1017/CHO9781139095099>.↵
25. As with the Bronze Age, rarer items from more distant lands usually
meant more status.↵
26. Belich, *The World the Plague Made*. Book production collapsed with
the plague, when far too many of the already scarce class of writers
and scholars died. This encouraged the development and uptake of the
1440 Gutenberg printing press. By 1500, Western Europe had
produced 20 million books.↵
27. Another key was fossil fuels. Around the same time, deforestation
made timber in England scarce, sending its price skyrocketing and
encouraging a move towards alternative energy sources such as coal.
See Di Muzio, *Carbon Capitalism*. Others argue that it was not
deforestation that drove the push towards coal, but rather active
attempts by elites to undermine labour power (Andreas Malm, *Fossil
Capital: The Rise of Steam-Power and the Roots of Global Warming*
(Verso, 2016).)↵
28. Jason Hickel, *Less Is More: How Degrowth Will Save the World*
(Windmill Books, 2021), 49. The ‘peasants are too rich’ quote comes
from Christopher Dyer, ‘A Redistribution of Incomes in England’,
Past and Present 39, no. 1 (1968): 11–33,
<https://doi.org/10.1093/past/39.1.11>. Malm, *Fossil Capital*.↵
29. Haldon, ‘The Political Economy of Empire’, 184.↵
30. Rein Taagepera, ‘Expansion and Contraction Patterns of Large
Polities: Context for Russia’, *International Studies Quarterly* 41, no. 3
(1997): 485–7, <https://doi.org/10.1111/0020-8833.00053>.↵
31. Based on data from Scheidel, ‘The Scale of Empire’. I’ve taken the
largest empire of every 500 years for the past 2,500 years, as well as

the first empire to approach 1 million km² (the Akkadian Empire). I've made one slight change by opting for the Roman Empire over the Macedonian Empire, mainly because the latter was so short-lived.↵

32. Daniel Headrick, 'A Double-Edged Sword: Communications and Imperial Control in British India', *Historical Social Research / Historische Sozialforschung* 35, no. 1 (2010): 53, <https://doi.org/10.12759/HSR.35.2010.1.51-65>.↵
33. Mohini Lal Majumdar, *Early History and Growth of Postal System in India* (RDDHI-India, 1995), 252.↵
34. Mortality estimates for the invasion of the Americas were taken from Alexander Koch et al., 'Earth System Impacts of the European Arrival and Great Dying in the Americas after 1492', *Quaternary Science Reviews* 207 (2019): 13–36, <https://doi.org/10.1016/j.quascirev.2018.12.004>. Estimates for the Black Death are from White, *The Great Big Book of Horrible Things*. These were then cross-referenced with global population estimates from Our World in Data, 'Population, 10,000 BCE to 2021', Our World in Data, 2022, <https://ourworldindata.org/grapher/population?yScale=log>. Luke Muelhauser also provides a useful dataset which compares different high-mortality events with global population estimates: <https://docs.google.com/spreadsheets/d/14I49e43PphzocDcZghYm6ekQZcJ3JeuicJ9ZafTa7GA/edit?gid=447443392#gid=447443392>. However, his estimates of the Mongol conquests under Chinggis Khan are outdated and too high (reaching almost 10 per cent under his calculations). Using more reliable data suggests that the Mongol Empire was closer to killing around 2 per cent of global population (around 10–14 million people, rather than 37–40 million). For a discussion of the problems with the initial estimates of Mongol casualties, and suggestions for more plausible numbers, see Michael Mann, 'Have Wars and Violence Declined?', *Theory and Society* 47, no. 1 (2018): 43–5, <https://doi.org/10.1007/s11186-018-9305-y>.↵
35. On global decimation threats, see Kemp et al., 'Climate Endgame'.↵
36. Johan Elverskog, 'Zünghar (Dzungar) Khanate', in *The Encyclopedia of Empire*, ed. Nigel Dalziel and John M. MacKenzie (Wiley, 2016), 1–6, <https://doi.org/10.1002/9781118455074.wbeoe065>.↵

37. Frankopan, *The Earth Transformed*, 383.↵
38. Malešević, *The Rise of Organised Brutality*. Chains of command also help to diffuse responsibility across numerous individuals, giving every person an easier way to rationalize immoral actions, such as ‘If I don’t do it, someone else will’.↵
39. Gary Uzonyi, ‘State Failure, Genocide and Politicide Reconsidered’, *British Journal of Politics and International Relations* 25, no. 1 (2023): 83–101, <https://doi.org/10.1177/13691481211044654>.↵

CHAPTER 14: COLONIZATION AND COLLAPSE

1. Not to be confused with the other Francisco Hernández de Córdoba, who founded Nicaragua.↵
2. The Columbian Exchange may have also introduced syphilis into Europe from the Americas, although this is a matter of ongoing debate. Others believe that syphilis had already been in Europe for a long period of time. See Marylynn Salmon, ‘Manuscripts and Art Support Archaeological Evidence that Syphilis Was in Europe Long Before Explorers Could Have Brought It Home from the Americas’, *The Conversation*, 13 July 2022, <https://theconversation.com/manuscripts-and-art-support-archaeological-evidence-that-syphilis-was-in-europe-long-before-explorers-could-have-brought-it-home-from-the-americas-182114>.↵
3. On the Aztecs, see Guido Alfani and Alfonso Carballo, ‘Income and Inequality in the Aztec Empire on the Eve of the Spanish Conquest’, *Nature Human Behaviour* 7 (2023): 1265–74, <https://doi.org/10.1038/s41562-023-01636-3>.↵
4. Michael E. Smith and Maëlle Sergheraert, ‘The Aztec Empire’, in *The Oxford World History of Empire*, vol. II: *The History of Empires*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), 276–88, <https://doi.org/10.1093/oso/9780197532768.003.0024>. On taxation, see Michael E. Smith, ‘The Aztecs Paid Taxes, Not Tribute’, *Mexicon* 36, no. 1 (2014): 19–22, [www.academia.edu/6578625/ The Aztecs Paid Taxes Not Tribute 2014](http://www.academia.edu/6578625/The_Aztecs_Paid_Taxes_Not_Tribute_2014). For an overview of how advanced Mesoamerican society was,

- see Charles C. Mann, *1491: New Revelations of the Americas before Columbus* (Vintage, 2006). Smith and Sergheraert, 'The Aztec Empire', provide an overview of the empire.↵
5. Mann, *1491*, 124.↵
 6. Matthew Restall, *When Montezuma Met Cortés: The True Story of the Meeting That Changed History* (ECCO, 2018).↵
 7. On Motecuhzoma's hesitation and the paralysis of the Aztecs, see Smith and Sergheraert, 'The Aztec Empire', 289–90.↵
 8. Ross Hassig, *Mexico and the Spanish Conquest* (University of Oklahoma Press, 2006); Nichols and Collins, 'Episodes of the Feathered Serpent', 177–9. For good general summaries of the 'New Conquest History' literature which emphasize indigenous autonomy and rebellion, see Camilla Townsend, *Fifth Sun: A New History of the Aztecs* (Oxford University Press, 2019); Matthew Restall, *Seven Myths of the Spanish Conquest* (Oxford University Press, 2021).↵
 9. Pizarro was a second cousin, once removed, of Cortés.↵
 10. This might seem an unlikely statistic, but it is corroborated by multiple sources. Bear in mind that we are talking about a dozen or more relay runners, while the horses weren't being relayed and were not at full gallop since they were on mountainous and often very winding roads.↵
 11. Chris Given-Wilson, 'Bureaucracy Without Alphabetic Writing: Governing the Inca Empire, c.1438–1532', in *Empires and Bureaucracy in World History: From Late Antiquity to the Twentieth Century*, ed. Peter Crooks and Timothy H. Parsons (Cambridge University Press, 2016), 22,
<https://doi.org/10.1017/CBO9781316694312.005>.↵
 12. The Incan Empire was an absolute, authoritarian monarchy with the ruling Sapa Incans claiming direct divine lineage. Each local polity had a landscape deity known as a *huaca*. All local inhabitants believed they had some ancestral link to the *huaca*, and a local leader, known as a *kuraka*, claimed direct and privileged lineage to this deity. These kin-line relationships, known as *ayllus*, extended up to the Incan supreme ruler. This divine web of relations justified the *mita* taxes (contributions of labour) that the ruling Inca exacted from their

territories. However, such a nested hierarchy was new for many peoples and often fragile.↵

13. The population data is from Koch et al., 'Earth System Impacts of the European Arrival and Great Dying in the Americas after 1492'. For a useful overview of the Incan economy and society, see Gordon Francis McEwan, *The Incas: New Perspectives* (ABC-CLIO, 2006).↵
14. That resistance continued even after the death of Atahualpa. Atahualpa's brother organized a rebellion against the Spaniards and almost succeeded in annihilating them. Despite heavy losses, the Spanish prevailed, pushing the Incan rebels out of the Andes. The Incan renegades moved inland to the edge of the Amazon and established a new hidden capital: Vilcabamba. From there they fought a thirty-six-year guerilla war against the Spanish. For an overview of the fall of the Incan Empire, see Kim MacQuarrie, *The Last Days of the Incas* (Simon & Schuster, 2008).↵
15. Nichols and Collins, 'Episodes of the Feathered Serpent', 184. See also Laura E. Matthew and Michel R. Oudijk, eds., *Indian Conquistadors: Indigenous Allies in the Conquest of Mesoamerica* (University of Oklahoma Press, 2007).↵
16. Camilla Townsend, 'Burying the White Gods: New Perspectives on the Conquest of Mexico', *American Historical Review* 108, no. 3 (2003), <https://doi.org/10.1086/ahr/108.3.659>.↵
17. Robert J. DiNapoli et al., 'Approximate Bayesian Computation of Radiocarbon and Paleoenvironmental Record Shows Population Resilience on Rapa Nui (Easter Island)', *Nature Communications* 12, no. 1 (June 2021): 3939, <https://doi.org/10.1038/s41467-021-24252-z>; Carl Lipo, Robert J. DiNapoli, and Terry Hunt, 'Claims and Evidence in the Population History of Rapa Nui (Easter Island)', in *The Prehistory of Rapa Nui (Easter Island): Towards an Integrative Interdisciplinary Framework*, ed. Valenti Rull and Christopher Stephenson (Springer, 2022), 565–85; Terry L. Hunt and Carl P. Lipo, *The Statues That Walked: Unraveling the Mystery of Easter Island* (Free Press, 2011).↵
18. The latest evidence suggests that the deforestation came from humans, not introduced rats gnawing at palm tree seeds. See Andreas Mieth and Hans-Rudolf Bork, 'Humans, Climate or Introduced Rats – Which Is

- to Blame for the Woodland Destruction on Prehistoric Rapa Nui (Easter Island)?', *Journal of Archaeological Science* 37, no. 2 (2010): 417, <https://doi.org/10.1016/j.jas.2009.10.006>.↵
19. Carl P. Lipo et al., 'Weapons of War? Rapa Nui *Mata'a* Morphometric Analyses', *Antiquity* 90, no. 349 (2016): 172–87, <https://doi.org/10.15184/aqy.2015.189>. The ecocide narrative largely rests on high estimates of a population of around 17,000 prior to European contact. The most recent evidence thoroughly debunks this. Numerous different methods, including shortwave infrared (SWIR) satellite scanning, have shown that the rock garden agriculture used by the Rapa Nui could never have supported such numbers. Instead, the island would have had a maximum population of around 3–4,000. That's also the rough number reported by the Europeans who first contacted the island. See Dylan S. Davis et al., 'Island-Wide Characterization of Agricultural Production Challenges the Demographic Collapse Hypothesis for Rapa Nui (Easter Island)', *Science Advances* 10, no. 25 (2024): eado1459, <https://doi.org/10.1126/sciadv.ado1459>; DiNapoli et al., 'Approximate Bayesian Computation of Radiocarbon and Paleoenvironmental Record Shows Population Resilience on Rapa Nui (Easter Island)'. For the most recent DNA analysis which upends the ecocide theory, see J. Víctor Moreno-Mayar et al., 'Ancient Rapanui Genomes Reveal Resilience and Pre-European Contact With the Americas', *Nature* 633, no. 8029 (2024): 389–97, <https://doi.org/10.1038/s41586-024-07881-4>.↵
20. Nichols and Collins, 'Episodes of the Feathered Serpent', 185. I've discounted one outdated, outlier estimate of 22 per cent. There are some suggestions that the particular severity of the introduction of Old World diseases into the Americas could be due to some kind of genetic vulnerability, including lower genetic diversity among indigenous groups. Increased diversity tends to lead to resistance to disease, while the Americas tended to have lower diversity since only a small outgroup migrated across the Bering Land Bridge into the Americas. However, this remains speculative. See I. King Jordan, 'The Columbian Exchange as a Source of Adaptive Introgression in Human Populations', *Biology Direct* 11, no. 1 (2016): 6, <https://doi.org/10.1186/s13062-016-0121-x>. On Australia, see Peter

Dowling, *Fatal Contact: How Epidemics Nearly Wiped Out Australia's First Peoples* (Monash University Publishing, 2021). For Hawaii, Roy Porter, *The Greatest Benefit to Mankind: A Medical History of Humanity* (W. W. Norton, 1999), 466.↵

21. Nichols and Collins, 'Episodes of the Feathered Serpent', 179–84.↵
22. On Tasmania, the population of 7,000 was nearly wiped out by disease, mass killings, and the frequent kidnapping of women and children. The remaining 220 survivors were shipped away from their homeland to Flinders Island. The overcrowded and poorly ventilated conditions made outbreaks of sickness even worse. The mission was closed when only forty-six remained. It is what many historians consider to be an early case of genocide. Henry Reynolds, *An Indelible Stain? The Question of Genocide in Australia's History* (Viking, 2001), 1; Dowling, *Fatal Contact*.↵
23. Robbie Ethridge, 'The Rise and Fall of Mississippian Ancient Towns and Cities, 1000–1700', in *Oxford Research Encyclopedia of American History* (Oxford University Press, 2018), 15–16, <https://doi.org/10.1093/acrefore/9780199329175.013.349>; Ben Raffield, 'Broken Worlds: Towards an Archaeology of the Shatter Zone', *Journal of Archaeological Method and Theory* 28, no. 3 (2021): 871–910, <https://doi.org/10.1007/s10816-021-09520-y>.↵
24. Ethridge, 'The Rise and Fall of Mississippian Ancient Towns and Cities, 1000–1700', 14–15.↵
25. The Haudenosaunee were more complicated. They lost thousands due to introduced epidemics and used captured peoples to replace them – quite literally. During a ritual ceremony known as the requickening, they would have the captive adopt the persona and position of a deceased member.↵
26. Robbie Franklin Ethridge, 'Introduction', in *Mapping the Mississippian Shatter Zone: The Colonial Indian Slave Trade and Regional Instability in the American South*, ed. Sheri Marie Shuck-Hall (University of Nebraska Press, 2009), 8, 14, 30; Matthew H. Jennings, 'Violence in a Shattered World', *ibid.*, 280.↵
27. Jennings, 'Violence in a Shattered World', 284.↵

28. Ethridge, 'Introduction', 39; Jennings, 'Violence in a Shattered World', 281–6.↵
29. Ronald Robinson, 'Non-European Foundations of European Imperialism: Sketch for a Theory of Collaboration', *Studies in the Theory of Imperialism* 120 (1972); Ronald Robinson, 'The Excentric Idea of Imperialism, With or Without Empire', in *Imperialism*, ed. Peter H. Cain and Mark Harrison (Routledge, 2023), 334–58, <https://doi.org/10.4324/9781003101529>.↵
30. Nichols and Collins, 'Episodes of the Feathered Serpent', 184–5; James Lockhart, *The Nahuas After the Conquest: A Social and Cultural History of the Indians of Central Mexico, Sixteenth Through Eighteenth Centuries* (Stanford University Press, 1993).↵
31. Henry Kamen, *Spain's Road to Empire: The Making of a World Power, 1492–1763* (Penguin, 2003).↵
32. Those that were not colonized often took on the shape of the colonizers regardless. Japan first encountered Portugal in 1543. Thereafter Japan started to look increasingly like a European colonial empire. It adopted their tactics and even their way of defending colonization. For instance, when Japan invaded Taiwan in 1874, it did so under a justification of bringing civilization to a savage, backward indigenous population. It established a powerful colonial empire that was influenced by Europeans. It's a process that scholars call 'imperial mimesis': one empire replicating the features of another successful one. It is, once again, an example of Goliath evolution. It was far safer to become a colonizer than risk being colonized. Robert Tierney, 'Japanese Imperialism', in *The Encyclopedia of Postcolonial Studies*, ed. Sangeeta Ray and Henry Schwarz (Wiley, 2016), 1–7, <https://doi.org/10.1002/9781119076506.wbeps184>. Robert Eskildsen, 'Of Civilization and Savages: The Mimetic Imperialism of Japan's 1874 Expedition to Taiwan', *American Historical Review* 107, no. 2 (2002), <https://doi.org/10.1086/ahr/107.2.388>, explores how Japan used a 'civilizing' narrative inspired by the European empires to justify its colonies, including its military expedition to Taiwan. Graeber and Sahlins have instead referred to this as 'galactic mimesis', based on their renaming of empires as 'galactic polities'. See Graeber and Sahlins, *On Kings*, 13.↵

33. Angus Maddison, *Contours of the World Economy, 1–2030 AD* (Oxford University Press, 2007), table A.1, table A.5. Dylan Sullivan and Jason Hickel, ‘Capitalism and Extreme Poverty: A Global Analysis of Real Wages, Human Height, and Mortality Since the Long 16th Century’, *World Development* 161 (2023): 12, <https://doi.org/10.1016/j.worlddev.2022.106026>. The figure of 50 million is based on India’s death rate in the 1880s. However, this rate was already high. When compared to mortality in sixteenth- and seventeenth-century England, the estimated excess deaths under British rule triples to 165 million.↵
34. Scott, *Against the Grain*, 113–15.↵
35. At the individual level, a greater degree of involvement with farming is strongly associated with higher fertility, but there is a lot of variation across cultures. For instance, stronger dependence on market labour seems to reduce fertility. See Page et al., ‘Women’s Subsistence Strategies Predict Fertility across Cultures, but Context Matters’. Some studies (H. Jabran Zahid, Erick Robinson, and Robert L. Kelly, ‘Agriculture, Population Growth, and Statistical Analysis of the Radiocarbon Record’, *Proceedings of the National Academy of Sciences* 113, no. 4 (2016): 931–5, <https://doi.org/10.1073/pnas.1517650112>) have found that modern North American hunter-gatherers have population growth rates that surpass early European farmers. Yet even if their fertility and growth rates were higher, their long-term population size was still constrained by the environment. This explains why population growth rates from the archaeological record are usually orders of magnitude lower than those observed in modern hunter-gatherers. For a discussion, see Miikka Tallavaara and Erlend Kirkeng Jørgensen, ‘Why Are Population Growth Rate Estimates of Past and Present Hunter-Gatherers So Different?’, *Philosophical Transactions of the Royal Society B: Biological Sciences* 376, no. 1816 (2021): 20190708, <https://doi.org/10.1098/rstb.2019.0708>.↵
36. The estimates of hunter-gatherer densities come from David Pimentel and Marcia Pimentel, *Food, Energy, and Society*, 3rd edn (CRC Press, 2008), 45–6. I’ve then translated these from hectares into square kilometres.↵

37. Carlo Martuscelli, 'The Populist Right Wants You to Make More Babies. The Question Is How', *Politico*, 11 September 2023, www.politico.eu/article/eu-populist-right-want-you-make-more-babies-viktor-orban/. Any slowing of population growth is also usually met with dismay by the powerful. It's a sentiment perhaps best expressed by Elon Musk's hyperbolic (as well as wrong and self-serving) opinion that 'population collapse due to low birth rates is a much bigger risk to civilization than global warming'.↵
38. Frank Götmark and Malte Andersson, 'Human Fertility in Relation to Education, Economy, Religion, Contraception, and Family Planning Programs', *BMC Public Health* 20, no. 1 (2020): 265, <https://doi.org/10.1186/s12889-020-8331-7>. Naturally we need to be careful here as there are multiple confounding factors (although this study does try to control for these by examining five of the most powerful factors: education, religiosity, economic strength (GDP and GDP per capita), the prevalence of contraceptives, and family planning).↵
39. Geography played an additional, crucial role. Geographical barriers such as mountains and swamps, which tended to block the spread of Goliaths and favour the art of not being governed, also tended to block the transmission of culture and disease (Chira, Gray, and Botero, 'Geography Is Not Destiny'). Note that the analysis here is largely for cultural transmission, not disease, although the two are certainly going to overlap; both are primarily transmitted by humans, after all. Notably, the study did not find support for Jared Diamond's hypothesis that the axis of continents would significantly help or hinder the spread of culture and disease. Geography also helped with the spread of disease between Goliaths. For instance, the Black Death that ravaged medieval Europe (1346–53) came from Kyrgyzstan, on the border of China, and travelled down the Silk Road; its spread across Europe correlates with traders moving down navigable rivers. See Maria A. Spyrou et al., 'The Source of the Black Death in Fourteenth-Century Central Eurasia', *Nature* 606, no. 7915 (2022): 718–24, <https://doi.org/10.1038/s41586-022-04800-3>. See also Ricci P. H. Yue, Harry F. Lee, and Connor Y. H. Wu, 'Trade Routes and Plague Transmission in Pre-Industrial Europe', *Scientific Reports* 7, no. 1 (2017): 12973, <https://doi.org/10.1038/s41598-017-13481-2>, and

- ‘Navigable Rivers Facilitated the Spread and Recurrence of Plague in Pre-Industrial Europe’, *Scientific Reports* 6, no. 1 (2016): 34867, <https://doi.org/10.1038/srep34867>.↵
40. The plague also played a role, with the Mongols particularly hard hit owing to their situation at the junction of China and Europe.↵
 41. The Chin and other examples are recounted in Scott, *The Art of Not Being Governed*, chap. 6.↵
 42. They are what the anthropologist David Graeber would have called ‘shit jobs’. These are not to be confused with ‘bullshit jobs’ like investment banking, which are socially pointless and even harmful but often well reimbursed. David Graeber, *Bullshit Jobs* (Simon & Schuster, 2018).↵
 43. The four ingredients discussed in this section are a combination of ideas from the three great thinkers referenced above: James C. Scott, Jared Diamond, and David Graeber. Scott points to fertility as the key advantage agriculturalists had over foragers. Diamond provides a compelling case for disease and military technology as being key to the triumph of the West. Graeber’s idea of shit and bullshit jobs, while never applied to this topic, provides a conceptual backbone.↵
 44. For a summary, see Mark Koyama and Jared Rubin, *How the World Became Rich: The Historical Origins of Economic Growth* (Polity Press, 2022), chap. 7.↵
 45. David Stasavage, ‘Was Weber Right? The Role of Urban Autonomy in Europe’s Rise’, *American Political Science Review* 108, no. 2 (2014): 337–54; Koyama and Rubin, *How the World Became Rich*, chap. 7.↵
 46. A similar point is raised in the framework of Hartley, ‘State Crisis Theory’. In short, conditions which improve ‘sufficiency’, such as positive technological change, economic development, and more open institutions can help ensure crisis doesn’t escalate into collapse.↵
 47. David Eltis and Lawrence C. Jennings, ‘Trade Between Western Africa and the Atlantic World in the Pre-Colonial Era’, *American Historical Review* 93, no. 4 (1988): 944, <https://doi.org/10.2307/1863530>. On the link between slave demand and malaria, see Elena Esposito, ‘The Side Effects of Immunity: Malaria and African Slavery in the United States’, *American Economic Journal: Applied Economics* 14, no. 3

(2022): 290, 325–6, <https://doi.org/10.1257/app.20190372>. Particular African ethnicities who appeared to be more malaria-resistant also fetched higher prices in the Louisiana slave market.↵

48. Hochschild, *Bury the Chains*, 67.↵

49. Sullivan and Hickel, ‘Capitalism and Extreme Poverty’. On life expectancy, see E. A. Wrigley and Roger Schofield, *The Population History of England, 1541–1871: A Reconstruction* (Cambridge University Press, 1989). On heights, see Galofré-Vilà, Hinde, and Guntupalli, ‘Heights Across the Last 2,000 Years in England’, fig. 5.↵

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CHAPTER 15: COLLAPSE IN THE MODERN WORLD

1. The 84 per cent estimate comes from Hoffman, *Why Did Europe Conquer the World?*, 2. The US also had Cuba as a territory until 1902.↵
2. The Portuguese held on longest to their official colonial empire, which ended in 1999 when it handed Macau back to the People's Republic of China (making Portugal the first and last maritime colonial empire).↵
3. Daniel Immerwahr, 'A New History of World War II', *The Atlantic*, 4 April 2022, www.theatlantic.com/magazine/archive/2022/05/world-war-ii-empire-colonialism/629371/; Richard Overy, *Blood and Ruins: The Last Imperial War, 1931–1945* (Viking, 2022); Daniel Immerwahr, 'How the US Has Hidden Its Empire', *Guardian*, 15 February 2019, www.theguardian.com/news/2019/feb/15/the-us-hidden-empire-overseas-territories-united-states-guam-puerto-rico-american-samoa; Daniel Immerwahr, *How to Hide an Empire: A History of the Greater United States* (Farrar, Straus and Giroux, 2019).↵
4. Although the White House had been trying to assist France and the UK through finance. It had also already imposed an oil embargo on the Japanese.↵
5. Sharanya Deepak, 'A "Forgotten Holocaust" Is Missing from Indian Food Stories', *Gastro Obscura*, 17 September 2020, www.atlasobscura.com/articles/indian-food-writing; Michael Safi, 'Churchill's Policies Contributed to 1943 Bengal Famine – Study', *Guardian*, 29 March 2019, www.theguardian.com/world/2019/mar/29/winston-churchill-policies-contributed-to-1943-bengal-famine-study.↵
6. John Darwin, *Decolonization: A History of Failure*, Library of Congress, 13 July 2011, www.youtube.com/watch?v=Y7bRx5fEJbU, provides an eloquent overview of the process of decolonization and why the disorderly process could even be considered 'a failure'. Stuart Ward, 'Decolonization and Neocolonialism', in *The Oxford World History of Empire*, vol. II: *The History of Empires*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press,

- 2021), 1161–86,
<https://doi.org/10.1093/oso/9780197532768.003.0042>, provides another useful summary.↵
7. Mohammed Hussein and Mohammed Haddad, ‘Infographic: US Military Presence around the World’, *Al Jazeera*, 9 October 2021, www.aljazeera.com/news/2021/9/10/infographic-us-military-presence-around-the-world-interactive. Figures come from the SIPRI Military Expenditure Database (1949–2023), www.sipri.org/databases/milex. Note that when making the defence spending comparison on a purchasing power parity basis (i.e. in real terms for each country), the US defence budget is only about as large as the next two countries (China and Russia) combined. In contrast to the 750 US military bases, China has only one such base overseas: a naval base in Djibouti.↵
 8. Sidita Kushi and Monica Duffy Toft, ‘Introducing the Military Intervention Project: A New Dataset on US Military Interventions, 1776–2019’, *Journal of Conflict Resolution* 67, no. 4 (2023): 752, <https://doi.org/10.1177/00220027221117546>.↵
 9. John Hall, ‘The End of Empires’, in *The Oxford World History of Empire*, vol. I: *The Imperial Experience*, ed. Peter Fibiger Bang, C. A. Bayly, and Walter Scheidel (Oxford University Press, 2021), <https://doi.org/10.1093/oso/9780199772360.003.0017>.↵
 10. The quotes come from Ron Suskind, ‘Without a Doubt’, *New York Times*, 17 October 2004. I unfortunately had to spend far too much time observing how the US shapes the world. My PhD thesis focused on how to manage US participation in international environmental treaties. While US participation tends to be critical to the success of international treaties, it also has the unfortunate habit of rarely ratifying (legally joining) them. During my PhD I repeatedly warned of the dangers of watering down the Paris Climate Agreement to encourage the US to join it. Alas, I was unsuccessful, and the US got its way in the end. Despite the Paris Agreement being weakened, the US still withdrew from it under President Trump in 2020, and is set to do so again at the time of writing. We now have an international climate agreement that is universal and useless. See Luke Kemp, ‘A Systems Critique of the 2015 Paris Agreement on Climate’, in *Pathways to a Sustainable Economy*, ed. Moazzem Hossain, Robert

- Hales, and Tapan Sarker (Springer International, 2018), 25–41, https://doi.org/10.1007/978-3-319-67702-6_3; Luke Kemp, ‘Better Out Than In’, *Nature Climate Change* 7, no. 7 (2017): 458–60, <https://doi.org/10.1038/nclimate3309>; Luke Kemp, ‘US-Proofing the Paris Climate Agreement’, *Climate Policy* 17, no. 1 (2017): 86–101, <https://doi.org/10.1080/14693062.2016.1176007>.↵
11. The Catalans instead have been folded into Spain (owing to previous imperial conquest), the Scots into the UK (owing to previous imperial conquest), while the Kurds are dispersed across Turkey, Iraq, and Syria (owing to the victors of the First World War breaking their promise to establish a Kurdish state). Strangely, many will still reflexively hate any attempt by these nations to secede. Kumar, *Empires*, chap. 4 makes the point about nations and states often not overlapping.↵
 12. There is a copious literature on this, usually under the banner of ‘dependency theory’ or ‘world-systems theory’. The essential idea is that we now operate under a single economic system which has a ‘core’ composed of former colonial empires and their closest allies. This exploits a ‘periphery’ of formerly conquered and dominated countries, primarily in Africa, Latin America, and Southeast Asia. Countries can move between the core and periphery, usually first by becoming a ‘semi-periphery’ country. Prominent examples of such mobility include China and Singapore. The classical text on world-systems theory is Wallerstein, *World-Systems Analysis*. For an accessible overview of how debt and unfair trade have led to underdevelopment in poorer countries, see Jason Hickel, *The Divide: A Brief Guide to Global Inequality and Its Solutions* (Heinemann, 2017).↵
 13. Jason Hickel et al., ‘Imperialist Appropriation in the World Economy: Drain from the Global South through Unequal Exchange, 1990–2015’, *Global Environmental Change* 73 (2022): 102467, <https://doi.org/10.1016/j.gloenvcha.2022.102467>.↵
 14. Jason Hickel, Morena Hanbury Lemos, and Felix Barbour, ‘Unequal Exchange of Labour in the World Economy’, *Nature Communications* 15, no. 1 (2024): 6298, <https://doi.org/10.1038/s41467-024-49687-y>. Note that I’m using ‘unequal exchange’ a little differently from how it is commonly defined in the ecological economics literature. That’s

because unequal exchange is such a broad term it better captures all forms of exchange that are done on an unequal power basis, including tribute and tax. I'm terming the transfer of resources from low-income to high-income countries due to structural economic differences – such as debt, wage differences, and trade agreements – unequal trade rather than unequal exchange. That is because trade is the primary conduit for the extraction of resources, not tax or tribute.↵

15. For a broader discussion of determinants of economic growth across different countries, see Koyama and Rubin, *How the World Became Rich*. They highlight the key factors as geography, demography, culture, institutions, and colonization. As we'll cover later, the importance of colonialism leaving more exclusive economic institutions in some countries has also been highlighted in the Nobel Prize-winning work of Daron Acemoglu and James Robinson: see especially Daron Acemoglu, Simon Johnson, and James A. Robinson, 'The Colonial Origins of Comparative Development: An Empirical Investigation', *American Economic Review* 91, no. 5 (2001): 1369–401; Acemoglu and Robinson, *Why Nations Fail*.↵
16. The potential sole exception here is North Korea. However, even North Korea still has many of the key ingredients of capitalism: private markets, private companies, wage labour, and a goal of economic growth. While private property is technically banned, it still often occurs in practice and the government still permits (with oversight) the transfer and inheritance of property. North Korea is more capitalist than most think.↵
17. Katharina Pistor, *The Code of Capital: How the Law Creates Wealth and Inequality* (Princeton University Press, 2020), chapter 1. Milanović, *Capitalism, Alone*, makes the compelling case that all countries are now capitalist. There are different flavours of capitalism, such as authoritarian and liberal, but they all have the same basic economic structure. On surveillance capitalism, see Shoshana Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power* (PublicAffairs, 2019).↵
18. And the Grand Duchy of Hesse: a tiny country in central Germany ruled by a duke.↵
19. Although it was essentially pre-empting Prussia taking it by force.↵

20. Tanisha M. Fazal, 'State Death in the International System', *International Organization* 58, no. 2 (2004): 311–44, <https://doi.org/10.1017/S0020818304582048>; Tanisha M. Fazal, *State Death: The Politics and Geography of Conquest, Occupation, and Annexation* (Princeton University Press, 2007).↵
21. Brandon Valeriano and John Van Benthuisen, 'When States Die: Geographic and Territorial Pathways to State Death', *Third World Quarterly* 33, no. 7 (2012): 1165–89, <https://doi.org/10.1080/01436597.2012.691826>.↵
22. Matthias Maass, 'The International States Systems Since 1648 and Small States' "Systemic Resilience"', *International Studies Review* 10, no. 2 (2009): 31–51, <https://doi.org/10.1163/2667078X-01002002>.↵
23. That respect for borders has birthed an entire literature on just how brutal and despotic a regime needs to be before another country is morally justified to undertake a military intervention (known as the 'responsibility to protect', or R2P).↵
24. Despite this, some states have ended and lost their sovereignty since the end of the Second World War. Some of these were through vicious civil wars, such as the break-up of Yugoslavia. Others were far more benign and temporary, such as when Syria combined with Egypt in 1958 to form the short-lived United Arab Republic.↵
25. For an overview, see Robert I. Rotberg, 'Failed States, Collapsed States, Weak States: Causes and Indicators', in *State Failure and State Weakness in a Time of Terror*, ed. Robert I. Rotberg (World Peace Foundation, 2003), chap. 1, 25.↵
26. Daniel Lambach, Eva Johais, and Markus Bayer, 'Conceptualising State Collapse: An Institutional Approach', *Third World Quarterly* 36, no. 7 (2015): 1299–1315, <https://doi.org/10.1080/01436597.2015.1038338>.↵
27. Jan C. Jansen and Jürgen Osterhammel, *Decolonization: A Short History*, trans. Jeremiah Riemer (Princeton University Press, 2017), 177. One counter here is that 'there have been numerous failures before colonization, so it can't be the key cause in the modern world'. One of the key points of this book is that not only does collapse change over time, it also depends on how the wider system is

structured. We can't fully understand the Late Bronze Age collapse without looking at the interconnections between different cities and empires; similarly we can't fully contemplate modern state failure without looking at the history of colonization. Note that the implication here is not that the borders are the only problem: clearly there are numerous African and Middle Eastern countries with borders drawn up by European powers that are not weak or failed states. However, such borders and the history they represent are often a major problem.↵

28. Rollin F. Tusalem, 'The Colonial Foundations of State Fragility and Failure', *Polity* 48, no. 4 (2016): 445–95, <https://doi.org/10.1057/s41279-016-0006-4>.↵
29. Robert Fisk, *The Great War for Civilization: The Conquest of the Middle East* (Fourth Estate, 2006), xxv. Thanks to Nathaniel Cooke for suggesting this quote.↵
30. Acemoglu, Johnson, and Robinson, 'The Colonial Origins of Comparative Development' and *Why Nations Fail*.↵
31. Hudson, Bowen, and Nielsen, *The First Political Order*, 'Introduction'.↵
32. The idea of state failure and collapse became prominent after 9/11. US policymakers blamed Afghanistan for allowing Al-Qaeda operations within its borders. The underlying fear was that weak, fragile, or failed states would be a breeding ground for instability and terrorism. As the 2002 US National Security Strategy declared, 'America is now threatened less by conquering states than by failing ones.' There is now a veritable industry of indices that measure state fragility and failure, including by the Brookings Institution and the World Bank. For an overview of the indices, including their similarities, differences, and problems, see Ines A. Ferreira, 'Measuring State Fragility: A Review of the Theoretical Groundings of Existing Approaches', *Third World Quarterly* 38, no. 6 (2017): 1291–1309, <https://doi.org/10.1080/01436597.2016.1257907>.↵
33. Each of these has some obvious rebuttals. Plenty of fragile states seem to suffer from too many resources rather than too few. Syria had a population density of 124 people per sq. km prior to its civil war, while Italy today has almost double that with 200. Countries such as

- Australia and the US have lovable resources like coal and oil but are far richer and more stable than similarly endowed Venezuela. For useful critiques of the different hypotheses for why states fail, see Jonathan Di John, 'Conceptualising the Causes and Consequences of Failed States: A Critical Review of the Literature', Crisis States Research Centre paper, 2008, <https://gsdrc.org/document-library/conceptualising-the-causes-and-consequences-of-failed-states-a-critical-review-of-the-literature/>; Jonathan Di John, 'The Concept, Causes and Consequences of Failed States: A Critical Review of the Literature and Agenda for Research with Specific Reference to Sub-Saharan Africa', *European Journal of Development Research* 22, no. 1 (2010): 10–30, <https://doi.org/10.1057/ejdr.2009.44>.↵
34. World Bank, 'GDP (Current US\$) – Russian Federation', 2023, <https://data.worldbank.org/indicator/NY.GDP.MKTP.CD?locations=RU>.↵
 35. Our World in Data, 'Population, 10,000 BCE to 2021'. Published under the Creative Commons License Version 4.0.↵
 36. Justin McCurry, 'Story of Cities #24: How Hiroshima Rose from the Ashes of Nuclear Destruction', *Guardian*, 18 April 2016, www.theguardian.com/cities/2016/apr/18/story-of-cities-hiroshima-japan-nuclear-destruction.↵
 37. For definitions, see Brian Walker, David Salt, and Walter Reid, *Resilience Thinking: Sustaining Ecosystems and People in a Changing World* (Island Press, 2012); C. S. Holling, 'Resilience and Stability of Ecological Systems', *Annual Review of Ecology and Systematics* 4, no. 1 (1973): 1–23, <https://doi.org/10.1146/annurev.es.04.110173.000245>; Graeme S. Cumming and Garry D. Peterson, 'Unifying Research on Social-Ecological Resilience and Collapse', *Trends in Ecology & Evolution* 32, no. 9 (2017): 695–713, <https://doi.org/10.1016/j.tree.2017.06.014>; Carl Folke et al., 'Social-Ecological Resilience and Biosphere-Based Sustainability Science', *Ecology and Society* 21, no. 3 (2016), <https://doi.org/10.5751/ES-08748-210341>. They all tend to converge around this rough formulation. Academics often summarize resilience as 'three Rs': robustness to shocks (such as disease), recovery from damage, and resistance to fundamental change. The three Rs comes from R. Quentin

- Grafton et al., 'Realizing Resilience for Decision-Making', *Nature Sustainability* 2, no. 10 (2019): 907–13,
<https://doi.org/10.1038/s41893-019-0376-1>.↵
38. Nassim Nicholas Taleb, *Antifragile: Things That Gain from Disorder* (Random House, 2014).↵
39. Philip Riris et al., 'Frequent Disturbances Enhanced the Resilience of Past Human Populations', *Nature* 629, no. 8013 (2024): 837–42,
<https://doi.org/10.1038/s41586-024-07354-8>. This is technically a meta-analysis of sixteen studies rather than a single study.↵
40. For instance, the Yandruwandha people of Australia eat a plant called nardoo that is partially indigestible and mildly toxic. Despite this, nardoo is a central part of the Yandruwandha diet because over time, and through trial and error, they have refined a way of cooking it that reduces the thiaminase concentration (involving grinding and leaching the flour, cooking it, and then eating it with mussel shells). If you aren't taught these traditions, they are difficult to develop suddenly, even in life-or-death situations.↵
41. Henrich, *The Secret of Our Success*, chap. 3.↵
42. Yitzchak Jaffe, Ari Caramanica, and Max D. Price, 'Towards an Antifragility Framework in Past Human–Environment Dynamics', *Humanities and Social Sciences Communications* 10, no. 1 (2023): 915, <https://doi.org/10.1057/s41599-023-02413-3>.↵
43. Others suggest that bigger populations allowed for more innovations and cultural evolution, although this is highly disputed.↵
44. Taagepera, 'Expansion and Contraction Patterns of Large Polities'. For a fuller analysis and discussion, see Scheidel, 'The Scale of Empire'.↵
45. There was some selection for more fast-growing grains but no new source of lootable resources.↵
46. This was part of a host of other governance measures. In China, the problem of elite competition was partly outmanoeuvred by relying on eunuchs. The working assumption was that the castrated would not seize power or seduce the women of the emperor's harem. That often backfired, with scheming eunuchs even playing a role in the overthrow of the Han Dynasty. Others tried depending on slaves instead. Both the

Mughal Empire of India and the Ottoman Empire filled their militaries and bureaucracies with slave soldiers. Slaves were seen as despised outsiders who were not fully human. This was because they'd faced 'social death', having been ripped from their homes and lineage. Since slaves were both widely disliked and isolated, they were seen to be less of a threat, and less likely to jostle for the throne. Empires tried many different types of governance structures, but the most successful, like China's reliance on big bureaucracy and coins, tended to be selected for and emulated by competitors and later regimes. See Orlando Patterson, *Slavery and Social Death: A Comparative Study: With a New Preface* (Harvard University Press, 2018).[↵](#)

47. Cline, *After 1177*, 2nd edn; Kemp and Cline, 'Systemic Risk and Resilience in the Bronze Age'.[↵](#)
48. Statistics on literacy, height, and infant mortality all come from Our World in Data.[↵](#)
49. See *Technology Eats History: Techno Metabolism and Time in the Anthropocene*, Institute of the Humanities and Global Culture, 2022, www.youtube.com/watch?v=57yQRnfiemw.[↵](#)
50. Tim Hatton, 'Air Pollution in Victorian-Era Britain – Its Effects on Health Now Revealed', *The Conversation*, 14 November 2017, <https://theconversation.com/air-pollution-in-victorian-era-britain-its-effects-on-health-now-revealed-87208>.[↵](#)
51. Sullivan and Hickel, 'Capitalism and Extreme Poverty'; Simon Szreter, 'Economic Growth, Disruption, Deprivation, Disease, and Death: On the Importance of the Politics of Public Health for Development', *Population and Development Review* 23, no. 4 (1997): 693, <https://doi.org/10.2307/2137377>.[↵](#)
52. Munira Z. Gunja, Evan D. Gumas, and Reginald D. Williams II, 'U.S. Health Care from a Global Perspective, 2022: Accelerating Spending, Worsening Outcomes', Commonwealth Fund, 31 January 2023, www.commonwealthfund.org/publications/issue-briefs/2023/jan/us-health-care-global-perspective-2022.[↵](#)
53. Frackowiak et al., 'Subjective Happiness Among Polish and Hadza People'; Martin and Cooper, 'Subjective Well-Being in a Remote Culture: The Himba'.[↵](#)

54. Carol R. Ember, 'Hunter-Gatherers (Foragers)', Human Relations Area Files (Yale), 1 June 2020, <https://hraf.yale.edu/ehc/summaries/hunter-gatherers>.↵
55. This is to say nothing about their cost in souls and suffering. Many modern atrocities dwarf historical collapses in their human toll.↵
56. It was originally a project commissioned by the CIA called 'The State Failure Task Force'. Academic contributors included Jack Goldstone, a prominent theorist of collapse and revolutions, and the forefather of structural-demographic theory.↵
57. Naomi Klein, *The Shock Doctrine: The Rise of Disaster Capitalism* (Picador, 2008).↵
58. Tom Mueller, *Crisis of Conscience: Whistleblowing in an Age of Fraud* (Riverhead Books, 2019).↵
59. There is some emerging empirical evidence that this is the case. See Kemp et al., '12,000 Years of Accelerating Social System Death' (forthcoming, 2025). This is based on a workshop held at the Santa Fe Institute in December 2024.↵

CHAPTER 16: MORS EX MACHINA

1. This vignette is based roughly on the details provided in Pete Pattison, "'Like Slave and Master': DRC Miners Toil for 30p an Hour to Fuel Electric Cars', *Guardian*, 8 November 2021, www.theguardian.com/global-development/2021/nov/08/cobalt-drc-miners-toil-for-30p-an-hour-to-fuel-electric-cars.↵
2. Roy Maconachie, "'We Miners Die a Lot.'" Appalling Conditions and Poverty Wages: The Lives of Cobalt Miners in the DRC', *The Conversation*, 30 January 2024, <https://theconversation.com/we-miners-die-a-lot-appalling-conditions-and-poverty-wages-the-lives-of-cobalt-miners-in-the-drc-220986>.↵
3. Frank Swain, 'The Forgotten Mine That Built the Atomic Bomb', BBC News, 4 August 2020, www.bbc.com/future/article/20200803-the-forgotten-mine-that-built-the-atomic-bomb. Technically, the Trinity and Fat Man bombs were plutonium bombs made from reprocessed uranium, although both also included pure uranium in the tamper.↵

4. Joseph Kahn, 'China's Coal Miners Risk Danger for a Better Wage', *New York Times*, 28 January 2003, www.nytimes.com/2003/01/28/world/china-s-coal-miners-risk-danger-for-a-better-wage.html.↵
5. For the typical hazard list approach, see Rees, *Our Final Century*; Nick Bostrom and Milan M. Ćirković, eds., *Global Catastrophic Risks* (Oxford University Press, 2012); Ord, *The Precipice*; MacAskill, *What We Owe the Future*. This means there is little study even on how risks such as nuclear weapons and climate change intersect, even if there are obvious connections. See Sterre van Buuren, Thomas Fraise, and Benoît Pelopidas, 'Armes Nucléaires et Environnement:', *Raison Présente* 230, no. 2 (2024): 97–104, <https://doi.org/10.3917/rpre.230.0097>.↵
6. Technically, such a collapse could occur even at or below the level of a decimation threat, owing to the vulnerability of the Global Goliath today (see Chapter 18 in particular). I've restricted the present analysis to above the decimation level both for simplicity and analytical clarity.↵
7. This is a slightly modified definition of global catastrophic risk from Kemp et al., 'Climate Endgame', table 1. This diverges from the typical definitions of global catastrophic risk, which are usually unhelpfully broad and loose, such as 'serious damage to human wellbeing on a global scale' (Bostrom and Ćirković, *Global Catastrophic Risks*, 1). Such a definition leaves it unclear whether air pollution or even cigarettes are a global catastrophic risk. Similarly, I diverge from the usual definitions of existential risk as 'a loss of humanity's potential' (Nick Bostrom, 'Existential Risk Prevention as Global Priority', *Global Policy* 4, no. 1 (2013): 15–31, <https://doi.org/10.1111/1758-5899.12002>; Nick Bostrom, 'Existential Risks: Analyzing Human Extinction Scenarios and Related Hazards', *Journal of Evolution and Technology* 9 (2002), <https://nickbostrom.com/existential/risks.pdf>; Ord, *The Precipice*). 'Potential' is either left frustratingly vague or specified to mean a techno-utopian future in which humans have maximum control over nature, that is, a spacefaring society which harvests stars. Hence, staying on earth for the next billion years in a borderline utopia, but

failing to take to space, would be an existential threat in this definition. It's a little silly and not analytically useful. For a full discussion and critique of the usual definitions, see Carla Zoe Cremer and Luke Kemp, 'Democratising Risk: In Search of a Methodology to Study Existential Risk', *SSRN Electronic Journal*, 2021, 3–13, <https://ssrn.com/abstract=3995225>. The definition I use here is a practical and descriptive one. When people speak of and study existential risk they are referring to the loss of the existence of either our global society or our species, that is, a lasting global societal collapse or human extinction.↵

8. There is the obvious exception of natural hazards such as asteroids and volcanic eruptions. However, the risk of these also depends on vulnerabilities, as we'll see in Chapter 18.↵
9. Estimates of the number of deaths range from 110,000 to 210,000: Alex Wellerstein, 'Counting the Dead at Hiroshima and Nagasaki', *Bulletin of the Atomic Scientists*, 4 August 2020, <https://thebulletin.org/2020/08/counting-the-dead-at-hiroshima-and-nagasaki/>.↵
10. Zia Mian and Benoît Pelopidas, 'Producing Collapse: Nuclear Weapons as Preparation to End Civilization', in *How Worlds Collapse*, ed. Miguel Centeno et al. (Routledge, 2023), 25–36, <https://doi.org/10.4324/9781003331384-21>.↵
11. Wallerstein, 'Nukemap'. The scenario imagines the use of a Topol SS-25 missile (a fairly standard Russian ICBM) in the middle of downtown New York, with an airburst.↵
12. Alex Wallerstein et al., 'Plan A', Science and Global Security Centre, 2019, <https://sgs.princeton.edu/the-lab/plan-a>.↵
13. Lili Xia et al., 'Global Food Insecurity and Famine from Reduced Crop, Marine Fishery and Livestock Production Due to Climate Disruption from Nuclear War Soot Injection', *Nature Food* 3, no. 8 (2022): table 1, <https://doi.org/10.1038/s43016-022-00573-0>.↵
14. It might also be better, since that prediction of 5 billion is based on some pessimistic assumptions, especially the modelling assumption that there would be no trade in the wake of a nuclear cataclysm. Yet, as

we'll see, there are even more reasons to expect that this is an underestimate.↵

15. Jessica Moersdorf et al., 'The Fragile State of Industrial Agriculture: Estimating Crop Yield Reductions in a Global Catastrophic Infrastructure Loss Scenario', *Global Challenges* 8, no. 1 (2024): 2300206, <https://doi.org/10.1002/gch2.202300206>; Christopher Cooper and Benjamin K. Sovacool, 'Not Your Father's Y2K: Preparing the North American Power Grid for the Perfect Solar Storm', *Electricity Journal* 24, no. 4 (2011): 47–61, <https://doi.org/10.1016/j.tej.2011.04.005>. I came across the idea of Border Gateway Protocol preventing a recovery of the internet while participating in the 2024 Global Catastrophic Risk Horizon Scan.↵
16. Florian Ulrich Jehn, 'Anthropocene Under Dark Skies: The Compounding Effects of Nuclear Winter and Overstepped Planetary Boundaries', in *Intersections, Reinforcements, Cascades: Proceedings of the 2023 Stanford Existential Risks Conference* (Stanford Existential Risks Initiative, 2024), <https://doi.org/10.25740/ZB109MZ2513>.↵
17. On the statistical debunking of the long peace, see Braumoeller, *Only the Dead*; Aaron Clauset, 'Trends and Fluctuations in the Severity of Interstate Wars', *Science Advances* 4, no. 2 (2018): eaao3580, <https://doi.org/10.1126/sciadv.aao3580>; Pasquale Cirillo and Nassim Nicholas Taleb, 'On the Statistical Properties and Tail Risk of Violent Conflicts', *Physica A: Statistical Mechanics and Its Applications* 452 (2016): 29–45, <https://doi.org/10.1016/j.physa.2016.01.050>.↵
18. The number of civil wars has also shown little connection to nuclear weapons and is now at its highest point in the past three decades. See Bastian Herre, Lucas Rodés-Guirao, and Max Roser, 'War and Peace', Our World in Data, 2024, <https://ourworldindata.org/war-and-peace>.↵
19. Pavel Aksenov, 'Stanislav Petrov: The Man Who May Have Saved the World', BBC News, 26 September 2013, www.bbc.co.uk/news/world-europe-24280831.↵
20. Richard Ned Lebow and Benoît Pelopidas, 'Facing Nuclear War: Luck, Learning, and the Cuban Missile Crisis', in *The Oxford Handbook of History and International Relations*, ed. Mlada Bukovansky et al.

- (Oxford University Press, 2023), 705–20,
<https://doi.org/10.1093/oxfordhb/9780198873457.013.47>.↵
21. Personal communication. This is expanded on in Pelopidas, *Betting Blind*. Note that practices over the past three to five decades are still classified even if older ones have been revealed. Luck is a matter of disobedience, technical failures, and random decisions: all factors that lie outside the control apparatus of the military.↵
 22. Julian E. Barnes and David E. Sanger, ‘U.S. Considers Expanded Nuclear Arsenal, a Reversal of Decades of Cuts’, *New York Times*, 7 June 2024, www.nytimes.com/2024/06/07/us/politics/us-nuclear-russia-china.html.↵
 23. Kai Kupferschmidt, ‘How Canadian Researchers Reconstituted an Extinct Poxvirus for \$100,000 Using Mail-Order DNA’, *Science*, 6 July 2017, <https://doi.org/10.1126/science.aan7069>.↵
 24. As noted by Martin Rees in *Our Final Century*.↵
 25. World Health Organization, ‘Emerging Technologies and Dual-Use Concerns: A Horizon Scan for Global Public Health’, World Health Organization horizon scan, 2021,
<https://apps.who.int/iris/handle/10665/346862>.↵
 26. There is no definitive answer to how likely a global catastrophic bioterror attack is. My perspective is largely derived from hearing discussions between biosecurity and bioengineering experts while facilitating horizon scans on both bioengineering and dual-use research of concern, as well as in-depth discussions with colleagues who work on biosecurity (such as Lalitha Sundaram, Freya Jephcott, and Tom Hobson) and on existential terrorism (Mel Cowans).↵
 27. For a useful summary of different estimates, see Piers Millett and Andrew Snyder-Beattie, ‘Existential Risk and Cost-Effective Biosecurity’, *Health Security*, 15, no. 4 (2017): 375,
<https://doi.org/10.1089/hs.2017.0028>.↵
 28. Jansen et al., ‘Biological Warfare, Bioterrorism, and Biocrime’, *Clinical Microbiology and Infection*, 20, no. 6 (2014): 494–5,
<https://doi.org/10.1111/1469-0691.12699>. The authors conclude that given the low (but not negligible) probability strengthening public

health generally may be wise, especially as they will benefit both bioterror response and general health.[↵]

29. Kemp, ‘Agents of Doom’. Many of these points have been gathered from expert discussions while running horizon scans on bioengineering, biosecurity and dual-use research of concern (Kemp et al., ‘Bioengineering Horizon Scan 2020’; Kemp et al., ‘80 Questions for UK Biological Security’, *PLOS ONE* 16, no. 1 (2021): e0241190, <https://doi.org/10.1371/journal.pone.0241190>); World Health Organization, ‘Emerging Technologies and Dual-Use Concerns’.[↵]
30. Centre for AI Safety, ‘Statement on AI Risk: AI Experts and Public Figures Express Their Concern About AI Risk’, 2023, www.safe.ai/statement-on-ai-risk.[↵]
31. Melissa Heikkilä, ‘A Short History of AI, and What It Is (and Isn’t)’, *MIT Technology Review*, 16 July 2024, www.technologyreview.com/2024/07/16/1095001/a-short-history-of-ai-and-what-it-is-and-isnt/.[↵]
32. For one collection of 71 different definitions, see Shane Legg and Marcus Hutter, ‘A Collection of Definitions of Intelligence’, SocArXiv, 25 June 2007: 1–8, <https://doi.org/10.48550/arXiv.0706.3639>. Perhaps the most common textbook definition is any machine system capable of functioning ‘appropriately and with foresight in its environment’ (Nils J. Nilsson, *The Quest for Artificial Intelligence* (Cambridge University Press, 2009), 13.[↵]
33. The computer science researcher Kate Crawford once remarked that ‘AI is neither artificial nor intelligent’ (Crawford, *Atlas of AI* (Yale University Press, 2022)). That might be overstating the case a little, but she is pointing out a genuine problem, namely that AI as a term can easily obscure all the materials that go into building an algorithm as well as the real constraints on their performance.[↵]
34. The idea of automated cognition is based on personal communications with Carla Zoe Cremer.[↵]
35. This is also probably where the idea of the ‘Butlerian Jihad’ – a successful crusade against thinking machines – in Frank Herbert’s novel *Dune* comes from.[↵]

36. Irving John Good, 'Speculations Concerning the First Ultraintelligent Machine', in *Advances in Computers*, 6 (1966), 31–88, [https://dio.org.10.1016.50065-2458\(08\)60418.0](https://dio.org.10.1016.50065-2458(08)60418.0).↵
37. There is, of course, a wider set of potential AI risks, including the slow and gradual disempowerment of humanity.↵
38. Nick Bostrom, *Superintelligence: Paths, Dangers, Strategies* (Oxford University Press, 2016).↵
39. For an overview of the different surveys, and other forecasting attempts, see Max Roser, 'AI Timelines: What Do Experts in Artificial Intelligence Expect for the Future?', Our World in Data, 7 February 2023, <https://ourworldindata.org/ai-timelines>. Note that similar surveys turn up similar results. A 2019 survey inquired when AI systems will be able to complete 90 per cent of the economically relevant tasks that humans are currently paid to do: 81 per cent said it would happen within a century, while 1.4 per cent responded that it would never occur. A 2018 survey asked when AI systems could complete 99 per cent of the tasks humans are paid to do with equal or superior competence: 75 per cent said it would happen within the next 100 years, while 22 per cent said it would take longer than 2160. The superforecasters' estimate is from Ezra Karger et al., 'Forecasting Existential Risks: Evidence from a Long-Run Forecasting Tournament', *Forecasting Research Institute*, 2023. Domain experts in this study put the risk of an AI-induced extinction a little higher, at around 3 per cent.↵
40. Philip E. Tetlock, *Expert Political Judgment: How Good Is It? How Can We Know?* (Princeton University Press, 2006).↵
41. Most of these surveys were also funded by Open Philanthropy, an outfit that is deeply worried about AI and that may have a motivation to fund work that exaggerates its threat. This led one researcher to refer to these surveys as a 'well-funded panic campaign': Michael Nolan, 'Weighing the Prophecies of AI Doom', *IEEE Spectrum*, 25 January 2024, <https://spectrum.ieee.org/ai-existential-risk-survey>.↵
42. Vaclav Smil, *Growth: From Microorganisms to Megacities* (MIT Press, 2020).↵

43. Pablo Villalobos et al., ‘Will We Run Out of Data? Limits of LLM Scaling Based on Human-Generated Data’, SocArXiv, 26 October 2022, <https://doi.org/10.48550/arXiv.2211.04325>. The authors are on the more optimistic side that innovations such as synthetic data and efficiency improvements will be able to overcome data limits, although they too note the uncertainty.↵
44. Gartner, Inc., ‘Gartner Predicts Power Shortages Will Restrict 40% of AI Data Centers By 2027’, 12 November 2024, www.gartner.com/en/newsroom/press-releases/2024-11-12-gartner-predicts-power-shortages-will-restrict-40-percent-of-ai-data-centers-by-2027.↵
45. Allison Morrow, ‘AI Is Hitting a Wall Just as the Hype Around It Reaches the Stratosphere’, CNN, 19 November 2024, <https://edition.cnn.com/2024/11/19/business/ai-chatgpt-nvidia-nightcap/index.html>. Perhaps the largest study on whether scaling can continue is Jaime Sevilla et al., ‘Can AI Scaling Continue Through 2030?’ (Epoch AI, 2024), <https://epoch.ai/blog/can-ai-scaling-continue-through-2030>. They are optimistic that chip manufacturing, latency, data, and energy constraints can be overcome, at least until 2030. Although this is an uncertain and underexplored area, especially for longer timeframes.↵
46. There are other issues. Large language models (LLMs) are really large text models. Language is about conveying meaning: if I wish you well, I’m often trying to convey that I genuinely hope you have a positive experience. If Chat GPT wishes you well, it gives you a statistically reasonable response based on its understanding of grammar and vocabulary. It’s not trying to elicit any particular reaction in you. The point on large text models comes from Ted Chiang (personal communication).↵
47. For the case against current large text models having emergent properties, see Rylan Schaeffer, Brando Miranda, and Sanmi Koyejo, ‘Are Emergent Abilities of Large Language Models a Mirage?’, SocArXiv, 22 May 2023, <https://arxiv.org/abs/2304.15004>; Sheng Lu et al., ‘Are Emergent Abilities in Large Language Models Just In-Context Learning?’, SocArXiv, 15 July 2024, <https://arxiv.org/abs/2309.01809>. For an overview of how experts

disagree on what cognitive abilities current deep-learning algorithms are displaying, see Carla Zoe Cremer, ‘Deep Limitations? Examining Expert Disagreement Over Deep Learning’, *Progress in Artificial Intelligence* 10, no. 4 (2021): 449–64, <https://doi.org/10.1007/s13748-021-00239-1>.↵

48. Richard Rhodes, *The Making of the Atomic Bomb* (Simon & Schuster, 1986).↵
49. Niamh Rowe, “‘It’s Destroyed Me Completely’: Kenyan Moderators Decry Toll of Training of AI Models’, *Guardian*, 2 August 2023, www.theguardian.com/technology/2023/aug/02/ai-chatbot-training-human-toll-content-moderator-meta-openai.↵
50. James Muldoon, Mark Graham, and Callum Cant, *Feeding the Machine: The Hidden Human Labour Powering AI* (Canongate, 2024); Ian Tucker, ‘James Muldoon, Mark Graham and Callum Cant: “AI Feeds Off the Work of Human Beings”’, *Guardian*, 6 July 2024, www.theguardian.com/technology/article/2024/jul/06/james-muldoon-mark-graham-callum-cant-ai-artificial-intelligence-human-work-exploitation-fairwork-feeding-machine.↵
51. In some cases, as with social media, one could argue that we are giving away our data in exchange for free access to services. That’s a deceptive argument since most people aren’t aware of how their data is used and have few other options or alternatives but to accept long and dry data-privacy terms (you could say that they have few exit options). One study calculated that it would take the average US citizen seventy-six full-time workdays to read the terms and conditions of privacy policies they agree to every year. Moreover, some initiatives, such as large language models, do rely on just direct data theft from sites across the internet, such as Reddit. OpenAI isn’t offering a service in exchange for data; it’s collecting data without permission to offer a commercialized algorithm.↵
52. On the case for compensating people for the production of data, and how this could be practically introduced in the near-term, read Imand Arrieta-Ibarra et al., ‘Should We Treat Data as Labor? Moving Beyond “Free”’, in *AEA Papers and Proceedings*, vol. 108 (2018), 38–42.↵
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www.hks.harvard.edu/sites/default/files/2024-05/Erica%20Chenoweth_2020-005.pdf.↵
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CHAPTER 17: GAIA VS GOLIATH

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Massive Gap between Rhetoric and Reality, Countries Draft New Climate Commitments (United Nations Environment Programme, 2024), <https://doi.org/10.59117/20.500.11822/46404>. Estimates that focus on either short- or long-term pledges tend to be more optimistic. Climate Action Tracker suggests that existing pledges and targets will limit warming to 2.1°C while existing policies put us on track for 2.5–2.9°C (Climate Action Tracker, ‘Temperatures’, 2024, <https://climateactiontracker.org/global/temperatures/>). Similarly, a recent large-scale analysis of climate pledges and policies found that long-term pledges might limit warming to below 2°C, while existing actions would see the planet warm well above 2°C (Dirk-Jan van de Ven et al., ‘A Multimodel Analysis of Post-Glasgow Climate Targets and Feasibility Challenges’, *Nature Climate Change* 13, no. 6 (2023): 570–78, <https://doi.org/10.1038/s41558-023-01661-0>). Existing actions are a far more reliable measurement than unenforceable, aspirational pledges, especially since delaying climate action is the single biggest factor that increases the costs of reducing emissions (Luke Kemp and Frank Jotzo, ‘Delaying Climate Action Would Be Costly for Australia and the World’, WWF-Australia National Office, 2015, www.academia.edu/23059430/Delaying_Climate_Action_Would_be_Costly_for_Australia_and_the_World). In other words, the longer actions remain insufficient the less likely we are to achieve ambitious long-term political promises. Moreover, all the issues of uncertainty mentioned later in this section apply to such forecasts.↵

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relationship is complicated and climate is not the strongest factor in predicting conflict. However, all these studies are, of course, just looking at recent decades. We don't know how climate change will influence conflict at 3°C of warming. It's reasonable to assume that, as impacts become more severe, climate change will become a stronger driver of combat.↵

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CHAPTER 18: THE DEATH-STAR SYNDROME

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CHAPTER 19: THE RUNGLESS LADDER

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CHAPTER 20: THE ROOTS OF OUR ENDGAME

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30. Garrett Hardin, 'The Tragedy of the Commons', *Science* 162, no. 3859 (1968): 1243–8, <https://doi.org/10.1126/science.162.3859.1243>.↵
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35. David Runciman, *The Handover: How We Gave Control of Our Lives to Corporations, States and AIs* (Liveright, 2023).↵
36. It was originally used for the legal territories of Rome (*status imperii*, or *status ecclesiasticus* for the papal states). It seems to have drifted since then from describing the legal standing of a territory to a territory's independence and self-governance.↵
37. The current Official Development Assistance aim is for rich countries to provide 0.7 per cent of their Gross National Income (GNI) as aid. The US has not agreed to the targets and in 2023 the US provided just 0.24 per cent of its GNI as aid.↵

38. Ironically, Machiavelli appears to have been intentionally giving bad advice. The book was originally intended for the Medici family, who killed Machiavelli's father. Machiavelli himself was a staunch republican, and *The Prince* diverges dramatically from his other work, such as *The Discourses*. There is a strong case to be made that Machiavelli was attempting to give the Medicis bad advice so they would anger the populace and start a rebellion. See Mary G. Dietz, 'Trapping the Prince: Machiavelli and the Politics of Deception', *American Political Science Review* 80, no. 3 (1986): 777–99, <https://doi.org/10.2307/1960538>.↵
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40. Paul Deutchman and Jessica Sullivan, 'The Dark Triad and Framing Effects Predict Selfish Behavior in a One-Shot Prisoner's Dilemma', *PLOS ONE* 13, no. 9 (2018): e0203891, <https://doi.org/10.1371/journal.pone.0203891>; Marta Malesza, 'The Effects of the Dark Triad Traits in Prisoner's Dilemma Game', *Current Psychology* 39, no. 3 (2020): 1055–62, <https://doi.org/10.1007/s12144-018-9823-9>.↵
41. Pieter D. Wezeman et al., 'Trends in International Arms Transfers, 2023', Stockholm International Peace Research Institute, 2024, www.sipri.org/publications/2024/sipri-fact-sheets/trends-international-arms-transfers-2023.↵
42. Others call this a 'multipolar trap' or 'Moloch trap'. Some have also pointed towards 'Darwinian traps': that existential risk and collective action problems are problems of game theory and human nature

(which, as covered, isn't the case). Other scholars have suggested that the current world is facing a set of fourteen 'Anthropocene traps', such as an over-pursuit of growth and short-termism (all fourteen traps fall into one of the three categories of Goliath trap I've outlined). The Goliath-traps framework suggested here is an improvement on these previous suggestions. It ties the trap dynamic back to specific, well-documented human behavioural patterns and can help explain why game theory seems to work in some contexts (and with some individuals) but not others. See Peter Søgaard Jørgensen et al., 'Evolution of the Polycrisis: Anthropocene Traps that Challenge Global Sustainability', *Philosophical Transactions of the Royal Society B: Biological Sciences* 379, no. 1893 (2024): 20220261, <https://doi.org/10.1098/rstb.2022.0261>; Rönn, *The Darwinian Trap*; Scott Alexander, 'Meditations on Moloch', *Slate Star Codex* (blog), 30 July 2014, <https://slatestarcodex.com/2014/07/30/meditations-on-moloch/>.↵

43. As you might have spotted, this is a play on the mythical Norse Tree of Life, 'Yggdrasil'. Feel free to call this 'Yggdrakill', the tree of death.↵

CHAPTER 21: THE FATES OF GOLIATH

1. All these numbers refer to irrecoverable collapse or a collapse leading to an extinction-level event. The figures quoted are from David Wallace-Wells, 'Jared Diamond: There's a 49 Percent Chance the World As We Know It Will End by 2050', *New York Magazine*, 10 May 2019, <https://nymag.com/intelligencer/2019/05/jared-diamond-on-his-new-book-upheaval.html>; Rees, *Our Final Century*; Bostrom, 'Existential Risks'; Anders Sandberg and Nick Bostrom, 'Global Catastrophic Risks Survey', FHI Technical Report #2008-1, Future of Humanity Institute, Oxford University, 2008, www.fhi.ox.ac.uk/reports/2008-1.pdf; Ord, *The Precipice*.↵
2. As superforecasting pioneer Philip Tetlock said, 'There is no evidence that geo-political or economic forecasters can predict anything ten years out.' Tetlock and Gardner, *Superforecasting*, 243. On this, and a

broader set of problems with forecasting, see Kemp, ‘Foreseeing Extreme Technological Risk’. It’s always possible to point to a few potential exceptions, even when it comes to predicting social crisis. One example is Peter Turchin, the pioneer of structural-demographic theory, who declared in a letter to *Nature* in 2010 that the 2020s would be a period of heightened political violence and instability in the US (Turchin, ‘Political Instability May Be a Contributor in the Coming Decade’). He’s right so far. Despite this success, it doesn’t mean we can reliably forecast the long-term future. His wasn’t a clear, quantitative forecast, and we’d need multiple examples to know that Turchin is consistently accurate, especially since much of the violence in the early 2020s was a reaction to the pandemic and lockdowns, something Turchin didn’t foresee. Generally speaking, however, we have no effective way to make detailed bets about future events beyond a year or two into the future. Turchin, to his credit, has made a more precise forecast of whether his theory is right: we should expect more than 100 political instability events (such as riots) in the period 2020–25 and fatalities of more than 5 per million citizens (Peter Turchin, ‘A Quantitative Prediction for Political Violence in the 2020s’, 21 January 2017, <https://peterturchin.com/quantitative-prediction-political-violence-2020s/>). Whether his bet proves to be right remains unknown.↵

3. Kemp, ‘Foreseeing Extreme Technological Risk’. We also currently have no evidence that superforecasters work for very low-probability, high-impact events, like 9/11 or the Russian Revolution.↵
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from the World Inequality Database, ‘Top 10% National Income Share, 2025’,

[https://wid.world/world/#\\$ptinc_p9op100_z/US;FR;DE;CN;GB/last/ek/p/yearly/s/false/24.98150000000004/80/curve/false/country](https://wid.world/world/#$ptinc_p9op100_z/US;FR;DE;CN;GB/last/ek/p/yearly/s/false/24.98150000000004/80/curve/false/country).

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19. Geoffrey West, *Scale: The Universal Laws of Life and Death in Organisms, Cities and Companies* (Weidenfeld & Nicolson, 2017), 563–7. Figure 49 is adapted from Martin Hilbert, ‘Digital Technology and Social Change: The Digital Transformation of Society from a Historical Perspective’, *Dialogues in Clinical Neuroscience* 22, no. 2 (2020): fig. 1, <https://doi.org/10.31887/DCNS.2020.22.2/mhilbert>. It is technically a schematic of Schumpeterian waves of economic development. The figures aren’t exact, but are correct in terms of orders of magnitude.↵
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 29. See Kemp, ‘Agents of Doom’.↵
 30. Nick Bostrom, ‘The Vulnerable World Hypothesis’, *Global Policy* 10, no. 4 (2019): 455–76, <https://doi.org/10.1111/1758-5899.12718>.↵
 31. There is an old idea of ‘Amara’s Law’ in foresight which suggests that we tend to overestimate the short-term effects of technology and underestimate its long-term effects. While not an iron law, it is a useful heuristic in this case.↵
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CHAPTER 22: AFTER THE FALL

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compelling, there are good reasons to be sceptical of this. Namely that it's a 10–20-hour exploration and was published on a blog. ↵

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19. For informative and evocative overviews, see Peter D. Ward, *Under a Green Sky: Global Warming, the Mass Extinctions of the Past, and What They Can Tell Us About Our Future* (HarperCollins, 2008); Lynas, *Our Final Warning*, chap. 6.↵
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www.theguardian.com/news/2018/feb/15/why-silicon-valley-billionaires-are-prepping-for-the-apocalypse-in-new-zealand.↵

22. Maya Yang, 'Bankman-Fried Planned to Buy Nauru and Build Apocalypse Bunker – Lawsuit', *Guardian*, 21 July 2023, www.theguardian.com/business/2023/jul/21/sam-bankman-fried-nauru-bunker-ftx. Effective Altruism is a movement that seeks to maximize the impact of charitable donations and careers through hard-nosed mathematics.↵
23. Douglas Rushkoff, *Survival of the Richest: Escape Fantasies of the Tech Billionaires* (W. W. Norton, 2022).↵
24. Chris Begley, *The Next Apocalypse: The Art and Science of Survival* (Basic Books, 2021).↵
25. Moersdorf et al., 'The Fragile State of Industrial Agriculture', 1, 7.↵
26. They might even benefit. The wealth of banks, living largely as digital numbers in an account rather than as the hard reality of gold and silver, would disappear alongside the power system. African nations would suddenly have an extra \$163 billion they wouldn't need to repay in debt.↵
27. Technically, Phoenician is an 'abjad', a system of writing in which only the consonants are represented. The Greeks modified it by adding symbols for the vowels, thus making it a complete alphabet.↵
28. Josiah Ober, *The Rise and Fall of Classical Greece* (Princeton University Press, 2015).↵
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30. There is an old cliché that crisis creates more draconian government. The Bolshevik Revolution begat the Gulag of the Soviet Union. Mao's communist revolution led to the mass deaths of the Great Leap Forward. Yet these are completely different from a global collapse. Indeed, none were collapses. Each one relied on highly centralized, bureaucratic, and effective power hierarchies. And we've encountered examples that went the other way: Somalia, post-Mycenaean Greece, and Egypt during the post-imperial epochs. ↵

31. Cline, *After 1177*, 216. Kemp and Cline, ‘Systemic Risk and Resilience in the Bronze Age’, 221. [↵](#)

CHAPTER 23: *FIAT JUSTITIA, NE PEREAT MUNDUS*

1. Macron complained that ‘you can’t say that just because 150 citizens wrote something, it’s the Bible or the Koran’. Well, nor is the legislation produced by less than a thousand of France’s richest and most powerful in the National Assembly. [↵](#)
2. Several others and I have suggested different ways of using open democracy to mitigate catastrophic risk. See Kemp et al., ‘Climate Endgame’; C. Z. Cremer and J. Whittlestone, ‘Canaries in Technology Mines: Warning Signs of Transformative Progress in AI’, Apollo – University of Cambridge Repository, 24 September 2020, <https://doi.org/10.17863/CAM.57790>; Cremer and Kemp, ‘Democratising Risk’. [↵](#)
3. The best-known usage was as the motto of the Holy Roman Emperor Ferdinand I (r. 1556–64). Immanuel Kant later used it as a retort to his utilitarian critics, although his translation was ‘Let justice reign even if all the rascals in the world should perish from it.’ [↵](#)
4. Bostrom, ‘The Vulnerable World Hypothesis’. [↵](#)
5. See Chapter 7 for a summary. [↵](#)
6. Roman Krznaric, *The Good Ancestor: A Radical Prescription for Long-Term Thinking* (Experiment, 2020). [↵](#)
7. Nassim Nicholas Taleb, *Fooled by Randomness*, 2nd edn (Random House, 2016). [↵](#)
8. Daniel Kahneman, *Thinking, Fast and Slow* (Farrar, Straus and Giroux, 2013). [↵](#)
9. Taleb, *Fooled by Randomness*. [↵](#)
10. George Eliot, *Middlemarch*. Thanks to Itzxul Moreno for introducing me to this beautiful quote. [↵](#)

11. While tragedy is typically invoked to describe history as well as nuclear arms races, it is rarely an apt framing and hides political responsibility. For a full discussion in the realm of nuclear weapons, see Benoît Pelopidas and Neil C. Renic, 'The Tragedy Trap: On the Tragicized Politics of Nuclear Weapons and Armed Drones and the Making of Unaccountability', *Ethics & International Affairs* 38, no. 2 (Summer 2024): 209–31, <https://doi.org/10.1017/S0892679424000145>.↵

EPILOGUE: SLAYING GOLIATH

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2. Karn Vohra et al., 'Global Mortality from Outdoor Fine Particle Pollution Generated by Fossil Fuel Combustion: Results from GEOS-Chem', *Environmental Research* 195 (2021): 110754, <https://doi.org/10.1016/j.envres.2021.110754>.↵
3. Simon Black et al., 'IMF Fossil Fuel Subsidies Data: 2023 Update', IMF Working Paper, 2023, www.imf.org/en/Publications/WP/Issues/2023/08/22/IMF-Fossil-Fuel-Subsidies-Data-2023-Update-537281.↵
4. A trillion equals one thousand billion. UNESCO estimates that providing sanitation and safe drinking water to all low- and middle-income countries would cost around US \$114 billion. The International Institute for Sustainable Development calculated that ending hunger would cost an additional US \$11 billion per year between 2016 and 2030 (for a total of \$176 billion). The development economist Jeffrey Sachs has calculated that ending extreme poverty would cost around \$175 billion. Together this is around \$465 billion. Even if these are underestimates and it costs three times as much, the revenue from fossil-fuel subsidies would still cover it, with over half a trillion to spare. See Jeffrey Sachs, *The End of Poverty: Economic Possibilities for Our Time*, new edn (Penguin, 2015); David Laborde and Livia Bizikova, 'Ending Hunger: What Would It Cost?', International Institute for Sustainable Development and the International Food

- Policy Research Institute, 2016, 3, www.iisd.org/system/files/publications/ending-hunger-what-would-it-cost.pdf; UNESCO, ‘Valuing Water Supply, Sanitation Services’, UN World Water Development Report, 2021, www.unesco.org/reports/wwdr/2021/en/valuing-water-supply-sanitation-services.↵
5. Intergovernmental Panel on Climate Change, ‘Working Group III Technical Summary’, in *Climate Change 2022 – Mitigation of Climate Change* (Cambridge University Press, 2023), 51–148, <https://doi.org/10.1017/9781009157926.002>. ↵
 6. One calculation for the US suggests that the health costs of coal are 0.8–5.6 times more than the value it adds to the economy. See Nicholas Z. Muller, Robert Mendelsohn, and William Nordhaus, ‘Environmental Accounting for Pollution in the United States Economy’, *American Economic Review* 101, no. 5 (2011): 1649–75, <https://doi.org/10.1257/aer.101.5.1649>. That’s just the health costs, and those would be far higher in many other countries. For a wider discussion of health co-benefits, read Kemp, ‘Ecological Breakdown and Human Extinction’, 162–5.↵
 7. For a fuller exploration of potential options, see Jason Hickel et al., ‘Degrowth Can Work – Here’s How Science Can Help’, *Nature* 612, no. 7940 (2022): 400–403, <https://doi.org/10.1038/d41586-022-04412-x>.↵
 8. Jason Hickel and Dylan Sullivan, ‘How Much Growth Is Required to Achieve Good Lives for All? Insights from Needs-Based Analysis’, *World Development Perspectives* 35 (2024): 100612, <https://doi.org/10.1016/j.wdp.2024.100612>.↵
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 10. Arrieta-Ibarra et al., ‘Should We Treat Data as Labor? Moving beyond “Free”’.↵
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- <https://doi.org/10.48550/arXiv.2402.08797>.↵
12. On company activism, see Haydn Belfield, ‘Activism by the AI Community: Analysing Recent Achievements and Future Prospects’, SocArXiv, 17 January 2020, <https://doi.org/10.48550/arXiv.2001.06528>.↵
 13. This list of interventions is based on advice from Matthijs Maas.↵
 14. Moritz Kütt and Zia Mian, ‘Setting the Deadline for Nuclear Weapon Removal from Host States under the Treaty on the Prohibition of Nuclear Weapons’, *Journal for Peace and Nuclear Disarmament* 5, no. 1 (2022): 148–61, <https://doi.org/10.1080/25751654.2022.2046405>.↵
 15. Tytti Erästö, ‘Revisiting “Minimal Nuclear Deterrence”: Laying the Ground for Multilateral Nuclear Disarmament’, Stockholm International Peace Research Institute (SIPRI), 2022, 5, www.sipri.org/sites/default/files/2022-06/sipriinsight2206_minimal_nuclear_deterrence_1.pdf. This figure of 300 is the lowest number covered in this review of minimal nuclear deterrence estimates.↵
 16. Jonas Jägermeyr et al., ‘A Regional Nuclear Conflict Would Compromise Global Food Security’, *Proceedings of the National Academy of Sciences* 117, no. 13 (2020): 7071, <https://doi.org/10.1073/pnas.1919049117>.↵
 17. For instance, in 1939 the US produced 2,141 planes. In 1944, three years after entering the Second World War, it was producing almost 100,000 per year and had completely transformed its economy and society, including conscripting over 10 million men and introducing a rationing system.↵
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 19. Both *Open Democracy* by Hélène Landemore and *Plurality* by E. Glen Weyl and Audrey Tang outline multiple ways in which old forms of governance and new technology can be combined to create bigger, more inclusive democracies than ever seen before. See Hélène

- Landemore, *Open Democracy: Reinventing Popular Rule for the Twenty-First Century* (Princeton University Press, 2020); E. Glen Weyl and Audrey Tang, *Plurality: The Future of Collaborative Technology and Democracy* (RadicalxChange, 2024).↵
20. For an overview, see Bokyoung Shin et al., ‘A Systematic Analysis of Digital Tools for Citizen Participation’, *Government Information Quarterly* 41, no. 3 (2024): 101954, <https://doi.org/10.1016/j.giq.2024.101954>. Notably, most existing tools are owned by big tech. That won’t be a sustainable situation if we want genuine democracy.↵
21. Emmanuel Saez and Gabriel Zucman, *The Triumph of Injustice: How the Rich Dodge Taxes and How to Make Them Pay* (W. W. Norton, 2019).↵
22. For a full suite of options, see Hickel, *The Divide*, chap. 8.↵
23. Matthew Connelly, *The Declassification Engine: What History Reveals About America’s Top Secrets* (Pantheon Books, 2023).↵
24. International Center for Not-for-Profit Law, ‘US Protest Law Tracker’, 2024, www.icnl.org/usprotestlawtracker/.↵
25. Cremer and Whittlestone, ‘Canaries in Technology Mines’.↵
26. A similar proposal is covered (with more depth) in Kemp et al., ‘Climate Endgame’, 2, 3, and 7.↵
27. Colleagues and I have referred to this as a financial oath of Maimonides (originally the traditional oath for physicians and pharmacists that is more expansive in its requirements, opening with ‘The eternal providence has appointed me to watch over the life and health of thy creatures’). For further details of how and why institutional investors should use their portfolios to reduce catastrophic risk, read Luke Kemp et al., ‘Financing Our Final Hour’, in *An Anthology of Global Risk*, ed. S. J. Beard and T. Hobson (Open Book Publishers, 2024), 682, <https://doi.org/10.11647/obp.0360.23>.↵

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Footnote

16: Mors ex Machina

[fn1](#) The miner is unlikely to possess wealth, but their annual income of \$1,460 (assuming a daily wage of \$4) is a millionth of Sam Altman's estimated wealth of \$1.5 billion.

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